CHAPTER 16
MISSION EQUIPMENT

CHAPTER OVERVIEW

SECTION 16-1

EQUIPMENT DESCRIPTION AND DATA

SECTION OVERVIEW

16-1-1 SIX MAN LITTER KIT DESCRIPTION AND DATA.

16-1-1.1 System Description.

The Six Man Litter Kit provides space for six ambulatory patients in the cabin section of the helicopter. The kit consisting of two litter support poles, a rear cabin ceiling support, six litters, and support straps and webbing. The litter kit is mounted along the longitudinal axis of the helicopter. The litter support poles and straps attach to studs and D-rings, respectively, on the floor and ceiling. The central support units for the litters are two litter support poles and connecting drag tubes. The support poles are mounted on studs in the cabin floor and ceiling.

Each litter pole holds the upper, the mid-level, and the lower litters in a support at the inboard pole of the litter. Outboard litter poles are vertically supported by straps connected by safety hooks and D-rings to each other and to the cabin ceiling. The tiedown assembly on the cabin floor attaches to the lowest of the outboard strap D-rings to complete the floor-to-ceiling support strap configuration.

Cross straps from each of the front pole and rear pole litter supports pass beneath each litter. The ends opposite the litter supports wrap around the outboard litter support feet. These ends are then clipped to D-rings on the floor-to-ceiling vertical support straps. Patient restraint belts are anchored to the front and rear litter pole supports and to each of the outboard litter support straps. Litters are loaded with the patient head first and forward in the cabin. The upper litter on either side of the litter support poles is loaded before the lower litters.

16-1-1.2 Cabin Installation.

The Six Man Litter Kit is loaded in the cabin by placing the inboard litter pole into the litter pole supports and holding the outboard litter pole while the litter support straps and cross straps are hooked to the supporting D-rings. To complete the loading sequence, the vertical support straps for the upper, the mid-level, and the lower litters are connected to each other and to the cabin ceiling with safety hooks and D-rings, while the tiedown assembly on the cabin floor is attached to the lowest outboard D-ring to complete the floor-to-ceiling strap support. When not in use, the litters can be folded inboard, with all litter poles laid against the front and rear litter pole supports and secured to the supports with restraint straps for stowage.

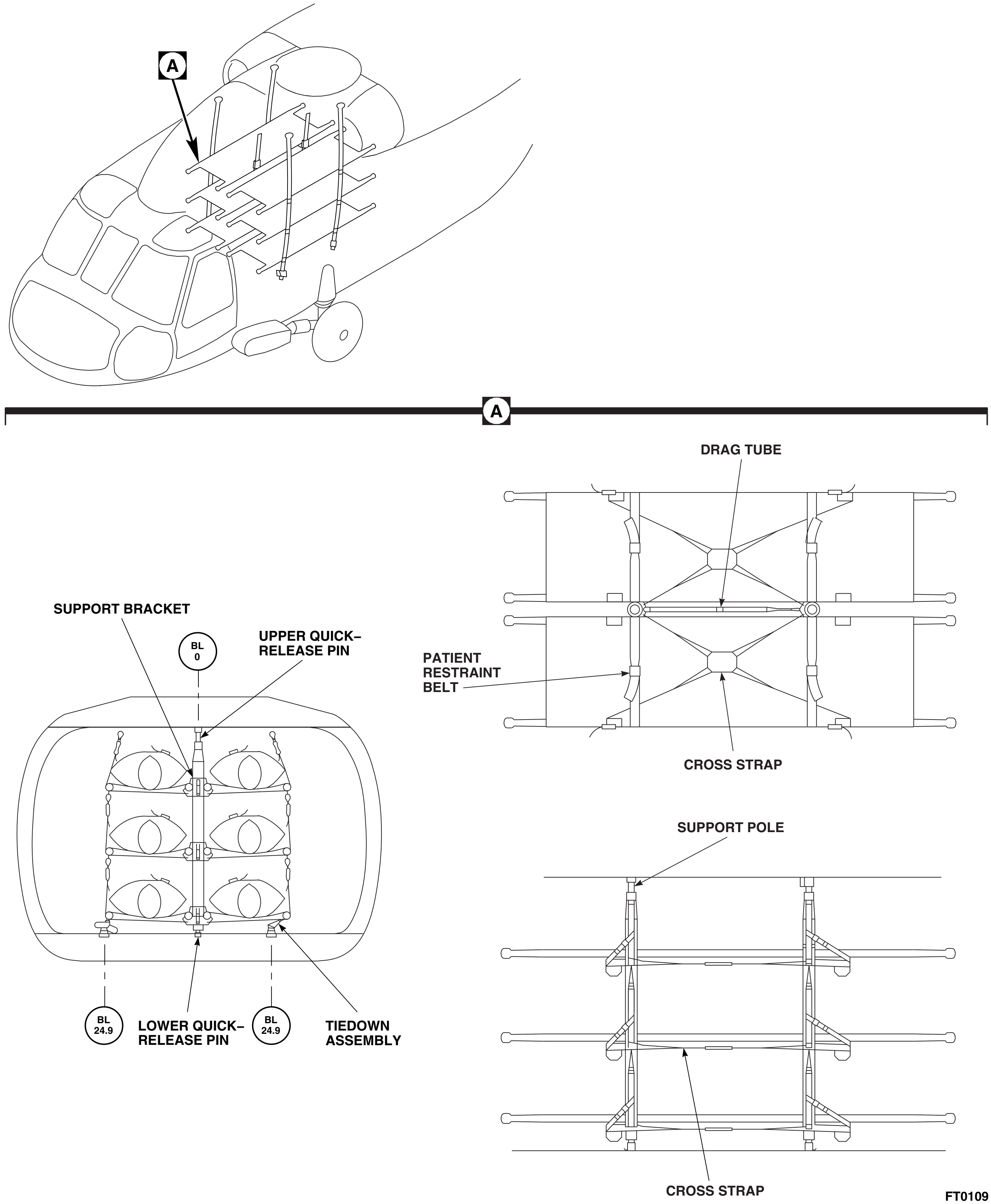


FIGURE 16-1-1-1. SIX MAN LITTER KIT LOCATION DIAGRAM

16-1-2 M-60D MACHINE GUN MOUNTS DESCRIPTION AND DATA.

16-1-2.1 System Description.

The M-60D machine gun mounts provide provisions for installing one M-60D machine gun at each gunner's window. The machine gun is pintle-mounted and held in place by a quick-release pin for easy removal and use on the ground. Each installed gun has an ammunition can mounted on the left side and an ejector control bag for spent cartridges mounted on the right. For complete instructions on the operation of the M-60D, refer to the Operator's Manual (Appropriate Flight Manual).

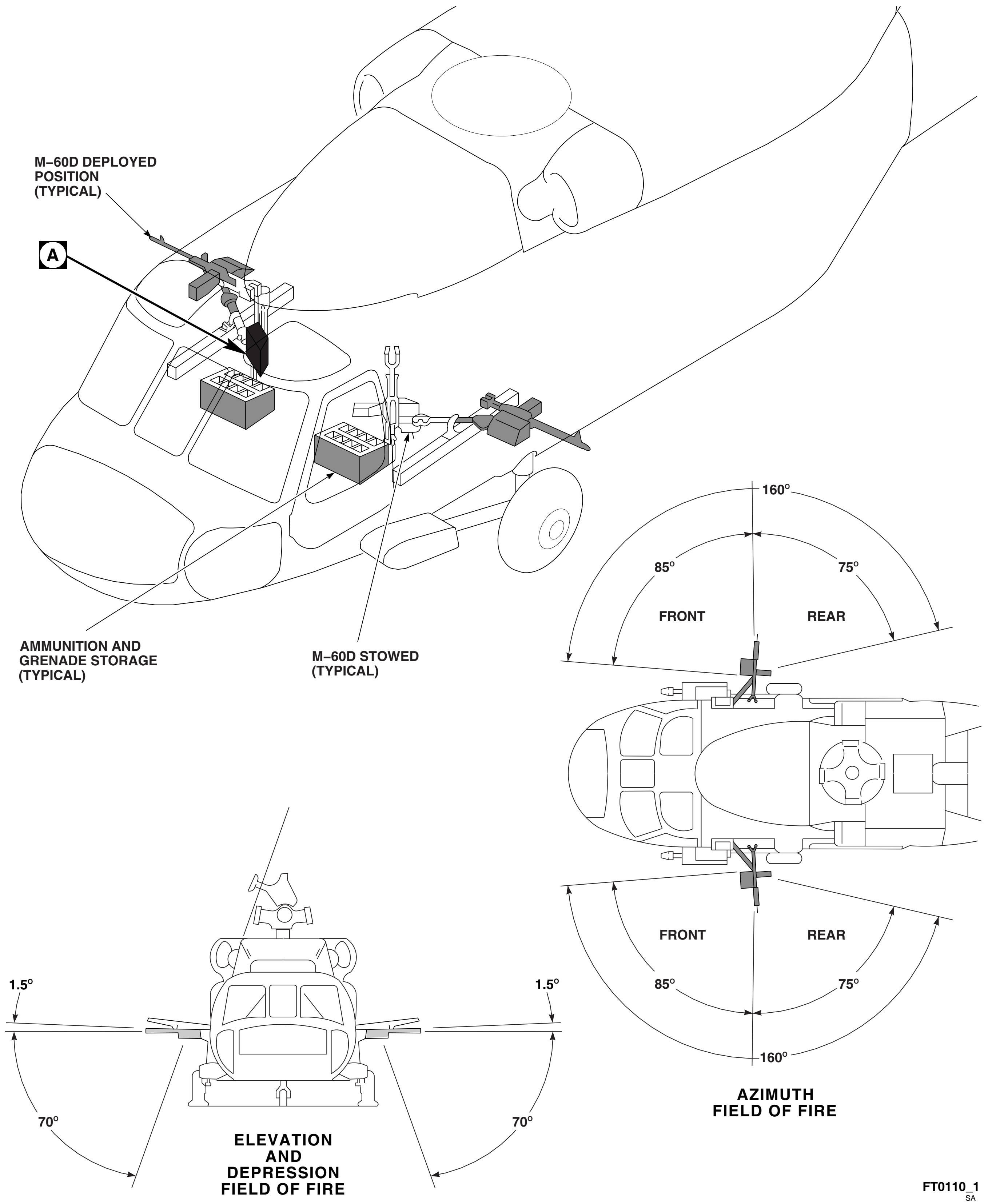


FIGURE 16-1-2-1. ARMAMENT (SHEET 1 OF 2)

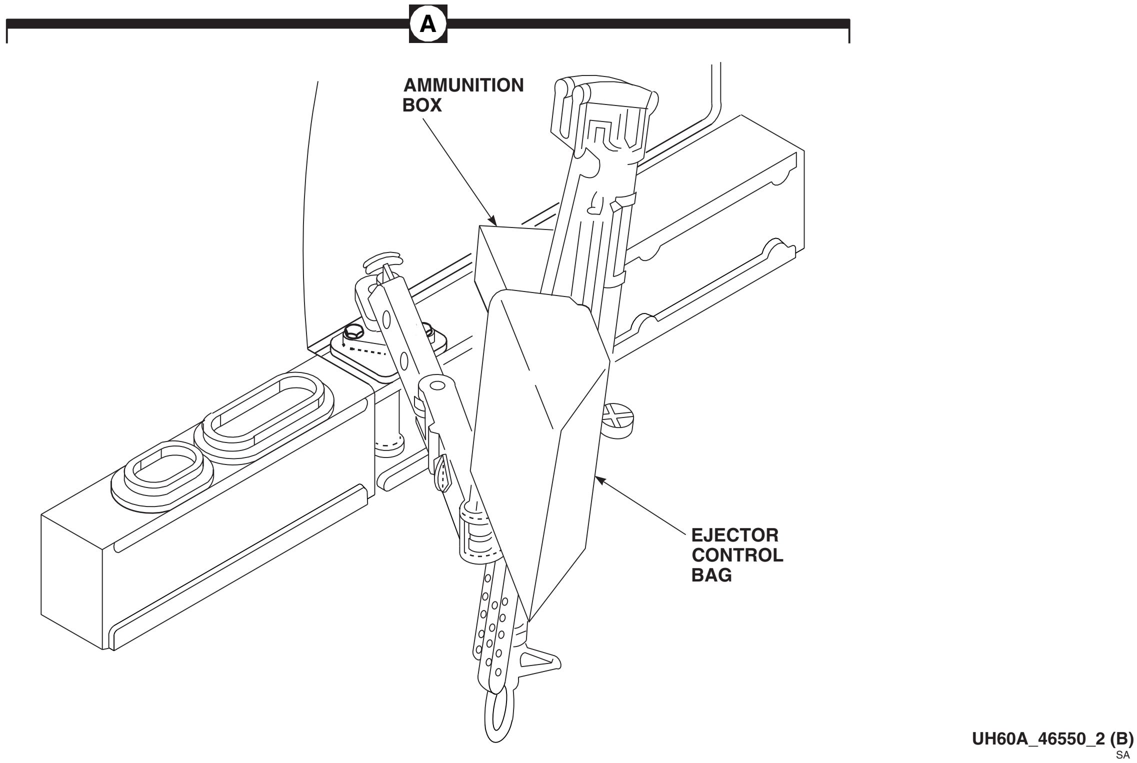


FIGURE 16-1-2-1. ARMAMENT (SHEET 2 OF 2)

16-1-3 BLACKOUT DEVICES KIT DESCRIPTION AND DATA.

16-1-3.1 System Description.

The blackout devices kit provides the means of covering the cabin windows and the opening between the pilot's compartment and the cabin. The kit consists of curtains and velcro tape which is used to attach the curtains to the cabin structure. Loops are attached to the curtains to make removal easy.

16-1-4 EXTERNAL STORES SUPPORT SYSTEM RANGE EXTENSION SYSTEM DESCRIPTION AND DATA.

16-1-4.1 System Description.

The external stores support system (ESSS) range extension system provides additional fuel from two 450 gallon external fuel tanks, and two 230 gallon external fuel tanks to extend the range of the helicopter's mission. The 450 gallon external fuel tanks are installed on the inboard ESSS horizontal stores support (HSS) and the 230 gallon external fuel tanks are installed on the outboard ESSS HSS. Major components of the system include auxiliary fuel management control panel, fuel overflow sensor, bleed air regulator valves, fuel shutoff gate valves, EXT fuel flow sensors, fuel temperature transducer, fuel flow transmitter, two ESSS HSS, and vertical stores pylons (VSP).

16-1-4.2 Principles Of Operation.

16-1-4.2.1 Power Distribution.

Electrical power for the ESSS range extension system is supplied from the No. 1 dc primary bus, No. 2 dc primary bus, and No. 2 ac primary bus. The auxiliary fuel management control panel is supplied 28 vdc from the No. 1 dc primary bus through the EXT FUEL LH circuit breaker, the No. 2 dc primary bus through the EXT FUEL RH circuit breaker and the NO. 2 XFER CONTROL circuit breaker. The auxiliary fuel management control panel also receives 115 vac from the No. 2 ac primary bus through the AUX FUEL QTY circuit breaker. Each of the above listed circuit breakers are located on the mission readiness circuit breaker panel.

16-1-4.2.2 Automatic Fuel Transfer.

The automatic fuel transfer circuits for the inboard and outboard auxiliary fuel tanks are identical in operation, therefore only the outboard auxiliary fuel tanks automatic fuel transfer circuit will be described.

Automatic fuel transfer can be initiated by placing the auxiliary fuel management control panel's MODE AUTO/MANUAL switch to AUTO, TANKS INBD/OUTBD switch to OUTBD, and PRESS OUTBD switch to ON (up). This will cause the left and right outboard bleed air regulator valves to open, and the left and right outboard fuel shutoff gate valves to open. With the bleed air regulator valves open and the fuel shutoff gate valves open, both outboard external fuel tanks will become pressurized allowing an immediate flow of fuel from the outboard auxiliary fuel tanks to the main fuel tanks when the fuel level in one of the main fuel tanks is below approximately 900 pounds of fuel. Fuel transfer will continue until the selected auxiliary fuel tanks are empty (EMPTY annunciator illuminated), or the fuel level in both main fuel cells reach approximately 1037 pounds, or the MODE AUTO/OFF/MANUAL switch is placed to OFF.

16-1-4.2.3 Manual Fuel Transfer.

The manual fuel transfer circuits for the inboard and outboard auxiliary fuel tanks are identical in operation, therefore only the outboard auxiliary fuel tanks manual fuel transfer circuit will be described.

Manual fuel transfer can be initiated by placing the auxiliary fuel management control panel's MODE AUTO/MANUAL switch to MANUAL, TANKS INBD/OUTBD switch to OUTBD, PRESS OUTBD switch to ON (up), and MANUAL XFR RIGHT ON/OFF and LEFT ON/OFF switches to ON. This will cause the left and right outboard bleed air regulator valves to open, and the left and right outboard fuel shutoff gate valves to open. With the bleed air regulator valves open and the fuel shutoff gate valves open, both outboard external fuel tanks will become pressurized, allowing an immediate flow of fuel from the outboard auxiliary fuel tanks to the main fuel tanks. Fuel transfer will continue until the main fuel tanks high level shutoff valve closes (1150 pounds), MANUAL XFR switches are turned OFF, or the MODE AUTO/OFF/MANUAL switch is placed to OFF.

16-1-4.2.4 Auxiliary Fuel Management Panel Self-Test.

Pressing the TEST button on the auxiliary fuel management panel initiates a system self-test routine. Once activated, the self-test verifies operation of the auxiliary fuel management panel, fuel flow transmitter by checking for an electrical ground or pulse, temperature sensor by checking for a temperature signal equivalent to -60°C to 65°C, signal conditioner, auxiliary fuel tank probes, and associated wiring interconnections. When the TEST button is pressed, the following will occur:

a. Auxiliary fuel management panel's indicator lights will go on and the digital display will indicate 8888.

b. When TEST button is released, digital display will indicate the digit 8 in sequence from left to right three times.

c. Then the auxiliary fuel management panel's digital display will indicate GOOD for approximately five seconds.

d. Then the auxiliary fuel management panel's digital display will indicate either 4, 5, or 8 (preset fuel type) in the left most bit for approximately three seconds.

e. Then the auxiliary fuel management panel's digital display shall indicate the amount of fuel remaining as selected by the AUX FUEL QTY OUTBD/INBD/TOTAL/CAL switch.

If for some reason the self-test should fail, one of the error codes listed in Auxiliary Fuel Management Panel Error Codes Table 16-1-4-1 will be displayed and the microprocessor will be disabled from doing any further computations.

If the test fails due to a malfunctioning temperature sensor input, the microprocessor can be placed in a degraded mode by doing two self-tests. In the degraded mode, a preselected value of temperature will be used as a constant for all calculations.

16-1-4.3 Component Description.

16-1-4.3.1 Auxiliary Fuel Management Control Panel.

The auxiliary fuel management control panel, located on the lower console, provides automatic and manual fuel transfer control of the auxiliary fuel tanks. For a more detail explanation of auxiliary fuel management control panel operation, refer to Auxiliary Fuel Management Control Panel Operation Table 16-1-4-2.

16-1-4.3.2 Fuel Overflow Sensor.

The fuel overflow sensor, located in the vent tube, is used to detect the presence of fuel in the vent tube. If fuel is detected, the sensor provides a signal to the auxiliary fuel management control panel which turns on the VENT SENSOR OVFL indicator.

16-1-4.3.3 Bleed Air Regulator Valves.

The bleed air regulator valves, one for each auxiliary fuel tank, control the amount of engine bleed air used to pressurize each auxiliary fuel tank. Each bleed air regulator valve is controlled by placing the auxiliary fuel management control panel's PRESS INBD and OUTBD ON/OFF switches to ON.

16-1-4.3.4 Fuel Shutoff Gate Valves.

The fuel shutoff gate valves, one for each auxiliary fuel tank, are solenoid-operated, normally open valves used to control the flow of fuel from the auxiliary fuel tanks. Each fuel shutoff gate valve is controlled by logic provided by the auxiliary fuel management control panel during auxiliary fuel transfer operations.

16-1-4.3.5 EXT Fuel Flow Sensors.

The EXT fuel flow sensor, one for both left and right auxiliary fuel tanks, is a float type switch used to monitor the flow of fuel in the auxiliary fuel tank lines.

16-1-4.3.6 Horizontal Stores Supports (HSS).

The HSS consists of a graphite epoxy box beam to which attachment fittings are bolted and bonded. Its fairings are made of aluminum. Fuel and pneumatic hoses and components are clamped to the rear side of the HSS, while electrical harnesses are routed on the front side. Each HSS has its own position light mounted on the tip cap.

16-1-4.3.7 Vertical Stores Pylons (VSP).

The VSP consist of ejector rack adapters, ejector racks, and aluminum fairings. MAU-40/A ejector racks are mounted on the inboard (buttline 80) VSP, while BRU-22A/A ejector racks are mounted on the outboard (buttline 112) VSP.

Table 16-1-4-1. Auxiliary Fuel Management Panel Error Codes.

FAULT CODE	MALFUNCTION
E01	Error with microprocessor
E02	Temperature sensor output out of tolerance
E03	Flow transmitter is not connected
E04	Error with fuel flow circuitry
E05	Error with fuel flow consumption
E06	Error with memory

Table 16-1-4-2. Auxiliary Fuel Management Control Panel Operation.

CONTROL/INDICATOR	FUNCTION
TANKS INBD/OUTBD Select Switch	Two position switch used to control which set (inboard/outboard) of auxiliary fuel tanks will transfer fuel to the main fuel tanks.
MANUAL XFR RIGHT/LEFT ON/OFF Select Switch	Two position ON/OFF switch used to manually control the flow of fuel from the associated fuel tanks to the main fuel tanks. Setting RIGHT and LEFT switch to ON opens the fuel shutoff gate valves for the left and right external fuel tanks. Setting the RIGHT and LEFT switch to OFF closes the fuel shutoff gate valves for the left and right external fuel tanks.

Table 16-1-4-2. Auxiliary Fuel Management Control Panel Operation. (Cont)

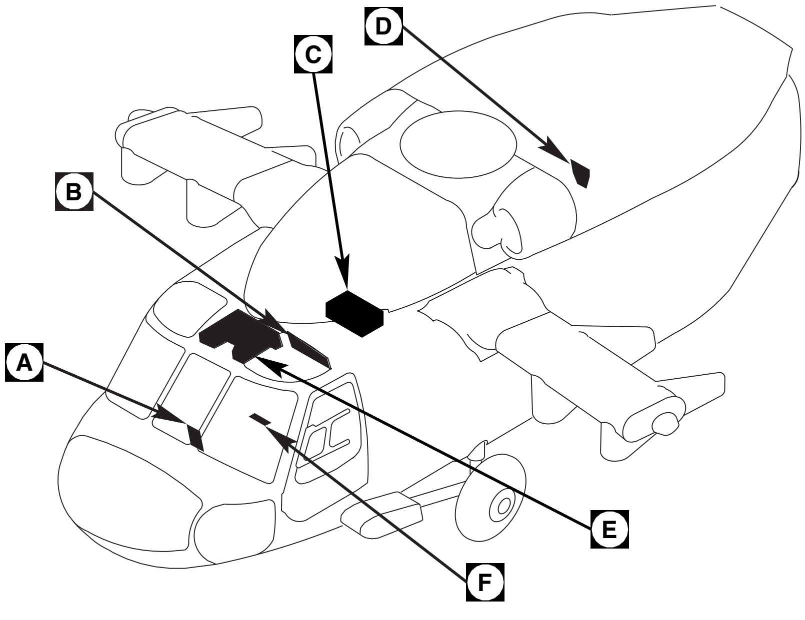
CONTROL/INDICATOR	FUNCTION
MODE AUTO/OFF/MANUAL Select Switch	Three position switch used to set mode of fuel transfer from auxiliary fuel tanks. Setting MODE switch to AUTO automatically enables fuel transfer from selected auxiliary fuel tanks to main fuel tanks when one or more main fuel tanks contain less than 900 pounds of fuel. Setting MODE switch to MANUAL manually transfers fuel from selected auxiliary fuel tanks to main fuel tanks until main fuel tanks high level fuel shutoff gate valves close (1150 pounds) or MODE switch is set to OFF. Setting MODE switch to OFF disables automatic and manual fuel transfer.
PRESS INBD/OUTBD Select Switch	Two position switch used to control which auxiliary fuel tanks (inboard/outboard) are pressurized. Setting PRESS INBD/OUTBD switch to ON (up) opens the associated left and right bleed air regulator valves. Setting PRESS INBD/OUTBD switch to OFF (down) removes electrical power from the associated left and right bleed air regulator valves in order for them to spring closed.
EXTERNAL INBD/OUTBD EMPTY Annunciators	Provides pilot with a visual indication when the associated external fuel tank is empty.
EXTERNAL RIGHT/LEFT NO FLOW Annunciators	Provides pilot with a visual indication when fuel flow from the selected external fuel tanks to the main fuel tanks has stopped.

Table 16-1-4-2. Auxiliary Fuel Management Control Panel Operation. (Cont)

CONTROL/INDICATOR	FUNCTION
VENT SENSOR FAIL/OVFL Annunciator	Two segment annunciator lights provide the pilot with a visual indication of the fuel overflow sensor status. With FAIL segment illuminated, indicates a failure (open) in the fuel overflow sensor circuit. With OVFL segment illuminated, indicates the presence of fuel in the overflow vent tube.
TEST Switch	Initiates self-test routine. During the self-test routine the digital display will indicate a good or an error code.
DEGRADED Indicator	Indicates a malfunction in a critical function of the control panel.
INCR/DECR Switch	Increases/decreases digital display for setting auxiliary fuel remaining and K FACTOR of fuel flow transmitter. Switch function is disabled in flight.
STATUS Switch	Resets AUX FUEL caution capsule, resets and stores conditions of NO FLOW and EMPTY indicators.
AUX FUEL QTY TOTAL/OUTBD/INBD/CAL Switch	Four position switch controls which tank information is displayed on the digital display. Setting switch to OUTBD displays total fuel remaining in outboard pair of fuel tanks. Setting switch to INBD displays total fuel remaining in inboard pair of fuel tanks. Setting switch to TOTAL, displays total fuel remaining in all external fuel tanks. Setting switch to CAL provides maintenance personnel a means of setting the K FACTOR value shown on fuel flow transmitter indicator plate into the control panel.

Table 16-1-4-2. Auxiliary Fuel Management Control Panel Operation. (Cont)

CONTROL/INDICATOR	FUNCTION
AUX FUEL QTY POUNDS Digital Display	Indicates in pounds, the amount of fuel remaining in symmetrical pairs of external auxiliary fuel tanks, or total auxiliary fuel remaining, as selected by AUX FUEL QTY switch. Provides maintenance personnel information needed to preset K FACTOR value from fuel flow transmitter when the AUX FUEL QTY switch is in CAL position.
QTY BRIGHTNESS Control	Provides a means of dimming and brightening AUX FUEL QTY POUNDS digital display.



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
J299	CABIN CEILING BL 10 LH, STA 302
J520	RIGHT EXT FUEL FLOW SENSOR
J521	LEFT EXT FUEL FLOW SENSOR
J522	INT AUX FUEL FLOW SENSOR
J524	FUEL TEMPERATURE TRANSDUCER
P1 / J1	RIGHT INBOARD FUEL TANK CONNECTOR
P1 / J1	LEFT INBOARD FUEL TANK CONNECTOR
P1 / J1	LEFT OUTBOARD FUEL TANK CONNECTOR
P1 / J1	RIGHT OUTBOARD FUEL TANK CONNECTOR
P2 / J835	LEFT INBOARD FUEL TANK CONNECTOR
P2 / J836	RIGHT INBOARD FUEL TANK CONNECTOR

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P2 / J849	LEFT OUTBOARD FUEL TANK CONNECTOR
P2 / J850	RIGHT OUTBOARD FUEL TANK CONNECTOR
P113 / J113	COCKPIT BL 6 LH, STA 197
P114 / J114	COCKPIT BL 8 RH, STA 200
P117 / J117	CAUTION / ADVISORY PANEL
P137	NO. 2 SIGNAL DATA CONVERTER
P138	NO. 1 SIGNAL DATA CONVERTER
P163 / J163	LEFT DRAG BEAM SWITCH
P217 / J217	MAIN ROTOR PYLON DECK BL 0, STA 284
P246 / J246	CABIN CEILING BL 5 RH, STA 247
P334 / J334	CABIN CEILING BL 16.5 LH, STA 390

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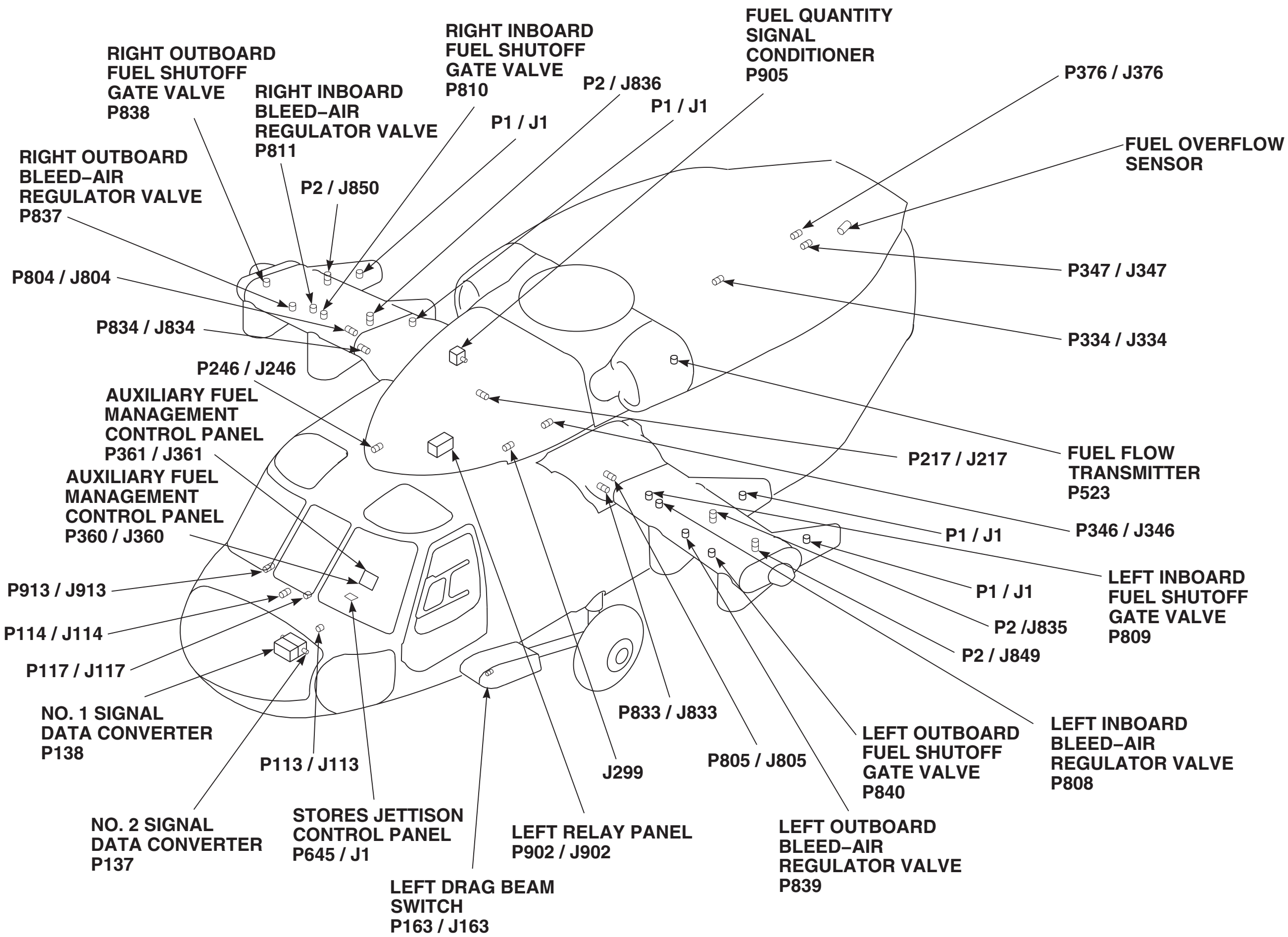
FIGURE 16-1-4-1. EXTERNAL STORES SUPPORT SYSTEM RANGE EXTENSION SYSTEM LOCATION DIAGRAM (SHEET 1 OF 4)



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P346 / J346	CABIN CEILING BL 20 LH, STA 304
P347 / J347	TRANSITION SECTION BL 10 LH, STA 425
P360 / J360	AUXILIARY FUEL MANAGE–MENT CONTROL PANEL
P361 / J361	AUXILIARY FUEL MANAGE–MENT CONTROL PANEL
P376 / J376	AFT TRANSITION BL 18.25 LH, STA 443.5
P523	FUEL FLOW TRANSMITTER
P645 / J1	STORES JETTISON CONTROL PANEL
P804 / J804	BL 59 RH, STA 295
P805 / J805	BL 59 LH, STA 295
P808	LEFT INBOARD BLEED–AIR REGULATOR VALVE
P809	LEFT INBOARD FUEL SHUT–OFF GATE VALVE

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P810	RIGHT INBOARD FUEL SHUTOFF GATE VALVE
P811	RIGHT INBOARD BLEED–AIR REGULATOR VALVE
P833 / J833	ATTACHMENT FITTING BL 59 LH, STA 295
P834 / J834	ATTACHMENT FITTING BL 59 LH, STA 295
P837	RIGHT OUTBOARD BLEED–AIR REGULATOR VALVE
P838	RIGHT OUTBOARD FUEL SHUTOFF GATE VALVE
P839	LEFT OUTBOARD BLEED–AIR REGULATOR VALVE
P840	LEFT OUTBOARD FUEL SHUTOFF GATE VALVE
P902 / J902	LEFT RELAY PANEL
P905	FUEL QUANTITY SIGNAL CONDITIONER
P913 / J913	COCKPIT BL 24 RH, STA 247

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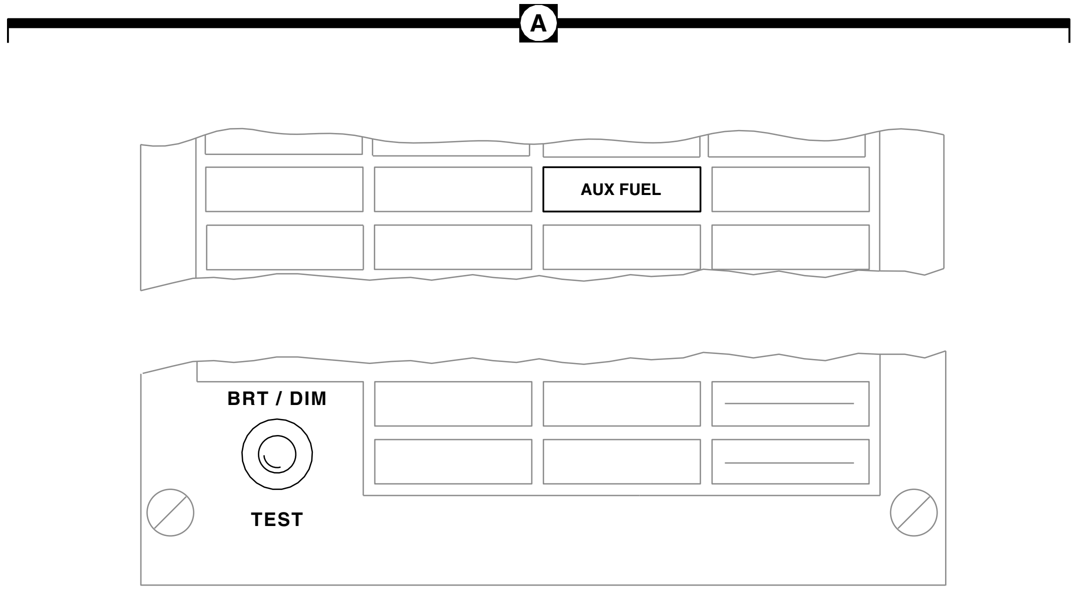
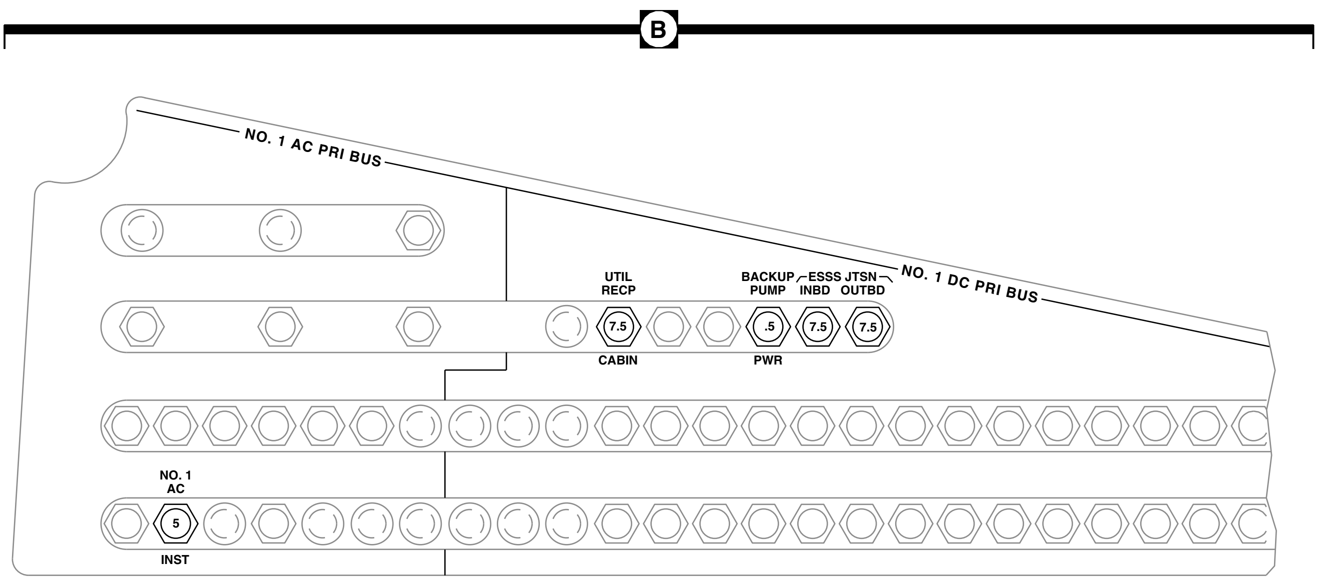
FIGURE 16-1-4-1. EXTERNAL STORES SUPPORT SYSTEM RANGE EXTENSION SYSTEM LOCATION DIAGRAM (SHEET 2 OF 4)



CAUTION/ADVISORY PANEL



COPILOT'S CIRCUIT BREAKER PANEL

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FIGURE 16-1-4-1. EXTERNAL STORES SUPPORT SYSTEM RANGE EXTENSION SYSTEM LOCATION DIAGRAM (SHEET 3 OF 4)

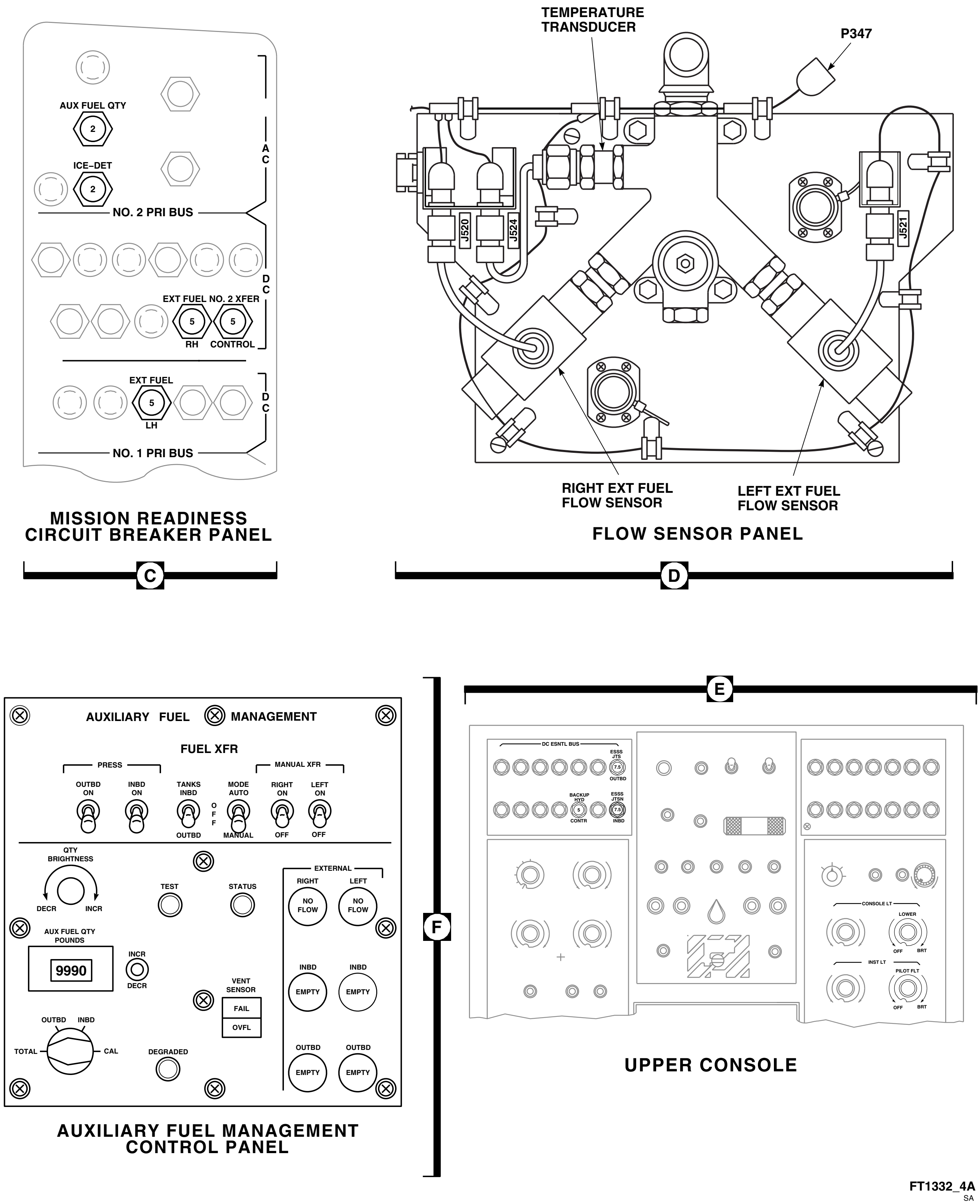


FIGURE 16-1-4-1. EXTERNAL STORES SUPPORT SYSTEM RANGE EXTENSION SYSTEM LOCATION DIAGRAM (SHEET 4 OF 4)

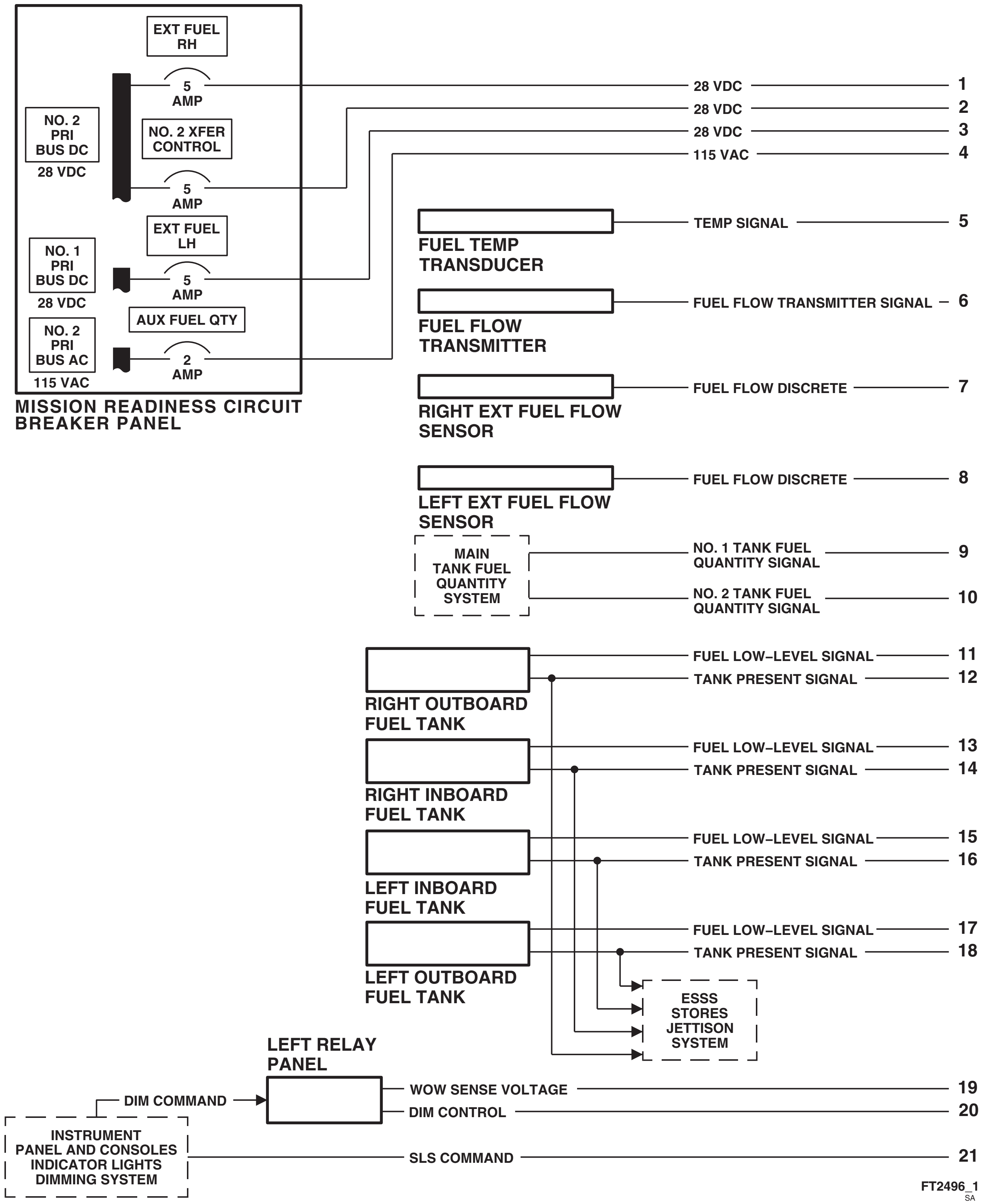


FIGURE 16-1-4-2. EXTERNAL STORES SUPPORT SYSTEM RANGE EXTENSION SYSTEM BLOCK DIAGRAM (SHEET 1 OF 2)

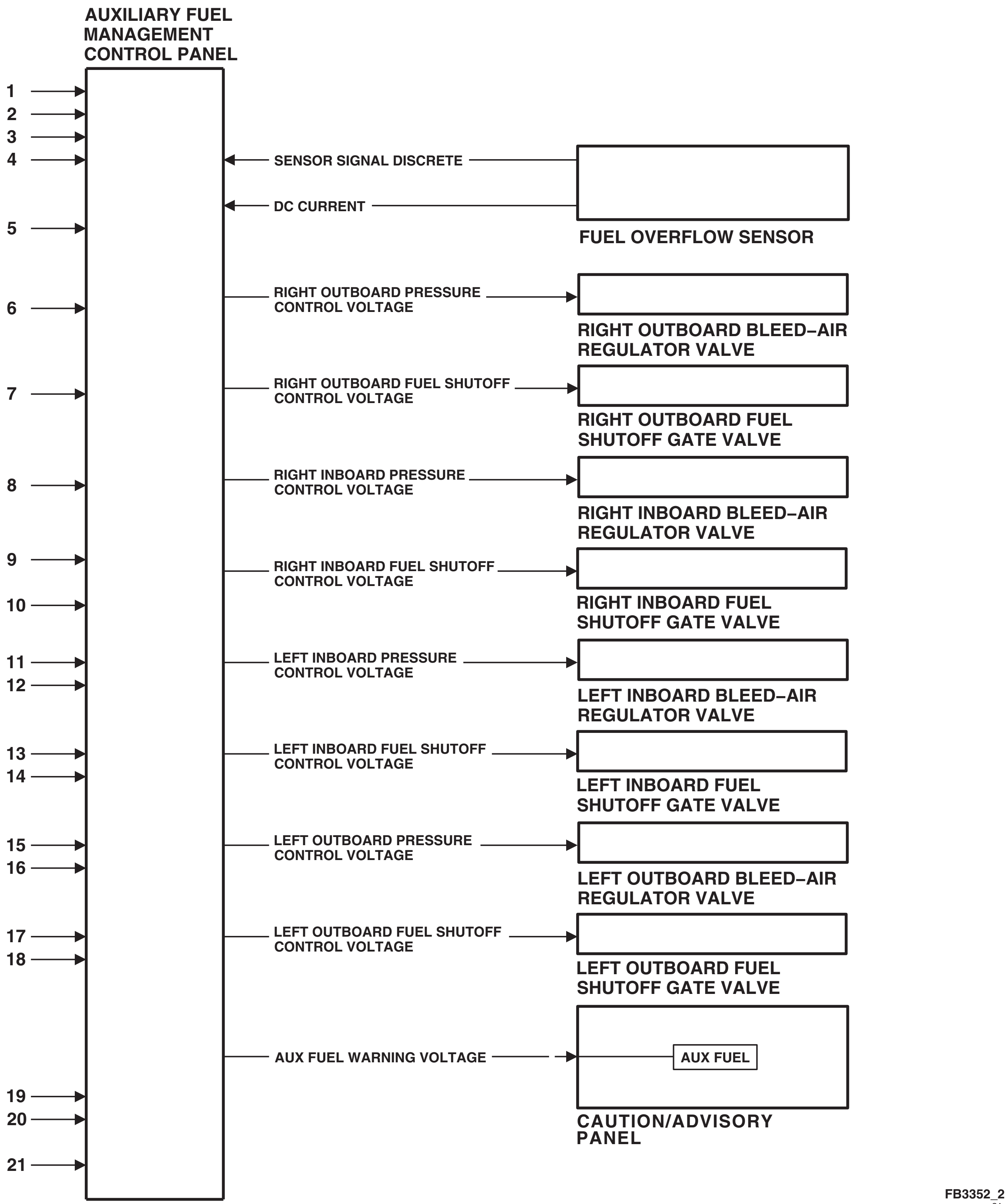


FIGURE 16-1-4-2. EXTERNAL STORES SUPPORT SYSTEM RANGE EXTENSION SYSTEM BLOCK DIAGRAM (SHEET 2 OF 2)

16-1-5 EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING DESCRIPTION AND DATA.

16-1-5.1 System Description.

The external stores support system (ESSS) range extension system provides additional fuel from four 230 gallon external fuel tanks to extend the range of the helicopter's mission. Major components of the system include ESSS control panel, fuel overflow sensor, bleed air regulators, fuel shutoff gate valves, signal conditioner, EXT fuel flow sensor, two horizontal stores supports (HSS), and vertical stores pylons (VSP).

16-1-5.2 Principles Of Operation.

16-1-5.2.1. Power Distribution.

Electrical power for the ESSS range extension system is supplied from the No. 1 dc primary bus and the No. 2 dc primary bus. The signal conditioner is supplied 28 vdc from the No. 1 dc primary bus through the #1 EXT FUEL XFER circuit breaker on the lower console circuit breaker panel. The ESSS control panel is supplied 28 vdc from the No. 2 dc primary bus through the #2 EXT FUEL XFER circuit breaker and #2 EXT FUEL CONTR circuit breaker on the lower console circuit breaker panel.

16-1-5.2.2. Automatic Fuel Transfer.

The automatic fuel transfer circuits for the inboard and outboard auxiliary fuel tanks are identical in operation, therefore only the outboard auxiliary fuel tanks automatic fuel transfer circuit will be described.

Automatic fuel transfer can be initiated by placing the ESSS control panel's AUTO XFER/MAN XFER switch to AUTO XFER, the LEFT and RIGHT INBD XFER/PRESS/OFF switch to PRESS, and the LEFT and RIGHT OUTBD XFER/PRESS/OFF switch to PRESS for approximately five seconds and then to XFER. This will cause both inboard and both outboard bleed air regulators to open, and the left and right outboard fuel shutoff gate valves to open. With the bleed air valves open and the fuel shutoff valves open, both outboard external fuel tanks will become pressurized allowing an automatic transfer of fuel from the outboard auxiliary fuel tanks to the main fuel tanks when the fuel level in one of the main fuel cells is 880 to 900 pounds or less and auxiliary fuel tanks are installed and contain fuel. While fuel is transferring, the left FLOW annunciator on the ESSS control panel will illuminate and the ESSS control panel's fuel quantity indicators for the selected tanks will start to countdown.

When the fuel level in one of the main fuel cells reaches approximately 1037 pounds or an auxiliary fuel tank fuel low signal is present (EMPTY annunciator illuminated), the ESSS control panel will terminate automatic fuel transfer by closing the left and right outboard fuel shutoff gate valves causing the flow of fuel to stop from the auxiliary fuel tanks to the main fuel tanks.

16-1-5.2.3. Manual Fuel Transfer.

The manual fuel transfer circuits for the inboard and outboard auxiliary fuel tanks are identical in operation, therefore only the left outboard auxiliary fuel tank manual fuel transfer circuit will be described.

Manual fuel transfer can be initiated by placing the ESSS control panel's AUTO XFER/MAN XFER switch to MAN XFER, the LEFT and RIGHT INBD XFER/PRESS/OFF switch to PRESS, and the LEFT XFER/PRESS/OFF switch to PRESS for approximately five seconds and then to XFER. This will cause both inboard and both outboard bleed air regulators to open, and the left outboard fuel shutoff gate valves to open, allowing an immediate flow if fuel from the left outboard auxiliary fuel tank. While fuel is transferring, the left FLOW annunciator on the ESSS control panel will illuminate and the ESSS control panel's fuel quantity indicators for the selected tanks will start to countdown.

Manual fuel transfer will continue until the selected auxiliary fuel tank is empty (EMPTY annunciator illuminated), or the fuel level in the main fuel cell reaches approximately 1150 pounds, or the XFER/PRESS/OFF switch is placed to OFF.

16-1-5.2.4. ESSS Control Panel Self-Test.

Pressing the TEST button on the ESSS control panel initiates a system self-test routine. Once activated, the self-test verifies the operation of the ESSS control panel, signal conditioner, auxiliary fuel tank probes, and associated wiring interconnections. When the TEST button is pressed, the following will occur:

(1) ESSS control panel's fuel quantity indicators will light for three seconds and then return to their original states. During this three second illumination of all indicators, the left outboard and right inboard indicators will display "Lamp", and the left inboard and right outboard indicators will display "Test".

(2) ESSS control panel's fuel quantity indicators will perform an integrity check in which each indicator will begin to countdown from 9999 to 0000.

(3) ESSS control panel will perform a system integrity check of the associated wiring, auxiliary fuel tank probes, and signal conditioner.

At the completion of the self-test, each installed auxiliary fuel tank fuel quantity indicator shall display PASS and then display the amount of usable fuel in the tank, and each non-installed auxiliary fuel tank fuel quantity indicator shall display N/A and then go blank. If for some reason the self-test should fail, one of the following error codes listed in ESSS Control Panel Error Codes Table 16-1-5-1 will be displayed.

16-1-5.3 Component Description.

16-1-5.3.1. ESSS Control Panel.

The ESSS control panel, located on the lower console, provides automatic and manual fuel transfer control of the auxiliary fuel tanks. For a more detail explanation of ESSS control panel operation, refer to ESSS Control Panel Operation Table 16-1-5-2.

16-1-5.3.2. Fuel Overflow Sensor.

The fuel overflow sensor, located in the vent tube, is used to detect the presence of fuel in the vent tube. If fuel is detected, the sensor provides a signal to the ESSS control panel which turns on the MN OVFL indicator.

16-1-5.3.3. Bleed Air Regulators.

The bleed air regulators, one for each auxiliary fuel tank, control the amount of engine bleed air used to pressurize each auxiliary fuel tank. Each bleed air regulator is controlled by placing the ESSS control panel's OUTBD and INBD XFER/PRESS/OFF switches to either XFER or PRESS.

16-1-5.3.4. Fuel Shutoff Gate Valves.

The fuel shutoff gate valves, one for each auxiliary fuel tank, are solenoid-operated, normally open valves used to control the flow of fuel from the auxiliary fuel tanks. Each fuel shutoff gate valve is controlled by logic provided by the ESSS control panel during auxiliary fuel transfer operations.

16-1-5.3.5. EXT Fuel Flow Sensor.

The EXT fuel flow sensor, one for both left and both right auxiliary fuel tanks, is a float type switch used to monitor the flow of fuel in the auxiliary fuel tank lines.

16-1-5.3.6. Horizontal Stores Supports (HSS).

The HSS consists of a graphite epoxy box beam to which attachment fittings are bolted and bonded. Its fairings are made of aluminum. Fuel and pneumatic hoses and components are clamped to the rear side of the HSS, while electrical harnesses are routed on the front side. Each HSS has its own position light mounted on the tip cap.

16-1-5.3.7. Vertical Stores Pylons (VSP).

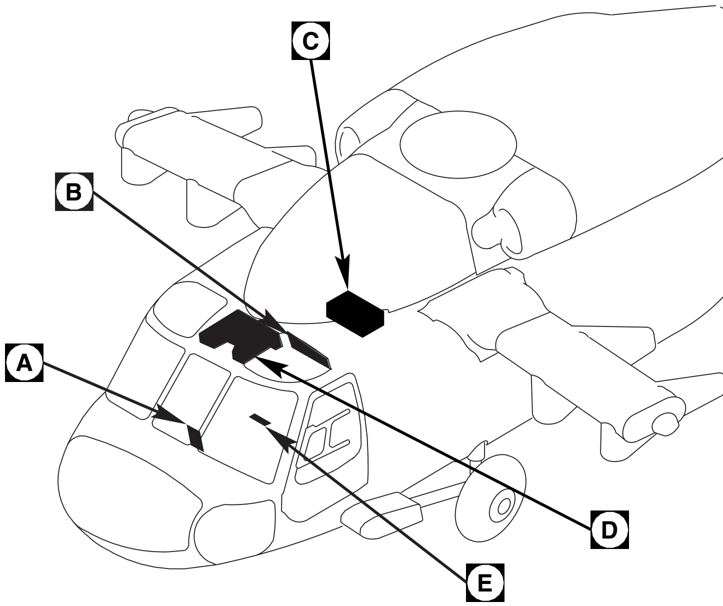
The VSP consist of ejector rack adapters, ejector racks, and aluminum fairings. MAU-40/A ejector racks are mounted on the inboard (buttline 80) VSP, while BRU-22A/A ejector racks are mounted on the outboard (buttline 112) VSP.

Table 16-1-5-1. ESSS Control Panel Error Codes.

FAULT CODE	MALFUNCTION
Fwd	Malfunctioning foward probe.
Aft	Malfunctioning aft probe.
LoZ1	LoZ side of inboard tank unit wiring open/shorted.
LoZ2	LoZ side of outboard tank unit wiring open/shorted.
HiZ1	HiZ side of left inboard or right outboard forward tank unit wiring open/shorted.
HiZ2	HiZ side of left inboard or right outboard aft tank unit wiring open/shorted.
HiZ3	HiZ side of left outboard or right inboard forward tank unit wiring open/shorted.
HiZ4	HiZ side of left outboard or right inboard aft outboard tank unit wiring open/shorted.
A/D	A/D converter is bad.
>Max	Calculated fuel mass is high.
T - TO	The processor has experienced a transmit timeout.

Table 16-1-5-2. ESSS Control Panel Operation.

CONTROL/INDICATOR	FUNCTION
Fuel XFER/PRESS/OFF Switches	Three position switch used to control transfer of fuel from the external fuel tanks to the main fuel tanks. Setting XFER/PRESS/OFF switch to PRESS, the associated fuel shutoff gate valve is powered closed and the associated bleed air regulator is powered open. Setting XFER/PRESS/OFF switch to XFER, opens both associated fuel shutoff gate and bleed air regulator valves. Setting XFER/PRESS/OFF switch to OFF, closes both associated fuel shutoff gate valve, and removes electrical power from the bleed air regulator valve in order for it to spring closed.
AUTO XFER/MAN XFER Transfer Switch	Two position switch used to select mode of transfer from external tanks. Setting AUTO XFER/MAN XFER switch to AUTO XFER, enables automatic fuel transfer from the external fuel tanks to the main fuel tanks when one main fuel tank is less then 1000 pounds. Setting AUTO XFER/MAN XFER switch to MAN XFER, provides operator manual control of fuel transfer from the external tanks to main fuel tanks through the fuel XFER/PRESS/OFF switches.
CAUT RESET Switch	Pushbutton switch used to turn off AUX FUEL caution capsule on caution/advisory panel.
DIM Switch	Control amount of brightness of the control panel’s fuel quantity indicators.
EMPTY Annunciators	Provides pilot with a visual indication when the associated external fuel tank is empty.
FLOW Annunciators	Provides pilot with a visual indication of fuel flow from the external fuel tanks to the main fuel tanks.
MN OVFL Annunciator	Provides pilot with a visual indication when a main tank overflow has occurred.
LAT IMBAL Annunciator	Provides pilot with a visual indication when a lateral imbalance (asymmetrical transfer) has occurred.



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
J299	CABIN CEILING BL 10 LH, STA 302
J520	RIGHT EXT FUEL FLOW SENSOR
J521	LEFT EXT FUEL FLOW SENSOR
P1 / J1	RIGHT INBOARD FUEL TANK CONNECTOR
P1 / J1	LEFT INBOARD FUEL TANK CONNECTOR
P1 / J1	LEFT OUTBOARD FUEL TANK CONNECTOR
P1 / J1	RIGHT OUTBOARD FUEL TANK CONNECTOR
P2 / J835	LEFT INBOARD FUEL TANK CONNECTOR
P2 / J836	RIGHT INBOARD FUEL TANK CONNECTOR

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P2 / J849	LEFT OUTBOARD FUEL TANK CONNECTOR
P2 / J850	RIGHT OUTBOARD FUEL TANK CONNECTOR
P113 / J113	COCKPIT BL 6 LH, STA 197
P114 / J114	COCKPIT BL 8 RH, STA 200
P117 / J117	CAUTION / ADVISORY PANEL
P137	NO. 2 SIGNAL DATA CONVERTER
P138	NO. 1 SIGNAL DATA CONVERTER
P163 / J163	LEFT DRAG BEAM SWITCH
P217 / J217	LOWER CONSOLE DISCONNECT
P246 / J246	CABIN CEILING BL 5 RH, STA 247
P334 / J334	CABIN DISCONNECT

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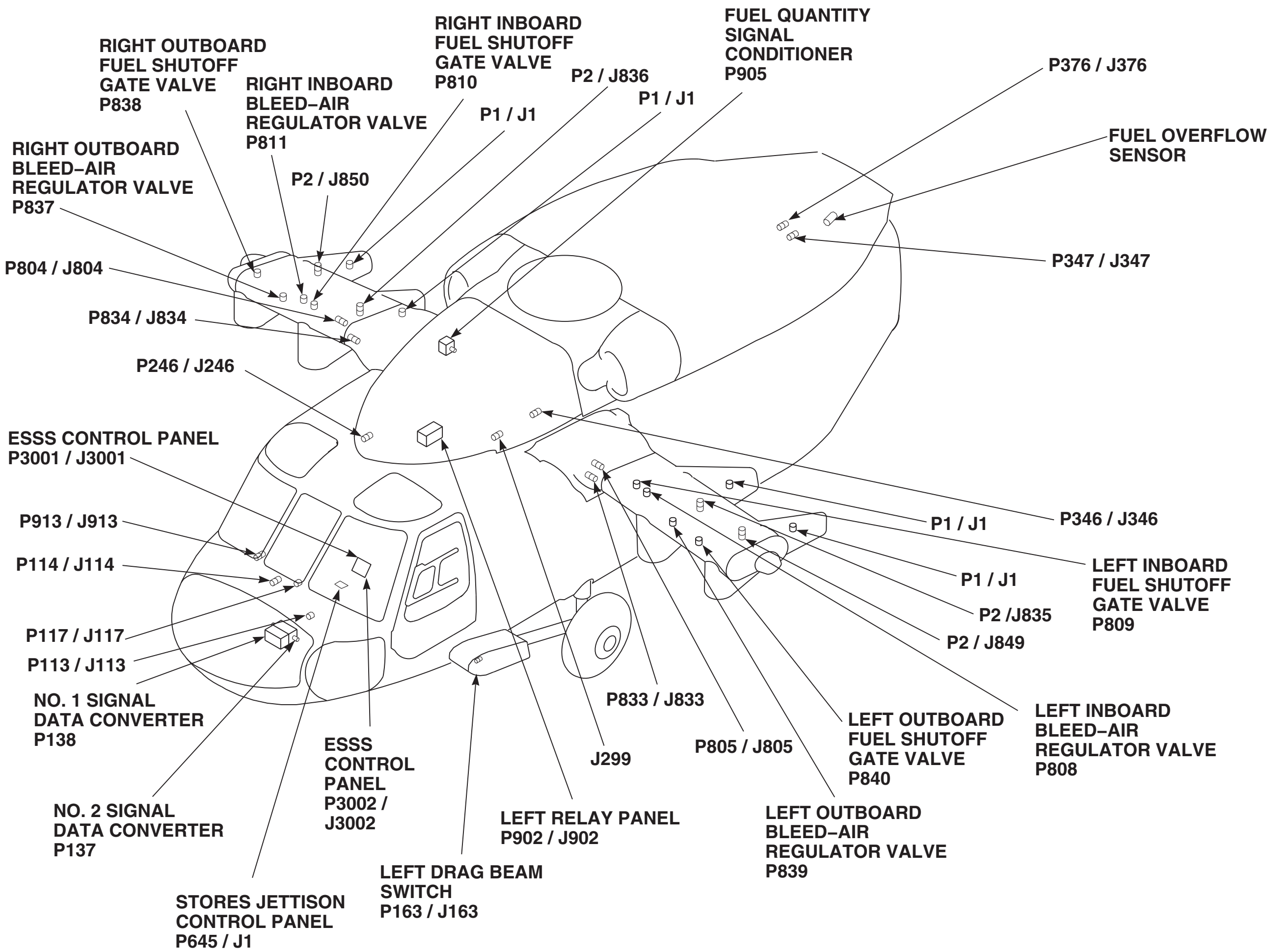
FIGURE 16-1-5-1. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING (SHEET 1 OF 4)



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P346 / J346	CABIN CEILING BL 20 LH, STA 304
P347 / J347	TRANSITION SECTION BL 10 LH, STA 425
P360 / J1	SIGNAL CONDITIONER
P361 / J2	SIGNAL CONDITIONER
P376 / J376	AFT TRANSITION BL 18.25 LH, STA 443.5
P645 / J1	STORES JETTISON CONTROL PANEL
P804 / J804	BL 59 RH, STA 295
P805 / J805	BL 59 LH, STA 295
P808	LEFT INBOARD BLEED–AIR REGULATOR VALVE
P809	LEFT INBOARD FUEL SHUT–OFF GATE VALVE
P810	RIGHT INBOARD FUEL SHUTOFF GATE VALVE
P811	RIGHT INBOARD BLEED–AIR REGULATOR VALVE

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P833 / J833	ATTACHMENT FITTING BL 59 LH, STA 295
P834 / J834	ATTACHMENT FITTING BL 59 LH, STA 295
P837	RIGHT OUTBOARD BLEED–AIR REGULATOR VALVE
P838	RIGHT OUTBOARD FUEL SHUTOFF GATE VALVE
P839	LEFT OUTBOARD BLEED–AIR REGULATOR VALVE
P840	LEFT OUTBOARD FUEL SHUTOFF GATE VALVE
P902 / J902	LEFT RELAY PANEL
P905	FUEL QUANTITY SIGNAL CONDITIONER
P913 / J913	COCKPIT BL 24 RH, STA 247
P3001 / J3001	ESSS CONTROL PANEL
P3002 / J3002	ESSS CONTROL PANEL

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FIGURE 16-1-5-1. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING (SHEET 2 OF 4)

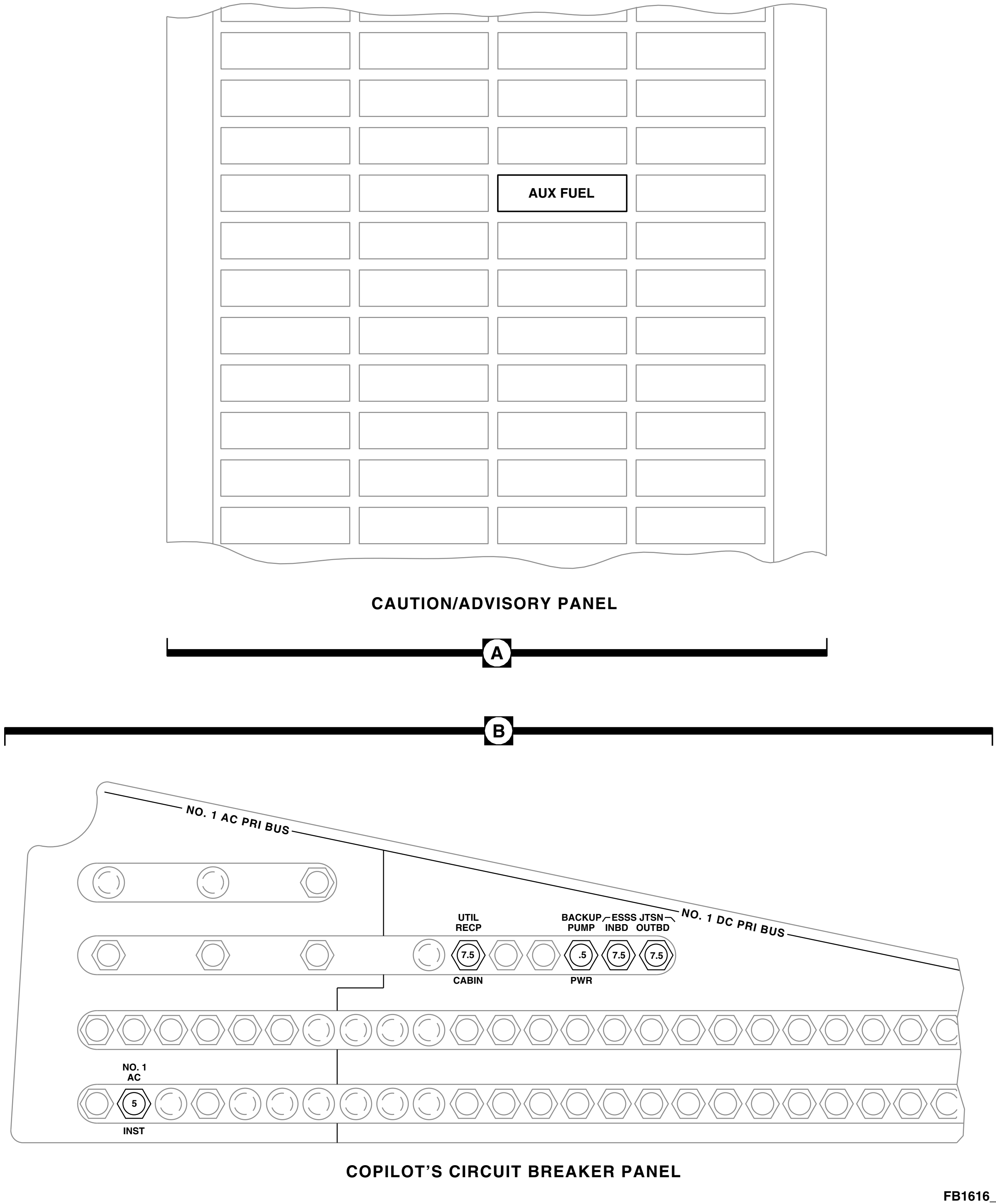

FIGURE 16-1-5-1. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING (SHEET 3 OF 4)

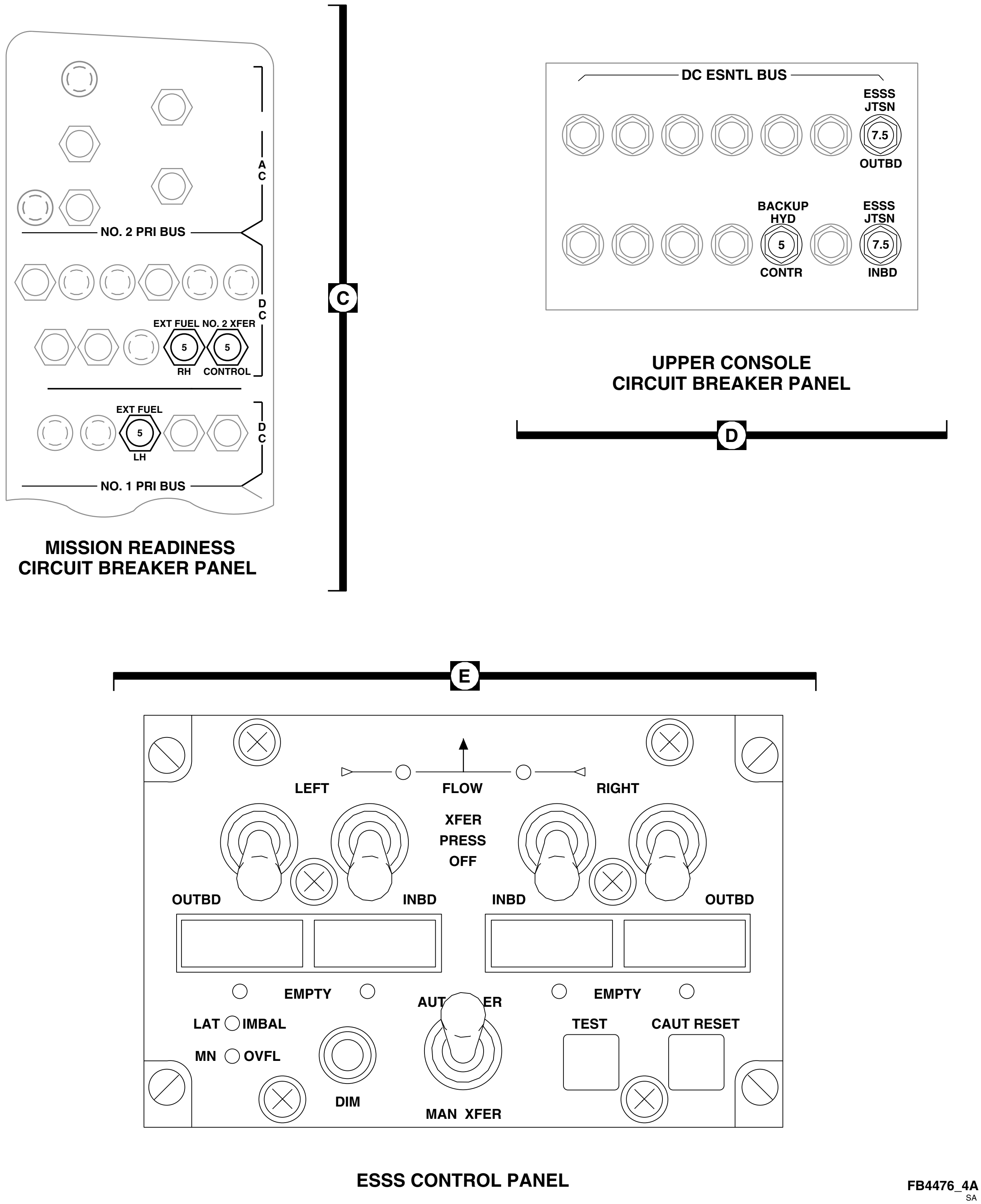

FIGURE 16-1-5-1. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING (SHEET 4 OF 4)

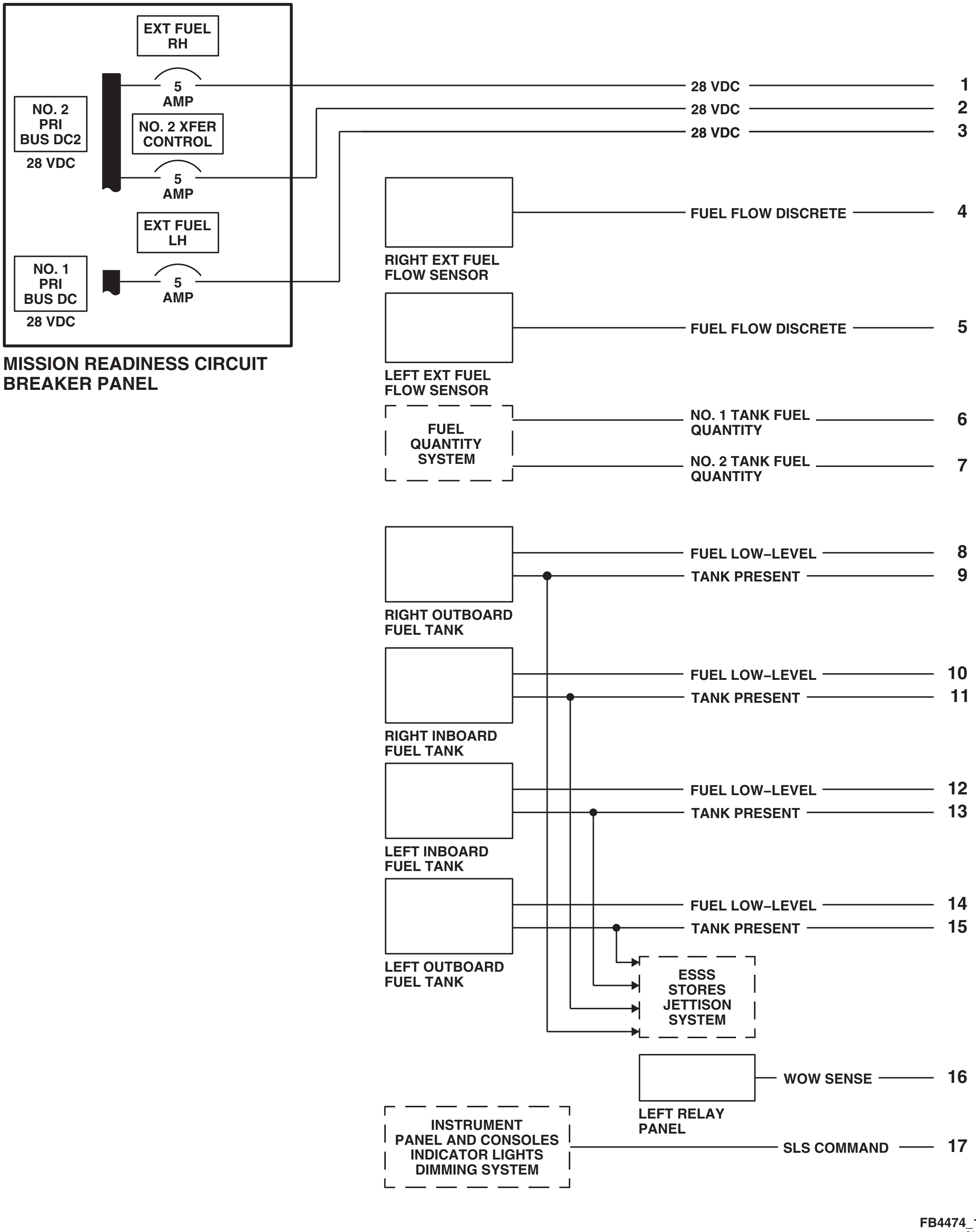

FIGURE 16-1-5-2. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING (SHEET 1 OF 3)

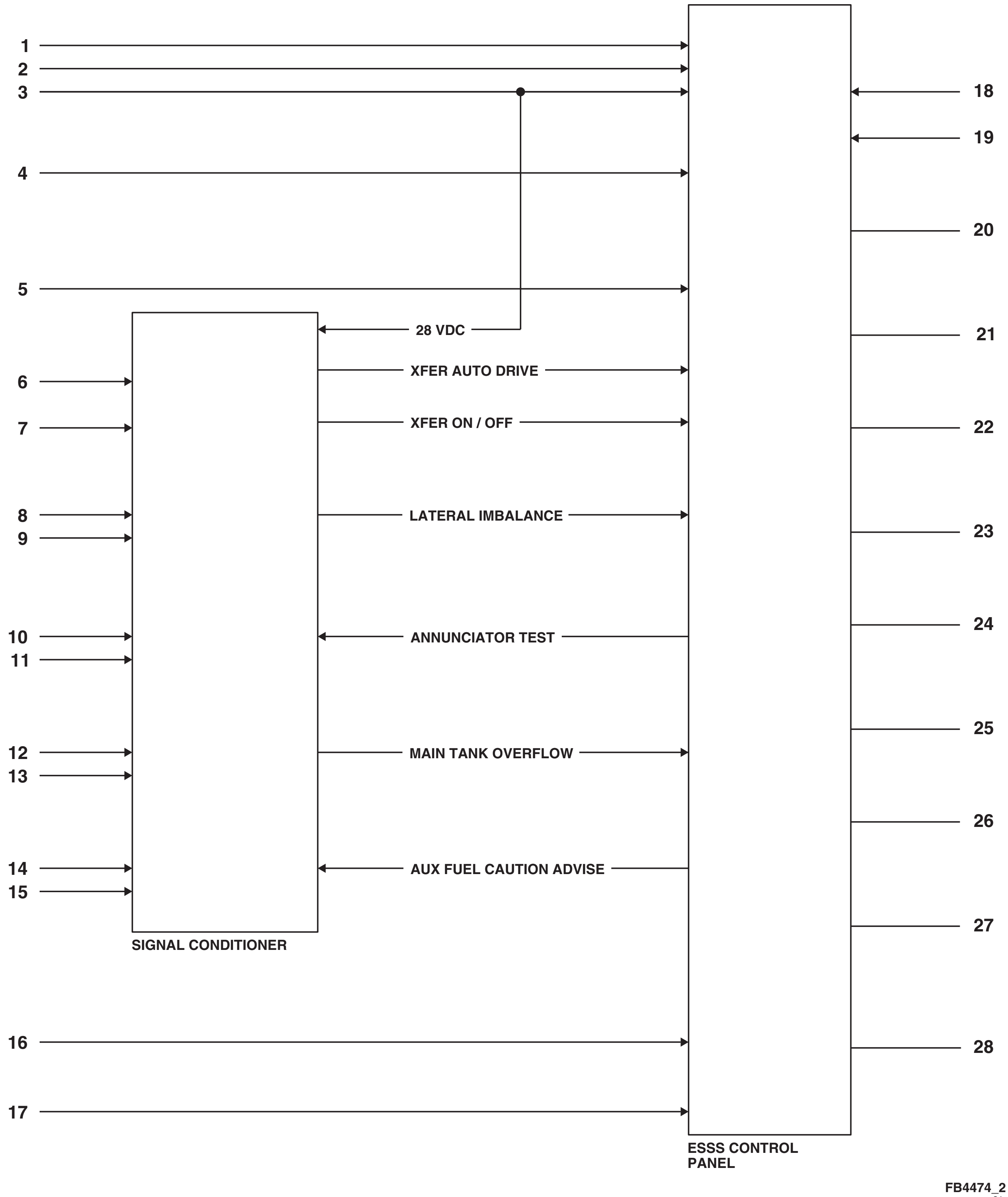

FIGURE 16-1-5-2. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING (SHEET 2 OF 3)

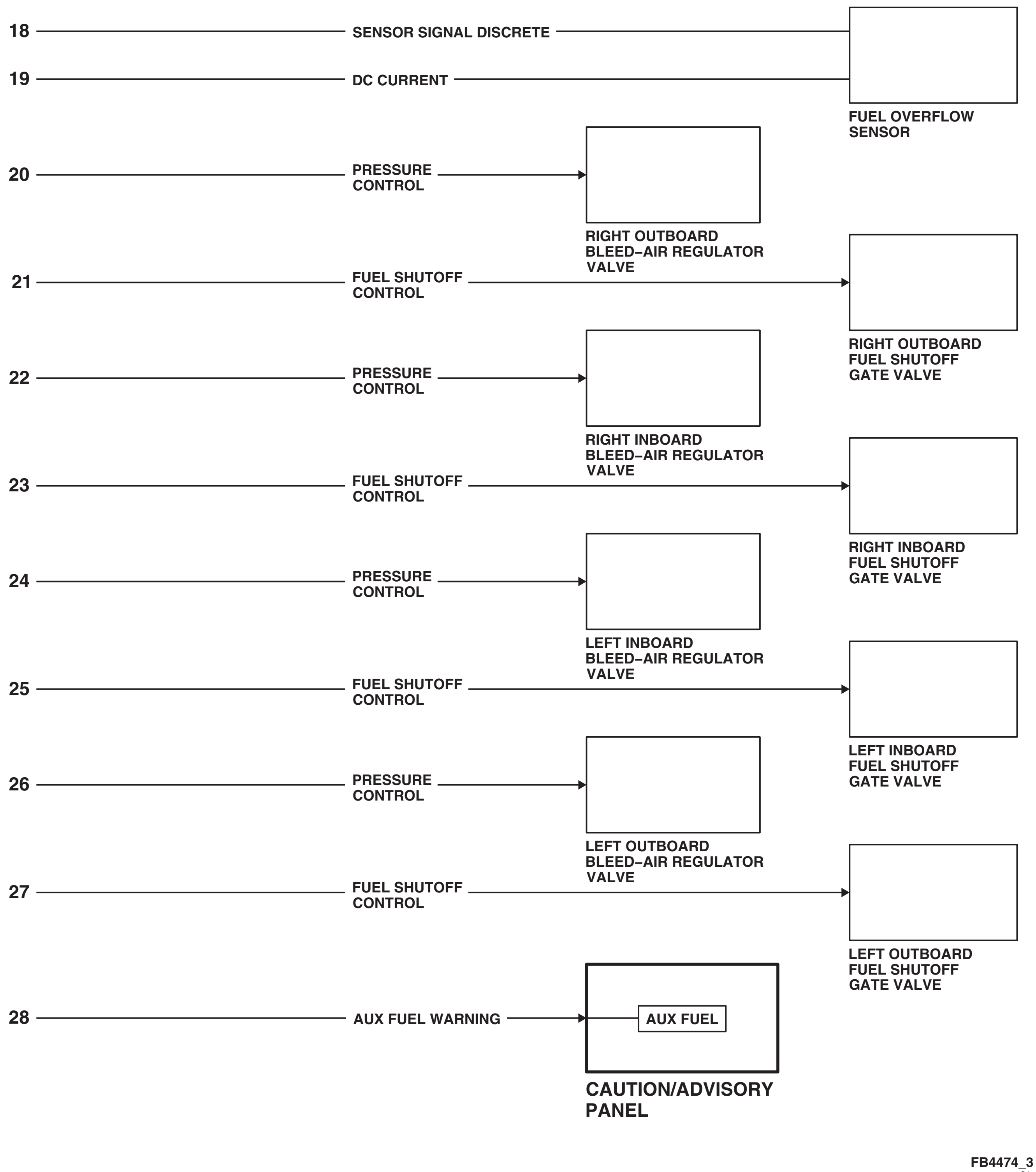

FIGURE 16-1-5-2. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING (SHEET 3 OF 3)

16-1-6 EXTERNAL STORES SUPPORT SYSTEM JETTISON SYSTEM DESCRIPTION AND DATA.

16-1-6.1 External Stores Support System (ESSS) Jettison System Description.

The ESSS jettison system provides two methods of jettisoning external stores: a primary jettison subsystem and an ESSS emergency jettison subsystem (Figure 16-1-6-1). The primary jettison subsystem provides selection of jettisoning individual external stores, a symmetrical pair of stores, or all external stores. An interlock circuit prevents individual stores jettison when external fuel tanks are installed. The ESSS emergency jettison subsystem provides immediate release of all external stores. The ESSS jettison system consists of a stores jettison control panel, on the lower console, and ejector racks attached to the two vertical stores pylons (VSP) on each horizontal stores support (HSS). The racks are used to attach fuel tanks or other external stores dispensers. The ESSS jettison system functionally interfaces with the ESSS fuel system. It also functionally interfaces with the right and left drag beam switches.

16-1-6.2 ESSS Jettison System Power Distribution.

Electrical power for the ESSS jettison system is supplied by the dc essential bus and the No. 1 dc primary bus (Figure 16-1-6-2). The 28 vdc from the dc essential bus is routed through the ESSS JTSN INBD and ESSS JTSN OUTBD circuit breakers on the upper console. Both the ESSS JTSN INBD and ESSS JTSN OUTBD circuit breakers supply 28 vdc to the stores jettison control panel and the left relay panel. The 28 vdc from the No. 1 dc primary bus is routed through the ESSS JTSN INBD and ESSS JTSN OUTBD circuit breakers on the copilot's circuit breaker panel. Both circuit breakers supply 28 vdc to the stores jettison control panel.

16-1-6.3 Stores Jettison Control Panel.

The stores jettison control panel provides two methods of jettisoning external stores systems (Figure 16-1-6-1). The stores jettison control panel consists of two toggle switches having switch guards, a rotary switch, and a lighted information plate on the front of the panel. Two electrical connectors are located on the rear of the panel. A relay bracket assembly, installed within the panel, provides mounting for seven relays and eight diodes.

16-1-6.4 Primary Jettison Subsystem.

Power to operate the primary jettison subsystem is supplied from the No. 1 dc primary bus through the ESSS JTSN INBD and ESSS JTSN OUTBD circuit breakers on the copilot's circuit breaker panel. A rotary switch, on the stores jettison control panel, allows selection of jettisoning individual external stores, a symmetrical pair of stores, or all external stores. With the rotary switch at ALL, the inboard stores will jettison one second after the outboard stores. When the fuel tanks are installed, tank present signals are supplied to the stores jettison control panel. These signals prevent individual stores jettison even if the rotary switch is at L or R. When the helicopter is on the ground, the right drag beam switch removes a ground from the jettison control panel to disable the primary jettison subsystem.

16-1-6.5 ESSS Emergency Jettison Subsystem.

The ESSS emergency jettison subsystem is electrically independent of the primary jettison subsystem. Power to operate the ESSS emergency jettison subsystem is supplied from the dc essential bus through the ESSS JTSN INBD and ESSS JTSN OUTBD circuit breakers on the upper console. When the EMER JETT ALL switch is actuated, all external stores are jettisoned, regardless of the rotary switch position. As with the primary jettison subsystem, the inboard stores will jettison one second after the outboard stores. When the helicopter is on the ground, the left drag beam switch removes a ground from the jettison control panel to disable the emergency jettison subsystem.

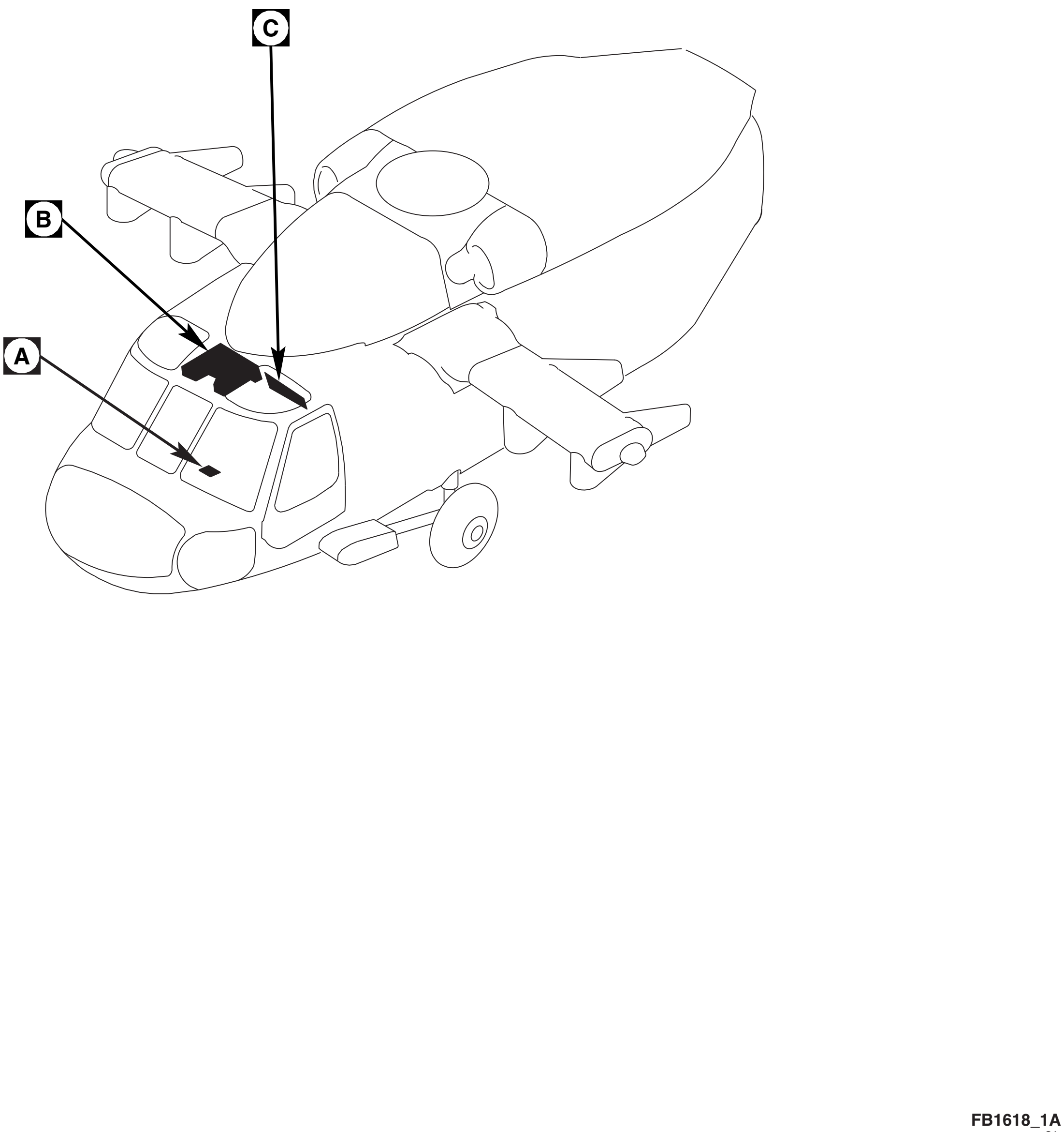

FIGURE 16-1-6-1. EXTERNAL STORES SUPPORT SYSTEM JETTISON SYSTEM LOCATION DIAGRAM (SHEET 1 OF 4)

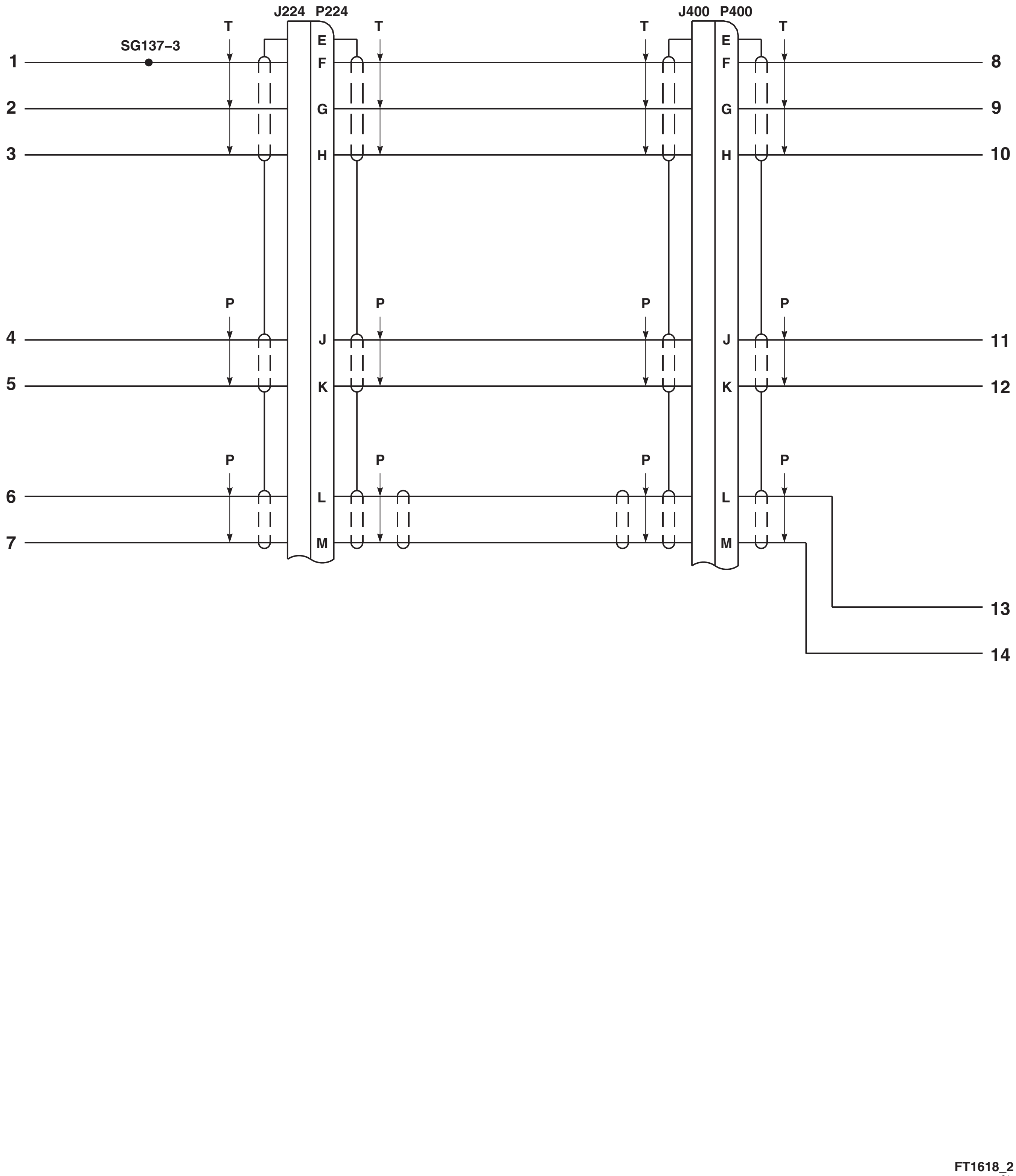

FIGURE 16-1-6-1. EXTERNAL STORES SUPPORT SYSTEM JETTISON SYSTEM LOCATION DIAGRAM (SHEET 2 OF 4)

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
J707	LEFT INBOARD EJECTOR RACK CONNECTOR
J708	RIGHT INBOARD EJECTOR RACK CONNECTOR
P1 / J1	LEFT INBOARD FUEL TANK CONNECTOR
P1 / J1	LEFT OUTBOARD FUEL TANK CONNECTOR
P1 / J1	RIGHT INBOARD FUEL TANK CONNECTOR
P1 / J1	RIGHT OUTBOARD FUEL TANK CONNECTOR
P1 / J705	LEFT OUTBOARD EJECTOR RACK CONNECTOR
P1 / J710	RIGHT OUTBOARD EJECTOR RACK CONNECTOR
P2 / J835	LEFT INBOARD FUEL TANK DISCONNECT
P2 / J836	RIGHT INBOARD FUEL TANK DISCONNECT
P2 / J849	LEFT OUTBOARD FUEL TANK DISCONNECT
P2 / J850	RIGHT OUTBOARD FUEL TANK DISCONNECT
P163 / J163	LEFT DRAG BEAM SWITCH
P164 / J164	RIGHT DRAG BEAM SWITCH
P219 / J219	CABIN FLOOR, BL 25 RH, STA 247
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P231 / J231	CABIN CEILING, BL 9 LH, STA 247
P246 / J246	CABIN CEILING, BL 5 RH
P247 / J247	CABIN, BL 40 LH, STA 248
P248 / J248	CABIN, BL 40 RH, STA 248

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P346 / J346	CABIN CEILING, BL 20 LH, STA 304
P645 / J1	STORES JETTISON CONTROL PANEL
P646 / J2	STORES JETTISON CONTROL PANEL
P660 / J660	LEFT SIDE LOWER CON– SOLE, BL 11 LH, STA 244
P661 / J661	LEFT SIDE LOWER CON– SOLE, BL 11 LH, STA 244
P677 / J677	ATTACHMENT FITTING BL 59 RH, STA 291
P678 / J678	ATTACHMENT FITTING BL 59 LH, STA 289
P679 / J679	ATTACHMENT FITTING BL 59 RH, STA 289
P680 / J680	ATTACHMENT FITTING BL 59 LH, STA 291
P706	RIGHT INBOARD EJECTOR RACK CONNECTOR
P709	LEFT INBOARD EJECTOR RACK CONNECTOR
P833 / J833	ATTACHMENT FITTING BL 59 LH, STA 295
P834 / J834	ATTACHMENT FITTING BL 59 RH, STA 295
P913 / J913	COCKPIT, BL 16 LH, STA 247

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FIGURE 16-1-6-1. EXTERNAL STORES SUPPORT SYSTEM JETTISON SYSTEM LOCATION DIAGRAM (SHEET 3 OF 4)

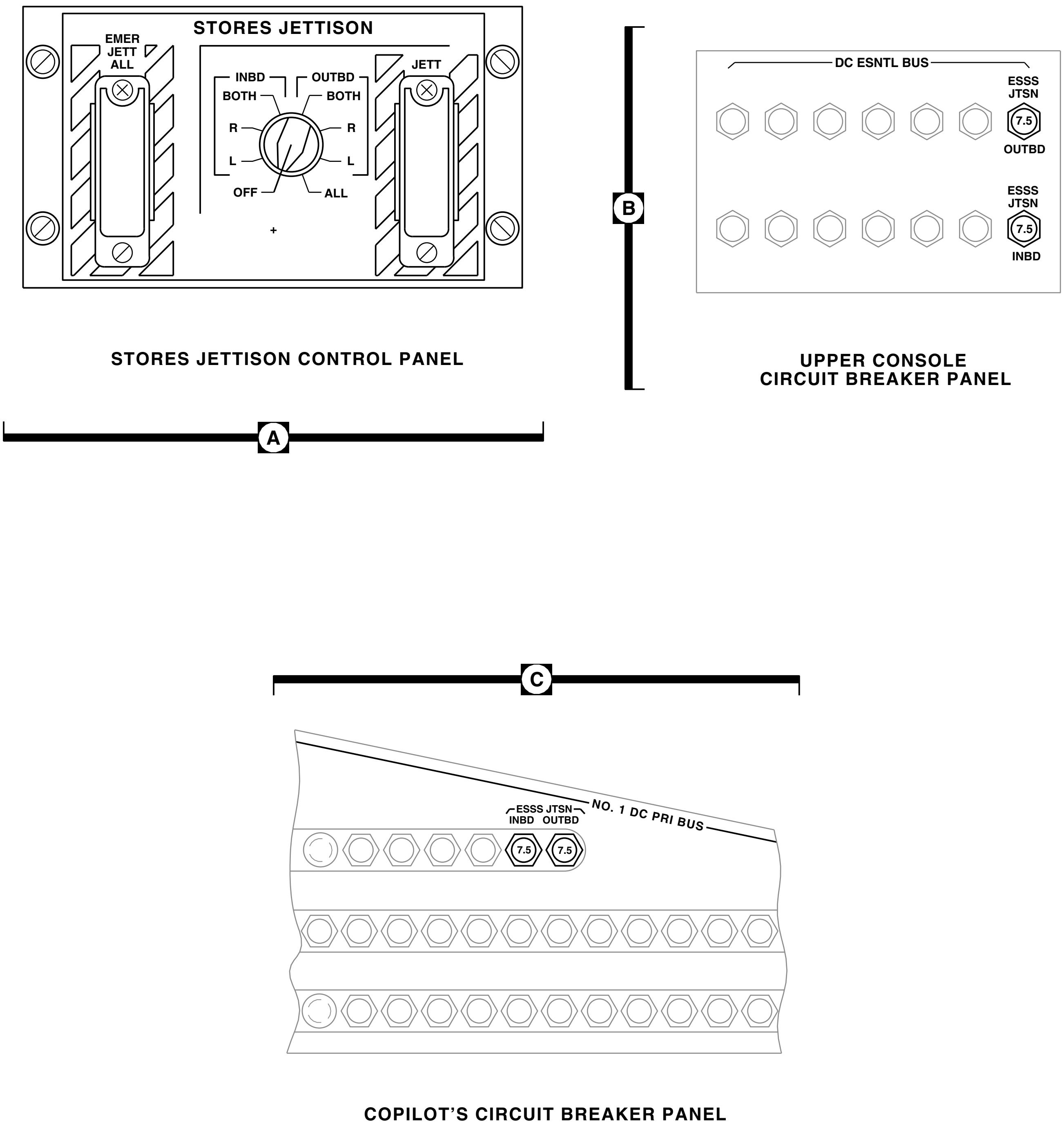

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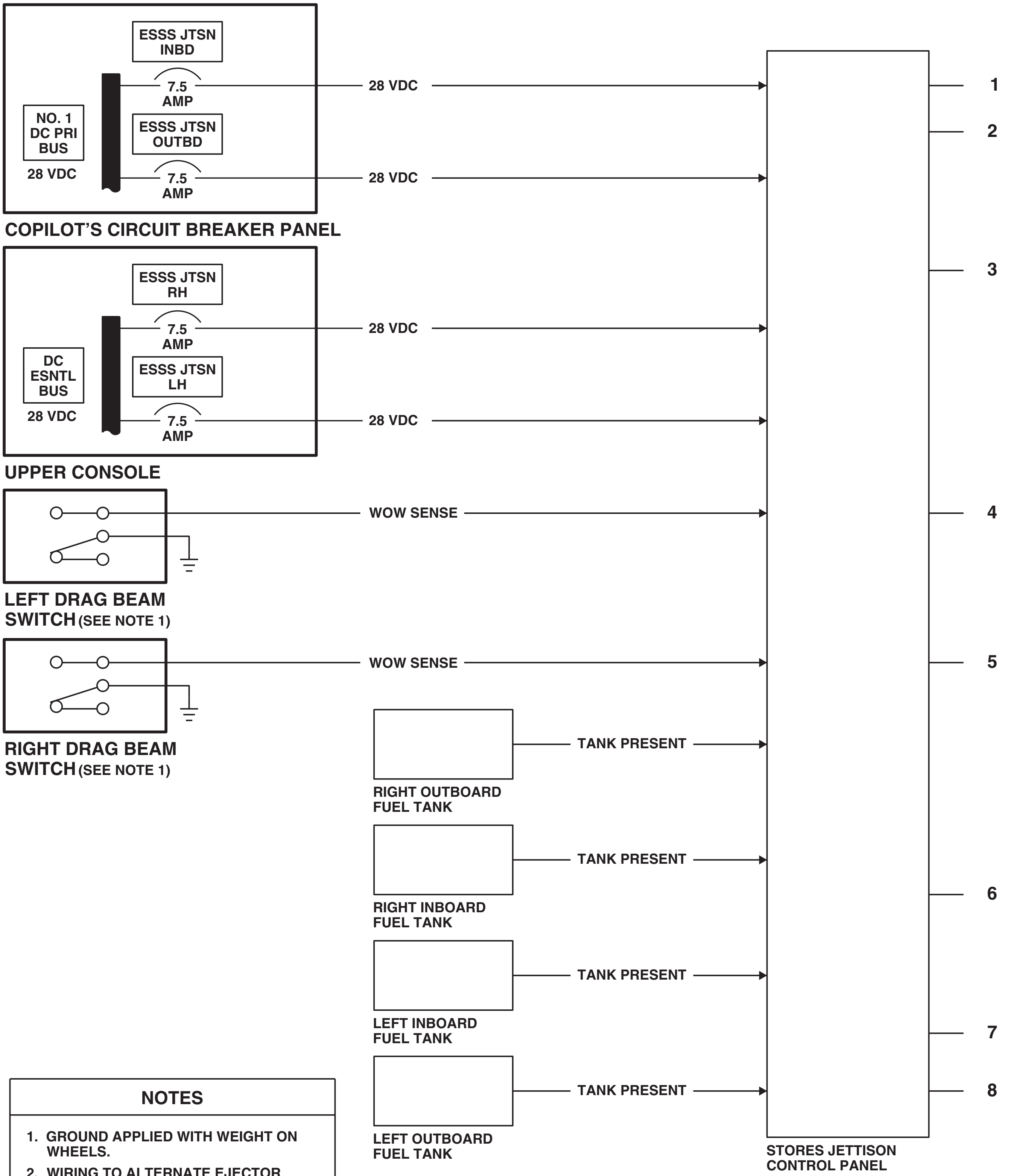
FIGURE 16-1-6-1. EXTERNAL STORES SUPPORT SYSTEM JETTISON SYSTEM LOCATION DIAGRAM (SHEET 4 OF 4)



NOTES

1. GROUND APPLIED WITH WEIGHT ON WHEELS.
2. WIRING TO ALTERNATE EJECTOR RACK IS PROVISIONAL AND NOT USED IN THIS CONFIGURATION.

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FIGURE 16-1-6-2. EXTERNAL STORES SUPPORT SYSTEM JETTISON SYSTEM BLOCK DIAGRAM (SHEET 1 OF 2)

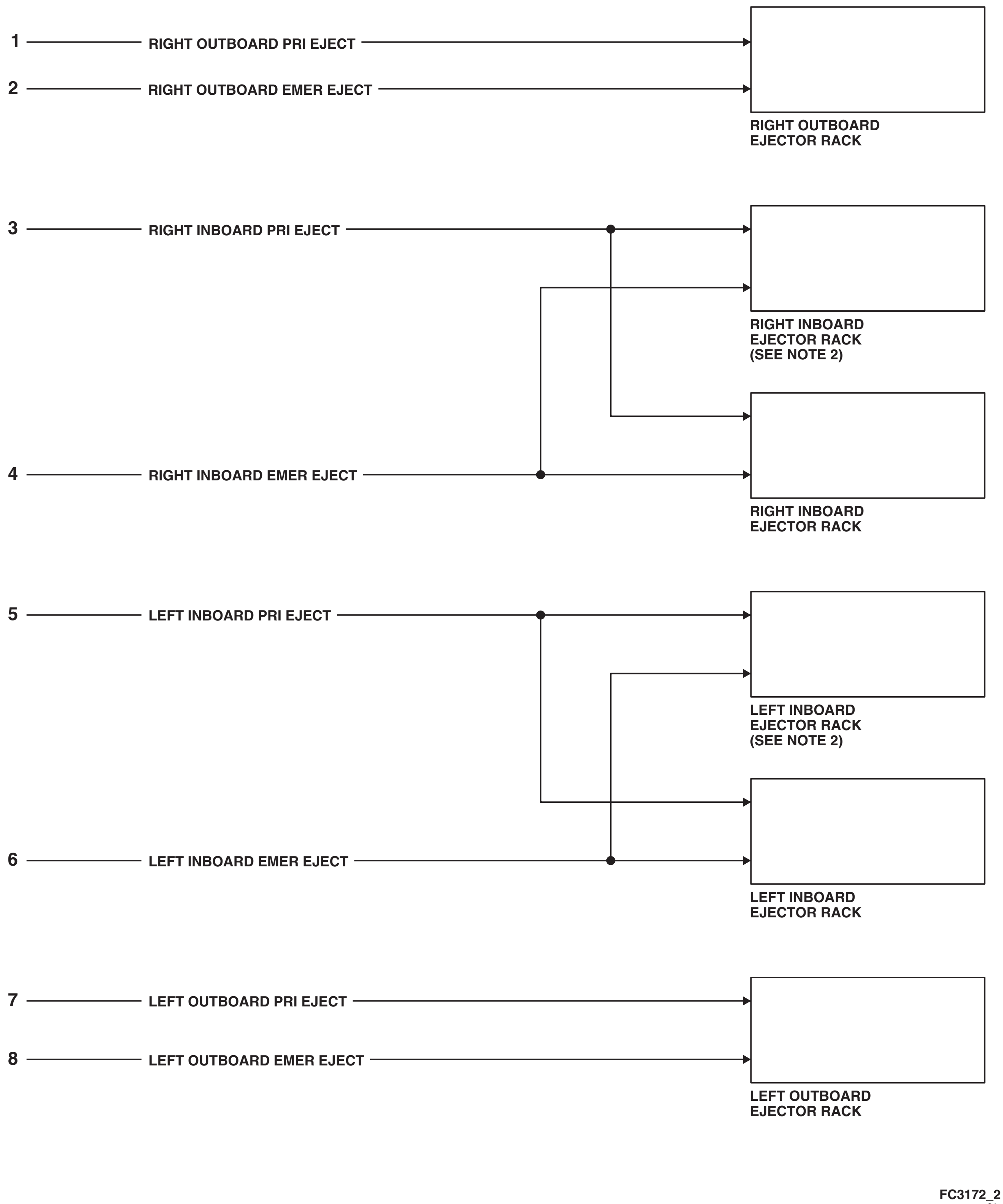


FIGURE 16-1-6-2. EXTERNAL STORES SUPPORT SYSTEM JETTISON SYSTEM BLOCK DIAGRAM (SHEET 2 OF 2)

16-1-7 MAIN AND TAIL ROTOR BLADE EROSION PROTECTION KIT DESCRIPTION AND DATA.

16-1-7.1 System Description.

The main and tail rotor blade erosion protection kit provides surface protection for the main and tail rotor blades when the helicopter is operating in desert environments. The main and tail rotor blades are protected by applying polyurethane tape to the leading edge of the blades. The main rotor blade tip caps are protected by applying either a polyurethane coating or polyurethane boots. The tail rotor blade tip caps are protected by applying a polyurethane coating.

SECTION 16-2

TROUBLESHOOTING

SECTION OVERVIEW

16-2-1 EXTERNAL STORES SUPPORT SYSTEM RANGE EXTENSION SYSTEM.

INITIAL SETUP

Applicable Configuration

- This Fault Isolation Procedure (FIP) Is Designed To Be Utilized When The Horizontal (Wings) And/Or Vertical (Pylons) Stores Are Initially Installed Or Replaced. When Only Tanks Are Removed Refer To PARA 16-2-3.

Test Equipment

- Compressor Unit, MIL-C-13874
- Locally-Made Fuel Quantity Simulator Test Set (simulator) Figure H-152, Appendix H
- Locally-Made External Stores Support System (ESSS) Gate Valve Test Harness, Figure H-156, Appendix H
- Multimeter, Fluke 27/AN
- Multimeter, Fluke 45 (2 required)

Tools

- Aircraft Mechanic's Toolkit
- Container, Suitable
- Electrical Repairer Toolkit
- CU Locally-Made Pneumatic Adapter, Figure H-151, Appendix H
- Locally-Made Drag Beam Wedge, Figure H-167, Appendix H
- Stopwatch, Hart 8500
- Torque Wrench, S-70 Torque Kit

References

- PARA 1-6-16
- PARA 16-4-21
- PARA 16-4-35
- PARA 16-4-47

Equipment Conditions

- External Tanks (Inboard or Outboard) **EMPTY**
- External Electrical Power Available or APU Operational
- Main Fuel Tanks, No More Than **650** Pounds Of Fuel In Each Tank
- Rear Fairings On All Vertical Stores (Pylons) Removed (PARA 16-4-21)
- All Explosive Cartridges Removed From Ejector Racks (PARA 16-4-35)

NOTE

Special icons used in this FIP:

(1) CU Steps that are to be done if compressor unit is used.

(2) INBD Steps that are to be done if inboard tanks are installed.

(3) OUTBD Steps that are to be done if outboard tanks are installed.

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

NOTE

As an alternate to using the compressor unit to pressurize the ESSS Range Extension System, the Aviation Ground Power Unit (AGPU) or the APU may be used. If the alternate is selected, steps with CU may be skipped.

NOTE

The fault isolation procedure will check the ESSS Range Extension System with any tank configuration, installed on the helicopter, authorized by the aircraft operator's manual.

16-2-1.1 Operational/Troubleshooting Procedure.

a. CU Unfasten fasteners on left inboard stores support rear fairing and hinge fairing open to gain access to bleed-air quick-disconnect (QD).

b. CU Disconnect hose from outboard side of bleed-air QD.

c. CU Using pneumatic adapter, tee air compressor into helicopter bleed-air QD and ESSS bleed-air hose.

d. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.

e. Pull out the following circuit breakers:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
ESSS JTSN OUTBD	Copilot's
ESSS JTSN INBD	Copilot's
UTIL RECP CABIN	Copilot's
ESSS JTSN OUTBD	Upper Console (copilot's side)
ESSS JTSN INBD	Upper Console (copilot's side)
EXT FUEL RH	Mission Readiness
EXT FUEL LH	Mission Readiness
AUX FUEL QTY	Mission Readiness
NO. 2 XFER CONTROL	Mission Readiness

NOTE

To ensure proper readings are obtained during operational check of the ESSS fuel system, dip switches on the back of the auxiliary fuel management control panel must be set to the type of fuel used.

f. On back of auxiliary fuel management control panel, verify dip switches are set to the type of fuel used.

g. Disconnect P360 from auxiliary fuel management control panel (control panel).

h. Connect P1 of ESSS gate valve test harness (test harness) to UTIL RCPT +28V receptacle J258 in cabin ceiling.

i. Turn on electrical power (PARA 1-6-16).

j. Push in UTIL RECP CABIN circuit breaker on copilot's circuit breaker panel.

k. Connect test harness pin E to P360-B.

l. Connect pins A and D of test harness to the following pairs of pins on P360, and alternately place test harness switch to OPEN and CLOSE.

TEST HARNESS PIN	P360-VALVE	OPERATION
A	F	LEFT OUTBD - OPEN
D	E	LEFT OUTBD - CLOSED
A	Y	LEFT INBD - OPEN
D	Z	LEFT INBD - CLOSED
A	g	RIGHT OUTBD - OPEN
D	P	RIGHT OUTBD - CLOSED
A	j	RIGHT INBD - OPEN
D	i	RIGHT INBD - CLOSED

RESULT: Gate valves shall open and close as indicated.

- If result is not as specified, connect test harness pins A, D, and E to pins A, D, and E of malfunctioning gate valve. Alternately place test harness switch to OPEN and CLOSED.

  (a) If gate valve operates normally, repair/replace wiring, as required, between P360 and gate valve's electrical connector (Appendix F).

  (b) If gate valve does not operate normally, replace gate valve (PARA 16-4-33).

m. Turn off electrical power (PARA 1-6-16).

n. Connect P360 to control panel.

o. Turn on electrical power (PARA 1-6-16).

p. Adjust upper console INST LT - PILOT FLT control to at least one quarter turn clockwise.

q. Adjust upper console CONSOLE LT - LOWER control to BRT.

RESULT: Information plate lighting on control panel shall go on at full brightness.

- If result is not as specified, troubleshoot upper and lower console lights (PARA 9-2-5).

r. Push in the following circuit breakers:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
AUX FUEL QTY	Mission Readiness
NO. 2 XFER CONTROL	Mission Readiness

s. Momentarily place caution/advisory panel BRT/DIM- TEST switch to BRT/DIM.

RESULT: AUX FUEL QTY POUNDS digital display (digital display) on control panel shall dim.

- If result is not as specified, troubleshoot instrument panel and consoles indicator lights dimming (PARA 9-2-10) or caution/advisory warning system (PARA 8-2-4), as required.

t. Adjust control panel QTY BRIGHTNESS control from DECR to INCR, then back to DECR.

RESULT: Digital display shall go to full brightness, then back to dim.

- If result is not as specified, replace control panel (PARA 16-4-38).

u. Adjust upper console CONSOLE LT - LOWER and INST LT - PILOT FLT controls to OFF.

v. Place control panel switches as indicated:

SWITCH	POSITION
FUEL XFR PRESS OUTBD	OFF
FUEL XFR PRESS INBD	OFF
TANKS	OUTBD
MODE	OFF

SWITCH	POSITION
MANUAL XFR RIGHT	OFF
MANUAL XFR LEFT	OFF
AUX FUEL QTY Rotary	TOTAL

w. Push in the following circuit breakers:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
EXT FUEL RH	Mission Readiness
EXT FUEL LH	Mission Readiness

x. Press and hold control panel TEST button.

RESULT:

(1) Digital display shall indicate 8888.

(2) DEGRADED and VENT SENSOR lights shall illuminate.

- If results (1) and (2) are not as specified, go to TABLE NO. 16-2-1-1.

y. Release TEST button.

RESULT:

(1) Digital display shall cycle the indication 8 from left to right 3 times.

(2) Digital display shall indicate the word GOOD for about 5 seconds.

(3) Digital display shall indicate either 4, 5, or 8 (preset fuel type) in left-most position for about 3 seconds.

(4) Digital display may indicate some value of auxiliary fuel remaining (unknown at this time).

- If result (1) and/or (3) are not as specified, replace control panel (PARA 16-4-38).

- If result (2) is not as specified, but display E01, E04, E05, or E06, replace control panel (PARA 16-4-38).

- If result (2) is not as specified, but displays E02, go to TABLE NO. 16-2-1-2.

- If result (2) is not as specified, but displays E03, go to TABLE NO. 16-2-1-3.

NOTE

Disregard any EXTERNAL indicator displays at this time.

z. Press and hold control panel STATUS button.

RESULT: EXTERNAL RIGHT INBD EMPTY, RIGHT OUTBD EMPTY, LEFT INBD EMPTY, and LEFT OUTBD EMPTY indicators shall go on.

- If result is not as specified, replace control panel (PARA 16-4-38).

aa. Release STATUS button.

RESULT: All control panel indicators and caution/advisory AUX FUEL capsule shall be off.

- If EXTERNAL RIGHT NO FLOW and/or EXTERNAL LEFT NO FLOW indicators are on, replace control panel (PARA 16-4-38).

- If EXTERNAL RIGHT INBD EMPTY, RIGHT OUTBD EMPTY, LEFT INBD EMPTY, or LEFT OUTBD EMPTY indicator is on (tank installed in that position), go to TABLE NO. 16-2-1-4.

- If result is not as specified, go to TABLE NO. 16-2-1-5.

ab. Turn off electrical power (PARA 1-6-16).

ac. OUTBD Fill each outboard tank, if installed, with **160 - 190** pounds of fuel.

ad. INBD Fill each inboard tank, if installed, with **220 - 250** pounds of fuel.

ae. Turn on electrical power (PARA 1-6-16).

af. CU Apply air source, regulated to 25 - 45 psi to helicopter's external tanks.

ag. **Record** CDU MAIN FUEL indication.

ah. OUTBD Place control panel PRESS - OUTBD switch to ON.

ai. INBD Place control panel PRESS - INBD switch to ON.

aj. **Record** time.

ak. Inspect fuel and pneumatic hoses and fittings for signs of leakage or damage.

RESULT: There shall be no signs of leakage or damage.

- If result is not as specified, replace hoses and fittings, as required.

NOTE

- While doing the following procedure, inspect fuel and pneumatic hoses and fittings as specified in step ak.
- The K FACTOR called out in the following steps is a calibration adjustment used for calibrating the fuel system and is stamped on the fuel flow transmitter. Record this value.
- Disregard digit to the right of decimal for K FACTOR entry.

al. **Record** K FACTOR.

am. Pull up and turn control panel AUX FUEL QTY switch to CAL.

an. Place control panel INCR-DECR switch to INCR then to DECR.

RESULT: Digital display indication shall increase, and then decrease.

- If result is not as specified, go to TABLE NO. 16-2-1-6.

ao. Place control panel INCR-DECR switch to either position to enter K FACTOR on digital display.

ap. Using drag beam wedge, push plunger down on left drag beam switch.

aq. Momentarily place control panel INCR-DECR switch to INCR, then DECR.

RESULT: Digital display indication shall not change.

- If result is not as specified, go to TABLE NO. 16-2-1-7.

ar. Remove drag beam wedge.

as. Turn control panel AUX FUEL QTY switch to OUTBD.

at. OUTBD Disconnect P2/J849 and P2/J850.

RESULT: Digital display shall indicate cccc.

- If result is not as specified, replace control panel (PARA 16-4-38).

au. OUTBD Connect P2/J849 and P2/J850 if outboard tanks are installed, or install insulated jumper wire between J1-1 and J1-2.

av. Place control panel INCR-DECR switch to either position to set digital display indication to 1000.

RESULT: Digital display shall indicate 1000 pounds.

- If result is not as specified, replace control panel (PARA 16-4-38).

- If trouble remains, replace left relay panel (PARA 9-4-2).

aw. Turn control panel AUX FUEL QTY switch to INBD.

ax. **INBD** Disconnect P2/J835 and P2/J836.

RESULT: Digital display shall indicate cccc.

- If result is not as specified, replace control panel (PARA 16-4-38).

ay. **INBD** Connect P2/J835 and P2/J836 if inboard tanks are installed, or install insulated jumper wire between inboard J1-1 and J1-2.

az. Place control panel INCR-DECR switch to either position to set digital display indication to 1000 pounds.

RESULT: Digital display shall indicate 1000 pounds.

- If result is not as specified, replace control panel (PARA 16-4-38).

  (a) If trouble remains, replace left relay panel (PARA 9-4-2).

ba. Turn control panel AUX FUEL QTY switch to TOTAL.

RESULT: Digital display shall indicate 2000 pounds.

- If result is not as specified, replace control panel (PARA 16-4-38).

bb. **OUTBD** Turn control panel AUX FUEL QTY switch to OUTBD.

bc. **OUTBD** Place control panel FUEL XFR - MODE switch to MANUAL.

bd. **OUTBD** Check CDU MAIN FUEL indication.

RESULT: CDU MAIN FUEL indication shall be the same as recorded in step ag.

- If result is not as specified, go to TABLE NO. 16-2-1-8.

be. **OUTBD** Before continuing, make sure that at least 5 minutes have elapsed from time recorded in step aj.

bf. **OUTBD** Place control panel MANUAL XFR - RIGHT switch to ON until CDU TOTAL FUEL indication increases by 20 pounds.

RESULT:

(1) Elapsed time required to increase indication by 20 pounds shall be less than 1 1/2 minutes.

NOTE

Disregard AUX FUEL capsule on the caution/advisory panel if it goes on momentarily.

(2) Control panel EXTERNAL RIGHT NO FLOW indicator shall momentarily go on, then go off.

(3) Remaining control panel EXTERNAL indicators shall remain off.

(4) Control panel digital display indication shall decrease.

- If results are not as specified, go to TABLE NO. 16-2-1-9.

bg. **OUTBD** Place control panel MANUAL XFR - RIGHT switch to OFF.

bh. **OUTBD** Place control panel MANUAL XFR - LEFT switch to ON until CDU TOTAL FUEL indication increases by 20 pounds.

RESULT:

(1) Elapsed time required to increase indication by 20 pounds shall be less than 1 1/2 minutes.

NOTE

Disregard AUX FUEL capsule on the caution/advisory panel if it goes on momentarily.

(2) Control panel EXTERNAL LEFT NO FLOW indicator shall momentarily go on, then go off.

(3) Remaining control panel EXTERNAL indicators shall remain off.

(4) Control panel digital display indication shall decrease.

- If results are not as specified, go to TABLE NO. 16-2-1-10.

bi. OUTBD Place control panel MANUAL XFER - LEFT switch to OFF.

bj. INBD Turn control panel AUX FUEL QTY switch to INBD.

bk. INBD Place control panel FUEL XFR - TANKS switch to INBD.

RESULT: Digital display indication shall not change.

- If result is not as specified, replace control panel (PARA 16-4-38).

bl. INBD Place control panel MANUAL XFR - RIGHT switch to ON until CDU MAIN FUEL indication increases by 20 pounds.

RESULT:

(1) Elapsed time required to increase indication by 20 pounds shall be less than 1 1/2 minutes.

NOTE

Disregard AUX FUEL capsule on the caution/advisory panel if it goes on momentarily.

(2) Control panel EXTERNAL RIGHT NO FLOW indicator shall momentarily go on, then go off.

(3) Remaining control panel EXTERNAL indicators shall remain off.

(4) Control panel digital display indication shall decrease.

- If CDU MAIN FUEL indication does not increase, and/or results are not as specified, go to TABLE NO. 16-2-1-11.

bm. INBD Place control panel MANUAL XFR - RIGHT switch to OFF.

bn. INBD Place control panel MANUAL XFR - LEFT switch to ON until CDU TOTAL FUEL indication increases by 20 pounds.

RESULT:

(1) Elapsed time required to increase indication by 20 pounds shall be less than 1 1/2 minutes.

NOTE

Disregard AUX FUEL capsule on the caution/advisory panel if it goes on momentarily.

(2) Control panel EXTERNAL LEFT NO FLOW indicator shall momentarily go on, then go off.

(3) Remaining control panel EXTERNAL indicators shall remain off.

(4) Control panel digital display indication shall decrease.

- If results are not as specified, go to TABLE NO. 16-2-1-12.

bo. INBD Place control panel MANUAL XFR - LEFT switch to OFF.

bp. OUTBD Place control panel FUEL XFR - TANKS and AUX FUEL QTY switches to OUTBD.

bq. OUTBD **Record** CDU MAIN FUEL indication.

br. OUTBD **Record** control panel digital display indication.

bs. OUTBD Place control panel MANUAL XFR - LEFT and RIGHT switches to ON.

bt. OUTBD When CDU MAIN FUEL indication is 100 pounds more than value recorded in step br., place control panel FUEL XFR - MODE switch to OFF.

bu. OUTBD Place control panel MANUAL XFR - LEFT and RIGHT switches to OFF.

bv. OUTBD **Record** control panel digital display indication.

bw. OUTBD Subtract recorded indication of step bv. from recorded indication of step br.

RESULT: The difference shall be between 90 and 110 pounds.

- If result is not as specified, go to TABLE NO. 16-2-1-13.

bx. Turn off electrical power (PARA 1-6-16).

by. Connect locally-made fuel quantity test simulator (simulator) power cord to UTIL RCPT +28V (J258) in cabin ceiling.

bz. Disconnect helicopter P905 from J905.

ca. Connect simulator J905 to helicopter P905.

cb. Turn simulator TANK 1 and TANK 2 controls fully counterclockwise.

cc. Connect No. 1 digital multimeter to simulator TANK 1 OUTPUT.

cd. Connect No. 2 digital multimeter to simulator TANK 2 OUTPUT.

ce. Turn on electrical power (PARA 1-6-16).

cf. Turn simulator TANK 1 control to obtain 6.0 vdc No. 1 multimeter indication.

cg. Turn simulator TANK 2 control to obtain 6.0 vdc No. 2 multimeter indication.

ch. Turn control panel AUX FUEL QTY switch to INBD.

ci. Place control panel INCR-DECR switch to INCR to obtain display of 6000 pounds.

cj. Turn control panel AUX FUEL QTY switch to OUTBD.

ck. Place control panel INCR-DECR switch to INCR to obtain display of 1500 pounds.

cl. Place control panel FUEL XFR - MODE switch to AUTO.

RESULT: Digital display indication shall not change.

- If result is not as specified, replace control panel (PARA 16-4-38).

NOTE

- Read steps cm. through cq. completely before continuing.
- Do steps cm. through cq. as quickly as possible.

cm. Slowly turn simulator TANK 1 control counterclockwise until control panel digital display indication begins to decrease.

cn. **Record** No. 1 multimeter indication.

co. Slowly turn simulator TANK 1 control clockwise until control panel digital display indication stops changing.

cp. **Record** No. 1 multimeter indication.

RESULT:

(1) Recorded indication of step cn. shall be between 4.65 and 4.85 vdc.

(2) Recorded indication of step cp. shall be between 4.9 and 5.1 vdc.

(3) EXTERNAL RIGHT NO FLOW indicator on control panel shall remain off during fuel transfer.

(4) EXTERNAL LEFT NO FLOW indicator on control panel shall remain off during fuel transfer.

- If result (1) and/or (2) are not as specified, go to TABLE NO. 16-2-1-14.
- If result (3) is not as specified, go to TABLE NO. 16-2-1-15.
- If result (4) is not as specified, go to TABLE NO. 16-2-1-16.

cq. Turn simulator TANK 1 control clockwise to obtain 6.0 vdc No. 1 multimeter indication.

NOTE

- Read steps cr. through cu. completely before continuing.
- Do steps cr. through cu. as quickly as possible.

cr. Slowly turn simulator TANK 2 control counterclockwise until control panel digital display indication begins to decrease.

cs. **Record** No. 2 multimeter indication.

ct. Slowly turn simulator TANK 2 control clockwise until control panel digital display indication stops changing.

cu. **Record** No. 2 multimeter indication.

RESULT:

(1) Recorded indication of step cs. shall be between 4.65 and 4.85 vdc.

(2) Recorded indication of step cu. shall be between 4.9 and 5.1 vdc.

(3) EXTERNAL RIGHT NO FLOW indicator on control panel shall remain off during fuel transfer.

(4) EXTERNAL LEFT NO FLOW indicator on control panel shall remain off during fuel transfer.

- If results (1) and/or (2) are not as specified, go to TABLE NO. 16-2-1-14.
- If result (3) is not as specified, go to TABLE NO. 16-2-1-15.
- If result (4) is not as specified, go to TABLE NO. 16-2-1-16.

cv. Turn simulator TANK 2 control to obtain 6.0 vdc No. 2 multimeter indication.

cw. Place control panel FUEL XFR - MODE switch to OFF.

cx. Place control panel INCR-DECR switch to either position to obtain digital display indication of 1000 pounds.

cy. Place control panel FUEL XFR - MODE switch to AUTO.

RESULT: Digital display indication shall not change.

- If result is not as specified, replace control panel (PARA 16-4-38).

NOTE

- Read steps cz. through dd. completely before continuing.
- Do steps cz. through dd. as quickly as possible.

cz. Slowly turn simulator TANK 1 control counterclockwise until control panel digital display indication begins to decrease.

da. **Record** No. 1 multimeter indication.

db. Slowly turn simulator TANK 1 control clockwise until control panel digital display indication stops changing.

dc. **Record** No. 1 multimeter indication.

RESULT:

(1) Recorded indication of step da. shall be between 3.65 and 3.85 vdc.

(2) Recorded indication of step dc. shall be between 4.9 and 5.1 vdc.

- If results are not as specified, replace control panel (PARA 16-4-38).

dd. Turn simulator TANK 1 control clockwise to obtain 6.0 vdc No. 1 multimeter indication.

NOTE

- Read steps de. through dh. completely before continuing.
- Do steps de. through dh. as quickly as possible.

de. Slowly turn simulator TANK 2 control counterclockwise until control panel digital display indication begins to decrease.

df. **Record** No. 2 multimeter indication.

dg. Slowly turn simulator TANK 2 counterclockwise until control panel digital display indication stops changing.

dh. **Record** No. 2 multimeter indication.

RESULT:

(1) Recorded indication of step df. shall be between 3.65 and 3.85 vdc.

(2) Recorded indication of step dh. shall be between 4.9 and 5.1 vdc.

- If results are not as specified, replace control panel (PARA 16-4-38).

di. Turn simulator TANK 2 control clockwise to obtain 6.0 vdc No. 2 multimeter indication.

dj. Place control panel FUEL XFR - MODE switch to OFF.

dk. Turn control panel AUX FUEL QTY switch to INBD.

dl. Place control panel INCR-DECR switch to either position to obtain digital display indication of 2000 pounds.

dm. Turn control panel AUX FUEL QTY switch to OUTBD.

dn. Place control panel FUEL XFR - MODE switch to AUTO.

RESULT: Digital display indication shall not change.

- If result is not as specified, replace control panel (PARA 16-4-38).

NOTE

- Read steps do. through ds. completely before continuing.
- Do steps do. through ds. as quickly as possible.

do. Slowly turn simulator TANK 1 control counterclockwise until control panel digital display indication begins to decrease.

dp. **Record** No. 1 multimeter indication.

dq. Slowly turn simulator TANK 1 control clockwise until control panel digital display indication stops changing.

dr. **Record** No. 1 multimeter indication.

RESULT:

(1) Recorded indication of step dp. shall be between 2.9 and 3.1 vdc.

(2) Recorded indication of step dr. shall be between 4.9 and 5.1 vdc.

- If results are not as specified, replace control panel (PARA 16-4-38).

ds. Slowly turn simulator TANK 1 control to obtain 6.0 vdc No. 1 multimeter indication.

NOTE

- Read steps dt. through dw. completely before continuing.
- Do steps dt. through dw. as quickly as possible.

dt. Slowly turn simulator TANK 2 control counterclockwise until control panel digital display indication begins to decrease.

du. **Record** No. 2 multimeter indication.

dv. Slowly turn simulator TANK 2 control clockwise until control panel digital display indication stops changing.

dw. **Record** No. 2 multimeter indication.

RESULT:

(1) Recorded indication of step du. shall be between 2.9 and 3.1 vdc.

(2) Recorded indication of step dw. shall be between 4.9 and 5.1 vdc.

- If results are not as specified, replace control panel (PARA 16-4-38).

dx. Turn simulator TANK 2 control to obtain 2.7 vdc No. 2 multimeter indication.

dy. Place control panel FUEL XFR - MODE switch to OFF.

dz. INBD Place control panel FUEL XFR - TANKS switch to INBD.

ea. INBD Place control panel FUEL XFR - MODE switch to AUTO until EXTERNAL RIGHT INBD EMPTY and LEFT INBD EMPTY indicators go on.

RESULT:

(1) EXTERNAL RIGHT INBD EMPTY and LEFT INBD EMPTY indicators shall go on.

(2) Caution/advisory panel AUX FUEL capsule shall go on immediately after either INBD EMPTY indicator goes on.

(3) Control panel EXTERNAL RIGHT NO FLOW and LEFT NO FLOW indicators shall go on immediately after the corresponding INBD EMPTY indicator goes on.

- If result (1) is not as specified, go to TABLE NO. 16-2-1-17.
- If result (2) and/or (3) is not as specified, go to TABLE NO. 16-2-1-18.

eb. INBD Turn simulator TANK 2 control to obtain 6.0 vdc No. 2 multimeter indication.

ec. INBD Turn simulator TANK 1 control to obtain 6.0 vdc No. 1 multimeter indication.

ed. INBD Place control panel FUEL XFR - MODE switch to MANUAL.

ee. INBD Place control panel MANUAL XFR - LEFT and RIGHT switches to ON.

RESULT:

(1) Control panel EXTERNAL RIGHT NO FLOW and LEFT NO FLOW indicators shall go off, then go on within 5 minutes.

(2) Control panel digital display indication shall not change when EXTERNAL RIGHT NO FLOW and LEFT NO FLOW indicators are on.

- If results are not as specified, replace corresponding fuel tank (PARA 16-4-54).

ef. INBD Momentarily press control panel STATUS button.

RESULT:

(1) Control panel EXTERNAL INBD EMPTY and OUTBD EMPTY indicators shall go off.

(2) Caution/advisory panel AUX FUEL capsule shall go off.

- If results are not as specified, go to TABLE NO. 16-2-1-5.

eg. INBD Turn simulator TANK 2 control to obtain 2.7 vdc No. 2 multimeter indication.

eh. OUTBD Make sure control panel FUEL XFR - MODE switch to OFF.

ei. OUTBD Place control panel FUEL XFR - TANKS switch to OUTBD.

ej. OUTBD Place control panel FUEL XFR - MODE switch to AUTO until EXTERNAL RIGHT OUTBD EMPTY and LEFT OUTBD EMPTY indicators go on.

RESULT:

(1) EXTERNAL RIGHT OUTBD EMPTY and LEFT OUTBD EMPTY indicators shall go on.

(2) Caution/advisory panel AUX FUEL capsule shall go on immediately after either OUTBD EMPTY indicator goes on.

(3) Control panel EXTERNAL RIGHT NO FLOW and LEFT NO FLOW indicators shall go on immediately after corresponding tank OUTBD EMPTY indicator goes on.

- If result (1) is not as specified, go to TABLE NO. 16-2-1-19.

- If result (2) and/or (3) is not as specified, go to TABLE NO. 16-2-1-18.

ek. OUTBD Momentarily press control panel STATUS button.

RESULT:

(1) Control panel EXTERNAL INBD EMPTY and OUTBD EMPTY indicators shall go off.

(2) Caution/advisory panel AUX FUEL capsule shall go off.

- If results are not as specified, go to TABLE NO. 16-2-1-5.

el. OUTBD Place control panel FUEL XFR - MODE switch to MANUAL.

em. OUTBD Place control panel MANUAL XFR - LEFT and RIGHT switches to ON.

RESULT:

(1) Control panel EXTERNAL RIGHT NO FLOW and LEFT NO FLOW indicators shall go off, then come back on within 5 minutes.

(2) Control panel digital display indication shall not change when the EXTERNAL RIGHT NO FLOW and LEFT NO FLOW indicators are on.

- If results are not as specified, replace corresponding fuel tank (PARA 16-4-54).

en. Place control panel FUEL XFR - MODE, MANUAL XFR - RIGHT and LEFT, and PRESS - INBD and OUTBD switches to OFF.

RESULT: All four ESSS bleed-air regulator valves shall be venting air.

- If result is not as specified, disconnect connector from bleed-air regulator.

(a) If bleed-air valve vents air, replace control panel (PARA 16-4-38).

(b) If bleed-air valve does not vent air, clean valve, then do steps CU af., OUTBD ah., INBD ai., and en.

1 If trouble remains, replace bleed-air regulator valve (PARA 16-4-32).

eo. Pull out NO. 2 XFER CONTROL circuit breaker on mission readiness circuit breaker panel.

ep. Remove overflow sensor from vent tube elbow in rear of main fuel tanks (PARA 16-4-47).

eq. Disconnect P376 from J376.

er. Push in circuit breaker pulled out in step eo.

RESULT: Control panel VENT SENSOR FAIL indicator shall go on.

- If result is not as specified, go to TABLE NO. 16-2-1-20.

es. Pull out circuit breaker pushed in step er.

et. Connect P376 to J376.

eu. Immerse thermistor portion of overflow sensor in container of water.

ev. Push in circuit breaker pulled out in step es.

RESULT: Control panel VENT SENSOR OVFL indicator shall go on.

- If result is not as specified, replace control panel (PARA 16-4-38).

ew. Install overflow sensor (PARA 16-4-47).

ex. Turn off electrical power (PARA 1-6-16).

ey. CU Turn off air source.

ez. Allow bleed-air regulator valves to vent air to ambient pressure.

fa. CU Disconnect air compressor, hoses, and locally-made pneumatic adapter.

fb. CU Connect pneumatic hose to outboard side of helicopter bleed-air QD.

fc. CU **TORQUE NUTS TO 135 - 150 INCH- POUNDS.**

fd. Disconnect test equipment.

fe. Connect helicopter electrical connectors.

ff. Install explosive cartridges (PARA 16-4-35).

fg. Install rear fairing on all vertical stores pylons (PARA 16-4-21).

TABLE NO. 16-2-1-1. When Control Panel TEST Button Is Pressed, All Control Panel Indicators Are Not On And/Or Digital Display Does Not Indicate 8888.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Does digital display indicate 8888?

Step 1. If 8888 is displayed, replace applicable auxiliary fuel management control panel indicator lamp (PARA 16-4-41). Go to **6.**

TABLE NO. 16-2-1-1. When Control Panel TEST Button Is Pressed, All Control Panel Indicators Are Not On And/Or Digital Display Does Not Indicate 8888. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If trouble remains, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **6.**
Step 3. If 8888 is not displayed, go to **2.**

2. Do some indicators go on?

Step 1. If indicators go on, replace auxiliary fuel management control panel digital display segment (PARA 16-4-41). Go to **6.**
Step 2. If trouble remains, replace auxiliary fuel management control panel (PARA 16-4-40). Go to **6.**
Step 3. If indicators do not go on, go to **3.**

3. Check for 28 vdc between P360-W and ground.

Step 1. If voltage is as specified, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **6.**
Step 2. If voltage is not as specified, go to **4.**

4. Check for 28 vdc between NO. 2 XFER CONTROL circuit breaker terminal 1 and ground.

Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **6.**

5. Check for 28 vdc between J346-35 and ground.

Step 1. If voltage is as specified, replace cabin fuel harness (PARA 16-4-43). Go to **6.**
Step 2. If voltage is not as specified, repair/replace wiring between J346-35 and circuit breaker terminal 1 (Appendix F). Go to **6.**

6. Procedure completed.

TABLE NO. 16-2-1-2. During Self-Test, Control Panel Digital Display Indicates E02 Instead Of GOOD.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for 115 vac between P360-f̲ and ground.

TABLE NO. 16-2-1-2. During Self-Test, Control Panel Digital Display Indicates E02 Instead Of GOOD. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, go to **2.**
Step 2. If voltage is not as specified, go to **4.**

2. Check continuity between:
P361-U and J524-2
P361-S and J524-1

Step 1. If continuity is present, replace fuel temperature transducer (PARA 16-4-30). Go to **6.**
Step 2. If continuity is not present, go to **3.**

3. Check continuity between:
P346-20 and P361-U
P346-21 and P361-S

Step 1. If continuity is present, repair/replace wiring between, then go to **6:**
J346-20 and J524-2
J346-21 and J524-1
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43). Go to **6.**

4. Check for 115 vac between AUX FUEL QTY circuit breaker terminal 1 and ground.

Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **6.**

5. Check for 115 vac between J346-38 and ground.

Step 1. If voltage is as specified, replace cabin fuel harness (PARA 16-4-43). Go to **6.**
Step 2. If voltage is not as specified, repair/replace wiring between J346-38 and circuit breaker terminal 1 (Appendix F). Go to **6.**

6. Procedure completed.

TABLE NO. 16-2-1-3. During Self-Test, Control Panel Digital Display Indicates E03 Instead Of GOOD.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check continuity between:**
   P361-X and P523-A
   P361-Y and P523-B

   Step 1. If continuity is present, go to **2.**
   Step 2. If continuity is not present, go to **3.**

2. **Check for 2,000 ohms to 3,000 ohms between fuel flow transmitter connector P523-A and P523-B.**

   Step 1. If resistance is as specified, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **4.**
   Step 2. If resistance is not as specified, replace fuel flow transmitter (PARA 16-4-31). Go to **4.**

3. **Check continuity between:**
   P361-X and P346-23
   P361-Y and P346-24

   Step 1. If continuity is present, repair/replace wiring between, then go to **4:**
   J346-23 and P523-A
   J346-24 and P523-B
   Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43). Go to **4.**

4. **Procedure completed.**

TABLE NO. 16-2-1-4. Control Panel EMPTY Indicator Is On With Adequate Fuel In Fuel Tank.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check for infinite resistance between these pins on applicable fuel tank connector:**
   3 and 5
   4 and 3

   Step 1. If resistance is as specified, go to **2.**
   Step 2. If resistance is not as specified, replace fuel quantity sensor harness (PARA 16-4-65). Go to **3.**

TABLE NO. 16-2-1-4. Control Panel EMPTY Indicator Is On With Adequate Fuel In Fuel Tank. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 3. If trouble remains, replace and send fuel tank to designated overhaul facility (PARA 16-4-54). Go to **3.**

2. Check continuity between:

J1-3 (Left Outboard) and P361-G
J1-4 (Left Outboard) and P361-F
J1-5 (Left Outboard) and P361-E
J1-3 (Left Inboard) and P361-D
J1-4 (Left Inboard) and P361-C
J1-5 (Left Inboard) and P361-B
J1-3 (Right Outboard) and P361-K
J1-4 (Right Outboard) and P361-L
J1-5 (Right Outboard) and P361-M
J1-3 (Right Inboard) and P361-N
J1-4 (Right Inboard) and P361-P
J1-5 (Right Inboard) and P361-R

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **3.**
Step 2. If continuity is not present, replace rear electrical harness (PARA 16-4-44) or cabin fuel harness (PARA 16-4-43), as required. Go to **3.**

3. Procedure completed.

TABLE NO. 16-2-1-5. When Control Panel STATUS Button Is Pressed And Then Released, Control Panel Indicators And/Or Caution/Advisory Panel AUX FUEL Capsule Is Not Off.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Are all control panel indicators off?

Step 1. If all indicators are off, go to **2.**
Step 2. If all indicators are not off, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **8.**

2. Check helicopter configuration.

Step 1. **EMEP** , go to **3.**
Step 2. **W/O EMEP** , go to **5.**

TABLE NO. 16-2-1-5. When Control Panel STATUS Button Is Pressed And Then Released, Control Panel Indicators And/Or Caution/Advisory Panel AUX FUEL Capsule Is Not Off. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

EMEP **3. Remove and check pin filtered adapter P117 (PARA 9-4-98).**

Step 1. If pin filtered adapter is good, go to **4.**
Step 2. If pin filtered adapter is not good, replace pin filtered adapter (PARA 9-4-98). Go to **8.**

EMEP **4. Install pin filtered adapter (PARA 9-4-98). Go to 5.**

5. Check for 28 vdc between P117-41 and ground.

Step 1. If voltage is as specified, replace caution/advisory panel (PARA 8-4-19). Go to **8.**
Step 2. If voltage is not as specified, go to **6.**

6. Check continuity between P117-41 and P360-X.

Step 1. If continuity is present, replace auxiliary fuel management contorl panel (PARA 16-4-38). Go to **8.**
Step 2. If continuity is not present, go to **7.**

7. Check continuity between P346-10 and P360-X.

Step 1. If continuity is present, repair/replace wiring between J346-10 and P117-41. Go to **8.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43). Go to **8.**

8. Procedure completed.

TABLE NO. 16-2-1-6. With Weight-On-Wheels, Auxiliary Fuel Management Control Panel CAL Indication Cannot Be Changed.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check continuity between P360-r and P360-n.

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **4.**

TABLE NO. 16-2-1-6. With Weight-On-Wheels, Auxiliary Fuel Management Control Panel CAL Indication Cannot Be Changed. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If continuity is not present, go to **2.**

2. Check continuity between:
P360-r and P902-C
P360-n and P902-D

Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to **4.**
Step 2. If continuity is not present, go to **3.**

3. Check continuity between:
P346-2 and P360-n
P346-3 and P360-r

Step 1. If continuity is present, repair/replace wiring (Appendix F), as required between, then go to **4:**
J346-2 and P902-D
J346-3 and P902-C
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43). Go to **4.**

4. Procedure completed.

TABLE NO. 16-2-1-7. Digital Display Indication Changes With Drag Beam Switch In In-flight Position.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check continuity between P360-r and P360-n.

Step 1. If continuity is present, go to **2.**
Step 2. If continuity is not present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **8.**

2. Check continuity between J163-A and ground.

Step 1. If continuity is present, go to **3.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **8.**

3. Remove drag beam wedge from left drag beam switch. Go to 4.

TABLE NO. 16-2-1-7. Digital Display Indication Changes With Drag Beam Switch In In-flight Position. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

4. Check continuity between P163-A and P163-C.

Step 1. If continuity is present, replace left drag beam switch (PARA 3-4-6). Go to **8.**
Step 2. If continuity is not present, go to **5.**

5. Check continuity between J163-C and P242-L.

Step 1. If continuity is present, go to **6.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **8.**

6. Check for 28 vdc between P902-CC and ground.

Step 1. If voltage is as specified, replace left relay panel (PARA 9-4-2). Go to **8.**
Step 2. If voltage is not as specified, go to **7.**

7. Check for 28 vdc between BACKUP HYD CONTR circuit breaker terminal 1 and ground.

Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P902-CC (Appendix F). Go to **8.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **8.**

8. Procedure completed.

TABLE NO. 16-2-1-8. CDU MAIN FUEL Indication Increases When Fuel Transfer Is Not Actuated.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Is the manual lever of all the four shutoff gate valves in the full CLSD position?

Step 1. If manual levers are in the full CLSD position, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **9.**
Step 2. If manual levers are not in the full CLSD position, go to **2.**

TABLE NO. 16-2-1-8. CDU MAIN FUEL Indication Increases When Fuel Transfer Is Not Actuated. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

2. For any valve that manual lever is not in full CLSD position, check for 28 vdc between:

P840-D and P840-E (Left Outboard)
P809-D and P809-E (Left Inboard)
P838-D and P838-E (Right Outboard)
P810-D and P810-E (Right Inboard)

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, go to **4.**

3. Check for 0 vdc between:

P840-A and P840-E (Left Outboard)
P809-A and P809-E (Left Inboard)
P838-A and P838-E (Right Outboard)
P810-A and P810-E (Right Inboard)

Step 1. If voltage is as specified, replace shutoff gate valve (PARA 16-4-33). Go to **9.**
Step 2. If voltage is not as specified, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **9.**

4. Check for which valve voltage was not as specified.

Step 1. If left outboard valve, go to **5.**
Step 2. If left inboard valve, go to **6.**
Step 3. If right inboard valve, go to **7.**
Step 4. If right outboard valve, go to **8.**

5. For left outboard valve, check continuity between:

P840-A and P360-F
P840-D and P360-E
P840-E and ground

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **9.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or rear electrical harness (PARA 16-4-44), as required. Go to **9.**

6. For left inboard valve, check continuity between:

TABLE NO. 16-2-1-8. CDU MAIN FUEL Indication Increases When Fuel Transfer Is Not Actuated. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

P809-A and P360-Y
P809-D and P360-Z
P809-E and ground

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **9.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or rear electrical harness (PARA 16-4-44), as required. Go to **9.**

7. **For right inboard valve, check continuity between:**
P810-A and P360-j
P810-D and P360-i
P810-E and ground

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **9.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or rear electrical harness (PARA 16-4-44), as required. Go to **9.**

8. **For right outboard valve, check continuity between:**
P838-A and P360-g
P838-D and P360-P
P838-E and ground

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **9.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or rear electrical harness (PARA 16-4-44), as required. Go to **9.**

9. **Procedure completed.**

TABLE NO. 16-2-1-9. With MANUAL Mode Selected, CDU MAIN FUEL Indication Does Not Increase And/Or Result Is Not As Specified When Attempting To Transfer Fuel From Right Outboard Fuel Tank.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Does CDU MAIN FUEL indication increase?**

TABLE NO. 16-2-1-9. With MANUAL Mode Selected, CDU MAIN FUEL Indication Does Not Increase And/Or Result Is Not As Specified When Attempting To Transfer Fuel From Right Outboard Fuel Tank. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If CDU MAIN FUEL indication does increase, go to **2.**
Step 2. If CDU MAIN FUEL indication does not increase, go to **7.**

2. Does control panel digital display indication decrease?

Step 1. If control panel digital display indication does decrease, go to **3.**
Step 2. If control panel digital display indication does not decrease, go to **6.**

3. Does control panel EXTERNAL RIGHT NO FLOW indicator and caution/ advisory panel AUX FUEL capsule momentarily go on, then off?

Step 1. If indicator and capsule momentarily go on, then off, go to **4.**
Step 2. If indicator and capsule do not momentarily go on, then off, go to **5.**

4. Do the remaining control panel EXTERNAL indicators remain off?

Step 1. If remaining indicators remain off, go to **TABLE NO. 16-2-1-13.**
Step 2. If remaining indicators do not remain off, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **21.**

5. Does control panel RIGHT NO FLOW indicator remain on?

Step 1. If indicator remains on, go to **TABLE NO. 16-2-1-15.**
Step 2. If indicator does not remain on, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **21.**

6. With control panel MANUAL XFR RIGHT switch at ON, check for more than 20 mvac between fuel flow transmitter connector pins A and B.

Step 1. If voltage is as specified, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **21.**
Step 2. If voltage is not as specified, replace fuel flow transmitter (PARA 16-4-31). Go to **21.**

7. Does control panel digital display indication decrease?

Step 1. If indication decreases, go to **8.**

TABLE NO. 16-2-1-9. With MANUAL Mode Selected, CDU MAIN FUEL Indication Does Not Increase And/Or Result Is Not As Specified When Attempting To Transfer Fuel From Right Outboard Fuel Tank. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If indication does not decrease, go to **9.**

8. Does RIGHT NO FLOW indicator go on?

Step 1. If indicator is on, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **21.**
Step 2. If indicator is not on, troubleshoot fuel quantity system (PARA 10-2-6 or PARA 10-2-5). Go to **21.**

9. Is manual lever of right outboard fuel shutoff gate valve in full OPEN position?

Step 1. If manual lever is in full OPEN position, go to **10.**
Step 2. If manual lever is not in full OPEN position, go to **12.**

10. Check for 28 vdc between P837-A and P837-B.

Step 1. If voltage is as specified, replace right outboard bleed-air regulator valve (PARA 16-4-32). Go to **21.**
Step 2. If voltage is not as specified, go to **11.**

11. Check continuity between:
P837-A and P360-R
P837-B and ground

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **21.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or rear electrical harness (PARA 16-4-44), as required. Go to **21.**

12. Is right outboard bleed-air regulator valve continuously venting air?

Step 1. If valve is continuously venting air, go to **13.**
Step 2. If valve is not continuously venting air, go to **18.**

13. Check for 28 vdc between P360-V and ground.

Step 1. If voltage is as specified, go to **14.**
Step 2. If voltage is not as specified, go to **16.**

TABLE NO. 16-2-1-9. With MANUAL Mode Selected, CDU MAIN FUEL Indication Does Not Increase And/Or Result Is Not As Specified When Attempting To Transfer Fuel From Right Outboard Fuel Tank. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

14. Check for 28 vdc between P837-A and P837-B.

Step 1. If voltage is as specified, replace right outboard bleed-air regulator valve (PARA 16-4-32). Go to **21.**
Step 2. If voltage is not as specified, go to **15.**

15. Check continuity between:
P837-A and P360-R
P837-B and ground

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **21.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or rear electrical harness (PARA 16-4-44), as required. Go to **21.**

16. Check for 28 vdc between EXT FUEL RH circuit breaker terminal 1 and ground.

Step 1. If voltage is as specified, go to **17.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **21.**

17. Check for 28 vdc between J346-33 and ground.

Step 1. If voltage is as specified, replace cabin fuel harness (PARA 16-4-43). Go to **21.**
Step 2. If voltage is not as specified, repair/replace wiring between J346-33 and EXT FUEL RH circuit breaker terminal 1 (Appendix F). Go to **21.**

18. Check for 28 vdc between P838-A and P838-E.

Step 1. If voltage is as specified, go to **19.**
Step 2. If voltage is not as specified, go to **20.**

19. Check for 0 vdc between P838-D and P838-E.

Step 1. If voltage is as specified, replace right outboard fuel shutoff gate valve (PARA 16-4-33). Go to **21.**

TABLE NO. 16-2-1-9. With MANUAL Mode Selected, CDU MAIN FUEL Indication Does Not Increase And/Or Result Is Not As Specified When Attempting To Transfer Fuel From Right Outboard Fuel Tank. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If voltage is not as specified, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **21.**

20. Check continuity between:
P838-A and P360-g
P838-D and P360-P
P838-E and ground

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **21.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or rear electrical harness (PARA 16-4-44), as required. Go to **21.**

21. Procedure completed.

TABLE NO. 16-2-1-10. With MANUAL Mode Selected, CDU MAIN FUEL Indication Does Not Increase And/Or Result Is Not As Specified When Attempting To Transfer Fuel From Left Outboard Fuel Tank.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Does CDU MAIN FUEL indication increase?

Step 1. If indication does increase, go to **2.**
Step 2. If indication does not increase, go to **7.**

2. Does control panel digital display indication decrease?

Step 1. If indication does decrease, go to **3.**
Step 2. If indication does not decrease, go to **6.**

3. Does control panel EXTERNAL LEFT NO FLOW indicator and caution/advisory panel AUX FUEL capsule momentarily go on, then off?

Step 1. If indicator and capsule momentarily go on, then off, go to **4.**
Step 2. If indicator and capsule do not momentarily go on, then off, go to **5.**

TABLE NO. 16-2-1-10. With MANUAL Mode Selected, CDU MAIN FUEL Indication Does Not Increase And/Or Result Is Not As Specified When Attempting To Transfer Fuel From Left Outboard Fuel Tank. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

4. Do the remaining control panel EXTERNAL indicators remain off?

Step 1. If remaining indicators remain off, go to **TABLE NO. 16-2-1-13.**
Step 2. If remaining indicators do not remain off, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **21.**

5. Does control panel LEFT NO FLOW indicator remain on?

Step 1. If indicator remains on, go to **TABLE NO. 16-2-1-16.**
Step 2. If indicator does not remain on, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **21.**

6. With control panel MANUAL XFR LEFT switch at ON, check for more than 20 mvac between fuel flow transmitter connector pins A and B.

Step 1. If voltage is as specified, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **21.**
Step 2. If voltage is not as specified, replace fuel flow transmitter (PARA 16-4-31). Go to **21.**

7. Does control panel digital display indication decrease?

Step 1. If indication decreases, go to **8.**
Step 2. If indication does not decrease, go to **9.**

8. Does LEFT NO FLOW indicator go on?

Step 1. If indicator is on, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **21.**
Step 2. If indicator is not on, troubleshoot fuel quantity system (PARA 10-2-6 or PARA 10-2-5). Go to **21.**

9. Is manual lever of left outboard fuel shutoff gate valve in full OPEN position?

Step 1. If manual lever is in full OPEN position, go to **10.**
Step 2. If manual lever is not in full OPEN position, go to **12.**

10. Check for 28 vdc between P839-A and P839-B.

TABLE NO. 16-2-1-10. With MANUAL Mode Selected, CDU MAIN FUEL Indication Does Not Increase And/Or Result Is Not As Specified When Attempting To Transfer Fuel From Left Outboard Fuel Tank. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, replace left outboard bleed-air regulator valve (PARA 16-4-32). Go to **21.**
Step 2. If voltage is not as specified, go to **11.**

11. Check continuity between:
P839-A and P360-G
P839-B and ground

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **21.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or rear electrical harness (PARA 16-4-44), as required. Go to **21.**

12. Is left outboard bleed-air regulator valve continuously venting air?

Step 1. If valve is continuously venting air, go to **13.**
Step 2. If valve is not continuously venting air, go to **18.**

13. Check for 28 vdc between P360-C and ground.

Step 1. If voltage is as specified, go to **14.**
Step 2. If voltage is not as specified, go to **16.**

14. Check for 28 vdc between P839-A and P839-B.

Step 1. If voltage is as specified, replace left outboard bleed-air regulator valve (PARA 16-4-32). Go to **21.**
Step 2. If voltage is not as specified, go to **15.**

15. Check continuity between:
P839-A and P360-G
P839-B and ground

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **21.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or rear electrical harness (PARA 16-4-44), as required. Go to **21.**

TABLE NO. 16-2-1-10. With MANUAL Mode Selected, CDU MAIN FUEL Indication Does Not Increase And/Or Result Is Not As Specified When Attempting To Transfer Fuel From Left Outboard Fuel Tank. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

16. Check for 28 vdc between EXT FUEL LH circuit breaker terminal 1 and ground.

Step 1. If voltage is as specified, go to **17.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **21.**

17. Check for 28 vdc between J346-45 and ground.

Step 1. If voltage is as specified, replace cabin fuel harness (PARA 16-4-43). Go to **21.**
Step 2. If voltage is not as specified, repair/replace wiring between J346-45 and circuit breaker terminal 1 (Appendix F). Go to **21.**

18. Check for 28 vdc between P840-A and P840-E.

Step 1. If voltage is as specified, go to **19.**
Step 2. If voltage is not as specified, go to **20.**

19. Check for 0 vdc between P840-D and P840-E.

Step 1. If voltage is as specified, replace left outboard fuel shutoff gate valve (PARA 16-4-33). Go to **21.**
Step 2. If voltage is not as specified, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **21.**

20. Check continuity between:
P840-A and P360-F
P840-D and P360-E
P840-E and ground

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **21.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or rear electrical harness (PARA 16-4-44), as required. Go to **21.**

21. Procedure completed.

TABLE NO. 16-2-1-11. With MANUAL Mode Selected, CDU MAIN FUEL Indication Does Not Increase And/Or Result Is Not As Specified When Attempting To Transfer Fuel From Right Inboard Fuel Tank.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Does CDU MAIN FUEL indication increase?

Step 1. If indication does increase, go to **2.**
Step 2. If indication does not increase, go to **7.**

2. Does control panel digital display indication decrease?

Step 1. If indication does decrease, go to **3.**
Step 2. If indication does not decrease, go to **6.**

3. Does control panel EXTERNAL RIGHT NO FLOW indicator and caution/ advisory panel AUX FUEL capsule momentarily go on, then off?

Step 1. If indicator and capsule momentarily go on, then off, go to **4.**
Step 2. If indicator and capsule does not momentarily go on, then off, go to **5.**

4. Do the remaining control panel EXTERNAL indicators remain off?

Step 1. If remaining indicators remain off, go to **TABLE NO. 16-2-1-13.**
Step 2. If remaining indicators do not remain off, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **18.**

5. Does control panel RIGHT NO FLOW indicator remain on?

Step 1. If indicator remains on, go to **TABLE NO. 16-2-1-15.**
Step 2. If indicator does not remain on, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **18.**

6. With control panel MANUAL XFR RIGHT switch at ON, check for more than 20 mvac between fuel flow transmitter connector pins A and B.

Step 1. If voltage is as specified, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **18.**
Step 2. If voltage is not as specified, replace fuel flow transmitter (PARA 16-4-31). Go to **18.**

7. Does control panel digital display indication decrease?

Step 1. If indication decreases, go to **8.**
Step 2. If indication does not decrease, go to **9.**

TABLE NO. 16-2-1-11. With MANUAL Mode Selected, CDU MAIN FUEL Indication Does Not Increase And/Or Result Is Not As Specified When Attempting To Transfer Fuel From Right Inboard Fuel Tank. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

8. Does RIGHT NO FLOW indicator go on?

Step 1. If indicator is on, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **18.**
Step 2. If indicator is not on, troubleshoot fuel quantity system (PARA 10-2-6 or PARA 10-2-5). Go to **18.**

9. Is manual lever of right inboard fuel shutoff gate valve in full OPEN position?

Step 1. If manual lever is in full OPEN position, go to **10.**
Step 2. If manual lever is not in full OPEN position, go to **12.**

10. Check for 28 vdc between P811-A and P811-B.

Step 1. If voltage is as specified, replace right inboard bleed-air regulator valve (PARA 16-4-32). Go to **18.**
Step 2. If voltage is not as specified, go to **11.**

11. Check continuity between:
P811-A and P360-s
P811-B and ground

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **18.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or rear electrical harness (PARA 16-4-44), as required. Go to **18.**

12. Is right inboard bleed-air regulator valve continuously venting air?

Step 1. If valve is continuously venting air, go to **13.**
Step 2. If valve is not continuously venting air, go to **15.**

13. Check for 28 vdc between P811-A and P811-B.

Step 1. If voltage is as specified, replace right inboard bleed-air regulator valve (PARA 16-4-32). Go to **18.**
Step 2. If voltage is not as specified, go to **14.**

TABLE NO. 16-2-1-11. With MANUAL Mode Selected, CDU MAIN FUEL Indication Does Not Increase And/Or Result Is Not As Specified When Attempting To Transfer Fuel From Right Inboard Fuel Tank. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

14. Check continuity between:
P811-A and P360-s
P811-B and ground

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **18.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or rear electrical harness (PARA 16-4-44), as required. Go to **18.**

15. Check for 28 vdc between P810-A and P810-E.

Step 1. If voltage is as specified, go to **16.**
Step 2. If voltage is not as specified, go to **17.**

16. Check for 0 vdc between P810-D and P810-E.

Step 1. If voltage is as specified, replace right inboard fuel shutoff gate valve (PARA 16-4-33). Go to **18.**
Step 2. If voltage is not as specified, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **18.**

17. Check continuity between:
P810-A and P360-j
P810-D and P360-i
P810-E and ground

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **18.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or rear electrical harness (PARA 16-4-44), as required. Go to **18.**

18. Procedure completed.

TABLE NO. 16-2-1-12. With MANUAL Mode Selected, CDU MAIN FUEL Indication Does Not Increase And/Or Result Is Not As Specified When Attempting To Transfer Fuel From Left Inboard Fuel Tank.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Does CDU MAIN FUEL indication increase?

Step 1. If indication increases, go to **2.**
Step 2. If indication does not increase, go to **7.**

2. Does control panel digital display indication decrease?

Step 1. If indication decreases, go to **3.**
Step 2. If indication does not decrease, go to **6.**

3. Does control panel EXTERNAL LEFT NO FLOW indicator and caution/advisory panel AUX FUEL capsule momentarily go on, then off?

Step 1. If indicator and capsule momentarily go on, then off, go to **4.**
Step 2. If indicator and capsule does not momentarily go on, then off, go to **5.**

4. Do the remaining control panel EXTERNAL indicators remain off?

Step 1. If remaining indicators remain off, go to **TABLE NO. 16-2-1-13.**
Step 2. If remaining indicators do not remain off, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **18.**

5. Does control panel LEFT NO FLOW indicator remain on?

Step 1. If indicator remains on, go to **TABLE NO. 16-2-1-16.**
Step 2. If indicator does not remain on, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **18.**

6. With control panel MANUAL XFR LEFT switch at ON, check for more than 20 mvac between fuel flow transmitter connector pins A and B.

Step 1. If voltage is as specified, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **18.**
Step 2. If voltage is not as specified, replace fuel flow transmitter (PARA 16-4-31). Go to **18.**

7. Does control panel digital display indication decrease?

Step 1. If indication decreases, go to **8.**
Step 2. If indication does not decrease, go to **9.**

TABLE NO. 16-2-1-12. With MANUAL Mode Selected, CDU MAIN FUEL Indication Does Not Increase And/Or Result Is Not As Specified When Attempting To Transfer Fuel From Left Inboard Fuel Tank. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

8. Does LEFT NO FLOW indicator go on?

Step 1. If indicator is on, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **18.**
Step 2. If indicator is not on, troubleshoot fuel quantity system (PARA 10-2-6 or PARA 10-2-5). Go to **18.**

9. Is manual lever of left inboard fuel shutoff gate valve in full OPEN position?

Step 1. If manual lever is in full OPEN position, go to **10.**
Step 2. If manual lever is not in full OPEN position, go to **12.**

10. Check for 28 vdc between P808-A and P808-B.

Step 1. If voltage is as specified, replace left inboard bleed-air regulator valve (PARA 16-4-32). Go to **18.**
Step 2. If voltage is not as specified, go to **11.**

11. Check continuity between:
P808-A and P360-m
P808-B and ground

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **18.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or rear electrical harness (PARA 16-4-44), as required. Go to **18.**

12. Is left inboard bleed-air regulator valve continuously venting air?

Step 1. If valve is continuously venting air, go to **13.**
Step 2. If valve is not continuously venting air, go to **15.**

13. Check for 28 vdc between P808-A and P808-B.

Step 1. If voltage is as specified, replace left inboard bleed-air regulator valve (PARA 16-4-32). Go to **18.**
Step 2. If voltage is not as specified, go to **14.**

TABLE NO. 16-2-1-12. With MANUAL Mode Selected, CDU MAIN FUEL Indication Does Not Increase And/Or Result Is Not As Specified When Attempting To Transfer Fuel From Left Inboard Fuel Tank. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

14. Check continuity between:
P808-A and P360-m
P808-B and ground

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **18.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or rear electrical harness (PARA 16-4-44), as required. Go to **18.**

15. Check for 28 vdc between P809-A and P809-E.

Step 1. If voltage is as specified, go to **16.**
Step 2. If voltage is not as specified, go to **17.**

16. Check for 0 vdc between P809-D and P809-E.

Step 1. If voltage is as specified, replace left inboard fuel shutoff gate valve (PARA 16-4-33). Go to **18.**
Step 2. If voltage is not as specified, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **18.**

17. Check continuity between:
P809-A and P360-Y
P809-D and P360-Z
P809-E and ground

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **18.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or rear electrical harness (PARA 16-4-44), as required. Go to **18.**

18. Procedure completed.

TABLE NO. 16-2-1-13. Fuel Transfer Time And/Or Gauging Accuracy Is Not As Specified.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **With control panel MANUAL XFR LEFT and/or RIGHT switch, as applicable, at ON, inspect fuel and pneumatic hoses on fittings for signs of leakage or damage.**

   Step 1. If leakage or damage is present, replace damaged or leaking hoses and/or fittings, as required. Go to **6.**
   Step 2. If leakage or damage is not present, go to **2.**

2. **Pull and turn control panel AUX FUEL QTY switch to CAL. Go to 3.**

3. **Does control panel digital display indication equal K FACTOR value on fuel flow transmitter?**

   Step 1. If indication is equal to K FACTOR, go to **4.**
   Step 2. If indication is not equal to K FACTOR, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **6.**

4. **With control panel MANUAL XFR LEFT and/or RIGHT switch, as applicable, at ON, check for more than 20 mvac between fuel flow transmitter connector pins A and B.**

   Step 1. If voltage is as specified, replace applicable bleed-air regulator valve (PARA 16-4-32). Go to **6.**
   Step 2. If trouble remains, go to **5.**
   Step 3. If voltage is not as specified, replace fuel flow transmitter (PARA 16-4-31). Go to **6.**

5. **Do operational/troubleshooting procedure of fuel quantity system (PARA 10-2-6 or PARA 10-2-5).**

   Step 1. If fuel quantity system is good, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **6.**
   Step 2. If fuel quantity system is not good, troubleshoot system (PARA 10-2-6 or PARA 10-2-5). Go to **6.**

6. **Procedure completed.**

TABLE NO. 16-2-1-14. Auto Mode Fuel Transfer Limits Are Not As Specified.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Did digital display indication increase or decrease?**

   Step 1. If indication changes, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **3.**
   Step 2. If indication does not change, go to **2.**

2. **Check continuity between:**
   P905-K (Tank 1) and J346-30
   P905-L (Tank 1) and J346-31
   P905-V (Tank 2) and J346-27
   P905-U (Tank 2) and J346-28

   Step 1. If continuity is present, replace cabin fuel harness (PARA 16-4-43). Go to **3.**
   Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **3.**

3. **Procedure completed.**

TABLE NO. 16-2-1-15. Control Panel EXTERNAL RIGHT NO FLOW Indicator Remains On During Fuel Transfer.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Disconnect J520 from right external fuel flow sensor. Go to 2.**

2. **Connect insulated jumper wire between J520-1 and J520-2. Go to 3.**

3. **Place control panel MANUAL XFR RIGHT switch to ON. Go to 4.**

4. **Does RIGHT NO FLOW indicator go on momentarily?**

   Step 1. If indicator goes on momentarily, replace right external fuel flow sensor (PARA 16-4-30). Go to **8.**
   Step 2. If indicator does not go on momentarily, go to **5.**

5. **Place control panel MANUAL XFR RIGHT switch to OFF. Go to 6.**

6. **Check continuity between:**

TABLE NO. 16-2-1-15. Control Panel EXTERNAL RIGHT NO FLOW Indicator Remains On During Fuel Transfer. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

J520-1 and P360-e
J520-2 and P360-d

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **8.**
Step 2. If continuity is not present, go to **7.**

7. Check continuity between:
J346-16 and J520-1
J346-17 and J520-2

Step 1. If continuity is present, replace cabin fuel harness (PARA 16-4-43). Go to **8.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **8.**

8. Procedure completed.

TABLE NO. 16-2-1-16. Control Panel EXTERNAL LEFT NO FLOW Indicator Remains On During Fuel Transfer.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Disconnect J521 from left external fuel flow sensor. Go to 2.

2. Connect insulated jumper wire between J521-1 and J521-2. Go to 3.

3. Place control panel MANUAL XFR LEFT switch to ON. Go to 4.

4. Does LEFT NO FLOW indicator go on momentarily?

Step 1. If indicator goes on momentarily, replace left external fuel flow sensor (PARA 16-4-30). Go to **8.**
Step 2. If indicator does not go on momentarily, go to **5.**

5. Place control panel MANUAL XFR LEFT switch to OFF. Go to 6.

6. Check continuity between:

TABLE NO. 16-2-1-16. Control Panel EXTERNAL LEFT NO FLOW Indicator Remains On During Fuel Transfer. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

J521-1 and P360-q
J521-2 and P360-d

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **8.**
Step 2. If continuity is not present, go to **7.**

7. Check continuity between:
J346-14 and J521-1
J346-15 and J521-2

Step 1. If continuity is present, replace cabin fuel harness (PARA16-4-43). Go to **8.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **8.**

8. Procedure completed.

TABLE NO. 16-2-1-17. Control Panel EXTERNAL RIGHT INBD EMPTY And/Or LEFT INBD EMPTY Indicators Do Not Go On With Inadequate Fuel In Tank.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Disconnect J1 from applicable inboard fuel tank. Go to 2.

2. Does applicable INBD EMPTY indicator go on?

Step 1. If indicator is on, replace corresponding fuel tank (PARA 16-4-54). Go to **4.**
Step 2. If indicator is not on, go to **3.**

3. Check continuity between:
J1-3 (Left) and P361-D
J1-4 (Left) and P361-C
J1-5 (Left) and P361-B
J1-3 (Right) and P361-N
J1-4 (Right) and P361-P
J1-5 (Right) and P361-R

TABLE NO. 16-2-1-17. Control Panel EXTERNAL RIGHT INBD EMPTY And/Or LEFT INBD EMPTY Indicators Do Not Go On With Inadequate Fuel In Tank. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **4.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or front electrical harness (PARA 16-4-45), as required. Go to **4.**

4. Procedure completed.

TABLE NO. 16-2-1-18. Caution/Advisory Panel AUX FUEL Capsule And/Or Control Panel EXTERNAL RIGHT NO FLOW And LEFT NO FLOW Indicators Do Not Go On Immediately After Corresponding Tank EMPTY Indicator Goes On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Do control panel EXTERNAL RIGHT NO FLOW and LEFT NO FLOW indicators go on?

Step 1. If indicators are on, go to **2.**
Step 2. If indicators are not on, go to **6.**

2. Momentarily place caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 3.

3. Does AUX FUEL capsule go on, then off?

Step 1. If capsule goes on, then off, go to **4.**
Step 2. If capsule does not go on, then off, troubleshoot caution/advisory warning system (PARA 8-2-4). Go to **10.**

4. Check for 28 vdc between P117-41 and ground.

Step 1. If voltage is as specified, replace caution/advisory panel (PARA 8-4-19). Go to **10.**
Step 2. If voltage is not as specified, go to **5.**

5. Check continuity between P346-10 and P360-X.

Step 1. If continuity is present, repair/replace wiring between J346-10 and P117-41 (Appendix F). Go to **10.**

TABLE NO. 16-2-1-18. Caution/Advisory Panel AUX FUEL Capsule And/Or Control Panel EXTERNAL RIGHT NO FLOW And LEFT NO FLOW Indicators Do Not Go On Immediately After Corresponding Tank EMPTY Indicator Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43). Go to **10.**

6. Disconnect applicable connector, J521 (Left) and J520 (Right), from external fuel flow sensor. Go to 7.

7. With control panel mode switch in AUTO, is correct corresponding NO FLOW indicator on?

Step 1. If indicator is on, replace external fuel flow sensor (PARA 16-4-30). Go to **10.**
Step 2. If indicator is not on, go to **8.**

8. Check for infinite resistance between:
J360-q (Left) and J360-d
J360-e (Right) and J360-d

Step 1. If resistance is as specified, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **10.**
Step 2. If resistance is not as specified, go to **9.**

9. Check for infinite resistance between:
J346-14 (Left) and J346-15
J346-16 (Right) and J346-17

Step 1. If resistance is as specified, replace cabin fuel harness (PARA 16-4-43). Go to **10.**
Step 2. If resistance is not as specified, repair/replace wiring (Appendix F), as required. Go to **10.**

10. Procedure completed.

TABLE NO. 16-2-1-19. Control Panel EXTERNAL RIGHT OUTBD EMPTY And/Or LEFT OUTBD EMPTY Indicators Do Not Go On With Inadequate Fuel In Tank.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Disconnect J1 from applicable outboard fuel tank. Go to 2.

2. Does applicable OUTBD EMPTY indicator go on?

Step 1. If indicator is on, replace corresponding fuel tank (PARA 16-4-54). Go to **4.**
Step 2. If indicator is not on, go to **3.**

3. Check continuity between:
J1-3 (Left) and P361-G
J1-4 (Left) and P361-F
J1-5 (Left) and P361-E
J1-3 (Right) and P361-K
J1-4 (Right) and P361-L
J1-5 (Right) and P361-M

Step 1. If continuity is present, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **4.**
Step 2. If continuity is not present, replace cabin fuel harness (PARA 16-4-43) or front electrical harness (PARA 16-4-45), as required. Go to **4.**

4. Procedure completed.

TABLE NO. 16-2-1-20. VENT SENSOR FAIL Indicator Does Not Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check that helicopter wiring harness J376 is connected to fuel overflow sensor P376.

Step 1. If J376 is connected to P376, go to **2.**
Step 2. If J376 is not connected to P376, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **4.**

2. Check fuel overflow sensor resistance:
P361-W and P361-Z (650-1500 ohms)
P361-Z and P361-V (360-440 ohms)

TABLE NO. 16-2-1-20. VENT SENSOR FAIL Indicator Does Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If resistance is as specified, replace auxiliary fuel management control panel (PARA 16-4-38). Go to **4.**

Step 2. If resistance is not as specified, go to **3.**

3. Check fuel overflow sensor resistance:
P376-C and P376-A (650-1500 ohms)
P376-C and P376-B (360-440 ohms)

Step 1. If resistance is as specified, repair/replace wiring, as required. Go to **4.**

Step 2. If resistance is not as specified, replace fuel overflow sensor (vent sensor) (PARA 16-4-47). Go to **4.**

4. Procedure completed.

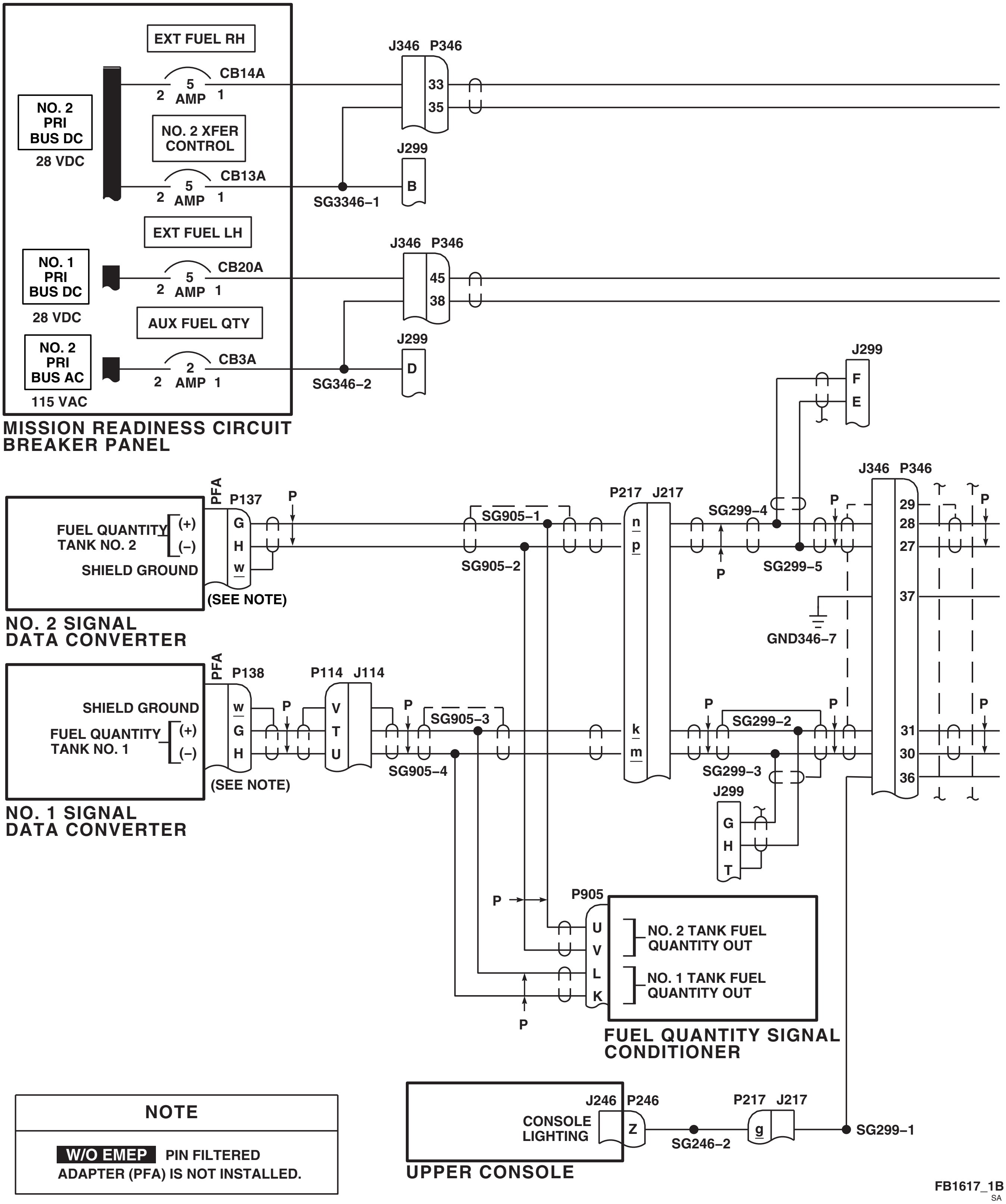


FIGURE 16-2-1-1. EXTERNAL STORES SUPPORT SYSTEM RANGE EXTENSION SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 6)

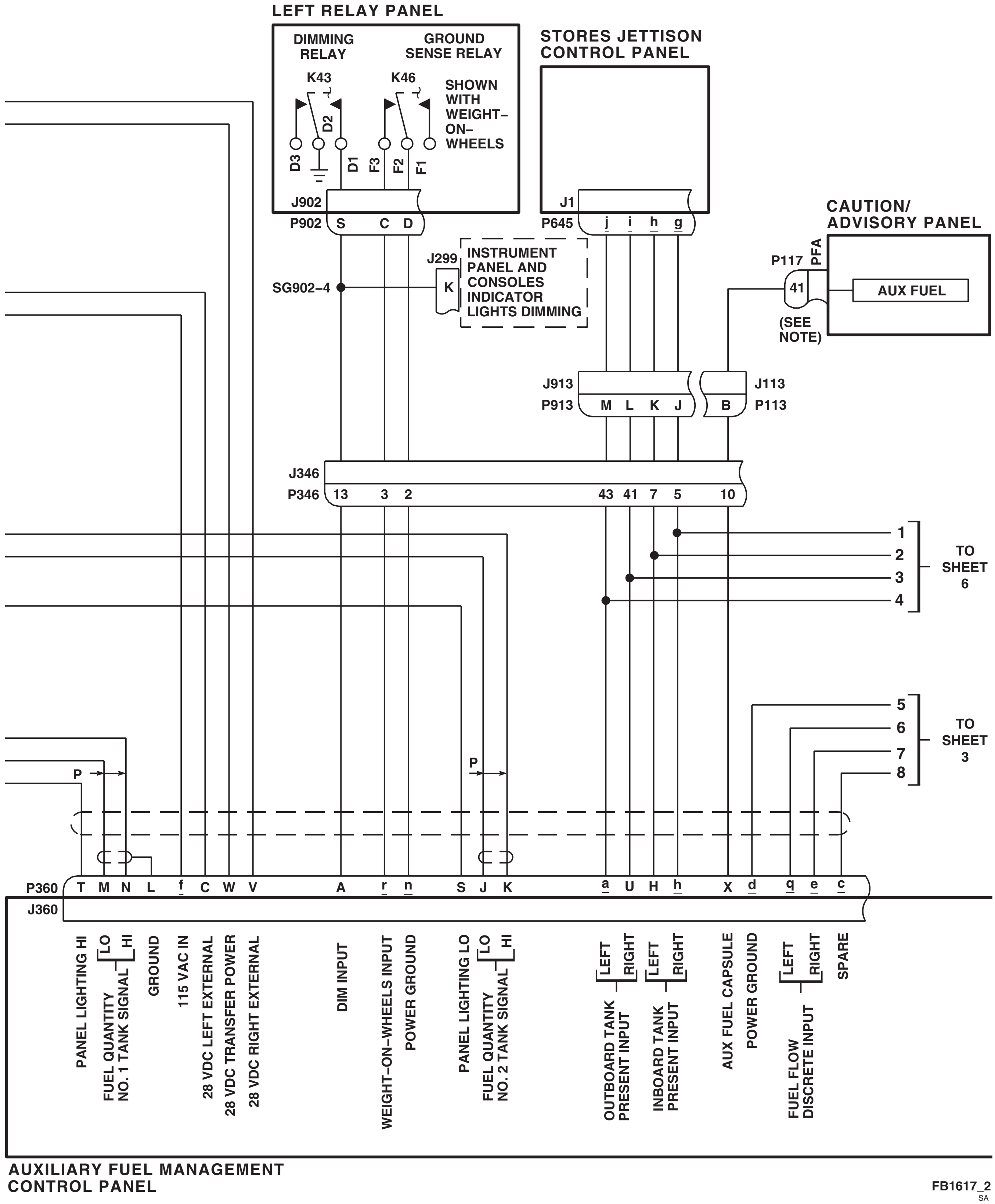


FIGURE 16-2-1-1. EXTERNAL STORES SUPPORT SYSTEM RANGE EXTENSION SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 6)

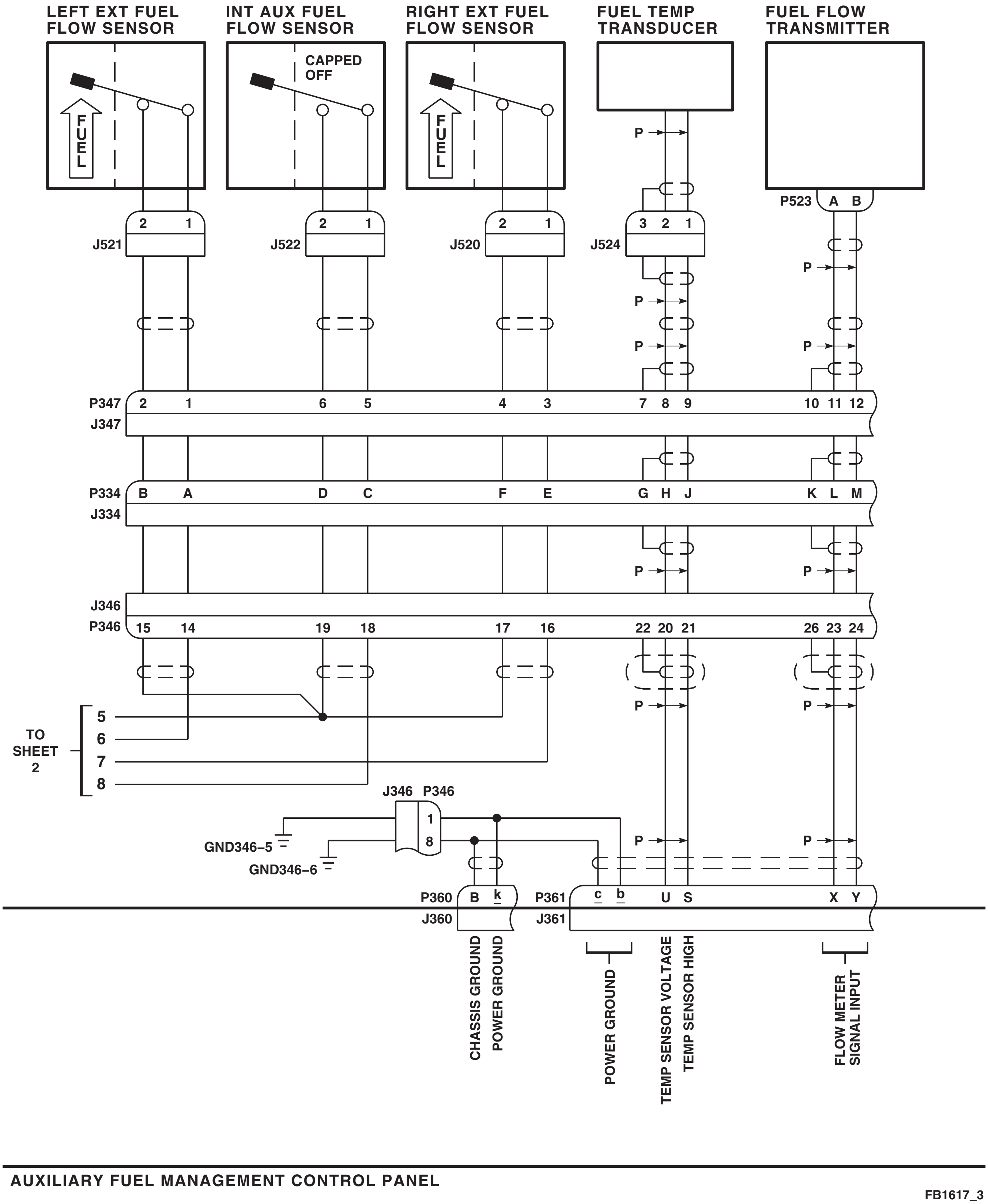


FIGURE 16-2-1-1. EXTERNAL STORES SUPPORT SYSTEM RANGE EXTENSION SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 6)

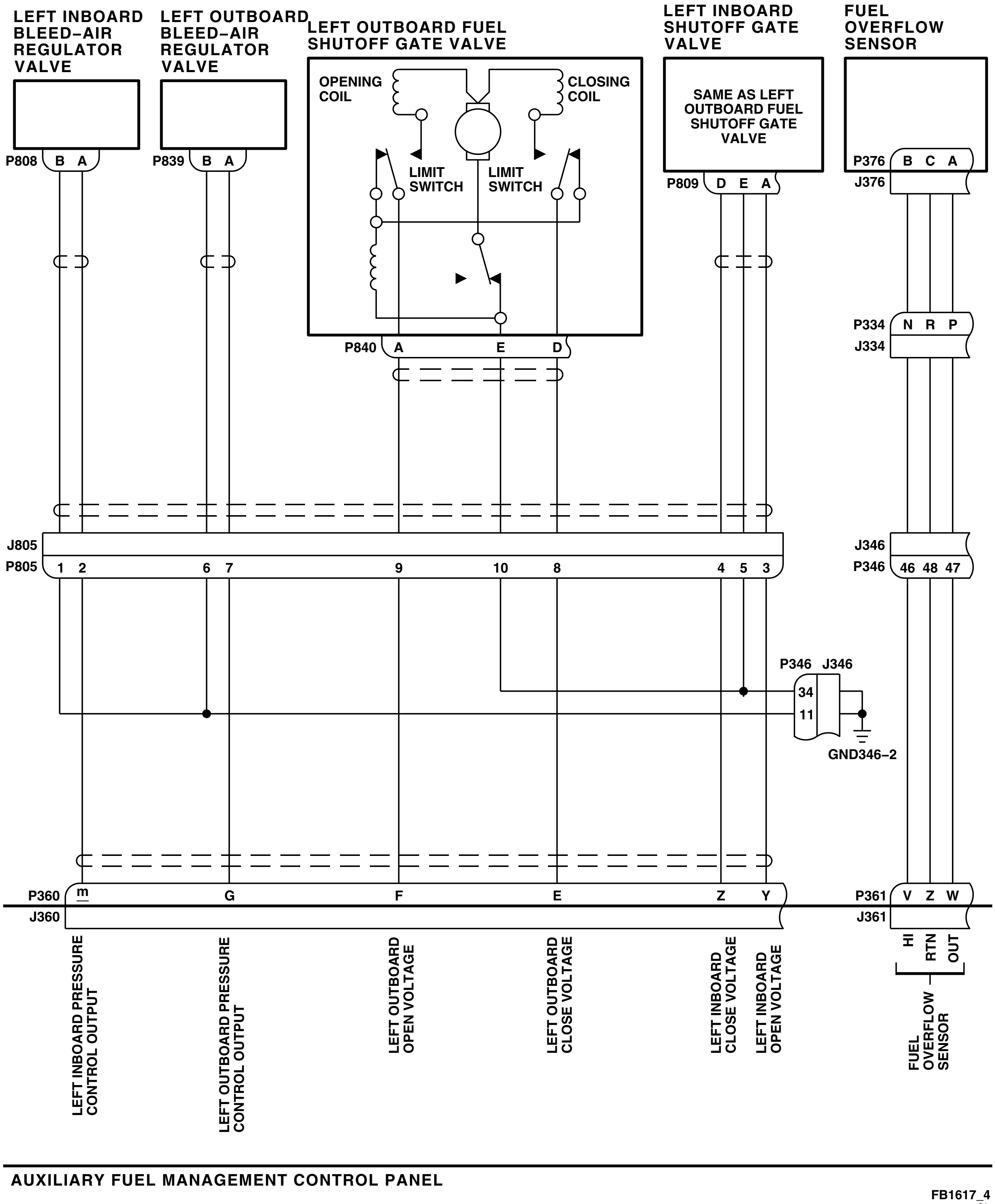


FIGURE 16-2-1-1. EXTERNAL STORES SUPPORT SYSTEM RANGE EXTENSION SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 6)

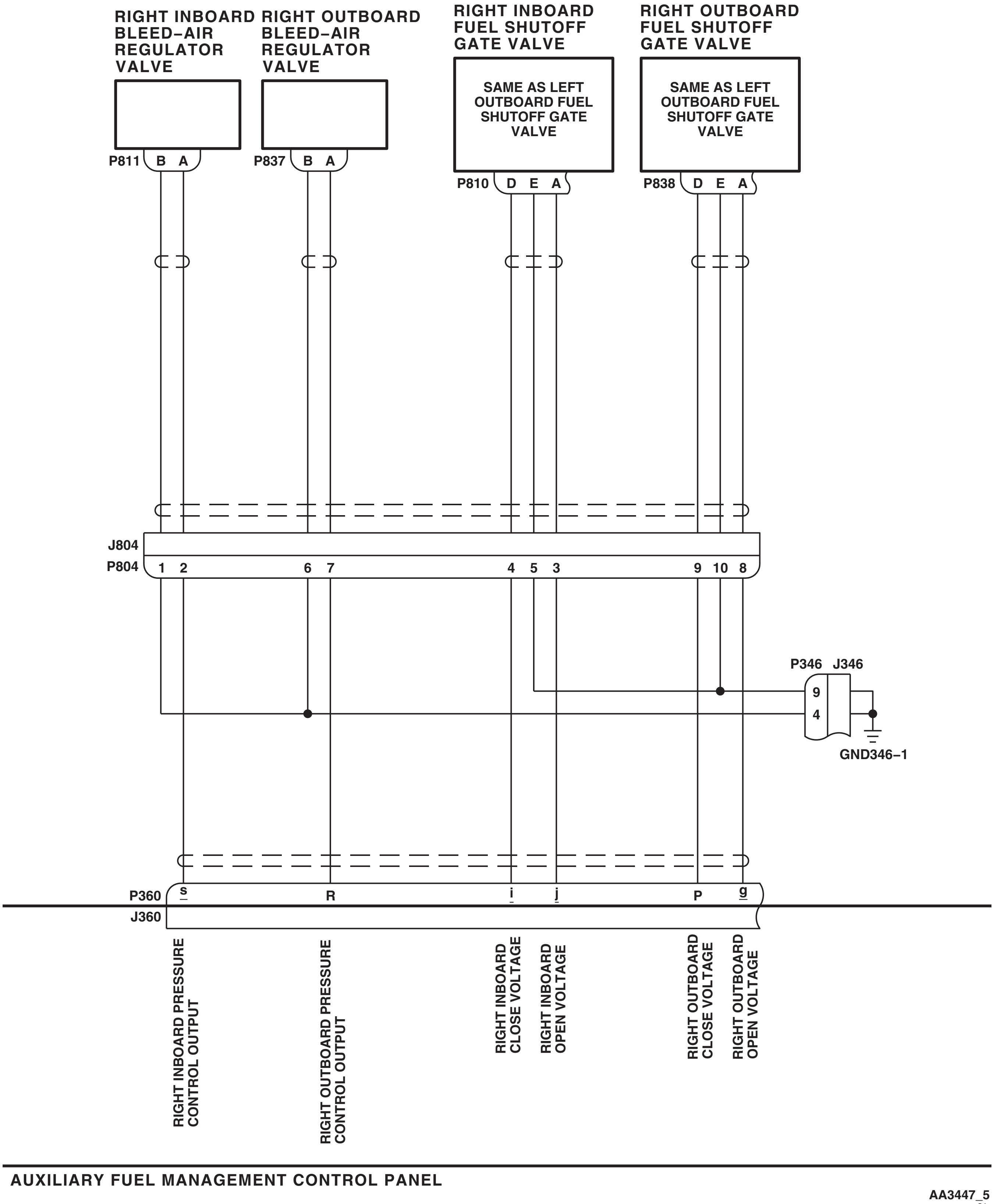


FIGURE 16-2-1-1. EXTERNAL STORES SUPPORT SYSTEM RANGE EXTENSION SYSTEM SCHEMATIC DIAGRAM (SHEET 5 OF 6)

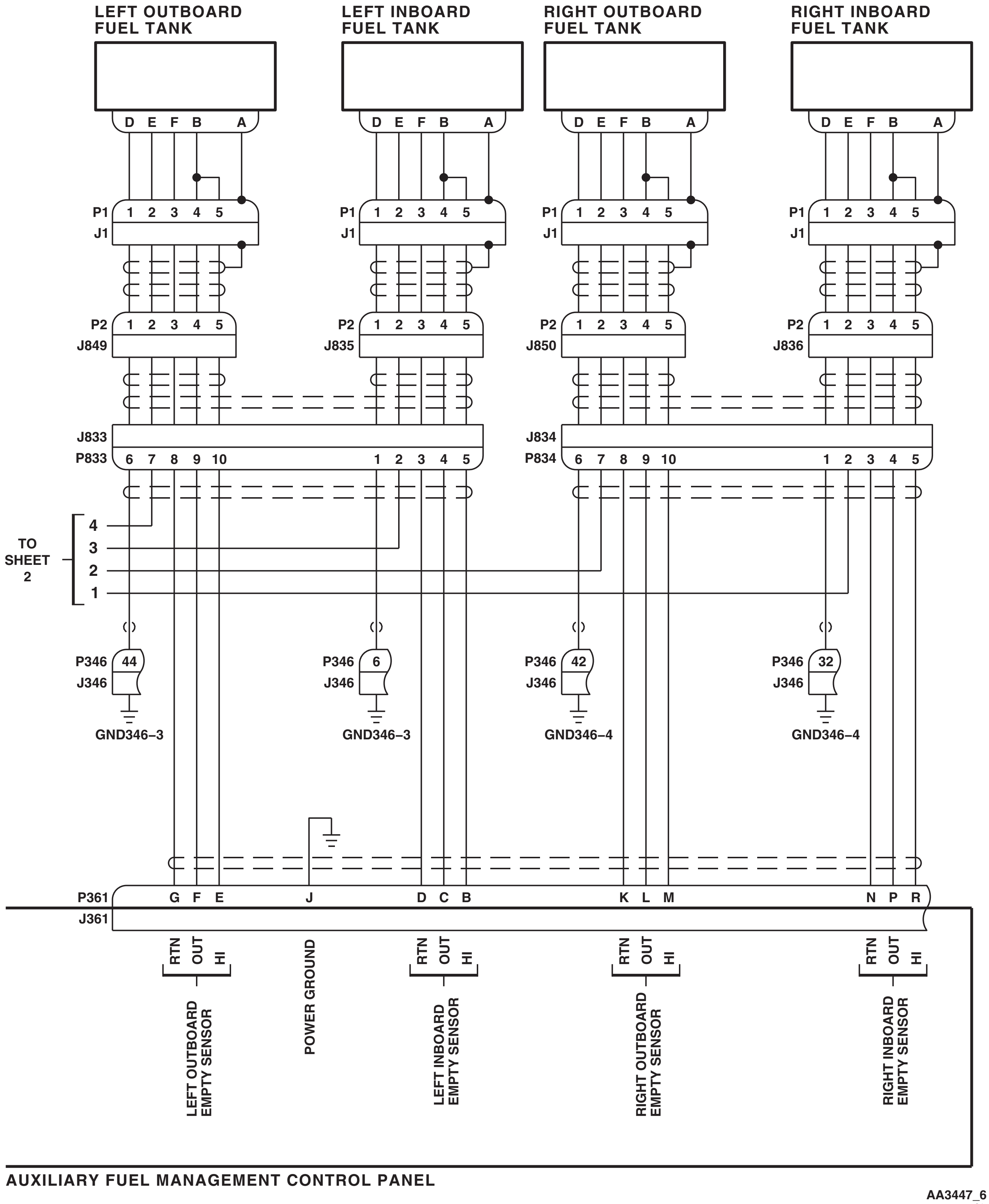


FIGURE 16-2-1-1. EXTERNAL STORES SUPPORT SYSTEM RANGE EXTENSION SYSTEM SCHEMATIC DIAGRAM (SHEET 6 OF 6)

16-2-2 EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING.

INITIAL SETUP

Test Equipment

- Multimeter Fluke 45

Tools

- Aircraft Mechanic's Toolkit
- Container, Suitable
- Electrical Repairer Toolkit

References

- PARA 1-3-17
- PARA 1-3-20
- PARA 1-3-24
- PARA 1-6-16
- PARA 16-4-21
- PARA 16-4-35
- PARA 16-4-47

Equipment Conditions

- External Range Extension Tanks (Inboard or Outboard) **EMPTY**
- Main Fuel Tanks **EMPTY**
- External Electrical Power and External Air Available
- Rear Fairings On All Vertical Stores (Pylons) Removed (PARA 16-4-21)
- All Explosive Cartridges Removed From Ejector Racks (PARA 16-4-35)

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

Special Instructions

This fault isolation procedure is designed to be used with the following configurations: inboard tanks only, outboard tanks only, and dual inboard/outboard tanks.

The ESSS operational/troubleshooting procedure is subdivided as follows:

(1) Self-Test Check

(2) Left Outboard/Inboard Manual Mode Transfer Check

(3) Right Outboard/Inboard Manual Mode Transfer Check

(4) Single Outboard Tank Auto Transfer And Lateral Imbalance Check

(5) Dual Outboard Tanks Auto Fuel Transfer Check

(6) Single Inboard Tank Auto Fuel Transfer Check And Lateral Imbalance Check

(7) Dual Inboard Tanks Auto Fuel Transfer Check

(8) Unbalanced Fuel Transfer Check

(9) Inboard And Outboard Tanks Fuel Transfer Check

(10) High-Level Auto Transfer Shutoff Check

(11) Main Tank Overflow Sensor Check

(12) Shutdown

16-2-2.1 Self-Test Check.

a. Make sure all circuit breakers are pushed in except those noted being pulled for maintenance or safety.

b. Turn on electrical power (PARA 1-6-16).

c. Rotate ESSS control panel DIM control fully clockwise then fully counterclockwise.

RESULT: ESSS control panel indicators shall brighten then dim.

- If result is not as specified, go to TABLE NO. 16-2-2-2.

d. Rotate ESSS control panel DIM knob 1/4 turn clockwise.

e. Rotate INST LT PILOT FLT switch on upper console 1/4 turn clockwise.

f. Rotate LIGHTED SWITCHES switch on upper console fully clockwise then counterclockwise.

RESULT: ESSS control panel indicators shall brighten then dim.

- If result is not as specified, troubleshoot upper and lower consoles lights (PARA 9-2-5).

g. Observe the following indicators on the ESSS control panel:

INDICATOR	DISPLAY
LEFT OUTBD EMPTY	On
LEFT INBD EMPTY	On
RIGHT INBD EMPTY	On
RIGHT OUTBD EMPTY	On
LEFT FLOW	Off
RIGHT FLOW	Off
LAT IMBAL	Off
MN OVFL	Off
LEFT OUTBD	0
RIGHT INBD	0
RIGHT OUTBD	0
LEFT INBD	0

RESULT: If displays are not as specified, perform step j.

NOTE

The AUX FUEL capsule on the caution/advisory system will come on upon initial power-up of system. Self-test function does not illuminate AUX FUEL capsule.

h. Observe AUX FUEL capsule on caution/advisory panel.

RESULT: AUX FUEL capsule shall be on.

- If result is not as specified, troubleshoot caution/advisory system (PARA 8-2-4).

i. Press and release ESSS control panel CAUT RESET switch.

RESULT: AUX FUEL capsule shall go off.

- If result is not as specified, replace ESSS control panel (PARA 16-4-49).

j. Press and release ESSS control panel TEST switch.

RESULT:

(1) The following indicators on control panel shall illuminate for 3 seconds and cycle back to displays listed in step g:

INDICATOR	DISPLAY
LEFT OUTBD EMPTY	On
LEFT INBD EMPTY	On
RIGHT INBD EMPTY	On
RIGHT OUTBD EMPTY	On
LEFT FLOW	On
RIGHT FLOW	On
LAT IMBAL	On
MN OVFL	On
LEFT OUTBD	LAMP
LEFT INBD	TEST
RIGHT INBD	LAMP
RIGHT OUTBD	TEST

(2) After displaying LAMP and TEST, fuel quantity indicators shall perform character sequencing test starting with 9999 and ending with 0000.

(3) At completion of character sequencing test, each fuel quantity indicator shall display PASS. If any external tank is not installed, fuel quantity indicator for that tank shall display N/A and then go blank.

(4) After PASS is displayed, each fuel quantity indicator shall display 0.

- If results (1), (2), and (4) are not as specified, replace ESSS control panel (PARA 16-4-49).
- If a fault code is displayed in place of PASS in result (3), refer to TABLE NO. 16-2-2-1 for fault codes and corrective actions.

TABLE NO. 16-2-2-1. ESSS Control Panel Fault Codes And Actions.

FAULT CODE	MALFUNCTION	ACTION
Fwd	Malfunctioning Fwd Probe	Replace indicated forward probe (PARA 16-4-65)
Aft	Malfunctioning Aft Probe	Replace indicated aft probe (PARA 16-4-65)
LoZ1	LoZ Side of Inboard Tank Unit Wiring Open/Shorted	TABLE NO. 16-2-2-3
LoZ2	LoZ Side of Outboard Tank Unit Wiring Open/Shorted	TABLE NO. 16-2-2-4
HiZ1	HiZ Side of Left Inboard or Right Outboard Foward Tank Unit Wiring Open/Shorted	TABLE NO. 16-2-2-5
HiZ2	HiZ Side of Left Inboard or Right Outboard Aft Tank Unit Wiring Open/Shorted	TABLE NO. 16-2-2-6

TABLE NO. 16-2-2-1. ESSS Control Panel Fault Codes And Actions. (Cont)

FAULT CODE	MALFUNCTION	ACTION
HiZ3	HiZ Side of Left Outboard or Right Inboard Foward Tank Unit Wiring Open/Shorted	TABLE NO. 16-2-2-7
HiZ4	HiZ Side of Left Outboard or Right Inboard Aft Tank Unit Wiring Open/Shorted	TABLE NO. 16-2-2-8
A/D	A/D Converter Malfunctioning	Replace fuel quantity signal conditioner (PARA 10-4-36)
>Max	Calculated Fuel Mass Too High	Replace fuel quantity signal conditioner (PARA 10-4-36)
T-TO	Processor Transmit Timeout	Replace fuel quantity signal conditioner (PARA 10-4-36)

k. Turn off electrical power (PARA 1-6-16).

16-2-2.2 Left Outboard/Inboard Manual Mode Transfer Check.

a. Fuel each inboard and outboard fuel tank with 50 pounds of fuel (PARA 1-3-24).

b. Turn on electrical power (PARA 1-6-16).

c. Press and release CAUT RESET switch on ESSS control panel to turn off AUX FUEL capsule.

d. Set ESSS control panel AUTO XFER/MAN XFER switch to MAN XFER.

e. Apply external air.

NOTE

- Have an assistant stand near ESSS wings to listen for bleed air regulator valves and fuel transfer gate valves opening.
- Allow at least 10 minutes for tanks to pressurize before attempting to transfer fuel.

f. In sequence, place each external tank (LEFT OUTBD, LEFT INBD, RIGHT OUTBD, RIGHT INBD) XFER/PRESS/OFF switch to PRESS.

RESULT: The bleed air regulator valve in each external tank shall be heard opening.

- If left outboard tank bleed air regulator valve does not open, go to TABLE NO. 16-2-2-9.
- If left inboard tank bleed air regulator valve does not open, go to TABLE NO. 16-2-2-10.
- If right outboard tank bleed air regulator valve does not open, go to TABLE NO. 16-2-2-11.
- If right inboard tank bleed air regulator valve does not open, go to TABLE NO. 16-2-2-12.

g. Place ESSS control panel LEFT OUTBD XFER/PRESS/OFF switch to XFER.

RESULT:

(1) Left outboard fuel transfer gate valve shall be heard opening.

(2) LEFT FLOW indicator shall go on.

- If results are not as specified, go to TABLE NO. 16-2-2-9.

h. Allow fuel transfer to occur until left outboard EMPTY indicator goes on.

RESULT: Within approximately two minutes, LEFT FLOW indicator shall go off.

- If result is not as specified, replace ESSS control panel (PARA 16-4-49).

i. Place ESSS control panel LEFT OUTBD XFER/PRESS/OFF switch to PRESS.

RESULT: Left outboard fuel transfer gate valve shall be heard closing.

- If result is not as specified, go to TABLE NO. 16-2-2-13.

j. Place ESSS control panel LEFT INBD XFER/PRESS/OFF switch to XFER.

RESULT:

(1) Left inboard fuel transfer gate valve shall be heard opening.

(2) LEFT FLOW indicator shall go on.

- If results are not as specified, go to TABLE NO. 16-2-2-10.

k. Allow left inboard tank to transfer fuel until EMPTY indicator goes on.

RESULT: Within approximately two minutes, LEFT FLOW indicator shall go off.

- If result is not as specified, replace ESSS control panel (PARA 16-4-49).

l. Place ESSS control panel LEFT INBD XFER/PRESS/OFF switch to PRESS.

RESULT: Left inboard fuel transfer gate valve shall be heard closing.

- If result is not as specified, go to TABLE NO. 16-2-2-14.

16-2-2.3 Right Outboard/Inboard Manual Mode Transfer Check.

a. Place ESSS control panel RIGHT OUTBD XFER/PRESS/OFF switch to XFER.

RESULT:

(1) Right outboard fuel transfer gate valve shall be heard opening.

(2) RIGHT FLOW indicator shall go on.

- If results are not as specified, go to TABLE NO. 16-2-2-11.

b. Allow fuel transfer to occur until right outboard EMPTY indicator goes on.

RESULT: Within approximately two minutes, RIGHT FLOW indicator shall go off.

- If result is not as specified, replace ESSS control panel (PARA 16-4-49).

c. Place ESSS control panel RIGHT OUTBD XFER/PRESS/OFF switch to PRESS.

RESULT: Right outboard fuel transfer gate valve shall be heard closing.

- If result is not as specified, go to TABLE NO. 16-2-2-15.

d. Place ESSS control panel RIGHT INBD XFER/PRESS/OFF switch to XFER.

RESULT:

(1) Right inboard fuel transfer gate valve shall be heard opening.

(2) RIGHT FLOW indicator shall go on.

- If results are not as specified, go to TABLE NO. 16-2-2-12.

e. Allow right inboard tank to transfer fuel until EMPTY indicator goes on.

RESULT: Within approximately two minutes, RIGHT FLOW indicator shall go off.

- If result is not as specified, replace ESSS control panel (PARA 16-4-49).

f. Place ESSS control panel RIGHT INBD XFER/PRESS/OFF switch to PRESS.

RESULT: Right inboard fuel transfer gate valve shall be heard closing.

- If result is not as specified, go to TABLE NO. 16-2-2-16.

g. Turn off electrical power (PARA 1-6-16).

h. Pressure defuel main fuel tanks (PARA 1-3-20).

16-2-2.4 Single Outboard Tank Auto Fuel Transfer and Lateral Imbalance Check.

a. Fuel main fuel tanks with 270 to 330 pounds of fuel per tank (PARA 1-3-17).

b. Fuel left outboard tank with 612 to 748 pounds of fuel (PARA 1-3-24).

c. Turn on electrical power (PARA 1-6-16).

d. Place all ESSS control panel XFER/PRESS/OFF switches to PRESS.

NOTE

Allow at least 10 minutes for tanks to pressurize before continuing checkout.

e. Press and release ESSS control panel TEST switch.

RESULT:

(1) At the completion of self-test, no fault codes shall be displayed.

(2) LEFT OUTBD fuel quantity indicator shall indicate between 612 and 748 pounds of fuel.

(3) LEFT OUTBD EMPTY indicator shall be off.

(4) LAT IMBAL indicator shall be on.

- If result (1) is not as specified, go to TABLE NO. 16-2-2-1.

- If results (2), (3), and (4) are not as specified, replace signal conditioner (PARA 16-4-48).

f. Place ESSS control panel AUTO XFER/MAN XFER switch to AUTO XFER.

g. Place LEFT OUTBD XFER/PRESS/OFF switch to XFER.

RESULT:

(1) LEFT FLOW indicator shall remain off.

(2) LEFT OUTBD fuel quantity indication shall not change from reading obtained in step e. (2).

- If results are not as specified, go to TABLE NO. 16-2-2-17.

h. Place ESSS control panel AUTO XFER/MAN XFER switch to MAN XFER.

RESULT:

(1) LEFT FLOW indicator shall go on.

(2) LEFT OUTBD fuel quantity indicator reading shall decrease.

(3) CDU MAIN FUEL quantity indicator shall increase.

- If results (1) and (2) are not as specified, replace ESSS control panel (PARA 16-4-49).

- If result (3) is not as specified, troubleshoot fuel quantity system (PARA 10-2-5 or PARA 10-2-6).

i. Place ESSS control panel LEFT OUTBD XFER/PRESS/OFF switch to OFF.

RESULT:

(1) LEFT FLOW indicator shall go off.

(2) LEFT OUTBD fuel quantity indicator reading shall remain constant.

- If results (1) and (2) are not as specified, replace ESSS control panel (PARA 16-4-49).

j. Record LEFT OUTBD fuel quantity.

k. Turn off electrical power (PARA 1-6-16).

16-2-2.5 Dual Outboard Tanks Auto Fuel Transfer Check.

a. Fuel right outboard tank with the quantity of fuel recorded in 16-2-2.4 step j.

b. Turn on electrical power (PARA 1-6-16).

c. Press and release ESSS control panel TEST switch.

RESULT:

(1) At completion of self-test, no fault codes shall be displayed.

(2) RIGHT OUTBD EMPTY indicator shall be off.

(3) RIGHT OUTBD fuel quantity indicator shall display fuel quantity obtained in step a.

(4) LEFT OUTBD fuel quantity indicator shall display value recorded in 16-2-2.4 step j.

(5) LAT IMBAL indicator shall be off.

- If result (1) is not as specified, go to TABLE NO. 16-2-2-1.
- If result (2) is not as specified, go to TABLE NO. 16-2-2-18.
- If results (3) and (4) are not as specified, replace ESSS control panel (PARA 16-4-49).
- If result (5) is not as specified, go to TABLE NO. 16-2-2-19.

d. Place all ESSS control panel XFER/PRESS/OFF switches to PRESS.

NOTE

Allow at least 10 minutes for tanks to pressurize before continuing checkout.

e. Place LEFT and RIGHT OUTBD XFER/PRESS/OFF switches to XFER.

f. Place AUTO XFER/MAN XFER switch to MAN XFER.

RESULT:

(1) LEFT FLOW indicator shall go on.

(2) RIGHT FLOW indicator shall go on.

(3) LEFT OUTBD fuel quantity indicator shall indicate a decreasing amount of fuel quantity from that displayed in step c. (4).

(4) RIGHT OUTBD fuel quantity indicator shall indicate a decreasing amount of fuel quantity from that displayed in step c. (3).

(5) CDU MAIN FUEL shall display an increase in fuel.

- If result (1), (2), (3), or (4) is not as specified, replace ESSS control panel (PARA 16-4-49).
- If result (5) is not as specified, troubleshoot fuel quantity system (PARA 10-2-5 or PARA 10-2-6).

g. Place AUTO XFER/MAN XFER switch to AUTO XFER.

RESULT: Fuel transfer shall continue.

- If result is not as specified, replace ESSS control panel (PARA 16-4-49).

h. Place LEFT OUTBD XFER/PRESS/OFF switch to OFF.

RESULT:

(1) LEFT OUTBD fuel quantity indicator shall display constant fuel quantity.

(2) RIGHT OUTBD fuel quantity indicator shall display constant fuel quantity.

- If results are not as specified, replace signal conditioner (PARA 16-4-48).

i. Place RIGHT OUTBD XFER/PRESS/OFF switch to OFF.

j. Turn off electrical power (PARA 1-6-16).

16-2-2.6 Single Inboard Tank Auto Fuel Transfer Check And Lateral Imbalance Check.

a. Fuel right inboard tank with 674 to 822 pounds of fuel (PARA 1-3-24).

b. Turn on electrical power (PARA 1-6-16).

c. Press and release ESSS control panel TEST switch.

RESULT:

(1) At completion of self-test, no fault codes shall be displayed.

(2) RIGHT INBD fuel quantity indicator shall indicate between 674 and 822 pounds of fuel.

(3) RIGHT INBD EMPTY indicator shall be off.

- If result (1) is not as specified, go to TABLE NO. 16-2-2-1.
- If result (2) is not as specified, replace signal conditioner (PARA 16-4-48).
- If result (3) is not as specified, go to TABLE NO. 16-2-2-20.

d. Place all ESSS control panel XFER/PRESS/OFF switches to PRESS.

NOTE

Allow at least 10 minutes for tanks to pressurize before continuing checkout.

e. Place ESSS control panel AUTO XFER/MAN XFER switch to AUTO XFER.

f. Place RIGHT INBD XFER/PRESS/OFF switch to XFER.

RESULT:

(1) RIGHT FLOW indicator shall remain off.

(2) RIGHT INBD fuel quantity indication shall not change from reading obtained in step c. (2).

- If result (1) is not as specified, replace ESSS control panel (PARA 16-4-49).
- If result (2) is not as specified, go to TABLE NO. 16-2-2-21.

g. Place ESSS control panel AUTO XFER/MAN XFER switch to MAN XFER.

RESULT:

(1) RIGHT FLOW indicator shall go on.

(2) RIGHT INBD fuel quantity indicator reading shall decrease.

(3) CDU MAIN FUEL quantity indicator shall increase.

- If results (1) and (2) are not as specified, go to TABLE NO. 16-2-2-12.
- If result (3) is not as specified, troubleshoot fuel quantity system (PARA 10-2-5 or PARA 10-2-6).

h. Place ESSS control panel RIGHT INBD XFER/PRESS/OFF switch to OFF.

RESULT:

(1) RIGHT FLOW indicator shall go off.

(2) RIGHT INBD fuel quantity indicator reading shall remain constant.

- If results are not as specified, replace ESSS control panel (PARA 16-4-49).

i. Record RIGHT INBD fuel quantity.

j. Turn off electrical power (PARA 1-6-16).

16-2-2.7 Dual Inboard Tanks Auto Fuel Transfer Check.

a. Fuel left inboard tank with the quantity of fuel recorded in 16-2-2.6 step i.

b. Turn on electrical power (PARA 1-6-16).

c. Press and release ESSS control panel TEST switch.

RESULT:

(1) At completion of self-test, no fault codes shall be displayed.

(2) LEFT INBD EMPTY indicator shall be off.

(3) LEFT INBD fuel quantity indicator shall display fuel quantity from step a.

(4) RIGHT INBD fuel quantity indicator shall display value recorded in 16-2-2.6 step i.

(5) LAT IMBAL indicator shall be off.

- If result (1) is not as specified, go to TABLE NO. 16-2-2-1.
- If results (2), (3), (4), and (5) are not as specified, replace ESSS control panel (PARA 16-4-49).

d. Place all ESSS control panel XFER/PRESS/OFF switches to PRESS.

NOTE

Allow at least 10 minutes for tanks to pressurize before continuing checkout.

e. Place LEFT and RIGHT INBD XFER/PRESS/OFF switch to XFER.

f. Place AUTO XFER/MAN XFER switch to MAN XFER.

RESULT:

(1) LEFT FLOW indicator shall go on.

(2) RIGHT FLOW indicator shall go on.

(3) LEFT INBD fuel quantity indicator reading shall indicate a decrease in fuel quantity from that displayed in step c. (3).

(4) RIGHT INBD fuel quantity indicator reading shall indicate a decrease in fuel quantity from that displayed in step c. (4).

(5) CDU MAIN FUEL shall display an increase in fuel.

- If result (1), (2), (3), or (4) is not as specified, replace ESSS control panel (PARA 16-4-49).
- If result (5) is not as specified, troubleshoot fuel quantity system (PARA 10-2-5 or PARA 10-2-6).

g. Place AUTO XFER/MAN XFER switch to AUTO XFER.

RESULT: Fuel transfer shall continue.

- If result is not as specified, replace ESSS control panel (PARA 16-4-49).

h. Place LEFT INBD XFER/PRESS/OFF switch to PRESS.

RESULT:

(1) LEFT INBD fuel quantity indicator shall display constant fuel quantity.

(2) RIGHT INBD fuel quantity indicator shall display constant fuel quantity.

- If results are not as specified, replace ESSS control panel (PARA 16-4-49).

i. Place RIGHT INBD XFER/PRESS/OFF switch to PRESS.

16-2-2.8 Unbalanced Fuel Transfer Check.

a. Place AUTO XFER/MAN XFER switch to AUTO XFER.

b. Place LEFT OUTBD and RIGHT INBD XFER/PRESS/OFF switches to XFER.

RESULT:

(1) LEFT FLOW indicator shall remain off.

(2) RIGHT FLOW indicator shall remain off.

(3) LEFT OUTBD fuel quantity indicator reading shall not change.

(4) RIGHT INBD fuel quantity indicator reading shall not change.

- If results are not as specified, replace ESSS control panel (PARA 16-4-49).

c. Place AUTO XFER/MAN XFER switch to MAN XFER.

RESULT:

(1) LEFT FLOW indicator shall go on.

(2) RIGHT FLOW indicator shall go on.

(3) LEFT OUTBD fuel quantity indicator shall display a decrease in fuel quantity.

(4) RIGHT INBD fuel quantity indicator shall display a decrease in fuel quantity.

- If results are not as specified, replace ESSS control panel (PARA 16-4-49).

d. Place LEFT OUTBD and RIGHT INBD XFER/PRESS/OFF switches to PRESS.

e. Place AUTO XFER/MAN XFER switch to AUTO XFER.

f. Place LEFT INBD and RIGHT OUTBD XFER/PRESS/OFF switches to XFER.

RESULT:

(1) LEFT FLOW indicator shall remain off.

(2) RIGHT FLOW indicator shall remain off.

(3) LEFT INBD fuel quantity indicator reading shall remain constant.

(4) RIGHT OUTBD fuel quantity indicator reading shall remain constant.

- If results are not as specified, replace ESSS control panel (PARA 16-4-49).

g. Place AUTO XFER/MAN XFER switch to MAN XFER.

RESULT:

(1) LEFT FLOW indicator shall go on.

(2) RIGHT FLOW indicator shall go on.

(3) LEFT INBD fuel quantity indicator reading shall decrease.

(4) RIGHT OUTBD fuel quantity indicator reading shall decrease.

- If results are not as specified, replace ESSS control panel (PARA 16-4-49).

h. Place LEFT INBD and RIGHT OUTBD XFER/PRESS/OFF switches to PRESS.

16-2-2.9 Inboard And Outboard Tanks Fuel Transfer Check.

a. Place AUTO XFER/MAN XFER switch to AUTO XFER.

b. Place LEFT INBD and RIGHT INBD XFER/PRESS/OFF switches to XFER.

c. Place LEFT OUTBD and RIGHT OUTBD XFER/PRESS/OFF switches to XFER.

RESULT:

(1) LEFT FLOW indicator shall go on.

(2) RIGHT FLOW indicator shall go on.

(3) LEFT INBD fuel quantity indicator shall display decrease in fuel quantity.

(4) RIGHT OUTBD fuel quantity indicator shall display decrease in fuel quantity.

(5) LEFT OUTBD fuel quantity indicator shall display decrease in fuel quantity.

(6) RIGHT INBD fuel quantity indicator shall display decrease in fuel quantity.

- If results are not as specified, replace ESSS control panel (PARA 16-4-49).

d. Place all XFER/PRESS/OFF switches to PRESS.

RESULT:

(1) LEFT FLOW indicator shall go off.

(2) RIGHT FLOW indicator shall go off.

(3) LEFT INBD fuel quantity indicator shall display a constant fuel quantity.

(4) RIGHT OUTBD fuel quantity indicator shall display a constant fuel quantity.

(5) LEFT OUTBD fuel quantity indicator shall display a constant fuel quantity.

(6) RIGHT INBD fuel quantity indicator shall display a constant fuel quantity.

- If results are not as specified, replace ESSS control panel (PARA 16-4-49).

16-2-2.10 High-Level Auto Transfer Shutoff Check.

a. Turn off electrical power (PARA 1-6-16).

NOTE

Automatic fuel transfer shall occur when either main fuel tank fuel quantity is less than 850 pounds.

b. Pressure defuel/refuel main fuel tanks as necessary to obtain a total fuel quantity of 900 pounds of fuel (PARA 1-3-20 or PARA 1-3-17).

c. Turn on electrical power (PARA 1-6-16).

NOTE

Be prepared to set LEFT OUTBD and RIGHT OUTBD XFER/PRESS/OFF switches to OFF in the event of a high-level shutoff valve failure. If high-level shutoff valves fail, fuel will enter vent lines and will be spilled overboard.

d. Place AUTO XFER/MAN XFER switch to AUTO XFER.

e. Place LEFT OUTBD and RIGHT OUTBD XFER/PRESS/OFF switches to XFER.

RESULT:

(1) LEFT and RIGHT FLOW indicators shall go on.

(2) All fuel quantity indicators shall display a decrease in fuel quantity.

f. Fuel transfer shall occur from external tanks into main fuel tanks. When high-level shutoff is reached, observe the following:

RESULT:

(1) LEFT and RIGHT FLOW indicators shall go off.

(2) All fuel quantity indicators shall display a constant fuel quantity.

(3) CDU MAIN FUEL indicator shall indicate between 2000 and 2500 pounds.

- If results (1) and (2) are not as specified, replace ESSS control panel (PARA 16-4-49).
- If result (3) is not as specified, troubleshoot fuel quantity system (PARA 10-2-5 or PARA 10-2-6).

g. Place LEFT OUTBD and RIGHT OUTBD XFER/PRESS/OFF switches to OFF.

h. Turn off electrical power (PARA 1-6-16).

i. Turn off external air.

16-2-2.11 Main Tank Overflow Sensor Check.

a. Disconnect electrical connector P376 from J376 fuel overflow sensor in vent tube.

b. Remove overflow sensor from vent tube (PARA 16-4-47).

c. Connect electrical connector P376 to J376.

d. Turn on electrical power (PARA 1-6-16).

e. Submerge overflow sensor in small container filled with water.

RESULT: MN OVFL indicator on ESSS control panel shall go on.

- If result is not as specified, go to TABLE NO. 16-2-2-22.

f. Turn off electrical power (PARA 1-6-16).

g. Disconnect electrical connector J376 from P376.

h. Install overflow sensor in vent tube (PARA 16-4-47).

i. Connect electrical connector P376 to J376 on overflow sensor.

j. If no other check is to be done, go to Shutdown.

16-2-2.12 Shutdown.

a. Install explosive cartridges (PARA 16-4-35).

b. Install rear fairing on all vertical stores pylons (PARA 16-4-21).

TABLE NO. 16-2-2-2. ESSS Control Panel INBD And OUTBD Indicators Do Not Dim Or Display 0000.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Do any ESSS control panel indicators go on?

Step 1. If indicators go on, go to **2.**
Step 2. If indicators do not go on, replace ESSS control panel (PARA 16-4-49). Go to **11.**

2. Check for 28 vdc between NO. 2 XFER CONTROL circuit breaker terminal 1 and ground.

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **11.**

3. Check continuity between:

TABLE NO. 16-2-2-2. ESSS Control Panel INBD And OUTBD Indicators Do Not Dim Or Display 0000. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

P3002-12 and P346-35
P3002-13 and P346-35
P3002-14 and P346-35

Step 1. If continuity is present, go to **4.**
Step 2. If continuity is not present, repair/replace wiring, as required. Go to **11.**

4. Check continuity between:
P3002-17 and ground
P3002-18 and ground
P3002-19 and ground
P3002-55 and ground

Step 1. If continuity is present, go to **5.**
Step 2. If continuity is not present, repair/replace wiring, as required. Go to **11.**

5. Check for 28 vdc between P3002-10 and P3002-17.

Step 1. If voltage is as specified, go to **6.**
Step 2. If voltage is not as specified, go to **8.**

6. Check for 115 vac between P3002-1 and ground.

Step 1. If voltage is as specified, go to **7.**
Step 2. If voltage is not as specified, go to **9.**

7. Check for 28 vdc between P3001-19 and ground.

Step 1. If voltage is as specified, replace ESSS control panel (PARA 16-4-49). Go to **11.**
Step 2. If voltage is not as specified, go to **10.**

8. Check for 28 vdc between EXT FUEL LH circuit breaker terminal 1 and ground.

Step 1. If voltage is as specified, repair/replace wiring (Appendix F), as required between, then go to **11:**
P361-1 and terminal 1 (EXT FUEL LH circuit breaker)
P361-2 and terminal 1 (EXT FUEL LH circuit breaker)
P361-3 and terminal 1 (EXT FUEL LH circuit breaker)

TABLE NO. 16-2-2-2. ESSS Control Panel INBD And OUTBD Indicators Do Not Dim Or Display 0000. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **11.**

9. Check for 115 vac between J246-Z and ground.

Step 1. If voltage is as specified, repair/replace wiring between P246-Z and P3002-1 (Appendix F). Go to **11.**
Step 2. If voltage is not as specified, troubleshoot upper and lower consoles lights (PARA 9-2-5). Go to **11.**

10. Check for 28 vdc between P902-X and ground.

Step 1. If voltage is as specified, repair/replace wiring between P902-X and P3001-19 (Appendix F). Go to **11.**
Step 2. If voltage is not as specified, troubleshoot instrument panel and consoles indicator lights dimming system (PARA 9-2-10). Go to **11.**

11. Procedure completed.

TABLE NO. 16-2-2-3. LoZ1 Side Of Inboard Tank Unit Wiring Open/Shorted.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Left or right inboard fuel quantity indicator displays fault code LoZ1.

Step 1. If LEFT INBD indicator displays fault code LoZ1, go to **2.**
Step 2. If RIGHT INBD indicator displays fault code LoZ1, go to **3.**

2. Check continuity between:
P2-D and P3-4
P3-1 and P360-12
P3-4 and ground
P3-7 and P360-9
P3-8 and P360-8

Step 1. If continuity is present, replace signal conditioner (PARA 16-4-48). Go to **4.**

TABLE NO. 16-2-2-3. LoZ1 Side Of Inboard Tank Unit Wiring Open/Shorted. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If continuity is not present, replace front electrical harness (PARA 16-4-45) or repair/replace wiring, as required between, then go to **4:**
P360-8 and P833-5
P360-9 and P833-14
P360-12 and P833-11

3. Check continuity between:
P2-D and P3-4
P3-1 and P360-16
P3-4 and ground
P3-7 and P360-22
P3-8 and P360-21

Step 1. If continuity is present, replace signal conditioner (PARA 16-4-48). Go to **4.**
Step 2. If continuity is not present, replace front electrical harness (PARA 16-4-45) or repair/replace wiring, as required between, then go to **4:**
P360-16 and P834-11
P360-21 and P834-15
P360-22 and P834-14

4. Procedure completed.

TABLE NO. 16-2-2-4. LoZ2 Side Of Outboard Tank Unit Wiring Open/Shorted.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Left or right outboard fuel quantity indicator displays fault code LoZ2.

Step 1. If LEFT OUTBD indicator displays fault code LoZ2, go to **2.**
Step 2. If RIGHT OUTBD indicator displays fault code LoZ2, go to **3.**

2. Check continuity between:
P2-D and P3-4
P3-1 and P360-6
P3-4 and ground
P3-7 and P360-3
P3-8 and P360-2

Step 1. If continuity is present, replace signal conditioner (PARA 16-4-48). Go to **4.**

TABLE NO. 16-2-2-4. LoZ2 Side Of Outboard Tank Unit Wiring Open/Shorted. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If continuity is not present, replace front electrical harness (PARA 16-4-45) or repair/replace wiring, as required between, then go to **4:**
P360-2 and P833-22
P360-3 and P833-21
P360-6 and P833-18

3. Check continuity between:
P2-D and P3-4
P3-1 and P360-19
P3-4 and ground
P3-7 and P360-15
P3-8 and P360-14

Step 1. If continuity is present, replace signal conditioner (PARA 16-4-48). Go to **4.**
Step 2. If continuity is not present, replace front electrical harness (PARA 16-4-45) or repair/replace wiring, as required between, then go to **4:**
P360-14 and P834-22
P360-15 and P834-21
P360-19 and P834-18

4. Procedure completed.

TABLE NO. 16-2-2-5. HiZ1 Side Of Left Inboard Or Right Outboard Forward Tank Unit Wiring Open/Shorted.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Left inboard or right outboard fuel quantity indicator displays fault code HiZ1.

Step 1. If LEFT INBD indicator displays fault code HiZ1, go to **2.**
Step 2. If RIGHT OUTBD indicator displays fault code HiZ1, go to **3.**

2. Check continuity between:
P3-2 and P360-11
P3-11 and P360-10

Step 1. If continuity is present, replace signal conditioner (PARA 16-4-48). Go to **4.**
Step 2. If continuity is not present, replace front electrical harness (PARA 16-4-45) or repair/replace wiring, as required between, then go to **4:**

TABLE NO. 16-2-2-5. HiZ1 Side Of Left Inboard Or Right Outboard Forward Tank Unit Wiring Open/Shorted. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

P360-10 and P833-13
P360-11 and P833-12

3. **Check continuity between:**
   P3-2 and P360-18
   P3-11 and P360-17

   Step 1. If continuity is present, replace signal conditioner (PARA 16-4-48). Go to **4.**
   Step 2. If continuity is not present, replace front electrical harness (PARA 16-4-45) or repair/replace wiring, as required between, then go to **4:**
   P360-17 and P834-20
   P360-18 and P834-19

4. **Procedure completed.**

TABLE NO. 16-2-2-6. HiZ2 Side Of Left Inboard Or Right Outboard Aft Tank Unit Wiring Open/Shorted.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Left inboard or right outboard fuel quantity indicator displays fault code HiZ2.**

   Step 1. If LEFT INBD indicator displays fault code HiZ2, go to **2.**
   Step 2. If RIGHT OUTBD indicator displays fault code HiZ2, go to **3.**

2. **Check continuity between:**
   P3-8 and P360-8
   P3-13 and P360-7

   Step 1. If continuity is present, replace signal conditioner (PARA 16-4-48). Go to **4.**
   Step 2. If continuity is not present, replace front electrical harness (PARA 16-4-45) or repair/replace wiring, as required between, then go to **4:**
   P360-7 and P833-16
   P360-8 and P833-15

3. **Check continuity between:**
   P3-8 and P360-14
   P3-13 and P360-13

TABLE NO. 16-2-2-6. HiZ2 Side Of Left Inboard Or Right Outboard Aft Tank Unit Wiring Open/Shorted. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, replace signal conditioner (PARA 16-4-48). Go to **4.**
Step 2. If continuity is not present, replace front electrical harness (PARA 16-4-45) or repair/replace wiring, as required between, then go to **4:**
P360-13 and P834-23
P360-14 and P834-22

4. **Procedure completed.**

TABLE NO. 16-2-2-7. HiZ3 Side Of Left Outboard Or Right Inboard Forward Tank Unit Wiring Open/Shorted.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Left outboard or right inboard fuel quantity indicator displays fault code HiZ3.**

Step 1. If LEFT OUTBD indicator displays fault code HiZ3, go to **2.**
Step 2. If RIGHT INBD indicator displays fault code HiZ3, go to **3.**

2. **Check continuity between:**
P3-2 and P360-5
P3-11 and P360-4

Step 1. If continuity is present, replace signal conditioner (PARA 16-4-48). Go to **4.**
Step 2. If continuity is not present, replace front electrical harness (PARA 16-4-45) or repair/replace wiring, as required between, then go to **4:**
P360-4 and P833-20
P360-5 and P833-19

3. **Check continuity between:**
P3-2 and P360-24
P3-11 and P360-23

Step 1. If continuity is present, replace signal conditioner (PARA 16-4-48). Go to **4.**
Step 2. If continuity is not present, replace front electrical harness (PARA 16-4-45) or repair/replace wiring, as required between, then go to **4:**
P360-23 and P834-13
P360-24 and P834-12

TABLE NO. 16-2-2-7. HiZ3 Side Of Left Outboard Or Right Inboard Forward Tank Unit Wiring Open/Shorted. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

4. **Procedure completed.**

TABLE NO. 16-2-2-8. HiZ4 Side Of Left Outboard Or Right Inboard Aft Tank Unit Wiring Open/ Shorted.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Left outboard or right inboard fuel quantity indicator displays fault code HiZ4.**

   Step 1. If LEFT OUTBD indicator displays fault code HiZ4, go to **2.**
   Step 2. If RIGHT INBD indicator displays fault code HiZ4, go to **3.**

2. **Check continuity between:**
   P3-8 and P360-2
   P3-13 and P360-1

   Step 1. If continuity is present, replace signal conditioner (PARA 16-4-48). Go to **4.**
   Step 2. If continuity is not present, replace front electrical harness (PARA 16-4-45) or repair/replace wiring, as required between, then go to **4:**
   P360-1 and P833-23
   P360-2 and P833-22

3. **Check continuity between P3-13 and P360-20.**

   Step 1. If continuity is present, replace signal conditioner (PARA 16-4-48). Go to **4.**
   Step 2. If continuity is not present, replace front electrical harness (PARA 16-4-45) or repair/replace wiring between P360-20 and P834-16, as required. Go to **4.**

4. **Procedure completed.**

TABLE NO. 16-2-2-9. Left Outboard MAN XFER Test Fails.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Is fuel flowing in external left line?**

TABLE NO. 16-2-2-9. Left Outboard MAN XFER Test Fails. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If fuel is flowing, go to **2.**
Step 2. If fuel is not flowing, go to **4.**

2. Disconnect J521 from left external fuel flow sensor. Observe LEFT FLOW indicator goes on.

Step 1. If LEFT FLOW indicator goes on, replace left external fuel flow sensor (flow transmitter) (PARA 16-4-31). Go to **10.**
Step 2. If LEFT FLOW indicator does not go on, go to **3.**

3. Check continuity between J521-1 and J521-2.

Step 1. If continuity is present, repair/replace wiring. Go to **10.**
Step 2. If continuity is not present, replace ESSS control panel (PARA 16-4-49). Go to **10.**

4. Visually check that manual lever of gate valve is in the OPEN position.

Step 1. If gate valve is in the OPEN position, go to **5.**
Step 2. If gate valve is in the CLSD position, go to **8.**

5. Check for 28 vdc between EXT FUEL RH circuit breaker terminal 1 and ground.

Step 1. If voltage is as specified, go to **6.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **10.**

6. Check continuity between P3002-11 and P346-33.

Step 1. If continuity is present, go to **7.**
Step 2. If continuity is not present, repair/replace wiring. Go to **10.**

7. Check for 28 vdc between:
P839-A and ground
P839-B and ground

Step 1. If voltage is as specified, replace left outboard bleed air regulator valve (PARA 16-4-32). Go to **10.**
Step 2. If trouble remains, replace ESSS control panel (PARA 16-4-49). Go to **10.**

TABLE NO. 16-2-2-9. Left Outboard MAN XFER Test Fails. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 3. If voltage is not as specified, replace rear electrical harness (PARA 16-4-44) or repair/replace wiring between P805-7 and P3002-36, as required. Go to **10.**

8. Check for 28 vdc between P840-A and P840-E.

Step 1. If voltage is as specified, replace left outboard fuel shutoff gate valve (PARA 16-4-33). Go to **10.**
Step 2. If voltage is not as specified, go to **9.**

9. Check continuity between:
P840-A and P3002-38
P840-E and ground

Step 1. If continuity is present, replace ESSS control panel (PARA 16-4-49). Go to **10.**
Step 2. If continuity is not present, replace rear electrical harness (PARA 16-4-44) or repair/replace wiring between P805-9 and P3002-38, as required. Go to **10.**

10. Procedure completed.

TABLE NO. 16-2-2-10. Left Inboard MAN XFER Test Fails.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Is fuel flowing in external left line?

Step 1. If fuel is flowing, go to **2.**
Step 2. If fuel is not flowing, go to **4.**

2. Disconnect J521 from left external fuel flow sensor. Observe LEFT FLOW indicator goes on.

Step 1. If LEFT FLOW indicator goes on, replace left external fuel flow sensor (flow transmitter) (PARA 16-4-31). Go to **8.**
Step 2. If LEFT FLOW indicator does not go on, go to **3.**

3. Check continuity between J521-1 and J521-2.

TABLE NO. 16-2-2-10. Left Inboard MAN XFER Test Fails. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, repair/replace wiring. Go to **8.**
Step 2. If continuity is not present, replace ESSS control panel (PARA 16-4-49). Go to **8.**

4. Visually check that manual lever of gate valve is in the OPEN position.

Step 1. If gate valve is in the OPEN position, go to **5.**
Step 2. If gate valve is in the CLSD position, go to **6.**

5. Check for 28 vdc between P808-A and ground.

Step 1. If voltage is as specified, replace left inboard bleed air regulator valve (PARA 16-4-32). Go to **8.**
Step 2. If trouble remains, replace ESSS control panel (PARA 16-4-49). Go to **8.**
Step 3. If voltage is not as specified, replace rear electrical harness (PARA 16-4-44) or repair/replace wiring between P805-2 and P3002-42, as required. Go to **8.**

6. Check for 28 vdc between P809-A and P809-E.

Step 1. If voltage is as specified, replace left outboard fuel shutoff gate valve (PARA 16-4-33). Go to **8.**
Step 2. If voltage is not as specified, go to **7.**

7. Check continuity between:
P809-A and P3002-45
P809-E and ground

Step 1. If continuity is present, replace ESSS control panel (PARA 16-4-49). Go to **8.**
Step 2. If continuity is not present, replace rear electrical harness (PARA 16-4-44) or repair/replace wiring between P805-3 and P3002-45, as required. Go to **8.**

8. Procedure completed.

TABLE NO. 16-2-2-11. Right Outboard MAN XFER Test Fails.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Is fuel flowing in external right line?

TABLE NO. 16-2-2-11. Right Outboard MAN XFER Test Fails. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If fuel is flowing, go to **2.**
Step 2. If fuel is not flowing, go to **4.**

2. Disconnect J520 from right fuel flow sensor. Observe RIGHT FLOW indicator goes on.

Step 1. If RIGHT FLOW indicator goes on, replace right external fuel flow sensor (flow transmitter) (PARA 16-4-31). Go to **9.**
Step 2. If RIGHT FLOW indicator does not go on, go to **3.**

3. Check continuity between J520-1 and J520-2.

Step 1. If continuity is present, replace fuel flow sensor harness (PARA 16-4-30) or repair/replace wiring, as required between, then go to **9:**
P346-16 and P3001-13
P346-17 and P3001-12
Step 2. If continuity is not present, replace ESSS control panel (PARA 16-4-49). Go to **9.**

4. Visually check that manual lever of gate valve is in the OPEN position.

Step 1. If gate valve is in the OPEN position, go to **5.**
Step 2. If gate valve is in the CLSD position, go to **6.**

5. Check for 28 vdc between P837-A and P837-B.

Step 1. If voltage is as specified, replace right outboard bleed air regulator valve (PARA 16-4-32). Go to **9.**
Step 2. If voltage is not as specified, go to **6.**

6. Check continuity between:
P837-A and P360-R
P837-B and ground

Step 1. If continuity is present, replace ESSS control panel (PARA 16-4-49). Go to **9.**
Step 2. If continuity is not present, replace rear electrical harness (PARA 16-4-44) or repair/replace wiring between P804-7 and P3002-51, as required. Go to **9.**

7. Check for 28 vdc between P838-A and P838-E.

TABLE NO. 16-2-2-11. Right Outboard MAN XFER Test Fails. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, replace right outboard fuel shutoff gate valve (PARA 16-4-33). Go to **9.**
Step 2. If voltage is not as specified, go to **8.**

8. Check continuity between:
P838-A and P3002-54
P838-E and ground

Step 1. If continuity is present, replace ESSS control panel (PARA 16-4-49). Go to **9.**
Step 2. If continuity is not present, replace rear electrical harness (PARA 16-4-44) or repair/replace wiring between P804-8 and P3002-54, as required. Go to **8.**

9. Procedure completed.

TABLE NO. 16-2-2-12. Right Inboard MAN XFER Test Fails.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Is fuel flowing in external right line?

Step 1. If fuel is flowing, go to **2.**
Step 2. If fuel is not flowing, go to **4.**

2. Disconnect J520 from right external fuel flow sensor (flow transmitter). Observe RIGHT FLOW indicator goes on.

Step 1. If RIGHT FLOW indicator goes on, replace right external fuel flow transmitter (PARA 16-4-31). Go to **8.**
Step 2. If RIGHT FLOW indicator does not go on, go to **3.**

3. Check continuity between J520-1 and J520-2.

Step 1. If continuity is present, repair/replace wiring. Go to **8.**
Step 2. If continuity is not present, replace ESSS control panel (PARA 16-4-49). Go to **8.**

4. Visually check that manual lever of gate valve is in the OPEN position.

TABLE NO. 16-2-2-12. Right Inboard MAN XFER Test Fails. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If gate valve is in the OPEN position, go to **5.**
Step 2. If gate valve is in the CLSD position, go to **6.**

5. Check for 28 vdc between:
P811-A and ground
P811-B and ground

Step 1. If voltage is as specified, replace right inboard bleed air regulator valve (PARA 16-4-32). Go to **8.**
Step 2. If trouble remains, replace ESSS control panel (PARA 16-4-49). Go to **8.**
Step 3. If voltage is not as specified, replace rear electrical harness (PARA 16-4-44) or repair/replace wiring between P804-2 and P3002-40, as required. Go to **8.**

6. Check for 28 vdc between P810-A and P810-E.

Step 1. If voltage is as specified, replace right inboard fuel shutoff gate valve (PARA 16-4-33). Go to **8.**
Step 2. If voltage is not as specified, go to **7.**

7. Check continuity between:
P810-A and P3002-49
P810-E and ground

Step 1. If continuity is present, replace ESSS control panel (PARA 16-4-49). Go to **8.**
Step 2. If continuity is not present, replace rear electrical harness (PARA 16-4-44) or repair/replace wiring between P804-3 and P3002-49, as required. Go to **8.**

8. Procedure completed.

TABLE NO. 16-2-2-13. Left Outboard Fuel Transfer Does Not Stop.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Is fuel flowing in external left line?

Step 1. If fuel is flowing, go to **2.**
Step 2. If fuel is not flowing, go to **5.**

TABLE NO. 16-2-2-13. Left Outboard Fuel Transfer Does Not Stop. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

2. Observe that manual lever on left outboard fuel shutoff gate valve is in the OPEN position.

Step 1. If manual lever is in the OPEN position, go to **3.**
Step 2. If manual lever is in the CLSD position, replace left outboard fuel shutoff gate valve (PARA 16-4-33). Go to **6.**

3. Check for 28 vdc between P840-D and P840-E.

Step 1. If voltage is as specified, replace left outboard fuel shutoff gate valve (PARA 16-4-33). Go to **6.**
Step 2. If voltage is not as specified, go to **4.**

4. Check continuity between:
P840-D and P3002-37
P840-E and ground

Step 1. If continuity is present, replace ESSS control panel (PARA 16-4-49). Go to **6.**
Step 2. If continuity is not present, replace rear electrical harness (PARA 16-4-44) or repair/replace wiring between P805-8 and P3002-37, as required. Go to **6.**

5. Connect jumper between P521-1 and P521-2. Observe that LEFT FLOW indicator goes off.

Step 1. If LEFT FLOW indicator goes off, remove jumper then replace left external fuel flow sensor (flow transmitter) (PARA 16-4-31). Go to **6.**
Step 2. If LEFT FLOW indicator stays on, remove jumper then replace ESSS control panel (PARA 16-4-49). Go to **6.**

6. Procedure completed.

TABLE NO. 16-2-2-14. Left Inboard Fuel Transfer Does Not Stop.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Is fuel flowing in external left line?

TABLE NO. 16-2-2-14. Left Inboard Fuel Transfer Does Not Stop. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If fuel is flowing, go to **2.**
Step 2. If fuel is not flowing, go to **5.**

2. Observe that manual lever on fuel transfer gate valve is in the OPEN position.

Step 1. If manual lever is in the OPEN position, go to **3.**
Step 2. If manual lever is in the CLSD position, replace fuel transfer gate valve (PARA 16-4-33). Go to **6.**

3. Check for 28 vdc between P809-D and P809-E.

Step 1. If voltage is as specified, replace left inboard fuel shutoff gate valve (PARA 16-4-33). Go to **6.**
Step 2. If voltage is not as specified, go to **4.**

4. Check continuity between:
P809-D and P3002-44
P809-E and ground

Step 1. If continuity is present, replace ESSS control panel (PARA 16-4-49). Go to **6.**
Step 2. If continuity is not present, replace rear electrical harness (PARA 16-4-44) or repair/replace wiring between P805-4 and P3002-44, as required. Go to **6.**

5. Connect jumper between P521-1 and P521-2. Observe that LEFT FLOW indicator goes off.

Step 1. If LEFT FLOW indicator goes off, remove jumper then replace left external fuel flow sensor (flow transmitter) (PARA 16-4-31). Go to **6.**
Step 2. If LEFT FLOW indicator stays on, remove jumper then replace ESSS control panel (PARA 16-4-49). Go to **6.**

6. Procedure completed.

TABLE NO. 16-2-2-15. Right Outboard Fuel Transfer Does Not Stop.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Is fuel flowing in external right line?

TABLE NO. 16-2-2-15. Right Outboard Fuel Transfer Does Not Stop. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If fuel is flowing, go to **2.**
Step 2. If fuel is not flowing, go to **5.**

2. Observe that manual lever on fuel transfer gate valve is in the OPEN position.

Step 1. If manual lever is in the OPEN position, go to **3.**
Step 2. If manual lever is in the CLSD position, replace fuel transfer gate valve (PARA 16-4-33). Go to **6.**

3. Check for 28 vdc between P838-D and P838-E.

Step 1. If voltage is as specified, replace right outboard fuel shutoff gate valve (PARA 16-4-33). Go to **6.**
Step 2. If voltage is not as specified, go to **4.**

4. Check continuity between:
P838-D and P3002-53
P838-E and ground

Step 1. If continuity is present, replace ESSS control panel (PARA 16-4-49). Go to **6.**
Step 2. If continuity is not present, replace rear electrical harness (PARA 16-4-44) or repair/replace wiring between P804-9 or P3002-53, as required. Go to **6.**

5. Connect jumper between P520-1 and P520-2. Observe that RIGHT FLOW indicator goes off.

Step 1. If RIGHT FLOW indicator goes off, remove jumper then replace right external fuel flow sensor (flow transmitter) (PARA 16-4-31). Go to **6.**
Step 2. If RIGHT FLOW indicator stays on, remove jumper then replace ESSS control panel (PARA 16-4-49). Go to **6.**

6. Procedure completed.

TABLE NO. 16-2-2-16. Right Inboard Fuel Transfer Does Not Stop.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Is fuel flowing in external right line?

Step 1. If fuel is flowing, go to **2.**
Step 2. If fuel is not flowing, go to **5.**

2. Observe that manual lever on fuel transfer gate valve is in the OPEN position.

Step 1. If manual lever is in the OPEN position, go to **3.**
Step 2. If manual lever is in the CLSD position, replace right inboard fuel shutoff gate valve (PARA 16-4-33). Go to **6.**

3. Check for 28 vdc between P810-D and P810-E.

Step 1. If voltage is as specified, replace right inboard fuel shutoff valve (PARA 16-4-33). Go to **6.**
Step 2. If voltage is not as specified, go to **4.**

4. Check continuity between:
P810-D and P3002-48
P810-E and ground

Step 1. If continuity is present, replace ESSS control panel (PARA 16-4-49). Go to **6.**
Step 2. If continuity is not present, replace rear electrical harness (PARA 16-4-44) or repair/replace wiring between P804-4 and P3002-48, as required. Go to **6.**

5. Connect jumper between P520-1 and P520-2. Observe that RIGHT FLOW indicator goes off.

Step 1. If RIGHT FLOW indicator goes off, remove jumper then replace right external fuel flow sensor (PARA 16-4-31). Go to **6.**
Step 2. If RIGHT FLOW indicator stays on, remove jumper then replace ESSS control panel (PARA 16-4-49). Go to **6.**

6. Procedure completed.

TABLE NO. 16-2-2-17. Left Outboard Test Fails In Auto Mode.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Is fuel flowing in external left line?

Step 1. If fuel is flowing, go to **2.**
Step 2. If fuel is not flowing, go to **4.**

2. Check for 28 vdc between P840-A and ground.

Step 1. If voltage is as specified, replace ESSS control panel (PARA 16-4-49). Go to **5.**
Step 2. If voltage is not as specified, go to **3.**

3. Check for 28 vdc between P840-D and ground.

Step 1. If voltage is as specified, replace left outboard fuel shutoff gate valve (PARA 16-4-33). Go to **5.**
Step 2. If voltage is not as specified, replace ESSS control panel (PARA 16-4-49). Go to **5.**

4. Connect jumper between J521-1 and J521-2.

Step 1. If FLOW indicator stays on, remove jumper then replace ESSS control panel (PARA 16-4-49). Go to **5.**
Step 2. If FLOW indicator goes off, remove jumper then replace left external fuel flow sensor (PARA 16-4-30). Go to **5.**

5. Procedure completed.

TABLE NO. 16-2-2-18. RIGHT OUTBD EMPTY Indicator Is On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check continuity between P2-B and P2-F on the right outboard tank.

Step 1. If continuity is present, replace right outboard float switch (PARA 16-4-68). Go to **2.**
Step 2. If continuity is not present, replace ESSS control panel (PARA 16-4-49). Go to **2.**

2. Procedure completed.

TABLE NO. 16-2-2-19. LAT IMBAL Indicator Is On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check continuity between:**
   P3-1 and P360-19
   P3-2 and P360-18
   P3-5 and P360-43
   P3-7 and P360-15
   P3-8 and P360-14
   P3-11 and P360-17
   P3-13 and P360-13
   P3001-17 and P360-34

   Step 1. If continuity is present, replace ESSS control panel (PARA 16-4-49). Go to **2.**
   Step 2. If trouble remains, replace signal conditioner (PARA 16-4-48). Go to **2.**
   Step 3. If continuity is not present, replace front electrical harness (PARA 16-4-45) or repair/replace wiring, as required between, then go to **2:**
   P360-13 and P834-23
   P360-14 and P834-22
   P360-15 and P834-21
   P360-17 and P834-20
   P360-18 and P834-19
   P360-19 and P834-18
   P360-34 and P3001-17
   P360-43 and P834-24

2. **Procedure completed.**

TABLE NO. 16-2-2-20. RIGHT INBD EMPTY Indicator Is On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check continuity between right inboard P2-B and P2-F.**

   Step 1. If continuity is present, replace right inboard float switch (PARA 16-4-68). Go to **2.**
   Step 2. If continuity is not present, replace ESSS control panel (PARA 16-4-49). Go to **2.**

TABLE NO. 16-2-2-20. RIGHT INBD EMPTY Indicator Is On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

2. **Procedure completed.**

TABLE NO. 16-2-2-21. Right Inboard Test Fails In Auto Mode.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Is fuel flowing in external right line?**

   Step 1. If fuel is flowing, go to **2.**
   Step 2. If fuel is not flowing, go to **4.**

2. **Check for 28 vdc between P810-A and ground.**

   Step 1. If voltage is as specified, replace ESSS control panel (PARA 16-4-49). Go to **5.**
   Step 2. If voltage is not as specified, go to **3.**

3. **Check for 28 vdc between P810-D and ground.**

   Step 1. If voltage is as specified, replace right inboard fuel shutoff gate valve (PARA 16-4-33). Go to **5.**
   Step 2. If voltage is not as specified, replace ESSS control panel (PARA 16-4-49). Go to **5.**

4. **Connect jumper between J520-1 and J520-2.**

   Step 1. If FLOW indicator remains on, remove jumper then replace ESSS control panel (PARA 16-4-49). Go to **5.**
   Step 2. If FLOW indicator goes off, remove jumper then replace right external fuel flow sensor (PARA 16-4-31). Go to **5.**

5. **Procedure completed.**

TABLE NO. 16-2-2-22. Main Tank Overflow Sensor Check Fails.

TEST OR INSPECTION
CORRECTIVE ACTION

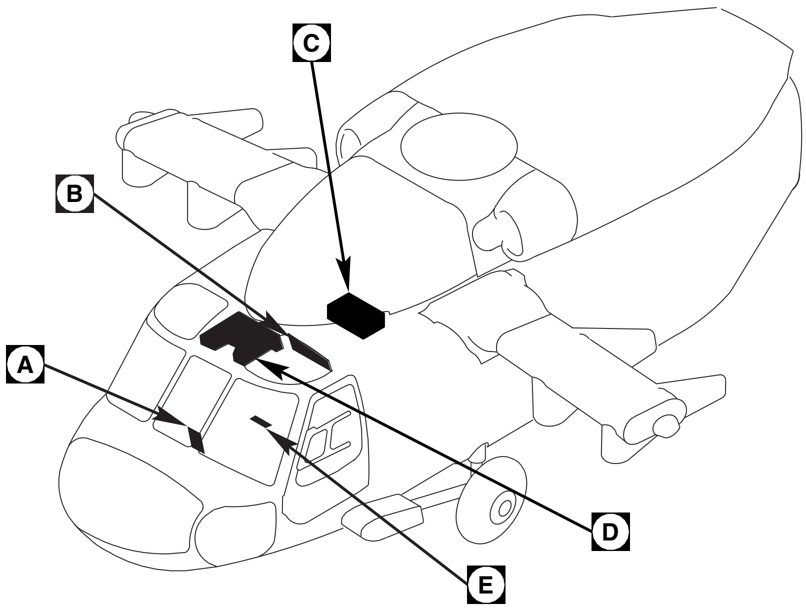
1. Check resistance of overflow sensor between:
P360-27 and P360-28 (650-1500 ohms)
P360-27 and P360-29 (360-440 ohms)

Step 1. If resistance is as specified, replace ESSS control panel (PARA 16-4-49). Go to **3.**
Step 2. If resistance is not as specified, go to **2.**

2. Check continuity between:
P360-27 and J376-C
P360-28 and J376-A
P360-29 and J376-B

Step 1. If continuity is present, replace overflow sensor (PARA 16-4-47). Go to **3.**
Step 2. If continuity is not present, repair/replace wiring, as required. Go to **3.**

3. Procedure completed.



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
J299	CABIN CEILING BL 10 LH, STA 302
J520	RIGHT EXT FUEL FLOW SENSOR
J521	LEFT EXT FUEL FLOW SENSOR
P1 / J1	RIGHT INBOARD FUEL TANK CONNECTOR
P1 / J1	LEFT INBOARD FUEL TANK CONNECTOR
P1 / J1	LEFT OUTBOARD FUEL TANK CONNECTOR
P1 / J1	RIGHT OUTBOARD FUEL TANK CONNECTOR
P2 / J835	LEFT INBOARD FUEL TANK CONNECTOR
P2 / J836	RIGHT INBOARD FUEL TANK CONNECTOR

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P2 / J849	LEFT OUTBOARD FUEL TANK CONNECTOR
P2 / J850	RIGHT OUTBOARD FUEL TANK CONNECTOR
P113 / J113	COCKPIT BL 6 LH, STA 197
P114 / J114	COCKPIT BL 8 RH, STA 200
P117 / J117	CAUTION / ADVISORY PANEL
P137	NO. 2 SIGNAL DATA CONVERTER
P138	NO. 1 SIGNAL DATA CONVERTER
P163 / J163	LEFT DRAG BEAM SWITCH
P217 / J217	LOWER CONSOLE DISCONNECT
P246 / J246	CABIN CEILING BL 5 RH, STA 247
P334 / J334	CABIN DISCONNECT

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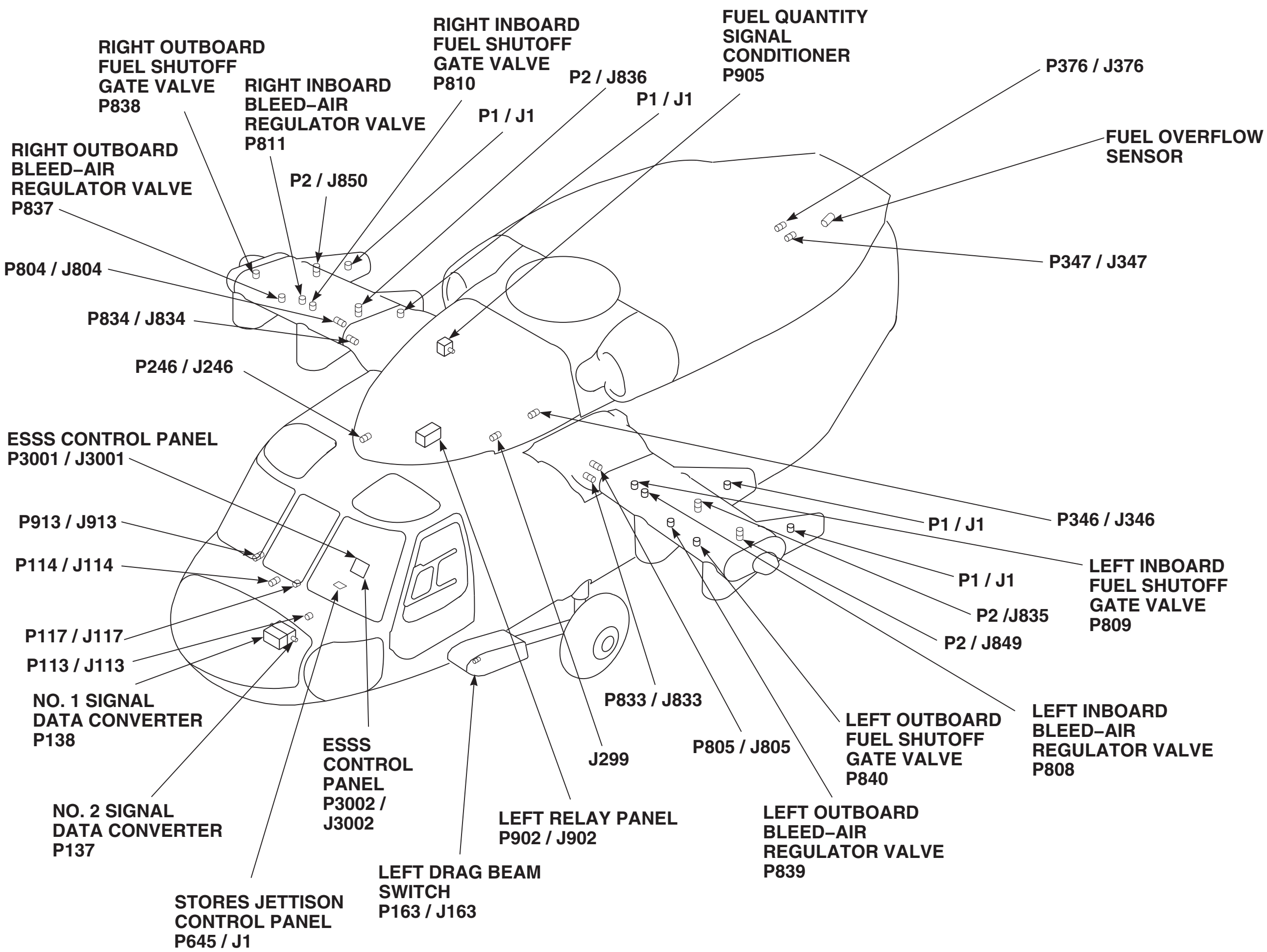
SA

FIGURE 16-2-2-1. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING LOCATION DIAGRAM (SHEET 1 OF 4)



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P346 / J346	CABIN CEILING BL 20 LH, STA 304
P347 / J347	TRANSITION SECTION BL 10 LH, STA 425
P360 / J1	SIGNAL CONDITIONER
P361 / J2	SIGNAL CONDITIONER
P376 / J376	AFT TRANSITION BL 18.25 LH, STA 443.5
P645 / J1	STORES JETTISON CONTROL PANEL
P804 / J804	BL 59 RH, STA 295
P805 / J805	BL 59 LH, STA 295
P808	LEFT INBOARD BLEED–AIR REGULATOR VALVE
P809	LEFT INBOARD FUEL SHUT– OFF GATE VALVE
P810	RIGHT INBOARD FUEL SHUTOFF GATE VALVE
P811	RIGHT INBOARD BLEED–AIR REGULATOR VALVE

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P833 / J833	ATTACHMENT FITTING BL 59 LH, STA 295
P834 / J834	ATTACHMENT FITTING BL 59 LH, STA 295
P837	RIGHT OUTBOARD BLEED– AIR REGULATOR VALVE
P838	RIGHT OUTBOARD FUEL SHUTOFF GATE VALVE
P839	LEFT OUTBOARD BLEED– AIR REGULATOR VALVE
P840	LEFT OUTBOARD FUEL SHUTOFF GATE VALVE
P902 / J902	LEFT RELAY PANEL
P905	FUEL QUANTITY SIGNAL CONDITIONER
P913 / J913	COCKPIT BL 24 RH, STA 247
P3001 / J3001	ESSS CONTROL PANEL
P3002 / J3002	ESSS CONTROL PANEL

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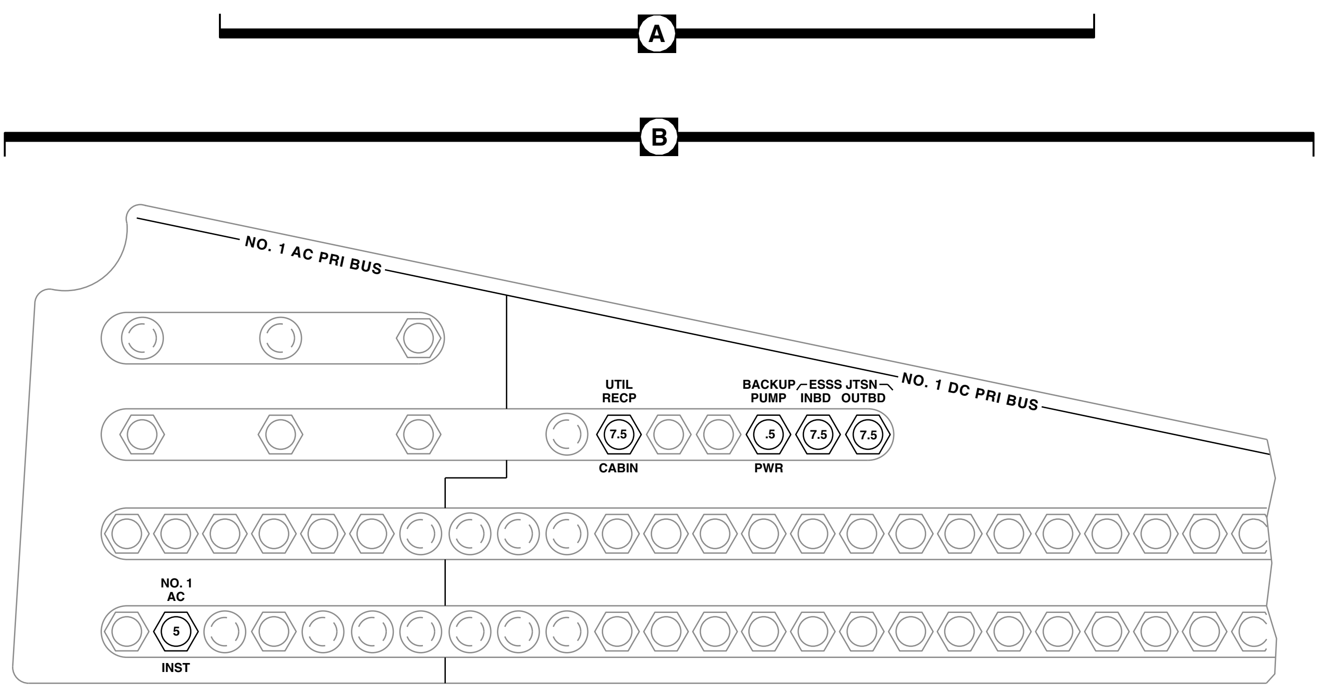
SA

FIGURE 16-2-2-1. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING LOCATION DIAGRAM (SHEET 2 OF 4)

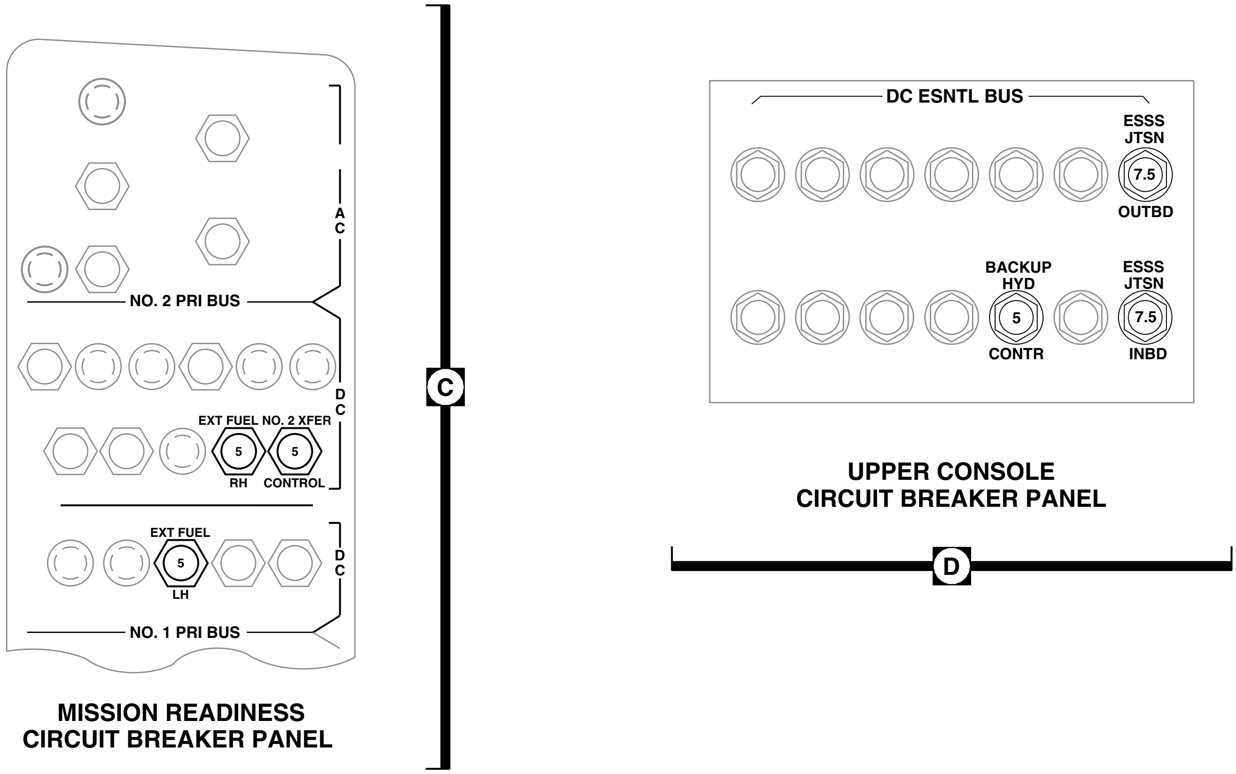
CAUTION/ADVISORY PANEL



COPILOT'S CIRCUIT BREAKER PANEL

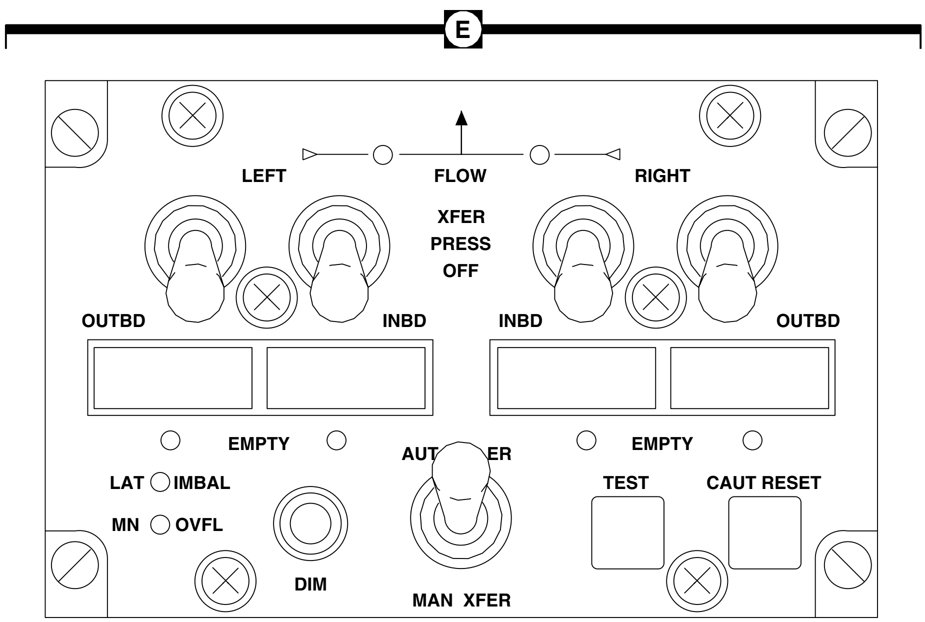
FB1616_3
SA

FIGURE 16-2-2-1. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING LOCATION DIAGRAM (SHEET 3 OF 4)



MISSION READINESS
CIRCUIT BREAKER PANEL

UPPER CONSOLE
CIRCUIT BREAKER PANEL



ESSS CONTROL PANEL

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SA

FIGURE 16-2-2-1. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING LOCATION DIAGRAM (SHEET 4 OF 4)

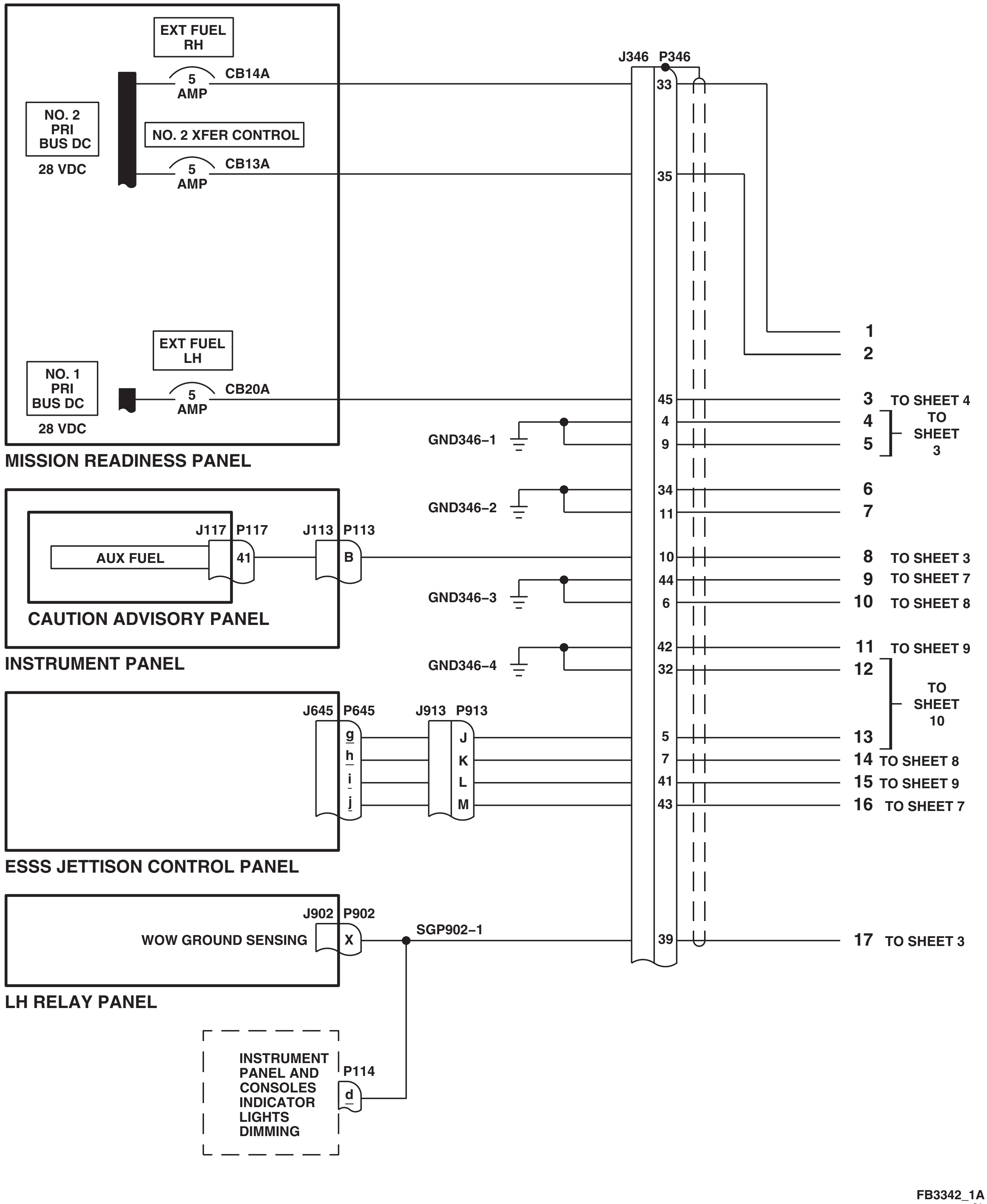


FIGURE 16-2-2-2. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING SCHEMATIC DIAGRAM (SHEET 1 OF 10)

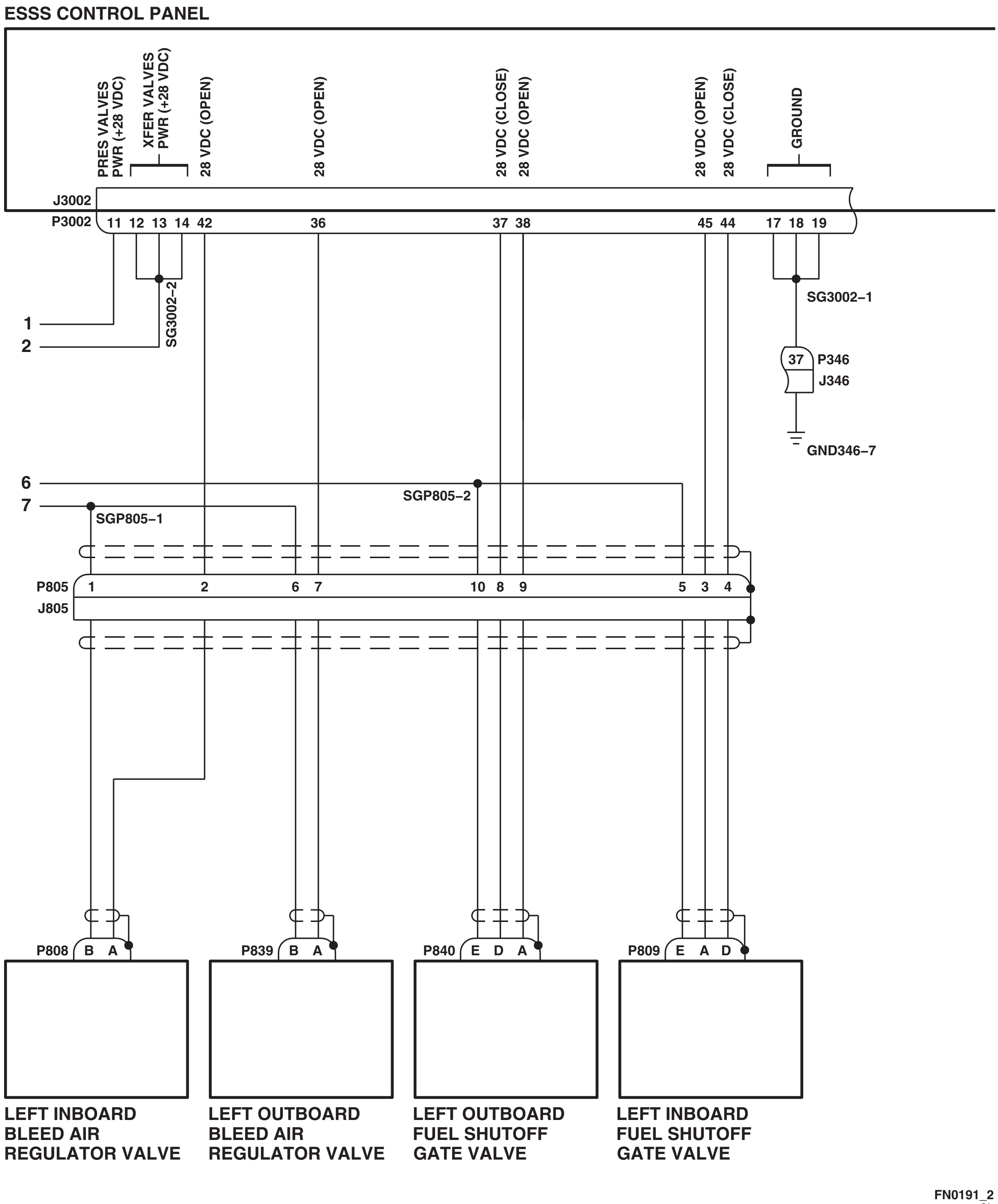


FIGURE 16-2-2-2. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING SCHEMATIC DIAGRAM (SHEET 2 OF 10)

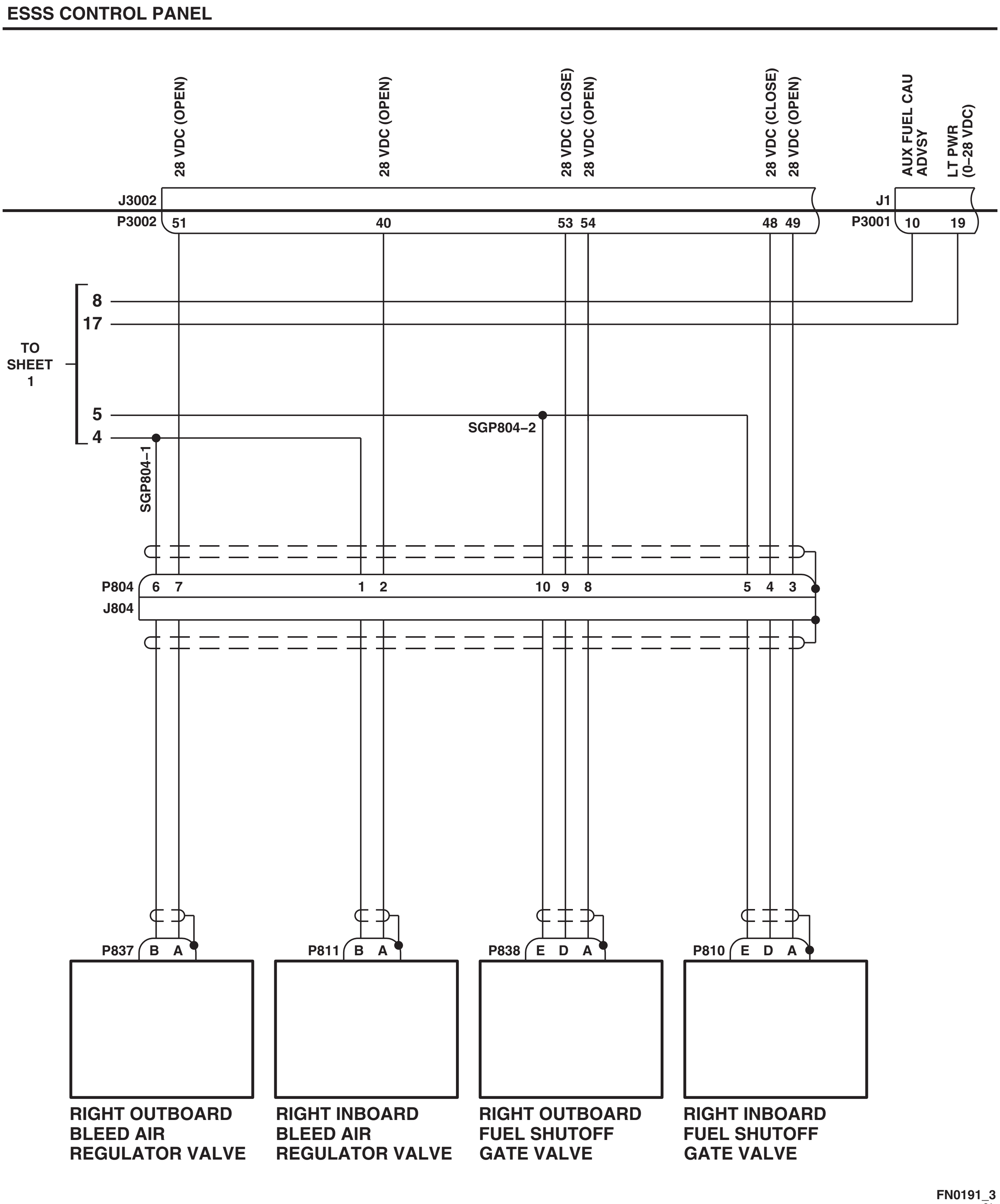


FIGURE 16-2-2-2. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING SCHEMATIC DIAGRAM (SHEET 3 OF 10)

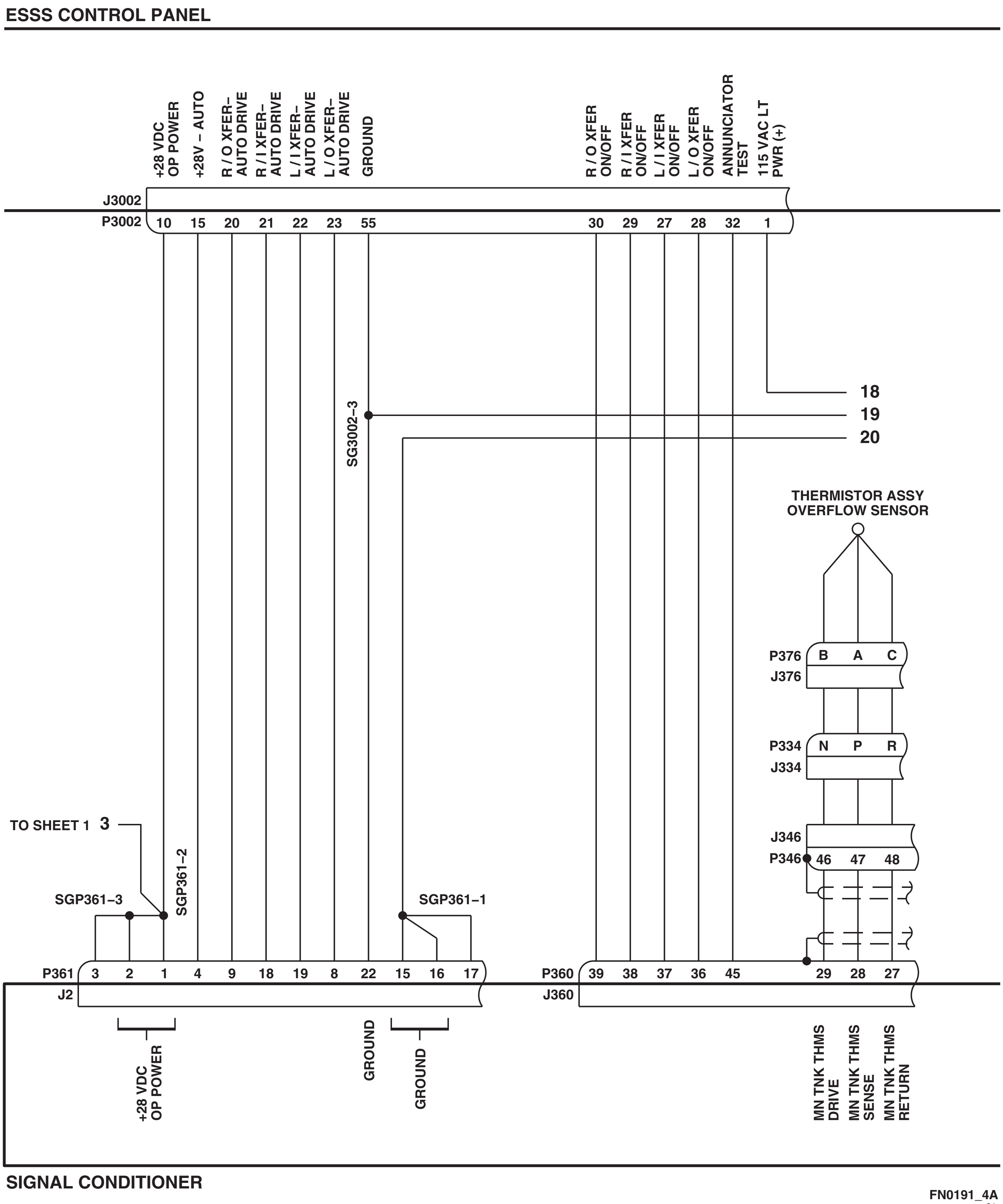


FIGURE 16-2-2-2. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING SCHEMATIC DIAGRAM (SHEET 4 OF 10)

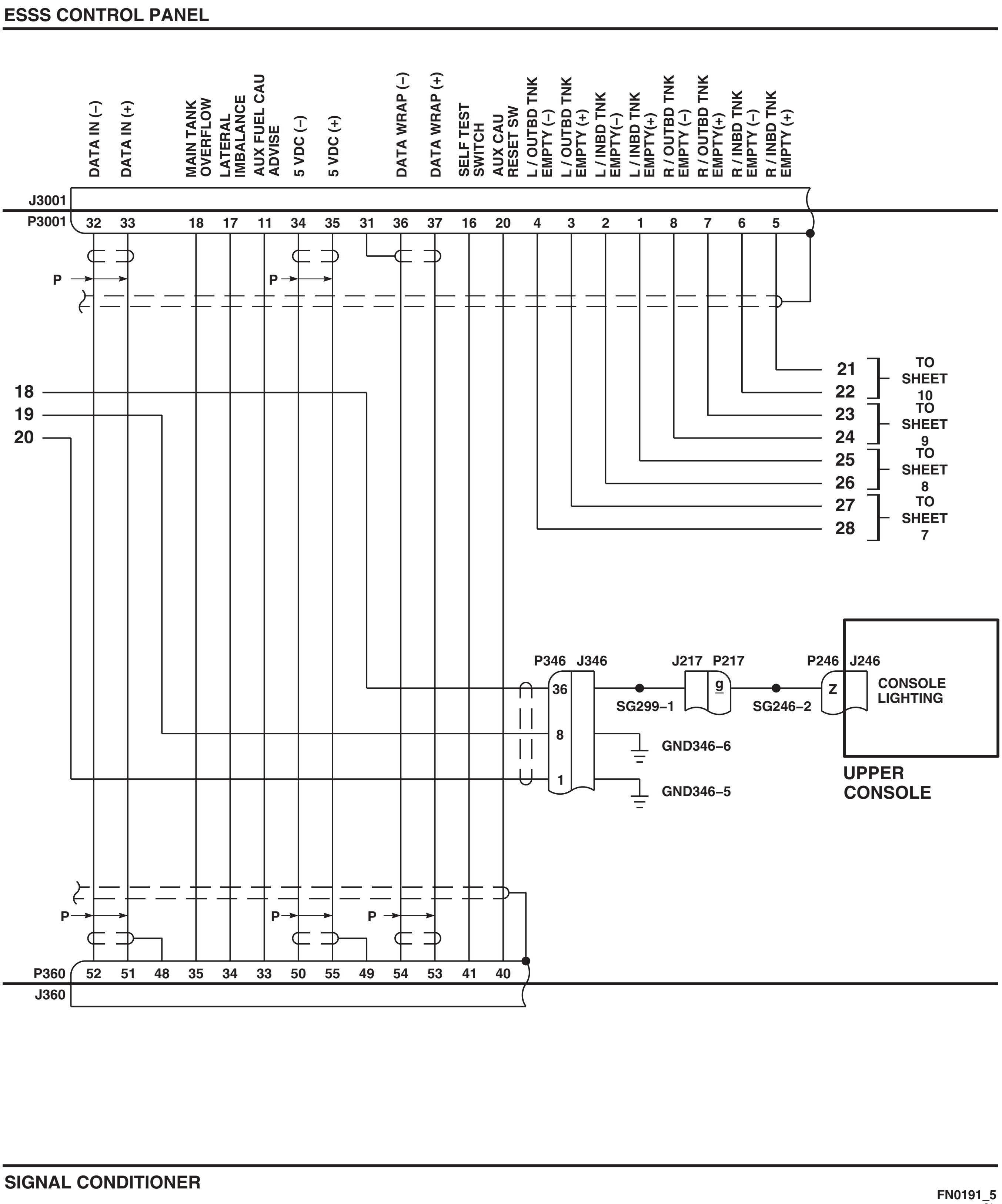


FIGURE 16-2-2-2. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING SCHEMATIC DIAGRAM (SHEET 5 OF 10)

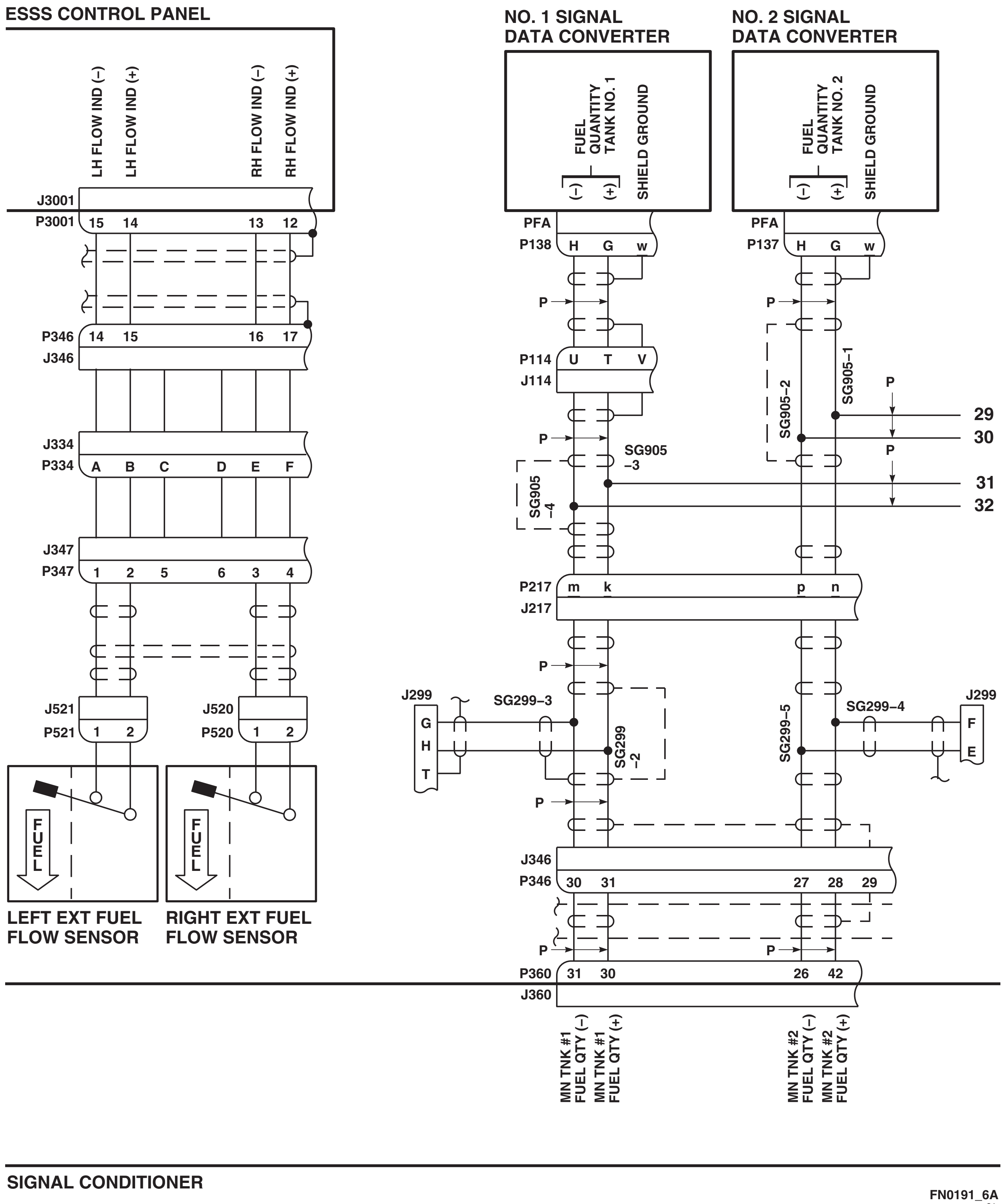
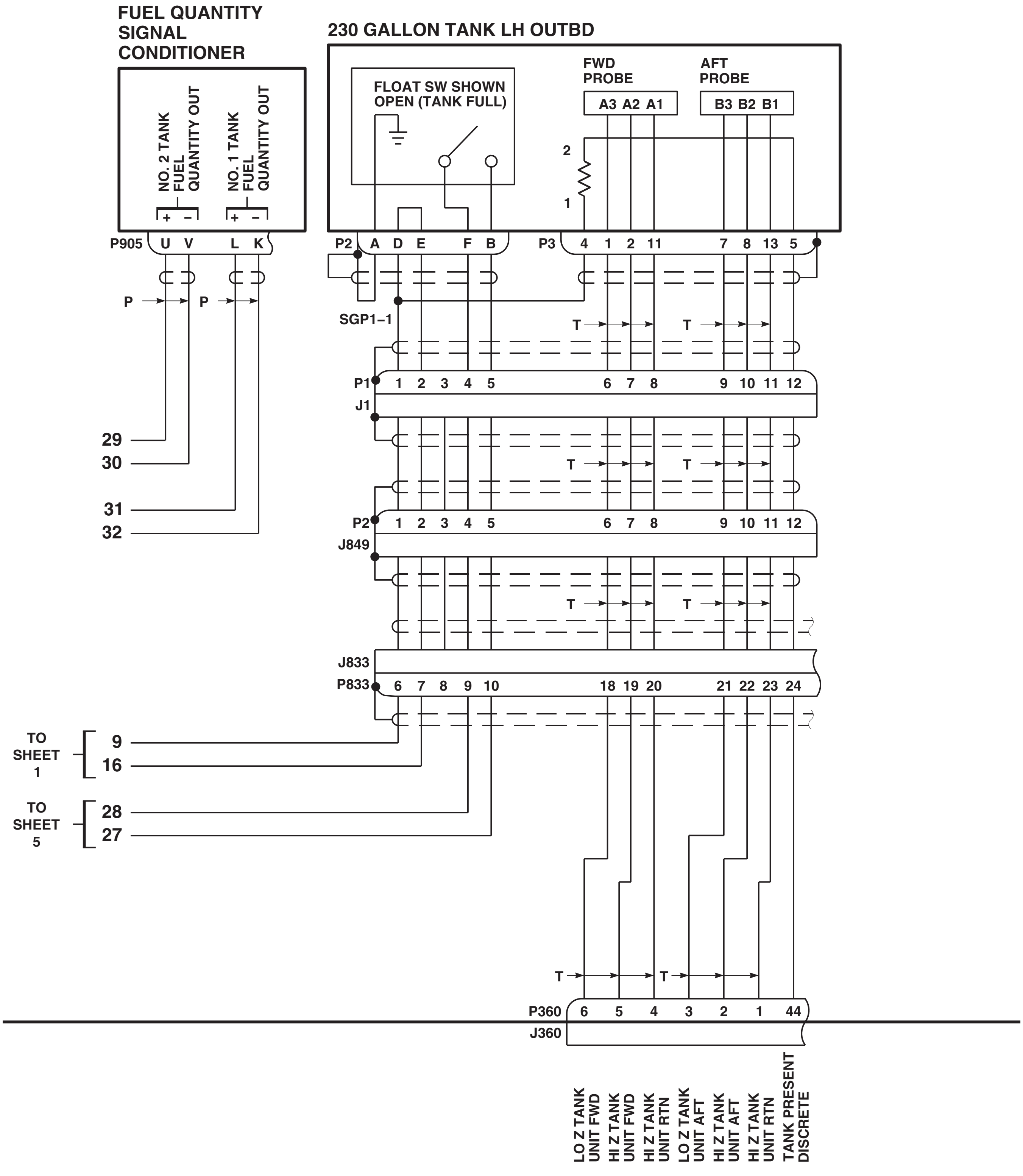


FIGURE 16-2-2-2. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING SCHEMATIC DIAGRAM (SHEET 6 OF 10)



SIGNAL CONDITIONER

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SA

FIGURE 16-2-2-2. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING SCHEMATIC DIAGRAM (SHEET 7 OF 10)

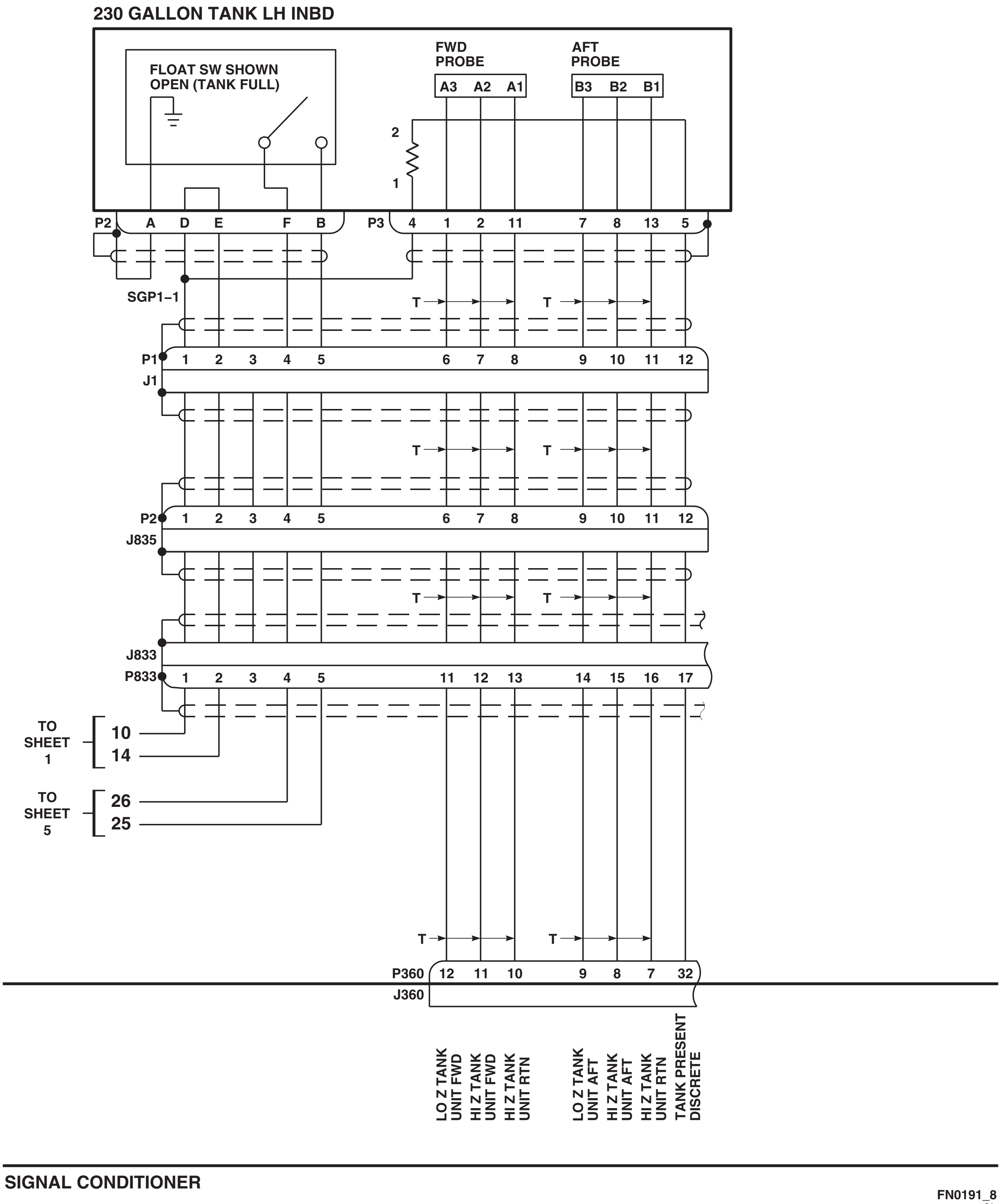


FIGURE 16-2-2-2. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING SCHEMATIC DIAGRAM (SHEET 8 OF 10)

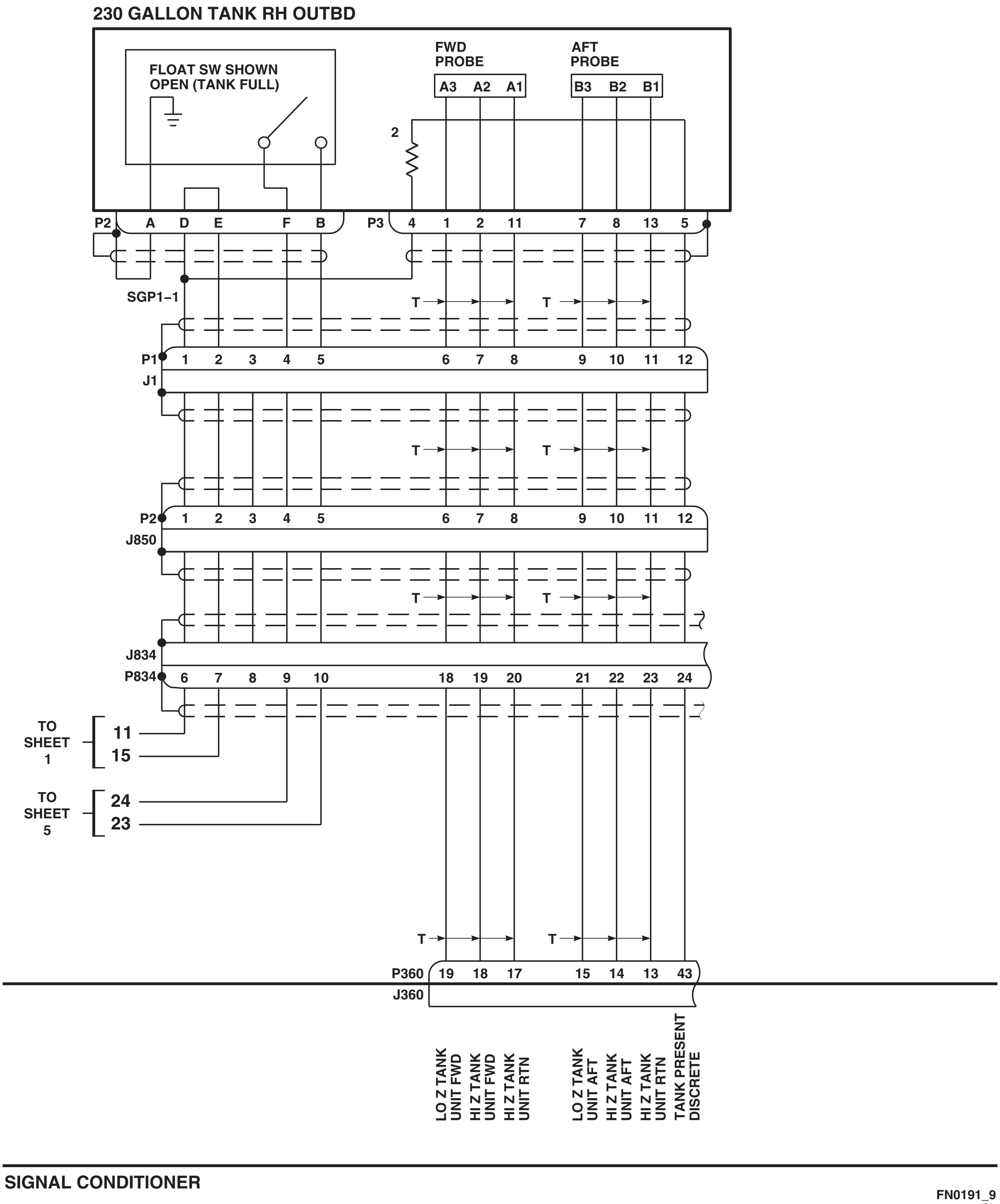


FIGURE 16-2-2-2. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING SCHEMATIC DIAGRAM (SHEET 9 OF 10)

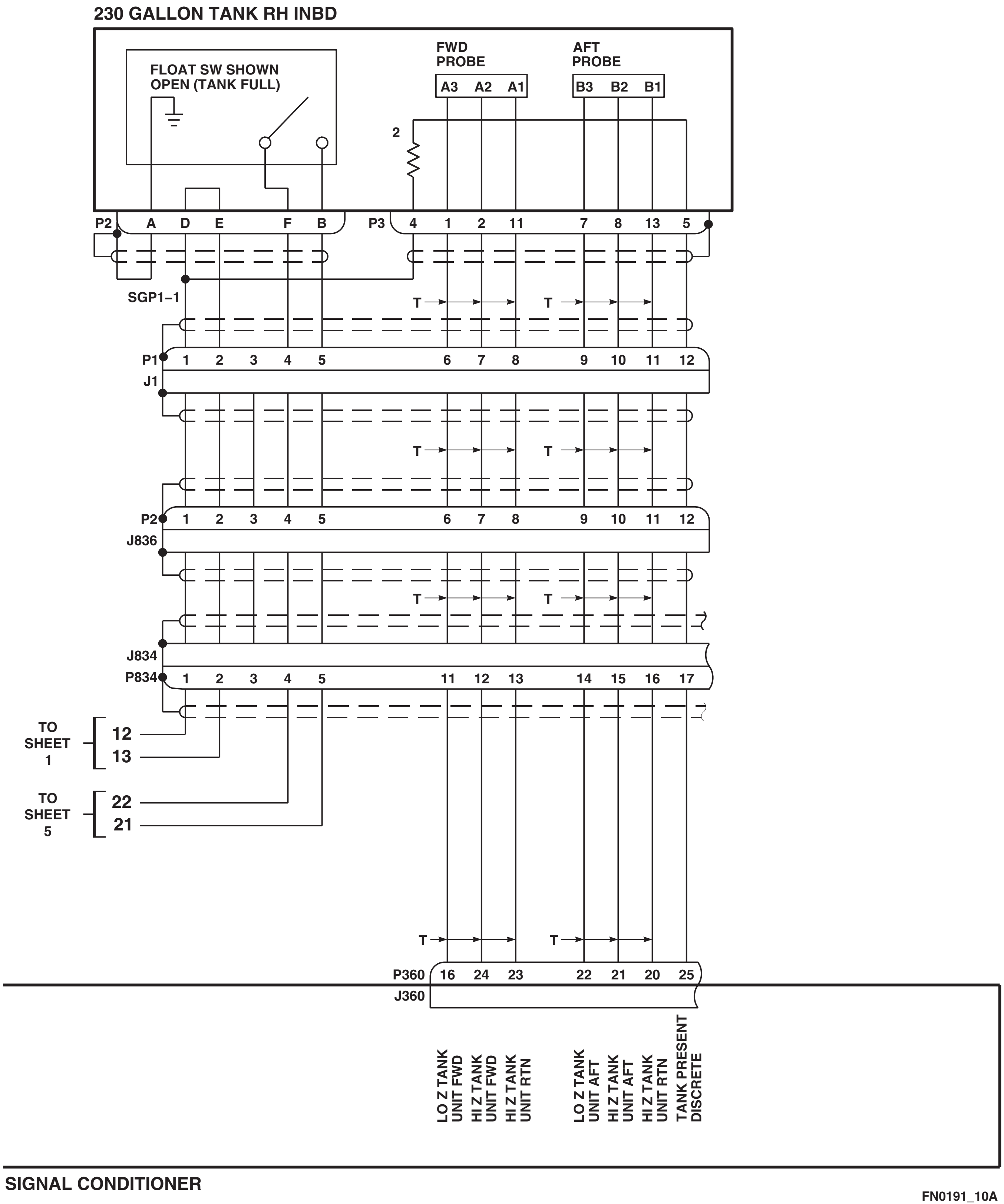


FIGURE 16-2-2-2. EXTERNAL STORES SUPPORT SYSTEM WITH FUEL GAUGING SCHEMATIC DIAGRAM (SHEET 10 OF 10)

16-2-3 EXTERNAL STORES SUPPORT SYSTEM RANGE EXTENSION EXTERNAL TANK CHECK.

INITIAL SETUP

Applicable Configuration

- This Fault Isolation Procedure Is Designed To Be Utilized When The Horizontal (Wings) And Vertical (Pylons) Stores Are Installed On A Permanent Basis, And Only The Inboard Or Outboard External Tanks Are Removed And Installed.

Test Equipment

- Compressor Unit, MIL-C-13874
- Locally-Made Fuel Quantity Simulator Test Set (simulator), Figure H-152, Appendix H
- Multimeter, Fluke 45 (2 required)

Tools

- Aircraft Mechanic's Toolkit
- Container, Suitable
- Electrical Repairer Toolkit
- Locally-Made Pneumatic Adapter, Figure H-151, Appendix H
- Torque Wrench, Item S-70 Torque Kit

References

- PARA 1-6-16
- PARA 16-4-21
- PARA 16-4-35
- TM 1-1500-204-23

Equipment Conditions

- External Range Extension Tanks **160-190** Pounds Of Fuel In Each Outboard, If Installed
- External Range Extension Tanks **220-250** Pounds Of Fuel In Each Inboard, If Installed
- External Electrical Power Available or APU Operational
- Main Fuel Tanks, No More Than **650** Pounds Of Fuel In Each Tank
- Rear Fairings On Vertical Stores (Pylons), Which Have Tanks Installed, Removed (PARA 16-4-21)
- All Explosive Cartridges Removed From Ejector Racks (PARA 16-4-35)

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

NOTE

As an alternate to using the compressor unit to pressurize the External Stores Support System (ESSS) Range Extension System, the APU may be used. If the alternate is selected, steps with CU may be skipped.

NOTE

The fault isolation procedure will check the ESSS Range Extension System with any tank configuration, installed on the helicopter, authorized by the aircraft operator's manual.

NOTE

If APU is selected for electrical power, it will provide air pressure.

NOTE

Special icons used in this fault isolation procedure:

CU Steps that are to be done if compressor unit is used.

INBD Steps that are to be done if inboard tanks are installed.

OUTBD Steps that are to be done if outboard tanks are installed.

Special Instructions

This fault isolation procedure is subdivided:

(1) Setup

(2) Outboard Tanks Installed

(3) Inboard Tanks Installed

(4) Shutdown

16-2-3.1 Setup.

a. CU Using pneumatic adapter, tee air compressor into helicopter bleed air QD and ESSS bleed air hose.

b. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.

c. Connect simulator power cord to UTIL RCPT +28V (J258) in cabin ceiling.

d. Disconnect helicopter connector P905 from fuel quantity signal conditioner connector J905.

e. Connect simulator connector J905 to helicopter connector P905.

f. Turn simulator TANK 1 and TANK 2 controls fully counterclockwise.

g. Connect No. 1 multimeter to simulator TANK 1 OUTPUT.

h. Connect No. 2 multimeter to simulator TANK 2 OUTPUT.

i. Turn on electrical power (PARA 1-6-16).

j. CU Apply air source, regulated at 35-45 psi, to helicopter's external tanks.

k. **Record** time.

l. INBD Place control panel PRESS INBD switch to ON.

m. OUTBD Place control panel PRESS OUTBD switch to ON.

n. Inspect fuel and pneumatic hoses and fittings for signs of leaking or damage (TM 1-1500-204-23).

o. Turn simulator TANK 1 control to get 6.0 vdc No. 1 multimeter indication.

p. Turn simulator TANK 2 control to get 6.0 vdc No. 2 multimeter indication.

q. INBD Turn control panel AUX FUEL QTY switch to INBD.

r. INBD Place control panel INCR-DECR switch to INCR to get display of 6000 pounds.

s. OUTBD Turn control panel AUX FUEL QTY switch to OUTBD.

t. OUTBD Place control panel INCR-DECR switch to INCR to get display of 1500 pounds.

16-2-3.2 Outboard Tanks Installed.

NOTE

If any trouble is encountered during this fault isolation procedure, go to PARA 16-2-1.

a. Do Setup, if not already done.

b. Turn simulator TANK 2 control to get 2.7 vdc No. 2 multimeter indication.

c. Place control panel FUEL XFR - MODE switch to OFF.

d. Place control panel FUEL XFR - TANKS switch to OUTBD.

e. Before continuing, make sure that at least 5 minutes have elapsed from time recorded in Setup step k.

f. Place control panel FUEL XFR - MODE switch to AUTO until EXTERNAL RIGHT OUTBD EMPTY and LEFT OUTBD EMPTY indicators go on.

g. Momentarily press control panel STATUS button.

h. Place control panel FUEL XFR - MODE switch to MANUAL.

i. Place control panel MANUAL XFR - LEFT and RIGHT switches to ON.

RESULT:

(1) Control panel EXTERNAL RIGHT NO FLOW and LEFT NO FLOW indicators shall go off, then come back on within 5 minutes.

(2) Control panel digital display indication shall not change when the EXTERNAL RIGHT NO FLOW and LEFT NO FLOW indicators are on.

j. Place control panel FUEL XFR - MODE, MANUAL XFR - RIGHT and LEFT, and PRESS - OUTBD switches to OFF.

k. Check for less than 50 mvac and between -50 and 50 mvdc between J705-A and J705-B (left outboard).

l. Check for less than 50 mvac and between -50 and 50 mvdc between J710-A and J710-B (right outboard).

m. If no other check is to be done, go to Shutdown.

16-2-3.3 Inboard Tanks Installed.

NOTE

If any trouble is encountered during this fault isolation procedure, go to PARA 16-2-1.

a. Do Setup, if not already done.

b. Turn simulator TANK 2 control to get 2.7 vdc No. 2 multimeter indication, if not already done.

c. Make sure control panel FUEL XFR - MODE switch is OFF.

d. Place control panel FUEL XFR - TANKS switch to INBD.

e. Before continuing, make sure that at least 5 minutes have elapsed from time recorded in Setup step k.

f. Place control panel FUEL XFR - MODE switch to AUTO until EXTERNAL RIGHT INBD EMPTY and LEFT INBD EMPTY indicators go on.

g. Momentarily press control panel STATUS button.

h. Place control panel FUEL XFR - MODE switch to MANUAL.

i. Place control panel MANUAL XFR - LEFT and RIGHT switches to ON.

RESULT:

(1) Control panel EXTERNAL RIGHT NO FLOW and LEFT NO FLOW indicators shall go off, then come back on within 5 minutes.

(2) Control panel digital display indication shall not change when the EXTERNAL RIGHT NO FLOW and LEFT NO FLOW indicators are on.

j. Place control panel FUEL XFR - MODE, MANUAL XFR - RIGHT and LEFT, and PRESS - INBD switches to OFF.

k. Check for less than 50 mvac and between -50 and 50 mvdc between J709-B and J709-A (left inboard).

l. Check for less than 50 mvac and between -50 and 50 mvdc between J709-C and J709-N (left inboard).

m. Check for less than 50 mvac and between -50 and 50 mvdc between J706-B and J706-A (right inboard).

n. Check for less than 50 mvac and between -50 and 50 mvdc between J706-C and J706-N (right inboard).

o. If no other check is to be done, go to Shutdown.

16-2-3.4 Shutdown.

a. Turn off electrical power (PARA 1-6-16).

b. CU Turn off air source.

c. Allow bleed air regulator valves to vent air to ambient pressure.

d. CU Disconnect air compressor, hoses, and pneumatic adapter.

e. CU Connect pneumatic hose to outboard side of helicopter bleed air QD.

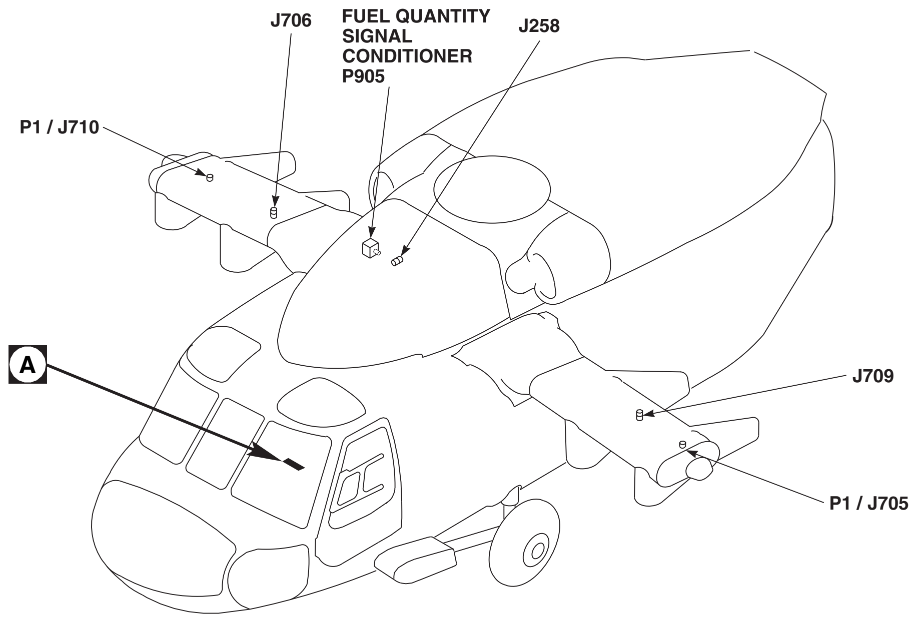
f. CU **TORQUE NUTS TO 135 - 150 INCH-POUNDS.**

g. Disconnect test equipment.

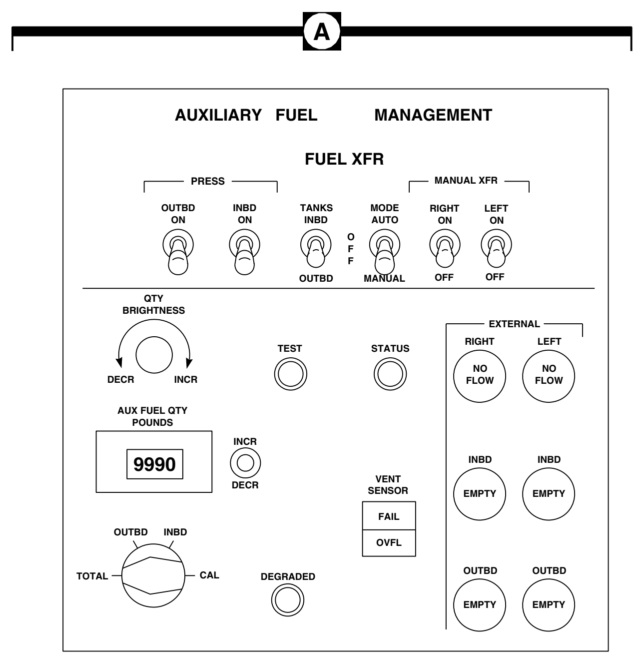
h. Connect helicopter electrical connectors.

i. Install all explosive cartridges in ejector racks (PARA 16-4-35).

j. Install all vertical stores (pylon) rear fairings (PARA 16-4-21).



TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
J258	UTILITY RECEPTACLE +28 VDC
J706	RIGHT INBOARD EJECTOR RACK CONNECTOR
J709	LEFT INBOARD EJECTOR RACK CONNECTOR
P1 / J705	LEFT OUTBOARD EJECTOR RACK CONNECTOR
P1 / J710	RIGHT OUTBOARD EJECTOR RACK CONNECTOR
P360 / J360	AUXILIARY FUEL MANAGE– MENT CONTROL PANEL
P361 / J361	AUXILIARY FUEL MANAGE– MENT CONTROL PANEL
P905	FUEL QUANTITY SIGNAL CONDITIONER



AUXILIARY FUEL MANAGEMENT CONTROL PANEL

FB2472
SA

FIGURE 16-2-3-1. EXTERNAL STORES SUPPORT SYSTEM RANGE EXTENSION EXTERNAL TANK CHECK LOCATION DIAGRAM

16-2-4 EXTERNAL STORES SUPPORT SYSTEM JETTISON SYSTEM.

INITIAL SETUP

Test Equipment

- Locally-Made Stores Jettison System Test Adapter (Test Lamps), Figure H-150, Appendix H (8 required)
- Multimeter, Fluke 27/AN
- Multimeter, Fluke 45

Tools

- Aircraft Mechanics Toolkit
- Electrical Repairer Toolkit
- Locally-Made Drag Beam Wedge, Figure H-167, Appendix H (2 required)
- Stopwatch, Hart 7500

References

- PARA 1-6-16
- PARA 16-4-21
- PARA 16-4-35

Equipment Conditions

- External Electrical Power Available or APU Operational

WARNING

Failure to do stray voltage check would result in injury if stray voltage is present when an explosive cartridge is installed.

To prevent accidental firing of an explosive cartridge, observe all safety precautions regarding the handling and storage of explosives.

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

If fuel tanks are connected to the left, right, or both stores stations, the BOTH mode of jettison is automatically selected even if the selector switch is at L or R. This fault isolation procedure isolates each tank, if installed. This enables the maintenance technician to check the selector switch in each position (INBD L, INBD R, INBD BOTH OUTBD BOTH, OUTBD R, OUTBD L, and ALL).

The fault isolation procedure checks all four stores stations, regardless of helicopter's external tank configuration.

16-2-4.1 Operational/Troubleshooting Procedure.

a. Remove all explosive cartridges installed in ejector racks (PARA 16-4-35).

b. Remove rear fairing on all vertical stores pylons (PARA 16-4-21).

c. Disconnect electrical connectors, if connected to rack harnesses:

J705	(left outboard)
J707	(left inboard)
P709	(left inboard)
P706	(right inboard)
J708	(right inboard)

J710	(right outboard)

d. Check continuity between:

J705-B and ground	(left outboard)
J707-B and ground	(left inboard)
P709-A and ground	(left inboard)
P709-N and ground	(left inboard)
P706-A and ground	(right inboard)
P706-N and ground	(right inboard)
J708-B and ground	(right inboard)
J710-B and ground	(right outboard)

RESULT: Continuity shall be present.

- If result is not as specified, repair/replace wiring (Appendix F), as required.

e. Disconnect P1 from J1 on installed external fuel tanks.

f. Connect test lamps between:

J705-A and J705-B (left outboard)
J707-A and J707-B (left inboard)
P709-B and P709-A (left inboard)
P709-C and P709-N (left inboard)
P706-B and P706-A (right inboard)
P706-C and P706-N (right inboard)
J708-A and J708-B (right inboard)
J710-A and J710-B (right outboard)

g. Make sure all circuit breakers are pushed in, except those noted being pulled out for maintenance or safety.

h. Pull out BACKUP HYD CONTR circuit breaker (upper console circuit breaker panel).

i. Turn on electrical power (PARA 1-6-16).

j. Turn upper console CONSOLE LT/LOWER control to BRT.

RESULT: Panel lights on stores jettison control panel (control panel) shall go on at full brightness.

- If result is not as specified, troubleshoot instrument panel and consoles indicator lights dimming (PARA 9-2-10).

k. Place control panel EMER JETT ALL switch up.

RESULT: All test lamps shall be off.

- If result is not as specified, replace control panel (PARA 16-4-50).

l. Place control panel EMER JETT ALL switch down.

m. Turn control panel selector switch to ALL.

n. Momentarily hold control panel JETT switch up.

RESULT: All test lamps shall be off.

- If result is not as specified, replace control panel (PARA 16-4-50).

o. Turn control panel selector switch to OFF.

p. Insert drag beam wedge into left drag beam switch to simulate in-flight condition.

q. Place control panel EMER JETT ALL switch up and start stopwatch.

RESULT:

(1) All outboard test lamps shall go on.

(2) All inboard test lamps shall go on 0.5 to 1.5 seconds later.

- If results (1) and (2) are not as specified, go to TABLE NO. 16-2-4-1.

- If only right outboard test lamps do not go on, go to TABLE NO. 16-2-4-2.

- If only left outboard test lamps do not go on, go to TABLE NO. 16-2-4-2.

- If result (1) is not as specified, check for 28 vdc between:

P646-h	and	P646-e
P646-j	and	P646-e

(a) If voltage is as specified, replace control panel (PARA 16-4-50).

(b) If voltage is not as specified, check for 28 vdc between ESSS JTSN OUTBD circuit breaker (upper console circuit breaker panel) and ground.

1 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

2 If voltage is as specified, repair/replace wiring between circuit breaker and P646-h, or circuit breaker and P646-j (Appendix F), as required.

- If only right inboard test lamps do not go on, replace control panel (PARA 16-4-50).

- If only left inboard test lamps do not go on, replace control panel (PARA 16-4-50).

- If inboard test lamps do not go on, check for 28 vdc between:

P646-f	and	P646-e
P646-g	and	P646-e

(a) If voltage is as specified, replace control panel (PARA 16-4-50).

(b) If voltage is not as specified, check for 28 vdc between ESSS JTSN INBD circuit breaker (upper console circuit breaker panel) and ground.

1 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

2 If voltage is as specified, repair/replace wiring between circuit breaker and P646-f, or circuit breaker and P646-g (Appendix F), as required.

- If inboard test lamps go on, but not in time specified, replace control panel (PARA 16-4-50).

r. Stop and reset stopwatch.

s. Place control panel EMER JETT ALL switch down.

t. Remove drag beam wedge from left drag beam switch.

u. Insert drag beam wedge into right drag beam switch to simulate in-flight condition.

v. Turn control panel selector switch to ALL.

w. Momentarily hold (for at least 2 seconds) control panel JETT switch up and start stopwatch.

RESULT:

(1) All outboard test lamps shall go on.

(2) All inboard test lamps shall go on 0.5 to 1.5 seconds later.

- If results (1) and (2) are not as specified, go to TABLE NO. 16-2-4-3.

- If only right outboard test lamps do not go on, go to TABLE NO. 16-2-4-4.

- If only left outboard test lamps do not go on, go to TABLE NO. 16-2-4-4.

- If result (1) is not as specified, check for 28 vdc between:

P645-f	and	P645-k
P645-q	and	P645-k

(a) If voltage is as specified, replace control panel (PARA 16-4-50).

(b) If voltage is not as specified, check for 28 vdc between ESSS JTSN OUTBD circuit breaker (copilot's circuit breaker panel) and ground.

1 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

2 If voltage is as specified, repair/replace wiring between circuit breaker and P645-f, or circuit breaker and P645-q (Appendix F), as required.

- If only right inboard test lamps do not go on, replace control panel (PARA 16-4-50).

- If only left inboard test lamps do not go on, replace control panel (PARA 16-4-50).

- If inboard test lamps do not go on, check for 28 vdc between:

P645-n	and	P645-k
P645-p	and	P645-k

(a) If voltage is as specified, replace control panel (PARA 16-4-50).

(b) If voltage is not as specified, check for 28 vdc between ESSS JTSN INBD circuit breaker (upper console circuit breaker panel) and ground.

1 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

2 If voltage is as specified, repair/replace wiring between circuit breaker and P645-n, or circuit breaker and P645-p (Appendix F), as required.

- If inboard test lamps go on, but not in time specified, replace control panel (PARA 16-4-50).

x. Stop and reset stopwatch.

y. Turn control panel selector switch to OUTBD L.

z. Momentarily hold control panel JETT switch up.

RESULT:

(1) Left outboard test lamps shall go on.

(2) All other test lamps shall be off.

- If results are not as specified, replace control panel (PARA 16-4-50).

aa. Turn control panel selector switch to OUTBD R.

ab. Momentarily hold control panel JETT switch up.

RESULT:

(1) Right outboard test lamps shall go on.

(2) All other test lamps shall be off.

- If results are not as specified, replace control panel (PARA 16-4-50).

ac. Turn control panel selector switch to OUTBD BOTH.

ad. Momentarily hold control panel JETT switch up.

RESULT:

(1) Right and left outboard test lamps shall go on.

(2) All other test lamps shall be off.

- If results are not as specified, replace control panel (PARA 16-4-50).

ae. Turn control panel selector switch to INBD BOTH.

af. Momentarily hold control panel JETT switch up.

RESULT:

(1) Right and left inboard test lamps shall go on.

(2) All other test lamps shall be off.

- If results are not as specified, replace control panel (PARA 16-4-50).

ag. Turn control panel selector switch to INBD R.

ah. Momentarily hold control panel JETT switch up.

RESULT:

(1) Right inboard test lamps shall go on.

(2) All other test lamps shall be off.

- If results are not as specified, replace control panel (PARA 16-4-50).

ai. Turn control panel selector switch to INBD L.

aj. Momentarily hold control panel JETT switch up.

RESULT:

(1) Left inboard test lamps shall go on.

(2) All other test lamps shall be off.

- If results are not as specified, replace control panel (PARA 16-4-50).

ak. Turn control panel selector switch to OFF.

al. Connect right outboard fuel tank connector P1 to J1, or if external fuel tank is not installed, install a jumper wire between J1-1 and J1-2.

am. Turn control panel selector switch to OUTBD R.

an. Momentarily hold control panel JETT switch up.

RESULT: Right and left outboard test lamps shall go on.

- If result is not as specified, go to TABLE NO. 16-2-4-5.

ao. Disconnect right outboard external fuel tank connector P1 from J1, or if external fuel tank is not installed, remove jumper wire.

ap. Connect right inboard external fuel tank connector P1 to J1, or if external fuel tank is not installed, install an insulated jumper wire between J1-1 and J1-2.

aq. Turn control panel selector switch to INBD R.

ar. Momentarily hold control panel JETT switch up, then release.

RESULT: 0

RESULT: Right and left inboard test lamps shall go on.

- If result is not as specified, go to TABLE NO. 16-2-4-6.

as. Disconnect right inboard external fuel tank connector P1 from J1, or if external fuel tank is not installed, remove jumper wire.

at. Connect left outboard external fuel tank connector P1 to J1, or if external fuel tank is not installed, install an insulated jumper wire between J1-1 and J1-2.

au. Turn control panel selector switch to OUTBD L.

av. Momentarily hold control panel JETT switch up.

RESULT: Right and left outboard test lamps shall go on.

- If result is not as specified, go to TABLE NO. 16-2-4-7.

aw. Disconnect left outboard external fuel tank connector P1 from J1, or if external fuel tank is not installed, remove jumper wire.

ax. Connect left inboard external fuel tank connector P1 to J1, or if external fuel tank is not installed, install an insulated jumper wire between J1-1 and J1-2.

ay. Turn control panel selector switch to INBD L.

az. Momentarily hold control panel JETT switch up.

RESULT: Right and left inboard test lamps go on.

- If result is not as specified, go to TABLE NO. 16-2-4-8.

ba. Disconnect left inboard external fuel tank connector P1 from J1, or if external fuel tank is not installed, remove jumper wire.

bb. Turn control panel selector switch to OFF.

bc. Install drag beam wedge into left drag beam switch. **DO NOT REMOVE DRAG BEAM WEDGE FROM RIGHT DRAG BEAM SWITCH.**

bd. Turn control panel selector switch to ALL.

be. Make sure control panel EMER JETT ALL and JETT switches are down and their switch guards are down.

bf. Check for ac and dc voltage between:

J705-AandJ705-B (left outboard)
J707-AandJ707-B (left inboard)
P709-BandP709-A (left inboard)
P709-CandP709-N (left inboard)
P706-BandP706-A (right inboard)
P706-CandP706-N (right inboard)
J708-AandJ708-B (right inboard)
J710-AandJ710-B (right outboard)

RESULT:

(1) Ac voltage shall be -50 to +50 mvac.

(2) Dc voltage shall be -10 to +10 mvdc.

- If results (1) and (2) are not as specified, go to TABLE NO. 16-2-4-9.

bg. Remove drag beam wedges from left and right drag beam switches.

bh. Check for ac and dc voltage between:

J705-AandJ705-B (left outboard)
J707-AandJ707-B (left inboard)
P709-BandP709-A (left inboard)
P709-CandP709-N (left inboard)
P706-BandP706-A (right inboard)
P706-CandP706-N (right inboard)
J708-AandJ708-B (right inboard)
J710-AandJ710-B (right outboard)

RESULT:

(1) Ac voltage shall be -50 to +50 mvac.

(2) Dc voltage shall be -10 to +10 mvdc.

- If result is not as specified, replace control panel (PARA 16-4-50).

bi. Install drag beam wedges in both left and right drag beam switches to simulate in-flight condition.

bj. Check the following:

CONNECTOR	**BREECH**	**RESULT**
P707–A	Center Pin	Continuity
P707–A	Shell	No Continuity
P707–B	Center Pin	No Continuity
P707–B	Shell	Continuity
P705–A	Center Pin	Continuity

P705–A	Shell	No Continuity
P705–B	Center Pin	No Continuity
P705–B	Shell	Continuity
P708–A	Center Pin	Continuity
P708–A	Shell	No Continuity
P708–B	Center Pin	No Continuity
P708–B	Shell	Continuity
P710–A	Center Pin	Continuity
P710–A	Shell	No Continuity
P710–B	Center Pin	No Continuity
P710–B	Shell	Continuity

RESULT: Result shall be as indicated.

- If result is not as specified, replace breech harness (PARA 16-4-53), as required.

bk. Turn off electrical power (PARA 1-6-16).

bl. Disconnect all test equipment.

bm. Remove drag beam wedges from drag beam switches.

bn. Remove test lamps.

bo. Remove all jumper wires.

bp. Connect all electrical connectors to their receptacles.

bq. Install rear fairings on all vertical stores pylons (PARA 16-4-21).

br. Install explosive cartridges in ejector racks (PARA 16-4-35).

bs. Push in BACKUP HYD CONTR circuit breaker (upper console circuit breaker panel).

TABLE NO. 16-2-4-1. None Of The Test Lamps Go On When EMER JETT ALL Switch Is Placed Up.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check continuity between P646-e and ground.

Step 1. If continuity is present, replace stores jettison control panel (PARA 16-4-50). Go to **9.**
Step 2. If continuity is not present, go to **2.**

2. Check continuity between J163-D and ground.

Step 1. If continuity is present, go to **3.**
Step 2. If continuity is not present, repair/replace ground wire (Appendix F). Go to **9.**

3. Install an insulated jumper wire between J163-F and J163-D. Go to 4.

4. Check continuity between P646-e and ground.

Step 1. If continuity is present, go to **5.**
Step 2. If continuity is not present, go to **7.**

TABLE NO. 16-2-4-1. None Of The Test Lamps Go On When EMER JETT ALL Switch Is Placed Up. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

5. **Remove jumper wire from J163. Go to 6.**

6. **Replace left drag beam switch (PARA 3-4-6). Go to 9.**

7. **Remove jumper wire from J163. Go to 8.**

8. **Repair/replace wiring between P646-e and J163-F (Appendix F). Go to 9.**

9. **Procedure completed.**

TABLE NO. 16-2-4-2. Either Outboard Test Lamp Does Not Go On When EMER JETT ALL Switch Is Placed Up.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Does right outboard test lamp go on?**

   Step 1. If test lamp is on, go to **2.**
   Step 2. If test lamp is not on, go to **6.**

2. **Check for 28 vdc between J705-A and ground.**

   Step 1. If voltage is as specified, repair/replace ground wire (Appendix F) or replace harness (PARA 16-4-53), as required. Go to **13.**
   Step 2. If voltage is not as specified, go to **3.**

3. **Check for 28 vdc between P680-8 and ground.**

   Step 1. If voltage is as specified, replace harness (PARA 16-4-53). Go to **13.**
   Step 2. If voltage is not as specified, go to **4.**

4. **Check for 28 vdc between J661-M and ground.**

   Step 1. If voltage is as specified, replace harness (PARA 16-4-53). Go to **13.**
   Step 2. If voltage is not as specified, go to **5.**

5. **Check continuity between J661-M and P646-G.**

TABLE NO. 16-2-4-2. Either Outboard Test Lamp Does Not Go On When EMER JETT ALL Switch Is Placed Up. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, replace stores jettison control panel (PARA 16-4-50). Go to **13.**
Step 2. If continuity is not present, replace harness (PARA 16-4-53). Go to **13.**

6. Check for 28 vdc between J710-A and ground.

Step 1. If voltage is as specified, repair/replace ground wire (Appendix F) or replace harness (PARA 16-4-53), as required. Go to **13.**
Step 2. If voltage is not as specified, go to **7.**

7. Check for 28 vdc between P677-1 and ground.

Step 1. If voltage is as specified, replace harness (PARA 16-4-53). Go to **13.**
Step 2. If voltage is not as specified, go to **8.**

8. Check for 28 vdc between J660-A and ground.

Step 1. If voltage is as specified, replace harness (PARA 16-4-53). Go to **13.**
Step 2. If voltage is not as specified, go to **9.**

9. Check continuity between J660-A and P645-R.

Step 1. If continuity is present, replace stores jettison control panel (PARA 16-4-50). Go to **13.**
Step 2. If continuity is not present, go to **10.**

10. Check continuity between P660-A and P677-1.

Step 1. If continuity is present, go to **11.**
Step 2. If continuity is not present, replace harness (PARA 16-4-53). Go to **13.**

11. Check continuity between J677-1 and J679-12.

Step 1. If continuity is present, go to **12.**
Step 2. If continuity is not present, replace harness (PARA 16-4-53). Go to **13.**

12. Check continuity between P679-12 and P646-R.

TABLE NO. 16-2-4-2. Either Outboard Test Lamp Does Not Go On When EMER JETT ALL Switch Is Placed Up. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, replace stores jettison control panel (PARA 16-4-50). Go to **13.**
Step 2. If continuity is not present, replace harness (PARA 16-4-53). Go to **13.**

13. Procedure completed.

TABLE NO. 16-2-4-3. None Of The Test Lamps Go On When Selector Switch Is Set To ALL And JETT Switch Is Held Up.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check continuity between P645-k and ground.

Step 1. If continuity is present, replace stores jettison control panel (PARA 16-4-50). Go to **4.**
Step 2. If continuity is not present, go to **2.**

2. Check continuity between J164-D and ground.

Step 1. If continuity is present, go to **3.**
Step 2. If continuity is not present, repair/replace ground wire (Appendix F). Go to **4.**

3. Check continuity between J164–F and P645-k.

Step 1. If continuity is present, adjust or replace right drag beam switch (PARA 3-4-6), as required. Go to **4.**
Step 2. If continuity is not present, repair/replace ground wire (Appendix F). Go to **4.**

4. Procedure completed.

TABLE NO. 16-2-4-4. Either Outboard Test Lamp Does Not Go On When Selector Switch Is Set To ALL And JETT Switch Is Held Up.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Does right outboard test lamp go on?

Step 1. If lamp is on, go to **2.**
Step 2. If lamp is not on, go to **5.**

2. Check for 28 vdc between P678-3 and ground.

Step 1. If voltage is as specified, replace cabin jettison harness (PARA 16-4-52). Go to **8.**
Step 2. If voltage is not as specified, go to **3.**

3. Check for 28 vdc between J660-L and ground.

Step 1. If voltage is as specified, replace cabin jettison harness (PARA 16-4-52). Go to **8.**
Step 2. If voltage is not as specified, go to **4.**

4. Check continuity between J660-L and P645-G.

Step 1. If continuity is present, replace stores jettison control panel (PARA 16-4-50). Go to **8.**
Step 2. If continuity is not present, replace cabin jettison harness (PARA 16-4-52). Go to **8.**

5. Check for 28 vdc between P677-1 and ground.

Step 1. If voltage is as specified, replace cabin jettison harness (PARA 16-4-52). Go to **8.**
Step 2. If voltage is not as specified, go to **6.**

6. Check for 28 vdc between J660-A and ground.

Step 1. If voltage is as specified, replace cabin jettison harness (PARA 16-4-52). Go to **8.**
Step 2. If voltage is not as specified, go to **7.**

7. Check continuity between J660-A and P645-R.

Step 1. If continuity is present, replace stores jettison control panel (PARA 16-4-50). Go to **8.**

TABLE NO. 16-2-4-4. Either Outboard Test Lamp Does Not Go On When Selector Switch Is Set To ALL And JETT Switch Is Held Up. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If continuity is not present, replace cabin jettison harness (PARA 16-4-52). Go to **8.**

8. Procedure completed.

TABLE NO. 16-2-4-5. Right And Left Outboard Test Lamps Do Not Go On When The Selector Switch Is Set To OUTBD R And JETT Switch Is Held Up.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check continuity between P645-i and ground.

Step 1. If continuity is present, replace stores jettison control panel (PARA 16-4-50). Go to **7.**
Step 2. If continuity is not present, go to **2.**

2. Is jumper wire being used in place of right outboard fuel tank?

Step 1. If jumper wire is being used, go to **3.**
Step 2. If jumper wire is not being used, go to **5.**

3. Remove jumper wire. Go to 4.

4. Check continuity between J1-1 and ground.

Step 1. If continuity is present, troubleshoot open wiring between J1-2 and P645-i. Replace cabin jettison harness (PARA 16-4-52). Go to **7.**
Step 2. If continuity is not present, troubleshoot open wiring between J1-1 and ground. Replace cabin jettison harness (PARA 16-4-52). Go to **7.**

5. Check continuity between right outboard fuel tank P1-1 and P1-2.

Step 1. If continuity is present, go to **6.**
Step 2. If continuity is not present, replace fuel tank (PARA 16-4-54). Go to **7.**

6. Check continuity between J1-1 and ground.

TABLE NO. 16-2-4-5. Right And Left Outboard Test Lamps Do Not Go On When The Selector Switch Is Set To OUTBD R And JETT Switch Is Held Up. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, troubleshoot open wiring between J1-2 and P645-i. Replace cabin jettison harness (PARA 16-4-52). Go to **7.**

Step 2. If continuity is not present, troubleshoot open wiring between J1-1 and ground. Replace cabin jettison harness (PARA 16-4-52). Go to **7.**

7. Procedure completed.

TABLE NO. 16-2-4-6. Right And Left Inboard Test Lamps Do Not Go On When The Selector Switch Is Set To INBD R And JETT Switch Is Held Up.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check continuity between P645-h and ground.

Step 1. If continuity is present, replace stores jettison control panel (PARA 16-4-50). Go to **7.**

Step 2. If continuity is not present, go to **2.**

2. Is jumper wire being used in place of right inboard fuel tank?

Step 1. If jumper wire is being used, go to **3.**

Step 2. If jumper wire is not being used, go to **5.**

3. Remove jumper wire. Go to 4.

4. Check continuity between J1-1 and ground.

Step 1. If continuity is present, troubleshoot open wiring between J1-2 and P645-h. Replace cabin jettison harness (PARA 16-4-52). Go to **7.**

Step 2. If continuity is not present, troubleshoot open wiring between J1-1 and ground. Replace cabin jettison harness (PARA 16-4-52). Go to **7.**

5. Check continuity between right inboard fuel tank P1-1 and P1-2.

Step 1. If continuity is present, go to **6.**

Step 2. If continuity is not present, replace fuel tank (PARA 16-4-54). Go to **7.**

6. Check continuity between J1-1 and ground.

TABLE NO. 16-2-4-6. Right And Left Inboard Test Lamps Do Not Go On When The Selector Switch Is Set To INBD R And JETT Switch Is Held Up. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, troubleshoot open wiring between J1-2 and P645-h. Replace cabin jettison harness (PARA 16-4-52). Go to **7.**
Step 2. If continuity is not present, troubleshoot open wiring between J1-1 and ground. Replace cabin jettison harness (PARA 16-4-52). Go to **7.**

7. Procedure completed.

TABLE NO. 16-2-4-7. Right And Left Outboard Test Lamps Do Not Go On When The Selector Switch Is Set To OUTBD L And JETT Switch Is Held Up.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check continuity between P645-j and ground.

Step 1. If continuity is present, replace stores jettison control panel (PARA 16-4-50). Go to **7.**
Step 2. If continuity is not present, go to **2.**

2. Is jumper wire being used in place of left outboard fuel tank?

Step 1. If jumper wire is being used, go to **3.**
Step 2. If jumper wire is not being used, go to **5.**

3. Remove jumper wire. Go to 4.

4. Check continuity between J1-1 and ground.

Step 1. If continuity is present, troubleshoot open wiring between J1-2 and P645-j. Replace cabin jettison harness (PARA 16-4-52). Go to **7.**
Step 2. If continuity is not present, troubleshoot open wiring between J1-1 and ground. Replace cabin jettison harness (PARA 16-4-52). Go to **7.**

5. Check continuity between left outboard fuel tank P1-1 and P1-2.

Step 1. If continuity is present, go to **6.**
Step 2. If continuity is not present, replace fuel tank (PARA 16-4-54). Go to **7.**

TABLE NO. 16-2-4-7. Right And Left Outboard Test Lamps Do Not Go On When The Selector Switch Is Set To OUTBD L And JETT Switch Is Held Up. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

6. **Check continuity between J1-1 and ground.**

   Step 1. If continuity is present, troubleshoot open wiring between J1-2 and P645-j. Replace cabin jettison harness (PARA 16-4-52). Go to **7.**
   Step 2. If continuity is not present, troubleshoot open wiring between J1-1 and ground. Replace cabin jettison harness (PARA 16-4-52). Go to **7.**

7. **Procedure completed.**

TABLE NO. 16-2-4-8. Right And Left Inboard Test Lamps Do Not Go On When The Selector Switch Is Set To INBD L And JETT Switch Is Held Up.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check continuity between P645-g and ground.**

   Step 1. If continuity is present, replace stores jettison control panel (PARA 16-4-50). Go to **7.**
   Step 2. If continuity is not present, go to **2.**

2. **Is jumper wire being used in place of left inboard fuel tank?**

   Step 1. If jumper wire is being used, go to **3.**
   Step 2. If jumper wire is not being used, go to **5.**

3. **Remove jumper wire. Go to 4.**

4. **Check continuity between J1-1 and ground.**

   Step 1. If continuity is present, troubleshoot open wiring between J1-2 and P645-g. Replace cabin jettison harness (PARA 16-4-52). Go to **7.**
   Step 2. If continuity is not present, troubleshoot open wiring between J1-1 and ground. Replace cabin jettison harness (PARA 16-4-52). Go to **7.**

5. **Check continuity between left inboard fuel tank P1-1 and P1-2.**

   Step 1. If continuity is present, go to **6.**
   Step 2. If continuity is not present, replace fuel tank (PARA 16-4-54). Go to **7.**

TABLE NO. 16-2-4-8. Right And Left Inboard Test Lamps Do Not Go On When The Selector Switch Is Set To INBD L And JETT Switch Is Held Up. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

6. Check continuity between J1-1 and ground.

Step 1. If continuity is present, troubleshoot open wiring between J1-2 and P645-g. Replace cabin jettison harness (PARA 16-4-52). Go to **7.**
Step 2. If continuity is not present, troubleshoot open wiring between J1-1 and ground. Replace cabin jettison harness (PARA 16-4-52). Go to **7.**

7. Procedure completed.

TABLE NO. 16-2-4-9. Multimeter Does Not Indicate Correctly.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check associated electrical circuit bonding jumpers for security and integrity. Go to 2.

2. Ensure that bonding jumpers and mounting surfaces are prepared for good electrical bonding.

Step 1. If jumpers and mounting surfaces are prepared, go to **3.**
Step 2. If jumpers and mounting surfaces are not prepared, repair bonding jumper or replace cabin jettison harness (PARA 16-4-52), as required. Go to **8.**

3. Check associated electrical circuit harnesses for damage to alphabraid shielding.

Step 1. If associated electrical circuit harness is damaged, replace malfunctioning harness (PARA 16-4-52). Go to **8.**
Step 2. If associated electrical circuit harness is not damaged, go to **4.**

4. Disconnect P645 and P646 from stores jettison control panel. Is voltage still greater than 10 mvdc or 50 mvac?

Step 1. If voltage is still greater than 10 mvdc or 50 mvac, go to **5.**
Step 2. If voltage is not still greater than 10 mvdc or 50 mvac, replace stores jettison control panel (PARA 16-4-50). Go to **8.**

TABLE NO. 16-2-4-9. Multimeter Does Not Indicate Correctly. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

5. **Starting from control panel, sequentially disconnect harness leading to the ejector rack disconnect where 10 mvdc or 50 mvac is exceeded. Go to 6.**

6. **After each cabin jettison harness is disconnected, recheck voltage with multimeter. Go to 7.**

7. **When voltage is less than 10 mvdc or 50 mvac, last harness disconnected is known to be malfunctioning. Replace cabin jettison harness (PARA 16-4-52). Go to 8.**

8. **Procedure completed.**

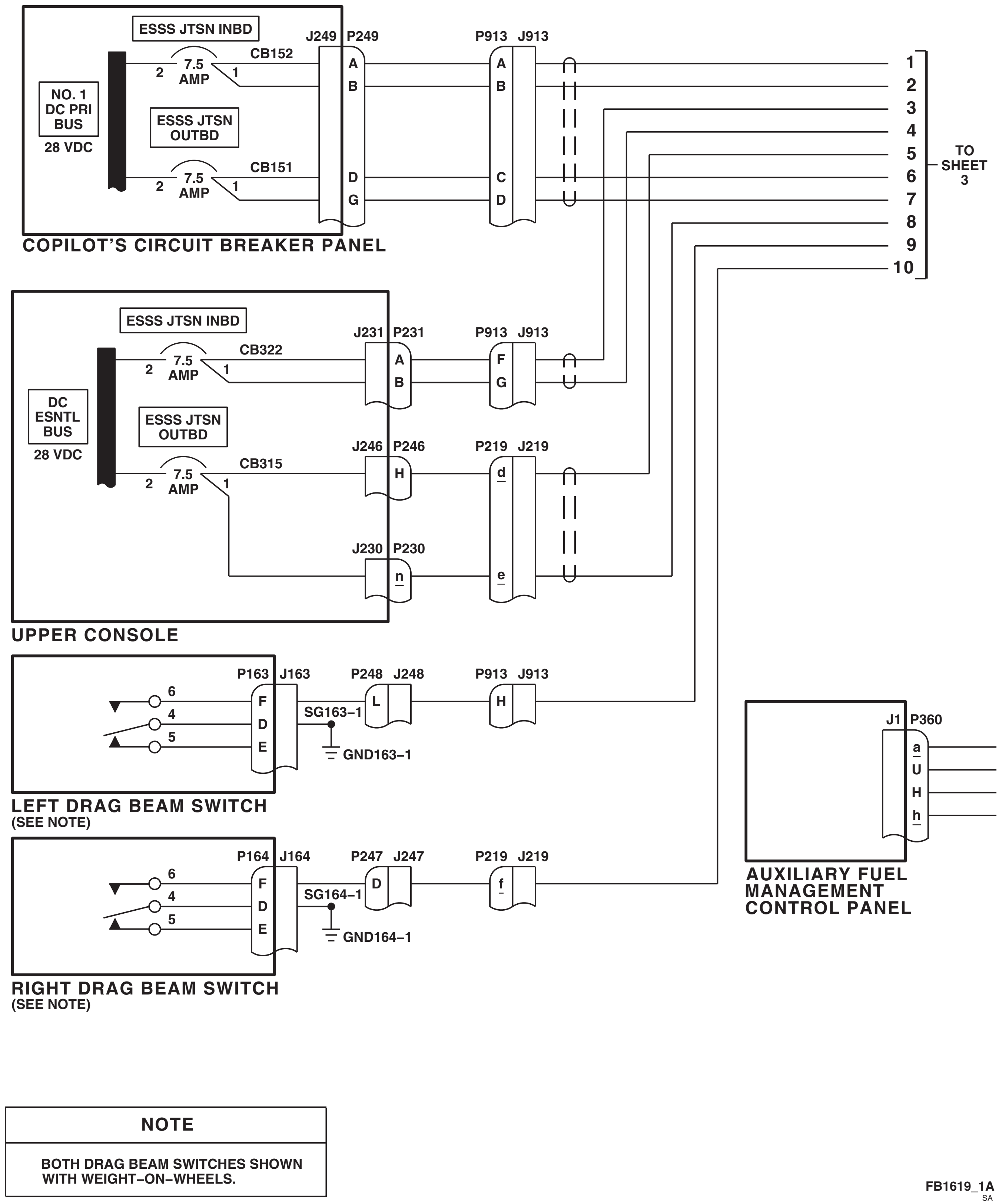


FIGURE 16-2-4-1. EXTERNAL STORES SUPPORT SYSTEM JETTISON SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 6)

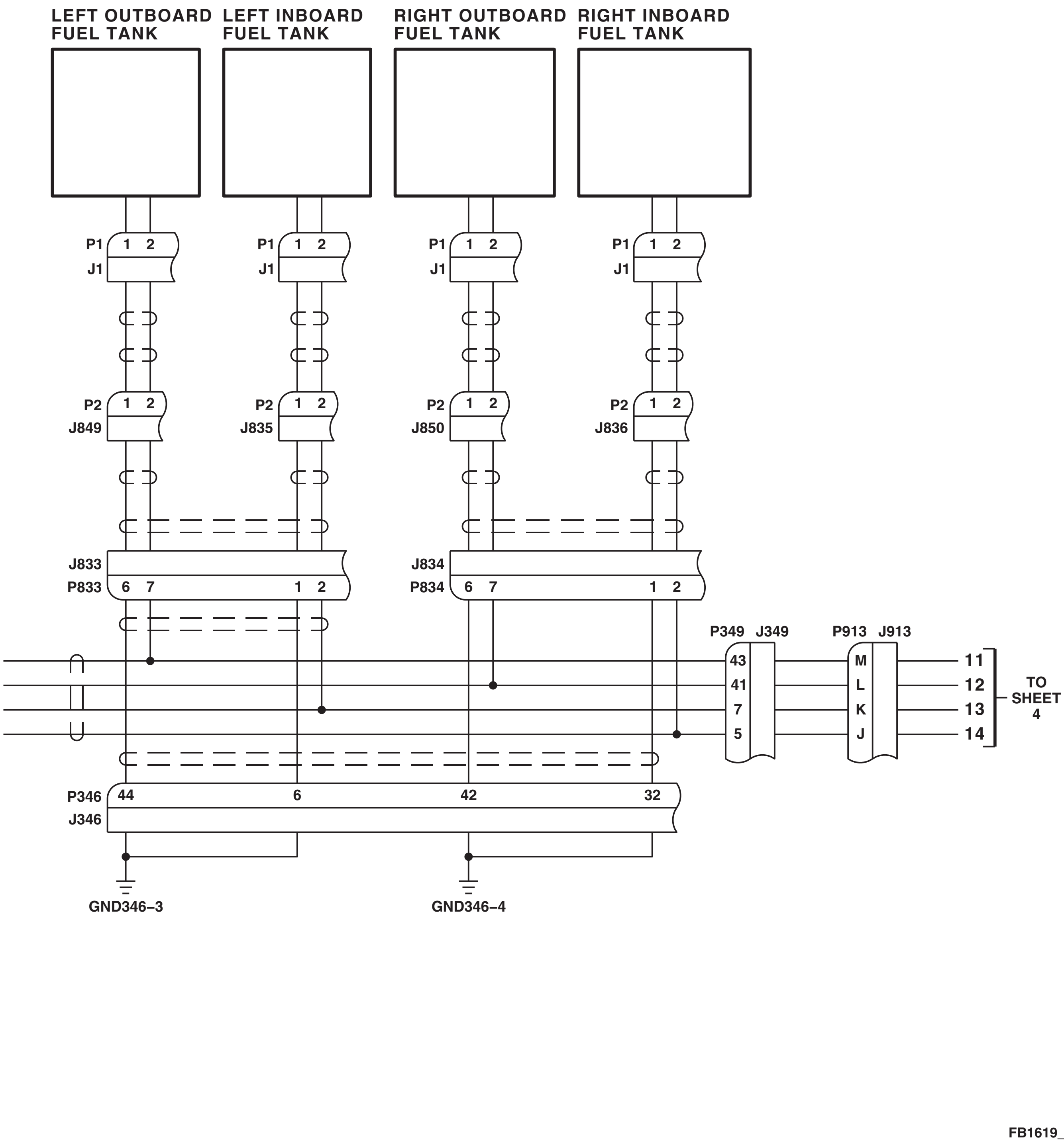


FIGURE 16-2-4-1. EXTERNAL STORES SUPPORT SYSTEM JETTISON SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 6)

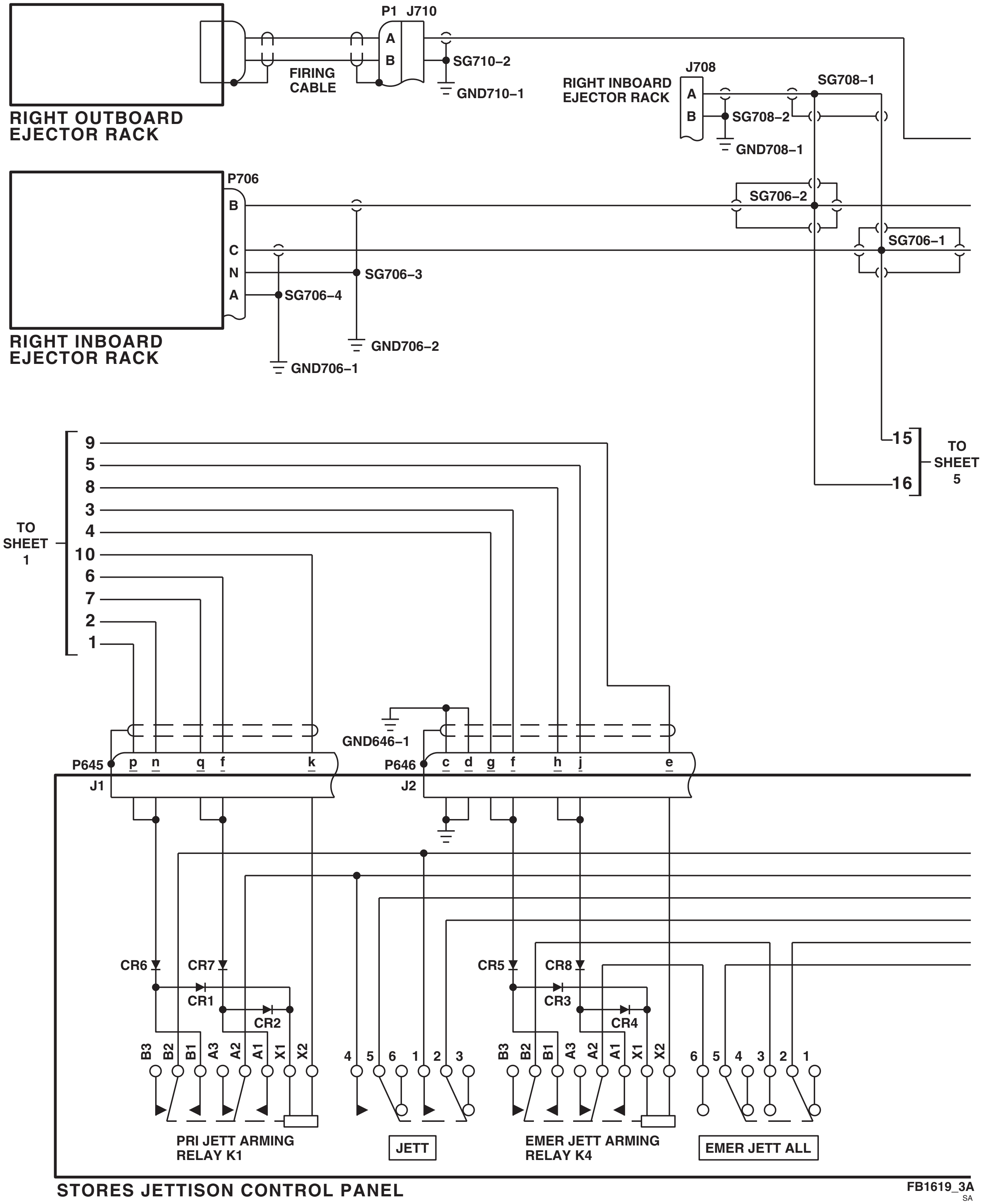


FIGURE 16-2-4-1. EXTERNAL STORES SUPPORT SYSTEM JETTISON SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 6)

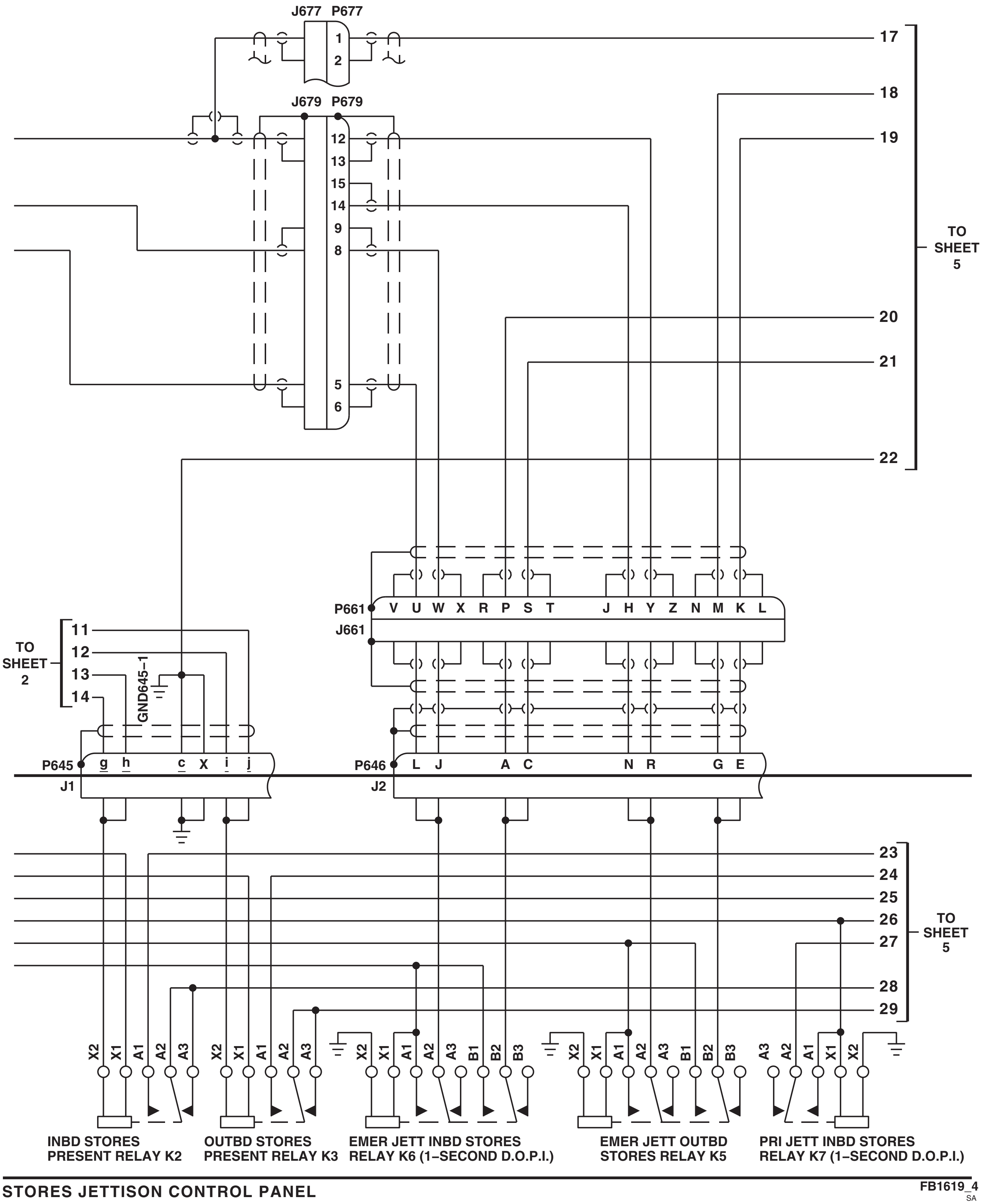


FIGURE 16-2-4-1. EXTERNAL STORES SUPPORT SYSTEM JETTISON SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 6)

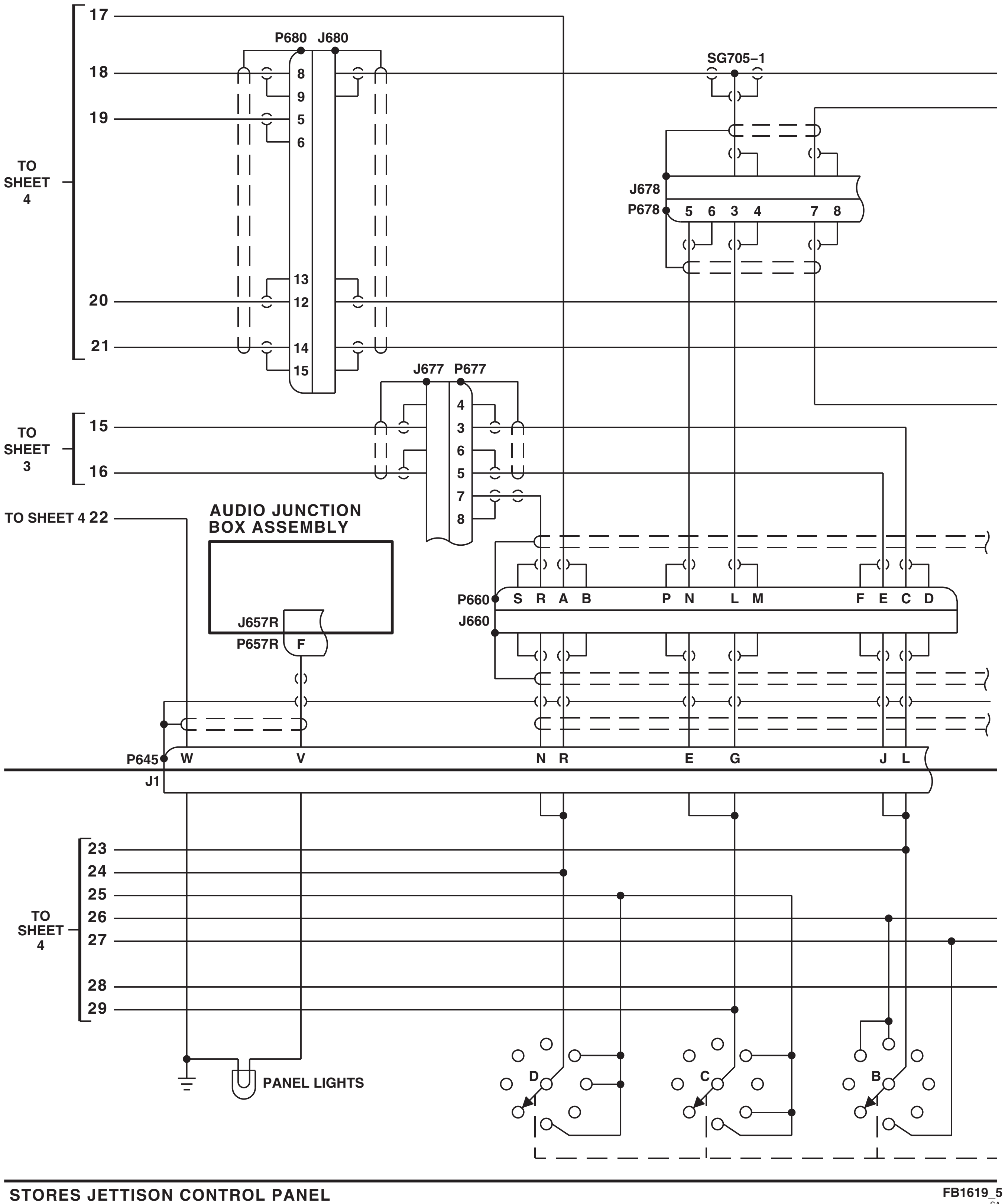


FIGURE 16-2-4-1. EXTERNAL STORES SUPPORT SYSTEM JETTISON SYSTEM SCHEMATIC DIAGRAM (SHEET 5 OF 6)

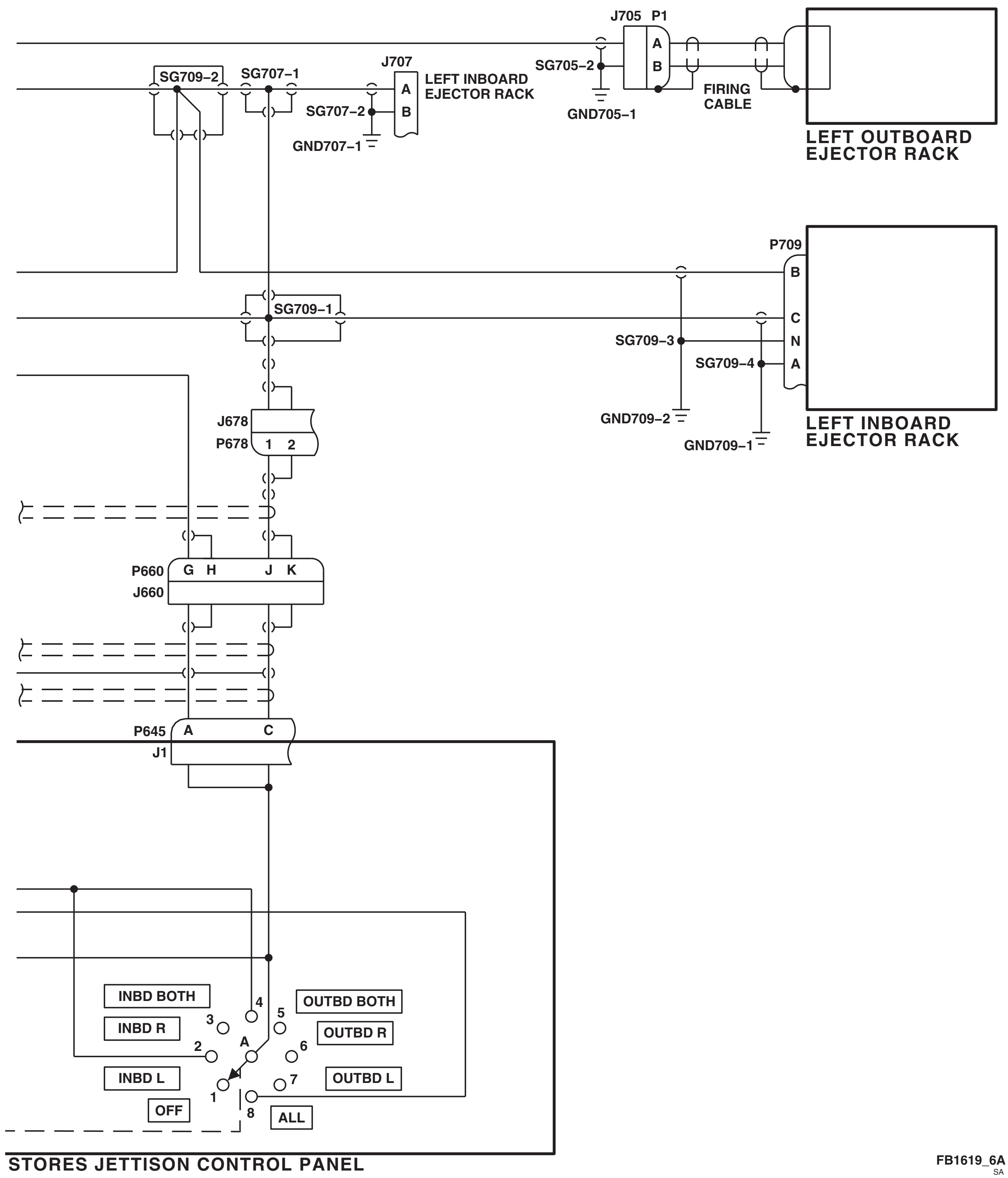


FIGURE 16-2-4-1. EXTERNAL STORES SUPPORT SYSTEM JETTISON SYSTEM SCHEMATIC DIAGRAM (SHEET 6 OF 6)

16-2-5 STORES JETTISON CONTROL PANEL (BENCH).

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

Equipment Conditions

- Between 27 and 28 vdc Bench Power Available

16-2-5.1 Operational/Troubleshooting Procedure.

a. Remove cover from stores jettison control panel (control panel) and inspect for broken or burnt wires, security of components, and general condition.

RESULT: Wires shall not be broken or burnt, and components shall be secure and in good condition.

- If result is not as specified, repair/replace wiring or components, as required.

b. Check continuity between:

J1-c	and	J1-X
J1-c	and	J2-c
J1-c	and	J2-d
J1-j	and	J1-i
J1-g	and	J1-h
J1-n	and	J1-p
J1-f	and	J1-q
J1-J	and	J1-L
J1-G	and	J1-E
J1-A	and	J1-C
J2-f	and	J2-g
J2-j	and	J2-h
J1-R	and	J1-N
J2-J	and	J2-L
J2-A	and	J2-C
J2-N	and	J2-R
J2-E	and	J2-G

RESULT: Continuity shall be present.

- If result is not as specified, repair/replace wiring, as required.

c. Connect 28 vdc between J1-V and J1-W.

RESULT: Information plate legend shall light evenly.

- If information plate lights, but not evenly, replace malfunctioning information plate lamp (PARA 16-4-51).

  (a) If trouble remains, replace information plate (PARA 16-4-51).

- If no information plate lamps go on, check continuity between J1-V and J3-(center contact), and J1-W and J3-(ground contact).

  (a) If continuity is not present, repair/replace wiring.

  (b) If continuity is present, replace information plate (PARA 16-4-51).

d. Disconnect 28 vdc from control panel.

e. Make sure control panel JETT and EMER JETT ALL switches are down.

f. Turn control panel rotary switch to INBD BOTH.

g. Connect 28 vdc between J1-n and J1-k, and J1-n and J1-X.

h. With JETT switch held up, check for 28 vdc between J1-A and J1-c. Release switch.

RESULT: Multimeter shall indicate 28 vdc.

- If result is not as specified, go to TABLE NO. 16-2-5-1.

i. Disconnect jumper wire from J1-k.

j. With JETT switch held up, check for 0 vdc between J1-A and J1-c. Release switch.

RESULT: Multimeter shall indicate 0 vdc.

- If result is not as specified, replace relay K1 (PARA 16-5-3).

k. Connect jumper wire between 28 vdc supply ground and J1-k.

l. Check for 0 vdc between J1-A and J1-c.

RESULT: Multimeter shall indicate 0 vdc.

- If result is not as specified, replace JETT switch S1 (PARA 16-5-2).

m. Turn rotary switch to ALL.

n. With JETT switch held up, check for 28 vdc between J1-L and J1-c. Release switch.

RESULT: Multimeter shall indicate 28 vdc in about 1 second after JETT switch is placed up.

- If multimeter indicates 28 vdc, but time delay is not as specified, replace relay K7 (PARA 16-5-3).

- If multimeter does not indicate 28 vdc, go to TABLE NO. 16-2-5-2.

o. Disconnect jumper wire from J1-X.

p. With JETT switch held up, check for 0 vdc between J1-L and J1-K. Release switch.

RESULT: Multimeter shall indicate 0 vdc.

- If result is not as specified, replace relay K7 (PARA 16-5-3).

q. Connect jumper wire between 28 vdc supply ground and J1-X.

r. With JETT switch held up, check for indicated voltage between the following points and J1-c while turning rotary switch as indicated. Release switch after taking measurement.

ROTARY SWITCH POSITION	**J1-A**	**MULTIMETER INDICATION AT J1-L**	**J1- G**
OFF	0 VDC	0 VDC	
INBD L	28 VDC	0 VDC	
INBD R	0 VDC	28 VDC	
INBD BOTH	28 VDC	28 VDC	
OUTBD BOTH	0 VDC	0 VDC	
OUTBD R	0 VDC	0 VDC	
OUTBD L	0 VDC	0 VDC	
ALL	28 VDC	28 VDC	0 VDC

RESULT: Multimeter shall indicate corresponding dc voltage.

- If result is not as specified at J1-A and J1-L, replace rotary switch S3 (PARA 16-5-1).

  (a) If trouble remains, repair/replace wiring.

- If result is not as specified at J1-G, replace diode CR2 (PARA 16-5-4).

s. Turn rotary switch to INBD L.

t. Connect jumper wire between 28 vdc supply ground and J1-h.

u. With JETT switch held up, check for indicated voltage between the following points. Release switch after taking measurement.

J1-C	and	J1-c
J1-J	and	J1-c

RESULT: Multimeter shall indicate 28 vdc.

- If multimeter does not indicate 28 vdc between J1-C and J1-c, repair/replace wiring between terminals A2 and A3 of relay K2.

- If multimeter does not indicate 28 vdc between J1-J and J1-c, measure dc voltage between terminals X1 and X2 of relay K2.

  (a) If voltage is as specified, replace relay K2 (PARA 16-5-3).

  (b) If voltage is not as specified, repair/replace wiring, as required.

v. Remove jumper wire from J1-h.

w. With JETT switch held up, check for 0 vdc between J1-J and J1-c. Release switch.

RESULT: Multimeter shall indicate 0 vdc.

- If result is not as specified, replace relay K2 (PARA 16-5-3).

x. Remove jumper wire from J1-n and connect it to J1-f.

y. Turn rotary switch to ALL.

z. With JETT switch held up, check for 28 vdc between J1-N and J1-c. Release switch.

RESULT: Multimeter shall indicate 28 vdc.

- If result is not as specified, go to TABLE NO. 16-2-5-3.

aa. Disconnect jumper wire from J1-k.

ab. With JETT switch held up, check for 0 vdc between J1-N and J1-c. Release switch.

RESULT: Multimeter shall indicate 0 vdc.

- If result is not as specified, replace relay K1 (PARA 16-5-3).

ac. Connect jumper wire between 28 vdc supply ground and J1-k.

ad. Check for 0 vdc between J1-N and J1-c.

RESULT: Multimeter shall indicate 0 vdc.

- If result is not as specified, replace JETT switch S1 (PARA 16-5-2).

ae. With JETT switch held up, check indicated voltage between the following points and J1-c while operating rotary switch as indicated. Release switch after taking measurement.

ROTARY SWITCH POSITION	**J1-N**	**MULTIMETER INDICATION AT J1-G**	**J1- A**
OFF	0 VDC	0 VDC	
INBD L	0 VDC	0 VDC	
INBD R	0 VDC	0 VDC	
INBD BOTH	0 VDC	0 VDC	0 VDC
OUTBD BOTH	28 VDC	28 VDC	
OUTBD R	28 VDC	0 VDC	
OUTBD L	0 VDC	28 VDC	
ALL	28 VDC	28 VDC	

RESULT: Multimeter shall indicate corresponding dc voltage.

- If result is not as specified for J1-N and J1-G, replace rotary switch S3 (PARA 16-5-1).

  (a) If trouble remains, repair/replace wiring.

- If result is not as specified for J1-A, replace diode CR1 (PARA 16-5-4).

af. Turn rotary switch to OUTBD L.

ag. Connect jumper wire between 28 vdc supply ground and J1-j.

ah. With JETT switch held up, check for 28 vdc between the following points. Release switch after taking measurement.

J1-E	and	J1-c
J1-R	and	J1-c

RESULT: Multimeter shall indicate 28 vdc.

- If multimeter does not indicate 28 vdc between J1-E and J1-c, repair/replace wiring between terminals A2 and A3 of relay K3.

- If multimeter does not indicate 28 vdc between J1-R and J1-c, check for 28 vdc between terminals X1 and X2 of relay K3.

  (a) If voltage is as specified, replace relay K3 (PARA 16-5-3).

  (b) If voltage is not as specified, repair/ replace wiring.

ai. Remove jumper wire from J1-j.

aj. With JETT switch held up, check for 0 vdc between J1-R and J1-c. Release switch.

RESULT: Multimeter shall indicate 0 vdc.

- If result is not as specified, replace relay K3 (PARA 16-5-3).

ak. Disconnect all jumper wires from control panel.

al. Connect 28 vdc positive lead to J2-f and negative lead to J2-c.

am. Place EMER JETT ALL switch up and measure dc voltage between:

J2-J	and	J2-d
J2-A	and	J2-d
J2-G	and	J2-d

RESULT:

(1) Multimeter shall indicate 28 vdc between J2-J and J2-d and J2-A and J2-d in about 1 second after EMER JETT ALL switch is placed up.

(2) Multimeter shall indicate 0 vdc between J2-G and J2-d.

- If multimeter indicates 28 vdc between either J2-J and J2-d and J2-A and J2-d, but time delay is not as specified, replace relay K6 (PARA 16-5-3).

- If multimeter does not indicate 28 vdc between J2-J and J2-d and J2-A and J2-d, go to TABLE NO. 16-2-5-4.

- If multimeter does not indicate 0 vdc between J2-G and J2-d, replace diode CR4 (PARA 16-5-4).

an. Place EMER JETT ALL switch down and check for 0 vdc between J2-J and J2-d.

RESULT: Multimeter shall indicate 0 vdc.

- If result is not as specified, replace EMER JETT ALL switch (PARA 16-5-2).

ao. Disconnect 28 vdc from J2-c and J2-f.

ap. Place EMER JETT ALL switch in the up position and check for 0 vdc between:

J2-J	and	J2-d
J2-A	and	J2-d

RESULT: Multimeter shall indicate 0 vdc.

- If result is not as specified, replace relay K6 (PARA 16-5-3).

aq. Disconnect jumper wire from J2-e and connect it to J2-c.

ar. Check for 0 vdc between J2-J and J2-d.

RESULT: Multimeter shall indicate 0 vdc.

- If result is not as specified, replace relay K4 (PARA 16-5-3).

as. Remove jumper wire from J2-f and connect 28 vdc between J2-h and J2-c, and between J2-h and J2-e.

at. Measure dc voltage between:

FROM	TO	MULTI-METER INDICATION
J2-N	J2-d	28 VDC
J2-E	J2-d	28 VDC
J2-A	J2-d	0 VDC

RESULT: Multimeter shall indicate corresponding dc voltage.

- If multimeter does not indicate 28 vdc between J2-N and J2-d and/or J2-E and J2-d, go to TABLE NO. 16-2-5-5.

- If multimeter does not indicate 0 vdc between J2-A and J2-d, replace diode CR3 (PARA 16-5-4).

au. Place EMER JETT ALL switch down and check for 0 vdc between J2-N and J2-d.

RESULT: Multimeter shall indicate 0 vdc.

- If result is not as specified, replace EMER JETT ALL switch (PARA 16-5-2).

av. Disconnect jumper wire from J2-c.

aw. Place EMER JETT ALL switch up and measure dc voltage between:

J2-N	and	J2-d
J2-E	and	J2-d

RESULT: Multimeter shall indicate 0 vdc.

- If result is not as specified, replace relay K5 (PARA 16-5-3).

ax. Disconnect jumper wire from J2-e and connect it to J2-c.

ay. Check for 0 vdc between J2-N and J2-d.

RESULT: Multimeter shall indicate 0 vdc.

- If result is not as specified, replace relay K4 (PARA 16-5-3).

az. Disconnect all jumper wires from control panel.

TABLE NO. 16-2-5-1. Multimeter Does Not Indicate 28 Vdc Between J1-A And J1-c.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for 28 vdc between JETT switch terminal 1 and J1-c.

Step 1. If voltage is as specified, go to **2.**
Step 2. If voltage is not as specified, go to **4.**

2. With JETT switch held in up position, check for 28 vdc between switch terminal 2 and J1-c.

TABLE NO. 16-2-5-1. Multimeter Does Not Indicate 28 Vdc Between J1-A And J1-c. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, replace switch (PARA 16-5-2). Go to **9.**

3. Check continuity between JETT switch terminal 2 and INBD/OUTBD switch terminal A4.

Step 1. If continuity is present, replace rotary switch S3 (PARA 16-5-1). Go to **9.**
Step 2. If continuity is not present, repair/replace wiring. Go to **9.**

4. Check for 28 vdc between relay K1-B1 and K1-X2.

Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, go to **8.**

5. Check for 28 vdc between K1-X1 and K1-X2.

Step 1. If voltage is as specified, go to **6.**
Step 2. If voltage is not as specified, go to **7.**

6. Check continuity between K1-B2 and JETT switch terminal 1.

Step 1. If continuity is present, replace relay K1 (PARA 16-5-3). Go to **9.**
Step 2. If continuity is not present, repair/replace wiring. Go to **9.**

7. Check diode CR1.

Step 1. If diode is good, repair/replace wiring. Go to **9.**
Step 2. If diode is not good, replace diode CR1 (PARA 16-5-4). Go to **9.**

8. Check diode CR6.

Step 1. If diode is good, repair/replace wiring. Go to **9.**
Step 2. If diode is not good, replace diode CR6 (PARA 16-5-4). Go to **9.**

9. Procedure completed.

TABLE NO. 16-2-5-2. Multimeter Does Not Indicate 28 Vdc Between J1-L And J1-c.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check continuity between:**
   S3-A4 and K7-X1
   S3-B8 and K7-A2
   J1-X and K7-X2

   Step 1. If continuity is present, go to **2.**
   Step 2. If continuity is not present, repair/replace wiring. Go to **3.**

2. **Check continuity between INBD/OUTBD switch terminals B8 and BC.**

   Step 1. If continuity is present, replace relay K7 (PARA 16-5-3). Go to **3.**
   Step 2. If continuity is not present, replace rotary switch S3 (PARA 16-5-1). Go to **3.**

3. **Procedure completed.**

TABLE NO. 16-2-5-3. Multimeter Does Not Indicate 28 Vdc Between J1-N And J1-c.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check for 28 vdc between JETT switch terminal 4 and J1-c.**

   Step 1. If voltage is as specified, go to **2.**
   Step 2. If voltage is not as specified, go to **4.**

2. **With JETT switch held in up position, check for 28 vdc between switch terminal 5 and J1-c.**

   Step 1. If voltage is as specified, go to **3.**
   Step 2. If voltage is not as specified, replace switch (PARA 16-5-2). Go to **9.**

3. **Check continuity between JETT switch terminal 5 and INBD/OUTBD switch terminal D8.**

   Step 1. If continuity is present, replace rotary switch S3 (PARA 16-5-1). Go to **9.**
   Step 2. If continuity is not present, repair/replace wiring. Go to **9.**

4. **Check for 28 vdc between relay K1-A1 and K1-X2.**

TABLE NO. 16-2-5-3. Multimeter Does Not Indicate 28 Vdc Between J1-N And J1-c. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, go to **8.**

5. Check for 28 vdc between K1-X1 and K1-X2.

Step 1. If voltage is as specified, go to **6.**
Step 2. If voltage is not as specified, go to **7.**

6. Check continuity between K1-A2 and JETT switch terminal 4.

Step 1. If continuity is present, replace relay K1 (PARA 16-5-3). Go to **9.**
Step 2. If continuity is not present, repair/replace wiring. Go to **9.**

7. Check diode CR2.

Step 1. If diode is good, repair/replace wiring. Go to **9.**
Step 2. If diode is not good, replace diode CR2 (PARA 16-5-4). Go to **9.**

8. Check diode CR7.

Step 1. If diode is good, repair/replace wiring. Go to **9.**
Step 2. If diode is not good, replace diode CR7 (PARA 16-5-4). Go to **9.**

9. Procedure completed.

TABLE NO. 16-2-5-4. Multimeter Does Not Indicate 28 Vdc Between J2-J And J2-d And/Or Between J2-A And J2-d.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for 28 vdc between:
J2-J and J2-d
J2-A and J2-d

Step 1. If voltage is as specified, go to **2.**
Step 2. If voltage is not as specified, go to **3.**

2. Check continuity between:

TABLE NO. 16-2-5-4. Multimeter Does Not Indicate 28 Vdc Between J2-J And J2-d And/Or Between J2-A And J2-d. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

K6-A1 and K6-B1
K6-A1 and K6-X1

Step 1. If continuity is present, replace relay K6 (PARA 16-5-3). Go to **11.**
Step 2. If continuity is not present, repair/replace wiring. Go to **11.**

3. Check for 28 vdc between EMER JETT ALL switch terminal 3 and J2-d.

Step 1. If voltage is as specified, go to **4.**
Step 2. If voltage is not as specified, go to **6.**

4. With EMER JETT ALL switch in up position, check for 28 vdc between switch terminal 2 and J2-d.

Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, replace EMER JETT ALL switch S2 (PARA 16-5-2). Go to **11.**

5. Check continuity between:
S2-2 and K6-X1
K6-A1 and K6-X1
J2-c and K6-X2

Step 1. If continuity is present, replace relay K6 (PARA 16-5-3). Go to **11.**
Step 2. If continuity is not present, repair/replace wiring. Go to **11.**

6. Check for 28 vdc between K4-B1 and K4-X2.

Step 1. If voltage is as specified, go to **7.**
Step 2. If voltage is not as specified, go to **10.**

7. Check for 28 vdc between K4-X1 and K4-X2.

Step 1. If voltage is as specified, go to **8.**
Step 2. If voltage is not as specified, go to **9.**

8. Check continuity between K4-B2 and EMER JETT ALL switch terminal 3.

Step 1. If continuity is present, replace relay K4 (PARA 16-5-3). Go to **11.**
Step 2. If continuity is not present, repair/replace wiring. Go to **11.**

TABLE NO. 16-2-5-4. Multimeter Does Not Indicate 28 Vdc Between J2-J And J2-d And/Or Between J2-A And J2-d. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

9. Check diode CR3.

Step 1. If diode is good, repair/replace wiring. Go to **11.**
Step 2. If diode is not good, replace diode CR3 (PARA 16-5-4). Go to **11.**

10. Check diode CR5.

Step 1. If diode is good, repair/replace wiring. Go to **11.**
Step 2. If diode is not good, replace diode CR5 (PARA 16-5-4). Go to **11.**

11. Procedure completed.

TABLE NO. 16-2-5-5. Multimeter Does Not Indicate 28 Vdc Between J2-N And J2-d And/Or Between J2-E And J2-d.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for 28 vdc between:
J2-N and J2-d
J2-E and J2-d

Step 1. If voltage is as specified, go to **2.**
Step 2. If voltage is not as specified, go to **3.**

2. Check continuity between:
K5-A1 and K5-B1
K5-A1 and K5-X1

Step 1. If continuity is present, replace relay K5 (PARA 16-5-3). Go to **11.**
Step 2. If continuity is not present, repair/replace wiring. Go to **11.**

3. Check for 28 vdc between EMER JETT ALL switch terminal 6 and J2-d.

Step 1. If voltage is as specified, go to **4.**
Step 2. If voltage is not as specified, go to **6.**

TABLE NO. 16-2-5-5. Multimeter Does Not Indicate 28 Vdc Between J2-N And J2-d And/Or Between J2-E And J2-d. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

4. **With EMER JETT ALL switch in up position, check for 28 vdc between switch terminal 5 and J2-d.**

   Step 1. If voltage is as specified, go to **5.**
   Step 2. If voltage is not as specified, replace EMER JETT ALL switch S2 (PARA 16-5-2). Go to **11.**

5. **Check continuity between:**
   S2-5 and K5-X1
   K5-A1 and K5-X1
   J2-c and K5-X2

   Step 1. If continuity is present, replace relay K5 (PARA 16-5-3). Go to **11.**
   Step 2. If continuity is not present, repair/replace wiring. Go to **11.**

6. **Check for 28 vdc between K4-A1 and K4-X2.**

   Step 1. If voltage is as specified, go to **7.**
   Step 2. If voltage is not as specified, go to **10.**

7. **Check for 28 vdc between K4-X1 and K4-X2.**

   Step 1. If voltage is as specified, go to **8.**
   Step 2. If voltage is not as specified, go to **9.**

8. **Check continuity between K4-A2 and EMER JETT ALL switch terminal 6.**

   Step 1. If continuity is present, replace relay K4 (PARA 16-5-3). Go to **11.**
   Step 2. If continuity is not present, repair/replace wiring. Go to **11.**

9. **Check diode CR4.**

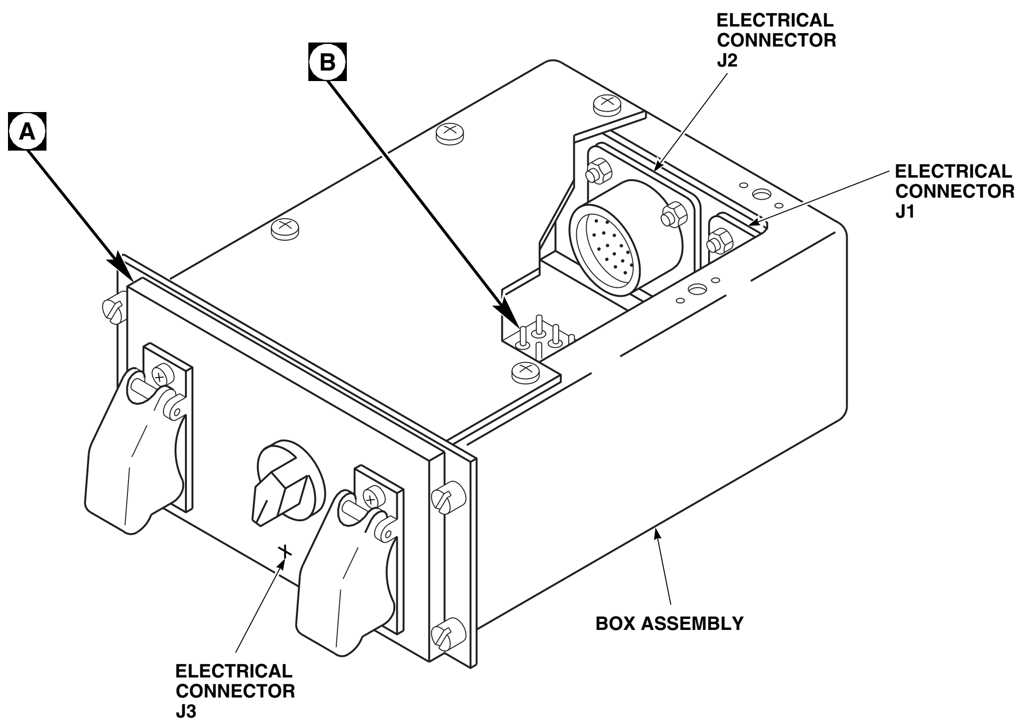
   Step 1. If diode is good, repair/replace wiring. Go to **11.**
   Step 2. If diode is not good, replace diode CR4 (PARA 16-5-4). Go to **11.**

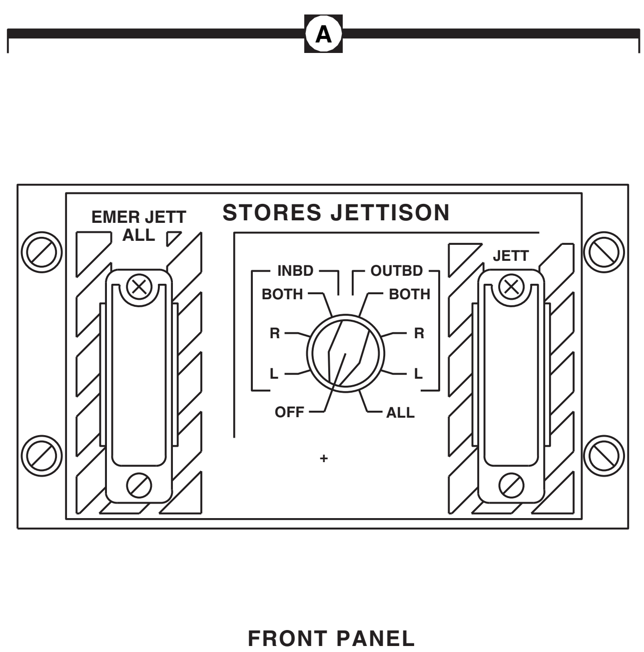
10. **Check diode CR8.**

    Step 1. If diode is good, repair/replace wiring. Go to **11.**
    Step 2. If diode is not good, replace diode CR8 (PARA 16-5-4). Go to **11.**

TABLE NO. 16-2-5-5. Multimeter Does Not Indicate 28 Vdc Between J2-N And J2-d And/Or Between J2-E And J2-d. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
11. Procedure completed.





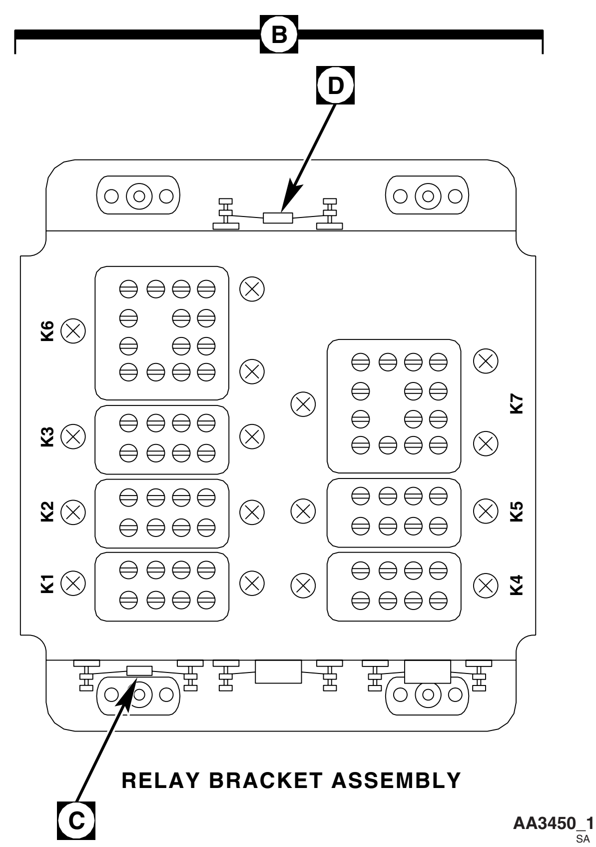


FIGURE 16-2-5-1. STORES JETTISON CONTROL PANEL COMPONENTS LOCATION DIAGRAM (SHEET 1 OF 2)

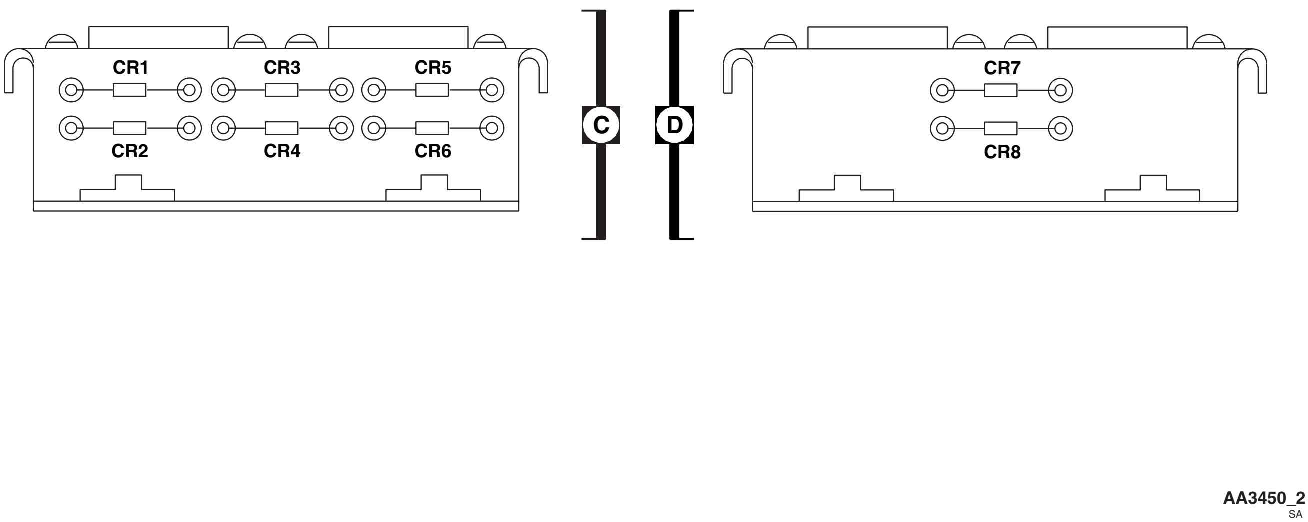


FIGURE 16-2-5-1. STORES JETTISON CONTROL PANEL COMPONENTS LOCATION DIAGRAM (SHEET 2 OF 2)

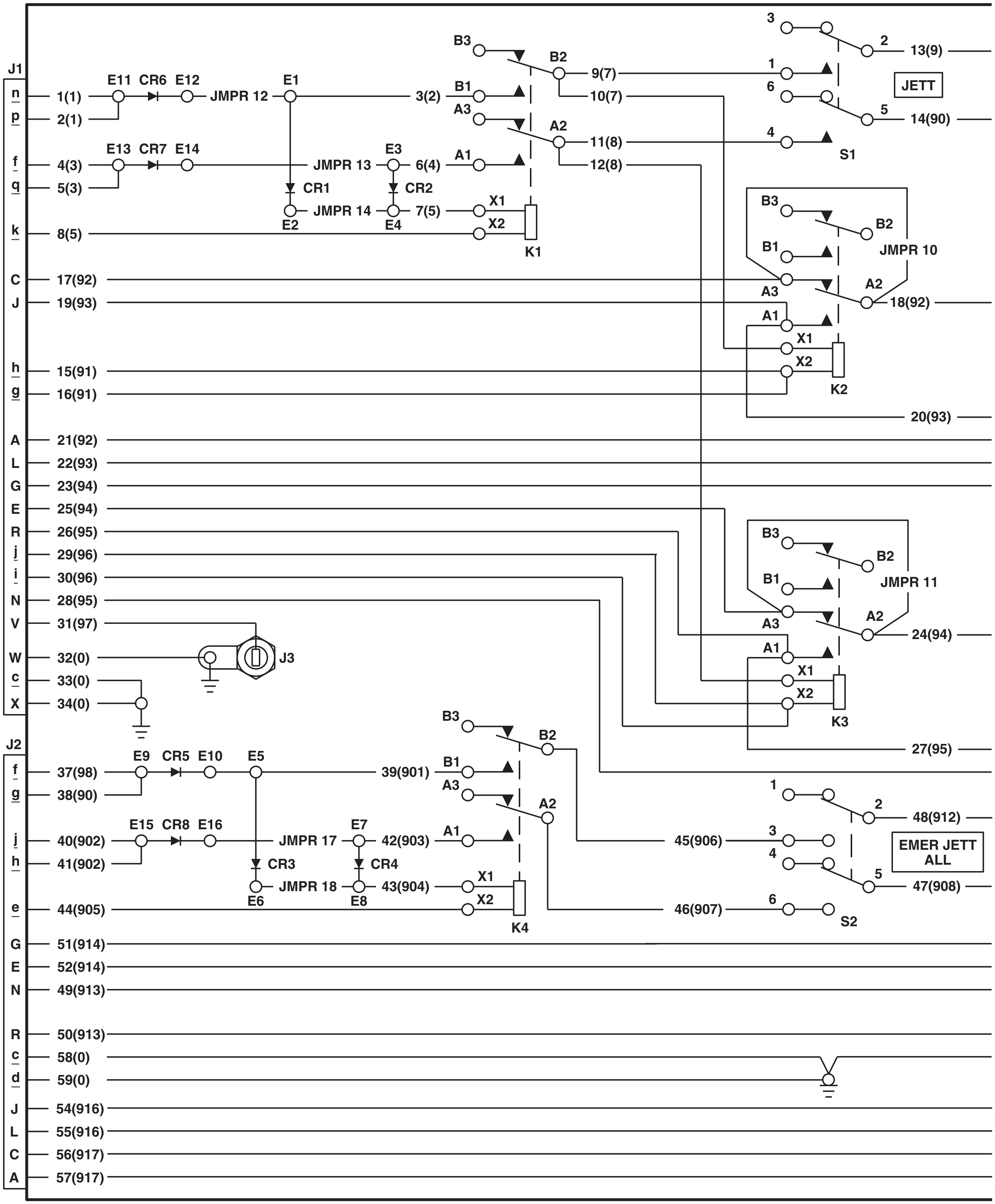


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FIGURE 16-2-5-2. STORES JETTISON CONTROL PANEL SCHEMATIC DIAGRAM (SHEET 1 OF 2)

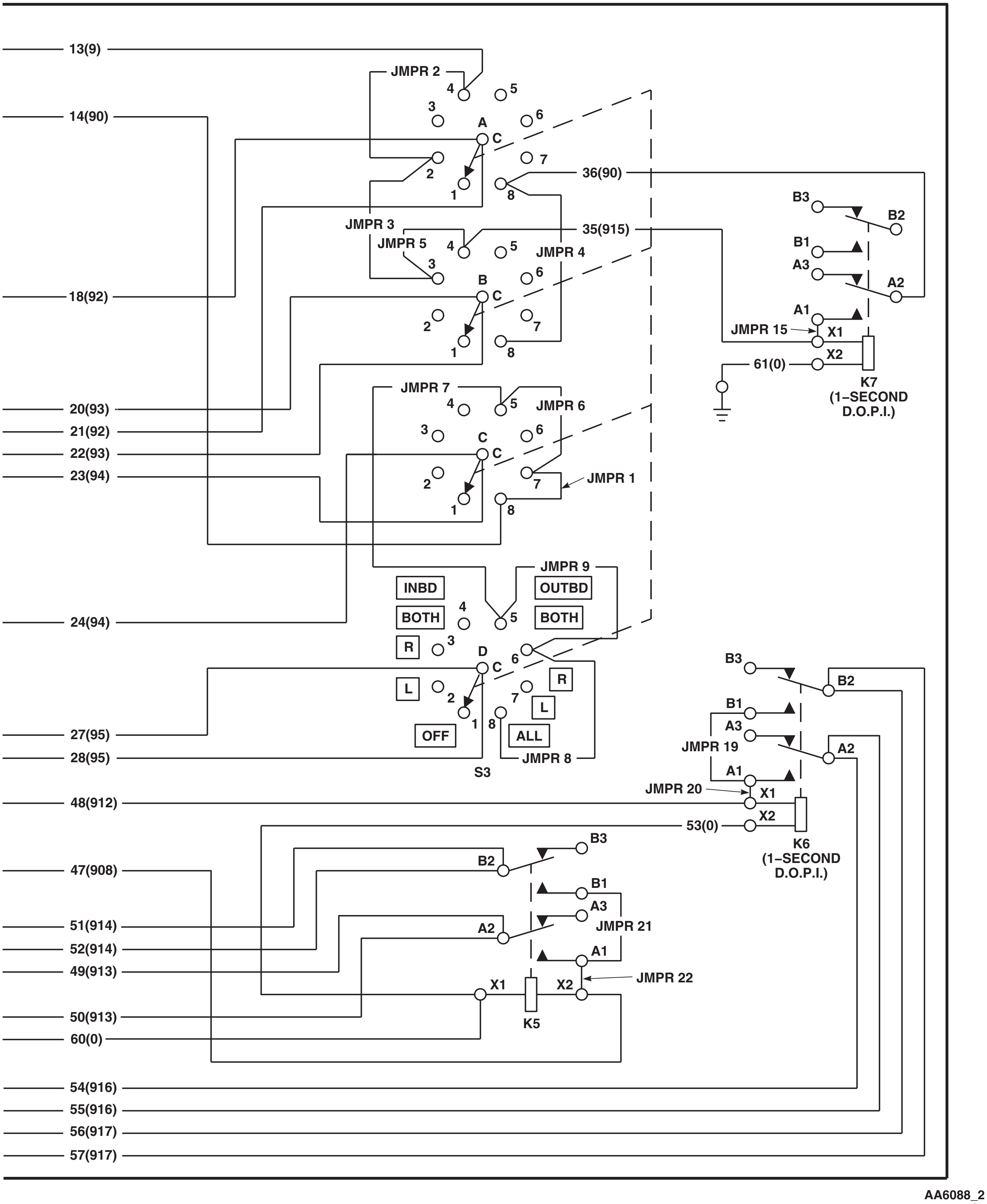


FIGURE 16-2-5-2. STORES JETTISON CONTROL PANEL SCHEMATIC DIAGRAM (SHEET 2 OF 2)

16-2-6 AIR TRANSPORT HYDRAULIC CART.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer Toolkit

Materials/Parts

- Hydraulic Fluid, Item 155, Appendix D

Equipment Conditions

- 28 vdc Electrical Power Available

CAUTION

Do not disturb factory set flow control valve settings. Adjustment could cause helicopter damage.

NOTE

If circuit breaker pops out during operational/troubleshooting procedure, a short circuit in hydraulic cart wiring is indicated.

16-2-6.1 Operational/Troubleshooting Procedure.

a. Connect hydraulic hoses from reservoir can to cart.

b. Connect all three landing gear hose assemblies to quick-disconnects on cart.

c. Connect ground strap to a suitable ground.

d. Layout electrical cable and connect to suitable 28 vdc electrical source.

e. **DO NOT TURN ELECTRICAL POWER ON AT THIS TIME.**

f. Make sure one reservoir can is filled with about 4 gallons of hydraulic fluid.

g. Open breather cap on top of can. If reservoir is not filled, fill with hydraulic fluid.

h. Crack open bleeder cap on top of the pump.

i. Bleed until air-free oil comes out.

j. Tighten bleeder cap and check for leaks.

k. Make sure MOTOR/PUMP POWER and CONTROL POWER circuit breakers on control box are pushed in.

NOTE

If red button on top of filter should pop up during pump operation, stop pump and replace filter element before doing next step.

l. On control pendant, press face of START button, then release.

RESULT:

(1) Pump shall start to operate when START button is pressed and continue to operate after button is released.

(2) POWER ON light on control panel and control pendant shall be on.

(3) Allow pump to operate for about 2 minutes. Visually check for any leaks.

- If cart motor does not operate, go to TABLE NO. 16-2-6-1.

- If cart motor stops after START button is released, go to TABLE NO. 16-2-6-2.

- If cart motor operates, but POWER ON light is not on, replace control pendant (PARA 16-4-7).

m. On control box, push EMERGENCY STOP button.

RESULT: Motor shall stop operating and both POWER ON lights shall go out.

- If result is not as specified, replace control box (PARA 16-4-8).

n. Disconnect return hose of reservoir can at SYSTEM RETURN quick-disconnect on cart frame.

o. On control pendant, press, then release START button.

RESULT:

(1) Both POWER ON lights shall be on and the pump shall momentarily start to operate.

(2) Then, the MOTOR STOP light shall go on, the POWER ON lights shall go off, and the pump shall stop operating.

(3) MOTOR STOP light shall go off once pressure has dropped.

- If results are not as specified, replace pressure switch (PARA 16-4-12).

p. Connect hose from reservoir can to cart.

q. On control pendant, press, then release START button to turn on cart.

r. On control pendant, push and hold RIGHT switch to UP.

s. Check cart and right main landing gear hose for leaks.

RESULT: Hose shall pressurize as indicated by sound of motor RPM decreasing.

- If result is not as specified, go to TABLE NO. 16-2-6-3.

t. Release switch to center position.

u. On control pendant, push and hold TAIL switch to UP.

v. Check cart and tail landing gear hose for leaks.

RESULT: Hose shall pressurize as indicated by sound of motor RPM decreasing.

- If result is not as specified, go to TABLE NO. 16-2-6-4.

w. Release switch to center position.

x. On control pendant, push and hold LEFT switch to UP.

y. Check cart and left main landing gear hose for leaks.

RESULT: Hose shall pressurize as indicated by sound of motor RPM decreasing.

- If result is not as specified, go to TABLE NO. 16-2-6-5.

z. Release switch to center position.

aa. Remove either quick-disconnect fitting from end of hose connected to RIGHT MAIN coupling on cart.

ab. Route hose into reservoir can.

ac. On control pendant, push and hold RIGHT switch to UP.

RESULT: Fluid shall be flowing into reservoir can.

ad. On control pendant, push and hold RIGHT switch to DOWN.

RESULT:

(1) Fluid shall stop flowing into reservoir can.

(2) On cart 70700-81650-041, sound of motor RPM increasing shall be heard.

- If result (1) is not as specified, go to TABLE NO. 16-2-6-6.
- If result (2) is not as specified, go to TABLE NO. 16-2-6-7.

ae. Release RIGHT switch to center position.

af. Remove hose from top of reservoir can.

ag. Install quick-disconnect fitting into end of hose.

ah. Remove either quick-disconnect fitting from end of hose coming from TAIL coupling on cart.

ai. Route hose into reservoir can.

aj. On control pendant, push and hold TAIL switch to UP.

RESULT: Fluid shall flow into reservoir can.

ak. On control pendant, push and hold TAIL switch to DOWN.

RESULT:

(1) Fluid shall stop flowing into reservoir can.

(2) On cart 70700-81650-041, sound of motor RPM increasing shall be heard.

- If result (1) is not as specified, go to TABLE NO. 16-2-6-8.
- If result (2) is not as specified, go to TABLE NO. 16-2-6-9.

al. Release TAIL switch to center position.

am. Remove hose from top of reservoir can.

an. Install quick-disconnect fitting into end of hose.

ao. Remove either quick-disconnect fitting from end of hose coming from LEFT coupling on cart.

ap. Route hose into reservoir can.

aq. On control pendant, push and hold LEFT switch to UP.

RESULT: Fluid shall flow into reservoir can.

ar. On control pendant, push and hold LEFT switch to DOWN.

RESULT:

(1) Fluid shall stop flowing into reservoir can.

(2) On cart 70700-81650-041, sound of motor RPM increasing shall be heard.

- If result (1) is not as specified, go to TABLE NO. 16-2-6-10.
- If result (2) is not as specified, go to TABLE NO. 16-2-6-11.

as. Release LEFT switch to center position.

at. Remove hose from top of reservoir can.

au. Install quick-disconnect fitting into end of hose.

av. On control pendant, press face of STOP switch.

RESULT: Pump motor shall stop operating.

- If result is not as specified, replace control pendant (PARA 16-4-7).

aw. If necessary, disconnect electrical power and ground strap.

TABLE NO. 16-2-6-1. Cart Motor Does Not Operate When START Button Is Pressed.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **With START button pressed, check for 28 vdc at terminals of motor.**

   Step 1. If voltage is as specified, replace pump and motor assembly (PARA 16-4-9). Go to **22.**
   Step 2. If voltage is not as specified, go to **2.**

2. **With START button pressed, check for 28 vdc between motor (+) terminal and cart ground.**

   Step 1. If voltage is as specified, go to **3.**
   Step 2. If voltage is not as specified, go to **5.**

3. **Check continuity between motor (-) terminal and P1-B.**

   Step 1. If continuity is present, go to **4.**
   Step 2. If continuity is not present, repair/replace wiring. Go to **22.**

4. **Check continuity between J1-B and TB1-2.**

   Step 1. If continuity is present, repair/replace wiring between TB1-1 and J2-C. Go to **22.**
   Step 2. If continuity is not present, repair/replace wiring. Go to **22.**

5. **With START button pressed, check for 28 vdc between J1-A and cart ground.**

   Step 1. If voltage is as specified, repair/replace wiring between P1-A and motor (+) terminal. Go to **22.**
   Step 2. If voltage is not as specified, go to **6.**

6. **Check continuity between J2-B and J4-B.**

   Step 1. If continuity is present, go to **7.**
   Step 2. If continuity is not present, go to **20.**

7. **With START button pressed, check continuity between P4-B and P4-D.**

   Step 1. If continuity is present, go to **8.**
   Step 2. If continuity is not present, go to **12.**

8. **Check continuity between J4-D and relay K1-X1.**

TABLE NO. 16-2-6-1. Cart Motor Does Not Operate When START Button Is Pressed. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, go to **9.**
Step 2. If continuity is not present, go to **11.**

9. Check continuity between relay K1-X2 and TB1-1.

Step 1. If continuity is present, go to **10.**
Step 2. If continuity is not present, repair/replace wiring. Go to **22.**

10. Check continuity between TB1-1 and J2-C.

Step 1. If continuity is present, replace control box (PARA 16-4-8). Go to **22.**
Step 2. If continuity is not present, repair/replace wiring. Go to **22.**

11. Check continuity between relay K1-A1 and K1-X1.

Step 1. If continuity is present, repair/replace wiring between J4-D and relay K1-A1. Go to **22.**
Step 2. If continuity is not present, repair/replace wiring. Go to **22.**

12. Check continuity between START button terminal 1 and P4-D.

Step 1. If continuity is present, go to **13.**
Step 2. If continuity is not present, go to **15.**

13. Check continuity between START button terminal 2 and P4-B.

Step 1. If continuity is present, replace control pendant (PARA 16-4-7). Go to **22.**
Step 2. If continuity is not present, go to **14.**

14. Check continuity between START button terminal 2 and STOP button terminal 2.

Step 1. If continuity is present, replace wiring between STOP button terminal 2 and P4-B. Go to **22.**
Step 2. If continuity is not present, repair/replace wiring. Go to **22.**

15. Check hydraulic cart configuration.

Step 1. If hydraulic cart is 70700-81650-041, go to **16.**
Step 2. If hydraulic cart is 70700-20437-041, go to **17.**

TABLE NO. 16-2-6-1. Cart Motor Does Not Operate When START Button Is Pressed. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

16. Check continuity between POWER ON terminal 1 and START button terminal 1.

Step 1. If continuity is present, repair/replace wiring between POWER ON terminal 1 and P4-D. Go to **22.**
Step 2. If continuity is not present, repair/replace wiring. Go to **22.**

17. Check continuity between START button terminal 1 and LEFT switch terminal 2.

Step 1. If continuity is present, go to **18.**
Step 2. If continuity is not present, repair/replace wiring. Go to **22.**

18. Check continuity between LEFT switch terminal 2 and TAIL switch terminal 2.

Step 1. If continuity is present, go to **19.**
Step 2. If continuity is not present, repair/replace wiring. Go to **22.**

19. Check continuity between RIGHT switch terminal 2 and TAIL switch terminal 2.

Step 1. If continuity is present, repair/replace wiring between RIGHT switch terminal 2 and P4-D. Go to **22.**
Step 2. If continuity is not present, repair/replace wiring. Go to **22.**

20. Check continuity between CONTROL POWER circuit breaker terminal 2 and J4-B.

Step 1. If continuity is present, go to **21.**
Step 2. If continuity is not present, repair/replace wiring. Go to **22.**

21. Check continuity between CONTROL POWER circuit breaker terminal 1 and J2-B.

Step 1. If continuity is present, replace control box (PARA 16-4-8). Go to **22.**
Step 2. If continuity is not present, repair/replace wiring. Go to **22.**

22. Procedure completed.

TABLE NO. 16-2-6-2. Cart Motor Stops After START Button Is Released.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check for 28 vdc between pressure switch terminal NC and terminal C.**

   Step 1. If voltage is as specified, replace pressure switch (PARA 16-4-12). Go to **10.**
   Step 2. If voltage is not as specified, go to **2.**

2. **Check for 28 vdc between pressure switch terminal NC and cart ground.**

   Step 1. If voltage is as specified, go to **3.**
   Step 2. If voltage is not as specified, go to **5.**

3. **Check continuity between pressure switch terminal C and P3-C.**

   Step 1. If continuity is present, go to **4.**
   Step 2. If continuity is not present, repair/replace wiring. Go to **10.**

4. **Check continuity between J3-C and relay K1-A2.**

   Step 1. If continuity is present, replace control box (PARA 16-4-8). Go to **10.**
   Step 2. If continuity is not present, repair/replace wiring. Go to **10.**

5. **Check continuity between pressure switch terminal NC and P3-D.**

   Step 1. If continuity is present, go to **6.**
   Step 2. If continuity is not present, repair/replace wiring. Go to **10.**

6. **Check continuity between J3-D and J4-C.**

   Step 1. If continuity is present, go to **7.**
   Step 2. If continuity is not present, go to **8.**

7. **Check continuity between P4-C and STOP switch terminal 1.**

   Step 1. If continuity is present, replace control pendant (PARA 16-4-7). Go to **10.**
   Step 2. If continuity is not present, repair/replace wiring. Go to **10.**

8. **Check continuity between EMERGENCY STOP switch terminal NC and J3-D.**

   Step 1. If continuity is present, go to **9.**
   Step 2. If continuity is not present, repair/replace wiring. Go to **10.**

9. **Check continuity between EMERGENCY STOP switch terminal NC and J4-C.**

TABLE NO. 16-2-6-2. Cart Motor Stops After START Button Is Released. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, replace control box (PARA 16-4-8). Go to **10.**
Step 2. If continuity is not present, repair/replace wiring. Go to **10.**

10. Procedure completed.

TABLE NO. 16-2-6-3. Hose Does Not Pressurize When RIGHT Switch Is Placed UP (Motor RPM Did Not Decrease).

TEST OR INSPECTION
CORRECTIVE ACTION

1. With cart motor operating, push right directional valve manual button on up side. Go to 2.

2. Hose shall pressurize and sound of motor RPM decreasing shall be heard.

Step 1. If hose does pressurize and motor RPM decreases, go to **3.**
Step 2. If hose does not pressurize and motor RPM does not decrease, replace directional control valve assembly (PARA 16-4-13). Go to **8.**

3. Check if TAIL and LEFT switches pressurize their hoses.

Step 1. If switches pressurize their hoses, go to **4.**
Step 2. If switches do not pressurize their hoses, repair/replace wiring between RIGHT switch terminal 2 and P4-D. Go to **8.**

4. With RIGHT switch placed UP, check for 28 vdc between UP solenoid terminal 1 and terminal 2.

Step 1. If voltage is as specified, replace directional control valve assembly (PARA 16-4-13). Go to **8.**
Step 2. If voltage is not as specified, go to **5.**

5. With RIGHT switch placed UP, check for 28 vdc between UP solenoid terminal 1 and cart ground.

Step 1. If voltage is as specified, go to **6.**
Step 2. If voltage is not as specified, go to **7.**

TABLE NO. 16-2-6-3. Hose Does Not Pressurize When RIGHT Switch Is Placed UP (Motor RPM Did Not Decrease). (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

6. Check continuity between UP solenoid terminal 2 and cart ground.

Step 1. If continuity is present, replace directional control valve assembly (PARA 16-4-13). Go to **8.**
Step 2. If continuity is not present, repair/replace wiring. Go to **8.**

7. Check continuity between UP solenoid terminal 1 and RIGHT switch terminal 3.

Step 1. If continuity is present, replace control pendant (PARA 16-4-7). Go to **8.**
Step 2. If continuity is not present, repair/replace wiring. Go to **8.**

8. Procedure completed.

TABLE NO. 16-2-6-4. Hose Does Not Pressurize When TAIL Switch Is Placed UP (Motor RPM Did Not Decrease).

TEST OR INSPECTION
CORRECTIVE ACTION

1. With cart motor operating, push tail wheel directional valve manual button on up side. Go to 2.

2. Hose shall pressurize and sound of motor RPM decreasing shall be heard.

Step 1. If hose does pressurize and motor RPM decreases, go to **3.**
Step 2. If hose does not pressurize and motor RPM does not decrease, replace directional control valve assembly (PARA 16-4-13). Go to **10.**

3. With TAIL switch placed UP, check for 28 vdc between UP solenoid terminal 1 and terminal 2.

Step 1. If voltage is as specified, replace directional control valve assembly (PARA 16-4-13). Go to **10.**
Step 2. If voltage is not as specified, go to **4.**

4. With TAIL switch placed UP, check for 28 vdc between UP solenoid terminal 1 and cart ground.

Step 1. If voltage is as specified, go to **5.**

TABLE NO. 16-2-6-4. Hose Does Not Pressurize When TAIL Switch Is Placed UP (Motor RPM Did Not Decrease). (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If voltage is not as specified, go to **6.**

5. **Check continuity between UP solenoid terminal 2 and cart ground.**

Step 1. If continuity is present, replace directional control valve assembly (PARA 16-4-13). Go to **10.**
Step 2. If continuity is not present, repair/replace wiring. Go to **10.**

6. **Check continuity between UP solenoid terminal 1 and TAIL switch terminal 3.**

Step 1. If continuity is present, go to **7.**
Step 2. If continuity is not present, repair/replace wiring. Go to **10.**

7. **Check hydraulic cart configuration.**

Step 1. If hydraulic cart is 70700-81650-041, go to **8.**
Step 2. If hydraulic cart is 70700-20437-041, repair/replace wiring. Go to **10.**

8. **Check continuity between TAIL switch terminal 2 and RIGHT switch terminal 5.**

Step 1. If continuity is present, go to **9.**
Step 2. If continuity is not present, repair/replace wiring. Go to **10.**

9. **Check continuity between TAIL switch terminal 2 and RIGHT switch terminal 2.**

Step 1. If continuity is present, replace control pendant (PARA 16-4-7). Go to **10.**
Step 2. If continuity is not present, repair/replace wiring. Go to **10.**

10. **Procedure completed.**

TABLE NO. 16-2-6-5. Hose Does Not Pressurize When LEFT Switch Is Placed UP (Motor RPM Did Not Decrease).

TEST OR INSPECTION
CORRECTIVE ACTION

1. **With cart motor operating, push left directional valve manual button on up side. Go to 2.**

TABLE NO. 16-2-6-5. Hose Does Not Pressurize When LEFT Switch Is Placed UP (Motor RPM Did Not Decrease). (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

2. Hose shall pressurize and sound of motor RPM decreasing shall be heard.

Step 1. If hose does pressurize and motor RPM decreases, go to **3.**
Step 2. If hose does not pressurize and motor RPM does not decrease, replace directional control valve assembly (PARA 16-4-13). Go to **10.**

3. With LEFT switch placed UP, check for 28 vdc between UP solenoid terminal 1 and terminal 2.

Step 1. If voltage is as specified, replace directional control valve assembly (PARA 16-4-13). Go to **10.**
Step 2. If voltage is not as specified, go to **4.**

4. With LEFT switch placed UP, check for 28 vdc between UP solenoid terminal 1 and cart ground.

Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, go to **6.**

5. Check continuity between UP solenoid terminal 2 and cart ground.

Step 1. If continuity is present, replace directional control valve assembly (PARA 16-4-13). Go to **10.**
Step 2. If continuity is not present, repair/replace wiring. Go to **10.**

6. Check continuity between UP solenoid terminal 1 and LEFT switch terminal 3.

Step 1. If continuity is present, go to **7.**
Step 2. If continuity is not present, repair/replace wiring. Go to **10.**

7. Check hydraulic cart configuration.

Step 1. If hydraulic cart is 70700-81650-041, go to **8.**
Step 2. If hydraulic cart is 70700-20437-041, repair/replace wiring. Go to **10.**

8. Check continuity between LEFT switch terminal 2 and TAIL switch terminal 5.

Step 1. If continuity is present, go to **9.**
Step 2. If continuity is not present, repair/replace wiring. Go to **10.**

TABLE NO. 16-2-6-5. Hose Does Not Pressurize When LEFT Switch Is Placed UP (Motor RPM Did Not Decrease). (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

9. **Check continuity between LEFT switch terminal 2 and TAIL switch terminal 2.**

   Step 1. If continuity is present, replace control pendant (PARA 16-4-7). Go to **10.**
   Step 2. If continuity is not present, repair/replace wiring. Go to **10.**

10. **Procedure completed.**

TABLE NO. 16-2-6-6. Fluid Does Not Stop Flowing When RIGHT Switch Is Placed DOWN.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **With cart motor operating, push right directional valve manual button on down side. Go to 2.**

2. **Fluid shall stop flowing into can.**

   Step 1. If fluid stops flowing, go to **3.**
   Step 2. If fluid does not stop flowing, replace directional control valve assembly (PARA 16-4-13). Go to **12.**

3. **With RIGHT switch placed DOWN, check for 28 vdc between DOWN solenoid terminal 1 and terminal 2.**

   Step 1. If voltage is as specified, replace directional control valve assembly (PARA 16-4-13). Go to **12.**
   Step 2. If voltage is not as specified, go to **4.**

4. **With RIGHT switch placed DOWN, check for 28 vdc between DOWN solenoid terminal 1 and cart ground.**

   Step 1. If voltage is as specified, go to **5.**
   Step 2. If voltage is not as specified, go to **6.**

5. **Check continuity between DOWN solenoid terminal 2 and cart ground.**

   Step 1. If continuity is present, replace directional control valve assembly (PARA 16-4-13). Go to **12.**
   Step 2. If continuity is not present, repair/replace wiring. Go to **12.**

TABLE NO. 16-2-6-6. Fluid Does Not Stop Flowing When RIGHT Switch Is Placed DOWN. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

6. Check continuity between DOWN solenoid terminal 1 and P3-J.

Step 1. If continuity is present, go to **7.**
Step 2. If continuity is not present, repair/replace wiring. Go to **12.**

7. Check hydraulic cart configuration.

Step 1. If hydraulic cart is 70700-81650-041, go to **8.**
Step 2. If hydraulic cart is 70700-20437-041, go to **10.**

8. Check continuity between J3-J and J4-N.

Step 1. If continuity is present, go to **9.**
Step 2. If continuity is not present, repair/replace wiring. Go to **12.**

9. Check continuity between RIGHT switch terminal 1 and P4-N.

Step 1. If continuity is present, replace control pendant (PARA 16-4-7). Go to **12.**
Step 2. If continuity is not present, repair/replace wiring. Go to **12.**

10. Check continuity between J3-J and J4-K.

Step 1. If continuity is present, go to **11.**
Step 2. If continuity is not present, repair/replace wiring. Go to **12.**

11. Check continuity between RIGHT switch terminal 1 and P4-K.

Step 1. If continuity is present, replace control pendant (PARA 16-4-7). Go to **12.**
Step 2. If continuity is not present, repair/replace wiring. Go to **12.**

12. Procedure completed.

TABLE NO. 16-2-6-7. Cart's Motor RPM Does Not Increase When RIGHT Switch Is Placed DOWN.

TEST OR INSPECTION
CORRECTIVE ACTION

1. With cart motor operating, push bleed valve manual button. Go to 2.

TABLE NO. 16-2-6-7. Cart's Motor RPM Does Not Increase When RIGHT Switch Is Placed DOWN. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

2. **Hose shall depressurize.**

   Step 1. If hose does depressurize, go to **3.**
   Step 2. If hose does not depressurize, replace directional control valve assembly (PARA 16-4-13). Go to **12.**

3. **Check hydraulic cart configuration.**

   Step 1. If hydraulic cart is 70700-81650-041, go to **4.**
   Step 2. If hydraulic cart is 70700-20437-041, go to **9.**

4. **With RIGHT switch placed DOWN, check for 28 vdc between BLEED solenoid terminal 1 and terminal 2.**

   Step 1. If voltage is as specified, replace directional control valve assembly (PARA 16-4-13). Go to **12.**
   Step 2. If voltage is not as specified, go to **5.**

5. **With RIGHT switch placed DOWN, check for 28 vdc between BLEED solenoid terminal 1 and cart ground.**

   Step 1. If voltage is as specified, repair/replace wiring between BLEED solenoid terminal 2 and cart ground. Go to **12.**
   Step 2. If voltage is not as specified, go to **6.**

6. **Check continuity between BLEED solenoid terminal 1 and P3-R.**

   Step 1. If continuity is present, go to **7.**
   Step 2. If continuity is not present, repair/replace wiring. Go to **12.**

7. **Check continuity between TB2-2 and J4-L.**

   Step 1. If continuity is present, go to **8.**
   Step 2. If continuity is not present, repair/replace wiring. Go to **12.**

8. **Check continuity between RIGHT switch terminal 6 and P3-L.**

   Step 1. If continuity is present, replace control pendant (PARA 16-4-7). Go to **12.**

TABLE NO. 16-2-6-7. Cart's Motor RPM Does Not Increase When RIGHT Switch Is Placed DOWN. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If continuity is not present, repair/replace wiring. Go to **12.**

9. Check continuity between DOWN solenoid terminal 1 and P3-J.

Step 1. If continuity is present, go to **10.**
Step 2. If continuity is not present, repair/replace wiring. Go to **12.**

10. Check continuity between J3-J and J4-K.

Step 1. If continuity is present, go to **11.**
Step 2. If continuity is not present, repair/replace wiring. Go to **12.**

11. Check continuity between RIGHT switch terminal 1 and P4-K.

Step 1. If continuity is present, replace control pendant (PARA 16-4-7). Go to **12.**
Step 2. If continuity is not present, repair/replace wiring. Go to **12.**

12. Procedure completed.

TABLE NO. 16-2-6-8. Fluid Does Not Stop Flowing When TAIL Switch Is Placed DOWN.

TEST OR INSPECTION
CORRECTIVE ACTION

1. With cart motor operating, push right directional valve manual button on down side. Go to 2.

2. Fluid shall stop flowing into can.

Step 1. If fluid stops flowing, go to **3.**
Step 2. If fluid does not stop flowing, replace directional control valve assembly (PARA 16-4-13). Go to **13.**

3. With TAIL switch placed DOWN, check for 28 vdc between DOWN solenoid terminal 1 and terminal 2.

Step 1. If voltage is as specified, replace directional control valve assembly (PARA 16-4-13). Go to **13.**

TABLE NO. 16-2-6-8. Fluid Does Not Stop Flowing When TAIL Switch Is Placed DOWN. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If voltage is not as specified, go to **4.**

4. **With TAIL switch placed DOWN, check for 28 vdc between DOWN solenoid terminal 1 and cart ground.**

Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, go to **6.**

5. **Check continuity between DOWN solenoid terminal 2 and cart ground.**

Step 1. If continuity is present, replace directional control valve assembly (PARA 16-4-13). Go to **13.**
Step 2. If continuity is not present, repair/replace wiring. Go to **13.**

6. **Check hydraulic cart configuration.**

Step 1. If hydraulic cart is 70700-81650-041, go to **7.**
Step 2. If hydraulic cart is 70700-20437-041, go to **10.**

7. **Check continuity between DOWN solenoid terminal 1 and P3-E.**

Step 1. If continuity is present, go to **8.**
Step 2. If continuity is not present, repair/replace wiring. Go to **13.**

8. **Check continuity between J3-E and J4-E.**

Step 1. If continuity is present, go to **9.**
Step 2. If continuity is not present, repair/replace wiring. Go to **13.**

9. **Check continuity between TAIL switch terminal 1 and P4-E.**

Step 1. If continuity is present, replace control pendant (PARA 16-4-7). Go to **13.**
Step 2. If continuity is not present, repair/replace wiring. Go to **13.**

10. **Check continuity between DOWN solenoid terminal 1 and P3-M.**

Step 1. If continuity is present, go to **11.**
Step 2. If continuity is not present, repair/replace wiring. Go to **13.**

11. **Check continuity between J3-M and J4-G.**

TABLE NO. 16-2-6-8. Fluid Does Not Stop Flowing When TAIL Switch Is Placed DOWN. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, go to **12.**
Step 2. If continuity is not present, repair/replace wiring. Go to **13.**

12. Check continuity between TAIL switch terminal 1 and P4-G.

Step 1. If continuity is present, replace control pendant (PARA 16-4-7). Go to **13.**
Step 2. If continuity is not present, repair/replace wiring. Go to **13.**

13. Procedure completed.

TABLE NO. 16-2-6-9. Cart's Motor RPM Does Not Increase When TAIL Switch Is Placed DOWN.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check continuity between TB2-3 and J4-G.

Step 1. If continuity is present, go to **2.**
Step 2. If continuity is not present, repair/replace wiring. Go to **3.**

2. Check continuity between TAIL switch terminal 6 and P4-G.

Step 1. If continuity is present, replace control pendant (PARA 16-4-7). Go to **3.**
Step 2. If continuity is not present, repair/replace wiring. Go to **3.**

3. Procedure completed.

TABLE NO. 16-2-6-10. Fluid Does Not Stop Flowing When LEFT Switch Is Placed DOWN.

TEST OR INSPECTION
CORRECTIVE ACTION

1. With cart motor operating, push tail wheel directional valve manual button on down side. Go to 2.

2. Fluid shall stop flowing into can.

Step 1. If fluid stops flowing, go to **3.**

TABLE NO. 16-2-6-10. Fluid Does Not Stop Flowing When LEFT Switch Is Placed DOWN. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If fluid does not stop flowing, replace directional control valve assembly (PARA 16-4-13). Go to **13.**

3. With LEFT switch placed DOWN, check for 28 vdc between DOWN solenoid terminal 1 and terminal 2.

Step 1. If voltage is as specified, replace directional control valve assembly (PARA 16-4-13). Go to **13.**
Step 2. If voltage is not as specified, go to **4.**

4. With LEFT switch placed DOWN, check for 28 vdc between DOWN solenoid terminal 1 and cart ground.

Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, go to **6.**

5. Check continuity between DOWN solenoid terminal 2 and cart ground.

Step 1. If continuity is present, replace directional control valve assembly (PARA 16-4-13). Go to **13.**
Step 2. If continuity is not present, repair/replace wiring. Go to **13.**

6. Check hydraulic cart configuration.

Step 1. If hydraulic cart is 70700-81650-041, go to **7.**
Step 2. If hydraulic cart is 70700-20437-041, go to **10.**

7. Check continuity between DOWN solenoid terminal 1 and P3-N.

Step 1. If continuity is present, go to **8.**
Step 2. If continuity is not present, repair/replace wiring. Go to **13.**

8. Check continuity between J3-N and J4-J.

Step 1. If continuity is present, go to **9.**
Step 2. If continuity is not present, repair/replace wiring. Go to **13.**

9. Check continuity between LEFT switch terminal 1 and P4-J.

Step 1. If continuity is present, replace control pendant (PARA 16-4-7). Go to **13.**
Step 2. If continuity is not present, repair/replace wiring. Go to **13.**

TABLE NO. 16-2-6-10. Fluid Does Not Stop Flowing When LEFT Switch Is Placed DOWN. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

10. Check continuity between DOWN solenoid terminal 1 and P3-K.

Step 1. If continuity is present, go to **11.**
Step 2. If continuity is not present, repair/replace wiring. Go to **13.**

11. Check continuity between J3-K and J4-E.

Step 1. If continuity is present, go to **12.**
Step 2. If continuity is not present, repair/replace wiring. Go to **13.**

12. Check continuity between LEFT switch terminal 1 and P4-E.

Step 1. If continuity is present, replace control pendant (PARA 16-4-7). Go to **13.**
Step 2. If continuity is not present, repair/replace wiring. Go to **13.**

13. Procedure completed.

TABLE NO. 16-2-6-11. Cart's Motor RPM Does Not Increase When LEFT Switch Is Placed DOWN.

TEST OR INSPECTION
CORRECTIVE ACTION

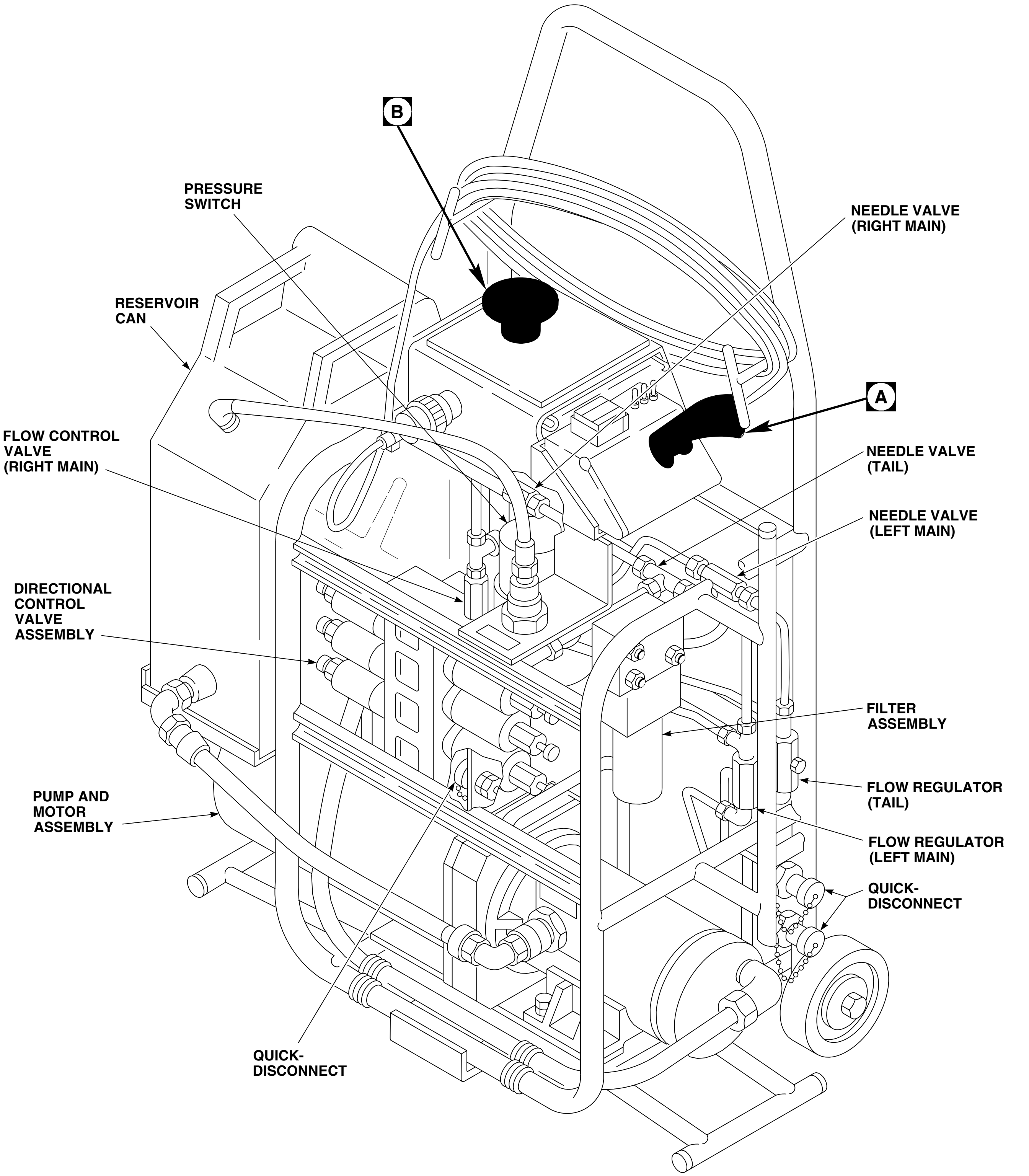
1. Check continuity between TB2-4 and J4-F.

Step 1. If continuity is present, go to **2.**
Step 2. If continuity is not present, repair/replace wiring. Go to **3.**

2. Check continuity between LEFT switch terminal 6 and P4-F.

Step 1. If continuity is present, replace control pendant (PARA 16-4-7). Go to **3.**
Step 2. If continuity is not present, repair/replace wiring. Go to **3.**

3. Procedure completed.



EFFECTIVITY

70700-20437-041.

AA8704_1
SA

FIGURE 16-2-6-1. AIR TRANSPORT HYDRAULIC CART LOCATION DIAGRAM (SHEET 1 OF 2)

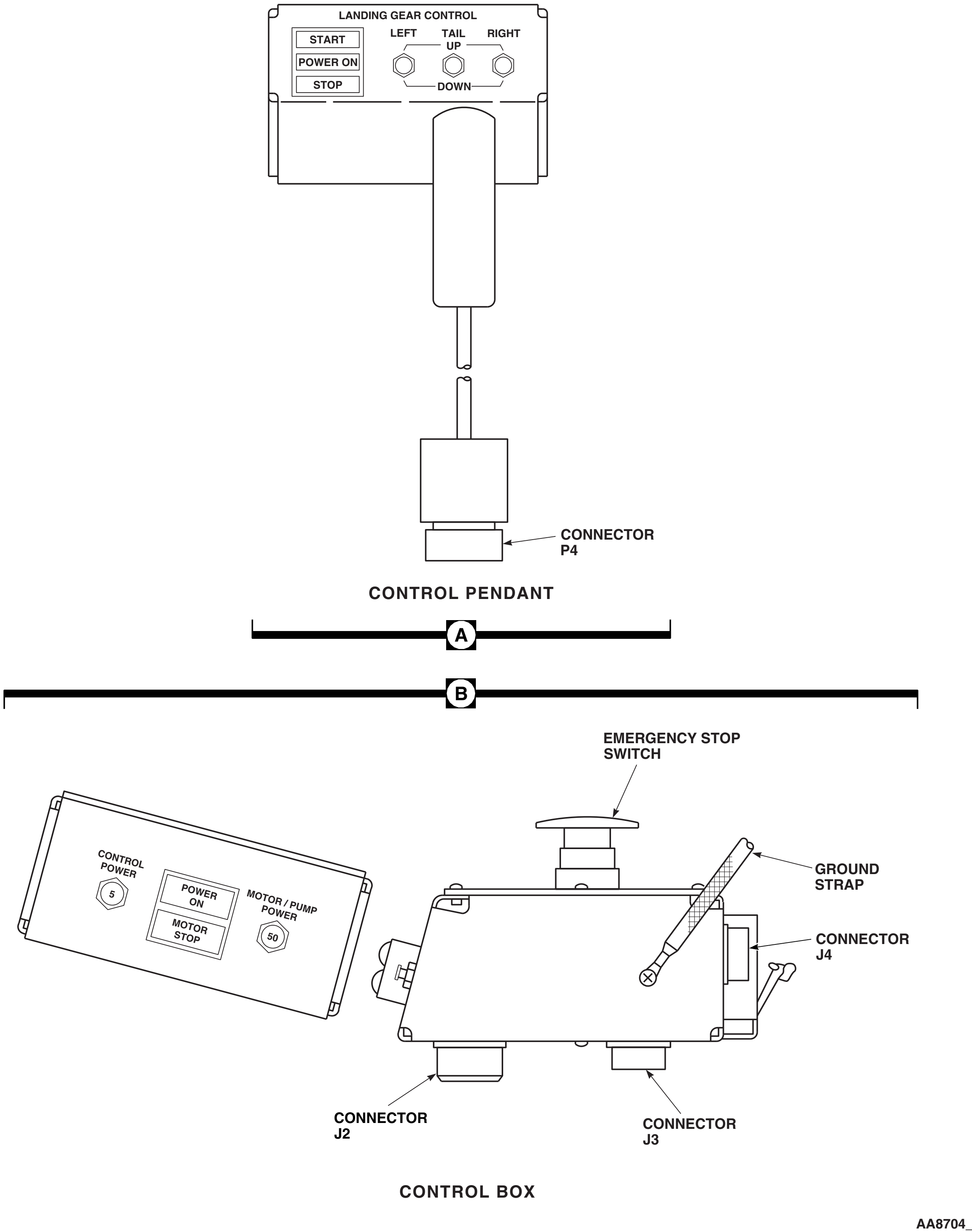
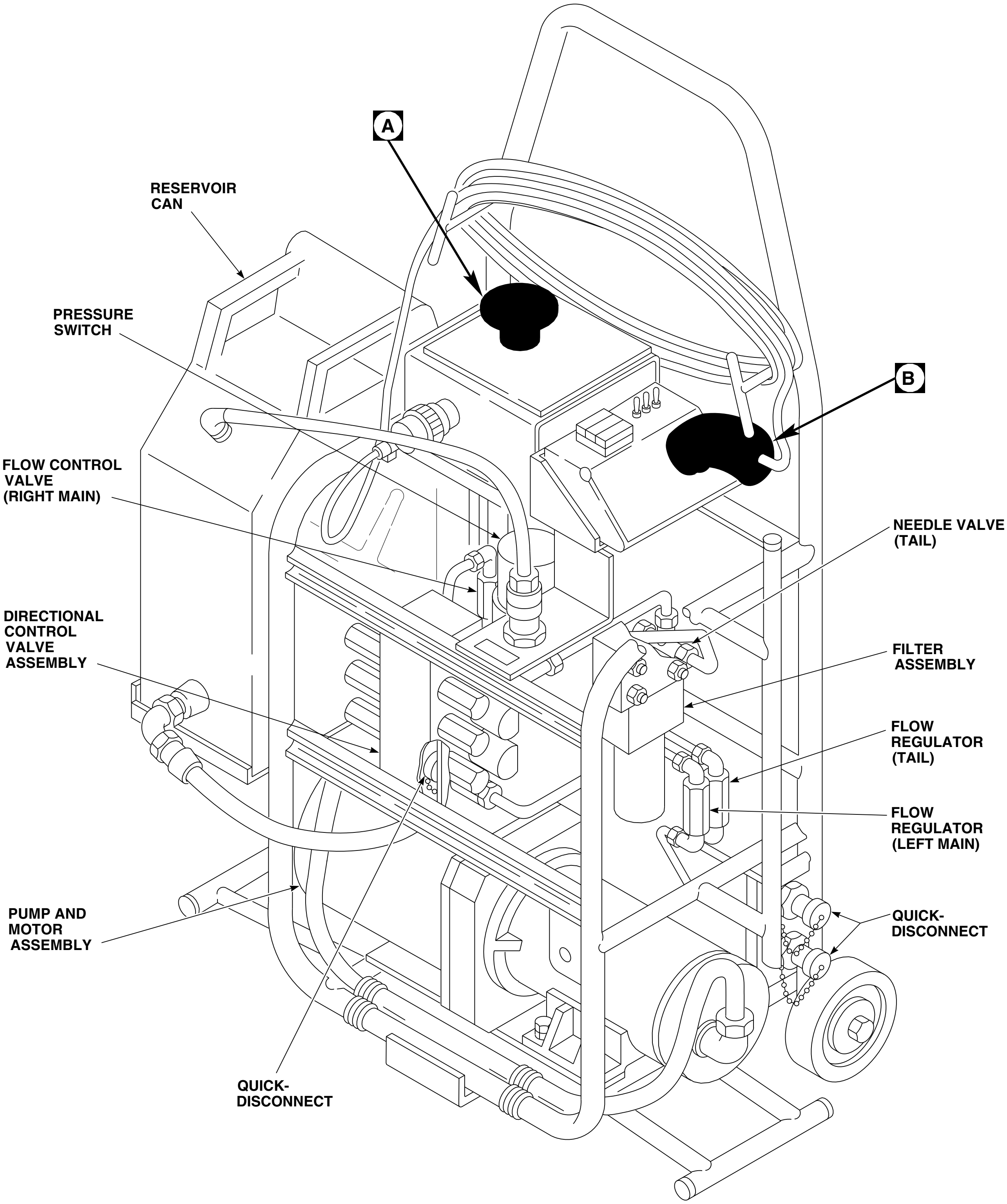


FIGURE 16-2-6-1. AIR TRANSPORT HYDRAULIC CART LOCATION DIAGRAM (SHEET 2 OF 2)

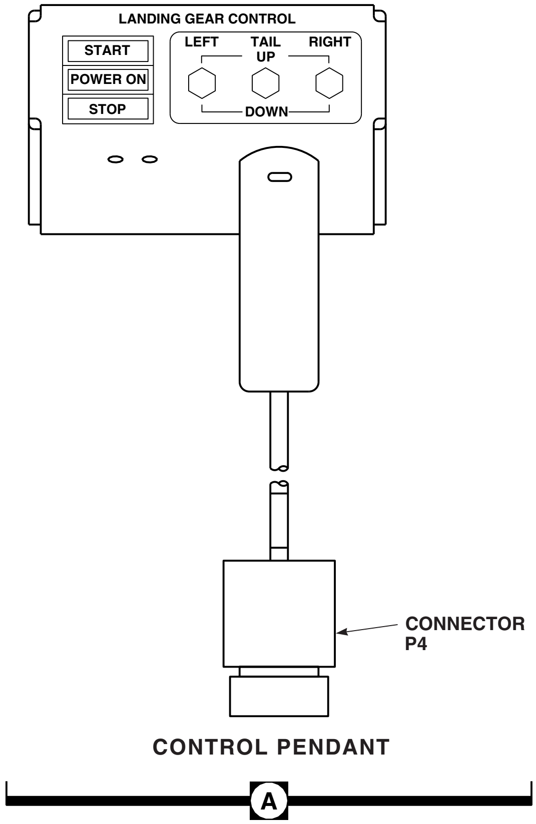


EFFECTIVITY

70700-81650-041.

AA8707_1
SA

FIGURE 16-2-6-2. AIR TRANSPORT HYDRAULIC CART LOCATION DIAGRAM (SHEET 1 OF 2)



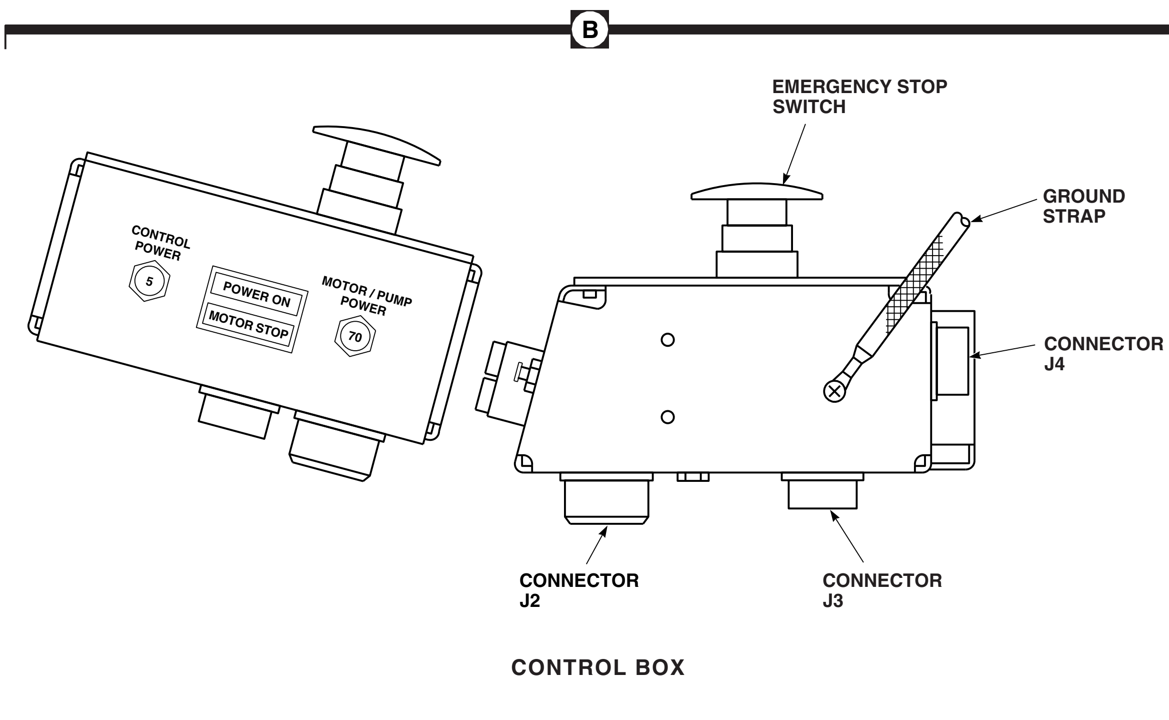
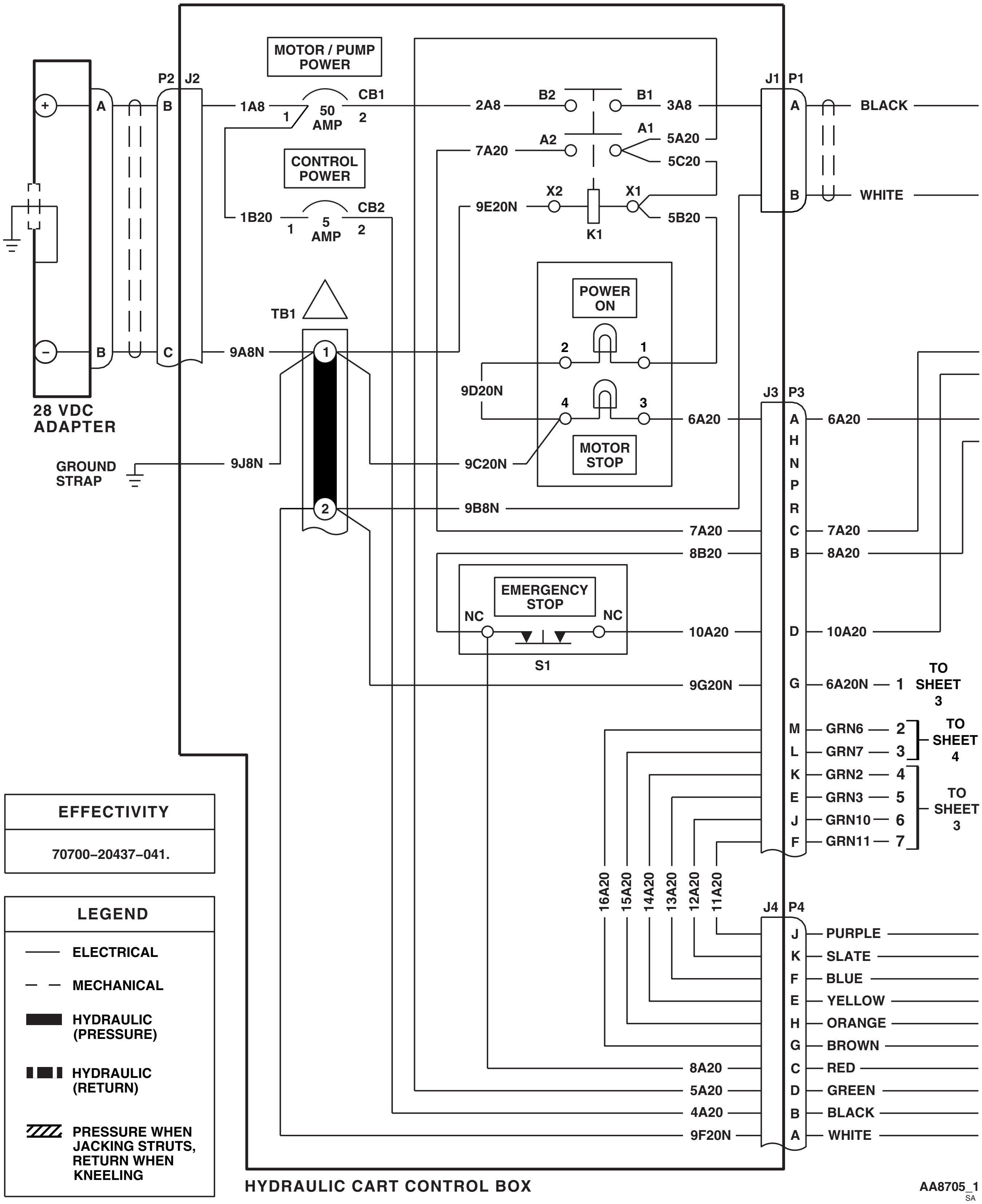


FIGURE 16-2-6-2. AIR TRANSPORT HYDRAULIC CART LOCATION DIAGRAM (SHEET 2 OF 2)



EFFECTIVITY
70700–20437–041.

LEGEND	
——	ELECTRICAL
— —	MECHANICAL
▬	HYDRAULIC (PRESSURE)
▮▬▮	HYDRAULIC (RETURN)
▨	PRESSURE WHEN JACKING STRUTS, RETURN WHEN KNEELING

FIGURE 16-2-6-3. AIR TRANSPORT HYDRAULIC CART SCHEMATIC DIAGRAM (SHEET 1 OF 4)

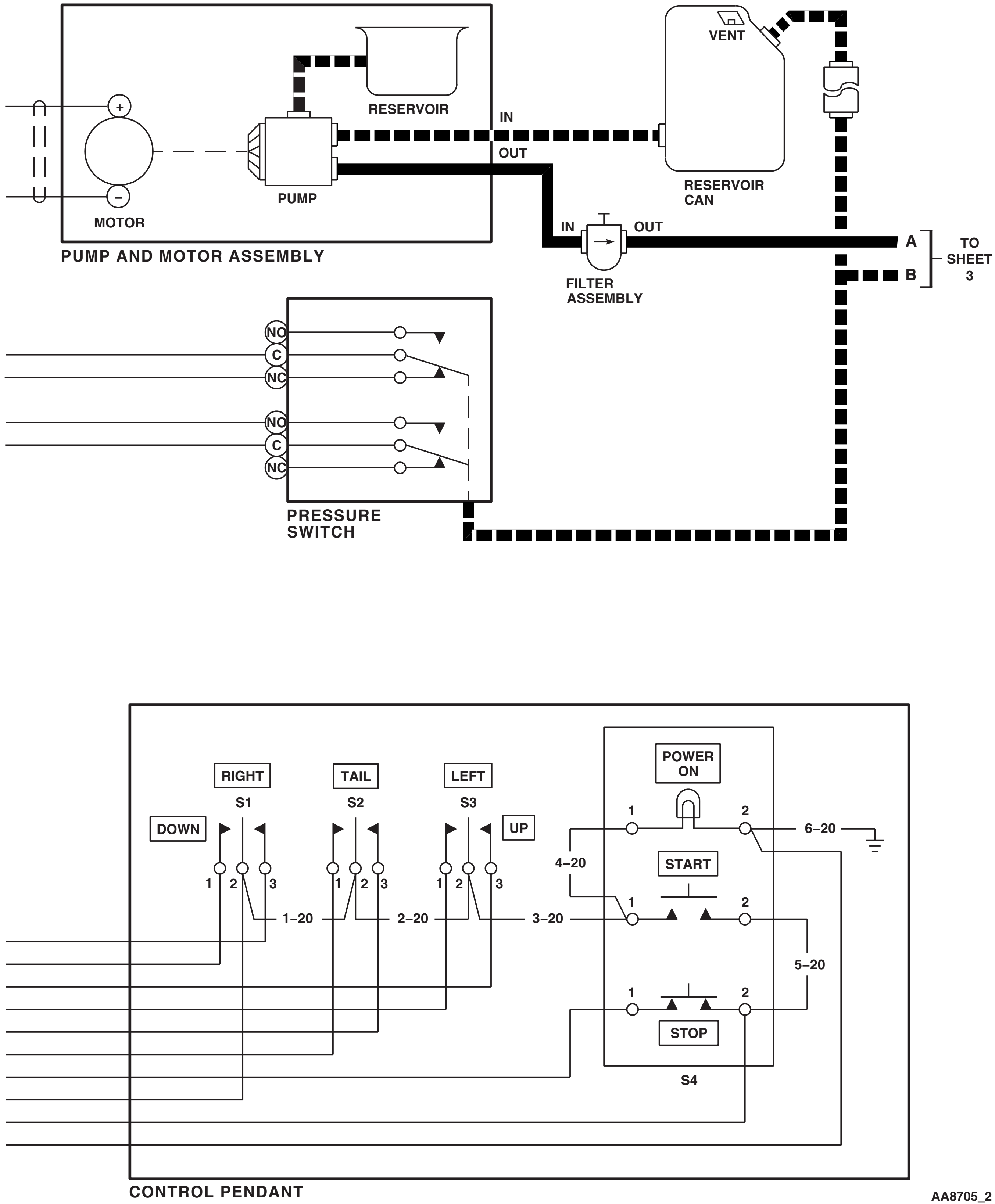


FIGURE 16-2-6-3. AIR TRANSPORT HYDRAULIC CART SCHEMATIC DIAGRAM (SHEET 2 OF 4)

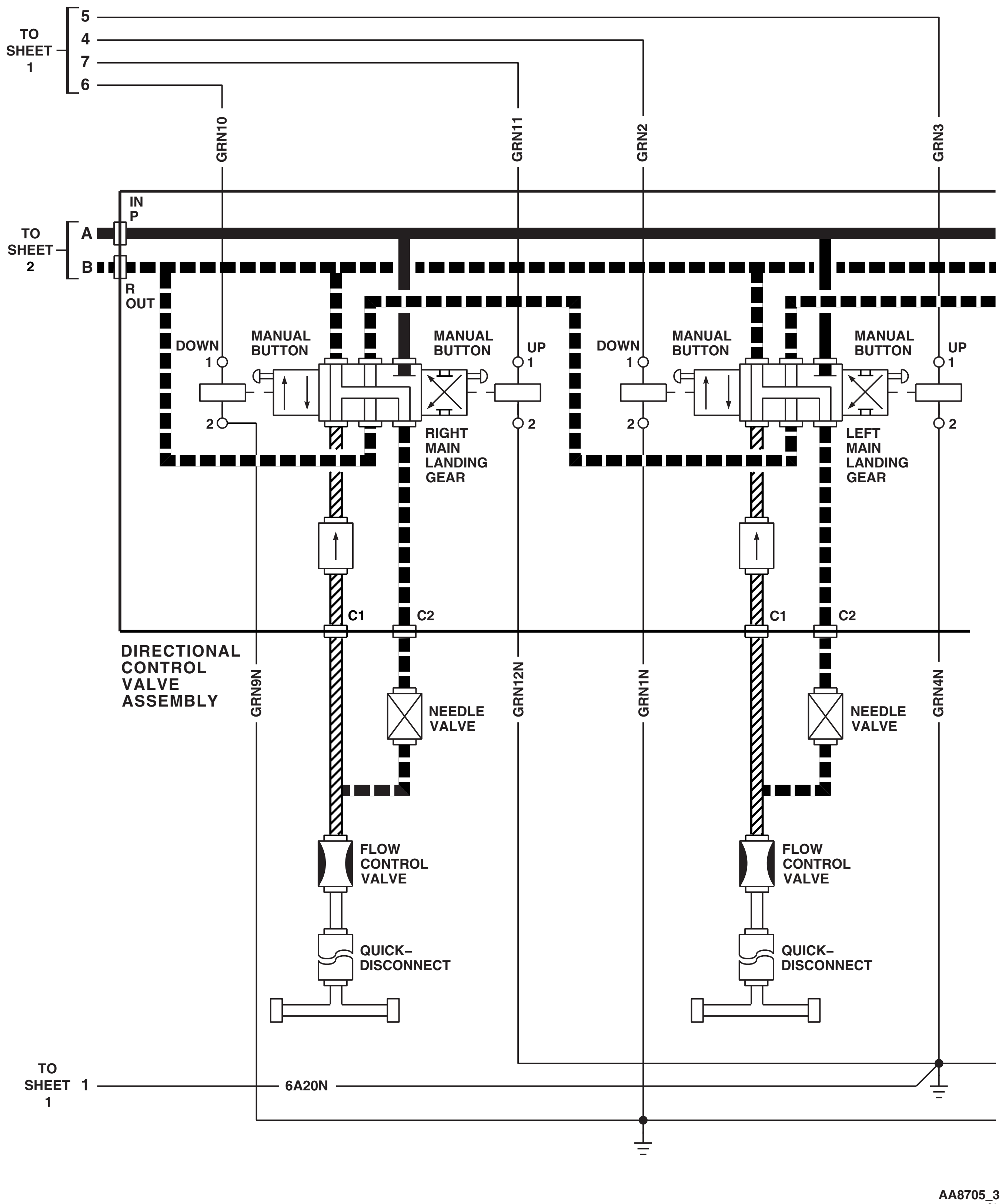


FIGURE 16-2-6-3. AIR TRANSPORT HYDRAULIC CART SCHEMATIC DIAGRAM (SHEET 3 OF 4)

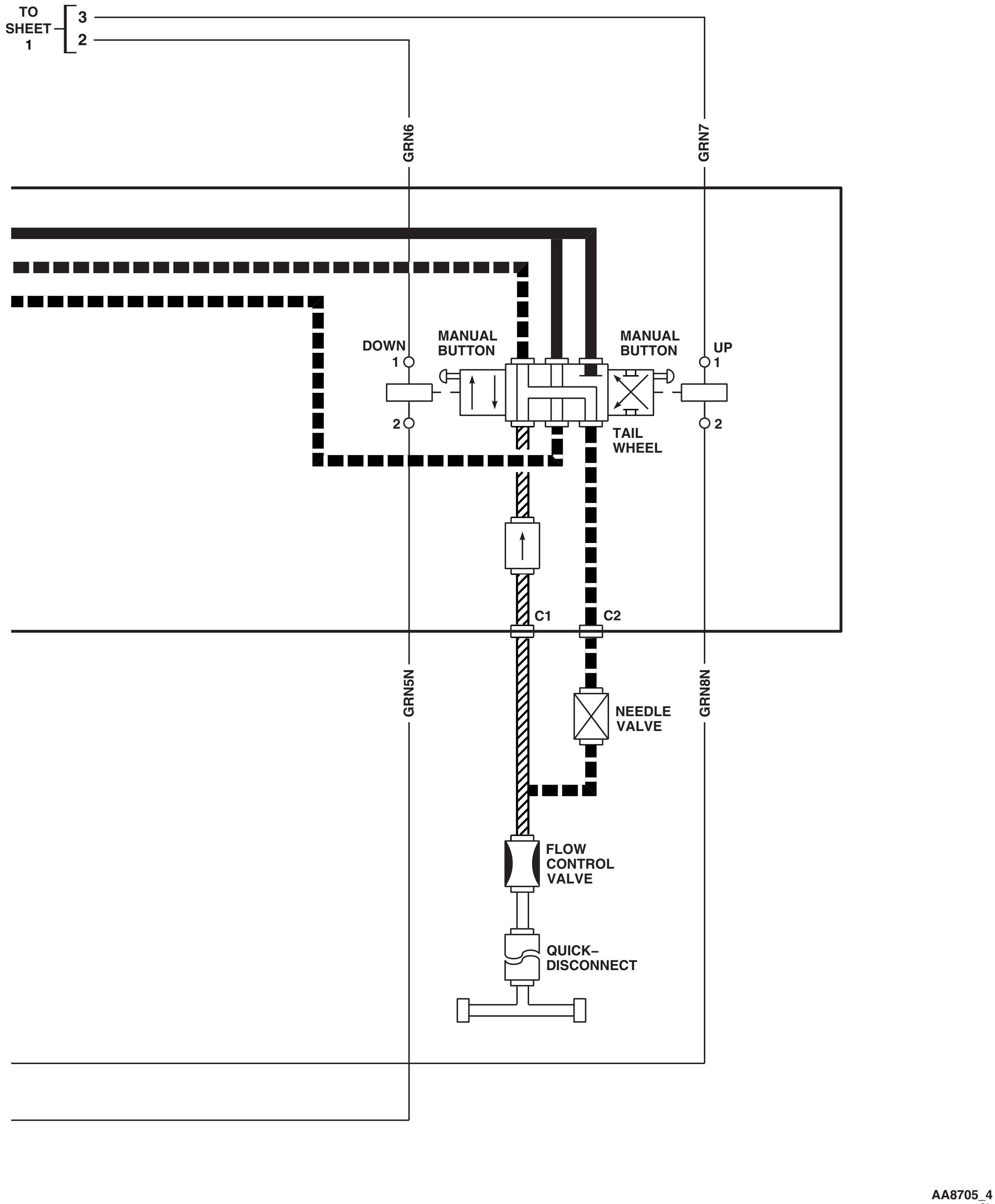


FIGURE 16-2-6-3. AIR TRANSPORT HYDRAULIC CART SCHEMATIC DIAGRAM (SHEET 4 OF 4)

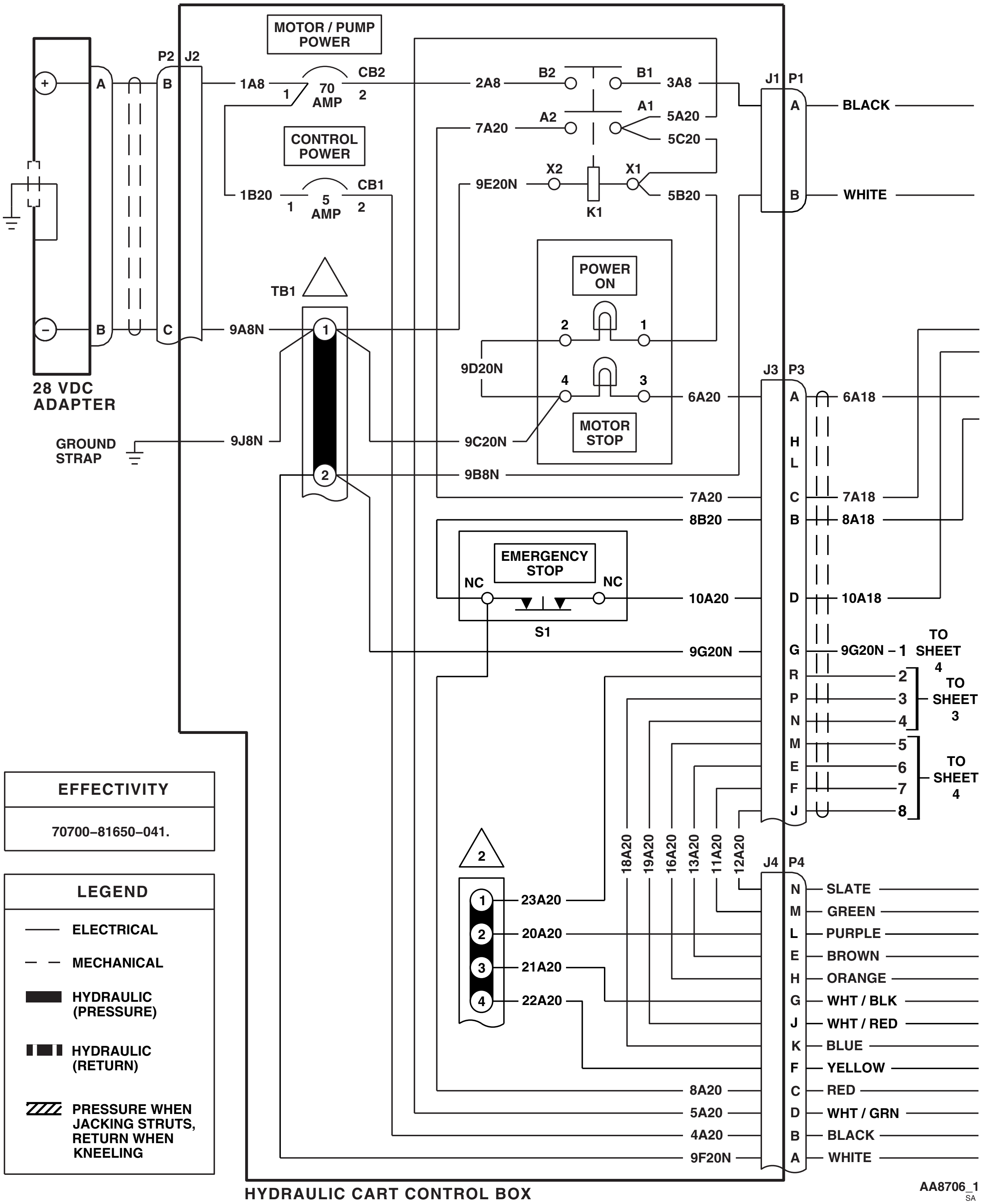


FIGURE 16-2-6-4. AIR TRANSPORT HYDRAULIC CART SCHEMATIC DIAGRAM (SHEET 1 OF 4)

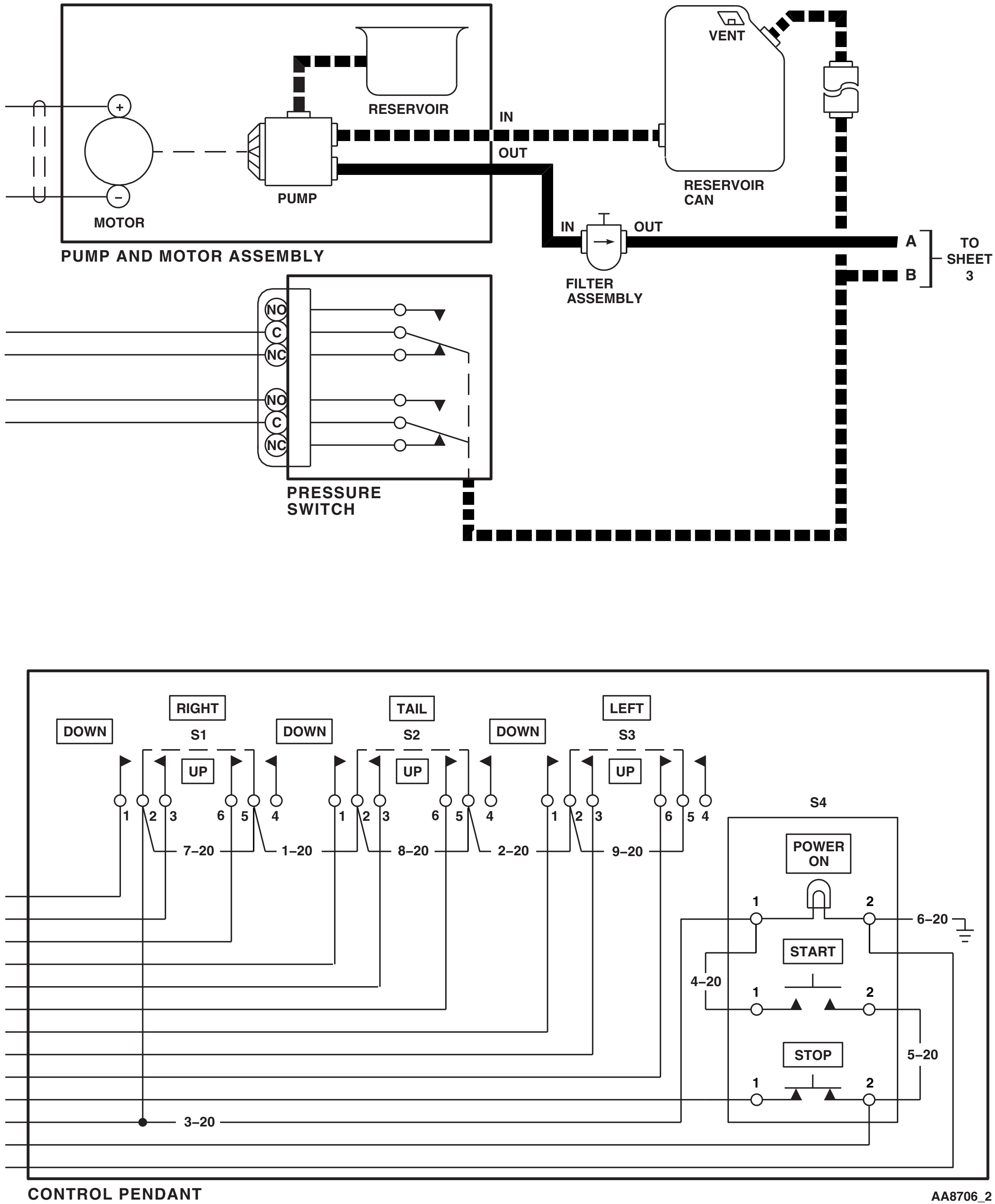


FIGURE 16-2-6-4. AIR TRANSPORT HYDRAULIC CART SCHEMATIC DIAGRAM (SHEET 2 OF 4)

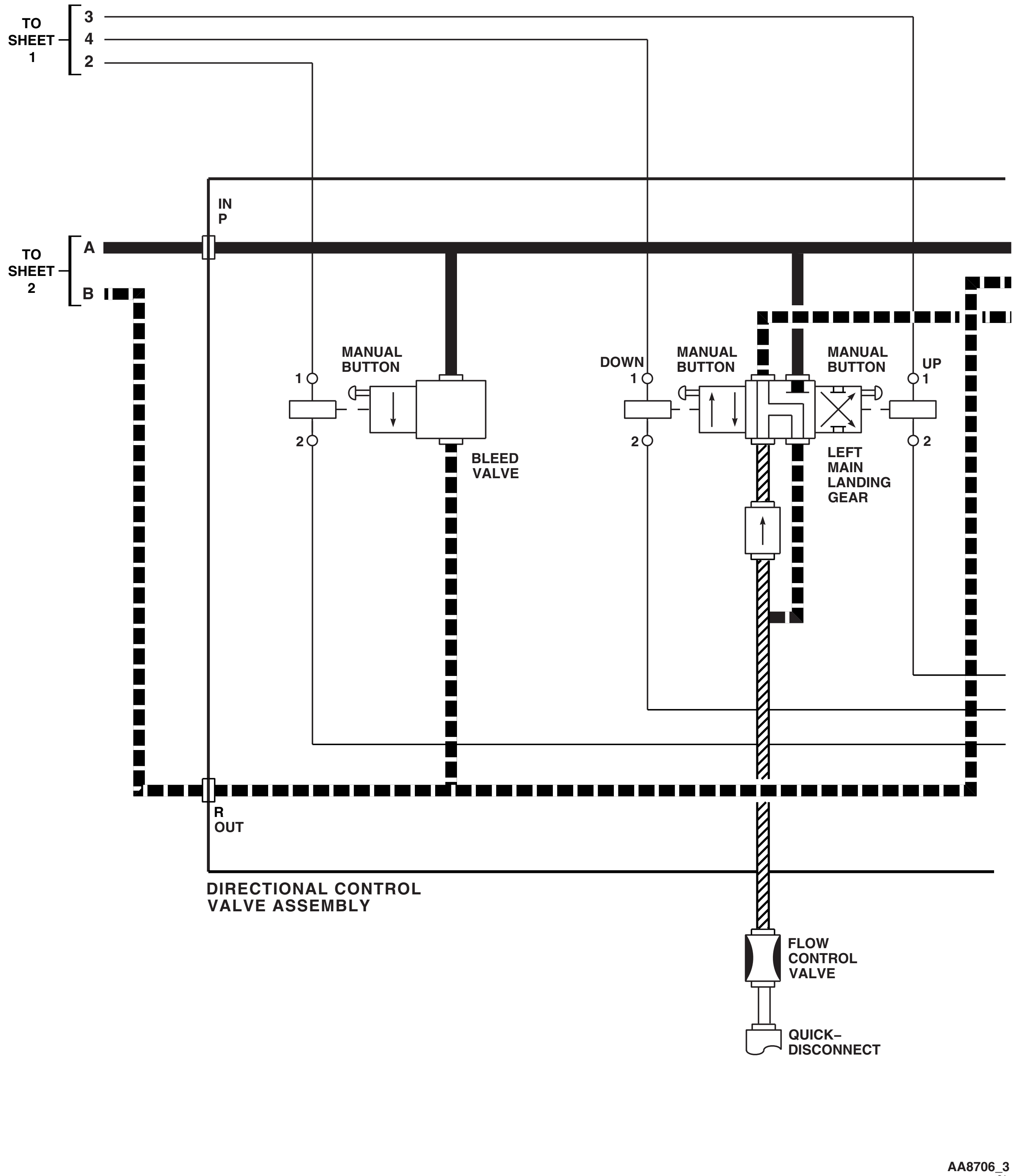


FIGURE 16-2-6-4. AIR TRANSPORT HYDRAULIC CART SCHEMATIC DIAGRAM (SHEET 3 OF 4)

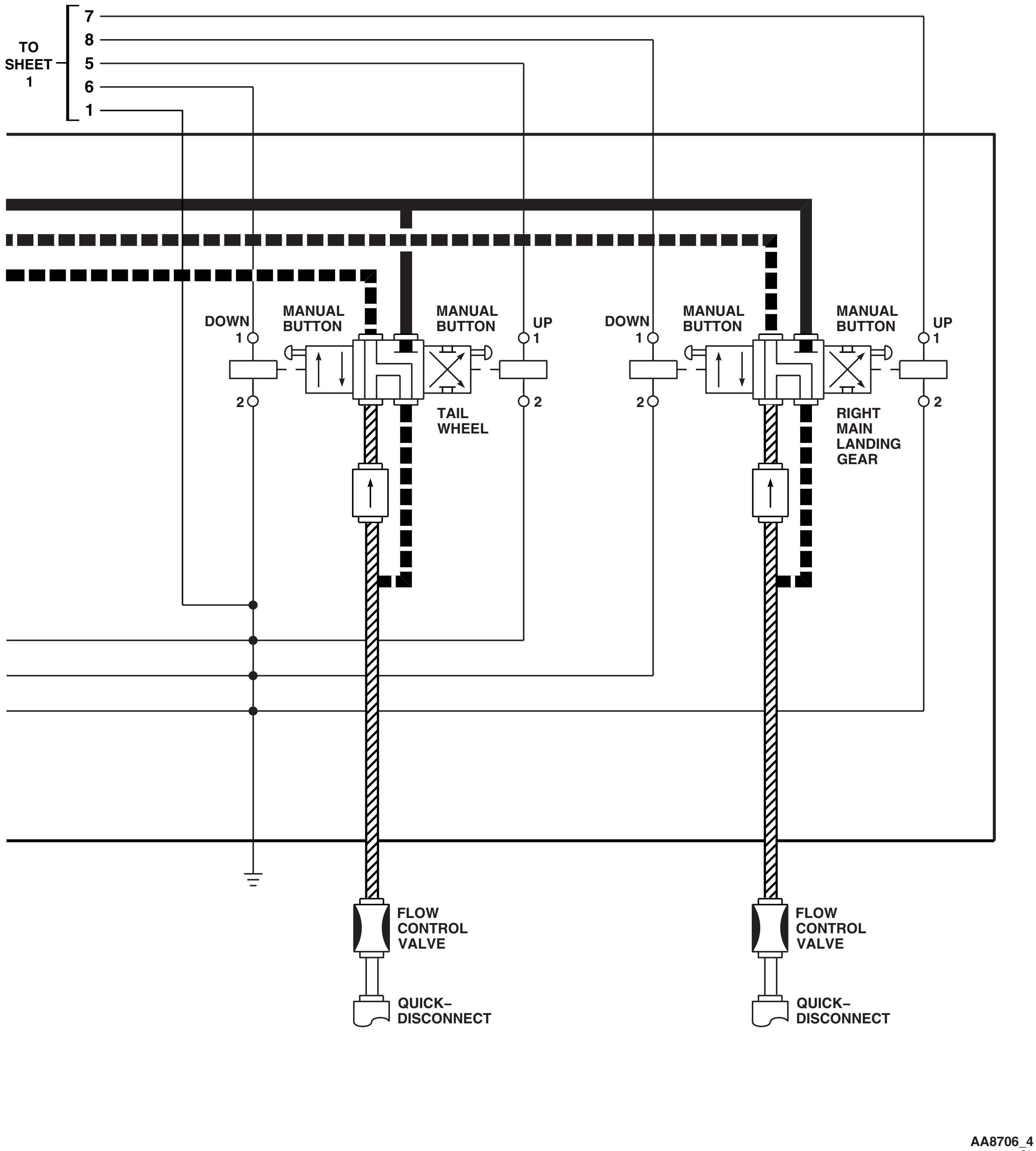


FIGURE 16-2-6-4. AIR TRANSPORT HYDRAULIC CART SCHEMATIC DIAGRAM (SHEET 4 OF 4)

SECTION 16-3

ON AIRCRAFT INSPECTIONS

SECTION OVERVIEW

16-3-1 EXTERNAL STORES SUPPORT SYSTEM (ESSS) INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Spring Scale
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Tie Wire

References

- PARA 16-4-37
- PARA 16-6-7
- PARA 16-6-8

16-3-1.1 Fuel Hose Inspection.

Inspect fuel hose, fuel tee unions, elbows, and connections for signs of leakage or damage. **NONE ALLOWED.** If leaking or damaged, replace component(s) as necessary.

16-3-1.2 Pneumatic Hose Inspection.

Inspect pneumatic hose, check valves, and connections for signs of leakage or damage. **NONE ALLOWED.** If leaking or damaged, replace component(s) as necessary.

16-3-1.3 Expandable Pin and Bolt Inspection.

a. Inspect expandable pin for missing or damaged segments and wedges (between segments and at top and bottom of pin) and for cracked or damaged washers. **NONE ALLOWED.** If any segment, wedges, or washers are damaged or missing, replace expandable pin.

b. Inspect expandable pin closing torque whenever new pin is installed or when component is replaced.

WARNING

FSCAP

Verification of proper torque required to lock pin or bolt is a critical characteristic.

c. To check expandable pins and bolts, perform the following:

(1) Insert unexpanded pin or bolt in its bushing holes.

(2) Using torque wrench, turn hexhead until locking pin snaps into locking tab hole. Note torque needed to lock pin or bolt.

NOTE

If earlier configuration pin without hexhead is installed, torque can be measured using spring scale and tie wire secured to hexhole in handle end. Force required to close with scale pulling at right angle to long section of handle shall be 20 - 40 pounds.

(3) **ON HORIZONTAL STORES SUPPORT (HSS) TO FUSELAGE PINS, TORQUE MUST BE BETWEEN 150 - 300 INCH-POUNDS. MEASURE TORQUE BY TORQUING HEXHEAD MACHINED INTO HANDLE PIVOT.** If pin is not within limits, replace pin.

NOTE

If earlier configuration pin without hexhead is installed, torque can be measured using spring scale and tie wire secured to hexhole in handle end. Force required to close with scale pulling at right angle to long section of handle shall be 36 - 72 pounds.

(4) Inspect support strut pins as follows:

(a) **TORQUE MUST BE BETWEEN 150 - 300 INCH-POUNDS. MEASURE TORQUE BY TORQUING HEXHEAD MACHINED INTO HANDLE PIVOT.**

(b) If torque is not within limits, measure support strut clevis bushings. **BUSHING DIAMETER TO BE 1.000 - 1.002-INCH.**

(c) If support strut clevis bushing diameter is not within limits, replace support strut (PARA 16-6-7 and PARA 16-6-8).

(d) If bushing diameter is within limits, replace pin(s).

(5) Inspect vertical support pylon (VSP) fitting to HSS fitting expandable bolts (BL 80.0 and BL 112.0) as follows:

(a) **TORQUE MUST BE BETWEEN 150 - 300 INCH-POUNDS.**

(b) If torque is not within limits, measure VSP to HSS stores attachment fitting bushings. **BUSHING DIAMETER TO BE 0.7525 - 0.7535-INCH.**

(c) If VSP to HSS stores attachment fitting bushing diameter is not within limits, replace VSP (PARA 16-4-37) or HSS (PARA 16-6-7 and PARA 16-6-8).

(d) If bushing diameter is within limits, replace bolt(s).

(6) Inspect elevation link expandable bolts as follows:

(a) **TORQUE MUST BE BETWEEN 80 - 150 INCH-POUNDS.**

(b) If torque is not within limits, measure VSP to elevation link bushings. **BUSHING DIAMETER TO BE 0.500 - 0.502-INCH.**

(c) If VSP to elevation link bushing diameter is not within limits, replace VSP or elevation link (PARA 16-4-37).

(d) If bushing diameter is within limits, replace bolt(s).

16-3-2 MAIN ROTOR BLADE EROSION PROTECTION KIT INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit

References

- PARA 16-6-10
- PARA 16-6-12
- PARA 16-6-14

16-3-2.1 Main Rotor Blade Polyurethane Tape Inspection.

a. Inspect 2-inch tape for holes. **NONE ALLOWED.** If holes are found, replace damaged polyurethane tape (PARA 16-6-10).

b. Inspect 8-inch tape for wear. **WEAR SHALL NOT EXCEED 1-INCH ON UPPER OR LOWER SIDE OF BLADE.** If limit is exceeded, replace damaged polyurethane tape (PARA 16-6-10).

c. Inspect upper and lower trailing edge of tape for disbonding. **DISBONDING SHALL NOT EXCEED 0.25-INCH WIDTH FOR ANY SPANWISE LENGTH, OR 0.5-INCH WIDTH FOR 6-INCH SPANWISE LENGTH, OR 1-INCH WIDTH FOR 2-INCH SPANWISE LENGTH.** If limits are exceeded, replace damaged polyurethane tape (PARA 16-6-10).

d. Inspect outboard end of tape segment for disbonding. **DISBONDING SHALL NOT EXCEED 1-INCH SPANWISE LENGTH.** If limit is exceeded, replace damaged polyurethane tape (PARA 16-6-10).

e. Inspect inboard end of tape segment for disbonding. **DISBONDING SHALL NOT EXCEED 0.5-INCH SPANWISE LENGTH.** If limit is exceeded, replace damaged polyurethane tape (PARA 16-6-10).

f. Inspect for internal disbonding of tape segment. **DISBONDING SHALL NOT EXCEED 6.0 SQUARE-INCHES.** There is no limit to the number of disbonds, providing material is not torn. If torn, replace polyurethane tape (PARA 16-6-10).

16-3-2.2 Main Rotor Blade Tip Cap Polyurethane Coating Inspection.

Inspect main rotor blade tip cap polyurethane coating. **WEAR THROUGH TO BLADE SURFACE NOT ALLOWED.** If polyurethane coating is worn, repair polyurethane coating (PARA 16-6-12).

16-3-2.3 Main Rotor Blade Tip Cap Polyurethane Boot Inspection.

a. Inspect main rotor blade tip cap polyurethane boot. **WEAR THROUGH TO BLADE SURFACE NOT ALLOWED.** If limit is exceeded, replace boot (PARA 16-6-14).

b. Inspect upper and lower trailing edge of boot for disbonding. **DISBONDING SHALL NOT EXCEED 0.25-INCH WIDTH FOR ANY SPANWISE LENGTH, OR 0.5-INCH WIDTH FOR 6-INCHES SPANWISE LENGTH, OR 1-INCH WIDTH FOR 2-INCHES SPANWISE LENGTH.** If limits are exceeded, replace boot (PARA 16-6-14).

c. Inspect inboard end of boot segment for disbonding. **DISBONDING SHALL NOT EXCEED 0.5-INCH SPANWISE LENGTH.** If limit is exceeded, replace boot (PARA 16-6-14).

d. Inspect for internal disbonding of boot segment. **DISBONDING SHALL NOT EXCEED 6 SQUARE-INCHES.** There is no limit to the number of disbonds, providing material is not torn. If torn, replace boot (PARA 16-6-14).

16-3-3 TAIL ROTOR BLADE EROSION PROTECTION KIT INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit

References

- PARA 16-6-16
- PARA 16-6-18

16-3-3.1 Tail Rotor Blade Polyurethane Tape Inspection.

a. Inspect 4 inch polyurethane tape for wear. **WEAR SHALL NOT EXCEED 0.5-INCH WIDTH ON TRAILING EDGE. NO HOLES ALLOWED ON ANY OTHER SURFACE OF TAPE.** If wear is found, replace damaged polyurethane tape (PARA 16-6-16).

b. Inspect outboard end of tape segment for disbonding. **DISBONDING SHALL NOT EXCEED 1 INCH.** If disbonding exceeds limits, replace damaged polyurethane tape (PARA 16-6-16).

c. Inspect upper and lower trailing edge of tape for disbonding. **DISBONDING SHALL NOT EXCEED 0.25-INCH WIDTH FOR ANY SPANWISE LENGTH OR 0.5-INCH WIDTH FOR 6 INCHES SPANWISE LENGTH OR 1 INCH WIDTH FOR 2 INCHES SPANWISE LENGTH.** If disbonding exceeds limits, replace damaged polyurethane tape (PARA 16-6-16).

d. Inspect inboard end of tape segment for disbonding. **DISBONDING SHALL NOT EXCEED 0.5-INCH SPANWISE LENGTH.** If disbonding exceeds limits, replace damaged polyurethane tape (PARA 16-6-16).

e. Inspect for internal disbonding of tape segment. **DISBONDING SHALL NOT EXCEED 6.0 SQUARE INCHES.** There is no limit to the number of disbonds providing material is not torn. If torn, replace polyurethane tape (PARA 16-6-16).

16-3-3.2 Tail Rotor Blade Tip Cap Polyurethane Coating Inspection.

Inspect tail rotor blade tip cap polyurethane coating. **WEAR THROUGH TO BLADE SURFACE NOT ALLOWED.** If worn, repair polyurethane coating (PARA 16-6-18).

SECTION 16-4

MAINTENANCE PROCEDURES

SECTION OVERVIEW

16-4-1 M-60D MACHINE GUN MOUNT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Lubricant, Solid Film, Item 201, Appendix D

References

- Appendix D
- Appropriate Flight Manual

16-4-1.1 Replace M-60D Machine Gun Mount.

NOTE

Replacement of left and right M-60D machine gun mounts is the same, except as noted.

16-4-1.1.1 Remove.

a. Remove machine gun (Appropriate Flight Manual).

b. Pull slide handles and fold inboard and outboard fittings (Figure 16-4-1-1, Detail A and Detail B).

CAUTION

Damage to external electrical power feeder cable may result if care is not used when removing right side cotter pin. For right side gun mount, use care when removing cotter pin as to not induce damage to external electrical power feeder cable.

c. Remove cotter pin from nut and remove nut securing gun mount to fuselage.

d. Lift and remove gun mount from fuselage support fittings.

16-4-1.1.2 Install.

a. Position gun mount in fuselage support fittings.

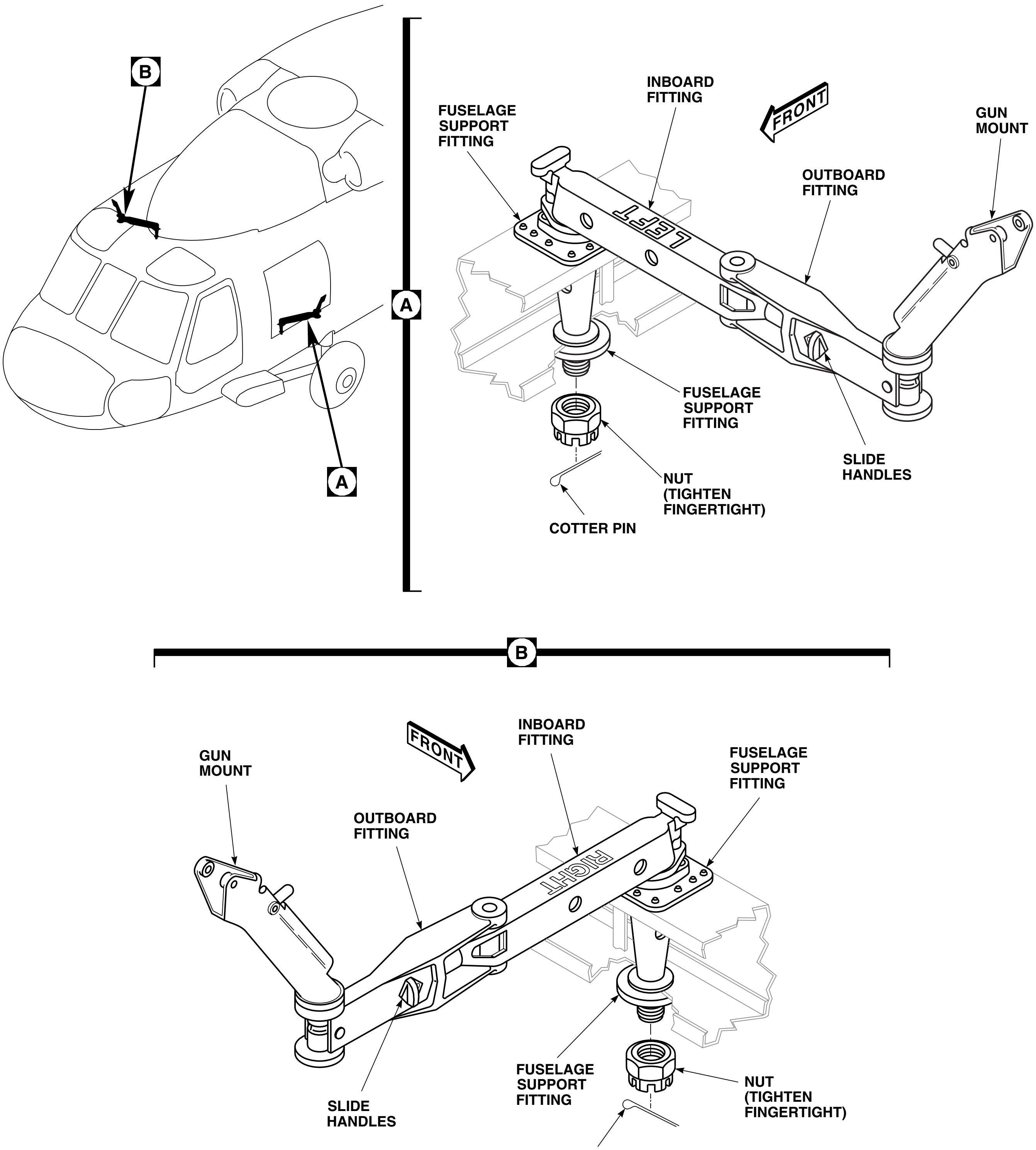
b. Lubricate face of nut with solid film lubricant, Item 201, Appendix D.

c. Install nut on gun mount (Figure 16-4-1-1, Detail A and Detail B). **TIGHTEN NUT FINGERTIGHT.**

d. For left side gun mount, back off nut enough to install cotter pin, and install cotter pin (Figure 16-4-1-1, Detail A). For right side gun mount, perform the following (Figure 16-4-1-1, Detail B):

(1) Back off nut enough to install cotter pin.

(2) Install cotter pin so that head of cotter pin faces rear of helicopter (right side of gun mount if viewed from inside helicopter, looking out window).



NOTE

FOR RIGHT GUN MOUNT, MAKE SURE COTTER PIN IS INSTALLED AS SHOWN TO PREVENT POSSIBLE CHAFING AGAINST EXTERNAL ELECTRICAL POWER FEEDER CABLE.

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FIGURE 16-4-1-1. REPLACE M-60D MACHINE GUN MOUNT

(3) Unfold gun mount and lock inboard to outboard fitting (deployed-lock position) outside of window.

(4) Carefully rotate gun mount through full range of motion (172°) while checking for cotter pin clearance. If clearance between cotter pin and external electrical power feeder cable is adequate, cotter pin installation is complete. If interference exists, perform the following:

(a) Trim and/or bend cotter pin prongs as necessary.

(b) Repeat step d. (4).

(c) Repeat step d. (4) (a) and step d. (4) (b) until adequate clearance is achieved.

e. For left side gun mount, unfold gun mount and lock inboard to outboard fitting.

f. Make sure area is clean and free of foreign material.

g. Install machine gun (Appropriate Flight Manual).

16-4-2 M-60D MACHINE GUN MOUNT DETENT SPRING.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 16-4-1

16-4-2.1 Replace M-60D Machine Gun Mount Detent Spring.

16-4-2.1.1. Remove.

a. Remove M-60D machine gun mount (PARA 16-4-1).

b. Remove spring pin (Figure 16-4-2-1).

c. Remove detent handle, spring, and detent pin.

16-4-2.1.2. Install.

a. Install spring on detent pin (Figure 16-4-2-1).

b. Install detent pin in inboard fitting.

c. Place detent handle over detent pin.

d. Install spring pin through detent handle and detent pin.

e. Make sure area is clean and free of foreign material.

f. Install M-60D machine gun mount (PARA 16-4-1).

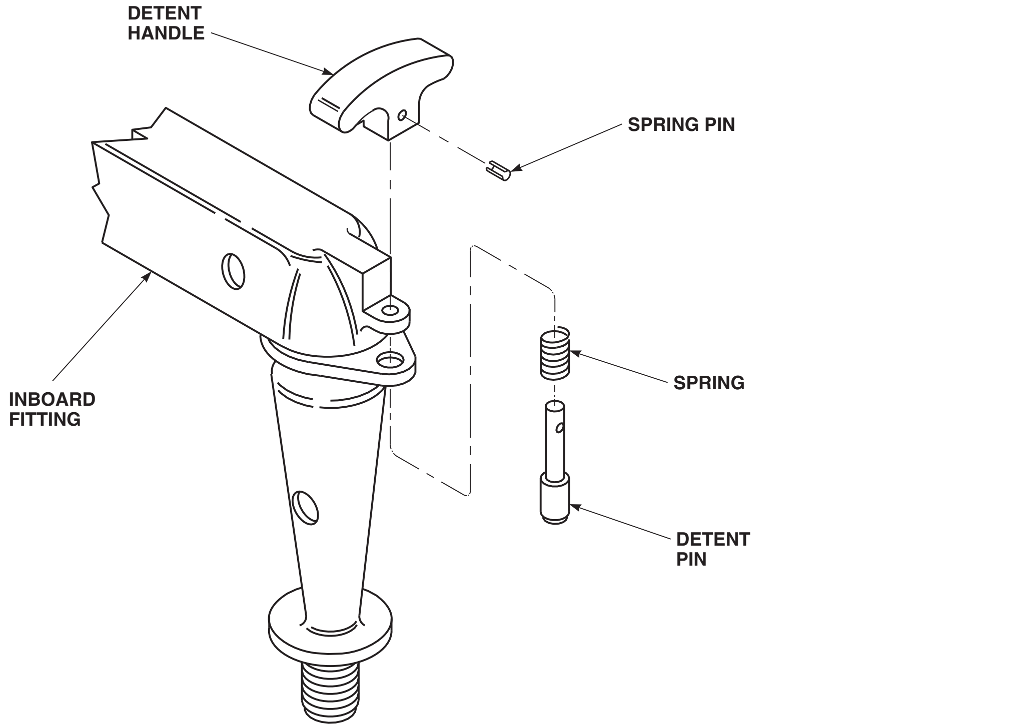


FIGURE 16-4-2-1. REPLACE M-60D MACHINE GUN MOUNT DETENT SPRING

16-4-3 M-60D MACHINE GUN MOUNT PINTLE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Micrometer

Materials/Parts

- Sealing Compound, Item 280, Appendix D

References

- Appendix D

16-4-3.1 Replace M-60D Machine Gun Mount Pintle.

16-4-3.1.1. Remove.

a. Remove clevis pin attaching pintle to outboard fitting (Figure 16-4-3-1).

b. Remove pintle from outboard fitting.

16-4-3.1.2. Inspect.

a. Loosen screw in pintle (Figure 16-4-3-1).

b. Move handle to position in slot nearest outboard fitting.

c. Unscrew handle from pintle.

d. Move downlock toward outboard fitting to expose retainer and spring pin.

e. Remove spring pin, retainer, and spring from pintle.

f. Inspect gun mounting holes in pintle using micrometer. **MAXIMUM INSIDE DIAMETER IS 0.3180 INCH. MAXIMUM OUT-OF-ROUND IS 0.010 INCH.** Replace pintle if limits are exceeded.

g. Slide downlock toward free end of pintle and remove downlock.

h. Inspect pintle for cracks or damage. **NONE ALLOWED.** Replace pintle if cracked or damaged.

i. Install downlock into pintle and seat fully to expose small shaft.

j. Insert spring and retainer over downlock small shaft.

k. Install spring pin through retainer and downlock shaft.

l. Position downlock handle hole in pintle slot, by compressing spring and turning retainer in pintle bore.

m. Coat threads on handle with sealing compound, Item 280, Appendix D, and install handle in pintle.

n. Move handle up slot and into locked position.

o. **TIGHTEN SCREW TO SECURE DOWNLOCK IN LOCKED POSITION.**

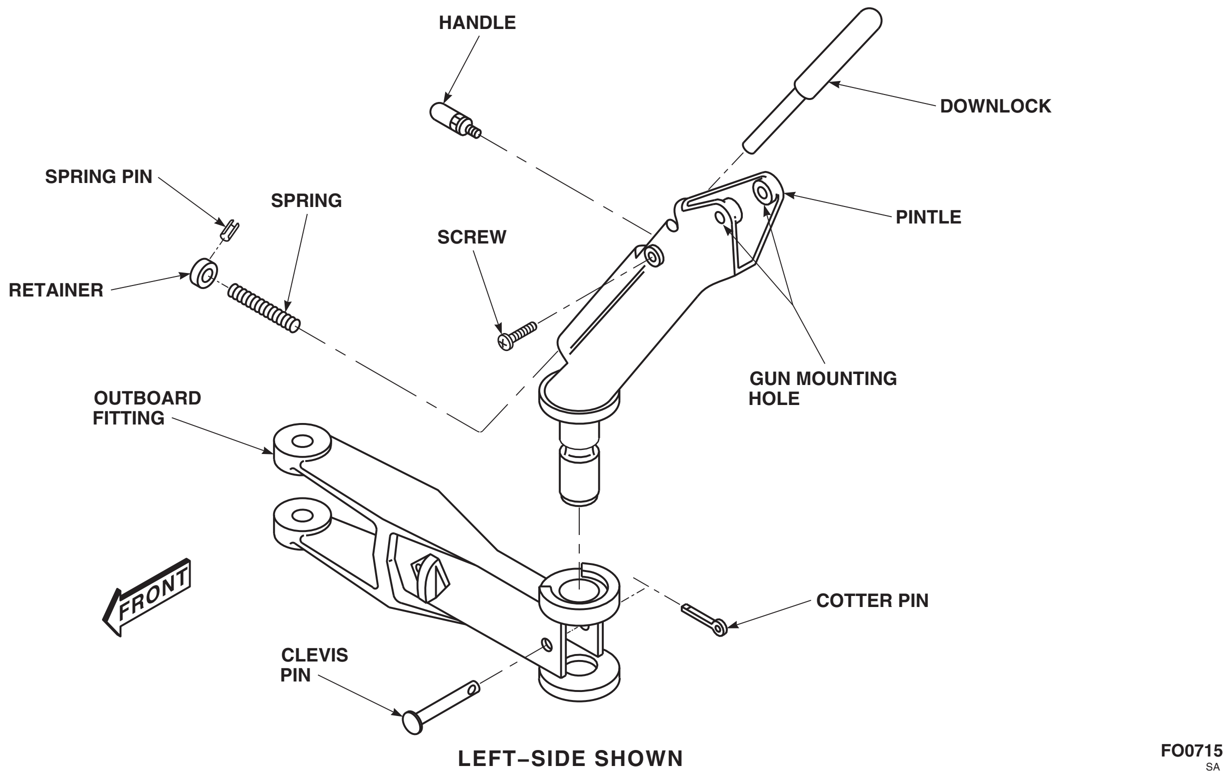


FIGURE 16-4-3-1. REPLACE M-60D MACHINE GUN MOUNT PINTLE

16-4-3.1.3. Install.

a. Place pintle in outboard fitting (Figure 16-4-3-1).

b. Install clevis pin and cotter pin.

c. Make sure area is clean and free of foreign material.

16-4-4 M-60D MACHINE GUN MOUNT OUTBOARD FITTING.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 16-4-3

16-4-4.1 Replace M-60D Machine Gun Mount Outboard Fitting.

16-4-4.1.1. Remove.

a. Remove M-60D machine gun mount pintle (PARA 16-4-3).

b. Remove outboard fitting from inboard fitting, by removing pin and spacer (Figure 16-4-4-1).

16-4-4.1.2. Disassemble.

CAUTION

Injury to personnel and damage to equipment will result if piston is not compressed and released carefully. Piston in outboard fitting is under spring pressure, carefully compress and release piston in outboard fitting when removing slide handles.

a. Compress piston into outboard fitting. Remove screws and slide handles (Figure 16-4-4-1).

b. Remove piston, spring, and disc from outboard fitting.

16-4-4.1.3. Assemble.

a. Install disc and spring in outboard fitting. Install piston with slots aligned with slots in outboard fitting (Figure 16-4-4-1).

CAUTION

Injury to personnel and damage to equipment will result if piston is not compressed and released carefully. Piston in outboard fitting is under spring pressure, carefully compress and release piston in outboard fitting when installing slide handles.

b. Compress piston to line up slots in piston with slots in outboard fitting.

c. Attach slide handles with screws.

16-4-4.1.4. Install.

a. Install outboard fitting on inboard fitting (Figure 16-4-4-1).

b. Install spacer, pin, and cotter pin.

c. Install M-60D machine gun mount pintle (PARA 16-4-3).

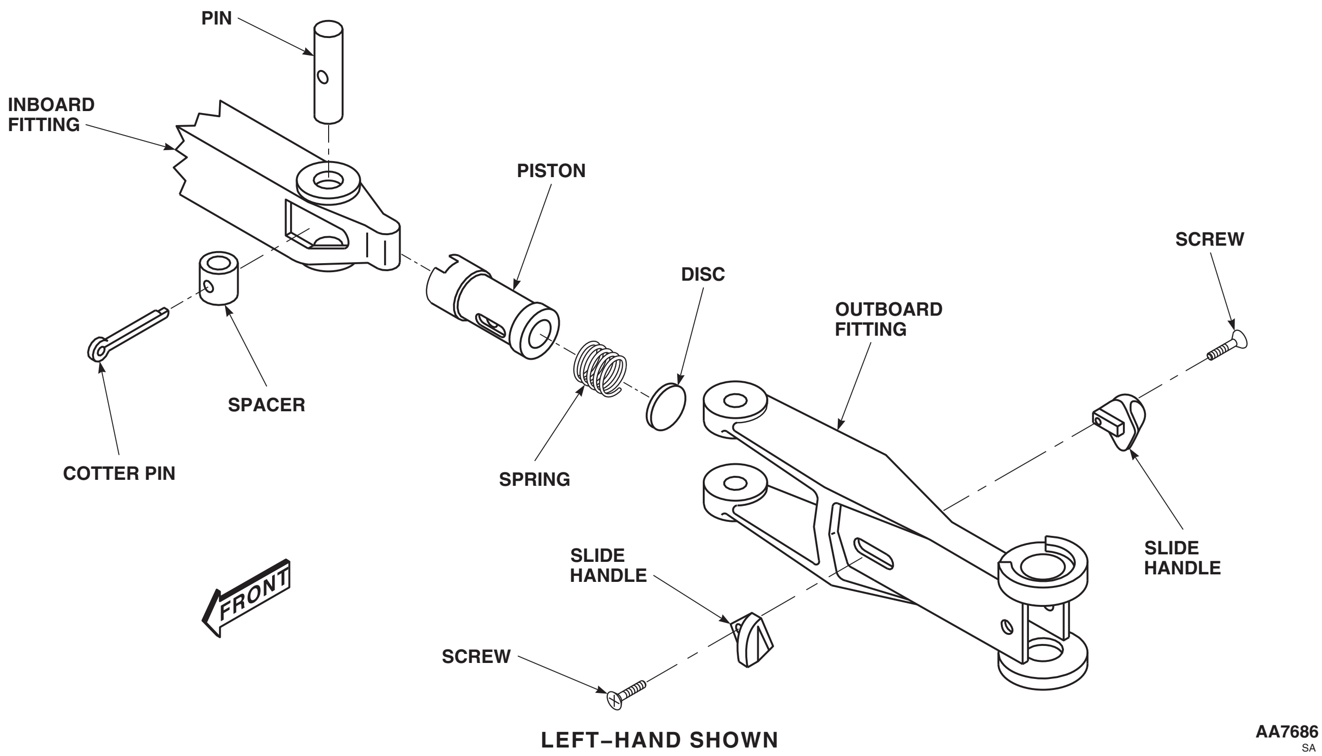


FIGURE 16-4-4-1. REPLACE M-60D MACHINE GUN MOUNT OUTBOARD FITTING

d. Make sure area is clean and free of foreign material.

16-4-5 CREW CHIEF'S COMMUNICATION CORD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20

16-4-5.1 Replace Crew Chief's Communication Cord.

16-4-5.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Unplug communication cord connectors from headset and from electrical connector, J132R (right side), or J133R (left side), at STA 265.0 and WL 258.0.

c. Install protective caps and plugs on electrical connectors.

16-4-5.1.2. Install.

a. Remove protective caps and plugs from electrical connectors.

b. Inspect and treat electrical connectors (PARA 1-7-20).

c. Plug communication cord connectors into headset and electrical connector, J132R (right side), or J133R (left side), at STA 265.0 and WL 258.0.

16-4-6 BLACKOUT CURTAINS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Needle
- Putty Knife

Materials/Parts

- Adhesive Primer, Item 67, Appendix D
- Brush, Acid, Item 84, Appendix D
- Cloth, Nylon, Item 104, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Fastener Tape, Hook, Item 137A, Appendix D
- Fastener Tape, Pile, Item 137B, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Thread, Item 325, Appendix D

References

- Appendix D
- PARA 16-6-4

16-4-6.1 Repair Blackout Curtain and Hook Fastener Tape.

a. If required, remove cockpit/cabin blackout kit (PARA 16-6-4).

b. Repair curtain by patching with nylon cloth, Item 104, Appendix D, or sewing, using needle and thread, Item 325, Appendix D.

c. Repair damaged or worn area of cabin crew chief's window or door window frame pile fastener tape as follows:

(1) Cut and strip pile fastener tape from area. Clean area using acid brush, Item 84, Appendix D, with dry-cleaning solvent, Item 124, Appendix D.

(2) Measure and cut new pile fastener tape, Item 137B, Appendix D, to size.

CAUTION

Crazing may occur if methyl ethyl ketone comes in contact with window. Do not allow methyl ethyl ketone to come in contact with window.

(3) Remove all paint, oil, and old adhesive from area with methyl ethyl ketone, Item 217, Appendix D, using machinery towel, Item 211, Appendix D.

(4) Within 15 minutes after cleaning, apply coat of adhesive primer, Item 67, Ap-

pendix D, to pile fastener tape and area of bulkhead/window frame with putty knife.

(5) Allow adhesive to become tacky, then press tape pile to bulkhead/window frame. Allow adhesive to cure for 16 hours before installing curtain.

d. Repair curtain hook fastener tape by sewing to cover using needle and thread, Item 325, Appendix D, or replace hook fastener tape, Item 137A, Appendix D, using adhesive, Item 67, Appendix D.

e. If required, install cockpit/cabin blackout kit (PARA 16-6-4).

16-4-7 AIR TRANSPORT HYDRAULIC CART CONTROL PENDANT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-7-20

16-4-7.1 Replace/Repair Air Transport Hydraulic Cart Control Pendant.

16-4-7.1.1 Remove.

a. If required, disconnect external electrical power from cart.

b. Disconnect electrical connector, P4, from electrical connector, J4, on control box (Figure 16-4-7-1).

c. Install protective caps and plugs on electrical connectors.

d. Unwrap cable from tees on cart handle.

e. Release fastener on side of support box.

f. Remove control pendant.

16-4-7.1.2 Install.

a. Place control pendant in support box and secure with fastener (Figure 16-4-7-1).

b. Wrap cable around tees on cart handle.

c. Remove protective caps and plugs from electrical connectors.

d. Inspect and treat electrical connectors (PARA 1-7-20).

e. Connect electrical connector, P4, to electrical connector, J4, on control box.

16-4-7.1.3 Repair.

a. Remove control pendant from cart (PARA 16-4-7.1.1).

b. Remove screws and washers securing cover to rear of control pendant. Remove cover (Figure 16-4-7-1, Detail A).

c. Repair or replace damaged or worn wiring or component(s).

d. Install cover on control pendant and secure with screws and washers.

e. Install control pendant on cart (PARA 16-4-7.1.2).

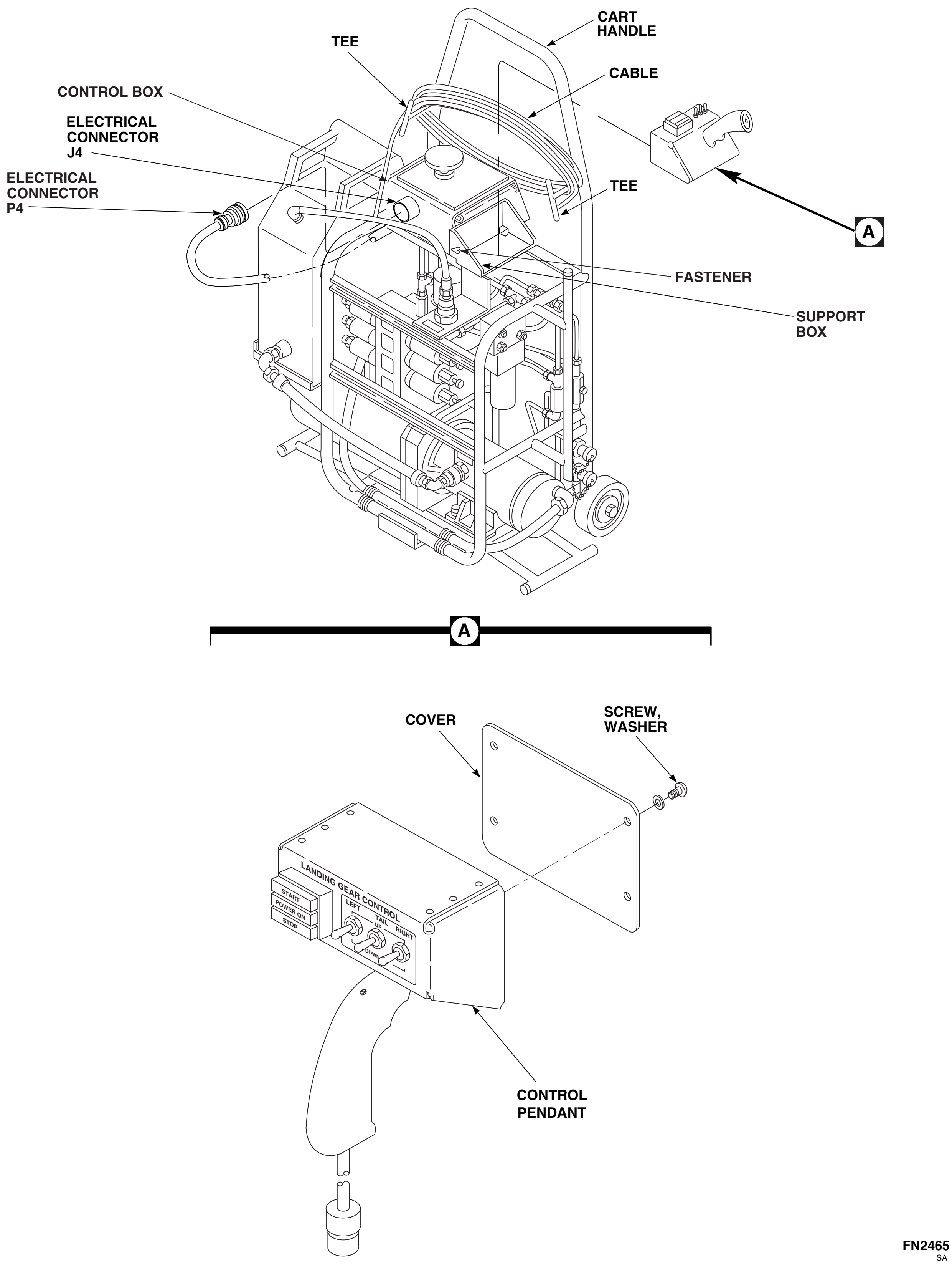


FIGURE 16-4-7-1. REPLACE/REPAIR AIR TRANSPORT HYDRAULIC CART CONTROL PENDANT

16-4-8 AIR TRANSPORT HYDRAULIC CART CONTROL BOX.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit

16-4-8.1 Repair Air Transport Hydraulic Cart Control Box.

a. If required, disconnect external electrical power from cart.

b. Remove screws securing cover to control box (Figure 16-4-8-1).

c. Repair or replace damaged or worn wiring or component.

d. Install cover on control box and secure with screws.

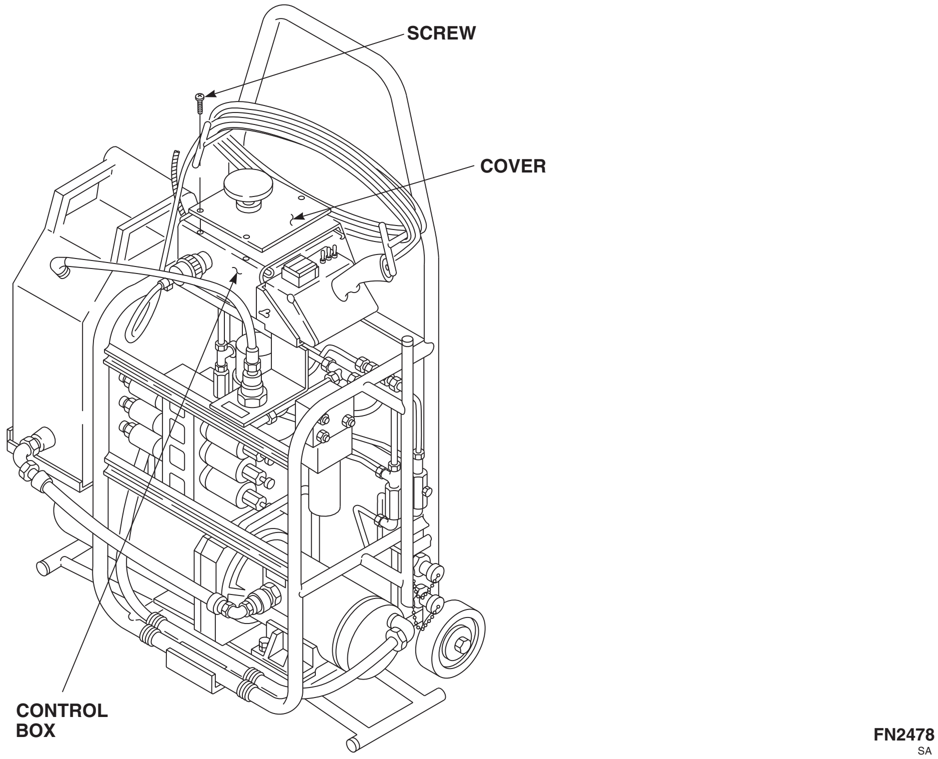


FIGURE 16-4-8-1. REPAIR TRANSPORT HYDRAULIC CART CONTROL BOX

16-4-9 AIR TRANSPORT HYDRAULIC CART PUMP AND MOTOR ASSEMBLY.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D

16-4-9.1 Replace Air Transport Hydraulic Cart Pump and Motor Assembly.

16-4-9.1.1. Remove.

a. If required, disconnect external electrical power from cart.

b. Unscrew sealing nut from elbow on motor cover. Remove nuts holding motor cover to end of motor. Slide cover back along power cable to get to wiring connections. Disconnect power cable from pump motor (Figure 16-4-9-1).

c. Remove locknut holding elbow to cover. Remove elbow from cover, leaving power cable in elbow. Replace cover on motor.

d. Place machinery towels, Item 211, Appendix D, under hydraulic connections to catch any spilled hydraulic fluid.

e. On hydraulic cart, 70700-20437-041, disconnect hydraulic lines from elbows on pump and motor assembly. Cap hydraulic lines.

f. On hydraulic cart, 70700-81650-041, disconnect tube assembly from pump output and hose assembly from pump input on pump and motor assembly. Cap hydraulic lines (Figure 16-4-9-1, Detail A).

g. Remove bolts, washers, and nuts and remove pump and motor assembly from cart (Figure 16-4-9-1).

NOTE

Note position of elbows.

h. On hydraulic cart, 70700-20437-041, remove elbows from pump and motor assembly.

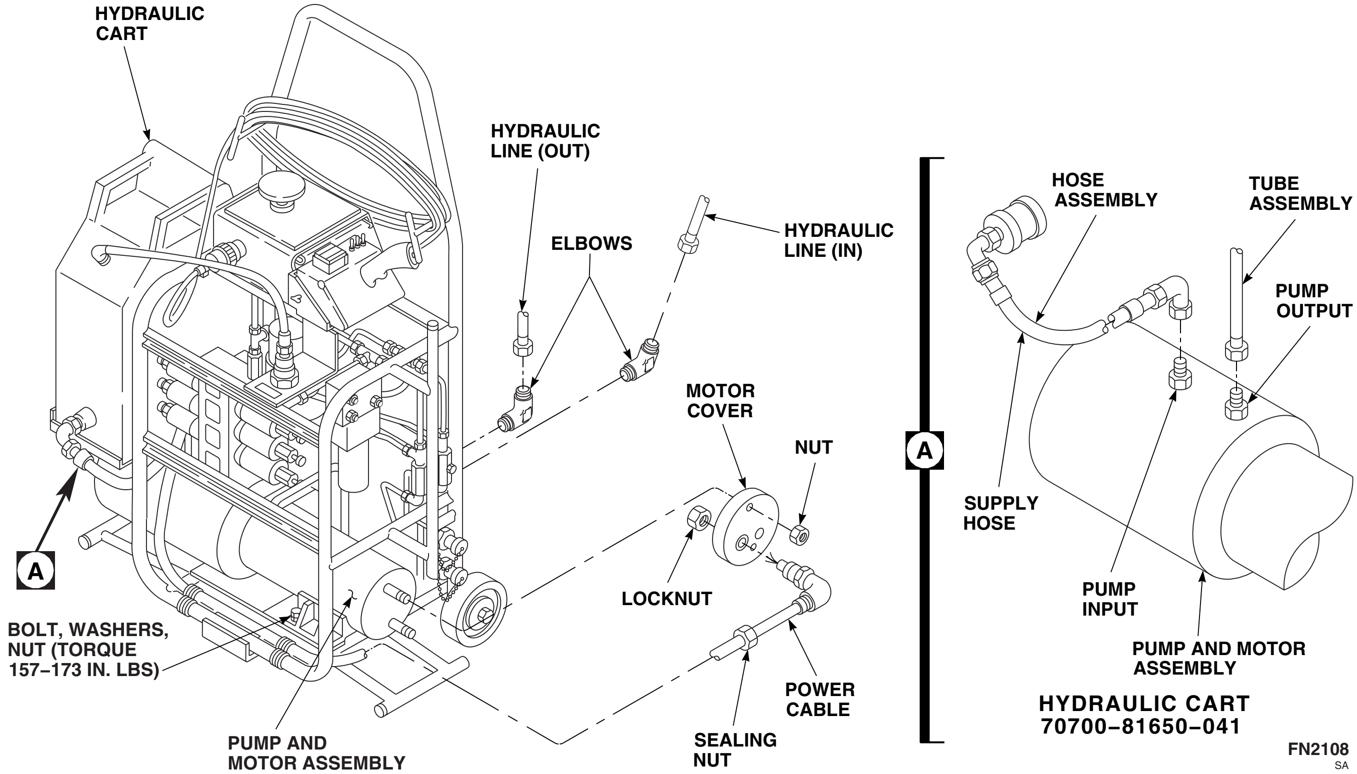


FIGURE 16-4-9-1. REPLACE AIR TRANSPORT HYDRAULIC CART PUMP AND MOTOR ASSEMBLY

16-4-9.1.2. Install.

a. On hydraulic cart, 70700-20437-041, install elbows in same position as noted during removal (Figure 16-4-9-1).

b. Place pump and motor assembly on cart and secure with bolts, washers, and nuts. **TORQUE NUTS TO 157 - 173 INCH-POUNDS.**

c. Remove motor cover from pump.

d. Place elbow in cover and secure with locknut. Make sure that power cable is still in elbow.

e. Connect power cable to motor. Slide cover onto motor. Secure cover to motor. Install sealing nut on elbow.

f. On hydraulic cart, 70700-20437-041, uncap and connect hydraulic lines to elbows on pump and motor assembly.

g. On hydraulic cart, 70700-81650-041, uncap and connect tube assembly to pump output (Figure 16-4-9-1, Detail A).

h. On hydraulic cart, 70700-81650-041, connect hose assembly to pump input on pump and motor assembly.

i. Fill pump reservoir with hydraulic fluid, Item 154, Appendix D.

j. Clean up any spilled hydraulic fluid.

16-4-10 AIR TRANSPORT HYDRAULIC CART FILTER ELEMENT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D

16-4-10.1 Replace Air Transport Hydraulic Cart Filter Element.

NOTE

Filter element should be replaced whenever red indicator button has popped out (extended position).

16-4-10.1.1. Remove.

a. Place machinery towels, Item 211, Appendix D, under filter bowl to catch any hydraulic fluid that might spill.

b. Unscrew filter bowl from filter assembly (Figure 16-4-10-1).

c. Drain fluid from filter bowl and remove filter element and packing. Discard packing.

16-4-10.1.2. Install.

a. Clean filter bowl with hydraulic fluid, Item 154, Appendix D.

b. Insert replacement filter element and packing in filter bowl (Figure 16-4-10-1).

c. Fill filter bowl with hydraulic fluid, Item 154, Appendix D.

d. Being careful not to spill fluid, screw bowl into filter assembly. Lockwire, Item 197, Appendix D, filter bowl to filter assembly.

e. Clean up any spilled fluid.

f. If necessary, reset filter button.

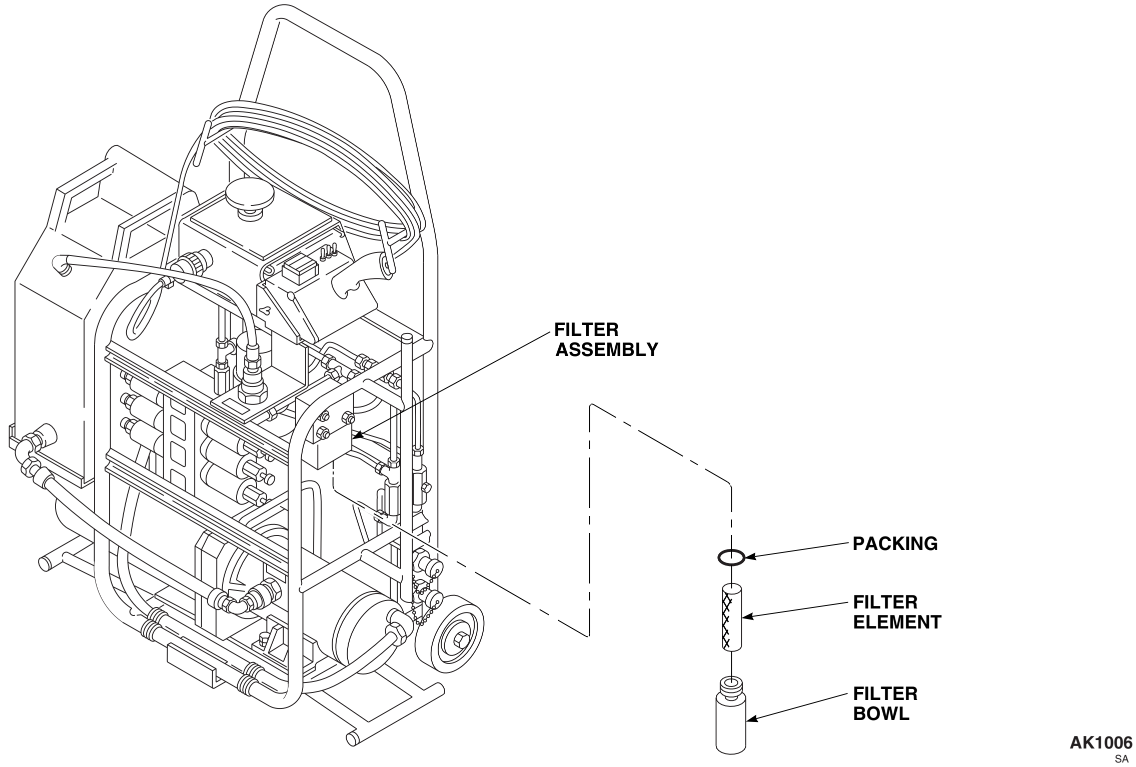


FIGURE 16-4-10-1. REPLACE AIR TRANSPORT HYDRAULIC CART FILTER ELEMENT

16-4-11 AIR TRANSPORT HYDRAULIC CART FILTER ASSEMBLY.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D

16-4-11.1 Replace Air Transport Hydraulic Cart Filter Assembly.

16-4-11.1.1 Remove.

a. Place machinery towel, Item 211, Appendix D, under filter assembly to catch any spilled hydraulic fluid.

b. Disconnect hydraulic tubes from elbows on filter assembly (Figure 16-4-11-1). Install protective caps on hydraulic tubes.

c. Remove bolts, washers, nuts, and filter assembly from hydraulic cart.

d. Loosen locknuts securing elbows to filter assembly. Remove elbows.

e. Remove packings, backup rings, and locknuts from elbows. Discard packings.

16-4-11.1.2 Install.

a. Coat all packings and threads with hydraulic fluid, Item 154, Appendix D, before installing.

b. Install locknuts, packings, and backup rings on elbows (Figure 16-4-11-1).

c. Install elbows on filter assembly and secure with locknuts.

d. Secure filter assembly to hydraulic cart with bolts, washers, and nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**

e. Remove protective caps from hydraulic tubes and connect tubes to elbows on filter assembly.

f. Clean up any spilled hydraulic fluid.

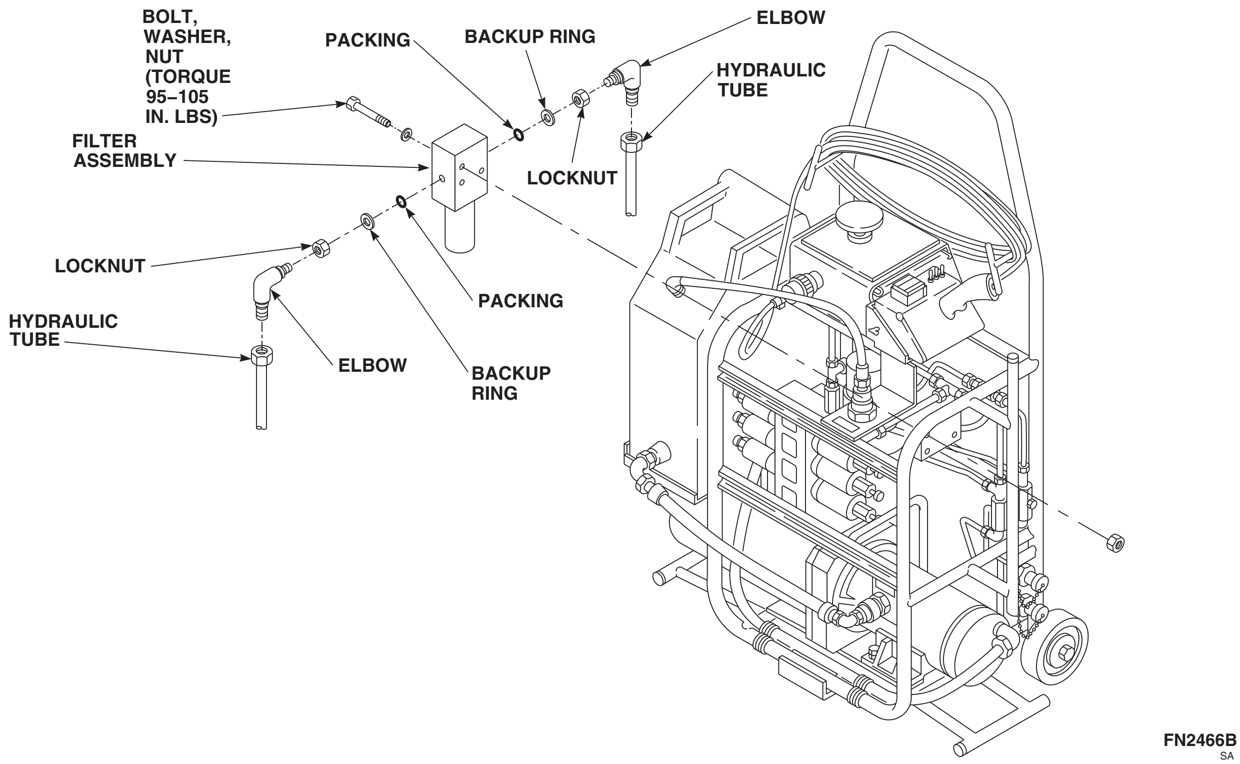


FIGURE 16-4-11-1. REPLACE AIR TRANSPORT HYDRAULIC CART FILTER ASSEMBLY

16-4-12 AIR TRANSPORT HYDRAULIC CART PRESSURE SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs
- Tags

References

- Appendix D

16-4-12.1 Replace Air Transport Hydraulic Cart Pressure Switch.

16-4-12.1.1 Remove.

a. If required, disconnect external electrical power from cart.

b. Tag and disconnect external electrical wiring from terminals on pressure switch.

c. Place machinery towel, Item 211, Appendix D, under pressure switch to catch any spilled hydraulic fluid.

d. Disconnect hydraulic tube from elbow on pressure switch (Figure 16-4-12-1). Install protective cap on hydraulic tube.

e. Remove bolts, washers, nuts, and pressure switch from hydraulic cart.

f. Remove elbow from pressure switch. Install protective plug in open port.

16-4-12.1.2 Install.

a. Coat all threads with hydraulic fluid, Item 154, Appendix D, before installing.

b. Remove protective plug from elbow and install elbow in pressure switch (Figure 16-4-12-1).

c. Position pressure switch on hydraulic cart and secure with bolts, washers, and nuts. **TORQUE NUTS TO 157 - 173 INCH-POUNDS.**

d. Remove protective cap from hydraulic tube and connect tube to pressure switch.

e. Connect electrical wiring to terminals on pressure switch. Remove tags.

f. Clean up any spilled hydraulic fluid.

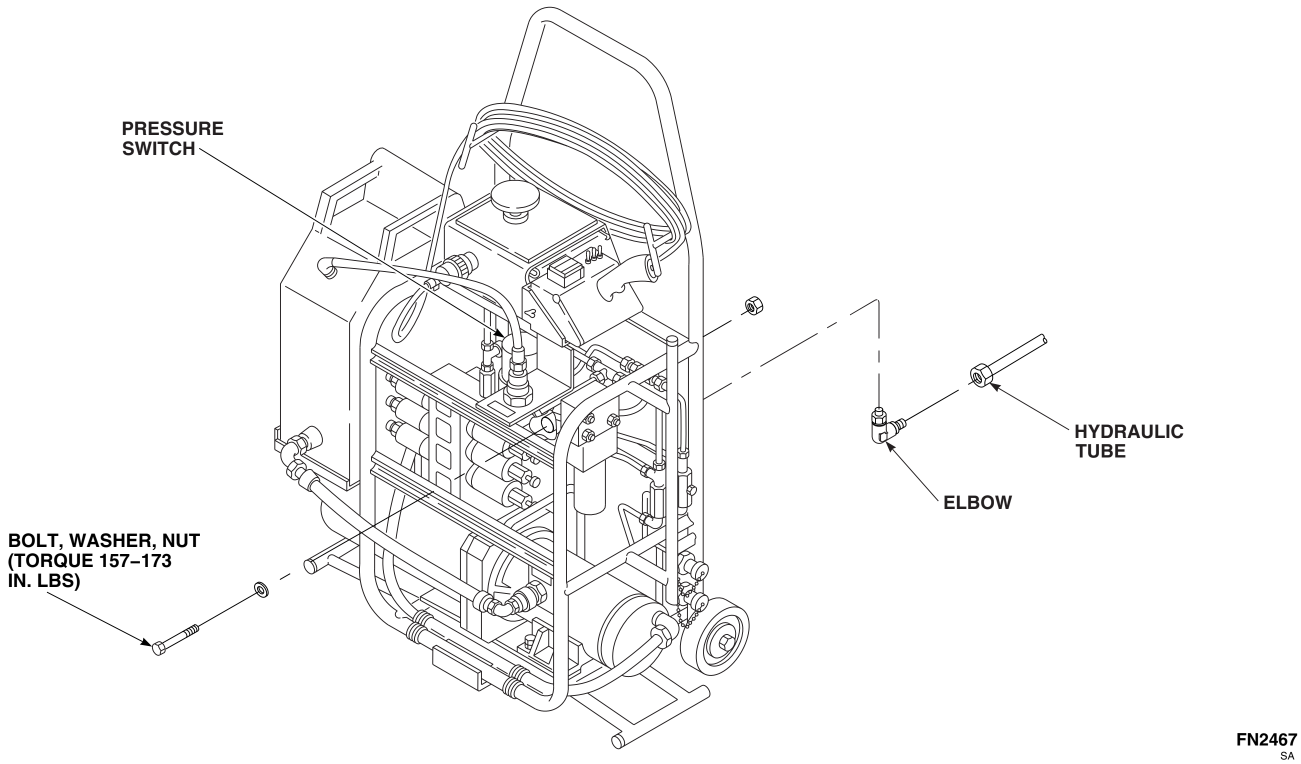


FIGURE 16-4-12-1. REPLACE AIR TRANSPORT HYDRAULIC CART PRESSURE SWITCH

16-4-13 AIR TRANSPORT HYDRAULIC CART DIRECTIONAL CONTROL VALVE/SPECIAL MANIFOLD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs
- Tags
- Tiedown Strap, Item 335, Appendix D

References

- Appendix D
- PARA 1-7-20

16-4-13.1 Replace Air Transport Hydraulic Cart Directional Control Valve/Special Manifold.

NOTE

The three valves that make up the directional control valve assembly (stack valve) are to be replaced as an assembly.

16-4-13.1.1 Remove.

a. If required, disconnect external electrical power from cart.

b. Disconnect electrical connector, P4, from electrical connector, J4, on control box (Figure 16-4-13-1).

c. Install protective caps and plugs on electrical connectors.

d. Remove strain relief clamp from electrical connector, P4.

e. On hydraulic cart, 70700-20437-041, perform the following:

(1) Tag and disconnect six wires from electrical connector, P4:

Wire	from	Pin No. (P4)
GRN 2		E
GRN 3		K
GRN 6		L
GRN 7		M

Wire	from	Pin No. (P4)
GRN 10		F
GRN 11		J

(2) Tag and disonnect the following valve wires from chassis ground on cart frame: GRN 1N, GRN 4N, GRN 5N, GRN8N, GRN 9N, and GRN 12N.

f. On hydraulic cart, 70700-81650-041, perform the following:

(1) Tag and disconnect seven wires from electrical connector, P4, pins, R, F, E, P, N, J, and M.

(2) Disconnect ground wires, number 149, from valves, V2, V4, and V6, from cart frame terminal, E2. Diconnect ground wires, number 149, from valves, V1, V3, V5, and V7, from cart frame terminal, E1.

g. Place machinery towel, Item 211, Appendix D, under directional control valve on hydraulic cart, 70700-20437-041, or special manifold on hydraulic cart, 70700-81650-041, to catch any spilled hydraulic fluid.

h. Disconnect hydraulic tubes from connectors, elbows, and unions on valve, or special manifold. Cap and plug lines and ports.

i. Remove bolt, washers, and nuts securing directional control valve, or special manifold to cart.

j. Remove connectors, elbows, nuts, packings, and unions. Discard packings.

16-4-13.1.2 Install.

a. Label wires on replacement directional control valve on hydraulic cart, 70700-20437-041, or special manifold on hydraulic cart, 70700-81650-041, to match those on valve or special manifold just removed. Cut wires on new valve or special manifold to same length as those on removed valve.

b. Install terminal lugs and plug pins on wires.

c. Coat all threads with hydraulic fluid, Item 154, Appendix D, before installing.

d. Install connectors and elbows in back of valve, or special manifold. Install nuts, packings, and unions in side of valve, or special manifold (Figure 16-4-13-1).

e. Position directional control valve assembly, or special manifold on cart. Make sure valve with wires, GRN 1N, GRN 2, GRN 3, and GRN 4N is at top.

f. Secure valve assembly or special manifold to cart with bolts, washers, and nuts. **TORQUE NUTS TO 262 - 288 INCH-POUNDS.**

g. Remove caps and plugs and connect hydraulic tubes to connectors, elbows, and unions on valve or special manifold.

h. On hydraulic cart, 70700-20437-041, connect the following wires to chassis ground on cart frame: GRN 1N, GRN 4N, GRN 5, GRN 8N, GRN 9N, and GRN 12N. Remove tags.

i. Remove protective caps and plugs from electrical connectors.

j. Inspect and treat electrical connectors (PARA 1-7-20).

k. On hydraulic cart, 70700-20437-041, connect the following six wires to electrical connector, P4 and remove tags:

Wire	to	Pin No. (P4)
GRN 2		E
GRN 3		K
GRN 6		L
GRN 7		M
GRN 10		F
GRN 11		J

l. On hydraulic cart, 70700-81650-041, perform the following:

(1) Connect seven wires to connector, P4, pins, R, F, E, P, N, J, and M. Remove tags.

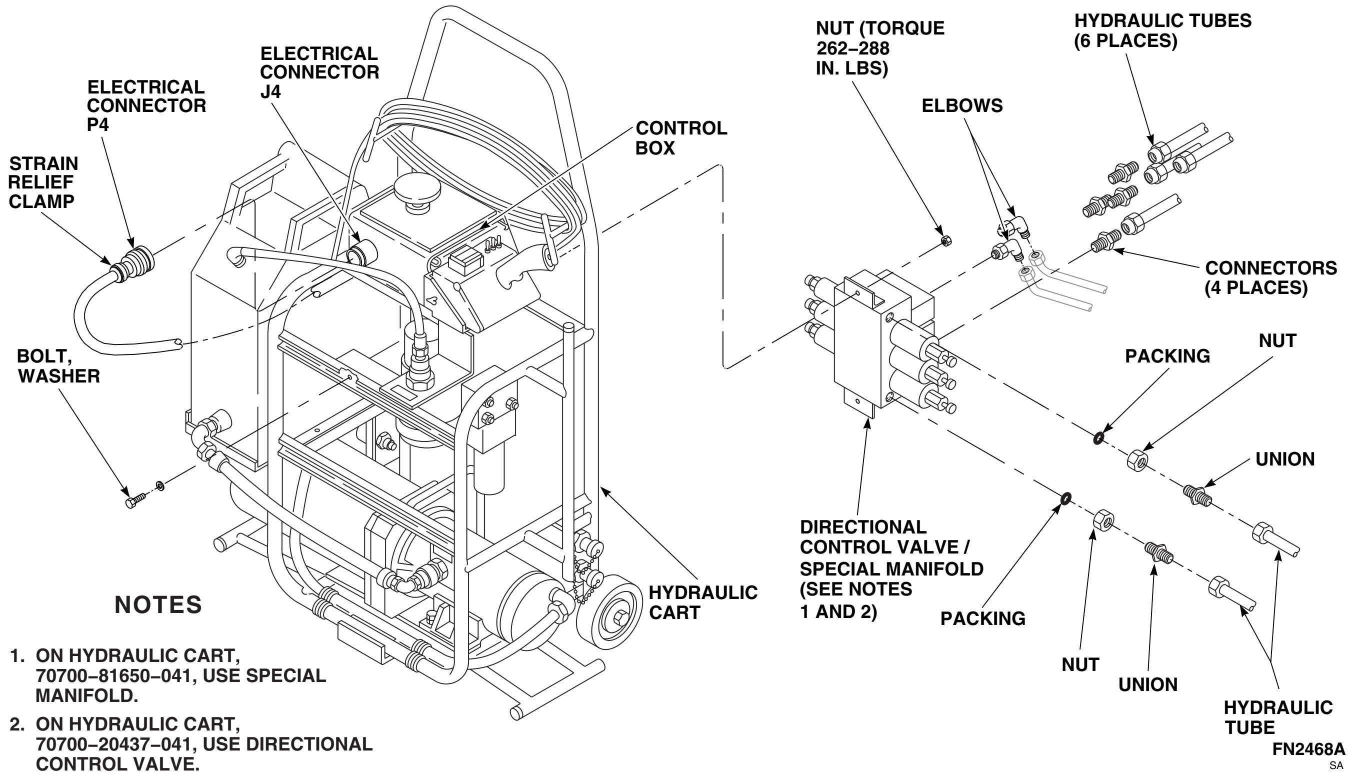


FIGURE 16-4-13-1. REPLACE AIR TRANSPORT HYDRAULIC CART DIRECTIONAL CONTROL VALVE/SPECIAL MANIFOLD

(2) Connect ground wires, number 149, from valves, V2, V4, and V6, to cart frame terminal, E2. Connect ground wires, number 149, from valves V1, V3, V5, and V7, to cart frame terminal, E1

m. Install strain relief clamp on electrical connector, P4. Secure wires in bundle with tiedown straps, Item 335, Appendix D.

n. Inspect and treat electrical connectors (PARA 1-7-20).

o. Connect electrical connector, P4, to electrical connector, J4, on control box.

p. Clean up any spilled hydraulic fluid.

16-4-14 AIR TRANSPORT HYDRAULIC CART FLOW REGULATORS/ NEEDLE VALVES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D

16-4-14.1 Replace Air Transport Hydraulic Cart Flow Regulators/Needle Valves.

16-4-14.1.1. Remove.

a. Place machinery towel, Item 211, Appendix D, under flow regulator/needle valve that will be removed, to catch any spilled hydraulic fluid (Figure 16-4-14-1).

NOTE

Note direction of flow regulators.

b. On hydraulic cart, 70700-20437-041, disconnect hydraulic tubes from each end of flow regulator/needle valve, and remove from cart. Cap lines.

NOTE

Note direction of flow regulators.

c. On hydraulic cart, 70700-81650-041, disconnect hydraulic tubes from each elbow on regulator and remove from cart. Cap lines. Remove elbows from regulator.

16-4-14.1.2. Install.

a. Coat all threads with hydraulic fluid, Item 154, Appendix D, before installing.

b. On hydraulic cart, 70700-20437-041, uncap lines and connect flow regulator/needle valve to hydraulic tubes. Make sure flow direction is the same as noted during removal (Figure 16-4-14-1).

c. On hydraulic cart, 70700-81650-041, install elbows on replacement flow regulator.

d. On hydraulic cart, 70700-81650-041, uncap lines and connect elbows to hydraulic tubes. Make sure flow direction is the same as noted during removal.

e. Clean up any spilled hydraulic fluid.

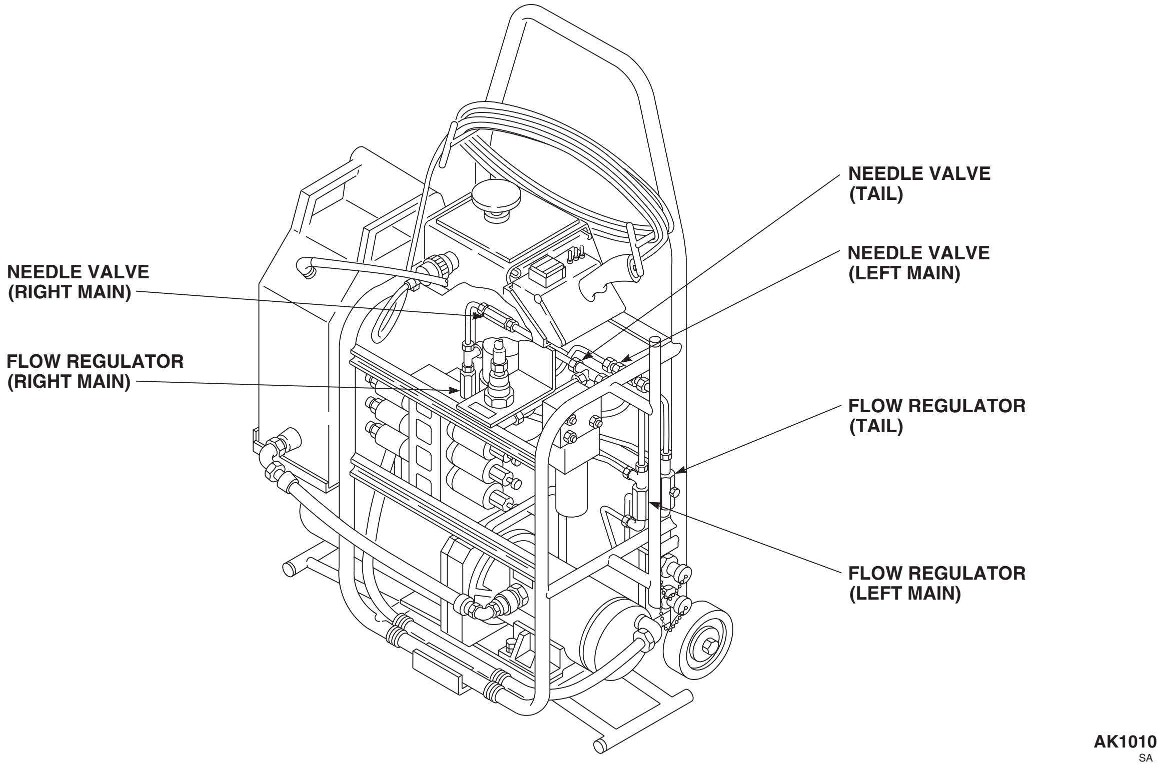


FIGURE 16-4-14-1. REPLACE AIR TRANSPORT HYDRAULIC CART FLOW REGULATORS/ NEEDLE VALVES

16-4-15 AIR TRANSPORT HYDRAULIC CART WHEEL.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

16-4-15.1 Replace Air Transport Hydraulic Cart Wheel.

16-4-15.1.1 Remove.

a. Remove nut securing wheel to axle (Figure 16-4-15-1).

b. Remove bushing and wheel.

16-4-15.1.2 Install.

a. Slide wheel onto axle (Figure 16-4-15-1).

b. Install bushing and nut. **TIGHTEN NUT UNTIL 0.015 TO 0.031-INCH CLEARANCE EXISTS BETWEEN AXLE BUSHING AND WHEEL BUSHING.**

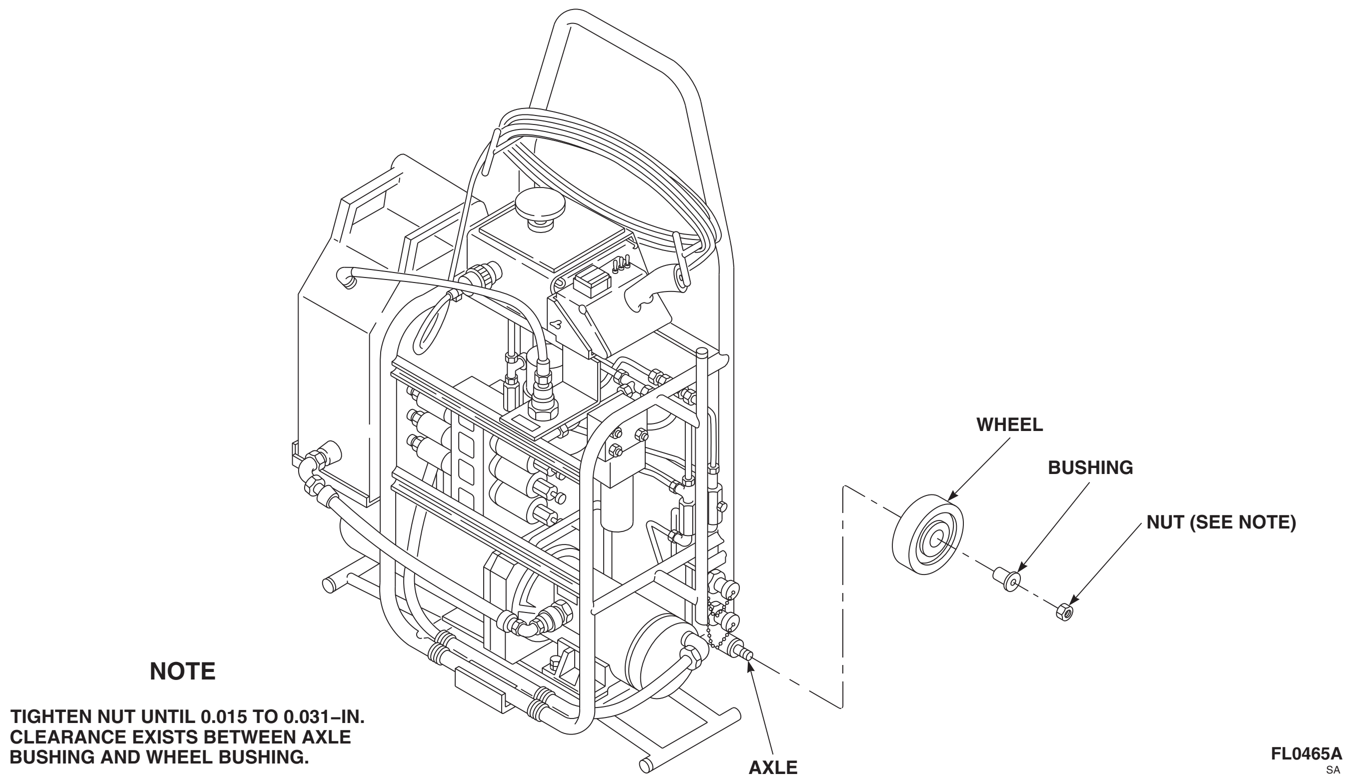


FIGURE 16-4-15-1. REPLACE AIR TRANSPORT HYDRAULIC CART WHEEL

16-4-16 AIR TRANSPORT HYDRAULIC CART HYDRAULIC LINES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D

16-4-16.1 Replace/Repair Air Transport Hydraulic Cart Hydraulic Lines.

16-4-16.1.1. Remove.

a. Place machinery towel, Item 211, Appendix D, under hydraulic lines that will be removed, to catch any spilled hydraulic fluid.

b. Disconnect and remove hydraulic tubes, hoses, or fittings as required (Figure 16-4-16-1, Sheet 1, Detail A, Detail B, Detail C, and Detail D and Sheet 2, Detail E and Detail F). Cap all open lines and fittings.

c. Repair hoses by replacing quick-disconnects.

16-4-16.1.2. Install.

a. Coat all packings and threads with hydraulic fluid, Item 154, Appendix D, before installing.

b. Uncap hydraulic lines and fittings and connect as necessary (Figure 16-4-16-1, Sheet 1, Detail A, Detail B, Detail C, and Detail D, and Sheet 2, Detail E and Detail F).

c. Clean up any spilled hydraulic fluid.

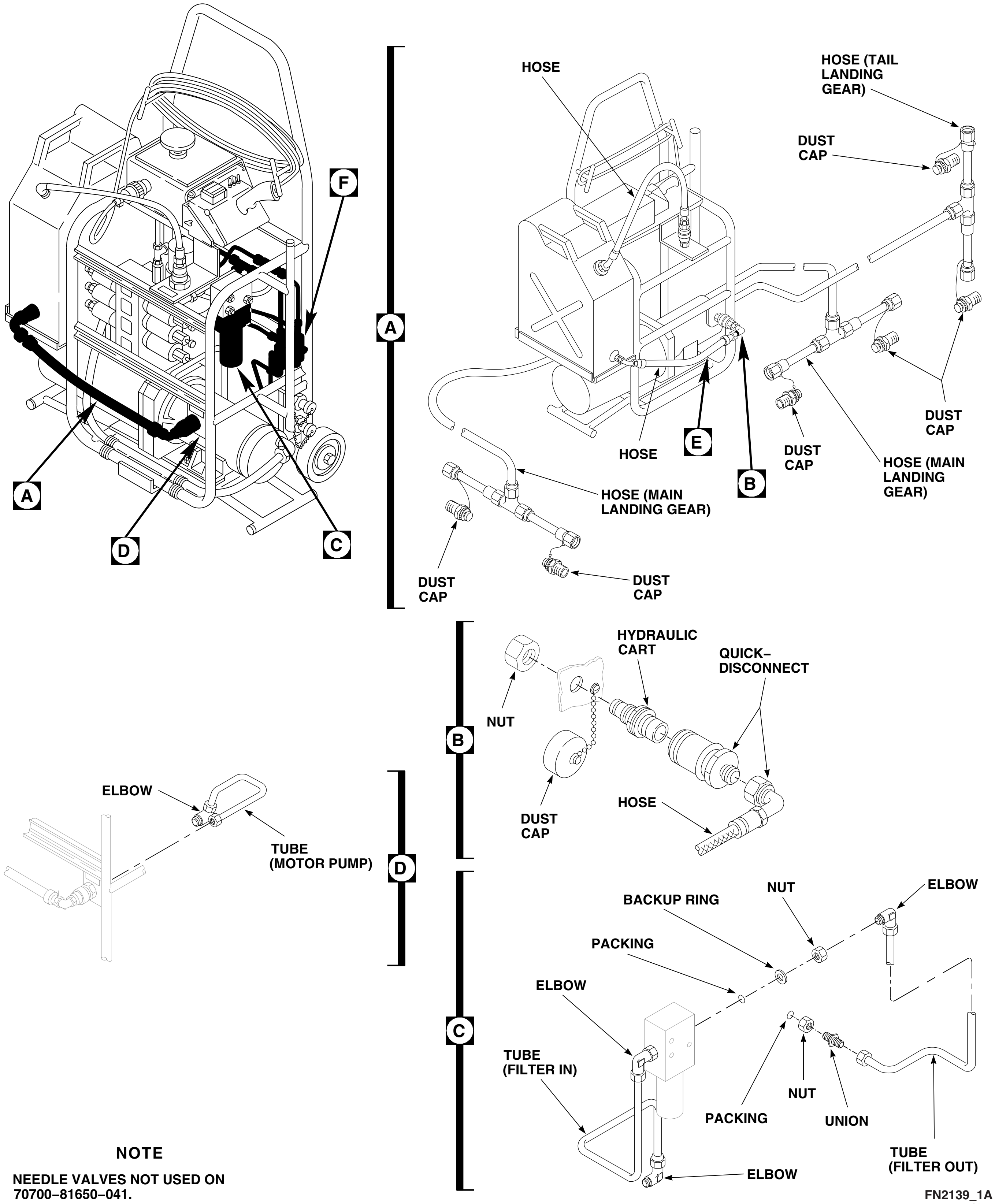
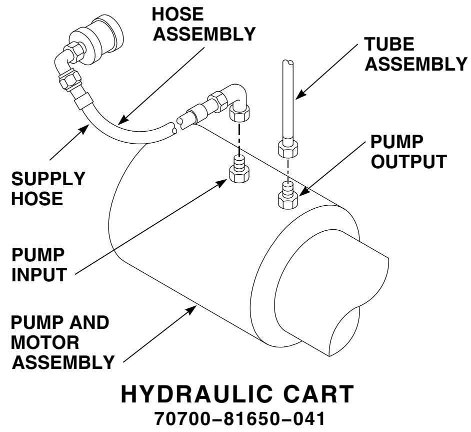


FIGURE 16-4-16-1. REPLACE/REPAIR AIR TRANSPORT HYDRAULIC CART HYDRAULIC LINES (SHEET 1 OF 2)



E

F

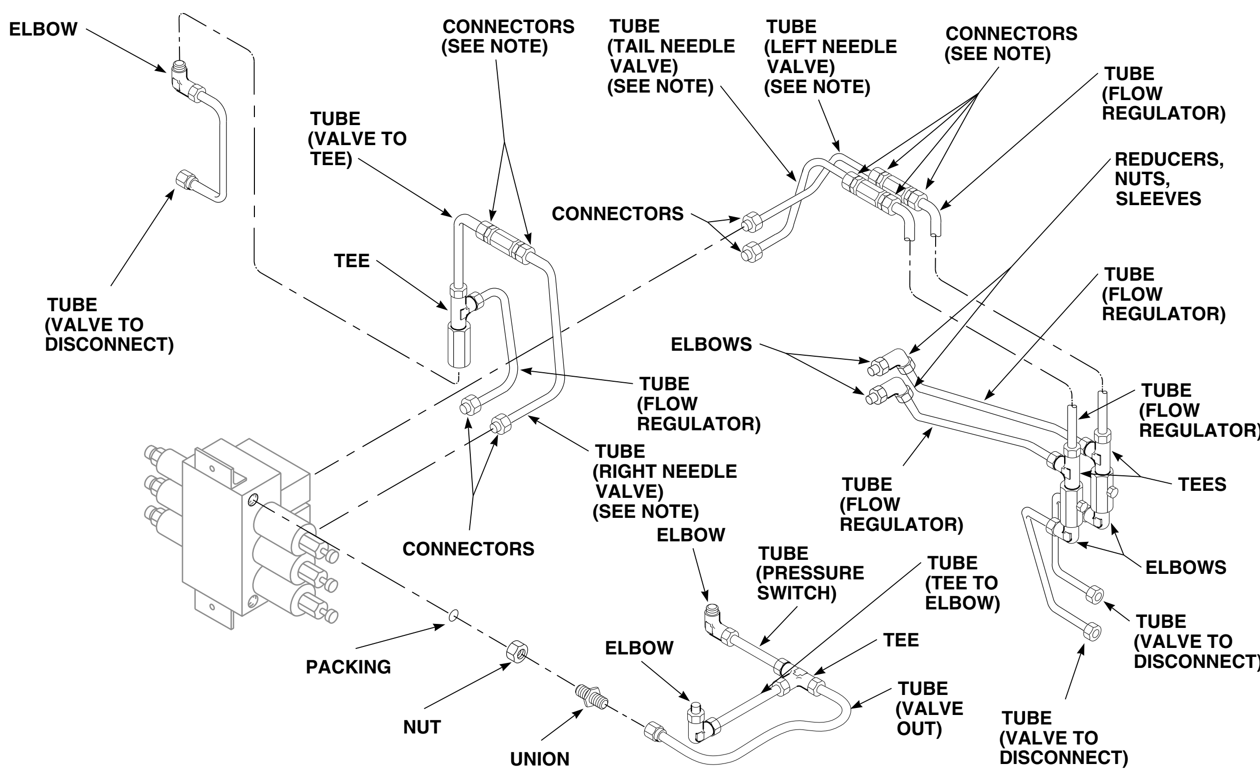


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FIGURE 16-4-16-1. REPLACE/REPAIR AIR TRANSPORT HYDRAULIC CART HYDRAULIC LINES (SHEET 2 OF 2)

16-4-17 AIR TRANSPORT HYDRAULIC CART RESERVOIR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Protective Caps and Plugs

References

- Appendix D

16-4-17.1 Repair Air Transport Hydraulic Cart Reservoir.

16-4-17.1.1 Remove.

a. On reservoir, 70700-20446-042 and 70700-20446-043, perform the following (Figure 16-4-17-1):

(1) Disconnect supply and return hoses from cart by releasing quick-disconnect fittings on hoses from couplings on cart.

(2) Install protective caps and plugs on couplings and quick-disconnect fittings.

(3) Remove pin from frame and bracket on reservoir.

(4) Remove reservoir, with return hoses attached, from cart. Install pin in bracket.

b. On reservoir, 70700-20446-046, perform the following (Figure 16-4-17-1):

(1) Disconnect supply hose from reservoir by releasing quick-disconnect fitting on hose from coupling at bottom of reservoir.

(2) Disconnect return hose from cart by releasing quick-disconnect fitting on hose from coupling on cart.

(3) Install protective caps and plugs on couplings and quick-disconnect fittings.

(4) Remove pin from frame and bracket on reservoir.

(5) Remove reservoir, with return hose attached, from cart. Install pin in bracket.

16-4-17.1.2 Disassemble.

a. On reservoir, 70700-20446-042, perform the following (Figure 16-4-17-2, Sheet 1):

(1) Remove screws and washers securing breather cap to reservoir and remove cap.

(2) Remove screws securing bracket to reservoir and remove bracket with cable and pin attached.

(3) Remove return hose from cap.

(4) Install protective plug in return hose.

(5) Remove cap from reservoir. Pour out any hydraulic fluid in reservoir.

(6) Remove supply hose from reducer.

(7) Install protective plug in supply hose.

(8) Remove reducer and packing from reservoir. Discard packing.

b. On reservoir, 70700-20446-043, perform the following (Figure 16-4-17-2, Sheet 2):

(1) Remove screws and washers securing breather cap to reservoir and remove cap.

(2) Remove screws securing bracket to reservoir and remove bracket with cable and pin attached.

(3) Remove cap from reservoir. Pour out any hydraulic fluid in reservoir.

(4) Remove return hose from union on cap.

(5) Install protective plug in return hose.

(6) Remove tube from union.

(7) Remove jamnut, packing, and union from cap. Discard packing.

(8) Remove supply hose from union on bottom of reservoir.

(9) Install protective plug in supply hose.

(10) Remove jamnut, union, and packing from reservoir. Discard packing.

c. On reservoir, 70700-20446-046, perform the following (Figure 16-4-17-2, Sheet 3):

(1) Remove screws and washers securing breather cap to reservoir and remove cap.

(2) Remove screws securing bracket to reservoir and remove bracket with cable and pin attached.

(3) Remove cap from reservoir. Pour out any hydraulic fluid in reservoir.

(4) Remove return hose from union on cap.

(5) Install protective plug in return hose.

(6) Remove tube from union.

(7) Remove jamnut, packing, and union from cap. Discard packing.

(8) Remove jamnut, coupling, and packing from reservoir. Discard packing.

16-4-17.1.3 Assemble.

a. On reservoir, 70700-20466-042, perform the following (Figure 16-4-17-2, Sheet 1):

(1) Lubricate packing with hydraulic fluid, Item 154, Appendix D.

(2) Install reducer and packing in reservoir.

(3) Remove protective plug from supply hose.

(4) Install supply hose on reducer. **TORQUE B-NUT TO 270 - 300 INCH-POUNDS.**

(5) Remove protective plug from return hose.

(6) Install return hose on cap. **TORQUE B-NUT TO 270 - 300 INCH-POUNDS.**

(7) Install cap on reservoir.

(8) Position bracket on reservoir, with cable and pin attached, and secure with screws.

(9) Install breather cap on reservoir and secure with screws and washers.

b. On reservoir, 70700-20446-043, perform the following (Figure 16-4-17-2, Sheet 2):

(1) Lubricate packings with hydraulic fluid, Item 154, Appendix D.

(2) Install union and packing on cap and secure with jamnut. **TORQUE JAMNUT TO 60 - 75 INCH-POUNDS.**

(3) Install union with packing on reservoir and secure with jamnut. **TORQUE JAMNUT TO 60 - 75 INCH-POUNDS.**

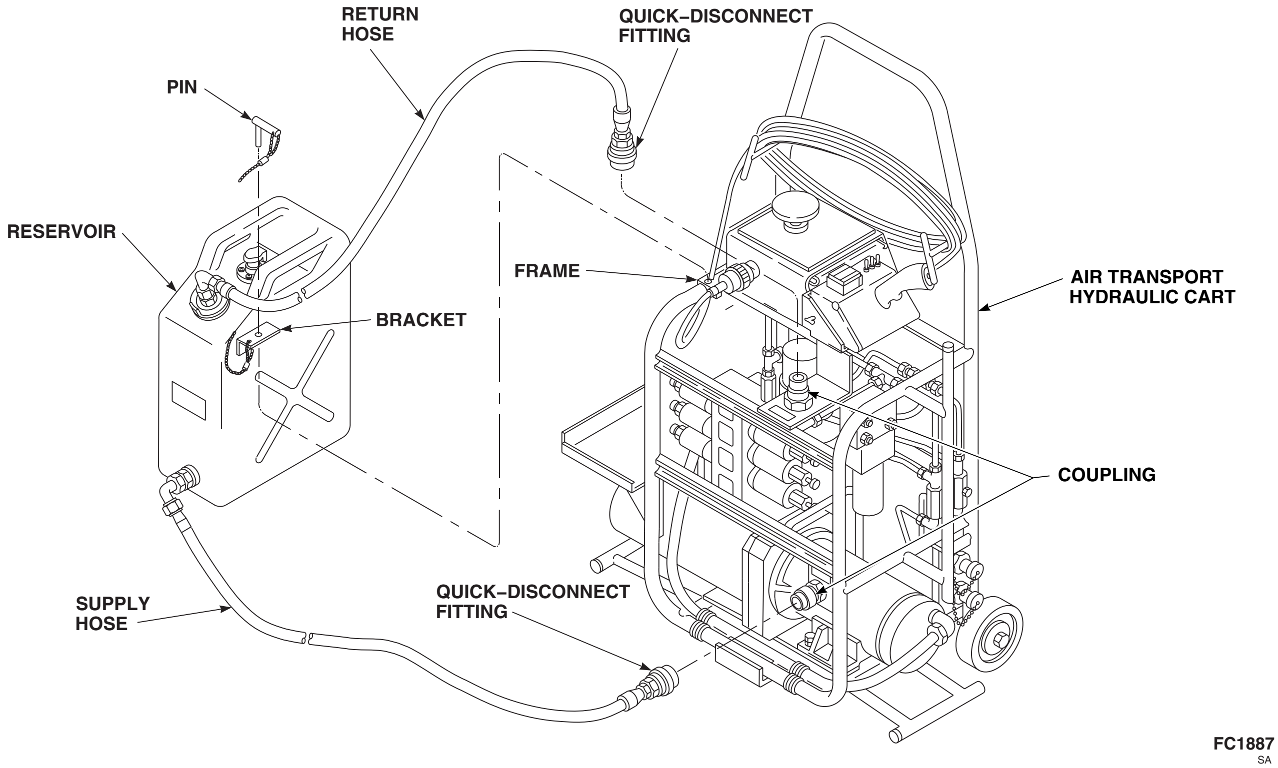


FIGURE 16-4-17-1. REPLACE AIR TRANSPORT HYDRAULIC CART RESERVOIR

(4) Install tube on union. **TORQUE B-NUT TO 270 - 300 INCH-POUNDS.**

(5) Remove protective plug from supply hose.

(6) Install supply hose on union. **TORQUE B-NUT TO 270 - 300 INCH-POUNDS.**

(7) Install cap on reservoir.

(8) Position bracket on reservoir, with cable and pin attached, and secure with screws.

(9) Install breather cap on reservoir and secure with screws and washers.

c. On reservoir, 70700-20446-046, perform the following (Figure 16-4-17-2, Sheet 3):

(1) Lubricate packings with hydraulic fluid, Item 154, Appendix D.

(2) Install union and packing on cap and secure with jamnut. **TORQUE JAMNUT TO 60 - 75 INCH-POUNDS.**

(3) Install tube on union. **TORQUE B-NUT TO 270 - 300 INCH-POUNDS.**

(4) Remove protective plug from return hose.

(5) Install return hose on union. **TORQUE B-NUT TO 270 - 300 INCH-POUNDS.**

(6) Install coupling with packing on reservoir and secure with jamnut. **TORQUE JAMNUT TO 60 - 75 INCH-POUNDS.**

(7) Install cap on reservoir.

(8) Position bracket on reservoir, with cable and pin attached, and secure with screws.

(9) Install breather cap on reservoir and secure with screws and washers.

16-4-17.1.4 Install.

a. On reservoir, 70700-20446-042 and 70700-20446-043, perform the following (Figure 16-4-17-1):

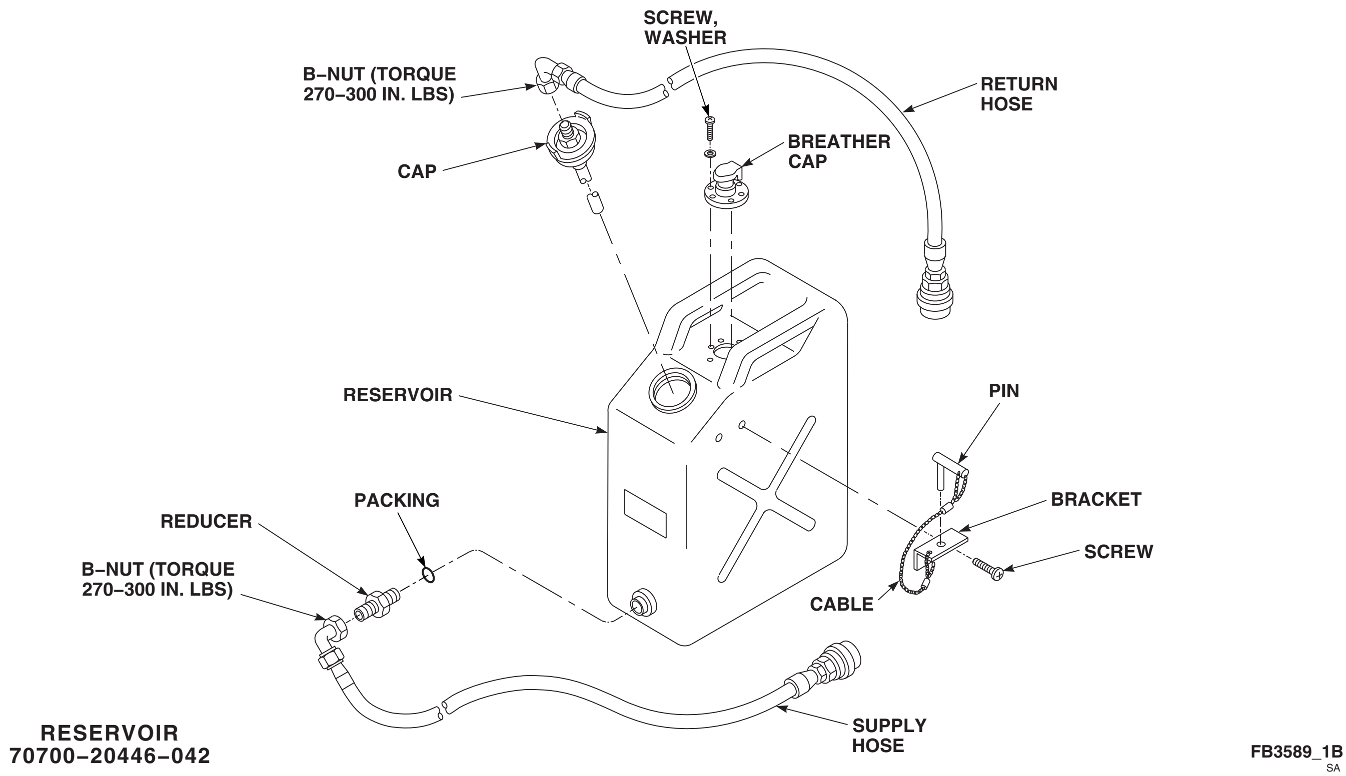


FIGURE 16-4-17-2. REPAIR AIR TRANSPORT HYDRAULIC CART RESERVOIR (SHEET 1 OF 3)

(1) Remove pin from bracket and install reservoir on cart. Secure by inserting pin in frame and bracket.

(2) Remove protective caps and plugs from couplings and quick-disconnect fittings.

(3) Connect supply and return hoses to cart by engaging quick-disconnect fitting on hose to coupling on cart.

(4) Fill reservoir with hydraulic fluid, Item 154, Appendix D.

b. On reservoir, 70700-20446-046, perform the following (Figure 16-4-17-1, Sheet 1):

(1) Remove pin from bracket and install reservoir on cart. Secure by inserting pin in frame and bracket.

(2) Remove protective caps and plug from coupling and quick-disconnect fitting.

(3) Connect supply hose to reservoir by engaging quick-disconnect fitting on hose to coupling at bottom of reservoir.

(4) Connect return hose to cart by engaging quick-disconnect fitting on hose to coupling on cart.

(5) Fill reservoir with hydraulic fluid, Item 154, Appendix D.

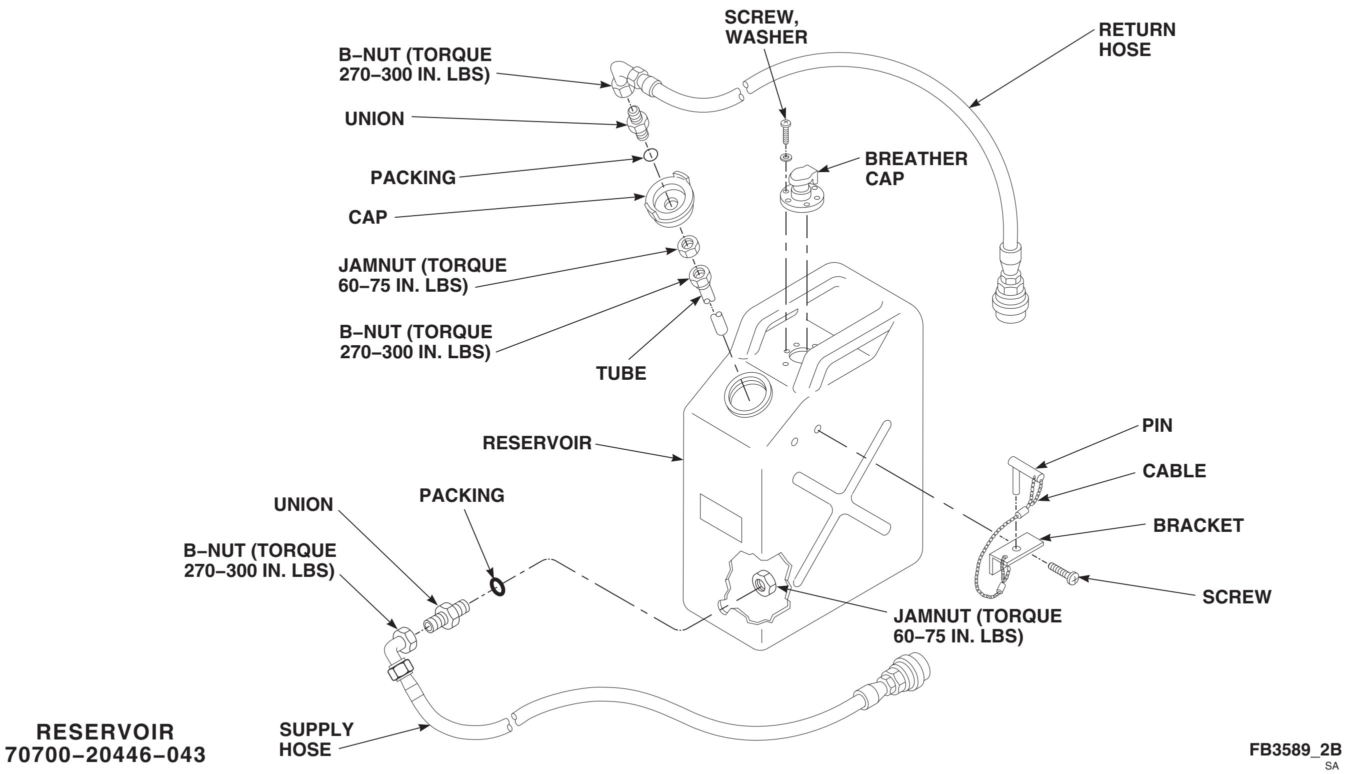


FIGURE 16-4-17-2. REPAIR AIR TRANSPORT HYDRAULIC CART RESERVOIR (SHEET 2 OF 3)

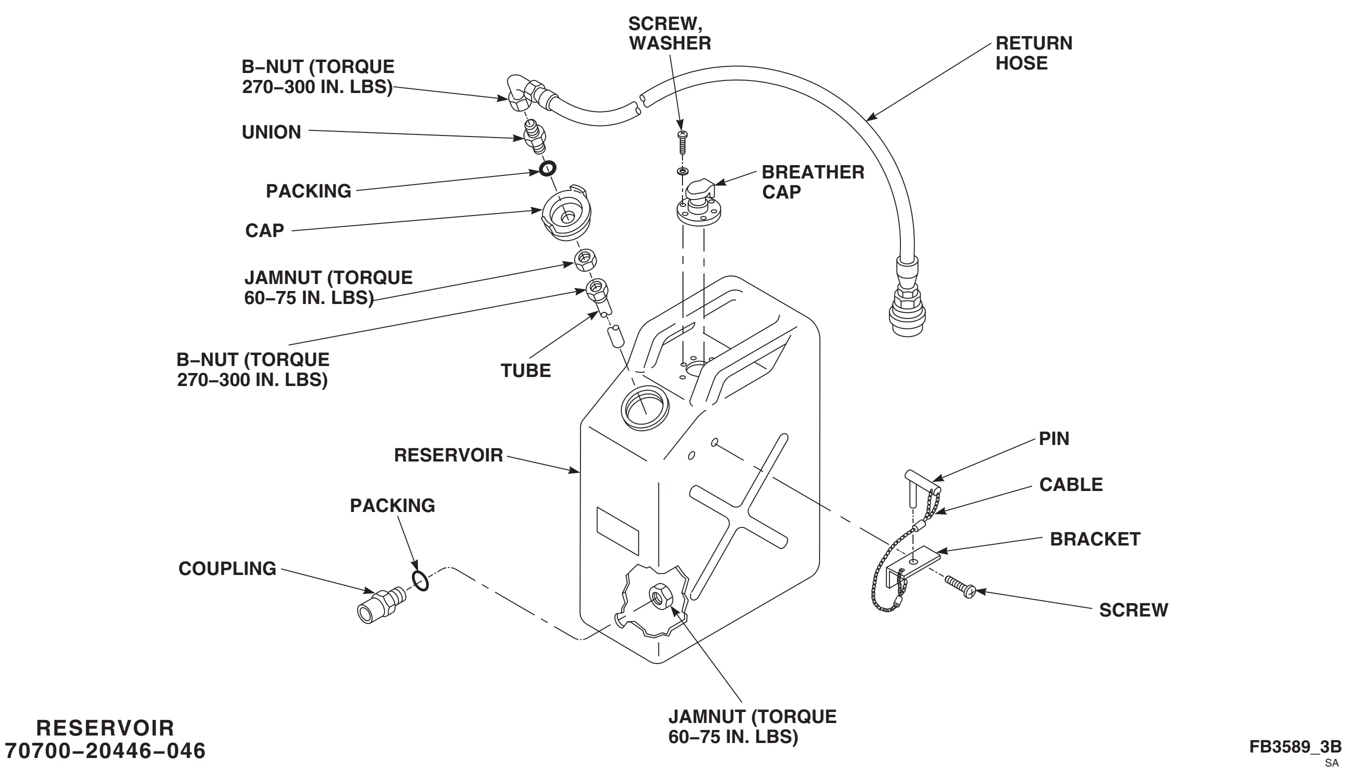


FIGURE 16-4-17-2. REPAIR AIR TRANSPORT HYDRAULIC CART RESERVOIR (SHEET 3 OF 3)

16-4-18 AIR TRANSPORT HYDRAULIC CART GROUNDING CABLE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Steel Wire Brush

Materials/Parts

- Acetone, Item 25, Appendix D
- Brush, Acid, Item 84, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D

16-4-18.1 Replace Air Transport Hydraulic Cart Grounding Cable.

16-4-18.1.1 Remove.

a. Remove screws securing cover to control box (Figure 16-4-18-1).

b. Move cover aside, making sure not to damage wiring.

c. Remove screw, washers, lockwasher, and nut securing grounding cable to side of control box (Figure 16-4-18-1, Detail A).

16-4-18.1.2 Install.

a. Clean area around screwhole using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D.

b. If new control box was installed, perform the following:

(1) Using steel wire brush, remove paint from area of screwhole.

(2) Using machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D, clean worked area.

(3) Apply corrosion resistant coating, Item 118, Appendix D, to area using acid brush, Item 84, Appendix D. Allow corrosion resistant coating to dry on area from 1 to 5 minutes or until area turns a golden iridescent color.

c. Connect grounding cable to box using screw, lockwasher, washers, and nut (Figure 16-4-18-1, Detail A).

d. Carefully replace cover on control and secure with screws (Figure 16-4-18-1).

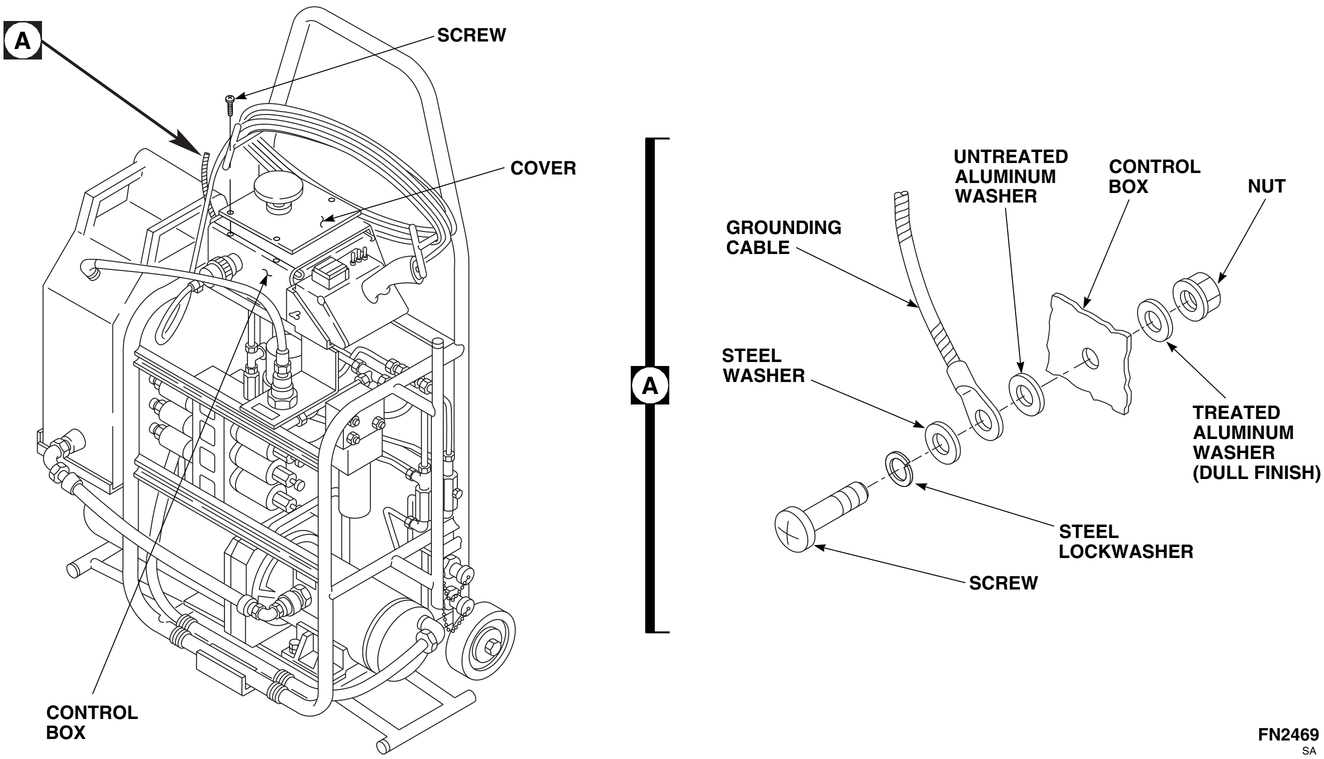


FIGURE 16-4-18-1. REPLACE AIR TRANSPORT HYDRAULIC CART GROUNDING CABLE

16-4-19 AIR TRANSPORT HYDRAULIC CART POWER CABLE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

References

- PARA 1-7-20

16-4-19.1 Replace Air Transport Hydraulic Cart Power Cable.

16-4-19.1.1 Remove.

a. If required, disconnect external power from cart.

b. Disconnect electrical connector from hydraulic cart 28 VDC adapter. Remove power cable.

16-4-19.1.2 Install.

a. Inspect and treat electrical connector (PARA 1-7-20).

b. Connect electrical connector on replacement power cable to hydraulic cart.

c. Connect power cable to external power and check for power to control box.

16-4-20 ESSS HORIZONTAL STORES SUPPORT (HSS) STRUTS REPAIR ESSS .

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit

References

- AMS 4049
- AMS-QQ-A-200
- Appendix D
- MIL-S-18729
- MIL-S-5059
- PARA 2-6-2
- PARA 2-6-3
- PARA 2-6-5

16-4-20.1 Repair ESSS Horizontal Stores Support (HSS) Struts.

NOTE

Repair of both ESSS HSS struts is the same.

a. Support struts are made of graphite epoxy. Support struts are primary structure. Repair support struts as follows:

(1) Negligible damage repairs are limited to scratches and gouges, of any area size, up to 0.005-inch deep.

(2) Touch up paint as required (PARA 2-6-5).

(3) Damage exceeding negligible damage must be reviewed by engineering authority for evaluation, repair, or replacement (Figure 16-4-20-1, Detail A).

b. Strut end fittings are 7075-T6511 or 7075-T7531, AMS-QQ-A-200/11, and are primary structure. Repair strut end fittings as follows (Figure 16-4-20-1, Detail B):

(1) Damage to strut end fittings is limited to nicks, scratches and corrosion no deeper than 5% of material thickness at point of damage.

(2) Blend damage smoothly to surrounding area, using long, even strokes.

(3) Polish out surface imperfections and restore protective finish (PARA 2-6-5).

(4) Damage exceeding limits must be reviewed by engineering authority for evaluation, repair, or replacement.

c. Strut fairings are 6061-T6, AMS 4049 Alclad, and are secondary structure. Repair strut fairings as follows (Figure 16-4-20-1, Detail B):

(1) Evaluate damage (PARA 2-6-2).

(2) If damage exceeds negligible limits, repair strut fairings (PARA 2-6-3).

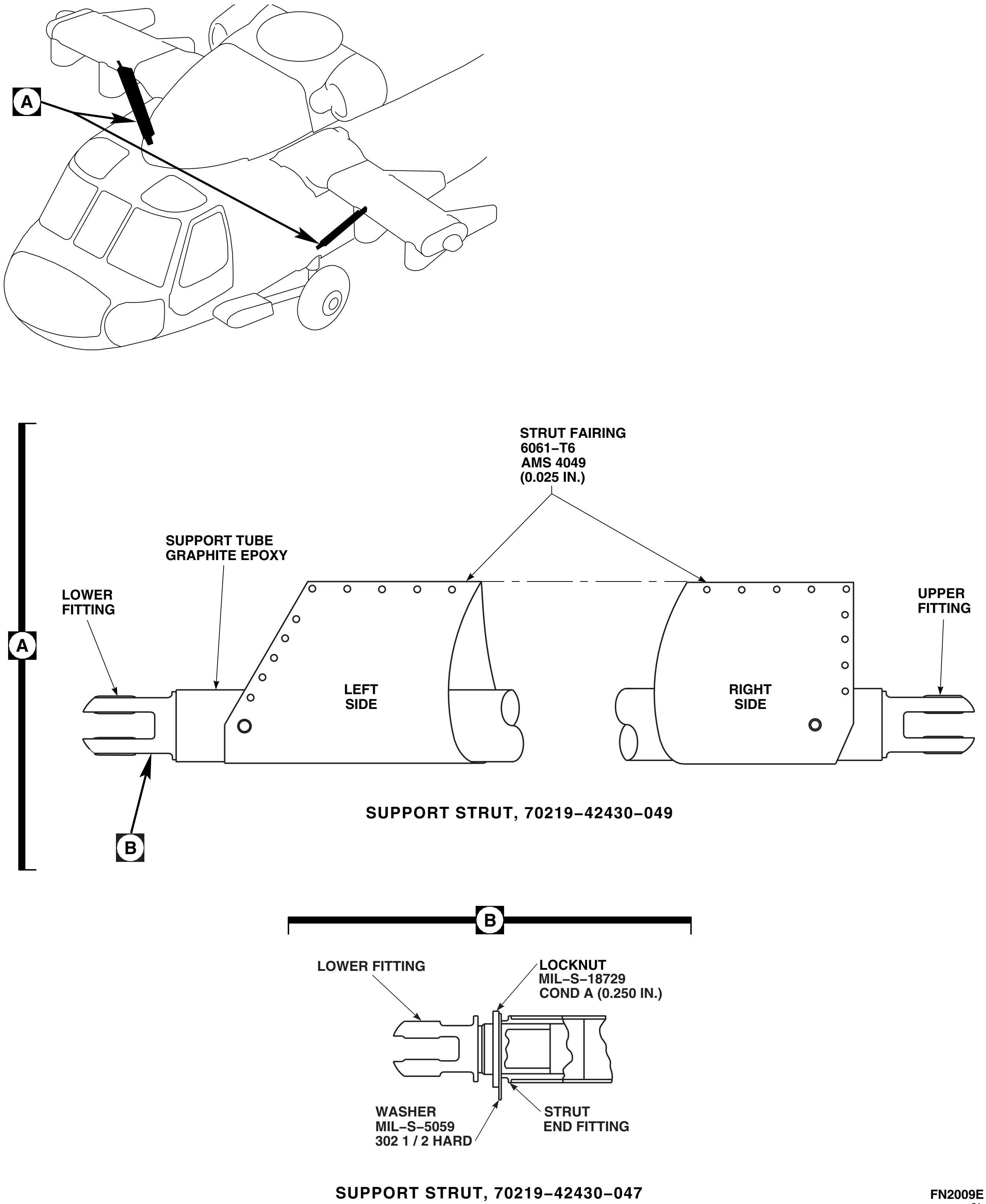


FIGURE 16-4-20-1. REPAIR ESSS HORIZONTAL STORES SUPPORT (HSS) STRUTS

16-4-21 ESSS HORIZONTAL STORES SUPPORT (HSS) FAIRINGS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made
- Nylon Wedges (or equivalent)
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Primer, Item 254, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 283, Appendix D
- Trichloroethane 1,1,1, Item 341, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20

16-4-21.1 Replace ESSS Horizontal Stores Support (HSS) Fairings.

NOTE

Replacement of left and right ESSS HSS fairings is the same, except as noted.

16-4-21.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Unlock quarter-turn fasteners and remove upper and lower root fairings from fuselage (Figure 16-4-21-1, Sheet 1, Detail A).

c. Unlock quarter-turn fasteners and remove front and rear vertical support pylon (VSP) fairings.

d. Remove screws and washers securing saddle assemblies to rear of HSS (Figure 16-4-21-1, Sheet 2, Detail B).

e. To remove leading edge fairing, perform the following:

(1) Remove screws and washers securing leading edge fairing on HSS.

(2) Remove bolts and washers at BL 80.0 and BL 112.0 securing fairing to HSS bracket.

(3) Using nonmetallic scraper and machinery towel, Item 211, Appendix D, dampened with trichloroethane 1,1,1, Item 341, Appendix D, remove sealing compound from HSS beam and fairing.

f. Remove screws, washers, and trailing edge fairings from HSS.

g. To remove tip cap fairing, perform the following:

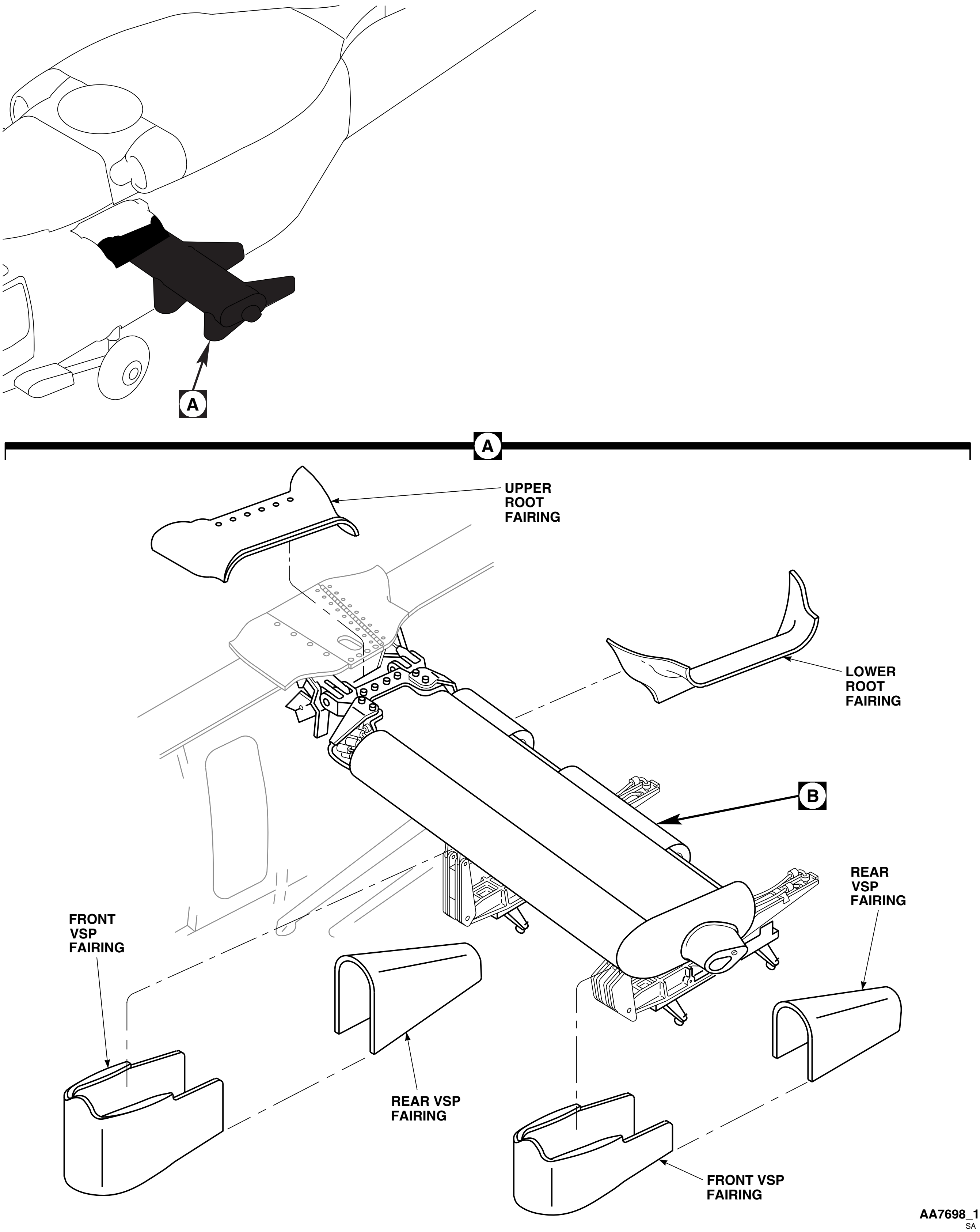


FIGURE 16-4-21-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) FAIRINGS (SHEET 1 OF 2)

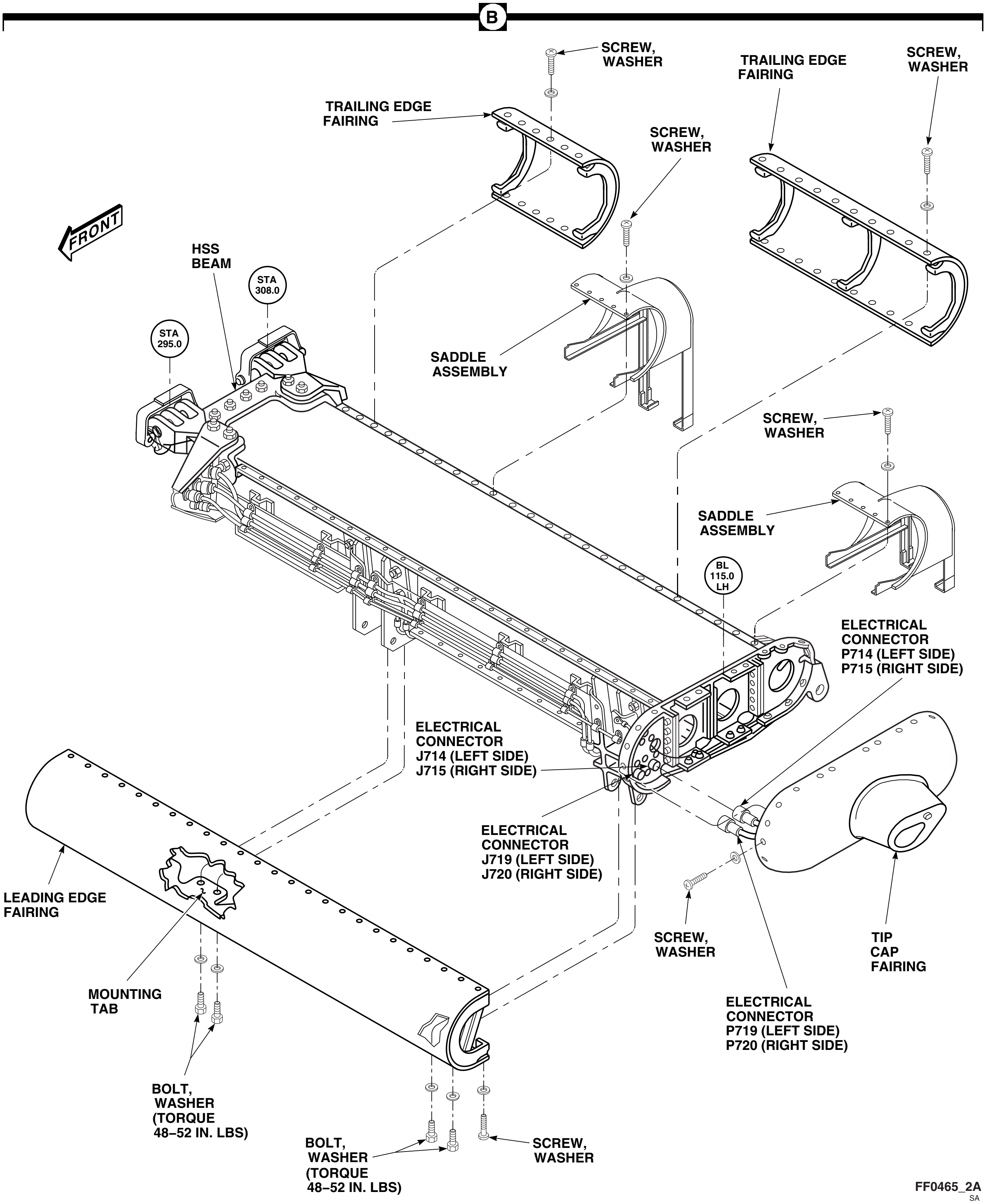


FIGURE 16-4-21-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) FAIRINGS (SHEET 2 OF 2)

NOTE

Note location of long screws.

(1) Remove screws and washers securing tip cap fairing to HSS.

(2) Using nylon wedges or equivalent, separate tip cap fairing from HSS.

(3) Disconnect incandescent and infrared position light electrical connectors, P714 and P719 (left side), or P715 and P720 (right side) from electrical connectors, J714 and J719 (left side), or J715 and J720 (right side), and remove tip cap fairing.

(4) Install protective caps and plugs on electrical connectors.

(5) Using nonmetallic scraper and clean cloth wet with trichloroethane 1,1,1, Item 341, Appendix D, remove sealing compound from HSS beam and tip cap fairing.

16-4-21.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Make sure area is clean and free of foreign material.

c. To install tip cap fairing, perform the following:

(1) Remove protective caps and plugs from electrical connectors.

(2) Inspect and treat electrical connectors (PARA 1-7-20).

(3) Connect incandescent and infrared position light electrical connectors, P714 and P719 (left side), or P715 and P720 (right side) to electrical connectors, J714 and J719 (left side), or J715 and J720 (right side) (Figure 16-4-21-1, Sheet 2, Detail B).

NOTE

Make sure contact areas on tip cap fairing and HSS are clean and dry.

(4) Apply sealing compound, Item 283, Appendix D, to HSS beam where tip cap fairing contacts beam.

(5) Install tip cap fairing on HSS with screws and washers wet with primer, Item 254, Appendix D.

d. Install trailing edge fairings with screws and washers wet with primer, Item 254, Appendix D.

e. To install leading edge fairing, perform the following:

NOTE

Make sure contact areas on leading edge fairing and HSS are clean and dry. Do not apply sealing compound to attaching edges of HSS beam or fairings.

(1) Apply sealing compound, Item 283, Appendix D, to HSS beam where mounting tabs of fairing rest against beam.

(2) Install leading edge fairing on HSS with screws and washers wet with primer, Item 254, Appendix D.

(3) Install leading edge fairing on HSS at BL 80.0 and BL 112.0 with bolts and washers wet with primer, Item 254, Appendix D. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

f. Install saddle assembly to trailing edge of HSS with screws and washers.

g. Install front and rear VSP fairings with quarter-turn fasteners (Figure 16-4-21-1, Sheet 1, Detail A).

h. Install and secure lower root fairing with quarter-turn fasteners.

i. Install and secure upper root fairings with quarter-turn fasteners.

j. Make sure area is clean and free of foreign material.

16-4-22 ESSS HORIZONTAL STORES SUPPORT (HSS) FAIRINGS REPAIR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit

References

- PARA 2-6-1
- PARA 2-6-2
- PARA 2-6-3
- PARA 2-6-5
- PARA 16-4-21
- QQ-A-250

16-4-22.1 Repair ESSS Horizontal Stores Support (HSS) Fairings.

NOTE

- All repairs are to be performed with fairings removed.
- Repair of left and right ESSS HSS fairings is the same.

a. If required, remove external stores support system (ESSS) HSS fairings (PARA 16-4-21).

b. Leading edge fairing, trailing edge fairings, and saddle assemblies consist of stringers, supports and ribs, made from aluminum alloy, 7075-T6, QQ-A-250/13 (Figure 16-4-22-1, Sheet 1, Detail A and Detail B, and Sheet 2, Detail C and Detail D). Fairing skin is aluminum alloy, 7075-T6, QQ-A-250/13. Saddle assembly skin is fiberglass composite. Fairings and saddle assemblies are secondary structure.

c. HSS tip cap consists of ribs and angles made from aluminum alloy, 7075-T6, QQ-A-250/13. HSS tip cap skin is aluminum alloy, 6061-T6, QQ-A-250/11. Standoff is fiberglass composite. Tip cap is secondary structure (Figure 16-4-22-1, Sheet 1, Detail B).

d. HSS root fairing is aluminum alloy, 6061-T6, QQ-A-250/11, honeycomb panel. Root fairing is secondary structure (Figure 16-4-22-1, Sheet 1).

e. Vertical stores pylon (VSP) front and rear fairings consist of angles and supports made from aluminum alloy, 7075-T6, QQ-A-250/13. Front VSP fairing skin is aluminum alloy, 7075-T6, QQ-A-250/13. Rear VSP fairing skin is aluminum alloy, 6061-T6, QQ-A-250/11. These are secondary structures (Figure 16-4-22-1, Sheet 2, Detail D).

f. Refer to PARA 2-6-1 when evaluating damage to aluminum skin. If damage exceeds negligible limits, repair aluminum skin (PARA 2-6-2).

g. Refer to PARA 2-6-1 when evaluating damage to supports and angles. If damage exceeds negligible limits, replace part.

h. Refer to PARA 2-6-3 when making composite repairs.

i. Touch up paint (PARA 2-6-5).

j. If required, install external stores support system (ESSS) HSS fairings (PARA 16-4-21).

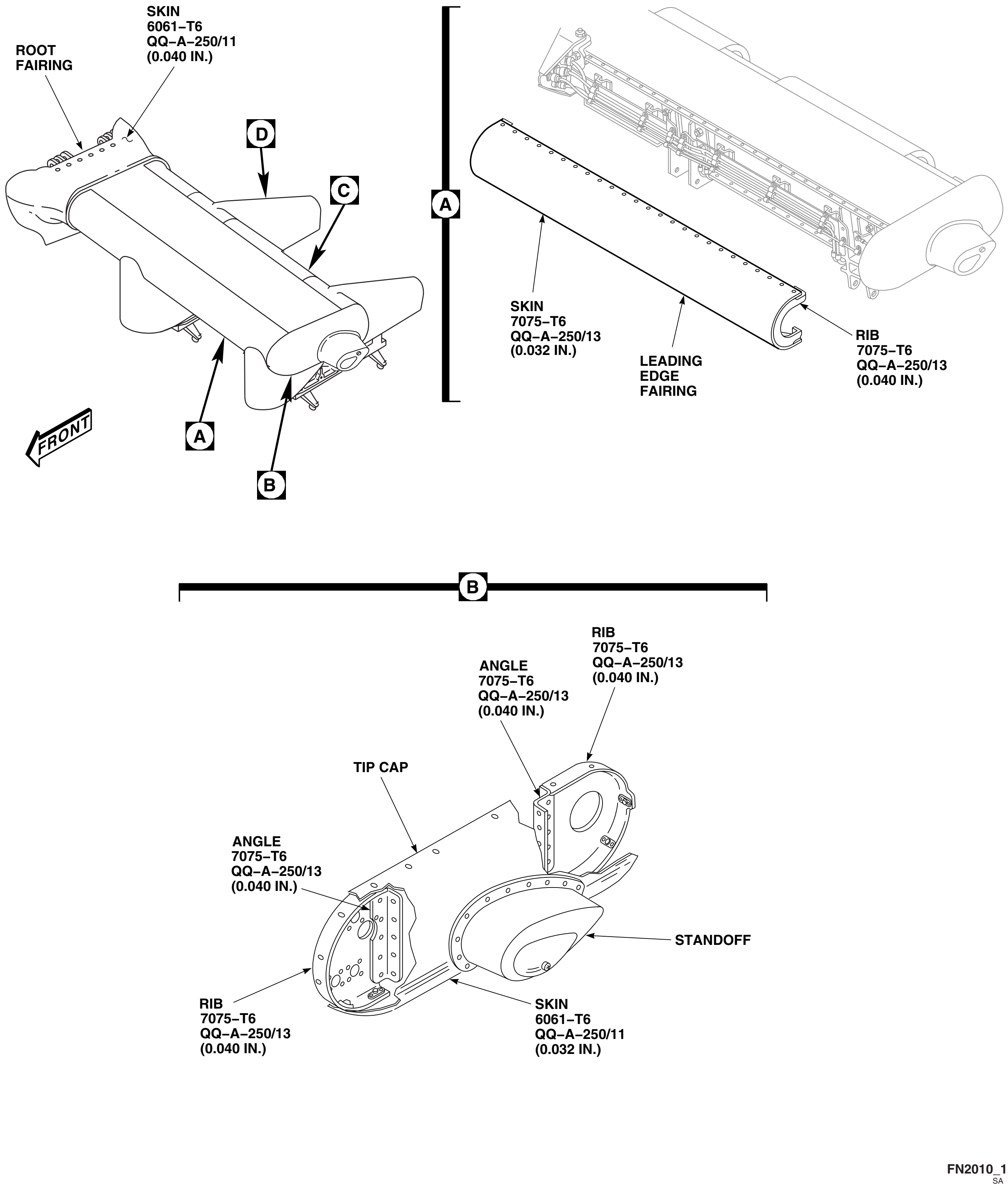
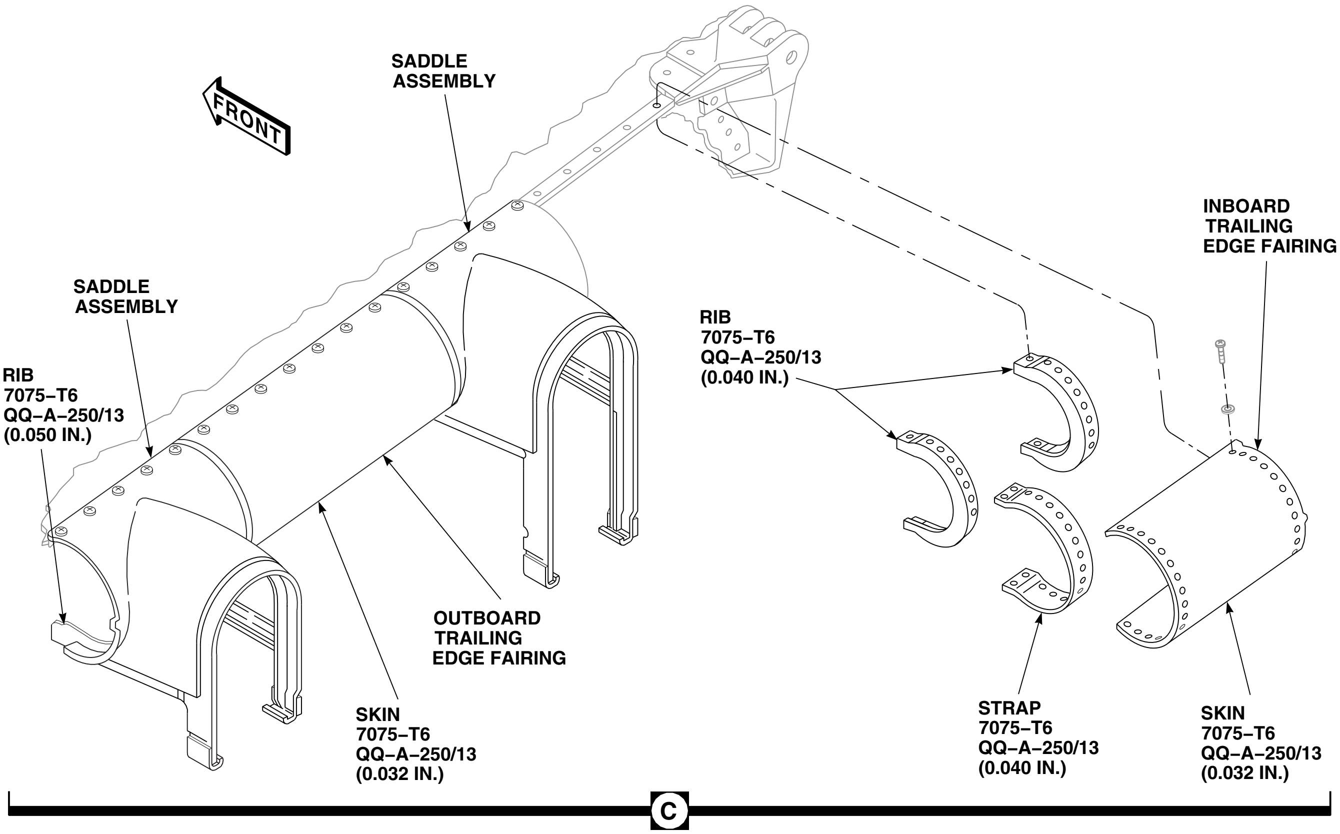


FIGURE 16-4-22-1. REPAIR ESSS HORIZONTAL STORES SUPPORT (HSS) FAIRINGS (SHEET 1 OF 2)



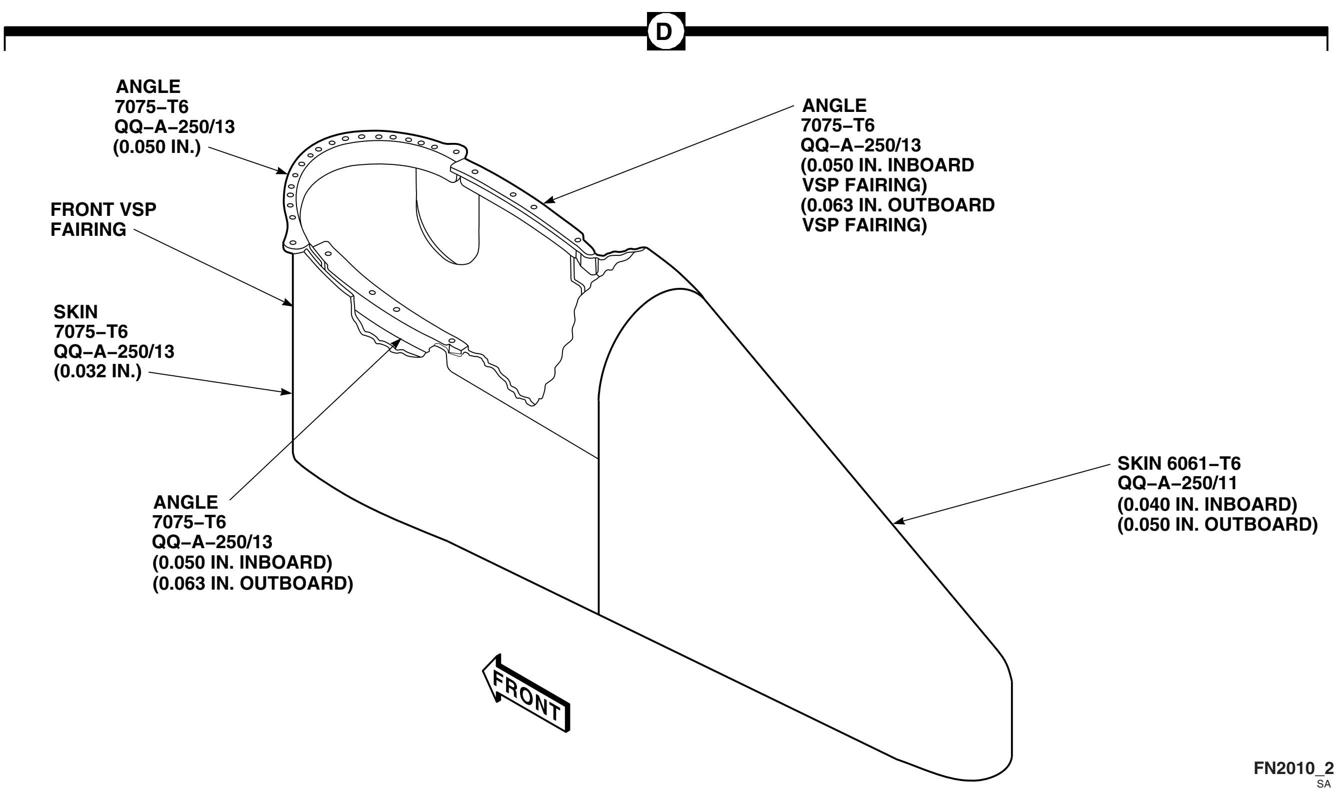


FIGURE 16-4-22-1. REPAIR ESSS HORIZONTAL STORES SUPPORT (HSS) FAIRINGS (SHEET 2 OF 2)

16-4-23 ESSS HORIZONTAL STORES SUPPORT (HSS) FRAMING REPAIR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Drill Set
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Adhesive, Item 42B, Appendix D
- Cheesecloth, Item 90, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- MIL-P-15035
- PARA 2-6-5
- PARA 16-4-21
- PARA 16-6-7
- PARA 16-6-8
- QQ-A-200
- QQ-A-250
- QQ-A-367

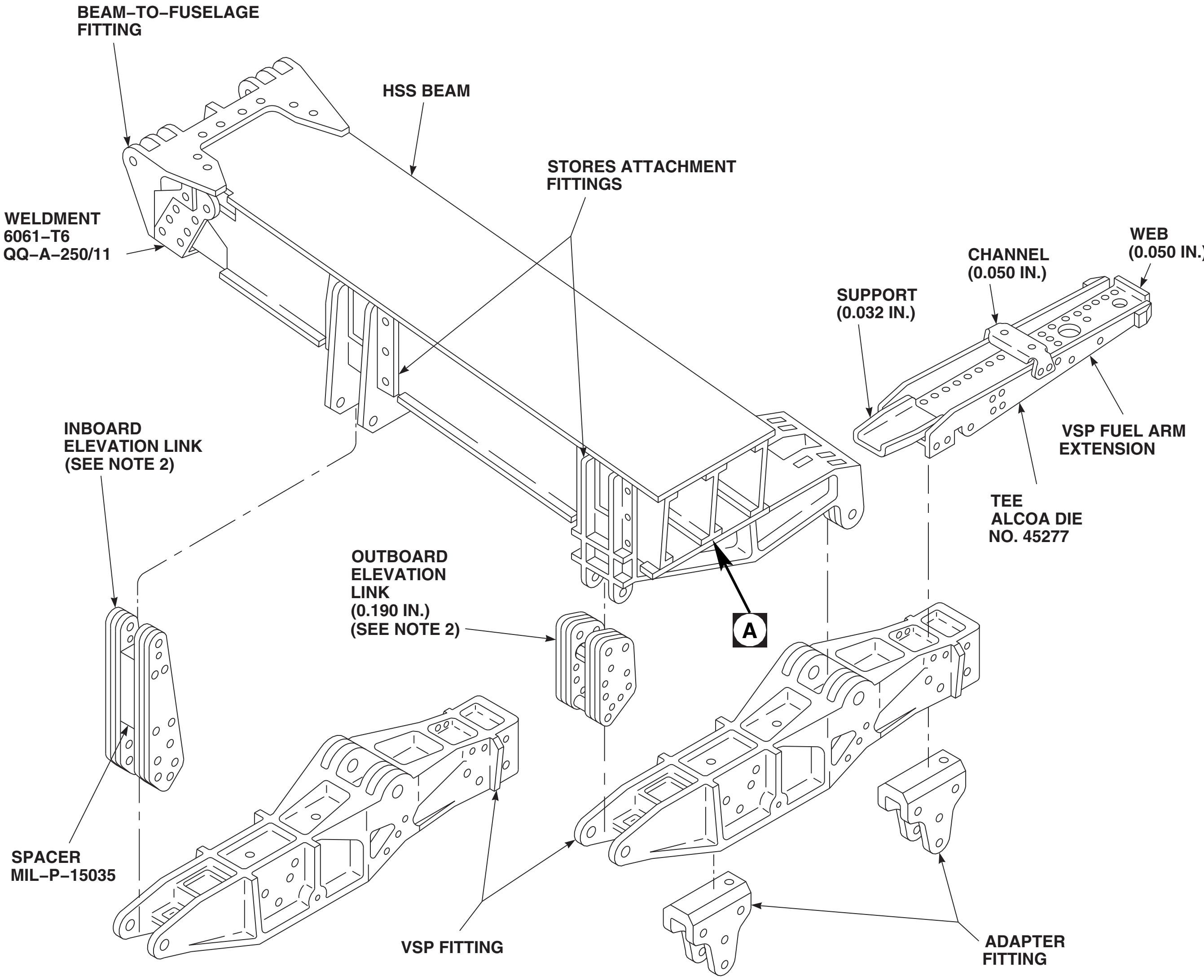
16-4-23.1 Repair ESSS Horizontal Stores Support (HSS) Framing.

NOTE

- All repairs are to be performed with HSS removed.
- Repair of left and right ESSS HSS framing is the same.

a. If required, remove horizontal stores support (HSS) (PARA 16-6-8).

b. HSS beam consists of graphite/epoxy laminate box assembly and its fittings (Figure 16-4-23-1,



NOTES

1. UNLESS OTHERWISE NOTED, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075–T6, QQ–A–250/13, AND ALL FORGED PARTS ARE 7075–T73, QQ–A–367.
2. LINKS ARE 7075–T6, QQ–A–250/12.
3. WRINKLES MAY BE FOUND IN THE LONGITUDINAL AND/OR TRANSVERSE DIRECTION.

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FIGURE 16-4-23-1. REPAIR ESSS HORIZONTAL STORES SUPPORT (HSS) FRAMING (SHEET 1 OF 2)

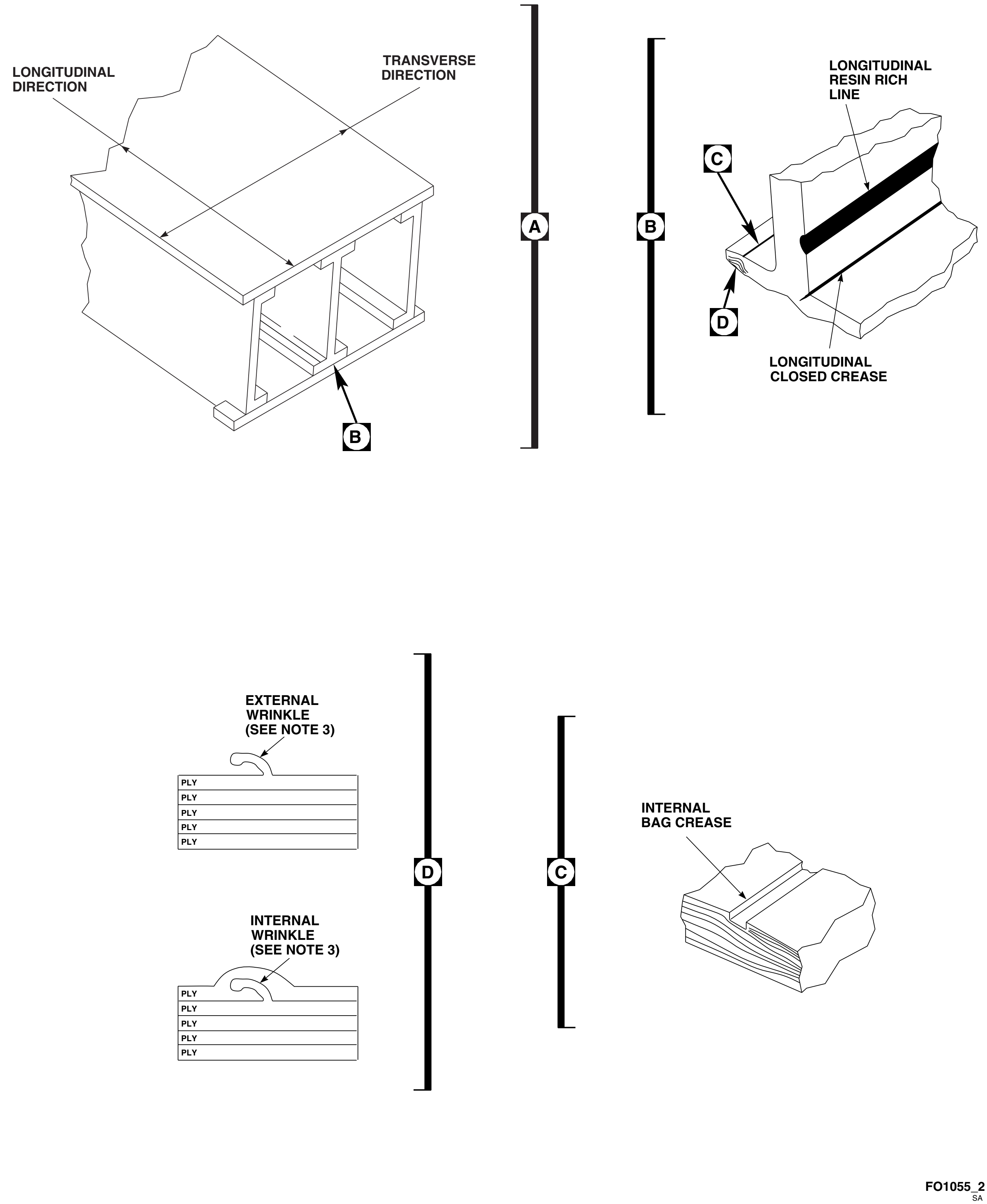


FIGURE 16-4-23-1. REPAIR ESSS HORIZONTAL STORES SUPPORT (HSS) FRAMING (SHEET 2 OF 2)

Sheet 1). Beam-to-fuselage fitting secures to fittings on cabin framing at STA 295.0 and STA 308.0. Stores attachment fittings are fixed to beam's lower surface at BL 80.0 and BL 112.0. These fittings, and vertical support pylon (VSP) fittings and adapters, are machined from 7075 aluminum alloy forgings, heat-treated to 7075-T73, QQ-A-367. The VSP fuel arm extension is made up of 7075-T6, QQ-A-200/11 extrusions and a web/support of 7075-T6, QQ-A-250/13. Inboard and outboard elevation links are made up of 7075-T6, QQ-A-250/12 links and MIL-P-15035 spacers.

c. HSS beam is primary structure. Scratches up to 0.005-inch deep and any length, or gouges and digs up to 0.005-inch and any area size, are considered negligible damage. Repair is limited to paint touchup (PARA 2-6-5).

d. Visually inspect HSS beam for material defects in the form of lateral wrinkles or creases. **WRINKLE/CREASE ON INTERNAL OR EXTERNAL SURFACE LESS THAN 0.030-INCH IN DEPTH/HEIGHT OR ANY LENGTH ALLOWED** (Figure 16-4-23-1, Sheet 2, Detail A, Detail C, and Detail D).

e. Visually inspect HSS beam for longitudinal wrinkles or creases. **WRINKLE/CREASE ON INTERNAL OR EXTERNAL SURFACE LESS THAN 0.060-INCH IN DEPTH/HEIGHT OR ANY LENGTH ALLOWED.**

f. Visually inspect longitudinal closed creases on HSS beam. **LONGITUDINAL CLOSED CREASES AT PREPLY INTERFACES LESS THAN 0.20-INCH IN DEPTH ALLOWED** (Figure 16-4-23-1, Sheet 2, Detail B).

g. Fittings, elevation links, VSP fittings, and VSP fuel arm extension are considered primary structure. Negligible damage to fittings, elevation links, VSP fittings, or VSP fuel arm extension is limited to nicks, scratches, and corrosion, no deeper than 5% of material thickness at point of damage. Blend damage smoothly to surrounding area, using long, even strokes. Polish out all surface imperfections and restore protective finish. Damage exceeded negligible limits in primary areas must be reviewed by engineering authority for damage evaluation, repair or replacement.

h. If required, install horizontal stores support (HSS) (PARA 16-6-7).

16-4-23.2 Replace ESSS Horizontal Stores Support (HSS) Beam Ground Straps and Tabs.

NOTE

Replacement of left and right ESSS HSS beam ground straps and tabs is the same, except as noted.

16-4-23.2.1 Remove.

NOTE

If possible, use care when removing ground straps or tabs. If strap or tab is removed without extensive damage, retain for use as a template for replacement.

a. Remove horizontal stores support (HSS) fairings as required (PARA 16-4-21).

b. Using nonmetallic scraper, remove sealing compound from edges of ground strap(s).

c. Remove ground strap, 70200-42401-134 or -139, as follows (Figure 16-4-23-2, Sheet 1, Detail A):

(1) Drill out rivets securing nutplates to strap and HSS beam flange. Remove nutplates.

(2) Remove strap from HSS beam flange using nonmetallic scraper.

d. Remove ground strap, 70200-42401-135, -136, -137, or 138, as follows:

(1) Using nonmetallic scraper, remove sealing compound from hardware securing bonding jumper.

(2) Remove screw, lockwasher, washers, and nut securing bonding jumper to clip on HSS beam (Figure 16-4-23-2, Sheet 2, Detail B or Detail C).

(3) Drill out rivets securing strap and clip to flange. Remove clip.

(4) If removing strap, 70200-42401-135 or -136, perform the following:

(a) Drill out rivets securing ground tab, 70200-42401-143, to strap and HSS beam flange (Figure 16-4-23-2, Sheet 1, Detail A).

(b) Using nonmetallic scraper, remove tab from strap.

(5) Drill out rivets securing nutplates to strap and HSS beam flange. Remove nutplates.

(6) Remove strap from HSS beam flange using nonmetallic scraper.

e. Using nonmetallic scraper, remove ground strap, 70200-42401-140 or -141, from HSS beam flange.

f. If required, remove ground tab, 70200-42401-143, as follows:

(1) Drill out rivets securing tab to strap, 70200-42401-135 or -136, on HSS beam flange.

NOTE

Excessive force may cause strap to become disbonded. Do not use excessive force when removing tab.

(2) Using nonmetallic scraper, carefully remove tab from strap.

g. Using nonmetallic scraper, remove adhesive and sealing compound from HSS beam. Remove remaining adhesive and sealing compound residue with cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean cheesecloth, Item 90, Appendix D.

16-4-23.2.2 Install.

a. Prepare replacement ground strap or ground tab for installation as follows:

NOTE

- When using HSS beam to mark holes, ground strap/tab will need to be positioned on top or below mating flange in order to access holes. Make sure strap/tab is properly positioned.
- If installing ground strap, 70200-42401-140 or -141, or ground tab, 70200-42401-143, do not drill outboard hole that mates with HSS tip cap. This hole must be drilled after strap/tab is installed.

(1) Using either removed ground strap/tab as a template or by positioning replacement strap on HSS beam, mark holes for drilling (Figure 16-4-23-2, Sheet 1, Detail A and Sheet 2, Detail B and Detail C).

(2) Using drill and suitable size drill bit, drill holes as required. Refer to Figure 16-4-23-3, Sheet 1, Sheet 2, Detail A and Detail B, and Sheet 3, Detail C and Detail D. Deburr holes.

(3) Countersink holes where required. Deburr holes.

b. Install ground strap, 70200-42401-134 or -139, as follows (Figure 16-4-23-2, Sheet 1, Detail A):

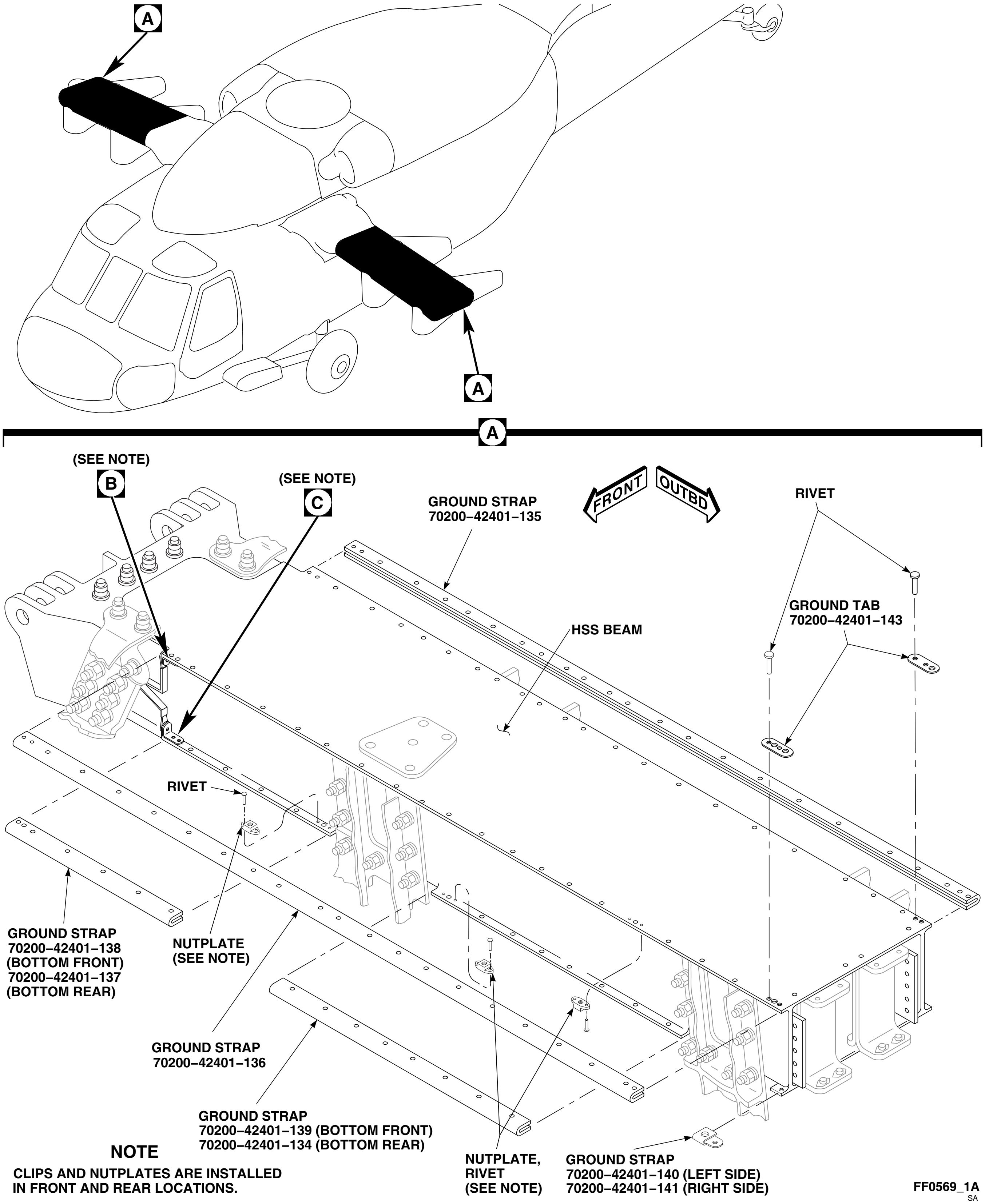
(1) Apply adhesive, 42B, Appendix D, to ground strap and flange mating surfaces.

(2) Install strap on HSS beam flange. Make sure strap makes good contact with flange and holes are aligned.

(3) Remove adhesive squeezeout from holes and around ground strap.

(4) Allow adhesive to cure for 2 hours at 140°F (60°C) or for 24 hours at room temperature.

(5) Position nutplates on strap and secure with rivets. Wet install rivets with sealing compound, Item 292, Appendix D.



NOTE

CLIPS AND NUTPLATES ARE INSTALLED IN FRONT AND REAR LOCATIONS.

FIGURE 16-4-23-2. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) BEAM GROUND STRAPS AND TABS (SHEET 1 OF 2)

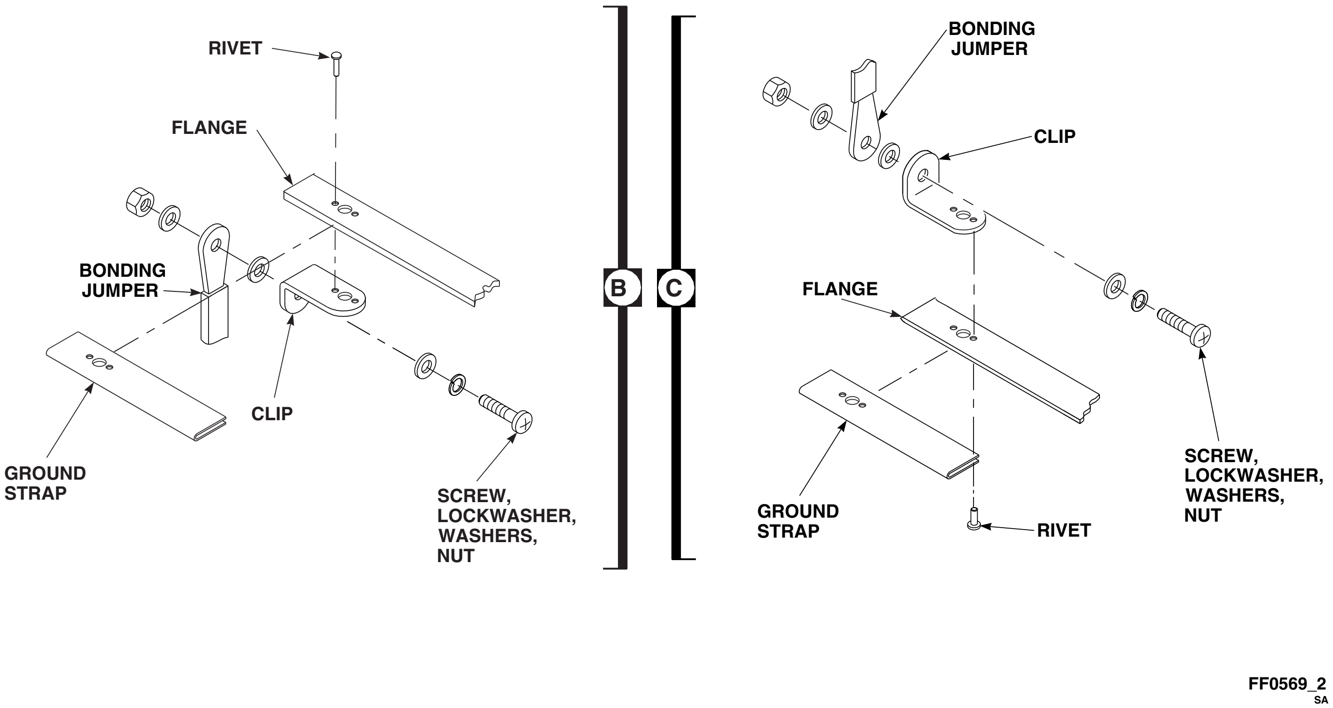


FIGURE 16-4-23-2. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) BEAM GROUND STRAPS AND TABS (SHEET 2 OF 2)

c. Install ground strap, 70200-42401-135, -136, -137, or -138, as follows:

(1) Apply adhesive, 42B, Appendix D, to ground strap and flange mating surfaces.

(2) Install strap on HSS beam flange. Make sure strap makes good contact with flange and holes are aligned.

(3) Remove adhesive squeezeout from holes and around ground strap.

(4) Allow adhesive to cure for 2 hours at 140°F (60°C) or for 24 hours at room temperature.

(5) Position nutplates on strap and secure with rivets. Wet install rivets with sealing compound, Item 292, Appendix D.

NOTE

Do not allow methyl ethyl ketone to dry on parts.

(6) Clean bonding surfaces with cleaning cloth, Item 102, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D. Wipe excess methyl ethyl ketone before it dries.

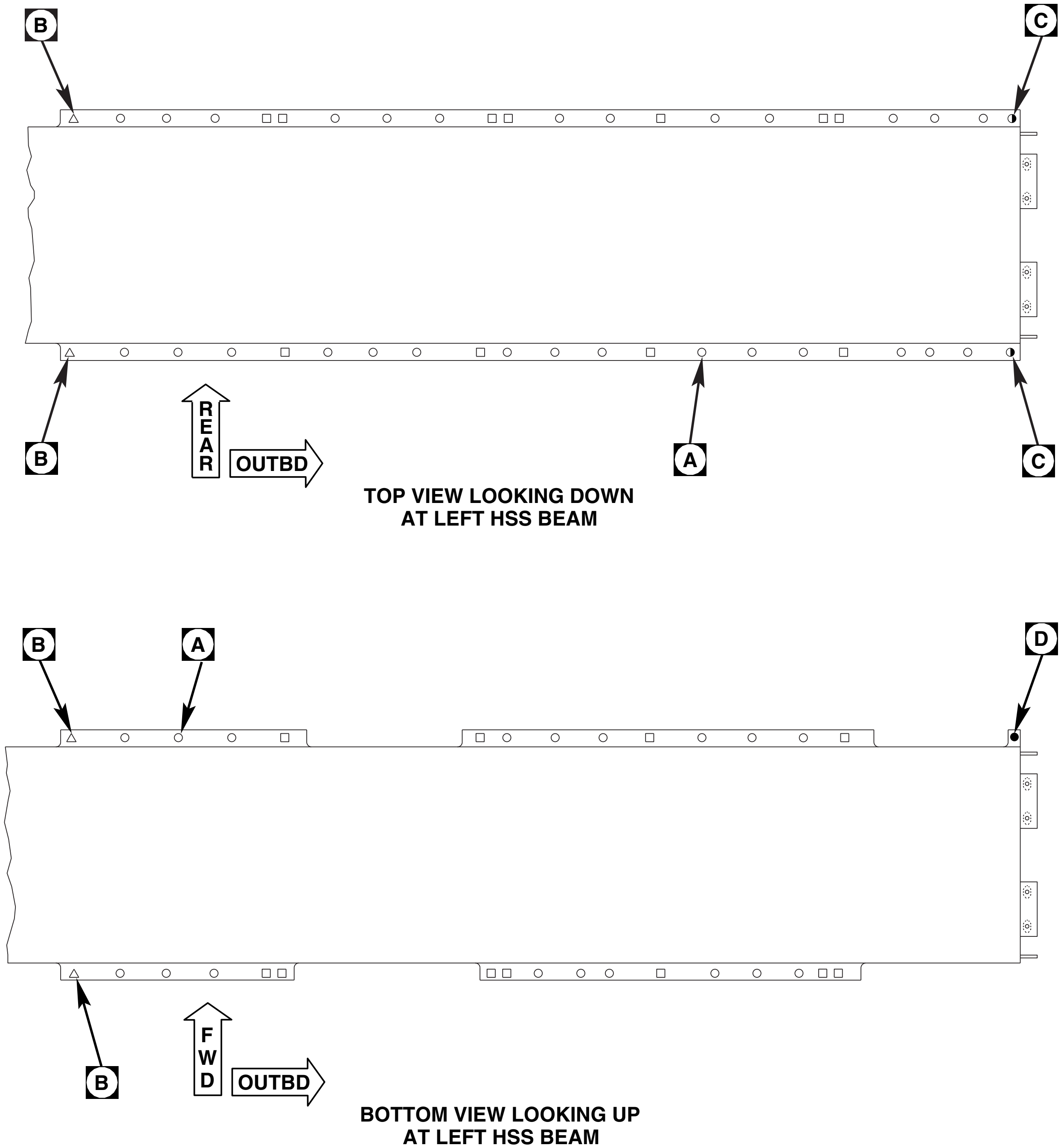
(7) Position clip on flange, aligning rivet and bonding jumper mounting holes, and secure with rivets (Figure 16-4-23-2, Sheet 2, Detail B or Detail C). Wet install rivets with sealing compound, Item 292, Appendix D.

(8) Install bonding jumper and washer on clip and secure with screw, lockwasher, washers, and nut.

(9) Apply sealing compound, Item 292, Appendix D, to completely cover screw, lockwasher, washers, and nut.

(10) If installing ground strap, 70200-42401-135 or -136, install ground tab. Proceed to step e.

d. Install ground strap, 70200-42401-140 or -141, as follows (Figure 16-4-23-2, Sheet 1, Detail A):



NOTES

1. COUNTERSINK NEAR SIDE ONLY.
2. 0.218 – 0.238 INCH–DIAMETER HOLES APPLIES TO GROUND STRAP/TAB IN FRONT LOCATIONS ONLY.

LEGEND

□ 0.218 – 0.238 INCH–DIAMETER
○ SEE DETAIL A
△ SEE DETAIL B
◑ SEE DETAIL C
● SEE DETAIL D

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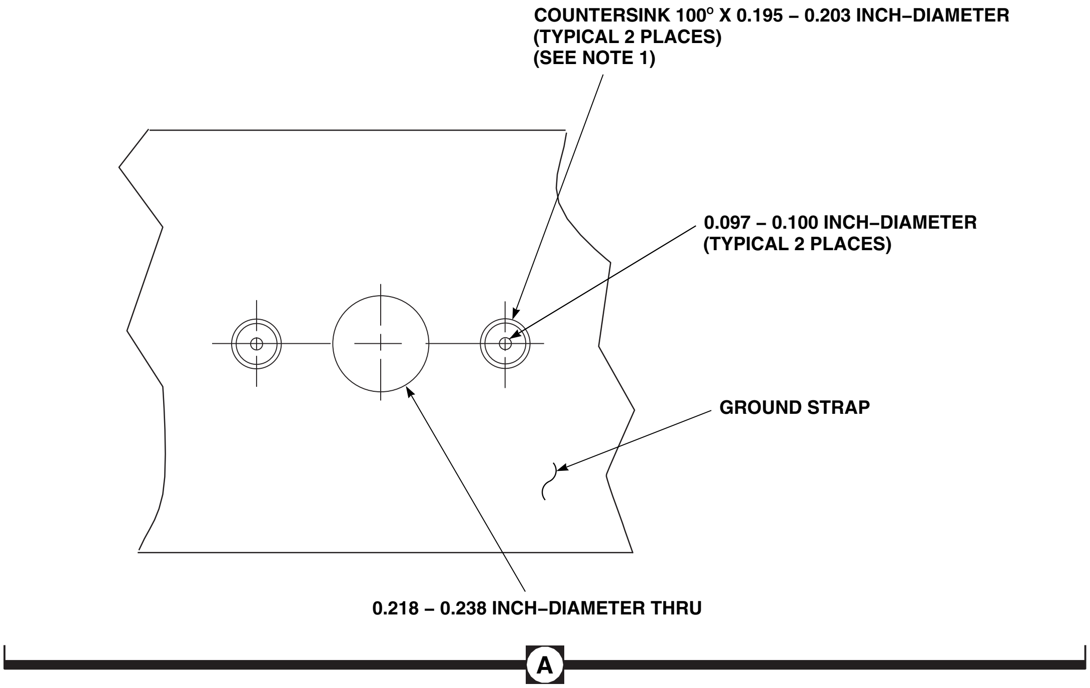
FIGURE 16-4-23-3. GROUND STRAPS AND TABS HOLE MARKING AND DRILLING (SHEET 1 OF 3)



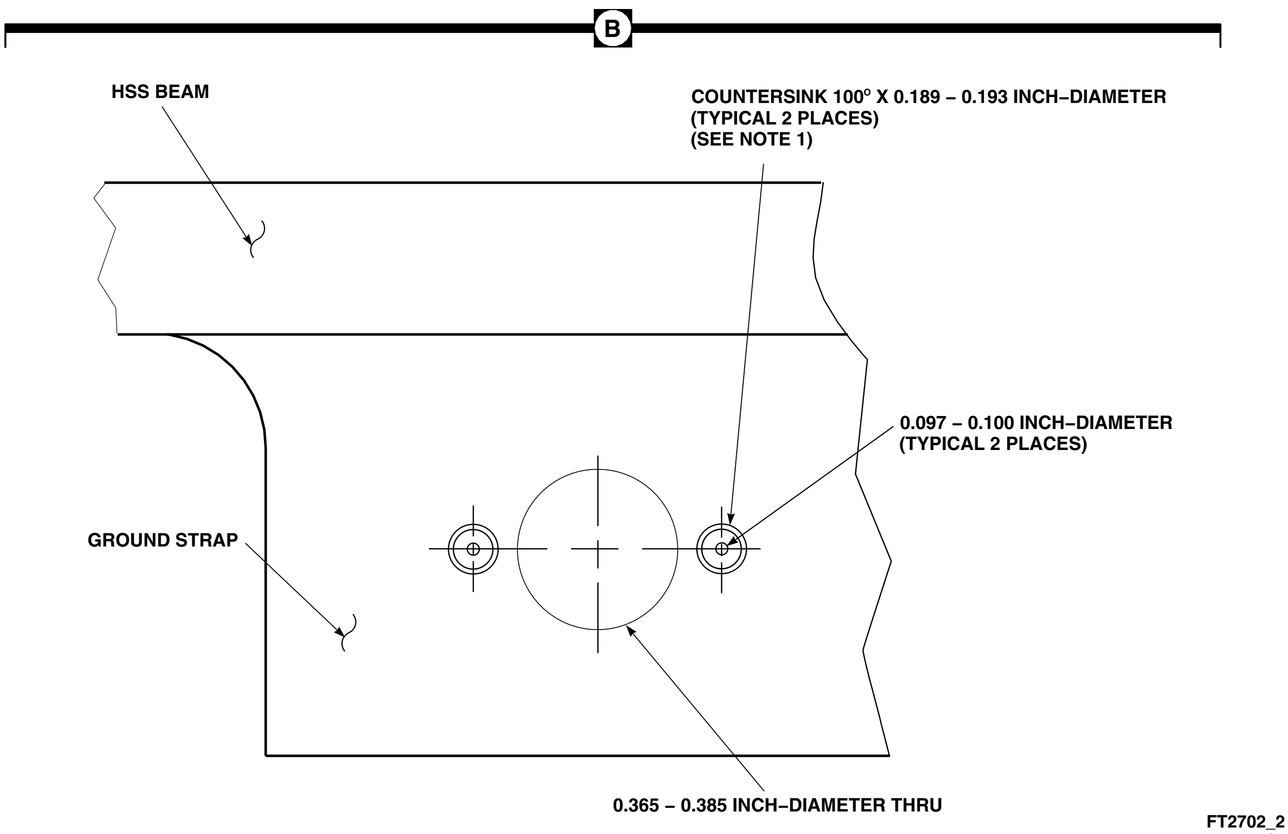


FIGURE 16-4-23-3. GROUND STRAPS AND TABS HOLE MARKING AND DRILLING (SHEET 2 OF 3)

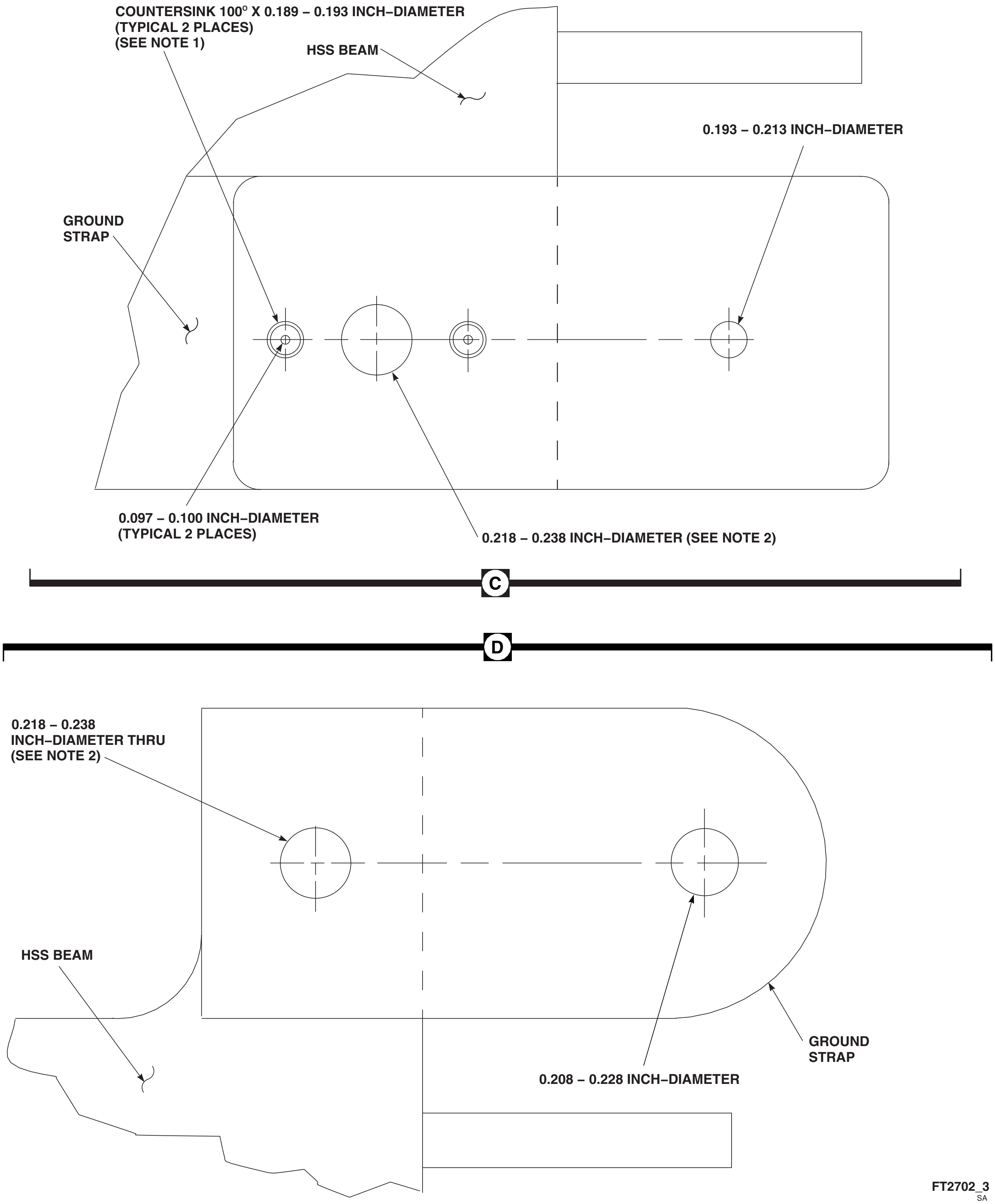


FIGURE 16-4-23-3. GROUND STRAPS AND TABS HOLE MARKING AND DRILLING (SHEET 3 OF 3)

(1) Apply adhesive, 42B, Appendix D, to ground strap and flange mating surfaces.

(2) Install strap on flange of HSS beam. Make sure strap makes good contact with flange and holes are aligned.

(3) Remove adhesive squeezeout from holes and around ground strap.

(4) Allow adhesive to cure for 2 hours at 140°F (60°C) or for 24 hours at room temperature.

(5) Position HSS tip cap fairing on HSS beam and mark hole on strap. Remove fairing.

NOTE

Use care not to bend strap when drilling hole.

(6) Drill marked hole using drill and suitable size drill bit. Refer to Figure 16-4-23-3, Sheet 3, Detail D for hole size.

e. If required, install ground tab, 70200-42401-143, as follows (Figure 16-4-23-2, Sheet 1, Detail A):

(1) Apply adhesive, 42B, Appendix D, to ground tab and ground strap mating surfaces.

(2) Install ground tab on strap, 70200-42401-135 or -136. Make sure tab makes good contact with strap and holes are aligned.

(3) Remove adhesive squeezeout from holes and around ground tab.

(4) Allow adhesive to cure for 2 hours at 140°F (60°C) or for 24 hours at room temperature.

(5) Secure tab to strap and HSS beam flange with rivets wet with sealing compound, Item 292, Appendix D.

(6) Position HSS tip cap fairing on HSS beam and mark hole on tab. Remove fairing.

NOTE

Use care not to bend tab when drilling hole.

(7) Drill marked hole using drill and suitable size drill bit. Refer to Figure 16-4-23-3, Sheet 3, Detail C for hole size.

NOTE

- If strap is bonded in place, make sure adhesive has fully cured before applying sealing compound.
- Sealing compound is not applied to ground tabs.

f. Apply fillet of sealing compound, 292, Appendix D, around edges of ground strap(s).

g. Install horizontal stores support (HSS) fairings as required (PARA 16-4-21).

16-4-24 ESSS HORIZONTAL STORES SUPPORT (HSS) ROOT-TO-FUSELAGE FUEL HOSE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 16-4-21

16-4-24.1 Replace ESSS Horizontal Stores Support (HSS) Root-to-Fuselage Fuel Hose.

NOTE

Replacement of left and right ESSS HSS root-to-fuselage fuel hoses is the same.

16-4-24.1.1 Remove.

a. Remove upper and lower root fairings and inboard trailing edge fairing (PARA 16-4-21).

b. Remove screws, washers, lockwasher, bonding jumper, nuts, and upper and lower clamps securing fuel hose to pneumatic hose (Figure 16-4-24-1, Sheet 1, Detail B).

c. Remove bolt, washers, lockwasher, and nut securing bonding jumpers to HSS root fitting (Figure 16-4-24-1, Sheet 2, Detail C).

NOTE

Install protective caps and plugs on fuel hoses and quick-disconnect valve halves as they are disconnected.

d. Disconnect quick-disconnect valve half on fuel hose from quick-disconnect valve half on fuselage fitting (Figure 16-4-24-1, Sheet 1, Detail B).

e. Loosen B-nut and remove quick-disconnect valve half from fuel hose.

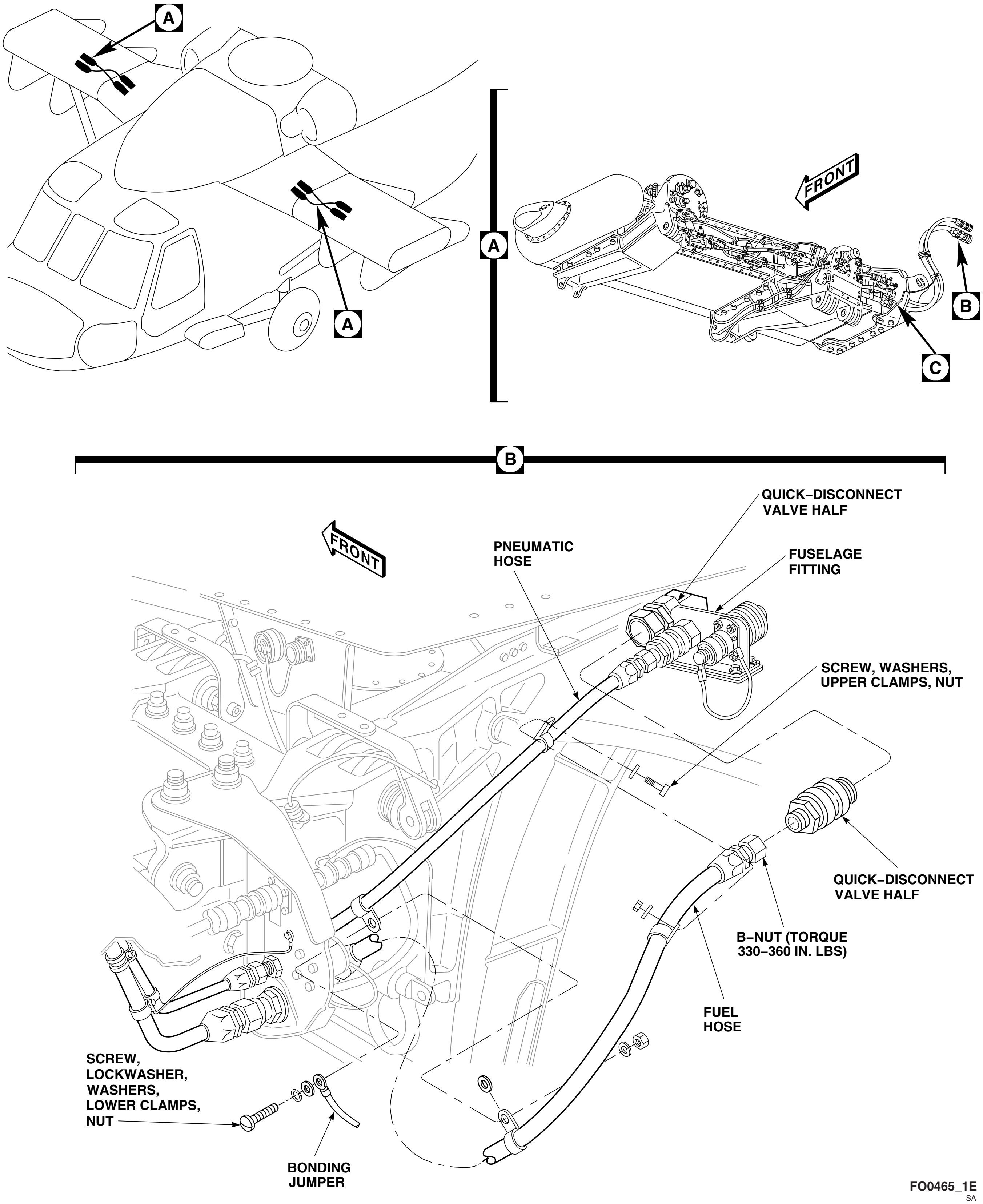


FIGURE 16-4-24-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) ROOT-TO-FUSELAGE FUEL HOSE (SHEET 1 OF 2)

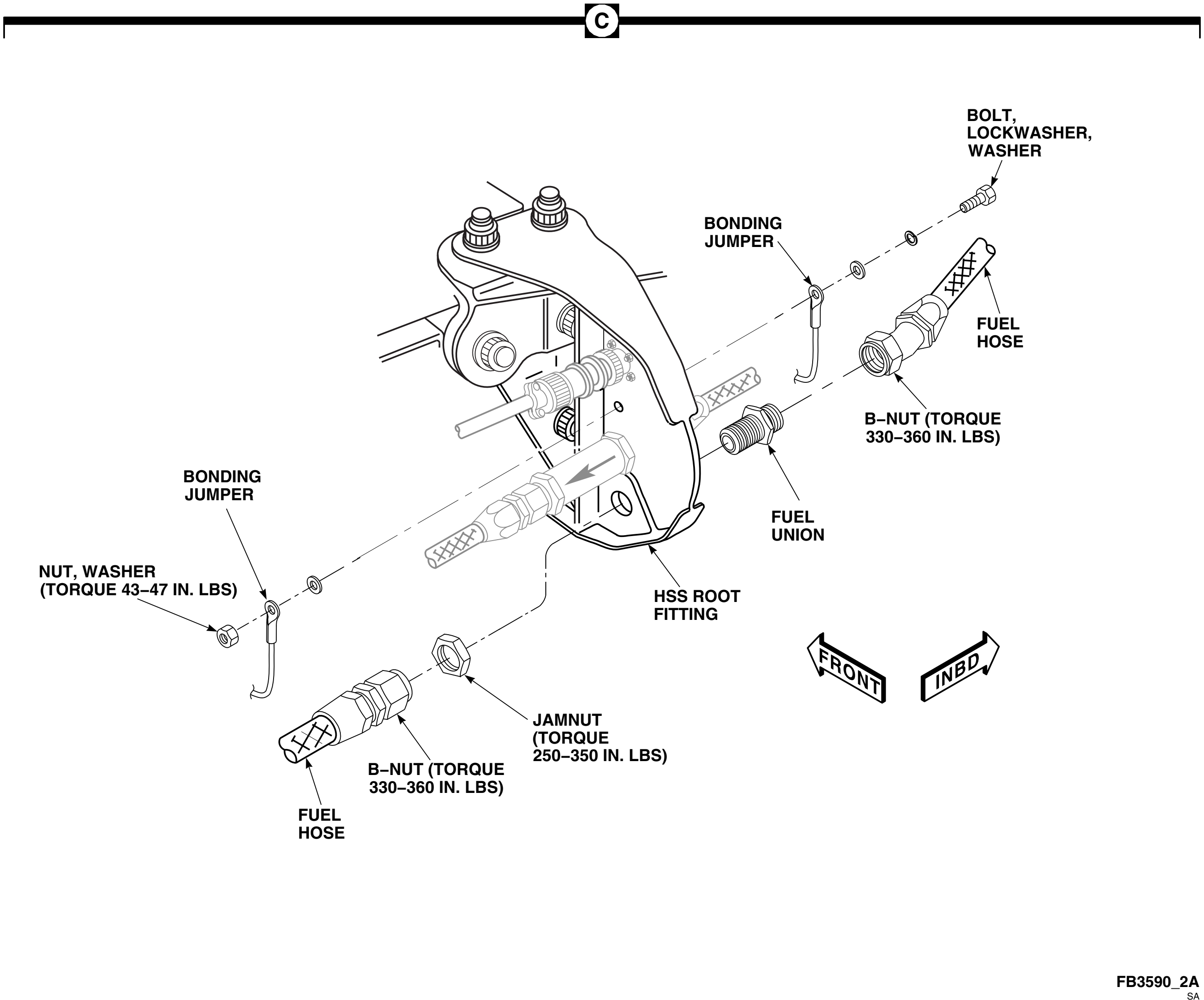


FIGURE 16-4-24-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) ROOT-TO-FUSELAGE FUEL HOSE (SHEET 2 OF 2)

f. Disconnect fuel hose from fuel union on outboard side of HSS root fitting (Figure 16-4-24-1, Sheet 2, Detail C).

g. Disconnect fuel hose from fuel union on inboard side of HSS root fitting.

h. Remove jamnut and remove fuel union from HSS root fitting.

16-4-24.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

NOTE

Remove protective caps and plugs from fuel hoses and quick-disconnect valve halves as they are connected.

b. Connect quick-disconnect valve half to fuel hose to quick-disconnect valve half on fuselage fitting (Figure 16-4-24-1, Sheet 1, Detail B).

c. Coat threads of fuel quick-disconnect valve half and fuel union with antiseize compound, Item 74, Appendix D.

d. Install fuel union to HSS root fitting with jamnut (Figure 16-4-24-1, Sheet 2, Detail C). **TORQUE JAMNUT TO 250 - 350 INCH-POUNDS.**

e. Connect fuel hose to fuel union on outboard side of HSS root fitting. **TORQUE B-NUT TO 330 - 360 INCH-POUNDS.**

f. Connect fuel hose to fuselage quick-disconnect valve half and fuel union. **TORQUE B-NUT TO 330 - 360 INCH-POUNDS** (Figure 16-4-24-1, Sheet 1, Detail B, and Sheet 2, Detail C).

g. Install bonding jumpers to HSS root fitting using bolt, washers, lockwasher, and nut. **TORQUE NUT TO 43 - 47 INCH-POUNDS** (Figure 16-4-24-1, Sheet 2, Detail C). Apply sealing compound, Item 292, Appendix D.

h. Install lower clamps and bonding jumper on pneumatic and fuel hoses using screws, washers, lockwasher, and nut (Figure 16-4-24-1, Sheet 1, Detail B).

i. Install upper clamps on pneumatic and fuel hoses using screw, washer, and nut.

j. Install upper and lower root fairing and inboard trailing edge fairing (PARA 16-4-21).

16-4-25 ESSS HORIZONTAL STORES SUPPORT (HSS) ROOT-TO-FUSELAGE PNEUMATIC HOSE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 16-4-21

16-4-25.1 Replace ESSS Horizontal Stores Support (HSS) Root-to-Fuselage Pneumatic Hose.

NOTE

Replacement of left and right ESSS HSS root-to-fuselage pneumatic hoses is the same.

16-4-25.1.1 Remove.

a. Remove upper and lower root fairings and inboard trailing edge fairing (PARA 16-4-21).

b. Remove screws, washers, lockwasher, bonding jumper, nuts, and upper and lower clamps securing fuel hose to pneumatic hose (Figure 16-4-25-1, Sheet 1, Detail B).

c. Remove bolt, washers, lockwasher, and nut securing bonding jumpers to HSS root fitting (Figure 16-4-25-1, Sheet 2, Detail C).

NOTE

Install protective caps and plugs on pneumatic check valve and quick-disconnect valve halves as they are disconnected.

d. Disconnect quick-disconnect valve half on pneumatic hose from quick-disconnect valve half on fuselage fitting (Figure 16-4-25-1, Sheet 1, Detail B).

e. Loosen B-nut and remove quick-disconnect valve half from pneumatic hose.

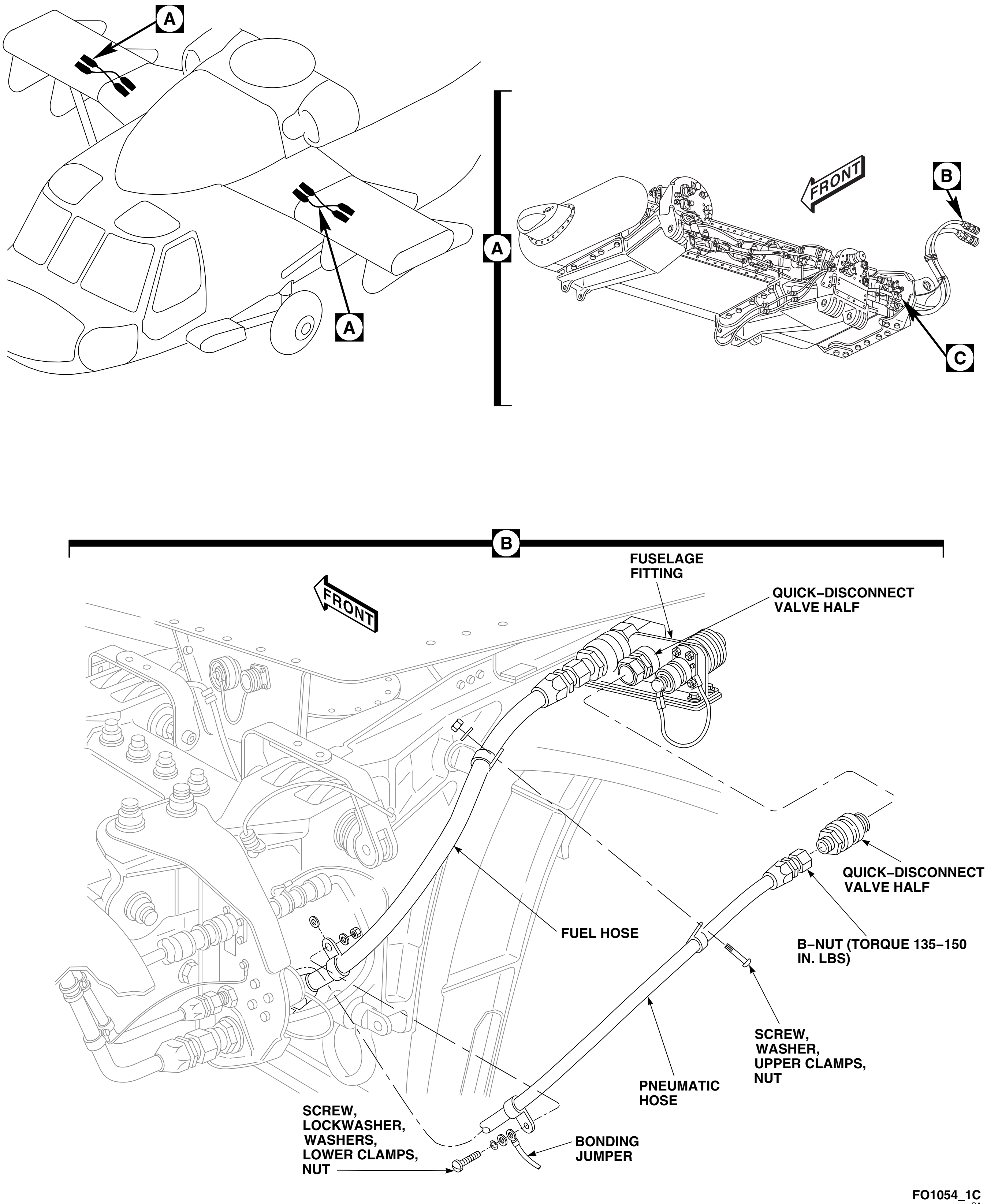


FIGURE 16-4-25-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) ROOT-TO-FUSELAGE PNEUMATIC HOSE (SHEET 1 OF 2)

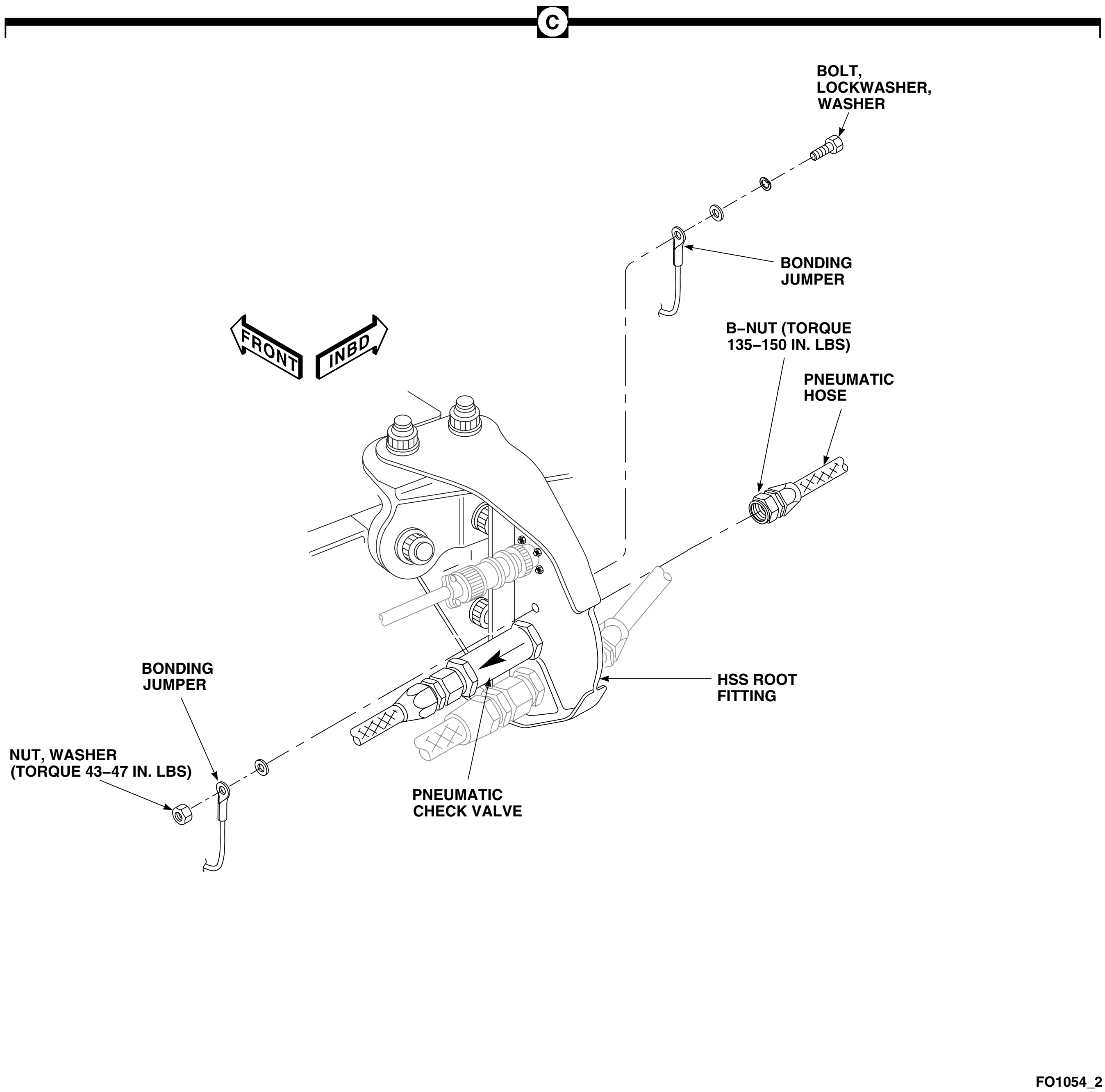


FIGURE 16-4-25-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) ROOT-TO-FUSELAGE PNEUMATIC HOSE (SHEET 2 OF 2)

f. Disconnect pneumatic hose from pneumatic check valve on inboard side of HSS root fitting (Figure 16-4-25-1, Sheet 2, Detail C).

16-4-25.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

NOTE

Remove protective caps and plugs from pneumatic hose, pneumatic check valve, and quick-disconnect valve halves as they are connected.

b. Connect quick-disconnect valve half to pneumatic hose and tighten B-nut (Figure 16-4-25-1, Sheet 1, Detail B). Do not torque B-nut now.

c. Coat threads of quick-disconnect valve half with antiseize compound, Item 74, Appendix D.

d. Connect pneumatic hose to fuselage quick-disconnect valve half and pneumatic check valve. **TORQUE B-NUTS TO 135 - 150 INCH-POUNDS** (Figure 16-4-25-1, Sheet 1, Detail B and Sheet 2, Detail C).

e. Install bonding jumpers to HSS root fitting using bolt, washers, lockwasher, and nut. **TORQUE NUT TO 43 - 47 INCH-POUNDS** (Figure 16-4-25-1, Sheet 2, Detail C). Apply sealing compound, Item 292, Appendix D.

f. Install lower clamps and bonding jumper on pneumatic and fuel hoses and secure with screws, washers, lockwasher, and nut (Figure 16-4-25-1, Sheet 1, Detail B).

g. Install upper clamps to pneumatic and fuel hoses using screw, washer, and nut.

h. Install upper and lower root fairing and inboard trailing edge fairing (PARA 16-4-21).

16-4-26 ESSS HORIZONTAL STORES SUPPORT (HSS) FUEL HOSES, FUEL TUBE ASSEMBLY, AND FUEL TEE UNION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 16-4-21

16-4-26.1 Replace ESSS Horizontal Stores Support (HSS) Fuel Hoses, Fuel Tube Assembly, and Fuel Tee Union.

NOTE

Replacement of left and right ESSS HSS fuel hoses, fuel tube assemblies, and fuel tee unions is the same.

16-4-26.1.1 Remove.

a. Remove inboard and outboard trailing edge fairings, inboard and outboard saddle assemblies, and upper and lower root fairings from HSS (PARA 16-4-21).

NOTE

Install protective caps and plugs on unions and fuel hoses as they are disconnected.

b. Remove screws, lockwashers, washers, bonding jumper, nuts, and clamps securing pneumatic hose to fuel hose (Figure 16-4-26-1, Sheet 1, Detail A).

c. Remove screws, washers, nuts, and clamps securing fuel hoses, fuel tee union, and fuel tube assembly to brackets on HSS and inboard bleed-air regulator valve (Figure 16-4-26-1, Sheet 2, Detail B).

d. Remove screw, lockwasher, washers, bonding jumper, nut, and clamp from fuel tube assembly.

e. Disconnect fuel hose from union on HSS root fitting (Figure 16-4-26-1, Sheet 1, Detail A).

f. Disconnect fuel hoses from inboard and outboard ends of fuel tee union (Figure 16-4-26-1, Sheet 2, Detail B).

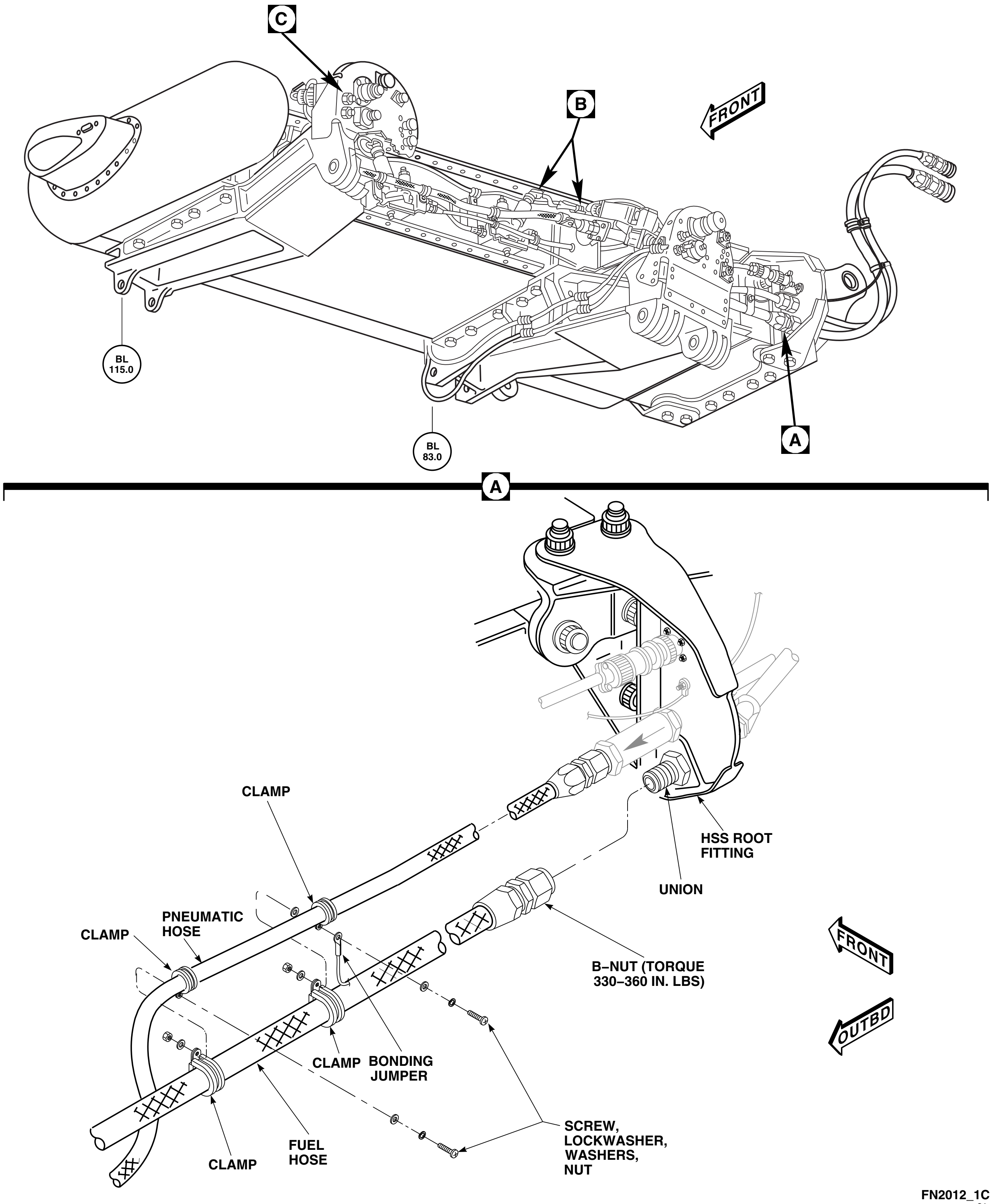


FIGURE 16-4-26-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) FUEL HOSES, FUEL TUBE ASSEMBLY, AND FUEL TEE UNION (SHEET 1 OF 2)

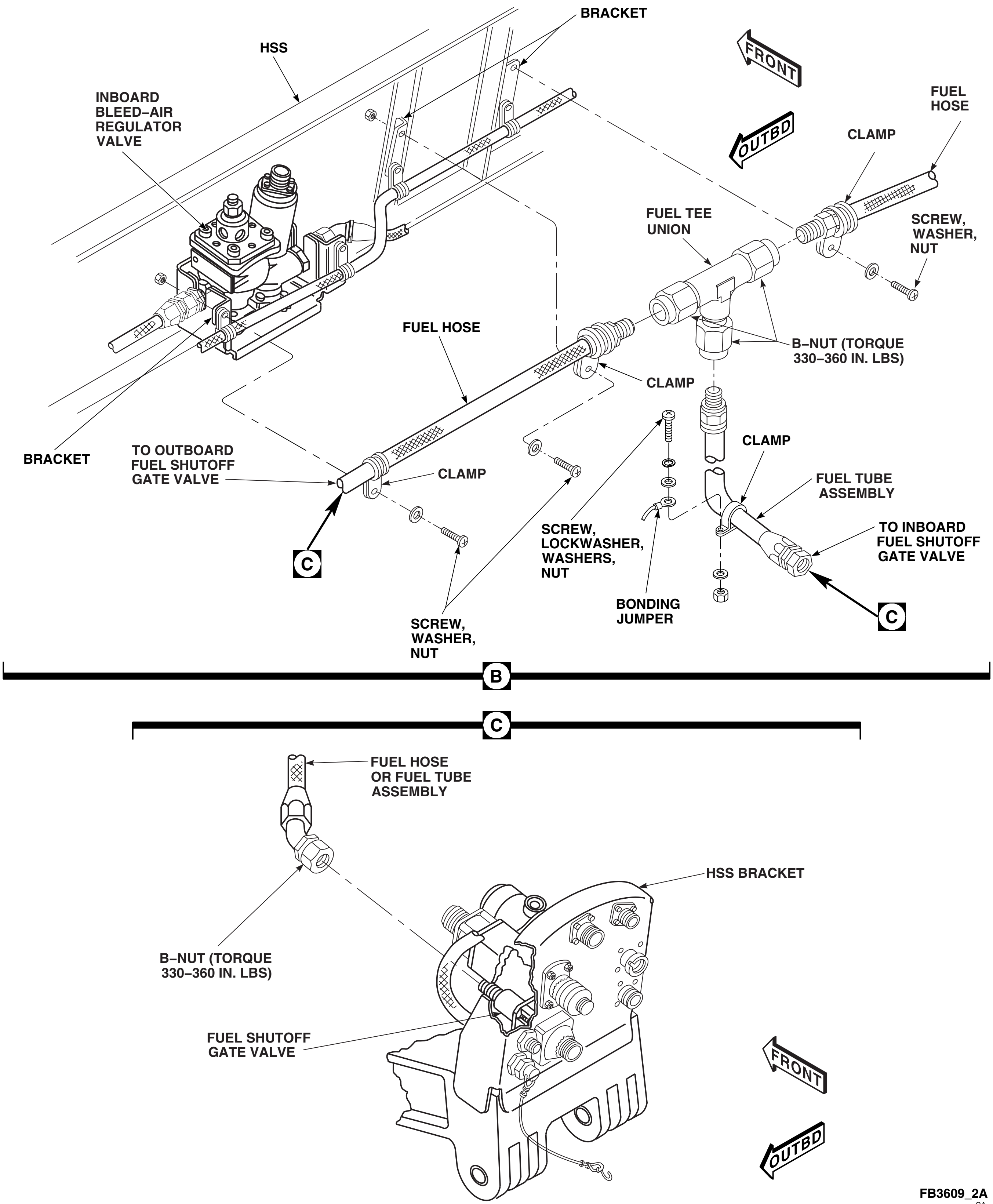


FIGURE 16-4-26-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) FUEL HOSES, FUEL TUBE ASSEMBLY, AND FUEL TEE UNION (SHEET 2 OF 2)

g. Disconnect fuel tube assembly from fuel tee union.

h. Disconnect fuel tube assembly from fuel shutoff gate valve on inboard HSS bracket (Figure 16-4-26-1, Sheet 2, Detail C).

i. Disconnect fuel hose from fuel shutoff gate valve on outboard HSS bracket.

16-4-26.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

NOTE

Remove protective caps and plugs from fuel hoses and unions as they are connected.

b. Coat threads of unions and fuel shutoff gate valves with antiseize compound, Item 74, Appendix D.

NOTE

Position fuel tee union so bottom points down before connecting fuel hoses.

c. Connect fuel hoses to inboard and outboard ends of fuel tee union (Figure 16-4-26-1, Sheet 2, Detail B). **TORQUE B-NUTS TO 330 - 360 INCH-POUNDS.**

d. Connect fuel hose to union on HSS root fairing. **TORQUE B-NUT TO 330 - 360 INCH-POUNDS** (Figure 16-4-26-1, Sheet 1, Detail A).

e. Connect fuel tube assembly to fuel tee union. **TORQUE B-NUT TO 330 - 360 INCH-POUNDS** (Figure 16-4-26-1, Sheet 2, Detail B).

f. Connect fuel tube assembly to fuel shutoff gate valve on inboard HSS bracket (Figure 16-4-26-1, Sheet 2, Detail C). **TORQUE B-NUT TO 330 - 360 INCH-POUNDS.**

g. Connect fuel hose to fuel shutoff gate valve on outboard HSS bracket. **TORQUE B-NUT TO 330 - 360 INCH-POUNDS.**

h. Install clamps on fuel hoses and secure to brackets on HSS and inboard bleed-air regulator valve using screws, washers, and nuts (Figure 16-4-26-1, Sheet 2, Detail B).

i. Install clamp and bonding jumper on fuel tube assembly with screw, lockwasher, washers, and nut.

j. Install clamps on pneumatic and fuel hoses and position bonding jumper to clamps nearest HSS root fitting. Secure pneumatic and fuel hoses with screws, washers, lockwashers, and nuts (Figure 16-4-26-1, Sheet 1, Detail A).

k. Install inboard and outboard trailing edge fairings, inboard and outboard saddle assemblies, and upper and lower root fairings to HSS (PARA 16-4-21).

16-4-27 ESSS HORIZONTAL STORES SUPPORT (HSS) PNEUMATIC HOSES AND PNEUMATIC CHECK VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Protection, Eye
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Cleaning Compound, Item 94, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 13-4-4
- PARA 16-4-21

16-4-27.1 Replace ESSS Horizontal Stores Support (HSS) Pneumatic Hoses and Pneumatic Check Valve.

NOTE

Replacement of left and right ESSS HSS pneumatic hoses and pneumatic check valves is the same.

16-4-27.1.1 Remove.

a. Remove inboard and outboard trailing edge fairings, inboard and outboard saddle assemblies, and upper and lower root fairings from HSS (PARA 16-4-21).

NOTE

Install protective caps and plugs on pneumatic check valve and pneumatic hoses as they are disconnected.

b. Remove screws, washers, lockwashers, bonding jumper, clamps, and nuts securing pneumatic hoses to HSS root fitting, inboard bleed-air regulator valve bracket, and fuel hose (Figure 16-4-27-1, Sheet 1, Detail A, and Sheet 2, Detail B).

c. Disconnect pneumatic hoses from inboard and outboard ends of pneumatic check valve on HSS root fitting (Figure 16-4-27-1, Sheet 1, Detail A).

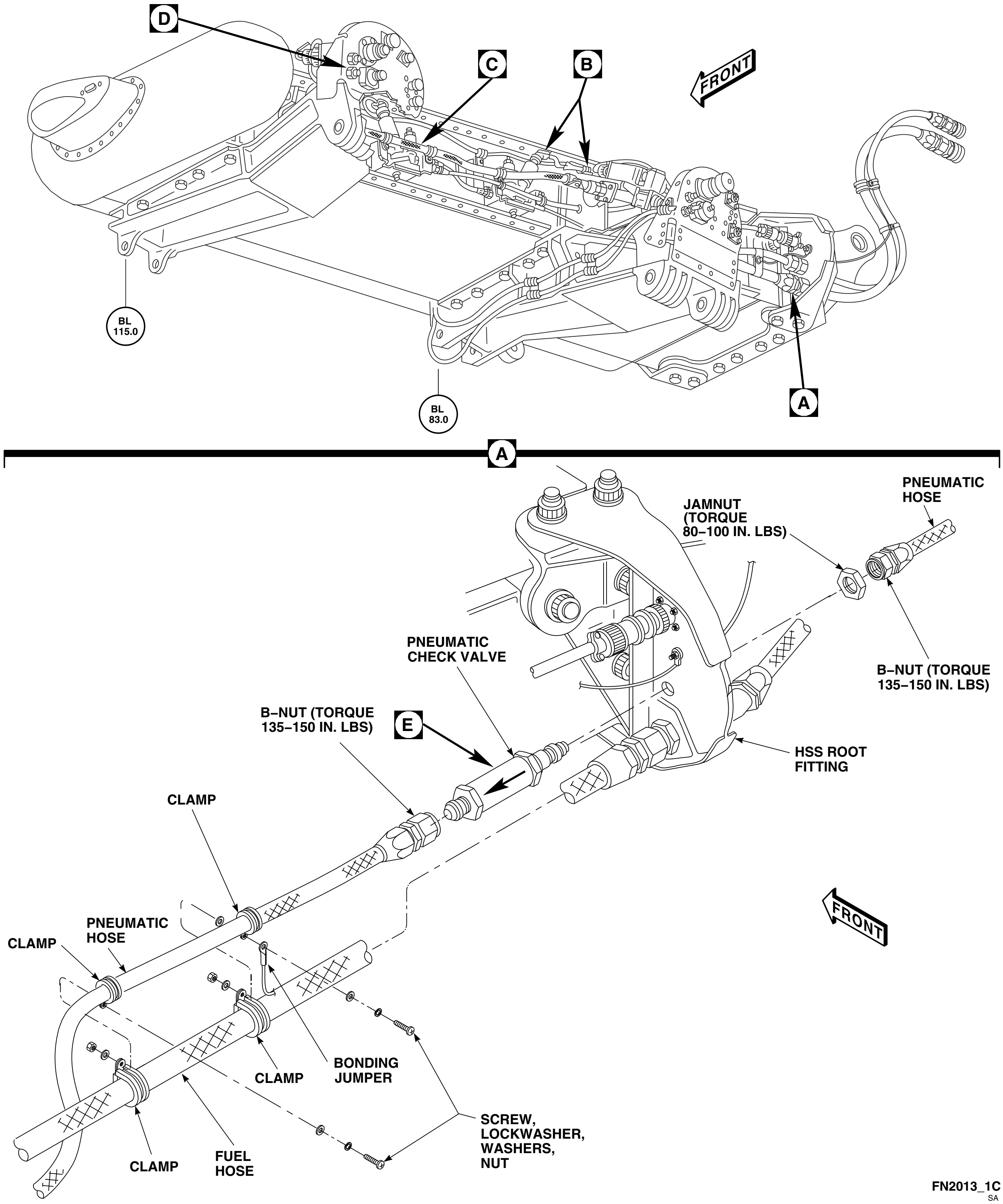


FIGURE 16-4-27-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) PNEUMATIC HOSES AND PNEUMATIC CHECK VALVE (SHEET 1 OF 3)

d. Remove jamnut and pneumatic check valve from HSS root fitting.

e. Disconnect pneumatic hose connecting inboard bleed-air regulator valve to union on inboard HSS bracket (Figure 16-4-27-1, Sheet 2, Detail B, and Sheet 3, Detail D).

f. Disconnect pneumatic hose connecting inboard bleed-air regulator valve and tee fitting on outboard bleed-air regulator valve (Figure 16-4-27-1, Sheet 2, Detail B and Detail C).

g. Disconnect pneumatic hose from tee fitting on outboard bleed-air regulator valve (Figure 16-4-27-1, Sheet 2, Detail C).

h. Disconnect pneumatic hose connecting outboard bleed-air regulator valve to union on outboard HSS bracket (Figure 16-4-27-1, Sheet 2, Detail C, and Sheet 3, Detail D).

16-4-27.1.2 Repair/Clean.

a. Repair pneumatic check valve as follows:

(1) Disassemble pneumatic check valve by unscrewing inner body housing from outer body housing (Figure 16-4-27-1, Sheet 3, Detail E).

(2) Remove spring from outer body housing.

(3) Remove poppet from inner body housing.

(4) Remove and discard packing from poppet.

(5) Clean parts using dry-cleaning solvent, Item 124, Appendix D, or cleaning compound, Item 94, Appendix D.

(6) Install packing on poppet.

(7) Install poppet in inner body housing.

(8) Install spring in outer body housing. Make sure spring is seated properly within spring guide of housing.

(9) Assemble inner and outer housings. **TORQUE HOUSING BODIES TO 137 - 151 INCH-POUNDS.**

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

(10) Make sure valve is working properly by using compressed air, 20 psi maximum. Air flow should only occur in direction of flow arrow on outer body housing.

b. Clean pneumatic hoses and tubes as follows:

(1) Disconnect bleed-air lines from tube tee fitting (PARA 13-4-4).

WARNING

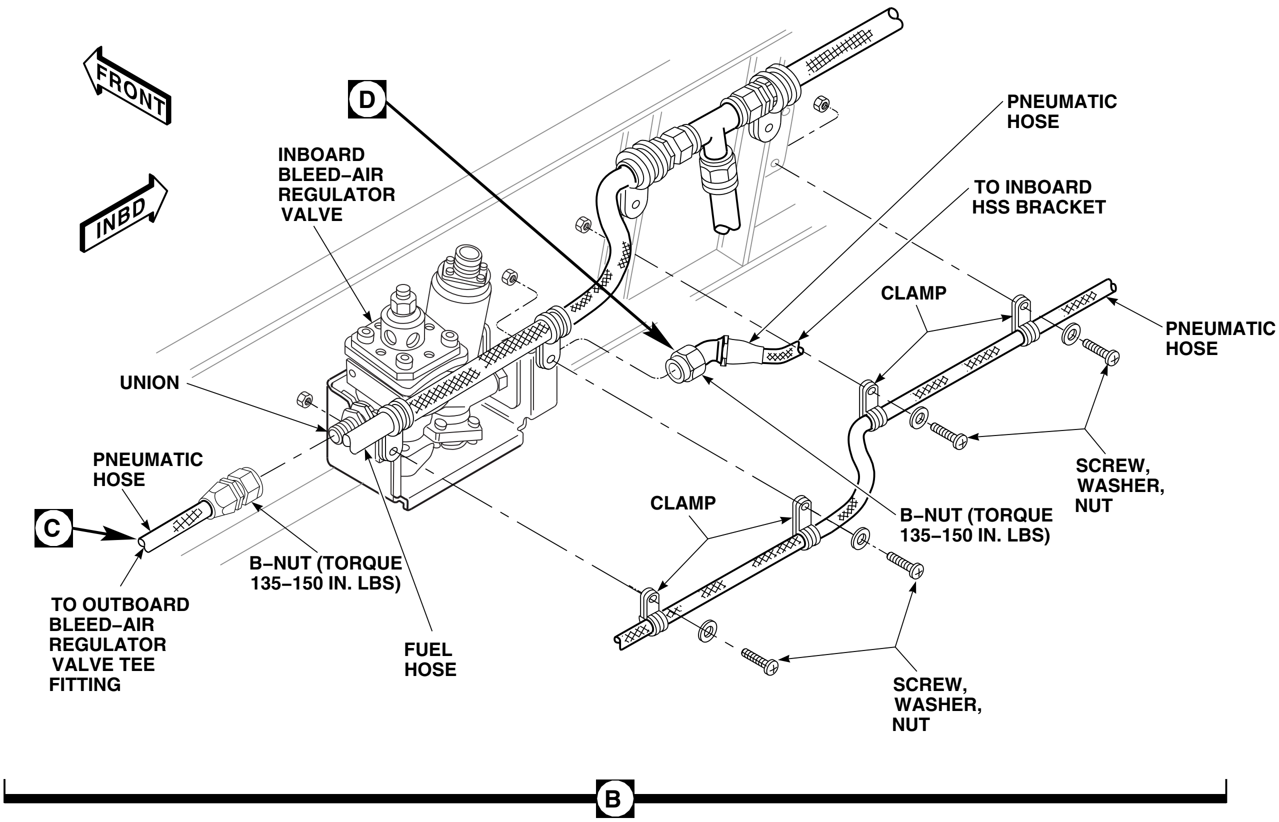
Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose toward personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

(2) Using compressed air set at 50 psi maximum, blow through hoses and tubes to remove dirt, sand, and debris.

(3) Connect bleed-air lines to tube tee fitting (PARA 13-4-4).

16-4-27.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).



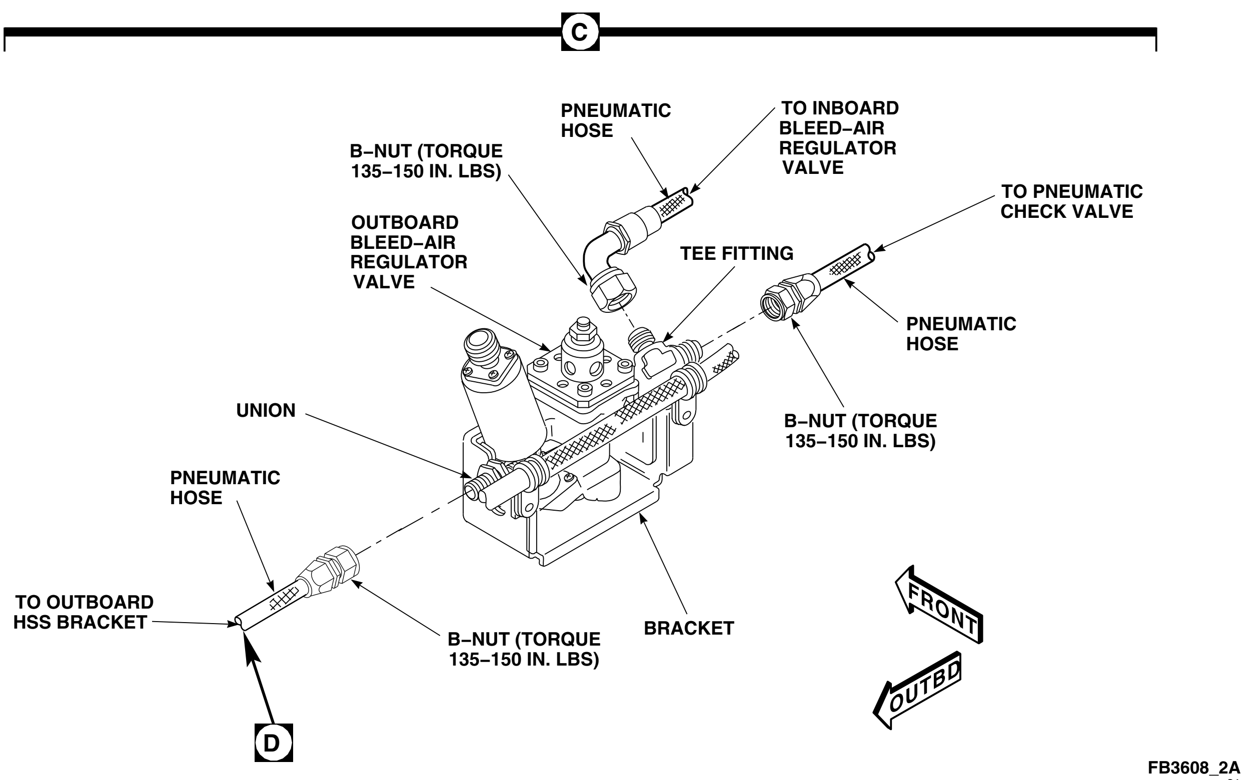
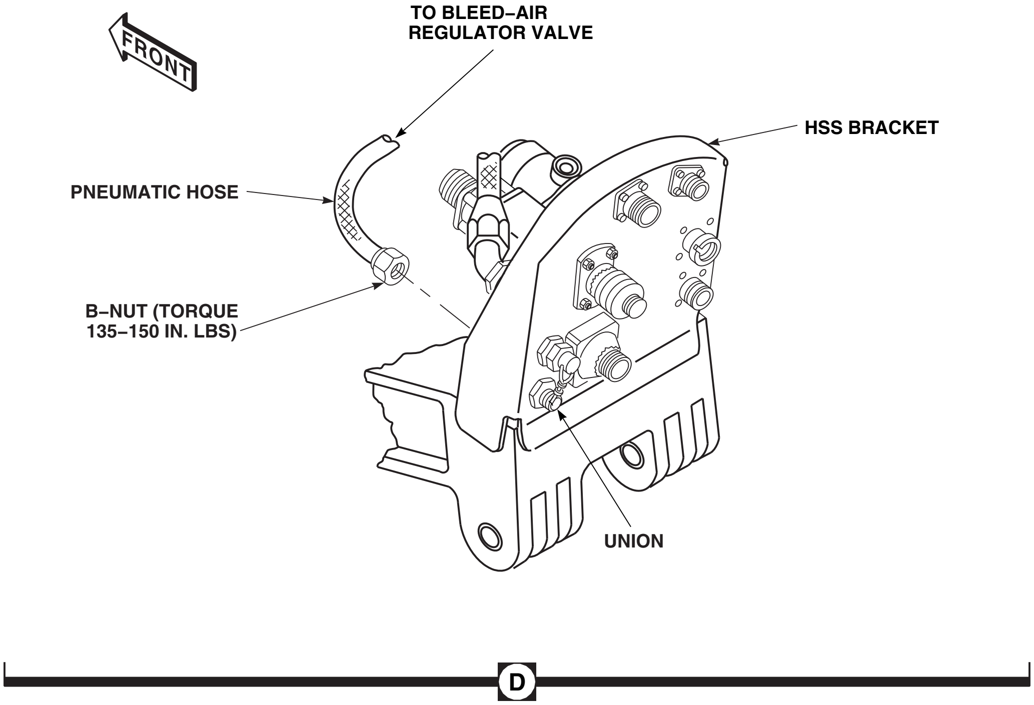


FIGURE 16-4-27-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) PNEUMATIC HOSES AND PNEUMATIC CHECK VALVE (SHEET 2 OF 3)



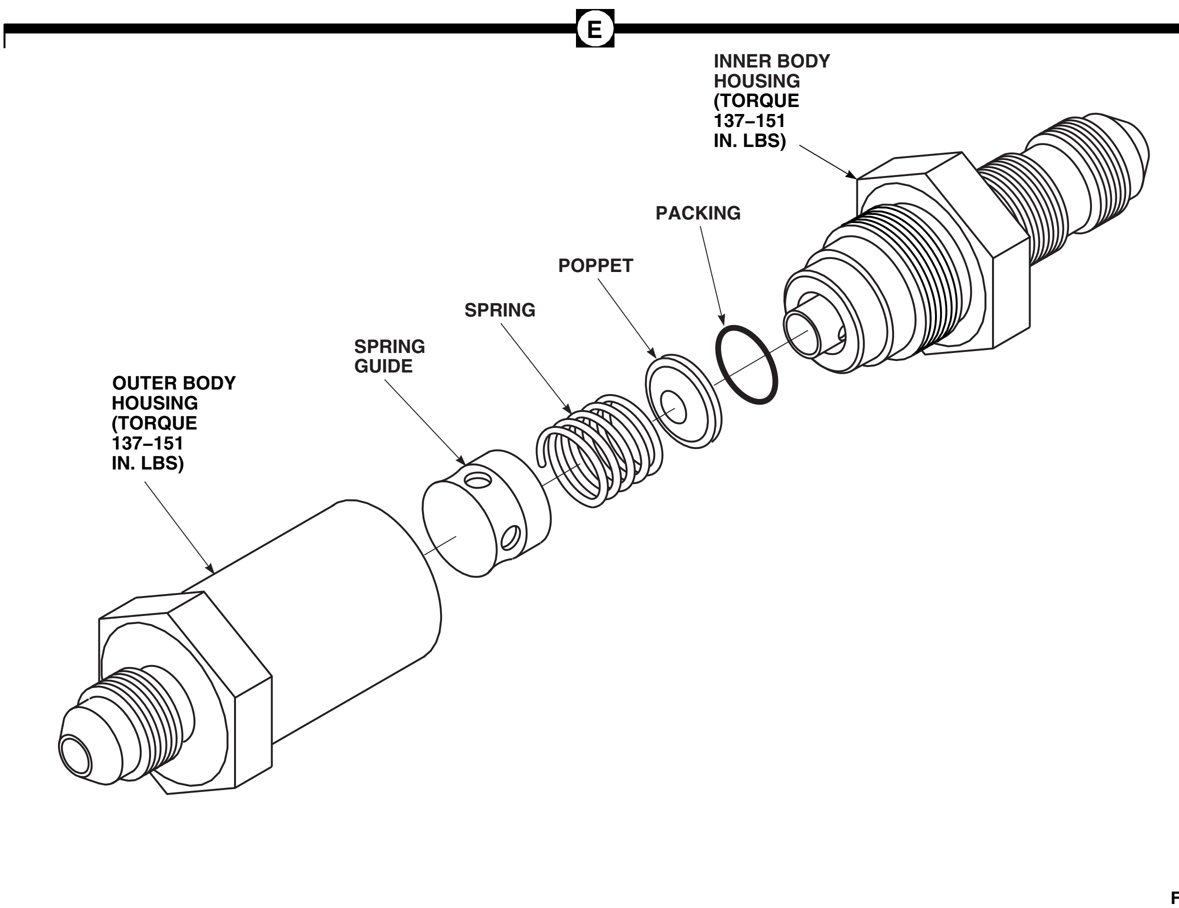


FIGURE 16-4-27-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) PNEUMATIC HOSES AND PNEUMATIC CHECK VALVE (SHEET 3 OF 3)

NOTE

- Remove protective caps and plugs from pneumatic check valves and pneumatic hoses as they are connected.
- Make sure arrow on pneumatic check valve is pointing outboard.

b. Attach pneumatic check valve to HSS root fitting and secure with jamnut (Figure 16-4-27-1, Sheet 1, Detail A). **TORQUE JAMNUT TO 80 - 100 INCH-POUNDS.**

c. Using antiseize compound, Item 74, Appendix D, coat threads of tee fitting, unions, and pneumatic check valve.

d. Install pneumatic hose to outboard bleed-air regulator valve union (Figure 16-4-27-1, Sheet 2, Detail C). **TORQUE B-NUT TO 135 - 150 INCH-POUNDS.**

e. Connect pneumatic hose from outboard bleed-air regulator valve to union on outboard HSS bracket (Figure 16-4-27-1, Sheet 3, Detail D). **TORQUE B-NUT TO 135 - 150 INCH-POUNDS.**

f. Connect pneumatic hose connecting inboard bleed-air regulator valve to tee fitting on outboard bleed-air regulator valve (Figure 16-4-27-1, Sheet 2, Detail B and Detail C). **TORQUE B-NUTS TO 135 - 150 INCH-POUNDS.**

g. Connect pneumatic hose connecting pneumatic check valve to tee fitting on outboard bleed-air regulator valve (Figure 16-4-27-1, Sheet 2, Detail C). **TORQUE B-NUT TO 135 - 150 INCH-POUNDS.**

h. Connect pneumatic hose to inboard bleed-air regulator valve union (Figure 16-4-27-1, Sheet 2, Detail B). **TORQUE B-NUT TO 135 - 150 INCH-POUNDS.**

i. Connect pneumatic hose from inboard bleed-air regulator valve to union on inboard bracket (Figure 16-4-27-1, Sheet 3, Detail D). **TORQUE B-NUT TO 135 - 150 INCH-POUNDS.**

j. Connect pneumatic hoses on pneumatic check valve on inboard and outboard side of HSS root fitting (Figure 16-4-27-1, Sheet 1, Detail A). **TORQUE B-NUTS TO 135 - 150 INCH-POUNDS.**

k. Install clamps securing pneumatic hoses to HSS and inboard bleed-air regulator valve bracket and secure with screws, washers and nuts (Figure 16-4-27-1, Sheet 2, Detail B and Detail C).

l. Install clamps securing pneumatic hose to fuel hose. Install bonding jumper to clamps nearest HSS root fitting. Secure clamps and bonding jumper with screws, washers, lockwashers, and nuts (Figure 16-4-27-1, Sheet 1, Detail A).

m. Install inboard and outboard trailing edge fairings, inboard and outboard saddle assemblies, and upper and lower root fairings to HSS (PARA 16-4-21).

16-4-28 ESSS VERTICAL SUPPORT PYLON (VSP) PNEUMATIC HOSE AND ELBOW.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 16-4-21

16-4-28.1 Replace ESSS Vertical Support Pylon (VSP) Pneumatic Hose and Elbow.

NOTE

Replacement of left and right ESSS VSP pneumatic hoses and elbows is the same.

16-4-28.1.1 Remove.

a. Remove rear VSP fairing and saddle assembly (PARA 16-4-21).

b. Remove screw, lockwasher, washers, bonding jumper, nut, and clamps securing fuel hose to pneumatic hose (Figure 16-4-28-1, Detail A).

c. Remove screw, washer, spacer, and clamp securing pneumatic hose, fuel hose, and isolation check valve to bracket on VSP fuel arm extension.

NOTE

Install protective caps and plugs on fittings as parts are removed and disconnected.

d. Disconnect pneumatic hose from union on horizontal stores support (HSS) bracket and at elbow on VSP fuel arm extension.

e. If ESSS external 230 gallon fuel tank is not installed, remove protective cap from elbow on VSP fuel arm extension. If ESSS external fuel tank is installed, disconnect air interface from elbow on VSP fuel arm extension.

f. Remove jamnuts, washer, and elbow from VSP fuel arm extension.

16-4-28.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

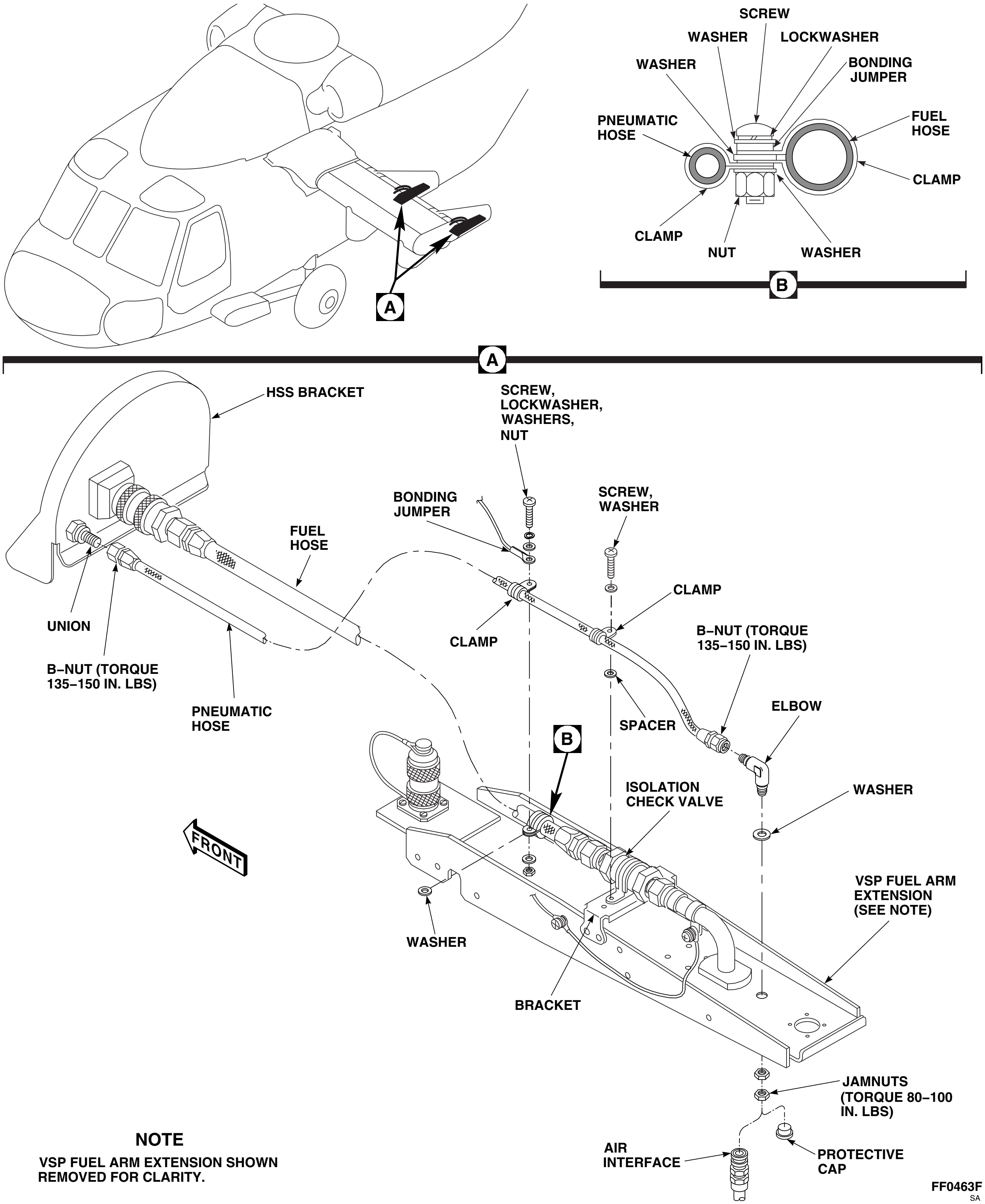


FIGURE 16-4-28-1. REPLACE ESSS VERTICAL SUPPORT PYLON (VSP) PNEUMATIC HOSE AND ELBOW

NOTE

Remove protective caps and plugs from hoses and fittings as parts are installed and connected.

b. Install washer and elbow in VSP fuel arm extension and secure with jamnuts (Figure 16-4-28-1, Detail A). **TORQUE JAMNUTS TO 80 - 100 INCH-POUNDS.**

c. Using antiseize compound, Item 74, Appendix D, coat threads of elbow and union.

d. Connect pneumatic hose to elbow on VSP fuel arm extension and to union on horizontal stores support (HSS) bracket. **TORQUE B-NUTS TO 135 - 150 INCH-POUNDS.**

e. Install clamp on pneumatic hose and secure pneumatic hose, fuel hose, isolation check valve to bracket on VSP fuel arm extension with screw, washer, and spacer.

f. Apply sealing compound, Item 292, Appendix D, to edges of clamps and attaching hardware.

g. Install clamp on pneumatic hose, and secure bonding jumper and clamp to fuel hose with screw, lockwasher, washers, and nut (Figure 16-4-28-1, Detail A and Detail B).

h. If ESSS external 230 gallon fuel tank is not installed, install protective cap on elbow in VSP fuel arm extension. If ESSS external 230 gallon fuel tank is installed, connect air interface to elbow on VSP fuel arm extension.

i. Install rear VSP fairing and saddle assembly (PARA 16-4-21).

16-4-29 ESSS VERTICAL SUPPORT PYLON (VSP) FUEL HOSE, FUEL TUBE, AND ISOLATION CHECK VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 16-4-21

16-4-29.1 Replace ESSS Vertical Support Pylon (VSP) Fuel Hose, Fuel Tube, and Isolation Check Valve.

NOTE

Replacement of left and right ESSS VSP fuel hoses, fuel tubes, and isolation check valves is the same.

16-4-29.1.1 Remove.

a. Remove rear VSP fairing and saddle assembly (PARA 16-4-21).

b. Remove screw, lockwasher, washers, bonding jumper, nut, and clamps securing fuel hose and pneumatic hose (Figure 16-4-29-1, Detail A and Detail B).

c. Remove screw, lockwasher, washers, nut, and clamp securing bonding jumper to fuel tube (Figure 16-4-29-1, Detail A).

NOTE

Install protective caps and plugs on hoses, tubes, and fittings as parts are removed and disconnected.

d. Disconnect fuel hose from fuel shutoff gate valve quick-disconnect on horizontal stores support (HSS) bracket and at isolation check valve on VSP fuel arm extension.

e. Disconnect fuel tube from isolation check valve.

f. Remove screw, washer, spacer, and clamp securing pneumatic hose and isolation check

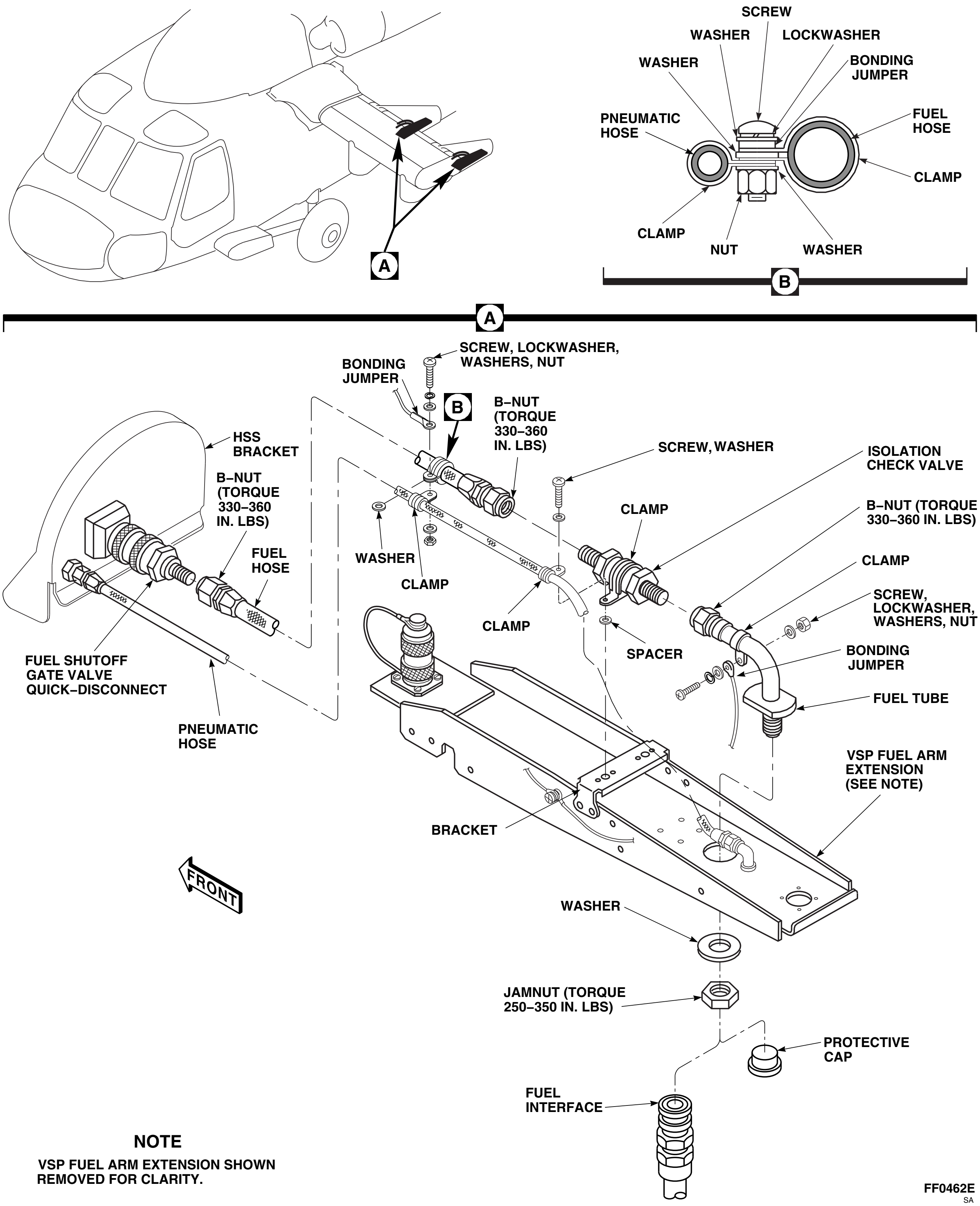


FIGURE 16-4-29-1. REPLACE ESSS VERTICAL SUPPORT PYLON (VSP) FUEL HOSE, FUEL TUBE, AND ISOLATION CHECK VALVE

valve to bracket on VSP fuel arm extension. Remove check valve.

g. If ESSS external 230 gallon fuel tank is not installed, remove protective cap from fuel tube on VSP fuel arm extension. If ESSS external 230 gallon fuel tank is installed, disconnect fuel interface from fuel tube on VSP fuel arm extension.

h. Remove jamnut and washers securing fuel tube to VSP fuel arm extension arm. Remove fuel tube.

16-4-29.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

NOTE

Remove protective caps and plugs from hoses, tubes, and fittings as parts are installed and connected.

b. Install fuel tube on VSP fuel arm extension and secure with washers and jamnut (Figure 16-4-29-1, Detail A). **DO NOT TIGHTEN JAMNUT AT THIS TIME.**

c. Using antiseize compound, Item 74, Appendix D, coat threads of isolation check valve.

NOTE

Make sure arrow on isolation check valve points forward.

d. Connect isolation check valve to fuel tube. **TORQUE B-NUT TO 330 - 360 INCH-POUNDS.**

e. Install clamp on isolation check valve and secure pneumatic hose and isolation check valve to bracket on VSP fuel arm extension with screw, washer, and spacer.

f. **TORQUE JAMNUT SECURING FUEL TUBE TO 250 - 350 INCH-POUNDS.**

g. Install clamp on fuel tube, and secure bonding jumper and clamp to fuel tube using screw, lockwasher, washers, and nut.

h. Connect fuel hose to isolation check valve. **TORQUE B-NUT TO 330 - 360 INCH-POUNDS.**

i. Coat threads of fuel shutoff gate valve quick-disconnect on horizontal stores support (HSS) bracket with antiseize compound, Item 74, Appendix D.

j. Connect fuel hose to fuel shutoff gate valve quick-disconnect on HSS bracket. **TORQUE B-NUT TO 330 - 360 INCH-POUNDS.**

k. Install clamp on fuel hose. Secure bonding jumper and clamp to pneumatic hose with screw, lockwasher, washers, and nut (Figure 16-4-29-1, Detail A and Detail B).

l. Apply sealing compound, Item 292, Appendix D, to edges of clamps and attaching hardware.

m. If ESSS external 230 gallon fuel tank is not installed, install protective cap on fuel tube in VSP fuel arm extension. If ESSS external 230 gallon fuel tank is installed, connect fuel interface to fuel tube on VSP fuel arm extension.

n. Install rear VSP fairing and saddle assembly (PARA 16-4-21).

16-4-30 ESSS FLOW SENSOR PANEL.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 1-Gallon
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-45
- PARA 2-4-69
- PARA 13-5-3
- PARA 13-5-6
- PARA 16-2-1
- PARA 16-2-2

16-4-30.1 Replace ESSS Flow Sensor Panel Components.

16-4-30.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Gain access to main fuel cell compartment as follows:

(1) Remove rear passenger's seats as required (PARA 2-4-45).

(2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).

(3) Remove cargo netting from stowage compartments.

(4) If air conditioning is installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

c. Unlock fasteners securing left and right panel enclosure on top of main fuel cells. Remove panels.

d. Position 1-gallon container to catch fuel.

e. Disconnect flow sensor fuel lines from quick-disconnect ends at STA 398.0. Remove protective caps from fittings and install on quick-disconnect ends at STA 398.0 (Figure 16-4-30-1, Sheet 1, Detail A).

f. Disconnect quick-disconnect bonding jumper between flow sensor panel and bracket.

g. Disconnect electrical connector, P347 from J347, on intercostal.

h. Disconnect electrical connector, P523, from flow transmitter on No. 2 main fuel cell.

i. Install protective caps and plugs on electrical connectors.

j. Remove screw, washer, spacer, nut, and bonding jumper from clamp on main fuel line.

k. Disconnect main fuel hose from elbow.

l. Unlock fasteners and remove flow sensor panel from brackets by sliding panel to right and removing through opening.

m. Remove screws, washers, spacers, nuts, and bonding jumpers from clamps on flow sensor fuel lines.

n. Disconnect flow sensor fuel lines from flow sensors.

o. Disconnect electrical connector, P520 from J520, and, P524 from J524, on right bracket (Figure 16-4-30-1, Sheet 2, Detail B).

p. Disconnect electrical connector, P521 from J521, on left bracket (Figure 16-4-30-1, Sheet 2, Detail C).

q. Install protective caps and plugs on electrical connectors.

r. Remove bolt, washer, and bonding jumpers from fuel fitting (Figure 16-4-30-1, Sheet 2, Detail D).

NOTE

Remove fuel fitting, flow sensors, and temperature transducer as an assembly.

s. Remove bolts and washers attaching fuel fitting to panel.

t. Loosen jamnut and remove temperature transducer from fuel fitting. Remove and discard packing from temperature transducer.

u. Loosen jamnuts and remove flow sensors and unions, as an assembly, from fuel fitting. Remove and discard packings from unions.

v. Loosen elbow jamnut and remove elbow from fuel fitting. Remove and discard packing from elbow.

w. Remove cap and jamnut securing bracket to union. Remove bracket.

x. Remove union from fuel fitting. Remove and discard packing from union.

16-4-30.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Coat threads of unions, elbow, temperature transducer, and flow sensors with a light coat of antiseize compound, Item 74, Appendix D.

NOTE

- Flow sensors and unions are to be installed as assemblies.
- Install flow sensors, with unions installed, with arrow on flow sensor pointing toward fuel fitting.

c. Install jamnuts on flow sensor unions. Lubricate packings with petrolatum, Item 234, Appendix D, and install on flow sensor unions. Install flow sensor/union assemblies in fuel fitting. **TORQUE JAMNUTS TO 250 - 350 INCH-POUNDS** (Figure 16-4-30-1, Sheet 2, Detail D).

d. Install jamnut on temperature transducer. Lubricate packing with petrolatum, Item 234, Appendix D, and install on transducer. Install transducer in fuel fitting. **TORQUE JAMNUT TO 250 - 350 INCH-POUNDS.**

e. Install jamnut on elbow. Lubricate packing with petrolatum, Item 234, Appendix D, and install on elbow. Install elbow in fuel fitting.

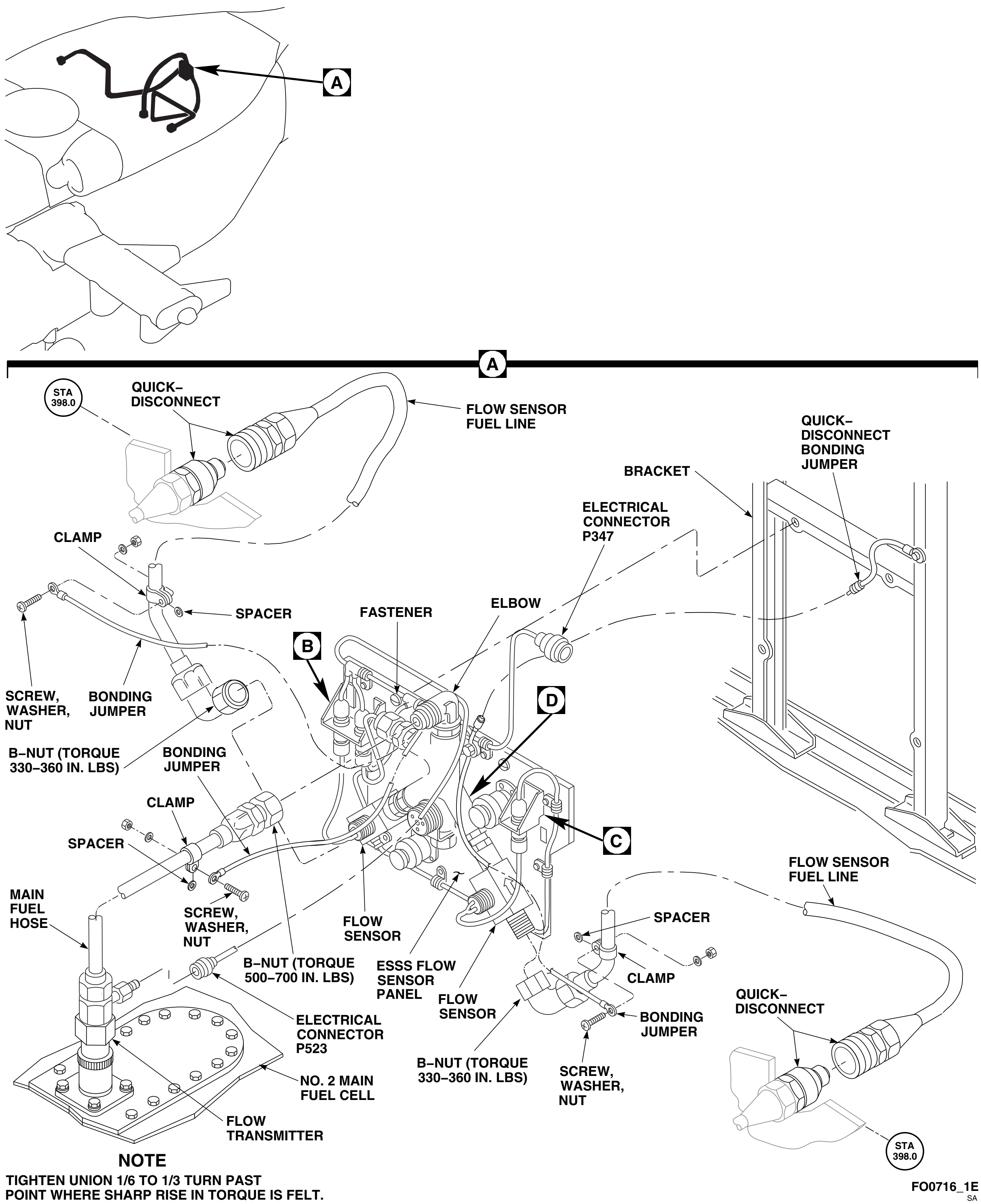


FIGURE 16-4-30-1. REPLACE ESSS FLOW SENSOR PANEL COMPONENTS (SHEET 1 OF 2)

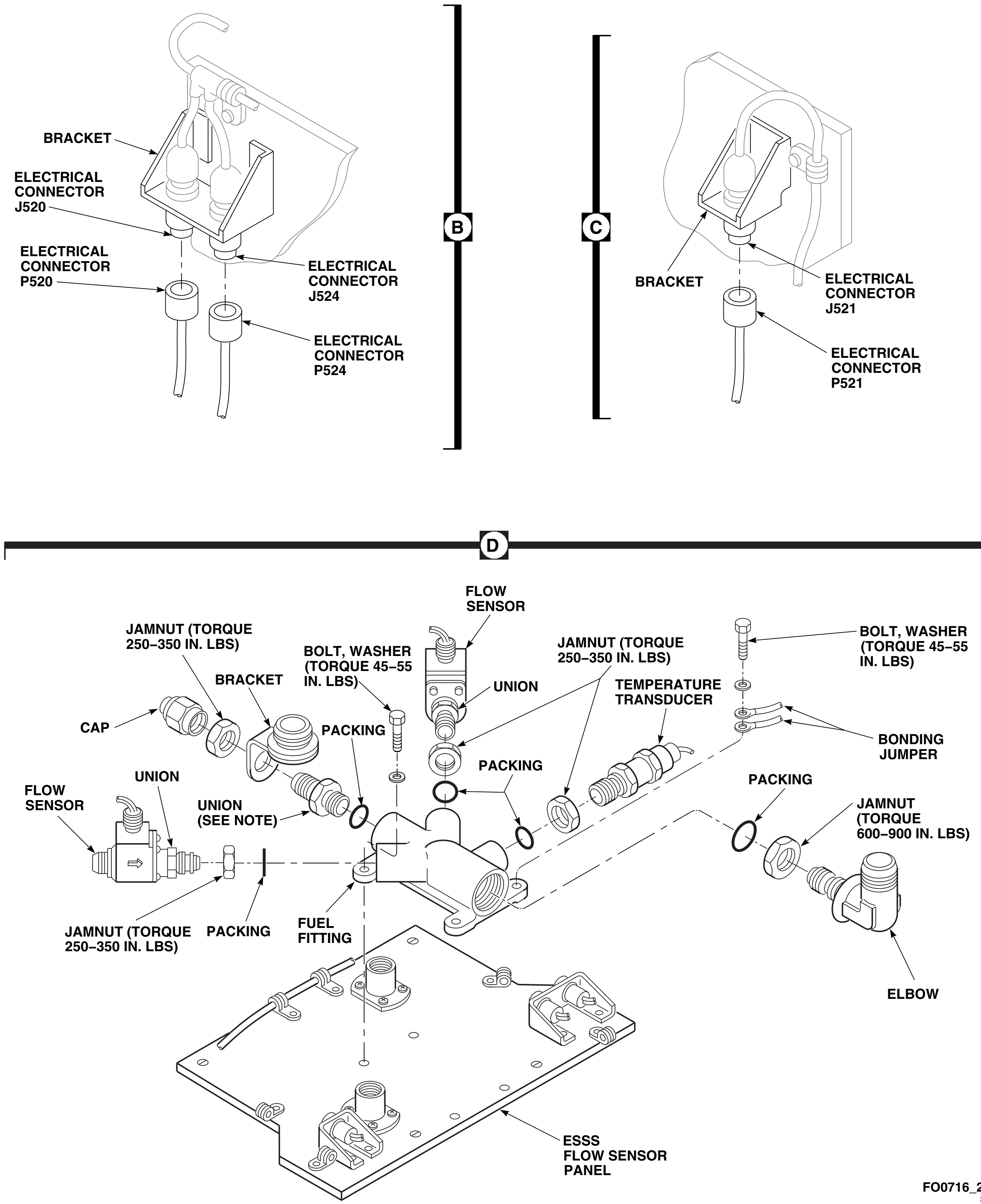


FIGURE 16-4-30-1. REPLACE ESSS FLOW SENSOR PANEL COMPONENTS (SHEET 2 OF 2)

TORQUE JAMNUT TO 600 - 900 INCH-POUNDS.

f. Lubricate packing with petrolatum, Item 234, Appendix D, and install on union. Install union in fuel fitting. **TIGHTEN UNION 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

g. Install jamnut and bracket on union in center port of fuel fitting. **TORQUE JAMNUT TO 250 - 350 INCH-POUNDS.** Install cap on union.

h. Position fuel fitting, temperature transducer, and flow sensors, as an assembly, over mounting holes in flow sensor panel.

i. Install bolts, washers, and bonding jumpers in fuel fitting and panel. **TORQUE BOLTS TO 45 - 55 INCH-POUNDS.**

j. Connect flow sensor fuel lines to flow sensors (Figure 16-4-30-1, Sheet 1, Detail A). **TORQUE B-NUTS TO 330 - 360 INCH-POUNDS.**

k. Install bonding jumpers to flow sensor fuel lines with screws, washers, spacers, and nuts.

l. Remove protective caps and plugs from electrical connectors.

m. Inspect and treat electrical connectors (PARA 1-7-20).

n. Connect electrical connector, P520 to J520, and, P524 to J524, on right bracket (Figure 16-4-30-1, Sheet 2, Detail B).

o. Connect electrical connector, P521 to J521, on left bracket (Figure 16-4-30-1, Sheet 2, Detail C).

p. Install flow sensor panel on bracket and secure with fasteners (Figure 16-4-30-1, Sheet 1, Detail A).

q. Connect main fuel hose to elbow. **TORQUE B-NUT TO 500 - 700 INCH-POUNDS.**

r. Install bonding jumper to clamp on main fuel line with screw, washer, spacer, and nut.

s. Remove protective caps and plugs from electrical connectors.

t. Inspect and treat electrical connectors (PARA 1-7-20).

u. Connect electrical connector, P523, to flow transmitter on No. 2 main fuel cell.

v. Remove protective caps and plugs from electrical connectors.

w. Inspect and treat electrical connectors (PARA 1-7-20).

x. Connect electrical connector, P347 to J347, on intercostal.

y. Connect quick-disconnect bonding jumper between flow sensor panel and bracket.

z. Remove caps from quick-disconnects at STA 398.0 and connect flow sensor fuel line quick-disconnects. Install caps on fittings.

aa. Perform operational check of ESSS range extension system and check for leaks (PARA 16-2-1 or PARA 16-2-2).

ab. Make sure area is clean and free of foreign objects.

ac. Install left and right panel enclosure on top of main fuel cells. Lock fasteners.

ad. Close fuel cell compartment as follows:

(1) If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

(2) Install cargo netting.

(3) Install soundproofing panels (PARA 2-4-69).

(4) Install rear passenger's seats (PARA 2-4-45).

16-4-31 ESSS FLOW TRANSMITTER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 1-Gallon
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-45
- PARA 2-4-69
- PARA 13-5-3
- PARA 13-5-6
- PARA 16-2-1
- PARA 16-2-2

16-4-31.1 Replace ESSS Flow Transmitter.

NOTE

ESSS flow transmitter is installed on No. 2 main fuel cell only.

16-4-31.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove right side rear passenger's seats (PARA 2-4-45).

c. Remove soundproofing panel from rear bulkhead covering right stowage compartment (PARA 2-4-69).

d. Remove cargo netting from stowage compartment.

e. If air conditioning is installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

f. Unlock fastener studs and remove left and right fuel enclosure panels.

g. Position 1-gallon container to catch fuel.

h. Disconnect main fuel hose from flow transmitter (Figure 16-4-31-1, Detail A).

i. Install protective plug in main fuel line.

j. Disconnect electrical connector, P523, from flow transmitter.

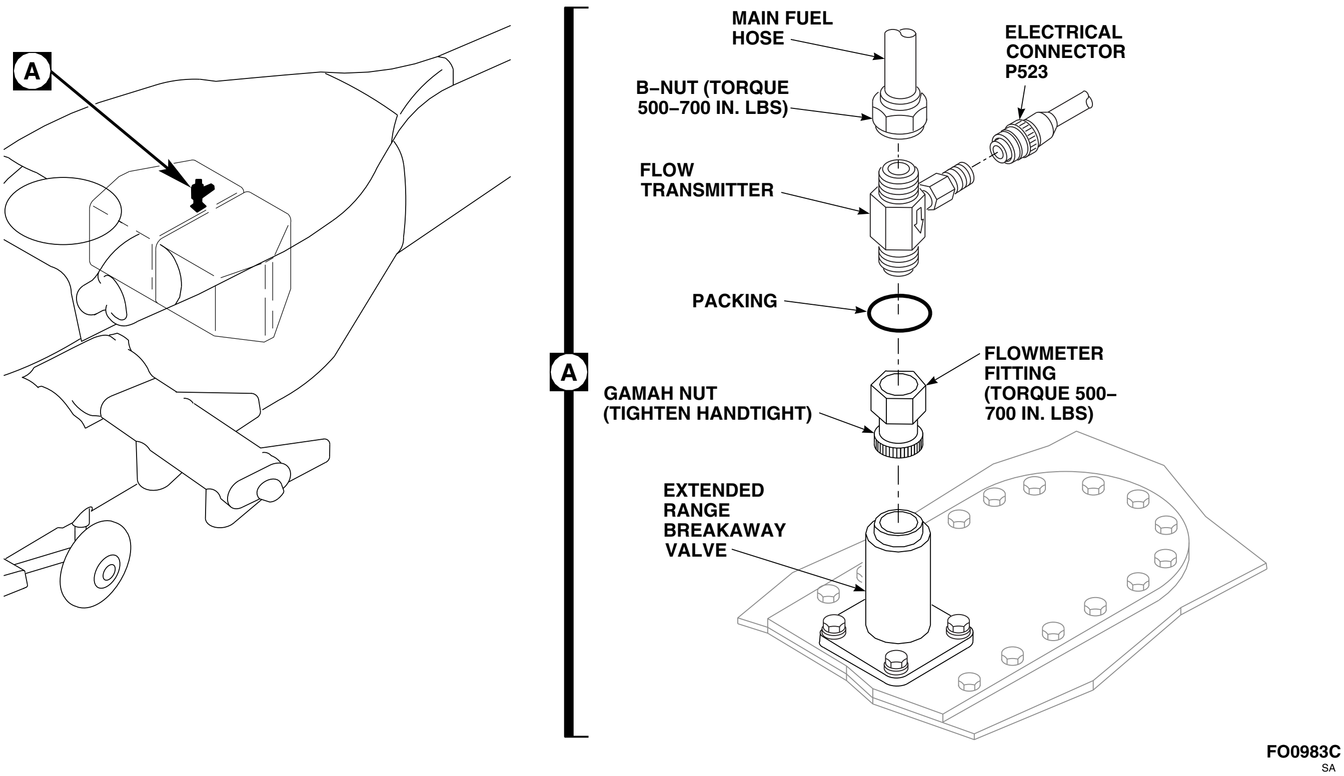


FIGURE 16-4-31-1. REPLACE ESSS FLOW TRANSMITTER

k. Install protective caps and plugs on electrical connectors.

l. Remove flowmeter fitting, with flow transmitter attached, from extended range breakaway valve.

m. Install protective cap on extended range breakaway valve.

n. Remove flow transmitter and packing from flowmeter fitting. Discard packing.

16-4-31.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Lubricate packing with petrolatum, Item 234, Appendix D, and install on flow transmitter.

c. Coat threads of flow transmitter with antiseize compound, Item 74, Appendix D.

d. Install flowmeter fitting on flow transmitter with arrow on transmitter pointing to fitting (Figure 16-4-31-1, Detail A).

e. **TORQUE FLOWMETER FITTING TO 500 - 700 INCH-POUNDS.**

f. Remove protective plug and connect main fuel hose to flow transmitter.

g. **TORQUE B-NUT TO 500 - 700 INCH-POUNDS.**

h. Remove protective plug from extended range breakaway valve.

CAUTION

Do not preload extended range breakaway valve when installing flowmeter fitting on extended range breakaway valve. Frangible section of breakaway valve is sensitive to bending and tension overload.

i. Install flowmeter fitting, with flow transmitter and main fuel hose attached, on extended range breakaway valve. **TIGHTEN GAMAH NUT ON FLOWMETER FITTING HANDTIGHT.**

j. Remove protective caps and plugs from electrical connectors.

k. Inspect and treat electrical connectors (PARA 1-7-20).

l. Connect electrical connector, P523, to flow transmitter.

NOTE

When replacing flow transmitter, K-FACTOR stamped on side must be calibrated to auxiliary fuel management control panel.

m. Perform fuel transfer check and check for leaks (PARA 16-2-1 or PARA 16-2-2).

n. Make sure area is clean and free of foreign objects.

o. Install left and right fuel enclosure panels and secure with fasteners.

p. If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

q. Install cargo netting.

r. Install soundproofing panel (PARA 2-4-69).

s. Install rear passenger's seats (PARA 2-4-45).

16-4-32 ESSS HORIZONTAL STORES SUPPORT (HSS) BLEED-AIR REGULATOR VALVES, UNIONS, AND TEE FITTING.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Protection, Eye
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Cleaning Compound, Item 94, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Lockwire, Item 197, Appendix D
- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs
- Tiedown Strap, Item 331, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 16-2-1
- PARA 16-2-2
- PARA 16-4-21

16-4-32.1 Replace ESSS Horizontal Stores Support (HSS) Bleed-Air Regulator Valves, Unions, and Tee Fitting.

NOTE

Replacement of left and right inboard and outboard bleed-air regulator valves, unions, and tee fittings is the same, except as noted.

16-4-32.1.1 Remove.

a. Remove outboard trailing edge fairing from ESSS horizontal support system (HSS) (PARA 16-4-21).

b. On inboard bleed-air regulator valves, perform the following (Figure 16-4-32-1, Sheet 1, Detail A):

(1) Disconnect electrical connector, P839 (left side), or P811 (right side).

(2) Install protective caps and plugs on electrical connectors.

c. On outboard bleed-air regulator valves, perform the following (Figure 16-4-32-1, Sheet 1, Detail B):

(1) Disconnect electrical connector, P808 (left side), or P837 (right side).

(2) Install protective caps and plugs on electrical connectors.

NOTE

Install protective caps and plugs on hoses and fittings as parts are removed and disconnected.

d. On inboard bleed-air regulator valve, disconnect pneumatic hoses from unions in inlet and outlet ports (Figure 16-4-32-1, Sheet 1, Detail A).

e. On outboard bleed-air regulator valve, disconnect pneumatic hose from union in outlet port and pneumatic hoses from tee fitting in inlet port (Figure 16-4-32-1, Sheet 1, Detail B).

f. Disconnect extension hose from relief port of bleed-air regulator valve (Figure 16-4-32-1, Sheet 1, Detail A and Detail B).

g. Remove bolts and washers and remove bleed-air regulator valve from mounting bracket.

h. Remove tiedown straps and insulation blanket from bleed-air regulator valve.

i. On outboard bleed-air regulator valve, perform the following (Figure 16-4-32-1, Sheet 1, Detail B):

(1) Loosen jamnut and remove tee fitting and packing from inlet port. Discard packing.

(2) Remove union and packing from outlet port. Discard packing.

j. On inboard bleed-air regulator valve, remove unions and packings from inlet and outlet ports (Figure 16-4-32-1, Sheet 1, Detail A). Discard packings.

16-4-32.1.2 Disassemble.

a. Remove screws securing solenoid valve and plunger assembly to control valve body. Remove solenoid valve and plunger assembly (Figure 16-4-32-1, Sheet 2, Detail C).

b. Remove packing from solenoid valve. Discard packing.

c. Remove pushrod from stop.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

CAUTION

Damage to equipment will result if high pressure is used to remove poppet and spacer assembly from valve body. Do not use high air pressure to remove poppet and spacer assembly from valve body.

d. Using clean, dry, low-pressure compressed air, carefully remove poppet and spacer assembly (upper and lower spacers). Place hand over top of control valve body and blow air into vent fitting to dislodge parts. Discard packings.

e. Remove shim and spring from control valve body. Inspect for corrosion between control valve body and spring. **NONE ALLOWED.** If corrosion is found, replace bleed-air regulator valve.

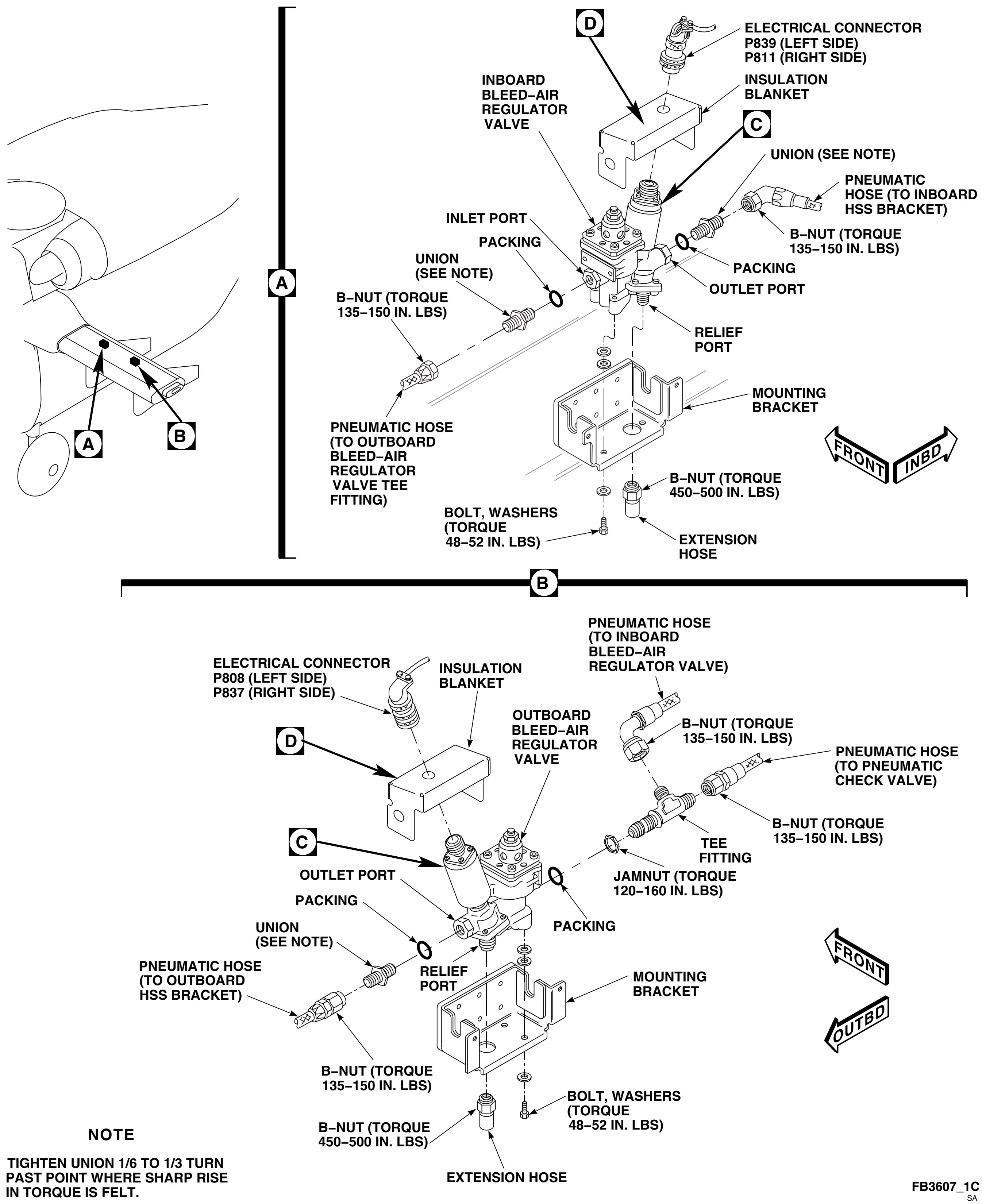


FIGURE 16-4-32-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) BLEED-AIR REGULATOR VALVES, UNIONS, AND TEE FITTING (SHEET 1 OF 3)

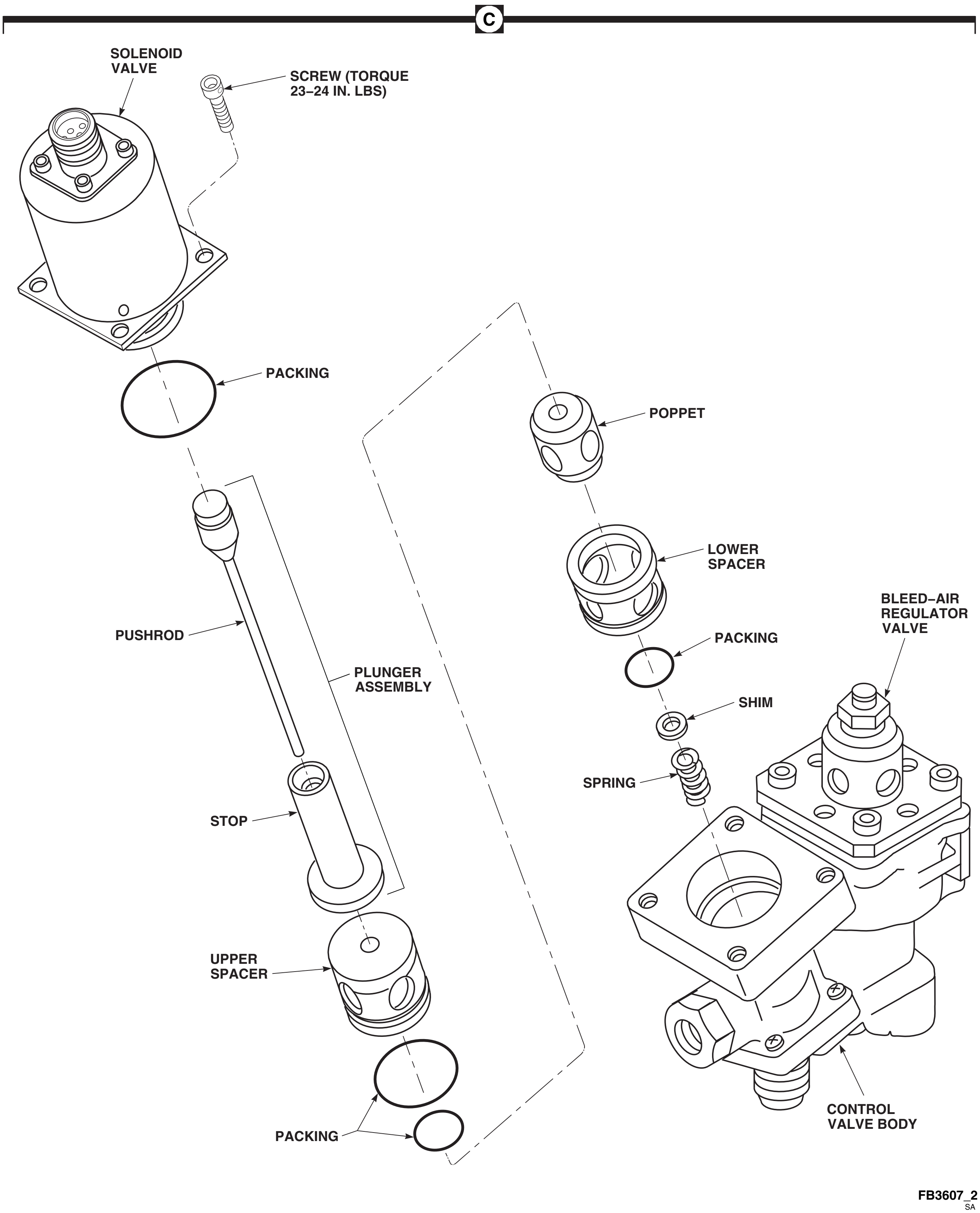


FIGURE 16-4-32-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) BLEED-AIR REGULATOR VALVES, UNIONS, AND TEE FITTING (SHEET 2 OF 3)

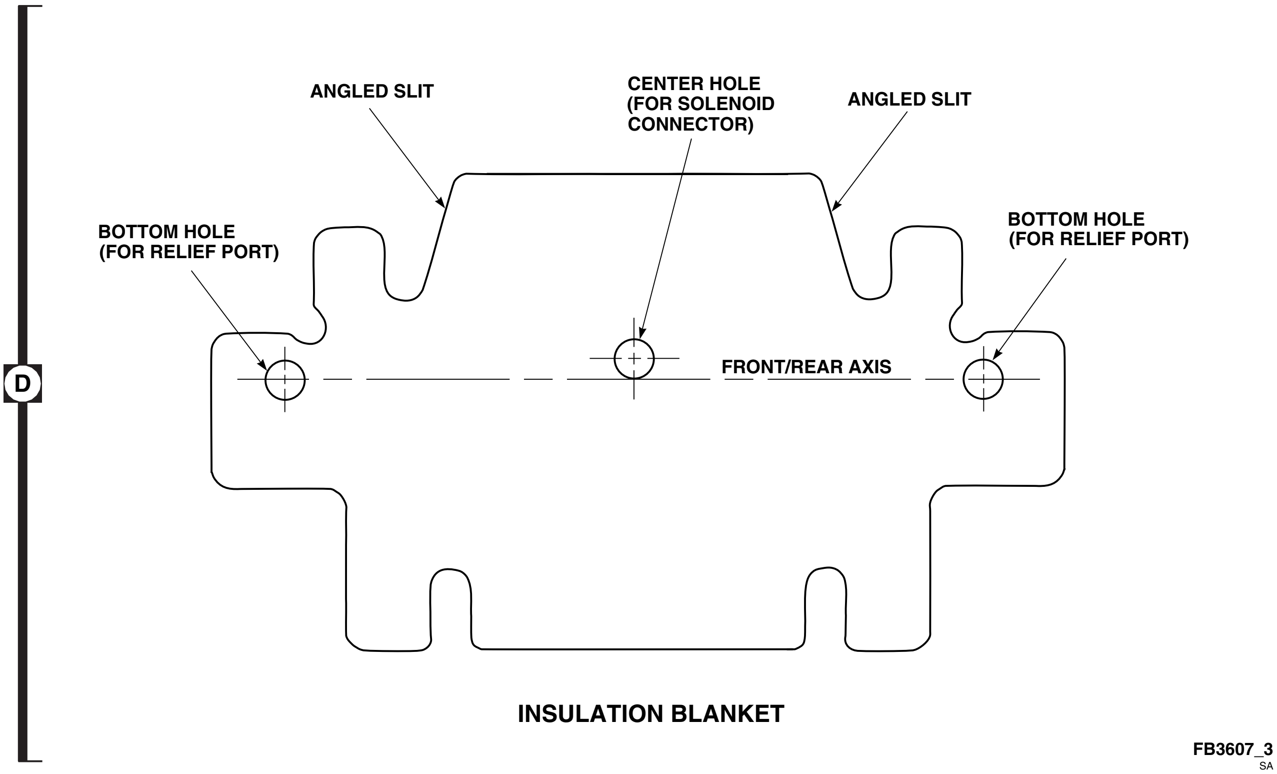


FIGURE 16-4-32-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) BLEED-AIR REGULATOR VALVES, UNIONS, AND TEE FITTING (SHEET 3 OF 3)

f. Using dry-cleaning solvent, Item 124, Appendix D, or cleaning compound, Item 94, Appendix D, clean plunger.

g. Remove sand and debris from bleed-air regulator valve by shaking.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

h. Remove any remaining sand or debris using compressed air (50 psi maximum), blown into valve through each opening.

i. Using dry-cleaning solvent, Item 124, Appendix D, or cleaning compound, Item 94, Appendix D, flush open ports.

16-4-32.1.3 Assemble.

a. Lubricate packings using petrolatum, Item 234, Appendix D. Install packing on bottom of lower spacer (Figure 16-4-32-1, Sheet 2, Detail C).

b. Install lower spacer in valve body with packing facing control valve body.

c. Install spring, shim, and poppet into control valve body.

d. Install packings on poppet, upper spacer, and solenoid valve.

e. Install upper spacer into valve body with packing facing control valve body.

f. Install pushrod into stop.

g. Install solenoid valve and plunger assembly on control valve body. **TORQUE SCREWS TO 23 - 24 INCH-POUNDS.**

16-4-32.1.4 Install.

a. Turn off all electrical power (PARA 1-6-16).

NOTE

Remove protective caps and plugs from hoses and fittings as parts are installed and connected.

b. On inboard bleed-air regulator valve, perform the following (Figure 16-4-32-1, Sheet 1, Detail A):

(1) Install packings lubricated with petrolatum, Item 234, Appendix D, on unions.

(2) Install union into inlet port. Thread union into boss of valve until face of union contacts valve. **TIGHTEN UNION ⅙ TO ⅓ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(3) Install union into outlet port. Thread union into boss of valve until face of union contacts valve. **TIGHTEN UNION ⅙ TO ⅓ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

c. On outboard bleed-air regulator valve, perform the following (Figure 16-4-32-1, Sheet 1, Detail B):

(1) Install jamnut on tee fitting.

(2) Install packing lubricated with petrolatum, Item 234, Appendix D, on tee fitting.

(3) Install tee fitting into inlet port. Make sure port of fitting is up and slightly forward. **TORQUE JAMNUT TO 120 - 160 INCH-POUNDS.**

(4) Install packing lubricated with petrolatum, Item 234, Appendix D, on union.

(5) Install union into outlet port. Thread union into boss of valve until face of union contacts valve. **TIGHTEN UNION ⅙ TO ⅓ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

d. Install insulation blanket to bleed-air regulator valve as follows (Figure 16-4-32-1, Sheet 1, Detail A and Detail B):

(1) Remove plastic sheet from insulation blanket.

(2) Position center hole of insulation blanket over electrical connector on regulator valve.

(3) Position insulation blanket so angled slits face pneumatic hose union side of solenoid (Figure 16-4-32-1, Sheet 1, Detail A and Detail B and Sheet 3, Detail D).

(4) Wrap insulation blanket around valve and secure outer holes in blanket over relief port union.

(5) Position slits in insulation blanket around both pneumatic fittings.

(6) Secure insulation blanket with two tiedown straps, Item 331, Appendix D, installed horizontally around bleed-air regulator valve.

e. Install bleed-air regulator valve on bracket with bolts and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

f. Using lockwire, Item 197, Appendix D, lockwire bolts.

g. Connect extension hose on relief port of bleed-air regulator valve. **TORQUE B-NUT TO 450 - 500 INCH-POUNDS.**

h. On inboard bleed-air regulator valve, connect pneumatic hoses as follows (Figure 16-4-32-1, Sheet 1, Detail A):

(1) Install pneumatic hose from inboard HSS bracket to union on inlet port of valve. **TORQUE B-NUT TO 135 - 150 INCH-POUNDS.**

(2) Install pneumatic hose from outboard bleed-air regulator valve tee fitting to union on outlet port of valve. **TORQUE B-NUT TO 135 - 150 INCH-POUNDS.**

i. On outboard bleed-air regulator valve, connect pneumatic hoses as follows (Figure 16-4-32-1, Sheet 1, Detail B):

(1) Install pneumatic hose from inboard bleed-air regulator valve to tee fitting on inlet port of valve. **TORQUE B-NUT TO 135 - 150 INCH-POUNDS.**

(2) Install pneumatic hose from pneumatic check valve to tee fitting on inlet port of valve. **TORQUE B-NUT TO 135 - 150 INCH-POUNDS.**

(3) Install pneumatic hose from outboard HSS bracket to union on outlet port of valve. **TORQUE B-NUT TO 135 - 150 INCH-POUNDS.**

j. On inboard bleed-air regulator valve, perform the following:

(1) Remove protective caps and plugs from electrical connectors.

(2) Inspect and treat electrical connectors (PARA 1-7-20).

(3) Connect electrical connector, P839 (left side), or P811 (right side) (Figure 16-4-32-1, Sheet 1, Detail A).

k. On outboard bleed-air regulator valves, perform the following:

(1) Remove protective caps and plugs from electrical connectors.

(2) Inspect and treat electrical connectors (PARA 1-7-20).

(3) Connect electrical connector, P808 (left side), or P837 (right side) (Figure 16-4-32-1, Sheet 1, Detail B).

l. Make sure area is clean and free of foreign material.

m. Perform operational check of ESSS range extension system and check for leaks (PARA 16-2-1 or PARA 16-2-2).

n. Install outboard trailing edge fairing on HSS (PARA 16-4-21).

16-4-33 ESSS HORIZONTAL STORES SUPPORT (HSS) FUEL SHUTOFF GATE VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 1-Gallon
- Digital Multimeter
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 16-2-1
- PARA 16-2-2
- PARA 16-4-21

16-4-33.1 Replace ESSS Horizontal Stores Support (HSS) Fuel Shutoff Gate Valve.

NOTE

Replacement of left and right ESSS HSS fuel shutoff gate valves is the same, except as noted.

16-4-33.1.1 Remove.

a. Remove rear vertical support pylon (VSP) fairing and saddle assembly (PARA 16-4-21).

NOTE

Install protective caps and plugs on hoses and valves as parts are disconnected and removed.

b. Remove inboard fuel shutoff gate valve as follows (Figure 16-4-33-1, Sheet 1, Detail A):

(1) Disconnect electrical connector, P809 (left side) or P810 (right side), from inboard fuel shutoff gate valve.

(2) Install protective caps and plugs on electrical connectors.

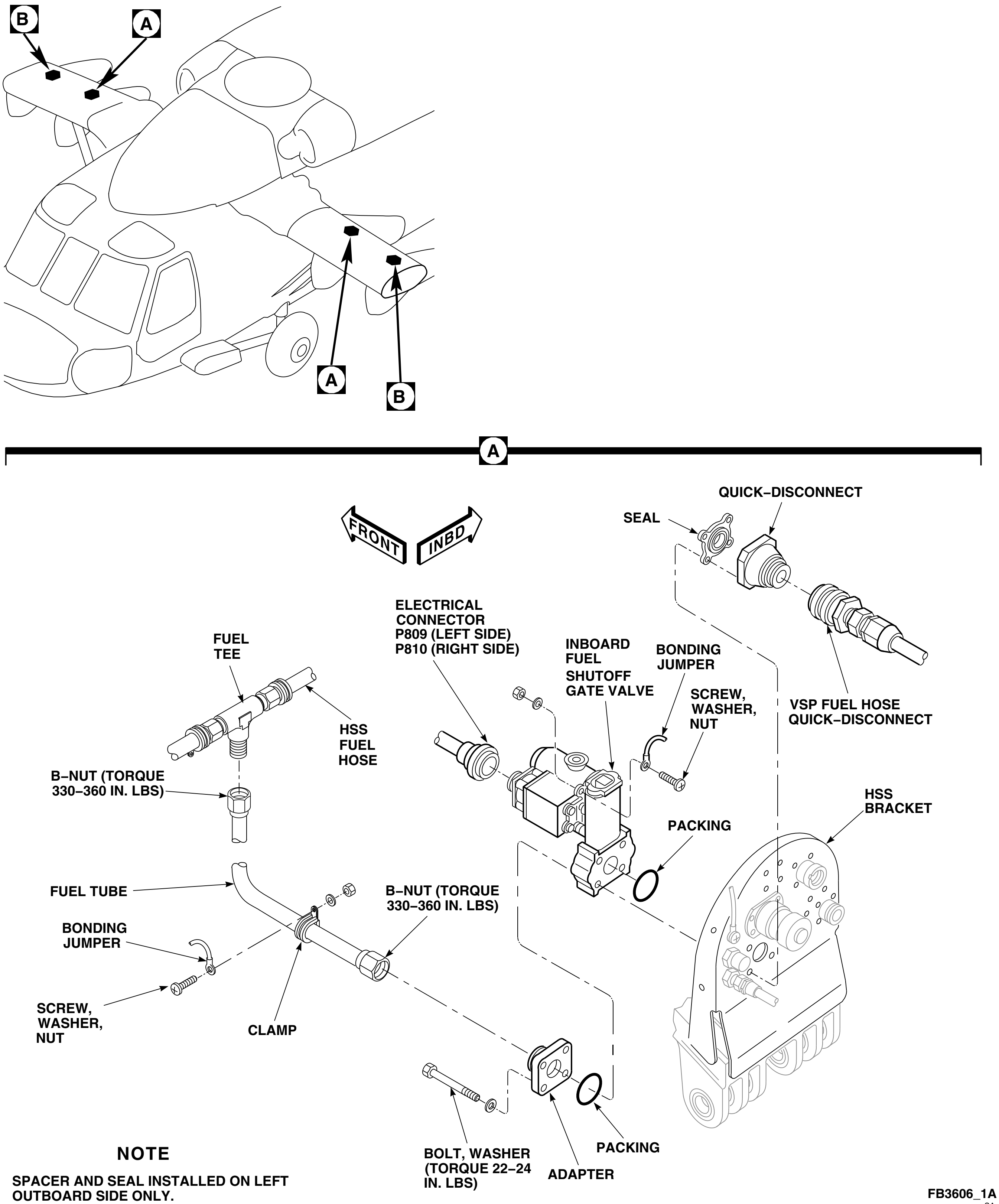


FIGURE 16-4-33-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) FUEL SHUTOFF GATE VALVE (SHEET 1 OF 2)

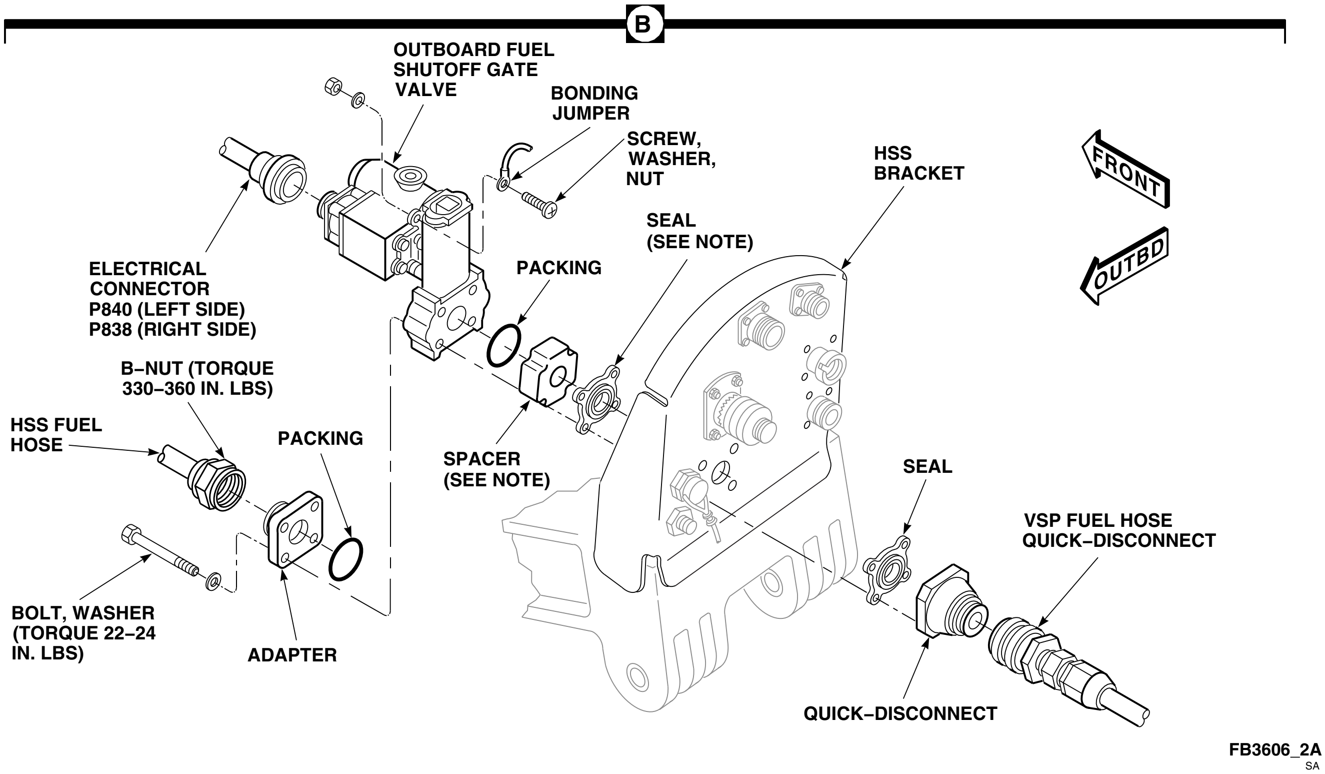


FIGURE 16-4-33-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) FUEL SHUTOFF GATE VALVE (SHEET 2 OF 2)

(3) Position 1-gallon container to catch fuel.

(4) Disconnect VSP fuel hose quick-disconnect from quick-disconnect on HSS bracket.

(5) Disconnect fuel tube from fuel tee on HSS fuel hose.

(6) Remove screw, washer, nut, clamp, and bonding jumper from fuel tube.

(7) Disconnect fuel tube from adapter on inboard fuel shutoff gate valve.

(8) Remove screw, washer, nut, and bonding jumper from fuel shutoff gate valve.

(9) Remove bolts and washers and remove adapter, inboard fuel shutoff gate valve, packings, seal, and quick-disconnect. Discard packings.

c. Remove outboard fuel shutoff gate valve, as follows (Figure 16-4-33-1, Sheet 2, Detail B):

(1) Disconnect electrical connector, P840 (left side), or P838 (right side) from outboard fuel shutoff gate valve.

(2) Install protective caps and plugs on electrical connectors.

(3) Position 1-gallon container to catch fuel.

(4) Disconnect VSP fuel hose quick-disconnect from quick-disconnect on HSS bracket.

(5) Disconnect HSS fuel hose from adapter on fuel shutoff gate valve.

(6) Remove screw, washer, nut, and bonding jumper from outboard fuel shutoff gate valve.

(7) Remove bolts and washers and remove adapter, outboard fuel shutoff gate valve, packings, spacer and seal (left side only), seal, and quick-disconnect. Discard packings.

16-4-33.1.2 Inspect.

NOTE

- The following resistance check is the same for all fuel shutoff gate valves.
- Allow approximately 20 seconds for capacitors to charge to attain a greater than 20 megohm reading.

a. Using digital multimeter, perform resistance test between fuel shutoff gate valve receptacles and valve body as follows:

(1) Measure resistance of left inboard fuel shutoff valve, receptacle, J809, by placing positive lead to pin -A of receptacle, J809, and common ground (valve body). Repeat procedure for pins -D and -E.

(2) Measure resistance of left outboard fuel shutoff valve, receptacle, J840, by placing positive lead to pin -A of receptacle, J840, and common lead to ground (valve body). Repeat procedure for pins -D and -E.

(3) Measure resistance of right inboard fuel shutoff valve, receptacle, J810, by placing positive lead to pin -A of receptacle, J810, and common lead to ground (valve body). Repeat procedure for pins -D and -E.

(4) Measure resistance of right outboard fuel shutoff valve, receptacle, J838, by placing positive lead to pin -A of receptacle, J838, and common lead to ground (valve body). Repeat procedure for pins -D and -E.

b. **MEASURED RESISTANCE AT ANY ONE OF THE INBOARD OR OUTBOARD FUEL SHUTOFF VALVES SHALL NOT BE LESS THAN 20 MEGOHMS WHEN TESTED.** If resistance is less than limit, proceed as follows:

(1) Replace affected shutoff valve.

(2) Repeat resistance test on each new fuel shutoff valve.

16-4-33.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).

NOTE

Remove protective caps and plugs from fuel hoses and valves as they are installed and connected.

b. Install inboard fuel shutoff gate valve, as follows (Figure 16-4-33-1, Sheet 1, Detail A):

(1) Coat packings with petrolatum, Item 234, Appendix D, and install in grooves of fuel shutoff gate valve.

(2) Install seal on quick-disconnect.

(3) Position fuel shutoff gate valve, adapter, and quick-disconnect on HSS bracket.

(4) Install bolts and washers. **TORQUE BOLTS IN A CRISSCROSS PATTERN TO 22 - 24 INCH-POUNDS.**

(5) Connect fuel tube to adapter on fuel shutoff gate valve. Do not torque nut.

(6) Connect fuel tube to fuel tee on HSS fuel hose.

(7) **TORQUE FUEL TUBE B-NUTS TO 330 - 360 INCH-POUNDS.**

(8) Install bonding jumper to fuel tube with clamp, screw, washer, and nut.

(9) Install bonding jumper to fuel shutoff gate valve with screw, washer, and nut.

(10) Connect vertical support pylon (VSP) fuel hose quick-disconnect to quick-disconnect on HSS bracket.

(11) Remove protective caps and plugs from electrical connectors.

(12) Inspect and treat electrical connectors (PARA 1-7-20).

(13) Connect electrical connector, P809 (left side), or P810 (right side).

c. Install outboard fuel shutoff gate valve as follows (Figure 16-4-33-1, Sheet 2, Detail B):

(1) Coat packings with petrolatum, Item 234, Appendix D, and install in grooves of shutoff gate valve.

(2) Install seal on quick-disconnect.

(3) On left outboard fuel shutoff gate valve, install seal on spacer.

(4) Position fuel shutoff gate valve, adapter, spacer with seal (left side only), seal, and quick-disconnect on HSS bracket.

(5) Install bolts and washers. **TORQUE BOLTS IN A CRISSCROSS PATTERN TO 22 - 24 INCH-POUNDS.**

(6) Connect HSS fuel hose to adapter on fuel shutoff gate valve. **TORQUE B-NUT TO 330 - 360 INCH-POUNDS.**

(7) Install bonding jumper to fuel shutoff gate valve with screw, washer, and nut.

(8) Connect VSP fuel hose quick-disconnect to quick-disconnect on HSS bracket.

(9) Remove protective caps and plugs from electrical connectors.

(10) Inspect and treat electrical connectors (PARA 1-7-20).

(11) Connect electrical connector, P840 (left side), or P838 (right side).

d. Make sure area is clean and free of foreign material.

e. Perform operational check of ESSS range extension system and check for leaks (PARA 16-2-1 or PARA 16-2-2).

f. Install rear VSP fairing and saddle assembly (PARA 16-4-21).

16-4-34 EJECTOR RACK, BRU-22A/A.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 16-4-21
- PARA 16-4-35
- PARA 16-4-54

16-4-34.1 Replace Ejector Rack, BRU-22A/A.

NOTE

Replacement of left and right ejector racks, BRU-22A/A, is the same, except as noted.

16-4-34.1.1 Remove.

a. Remove front and rear vertical support pylon fairings and saddle assembly (PARA 16-4-21).

NOTE

After removal of explosive cartridge, do not connect breech connector cap to breech.

b. Remove ejector rack, BRU-22A/A, explosive cartridge (PARA 16-4-35). Do not connect breech connector cap to breech.

c. Install protective cap and plug on breech connector cap and breech (Figure 16-4-34-1, Detail A).

d. If installed, remove ESSS external 230 gallon fuel tank (PARA 16-4-54).

e. Disconnect electrical connector, P1, from applicable electrical connector on horizontal stores support (HSS) bracket as follows (Figure 16-4-34-1, Detail B):

(1) J705 (left side - inboard)

(2) J707 (left side - outboard)

(3) J708 (right side - inboard)

(4) J710 (right side - outboard)

f. Remove cap from dummy receptacle on HSS bracket and install on electrical connector (J705, J707, J708, or J710) on HSS bracket.

Connect electrical connector, P1, to dummy receptacle from which cap was removed.

g. Remove cotter pins, bolts, countersunk washers, bushings, washers, and nuts securing ejector rack to adapters. Remove ejector rack (Figure 16-4-34-1, Detail A).

16-4-34.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

WARNING

FSCAP

Verification of proper hardware installation, torque, and safety installation of bolts are critical characteristics.

NOTE

- Make sure countersunk side of washer faces bolthead.
- Make sure adapters are installed on vertical support pylon (VSP) before installing ejector rack, BRU-22A/A.

b. Position ejector rack on adapters located on VSP (Figure 16-4-34-1, Detail A). Secure ejector rack as follows:

(1) Install bolts, countersunk washers, washers, bushings, and nuts in front and rear holes.

(2) **TIGHTEN NUTS FINGERTIGHT. ADVANCE NUTS TO NEXT CASTELLATION TO LINE UP COTTER PIN HOLES IN BOLTS.**

(3) Install cotter pins.

(4) Apply torque stripes (PARA 2-6-5).

c. Disconnect electrical connector, P1, from dummy receptacle on horizontal stores support (HSS) bracket. Remove cap from electrical connector (J705, J707, J708, or J710) on HSS bracket (Figure 16-4-34-1, Detail B).

d. Install cap on dummy receptacle.

e. Inspect and treat electrical connectors (PARA 1-7-20).

f. Connect electrical connector, P1, to applicable electrical connector on HSS bracket as follows:

(1) J705 (left side - inboard)

(2) J707 (left side - outboard)

(3) J708 (right side - inboard)

(4) J710 (right side - outboard)

g. Remove protective cap and plug from breech connector cap and breech.

h. Install ejector rack, BRU-22A/A, explosive cartridge (PARA 16-4-35).

NOTE

Do not install rear VSP fairings if external stores support system (ESSS) external fuel tanks are to be installed.

i. Install front and rear VSP fairings and saddle assembly (PARA 16-4-21).

j. If removed, install ESSS external 230 gallon fuel tank (PARA 16-4-54).

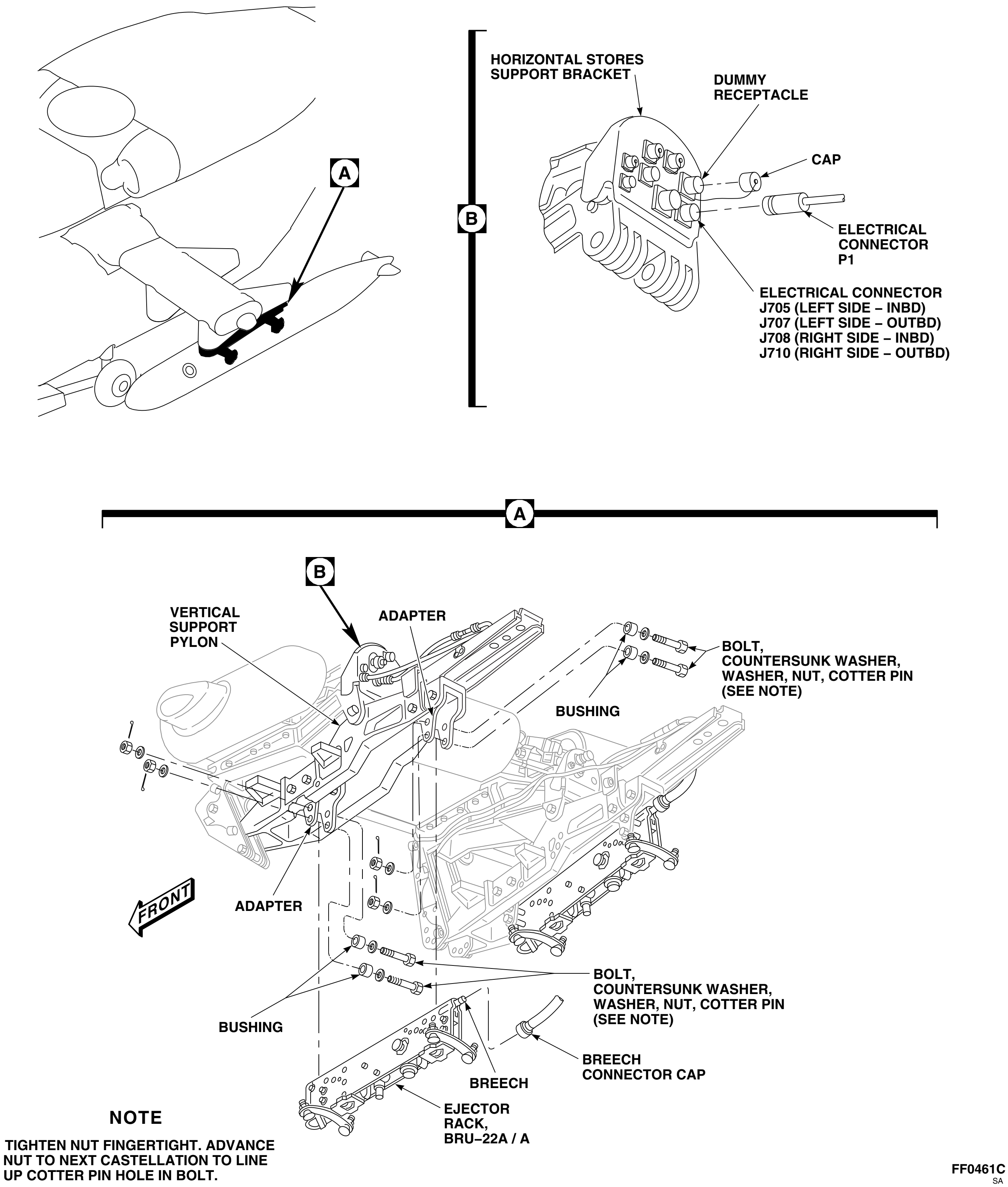


FIGURE 16-4-34-1. REPLACE EJECTOR RACK, BRU-22A/A

16-4-35 EJECTOR RACK, BRU-22A/A, EXPLOSIVE CARTRIDGE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Indelible Ink Marker

References

- PARA 1-4-16
- PARA 1-6-2
- PARA 1-6-16
- PARA 16-2-4

16-4-35.1 Replace Ejector Rack, BRU-22A/A, Explosive Cartridge.

NOTE

Replacement of left and right ejector racks, BRU-22A/A, explosive cartridges is the same.

16-4-35.1.1 Remove.

WARNING

Cartridges are explosive devices and may explode, causing injury to personnel or damage to equipment, if not handled with care. Use extreme caution when handling cartridges. Helicopter must not be in hangar when removing explosive cartridges. Make sure all electrical power to helicopter is turned off.

a. Turn off all electrical power (PARA 1-6-16).

b. Pull out ESSS JTSN INBD circuit breaker, DC ESNTL BUS - upper console circuit breaker panel and ESSS JTSN OUTBD circuit breaker, DC ESNTL BUS - upper console circuit breaker panel.

c. Pull out ESSS JTSN INBD circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel and ESSS JTSN OUTBD circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel.

d. Make sure jettison control panel switches are OFF.

e. Make sure safety stop lever is in locked position.

f. If necessary, tow helicopter to safe location outside hangar (PARA 1-6-2).

g. Disconnect breech connector cap from breech and remove explosive cartridge (Figure 16-4-35-1, Detail A).

h. If explosive cartridge has been fired, remove and clean breech (PARA 1-4-16).

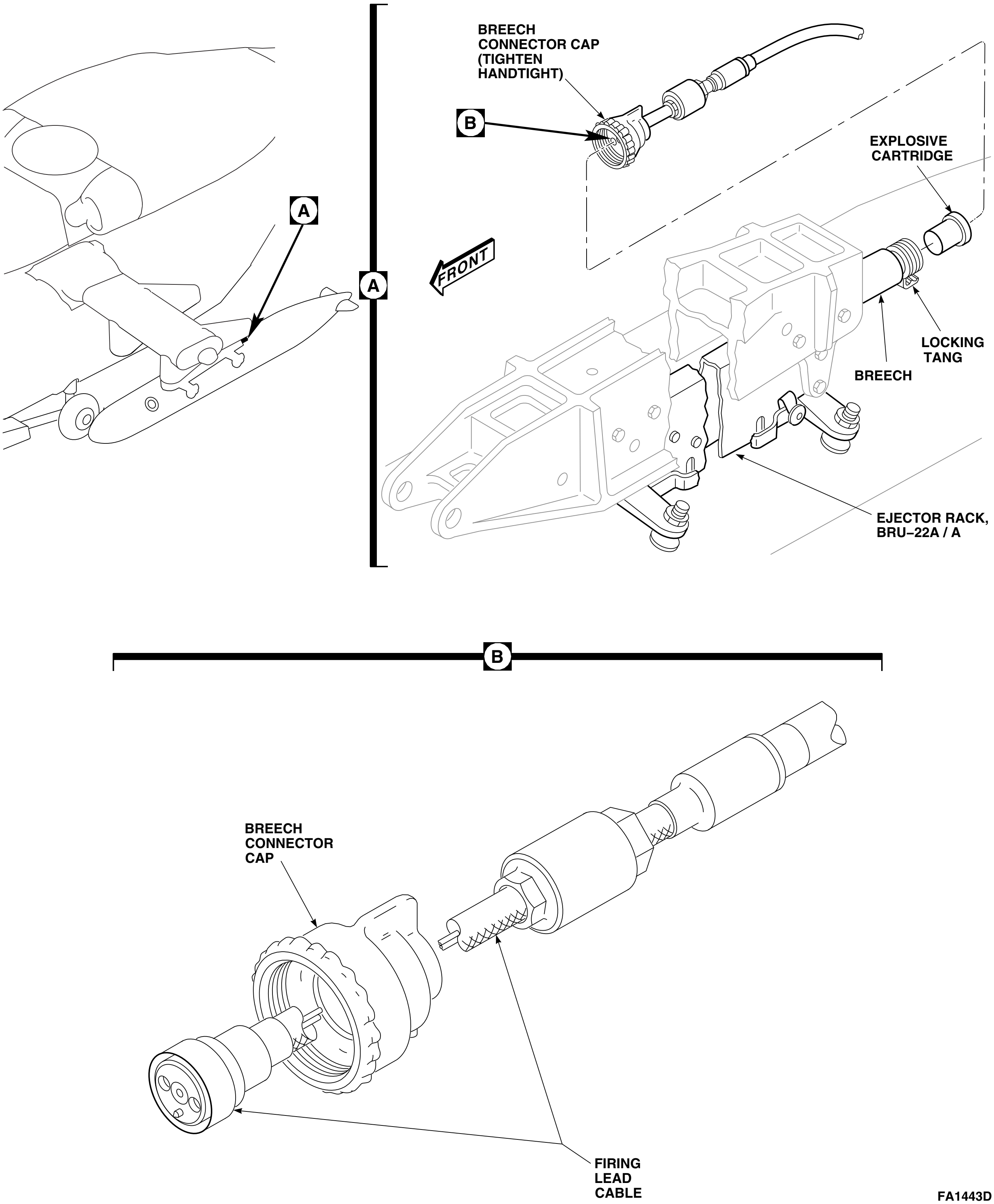


FIGURE 16-4-35-1. REPLACE EJECTOR RACK, BRU-22A/A, EXPLOSIVE CARTRIDGE

i. Perform stray voltage check of external stores support system jettison system (PARA 16-2-4).

j. Connect breech connector cap to breech. **TIGHTEN CAP HANDTIGHT. MAKE SURE LOCKING TANG ON BREECH CONTACTS BREECH CONNECTOR CAP.**

16-4-35.1.2 Inspect.

NOTE

- Ejector rack, BRU-22A/A, explosive cartridge is considered unserviceable after ten insertions and removals from ejector rack, or after expiration date marked on side of cartridge. Each time a cartridge is removed, a radial mark is placed on base of cartridge with indelible ink.

- Each radial mark on base of explosive cartridge represents one removal and installation.

a. Inspect explosive cartridge for expiration date. Proceed as follows:

(1) If expiration date is valid, proceed to step b.

(2) If expiration date has expired, properly discard explosive cartridge and obtain new cartridge.

b. Inspect base of explosive cartridge for presence of radial mark(s). Proceed as follows:

(1) If less than 10 radial marks are present, using indelible ink marker, place a radial mark on base of cartridge and retain for installation.

(2) If 10 radial marks are present, properly discard explosive cartridge and obtain new cartridge.

16-4-35.1.3 Install.

WARNING

Cartridges are explosive devices and may explode, causing injury to personnel or damage to equipment, if not handled with care. Use extreme caution when handling cartridges. Helicopter must not be in hangar when removing explosive cartridges. Make sure all electrical power to helicopter is turned off.

a. Turn off all electrical power (PARA 1-6-16).

b. Pull out ESSS JTSN INBD circuit breaker, DC ESNTL BUS - upper console circuit breaker panel and ESSS JTSN OUTBD circuit breaker, DC ESNTL BUS - upper console circuit breaker panel.

c. Pull out ESSS JTSN INBD circuit breaker, NO. 1 DC PRI BUS - copilot’s circuit breaker panel and ESSS JTSN OUTBD circuit breaker, NO. 1 DC PRI BUS - copilot’s circuit breaker panel.

d. Make sure jettison control panel switches are OFF.

CAUTION

Damage to ejector rack piston may occur if external stores are not installed on racks before arming ejector racks. Do not arm ejector rack before installing external stores.

e. Make sure safety stop lever is in locked position.

f. Disconnect breech connector cap from breech (Figure 16-4-35-1, Detail A).

g. Perform stray voltage check of external stores support system jettison system (PARA 16-2-4).

h. Install explosive cartridge in breech, making sure flange of cartridge is tight against end of breech.

i. Connect breech connector cap to breech. **TIGHTEN CAP HANDTIGHT. MAKE SURE LOCKING TANG ON BREECH CONTACTS BREECH CONNECTOR CAP.**

j. Check firing lead cable for binding in breech connector cap (Figure 16-4-35-1, Detail B). If cable is binding, loosen cap to free cable. **RETIGHTEN CAP.**

k. Push in ESSS JTSN INBD circuit breaker, DC ESNTL BUS - upper console circuit breaker panel and ESSS JTSN OUTBD circuit breaker, DC ESNTL BUS - upper console circuit breaker panel.

l. Push in ESSS JTSN INBD circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel and ESSS JTSN OUTBD circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel.

m. If necessary, tow helicopter into hangar (PARA 1-6-2).

16-4-36 EJECTOR RACK, BRU-22A/A, REPAIR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Feeler Gage
- Pin Punch
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Abrasive Cloth, Item 9, Appendix D
- Aluminum Oxide Abrasive Material
- Corrosion Resistant Coating, Item 118, Appendix D

References

- Appendix D
- PARA 16-4-35
- PARA 16-5-8

16-4-36.1 Repair Ejector Rack, BRU-22A/A.

NOTE

Repair of left and right ejector racks, BRU-22A/A, is the same.

16-4-36.1.1 Disassemble.

WARNING

Injury to personnel and damage to equipment will result if explosive cartridge is installed when repairing ejector rack. Cartridge could explode. Make sure explosive cartridge is removed before performing any repairs.

a. Remove ejector rack, BRU-22A/A, explosive cartridge (PARA 16-4-35).

NOTE

Perform the following procedures to completely disassemble ejector rack, BRU-22A/A. When complete disassembly is not required, refer to appropriate step.

b. Remove breech and gun as follows:

(1) Place safety stop lever in unlocked position.

(2) Pull manual release lever front and hold.

(3) Rotate breech 90° and pull rear to remove from ejector rack housing, then release manual release lever.

(4) If required, remove pressure rings from breech.

(5) Remove gun retaining pin assembly from ejector rack housing (Figure 16-4-36-2).

(6) Rotate gun front and down until clear of ejector rack housing.

(7) Rotate retainer 90° and pull down to remove piston from gun.

(8) Remove bearing ring and pressure rings from piston. Remove threaded foot from piston.

c. Remove sway brace as follows:

(1) Remove front sway brace as follows (Figure 16-4-36-1):

(a) Punch out spring pin securing front sway brace to ejector rack housing. Pull front sway brace down until clear of housing.

(b) Remove screws, washers, nose fairing, and rocket blast shield from ejector rack housing.

(2) Remove rear sway brace as follows (Figure 16-4-36-1, Detail A):

(a) Punch out spring pin and remove bolts securing rear sway brace to ejector rack housing. Pull rear sway brace to rear until clear of housing.

(b) Remove nut and washer from bolt securing safety stop lever and safety stop bushing to rear sway brace. Remove bolt, safety stop lever, safety stop bushing, and washer(s), if installed, from rear sway brace.

d. Remove hook assembly as follows:

(1) Remove front hook assembly as follows (Figure 16-4-36-2):

WARNING

Injury to personnel will result if caution is not used when removing front lug guide plate. Front lug guide plate is under spring tension. Use care when removing bolts.

(a) Remove bolts, washers, and nuts securing front lug guide plate to ejector rack housing.

(b) Remove front lug guide plate, spring, and hook spring from bottom of ejector rack housing.

(c) Remove pins and washers from ejector rack housing, and remove front hook assembly from bottom of ejector rack housing.

(d) Remove pins and disassemble hook assemblies as required for replacement of toggle rod roller, toggle spring rod, hook toggle lever, hook, or hook toggle pawl (Figure 16-4-36-2, Detail A).

(2) Remove rear hook assembly as follows (Figure 16-4-36-2):

WARNING

Injury to personnel will result if caution is not used when removing rear lug guide plate. Rear lug guide plate is under spring tension. Use care when removing bolts.

(a) Remove bolts, washers, and nuts securing rear lug guide plate to ejector rack housing.

(b) Remove rear lug guide plate, spring, and hook spring from bottom of ejector rack housing.

(c) Remove pin, washer, and gun retaining pin assembly from ejector rack housing. Remove gun retaining pin assembly from pin.

(d) Remove pins and washers from ejector rack housing, and remove rear

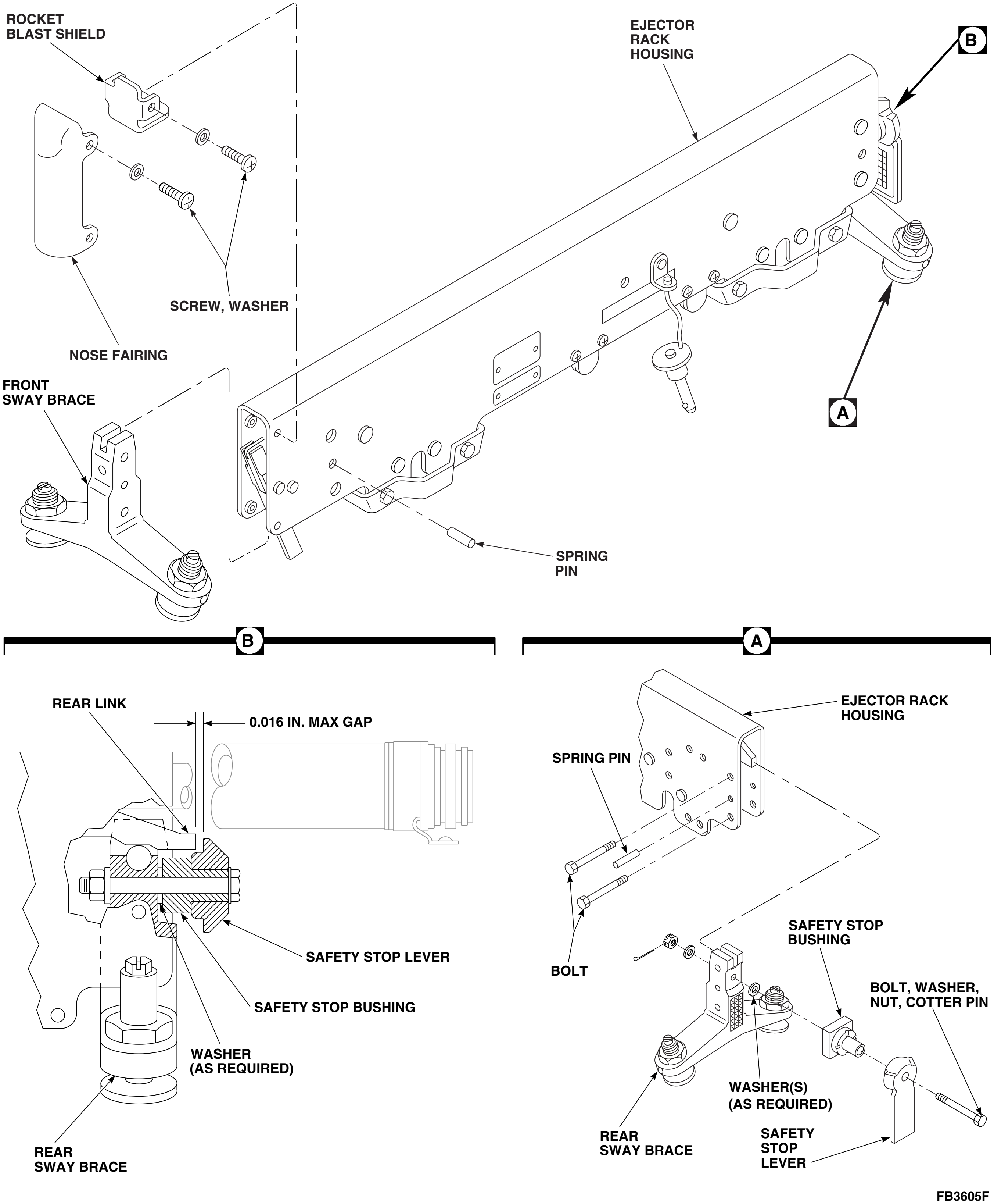


FIGURE 16-4-36-1. REPLACE SWAY BRACE

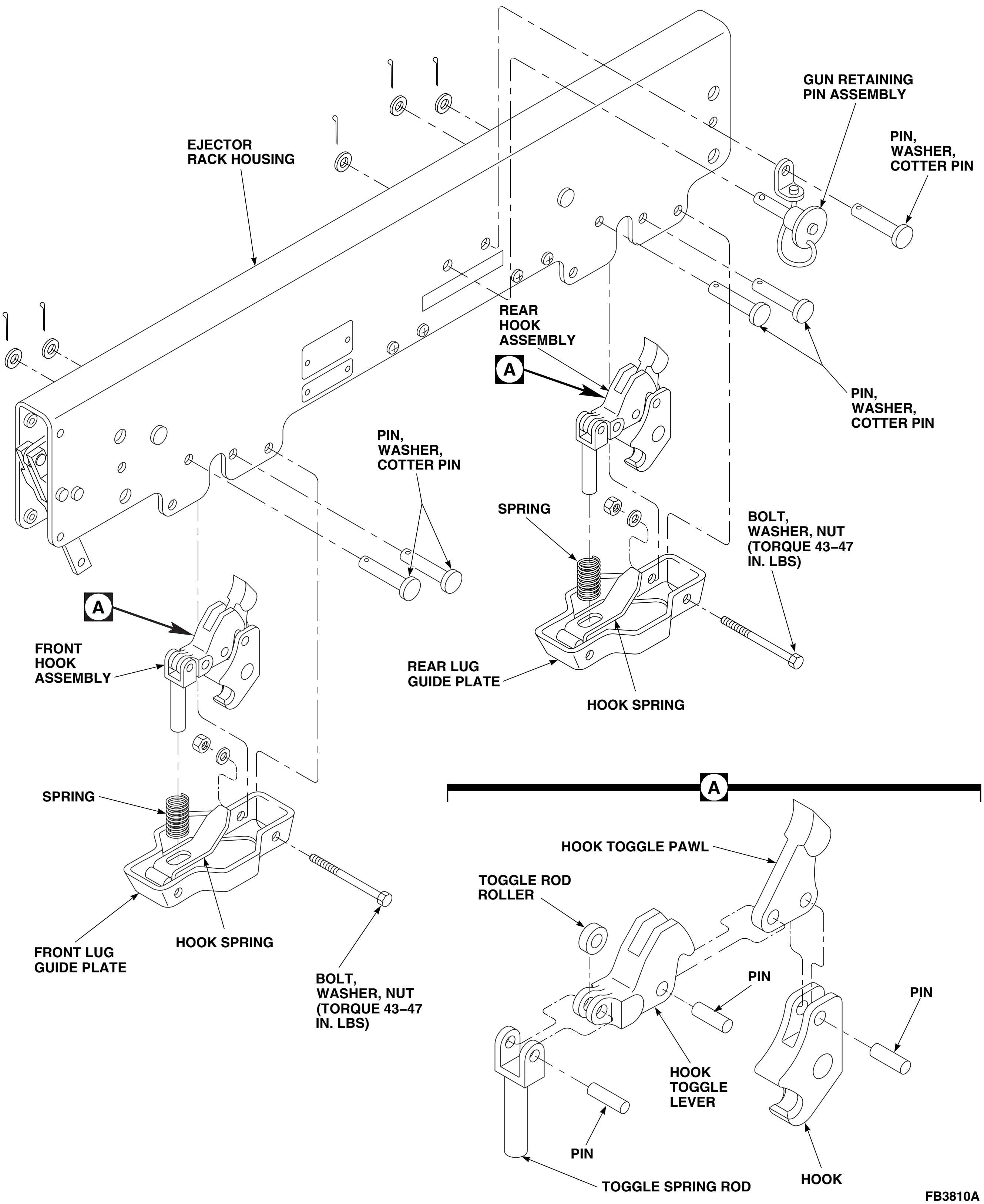


FIGURE 16-4-36-2. REPLACE HOOK ASSEMBLY

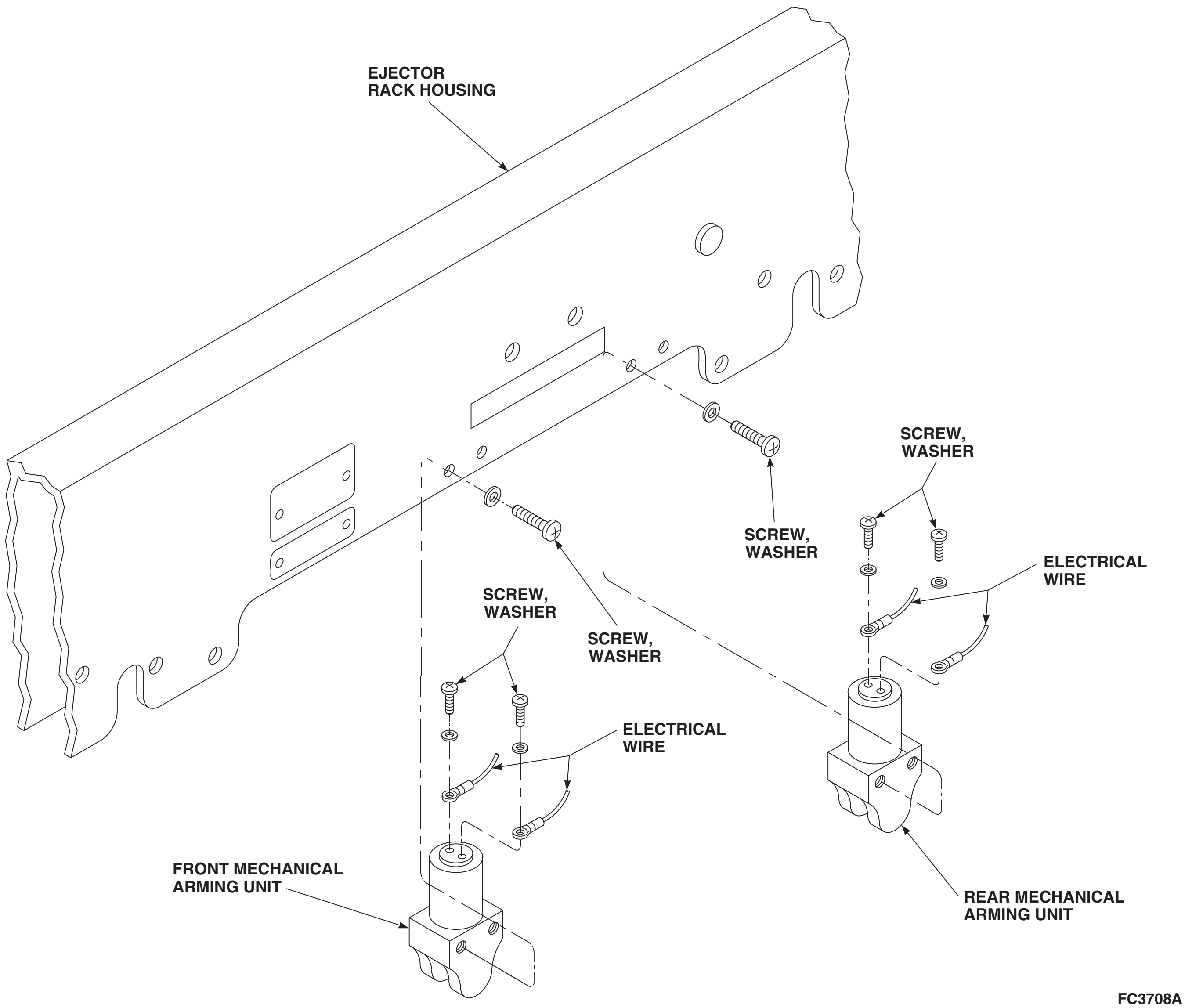


FIGURE 16-4-36-3. REPLACE MECHANICAL ARMING UNIT

hook assembly from bottom of ejector rack housing.

(3) Remove pins and disassemble hook assemblies as required for replacement of toggle rod roller, toggle spring rod, hook toggle lever, hook, or hook toggle pawl (Figure 16-4-36-2, Detail A).

e. Remove mechanical arming unit as follows (Figure 16-4-36-3):

NOTE

Removal of front and rear mechanical arming units is the same.

(1) Remove screws, washers, and lower front or rear mechanical arming unit until clear of ejector rack housing.

(2) Remove screws, washers, and remove electrical wires from mechanical arming unit.

f. Remove hook release assembly and wiring harness as follows (Figure 16-4-36-4, Sheet 1, Detail A):

(1) Remove nuts and washers from screws and remove arming unit wiring clip from ejector rack housing.

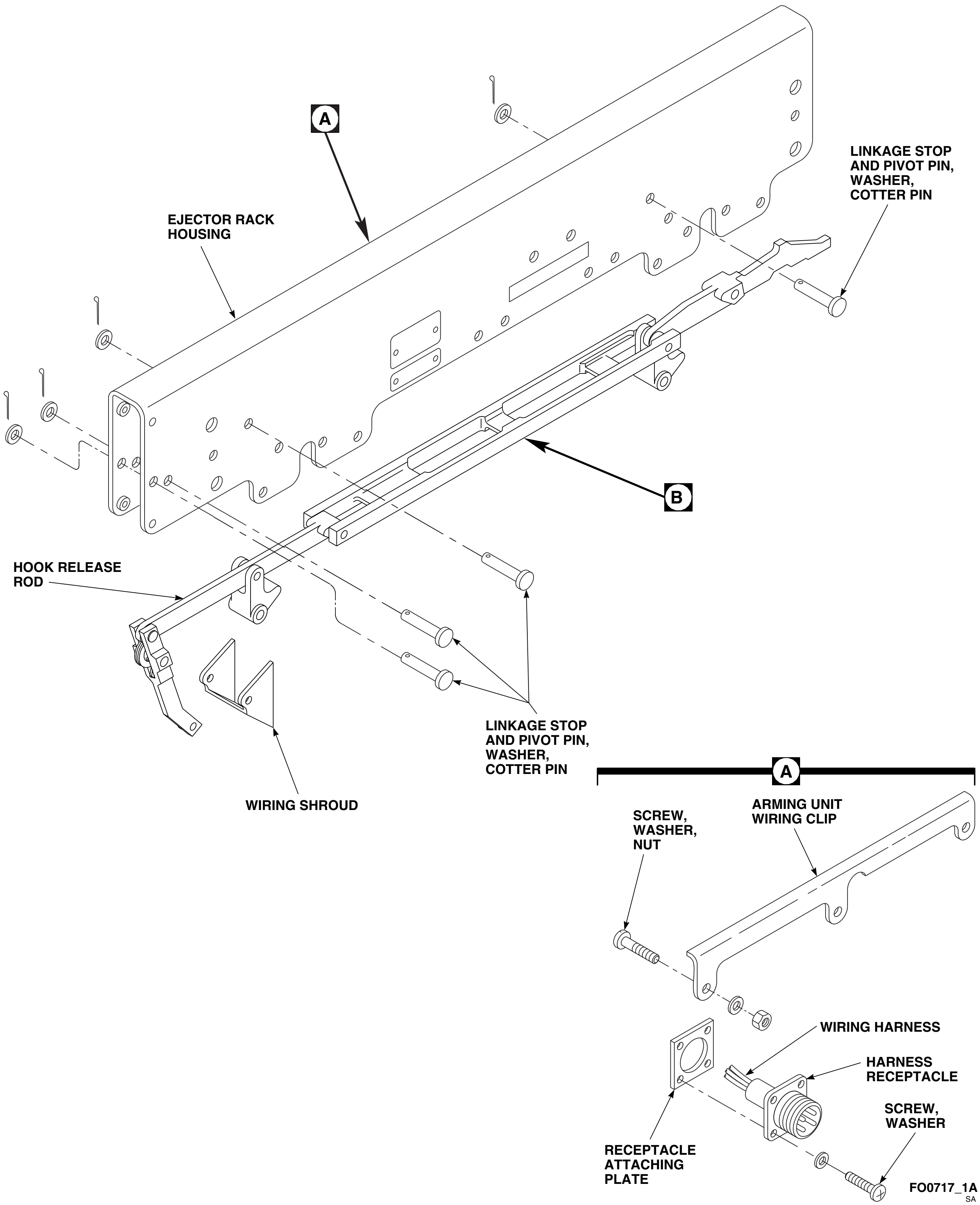


FIGURE 16-4-36-4. REPLACE HOOK RELEASE ASSEMBLY AND WIRING HARNESS (SHEET 1 OF 2)

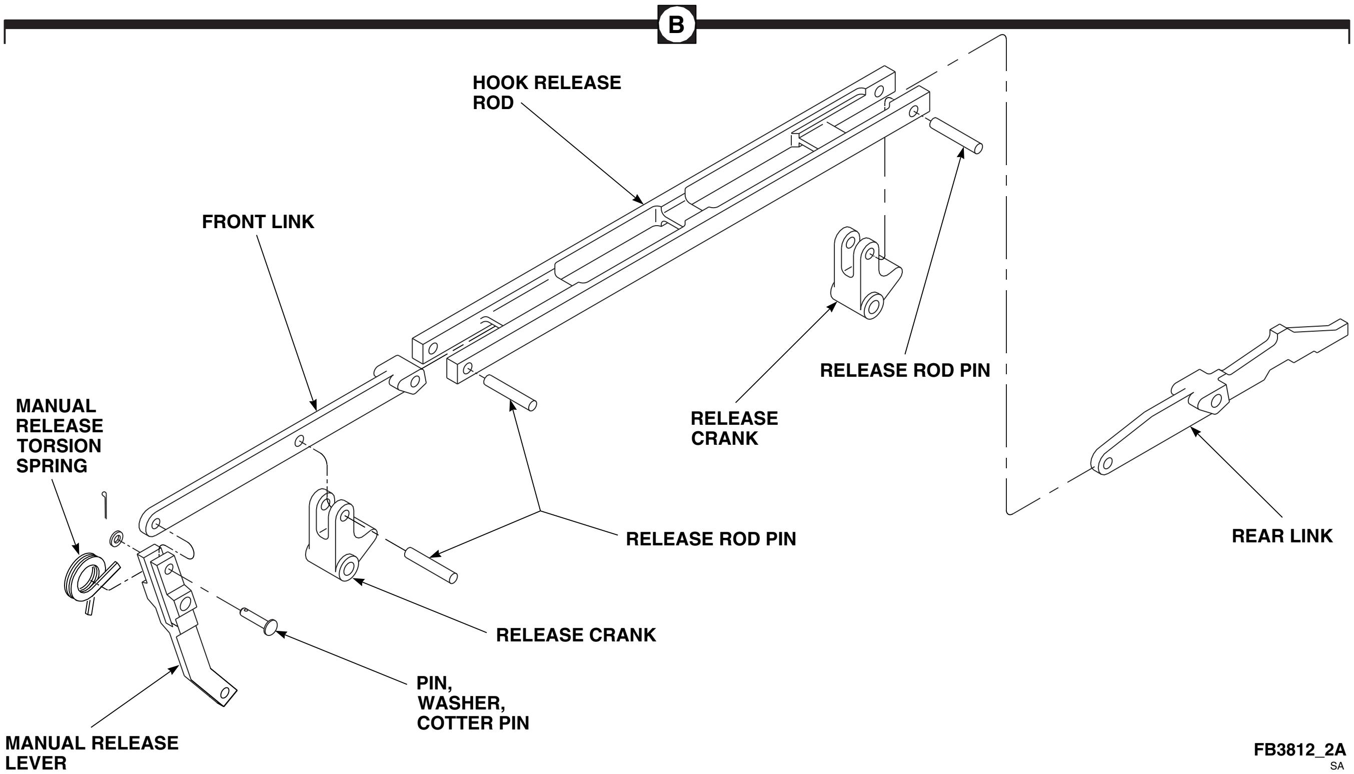


FIGURE 16-4-36-4. REPLACE HOOK RELEASE ASSEMBLY AND WIRING HARNESS (SHEET 2 OF 2)

(2) Remove screws, washers, receptacle attaching plate, and remove wiring harness from ejector rack housing.

(3) Remove linkage stops, pivot pins, and washers securing hook release rod to ejector rack housing (Figure 16-4-36-4, Sheet 1).

(4) Remove hook release rod from ejector rack housing.

(5) Remove manual release torsion spring and wiring shroud from hook release rod (Figure 16-4-36-4, Sheet 1 and Sheet 2, Detail B).

(6) Remove pin, washer, and manual release lever from front link (Figure 16-4-36-4, Sheet 2, Detail B).

(7) Remove release rod pins and separate rear link, front link, and release cranks from hook release rod.

16-4-36.1.2 Inspect Ejector Rack, BRU-22A/A, Components.

NOTE

- Ejector racks are designed to extremely close tolerances. Use of rack and its exposure to various atmospheric or storage conditions, particularly humid, salt-laden air, and stack gases, erodes both moving and stationary parts.

- Parts that do not meet dimensional limits must be discarded.

- Sliding surfaces (shafts, pins, holes, slots, etc.) may have slightly roughened surface because of wear. Raised roughness or corrosion that will act as an abrasive on mating surfaces is not acceptable.

- Figure 16-4-36-5, Sheet 1, Sheet 2, Sheet 3, Sheet 4, Sheet 5, Sheet 6, and

Sheet 7, illustrate maximum wear limits on ejector rack parts.

a. Visually inspect all internal linkages, springs, and pins that are visible without further disassembly for evidence of cracked or broken parts. **NONE ALLOWED.** If cracked or broken parts are found, reject ejector rack.

b. Inspect pin hole areas of ejector rack housing for distortion and corrosion. **NONE ALLOWED.** If distortion or corrosion is found, reject ejector rack housing.

c. Inspect piston for bends and condition of retainer. If piston is bent or condition of retainer cannot be determined, discard piston.

d. Inspect pressure rings and bearing ring in piston and breech for cracks, breaks, and corrosion.

e. Inspect rings for proper installation. Rings are properly installed when spring action keeps piston and breech in gun with binding (Figure 16-4-36-5, Sheet 3). If not properly installed, replace rings.

16-4-36.1.3 Corrosion Inspection and Treatment.

NOTE

Minor corrosion in areas other than hook pivot pin holes should be removed with mildest technique feasible.

a. Using abrasive cloth, Item 9, Appendix D, or aluminum oxide abrasive material with water, remove corrosion. Finish with abrasive cloth, Item 8, Appendix D.

CAUTION

The use of vacu-blast using glass beads is permissible only by personnel trained in the technique. Do not use abrasives in hook pivot pin holes or hook safety pin holes in housing, link pin holes in hooks, link holes in links, or toggle lever pin holes in housing. Do not use abrasives on hook pivot pins, toggle lever pins, link pins, or mechanical safety pins.

b. Rework limits for corrosion removal are provided in Figure 16-4-36-5, Sheet 1.

c. Treat aluminum surfaces with corrosion resistant coating, Item 118, Appendix D, following corrosion removal and prior to painting.

16-4-36.1.4 Assemble.

a. Install hook release assembly and wiring harness as follows:

(1) Install rear link, front link, and release cranks on hook release rod using release rod pins (Figure 16-4-36-4, Sheet 2, Detail B).

(2) Position manual release lever on front link and secure with pin and washer. Install cotter pin.

(3) Place manual release torsion spring in slot on manual release lever.

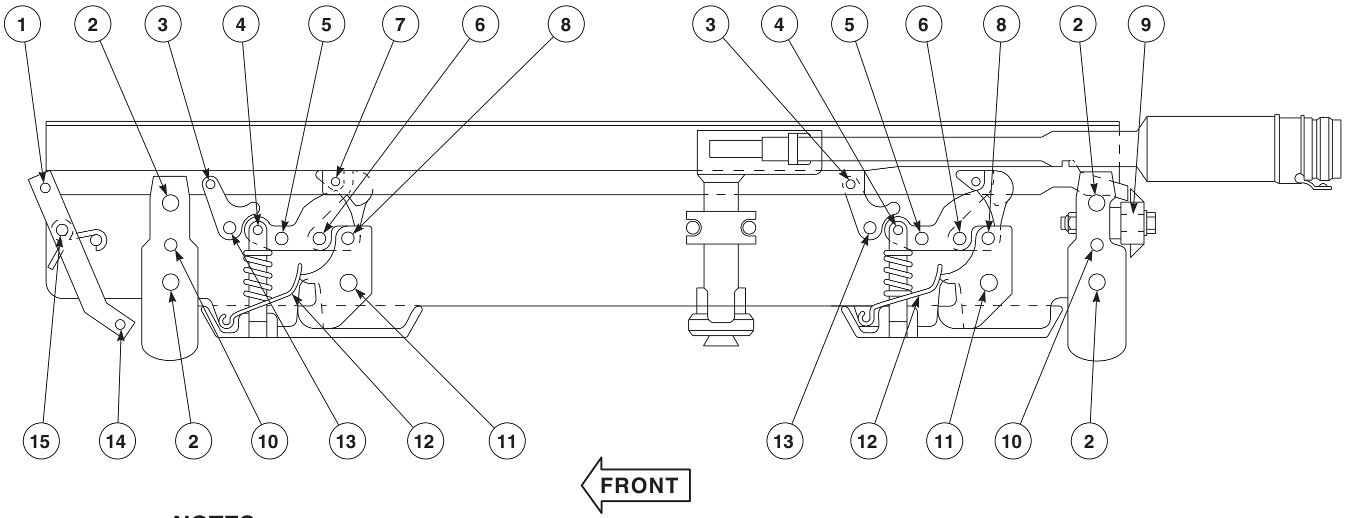
(4) Install wiring shroud into ejector rack housing (Figure 16-4-36-4, Sheet 1).

(5) Install hook release rod into ejector rack housing. Secure hook release rod and wiring shroud to housing with linkage stops, pivot pins, and washers. Install cotter pins.

(6) Check action of hook release assembly by pulling manual release lever front, then releasing; spring action must return manual release lever to full rear position.

(7) Position wiring harness on ejector rack housing with harness receptacle on right side of housing and receptacle key at 6 o'clock position.

(8) Secure harness receptacle to ejector rack housing with receptacle attaching plate, screws, and washers (Figure 16-4-36-4, Sheet 1, Detail A).



NOTES

1. ALL PARTS TO BE WITHIN DIMENSION LIMITS GIVEN AFTER CLEANING, CORROSION REMOVAL, OR REPAIR.
2. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.
3. INDEX NUMBERS ARE KEYED TO PARTS AT THAT LOCATION.
4. 0.047 IN. MAXIMUM TOTAL FREE PLAY FOR ASSEMBLY. (THIS DIMENSION DOES NOT PERMIT MAXIMUM ADVERSE ACCUMULATION OF TOLERANCES).
5. RINGS TO BE CHECKED IN GROOVE OF PISTON OR BREECH AND IN FREE (UNCOMPRESSED) CONDITION. MEASURE DIA 2 PLACES.
6. ORIGINAL SURFACE FINISH ON GUN CYLINDER BORES IS 16 MICROINCHES. SURFACE FINISH CHANGE TO APPROXIMATELY 63 MICROINCHES IS ACCEPTABLE. DEEP SCRATCHES OR ROUGH AREAS WHICH WILL AFFECT RINGS SEAL ARE NOT ACCEPTABLE.
7. TEST OVER 3/8 IN. AND 1/4 IN. DIA RODS.
8. MINIMUM SPRING–TORSION IS 10.8 IN. LBS TORQUE AND MAXIMUM IS 13.2 IN. LBS TORQUE AT INITIAL POSITION.
9. MINIMUM SPRING–TORSION IS 27.0 IN. LBS TORQUE AND MAXIMUM IS 33.0 IN. LBS TORQUE AT FINAL POSITION.
10. HOLES MAY HAVE SLIGHTLY ROUGHENED SURFACE DUE TO WEAR. RAISED ROUGHNESS WHICH WILL ACT AS ABRASIVE ON MATING SURFACE IS NOT ACCEPTABLE.
11. ○ DENOTES INTERNAL PINS.
12. ● DENOTES EXTERNAL PINS.
13. MINIMUM LOAD IS 14.5 POUNDS AT 15/16 IN. LENGTH. MAXIMUM LOAD IS 18.5 POUNDS.
14. SPRING TO COMPRESS SOLID WITHOUT PERMANENT SET.
15. REPLACE PIN WHEN ANY WEAR APPEARS LIKELY TO IMPAIR THE FUNCTION OF THE BALL–LOCKING MECHANISM.
16. MATERIAL MAY BE REMOVED ON OUTER SURFACE IN THIS AREA TO A MAXIMUM DEPTH OF 0.010 IN.
17. MATERIAL MAY BE REMOVED ON OUTER SURFACES IN THIS AREA. DO NOT EXCEED 0.010 IN. DEEP BY 1.0 IN. MEASURED NORMAL TO LONGITUDINAL AXIS FOR ANY ONE AREA. REMOVAL AREA WHEN LENGTH IS GREATER THAN WIDTH SHALL RUN PARALLEL TO LONGITUDINAL AXIS WITHIN ± 30 DEGREES. MATERIAL SHALL NOT BE REMOVED WITHIN 3/8 IN. OF HOLES. MINIMUM RADIUS AT BOTTOM OF GROOVES OR REMOVED AREA TO BE 1/4 IN.

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FIGURE 16-4-36-5. INSPECT EJECTOR RACK, BRU-22A/A, COMPONENTS (SHEET 1 OF 7)

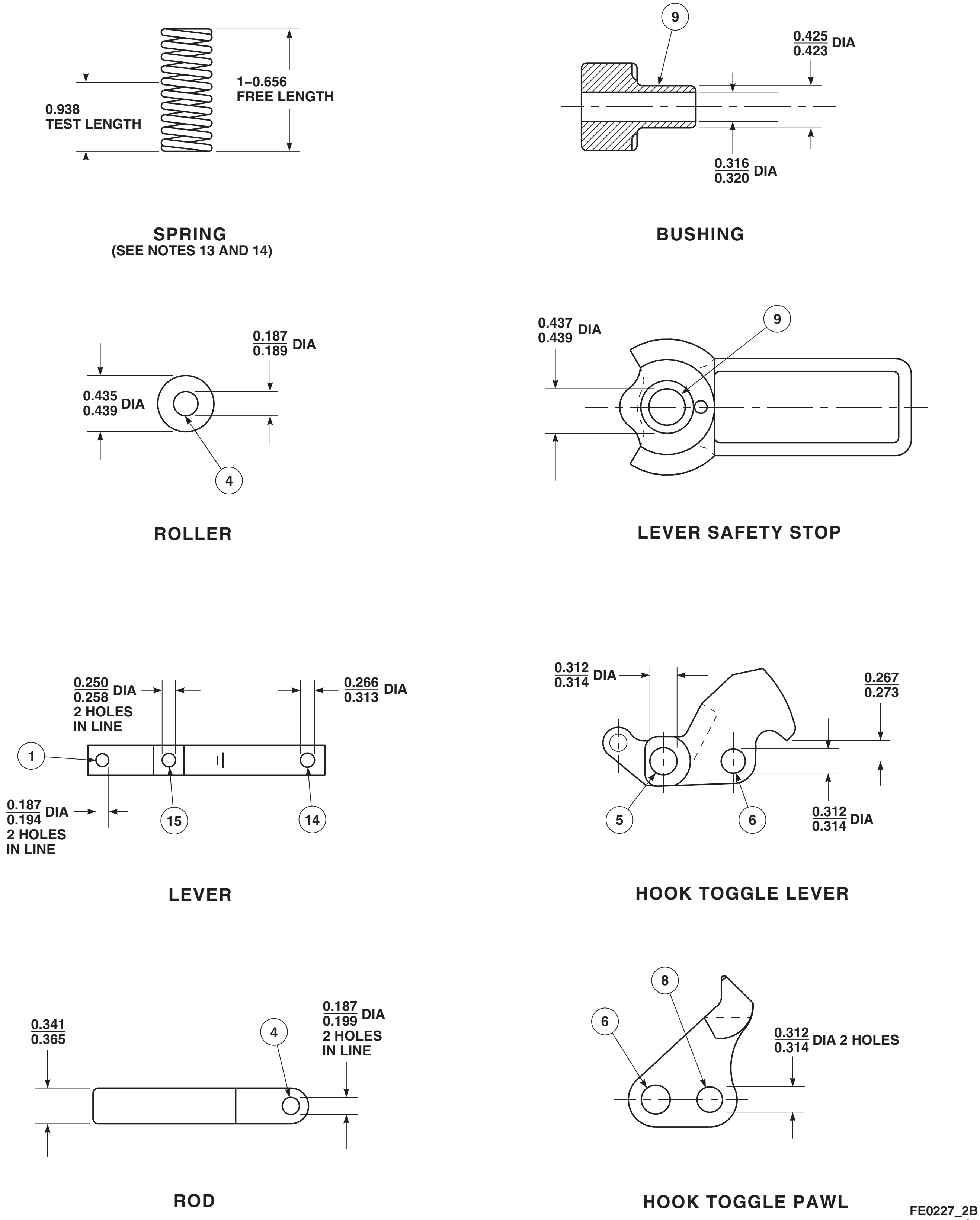


FIGURE 16-4-36-5. INSPECT EJECTOR RACK, BRU-22A/A, COMPONENTS (SHEET 2 OF 7)

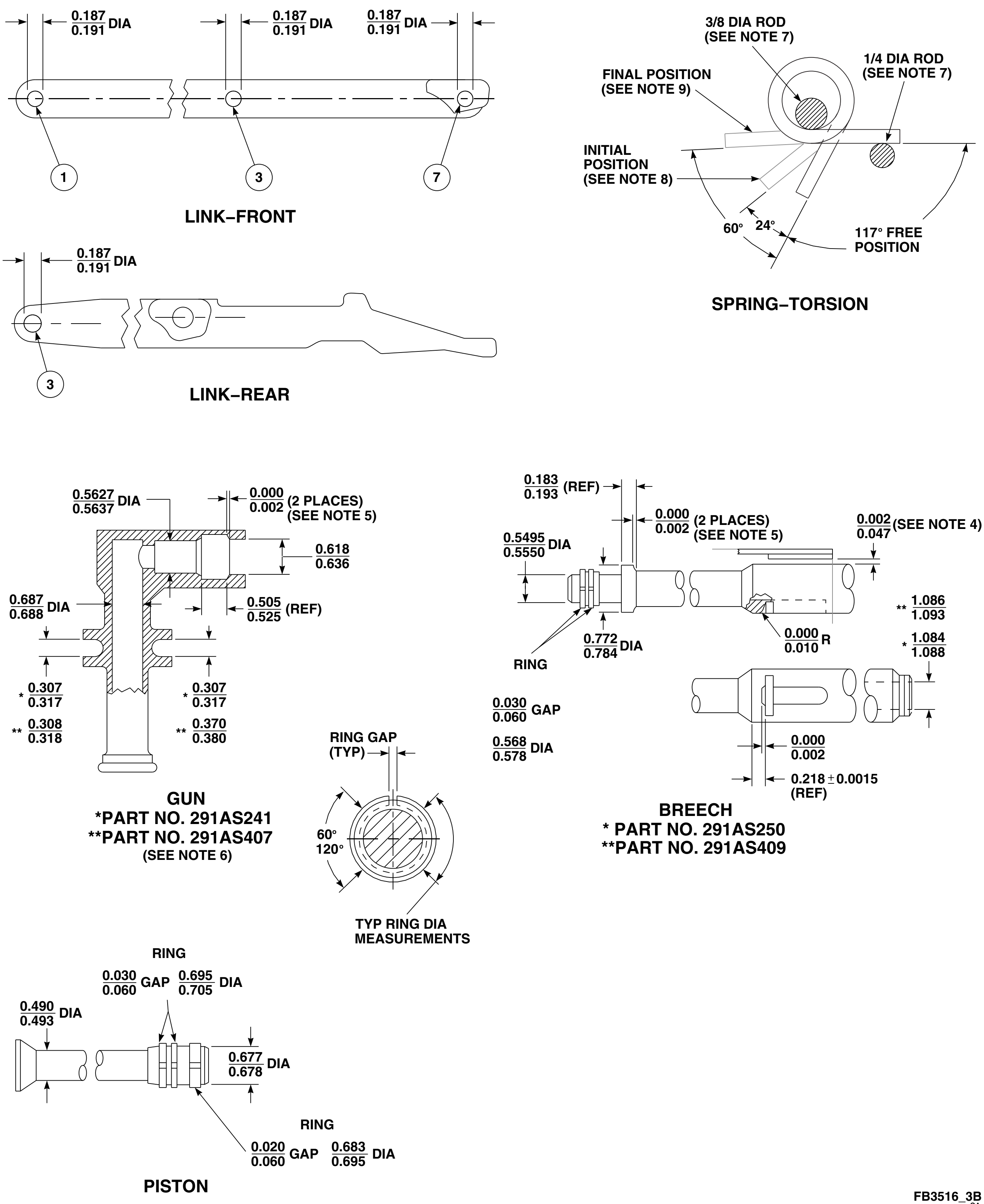


FIGURE 16-4-36-5. INSPECT EJECTOR RACK, BRU-22A/A, COMPONENTS (SHEET 3 OF 7)

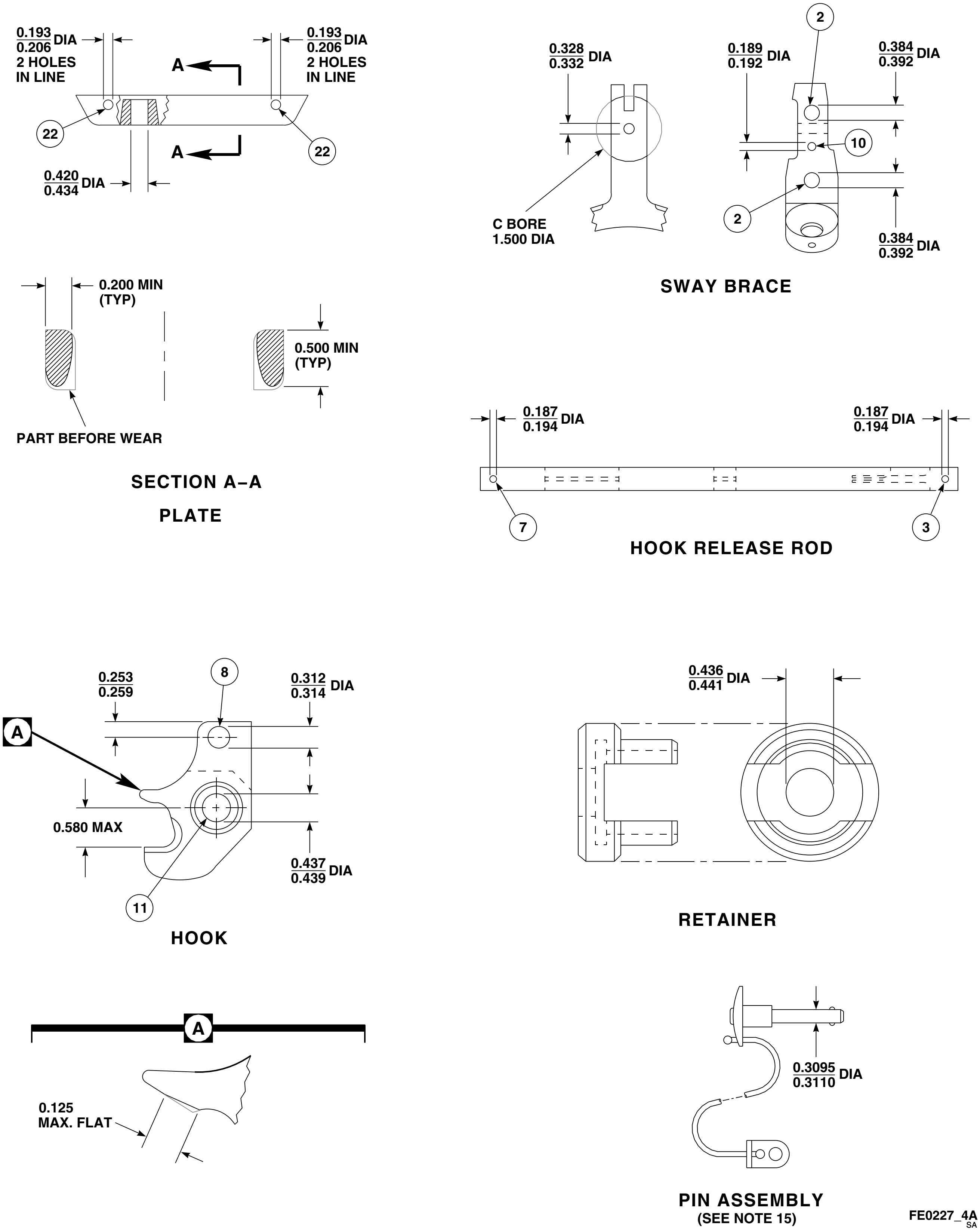


FIGURE 16-4-36-5. INSPECT EJECTOR RACK, BRU-22A/A, COMPONENTS (SHEET 4 OF 7)

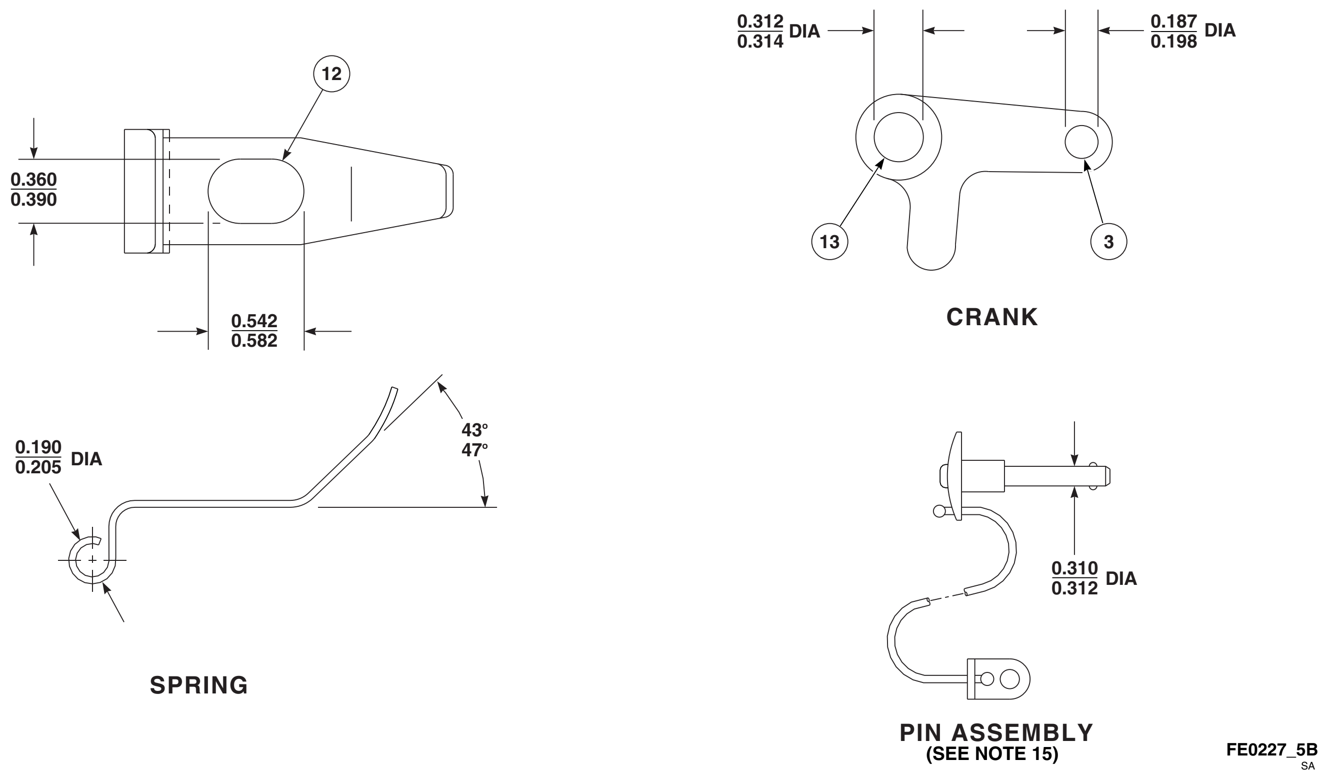


FIGURE 16-4-36-5. INSPECT EJECTOR RACK, BRU-22A/A, COMPONENTS (SHEET 5 OF 7)

(9) Install arming unit wiring clip on ejector rack housing with screws, washers, and nuts. Route wiring harness along wiring clip.

b. Install mechanical arming unit as follows (Figure 16-4-36-3):

NOTE

Installation of front and rear mechanical arming units is the same.

(1) Secure electrical wires to front or rear mechanical arming unit with screws and washers.

(2) Install mechanical arming unit in ejector rack and secure with screws and washers.

c. Install hook assembly as follows:

(1) Assemble hook assemblies as required for replacement of toggle rod roller, toggle spring rod, hook toggle lever, hook, or hook toggle pawl using pins (Figure 16-4-36-2, Detail A).

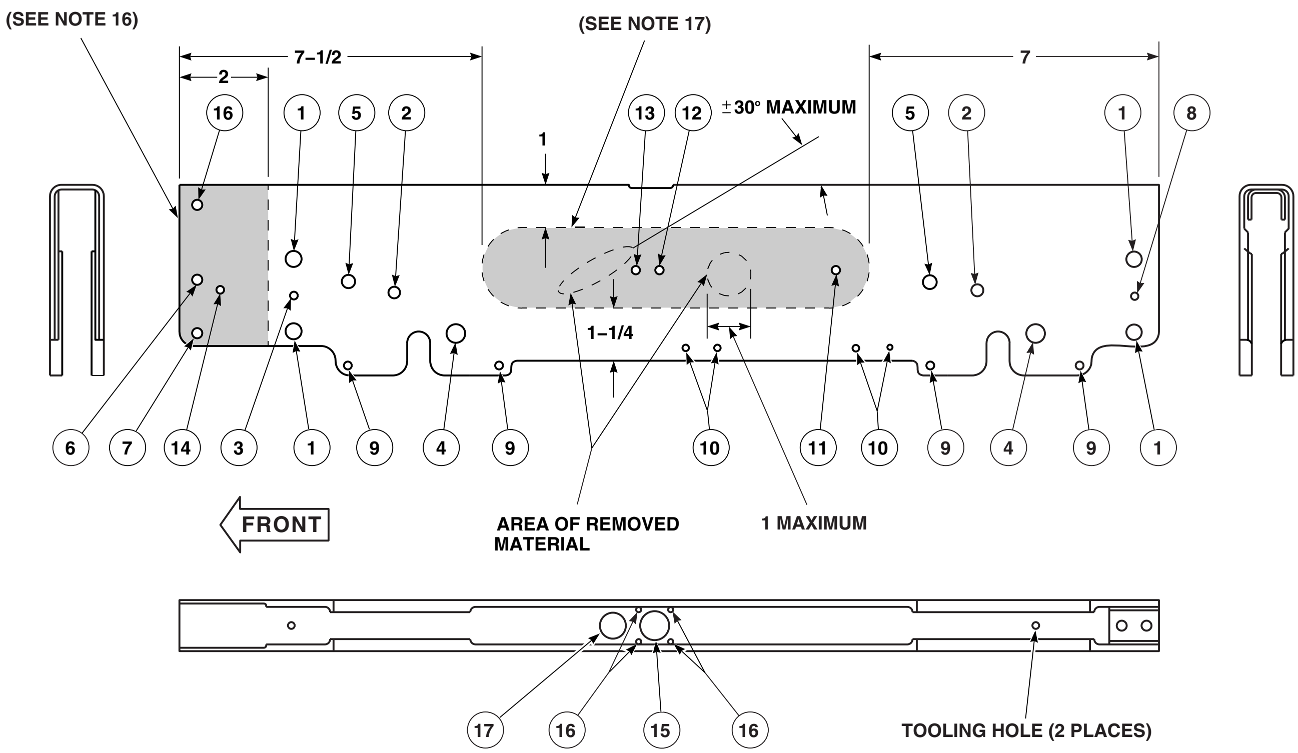
(2) Install front hook assembly as follows (Figure 16-4-36-2):

(a) Install hook assembly in front end of ejector rack housing and secure with pins and washers. Install cotter pins.

WARNING

Injury to personnel and damage to equipment will result if hook spring is not installed correctly. Make sure hook spring is installed between hook and hook toggle lever.

(b) Install hook spring and spring on front lug guide plate.



HOLE DIAMETERS–EJECTOR RACK HOUSING						
INDEX NUMBER	MINIMUM SERVICE DIAMETER	MAXIMUM SERVICE DIAMETER		INDEX NUMBER	MINIMUM SERVICE DIAMETER	MAXIMUM SERVICE DIAMETER
1	0.3950	0.4120		9	0.2010	0.2190
2	0.3120	0.3140		10	0.1940	0.2190
3	0.2320	0.2470		11 (FAR SIDE)	0.2190	0.2290
4	0.4370	0.4385		12	0.3730	0.3770
5	0.3120	0.3180		13	0.3120	0.3140
6	0.2485	0.2555		14	0.2500	0.2580
7	0.2500	0.2540		15	0.8020	0.8270
8 (FAR SIDE)	0.2190	0.2290		16	0.1540	0.1670
				17	0.6570	0.7190

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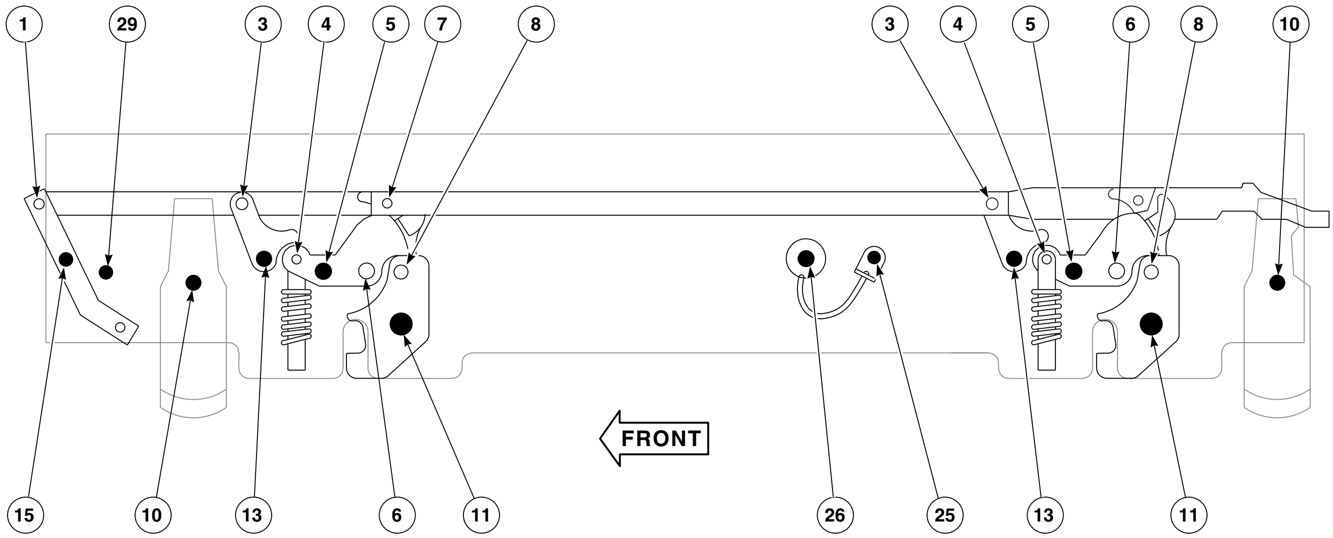
FIGURE 16-4-36-5. INSPECT EJECTOR RACK, BRU-22A/A, COMPONENTS (SHEET 6 OF 7)



INTERNAL PIN TOLERANCE DATA		
INDEX NUMBER	MINIMUM SERVICE DIAMETER	MAXIMUM SERVICE DIAMETER
1	0.1810	0.1860
3	0.1855	0.1860
4	0.1855	0.1860
6	0.3105	0.3110
7	0.1855	0.1860
8	0.3105	0.3110

EXTERNAL PIN TOLERANCE DATA		
INDEX NUMBER	MINIMUM SERVICE DIAMETER	MAXIMUM SERVICE DIAMETER
5	0.3105	0.3110
10	NOT APPLICABLE	
11	0.4360	0.4365
13	0.3105	0.3110
15	0.2480	0.2485
25	0.3105	0.3110
25	0.3710	0.3730
26	0.3095	0.3110
26	0.3095	0.3120
29	0.2480	0.2485

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FIGURE 16-4-36-5. INSPECT EJECTOR RACK, BRU-22A/A, COMPONENTS (SHEET 7 OF 7)

WARNING

Injury to personnel will result if caution is not used when installing front lug guide plate. Front lug guide plate will be under spring tension. Use care when installing bolts.

(c) Install front lug guide plate on ejector rack housing and secure with bolts, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

(3) Install rear hook assembly as follows:

(a) Install hook assembly in rear end of ejector rack housing and secure with pin and washer. Install cotter pin.

(b) Position hook assembly within housing and gun retaining pin assembly on housing, and install pins and washers. Install cotter pins.

WARNING

Injury to personnel and damage to equipment will result if hook spring is not installed correctly. Make sure hook spring is installed between hook and hook toggle lever.

(c) Install hook spring and spring on rear lug guide plate.

WARNING

Injury to personnel will result if caution is not used when installing rear lug guide plate. Rear lug guide plate will be under spring tension. Use care when installing bolts.

(d) Install rear lug guide plate on ejector rack housing and secure with bolts, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

d. Install sway brace as follows:

(1) Install front sway brace as follows (Figure 16-4-36-1):

(a) Install front sway brace in ejector rack housing. Using pin punch, install spring pin to secure sway brace to housing. Install pin flush to 0.030-inch below surface of housing.

(b) Position rocket blast shield and nose fairing on ejector rack housing and secure with screws and washers.

(2) Install rear sway brace as follows (Figure 16-4-36-1, Detail A):

(a) Position safety stop lever on safety stop bushing and install bolt. Install assembly on rear sway brace and secure with washer and nut.

(b) Install rear sway brace in ejector rack housing with bolts and spring pin.

(c) Using feeler gage, measure gap between safety stop lever and rear link (Figure 16-4-36-1, Detail B). **GAP MUST NOT EXCEED 0.016-INCH.** If limit is exceeded, add washer(s), as required, under safety stop bushing to obtain proper gap. Install cotter pin.

(d) Install rear sway brace in ejector rack housing (Figure 16-4-36-1, Detail A). Using pin punch, install spring pin to secure sway brace to housing. Install pin flush to 0.030-inch below surface of housing.

(e) Install bolts in housing.

e. Install breech and gun as follows:

(1) Install retainer on piston and secure with foot. **TORQUE FOOT TO 90 - 140 INCH-POUNDS.**

(2) Install pressure rings and bearing ring on piston.

(3) Rotate retainer and install piston into gun.

(4) Rotate gun rear and up and install in ejector rack housing. Install gun retaining pin assembly in ejector rack housing (Figure 16-4-36-2).

(5) If removed, install pressure rings on breech.

(6) Place breech in ejector rack housing in installed position. Do not force into housing.

(7) Place safety stop lever in unlocked position.

(8) Pull manual release lever front and hold. Rotate breech 90° to insert in unit housing, then rotate breech 90° in opposite direction to place breech in installed position.

(9) Release manual release lever and place safety stop lever in locked position.

(10) Make sure rear attachment bolts are installed.

f. Perform ejector rack, BRU-22A/A, proof load test (PARA 16-5-8).

16-4-37 ESSS VERTICAL SUPPORT PYLON (VSP).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 16-3-1
- PARA 16-4-34

16-4-37.1 Replace ESSS Vertical Support Pylon (VSP).

NOTE

Replacement procedures for left and right, inboard and outboard VSPs are the same, except as noted.

16-4-37.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. If installed, remove ejector rack, BRU-22A/A (PARA 16-4-34).

c. Disconnect VSP pneumatic hose from union on horizontal stores support (HSS) bracket (Figure 16-4-37-1, Sheet 2, Detail B or Detail C).

d. Install protective cap on pneumatic hose. Remove cap from dummy receptacle on HSS bracket and install on union on HSS bracket.

e. Disconnect VSP fuel hose from fuel shutoff gate valve quick-disconnect on HSS bracket.

f. Install protective cap on fuel hose. Remove cap from dummy receptacle on HSS bracket and install on fuel shutoff gate valve quick-disconnect on HSS bracket.

g. Disconnect electrical connector, P2, from applicable electrical connector on HSS bracket as follows:

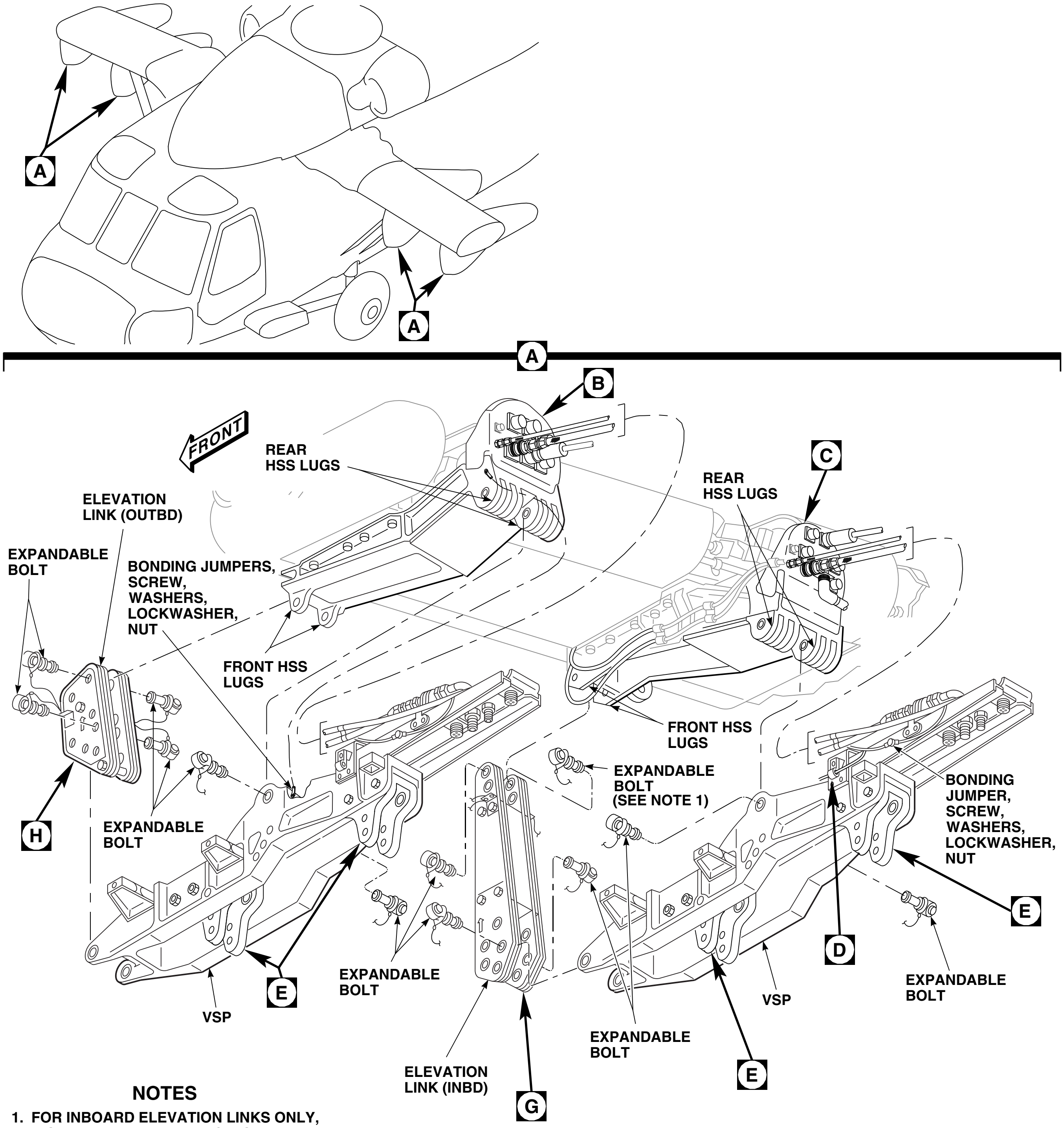
(1) J835 (left side - inboard)

(2) J849 (left side - outboard)

(3) J836 (right side - inboard)

(4) J850 (right side - outboard)

h. Remove cap from dummy receptacle on HSS bracket and install on electrical connector



NOTES

1. FOR INBOARD ELEVATION LINKS ONLY, INSTALL EXPANDABLE BOLTS SECURING LINK TO HSS WITH BOLTHEADS FACING OUTBOARD.
2. IF EJECTOR RACK, BRU–22A/A, IS TO BE INSTALLED, ADAPTERS MUST BE INSTALLED ON VSP.
3. ADD SUFFICIENT QUANITY OF WASHERS (MINIMUM: 1) TO MAKE SURE FINGERTIGHTENED NUT WILL ALLOW COTTER PIN INSTALLATION.

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FIGURE 16-4-37-1. REPLACE ESSS VERTICAL SUPPORT PYLON (VSP) (SHEET 1 OF 3)

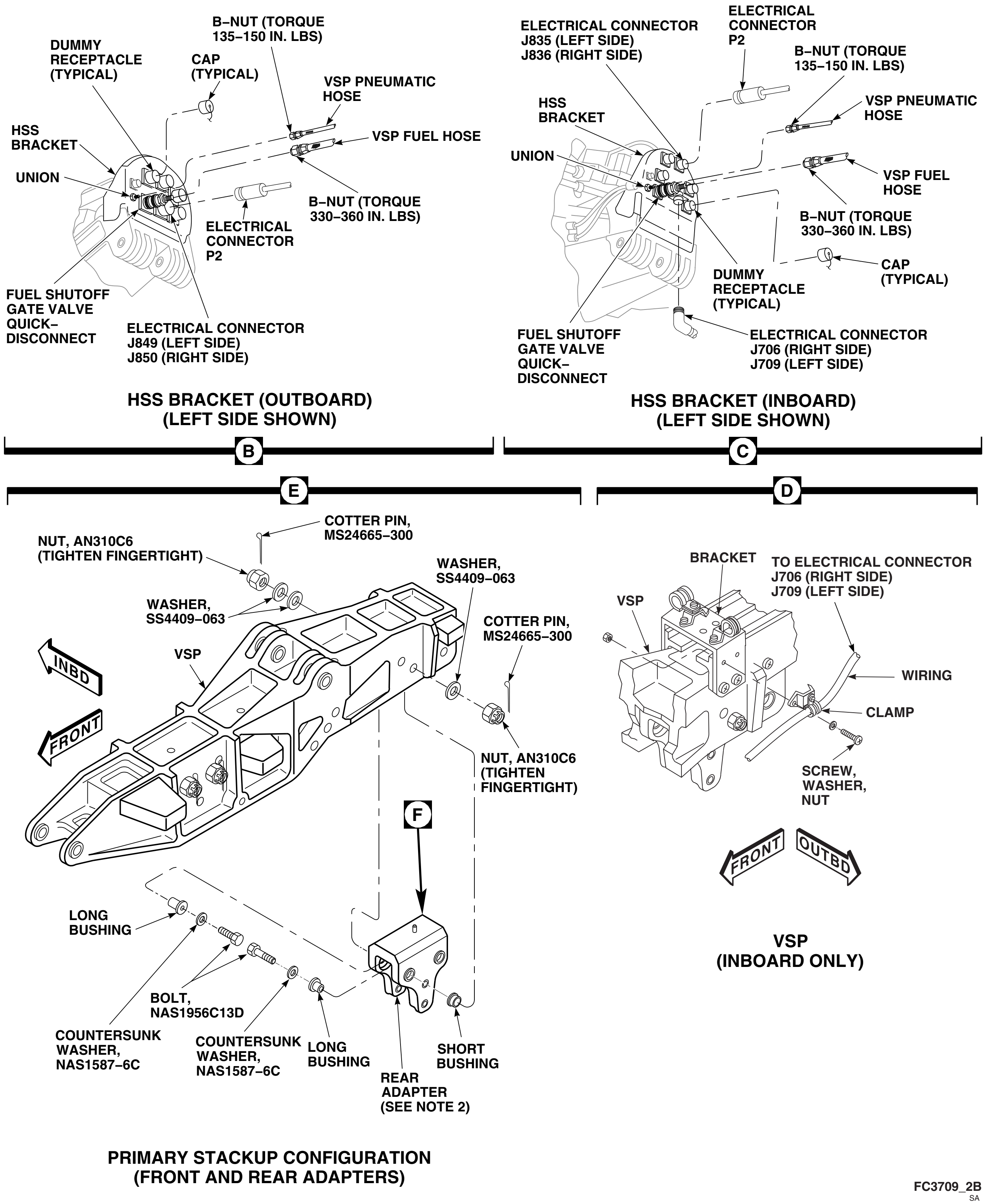


FIGURE 16-4-37-1. REPLACE ESSS VERTICAL SUPPORT PYLON (VSP) (SHEET 2 OF 3)

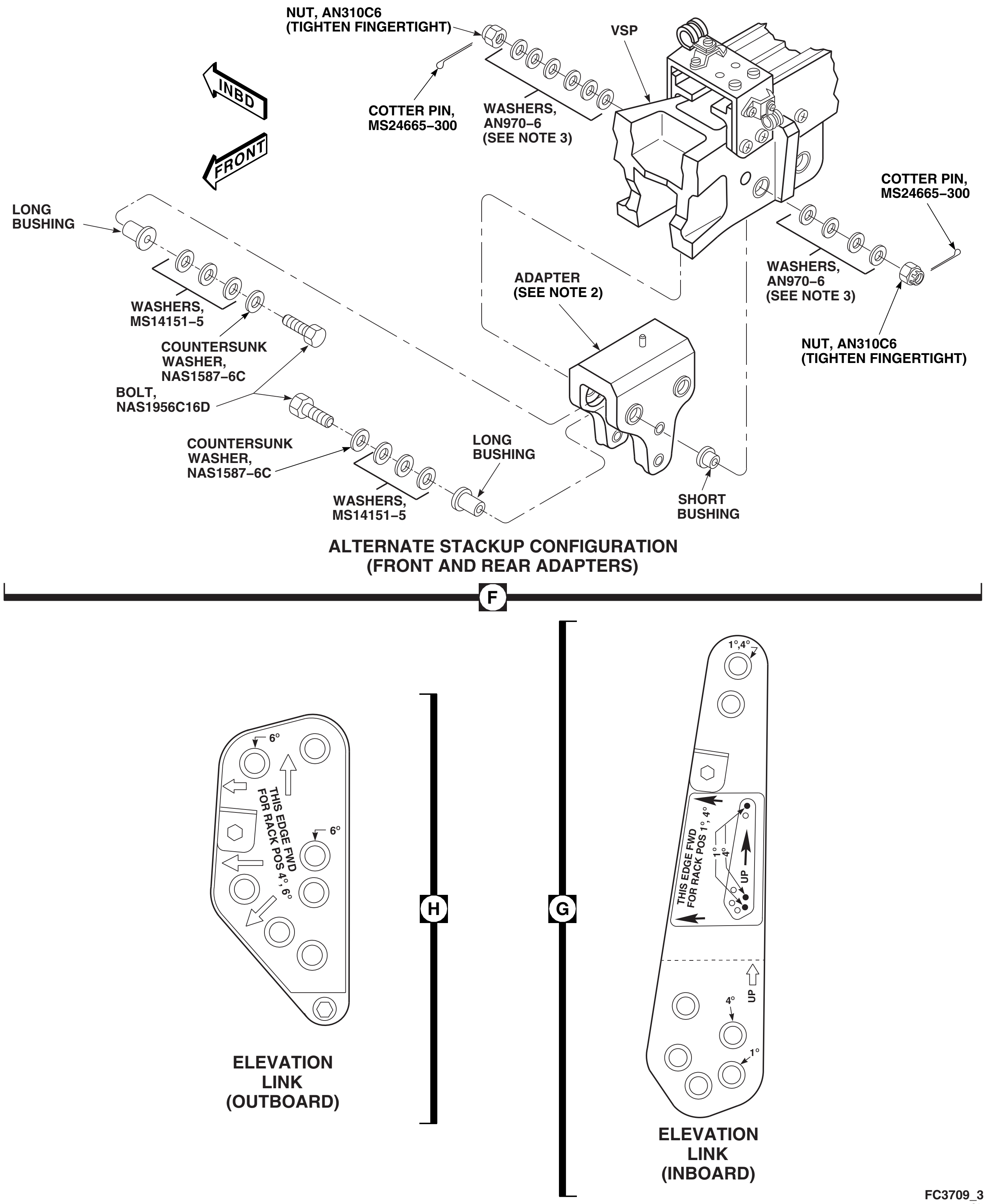


FIGURE 16-4-37-1. REPLACE ESSS VERTICAL SUPPORT PYLON (VSP) (SHEET 3 OF 3)

(J835, J836, J849, or J850) on HSS bracket. Connect electrical connector, P2, to dummy receptacle from which cap was removed.

i. If removing inboard VSP, perform the following:

(1) Disconnect electrical connector, J709 (left side) or J706 (right side), from HSS bracket (Figure 16-4-37-1, Sheet 2, Detail C).

(2) Install protective caps and plugs on electrical connectors.

(3) Remove screws, washers, and nuts securing clamp and wiring to bracket on VSP (Figure 16-4-37-1, Sheet 2, Detail D).

(4) Remove screw, washers, lockwasher, and nut securing bonding jumpers to VSP (Figure 16-4-37-1, Sheet 1, Detail A).

j. If removing outboard VSP, perform the following:

(1) Locate bonding jumpers connecting from outboard side of rear HSS lugs to immediately behind rear connection point on VSP.

(2) Remove screw, washers, lockwasher, and nut securing bonding jumper to VSP.

k. Remove expandable bolts and VSP as follows (Figure 16-4-37-1, Sheet 1, Detail A):

NOTE

Remove front expandable bolts first.

(1) Remove expandable bolts securing elevation link to front HSS lugs and front of VSP.

(2) Remove expandable bolts from rear HSS lugs, securing VSP to HSS.

(3) Remove VSP.

l. If ejector racks, BRU-22A/A, were removed, remove cotter pins, bolts, countersunk washers, washers, nuts, and long bushings securing adapters to VSP (Figure 16-4-37-1, Sheet 3, Detail F). Remove adapters.

m. Remove short bushings from VSP.

16-4-37.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

NOTE

- Decals on elevation links show multiple degree settings. Make sure that links are mounted to reflect 4° setting.
- For inboard elevation links only, install expandable bolts securing link to HSS with boltheads facing outboard.

b. Position elevation link on front HSS lugs set link to 4° UP. Install expandable bolts (Figure 16-4-37-1, Sheet 1, Detail A). Do not lock bolts now.

c. Position VSP on rear HSS lugs and install expandable bolts (Figure 16-4-37-1, Sheet 1, Detail A). Do not lock bolts now.

d. Position front of VSP on elevation link and install expandable bolts. Do not lock bolts now.

e. Lock all expandable bolts. Verify locking as follows:

(1) Make sure that locking pins are snapped into tabs.

(2) Make sure that locking rings are fully visible and expanded.

(3) Inspect expandable bolts (PARA 16-3-1).

(4) Apply torque stripes (PARA 2-6-5).

f. If ejector racks, BRU-22A/A, are to be installed, install adapters as follows:

(1) Using primary stackup configuration, install adapters as follows (Figure 16-4-37-1, Sheet 2, Detail E):

(a) Install short bushings on left side of VSP.

(b) Position adapter in VSP. From inside adapter, install long bushings through adapters and short bushings.

WARNING

FSCAP

Verification of proper hardware installation, torque, and safety installation of bolts are critical characteristics.

NOTE

Make sure countersunk washers face boltheads.

(c) From inside adapters, install bolts, NAS1956C13D, countersunk washers, NAS1587-6C, washers, SS4409-063 (2 inboard/1 outboard, under nuts), and nuts, AN310C6.

CAUTION

Damage to ejector rack will result if nuts are overtightened. Flange bushings may shear. Do not overtighten nuts.

(d) **TIGHTEN NUTS FINGERTIGHT.**

(e) Install cotter pins, MS24665-300. If cotter pins cannot be installed, proceed to step f. (2).

(2) If cotter pins cannot be installed after fingertightening nuts, using alternate stackup configuration, as follows (Figure 16-4-37-1, Sheet 3, Detail F):

(a) Remove bolts, countersunk washers, washers, and nuts securing adapter(s) to VSP.

NOTE

- Bolt and plain washers will be replaced with different part numbers.
- Make sure countersunk washers face boltheads.
- A minimum of one washer, AN970-6, must be installed under each nut.

(b) From inside adapter(s), install bolts, NAS1956C16D, countersunk washers, NAS1587-6C, washers, MS14151-5 (3 under countersunk washers), washers, AN970-6 (6 inboard max./4 outboard max., under nuts), and nuts, AN310C6.

CAUTION

Damage to ejector rack will result if nuts are overtightened. Flange bushings may shear. Do not overtighten nuts.

(c) **TIGHTEN NUTS FINGERTIGHT TO LINE UP CASTELLATION IN NUTS WITH COTTER PIN HOLE IN BOLTS.**

(d) Install cotter pins, MS24665-300. If cotter pins still cannot be installed, remove nuts. Add or subtract washers, AN970-6, as necessary (maintaining at least one under each nut), then install nut. Repeat step f. (2) (c). Repeat step f. (2) (d) until cotter pin can be installed.

g. Connect VSP fuel hose as follows (Figure 16-4-37-1, Sheet 2, Detail B or Detail C):

(1) Remove cap from fuel shutoff gate valve on horizontal stores support (HSS) bracket. Install cap on dummy receptacle on HSS bracket.

(2) Remove protective cap from VSP fuel hose.

(3) Coat threads of fuel shutoff gate valve quick-disconnect with antiseize compound, Item 74, Appendix D.

(4) Connect VSP fuel hose to fuel shutoff gate valve quick-disconnect. **TORQUE B-NUT TO 330 - 360 INCH-POUNDS.**

h. Connect VSP pneumatic hose as follows:

(1) Remove cap from union on HSS bracket. Install cap on dummy receptacle on HSS bracket.

(2) Remove protective cap from VSP pneumatic hose.

(3) Coat threads of union with antiseize compound, Item 74, Appendix D.

(4) Connect VSP pneumatic line to union. **TORQUE B-NUT TO 135 - 150 INCH-POUNDS.**

i. Disconnect electrical connector, P2, from dummy receptacle on HSS bracket. Remove cap from electrical connector (J835, J836, J849, or J850) on HSS bracket.

j. Install cap on dummy receptacle.

k. Inspect and treat electrical connectors (PARA 1-7-20).

l. Connect electrical connector, P2, to applicable electrical connector on HSS bracket as follows:

(1) J835 (left side - inboard)

(2) J849 (left side - outboard)

(3) J836 (right side - inboard)

(4) J850 (right side - outboard)

m. If inboard VSP was installed, perform the following:

(1) Remove protective caps and plugs from electrical connector, J709 (left side) or J706 (right side) (Figure 16-4-37-1, Sheet 2, Detail C).

(2) Inspect and treat electrical connectors (PARA 1-7-20).

(3) Connect electrical connector, J709 (left side) or J706 (right side), to HSS bracket.

(4) Position clamp and wiring on bracket, located on VSP, and secure with screws, washers, and nuts (Figure 16-4-37-1, Sheet 2, Detail D).

(5) Position bonding jumpers on VSP and secure with screw, lockwasher, washers, and nut (Figure 16-4-37-1, Sheet 1, Detail A).

n. If outboard VSP was installed, position bonding jumper on VSP and secure with screw, washers, lockwasher, and nut (Figure 16-4-37-1, Sheet 1, Detail A).

o. If required, install ejector rack, BRU-22A/A (PARA 16-4-34).

16-4-38 ESSS AUXILIARY FUEL MANAGEMENT CONTROL PANEL W/O AUX FUEL QTY .

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 16-2-1
- PARA 16-4-42

16-4-38.1 Replace ESSS Auxiliary Fuel Management Control Panel.

16-4-38.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Disconnect electrical connectors, P360 from J360, and P361 from J361, on auxiliary fuel management control panel (Figure 16-4-38-1, Detail A).

c. Install protective caps and plugs on electrical connectors.

d. Remove screws, lockwashers, and washers and remove auxiliary fuel management control panel from mounting bracket on lower console.

16-4-38.1.2 Install.

a. Adjust density switch settings (PARA 16-4-42).

b. Turn off all electrical power (PARA 1-6-16).

c. Prepare bonding surfaces.

d. Install auxiliary fuel management control panel to mounting bracket on lower console and secure with screws, lockwashers, and washers (Figure 16-4-38-1, Detail A).

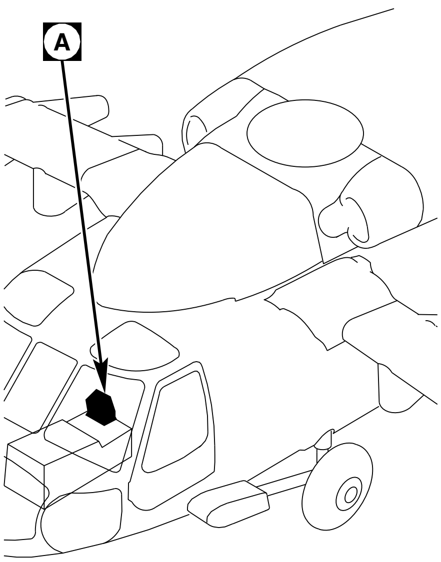
e. Remove protective caps and plugs from electrical connectors.

f. Inspect and treat electrical connectors (PARA 1-7-20).

g. Connect electrical connectors, P360 to J360, and P361 to J361, on auxiliary fuel management control panel.

h. Make sure area is clean and free of foreign material.

i. Perform operational check of ESSS range extension system (PARA 16-2-1).



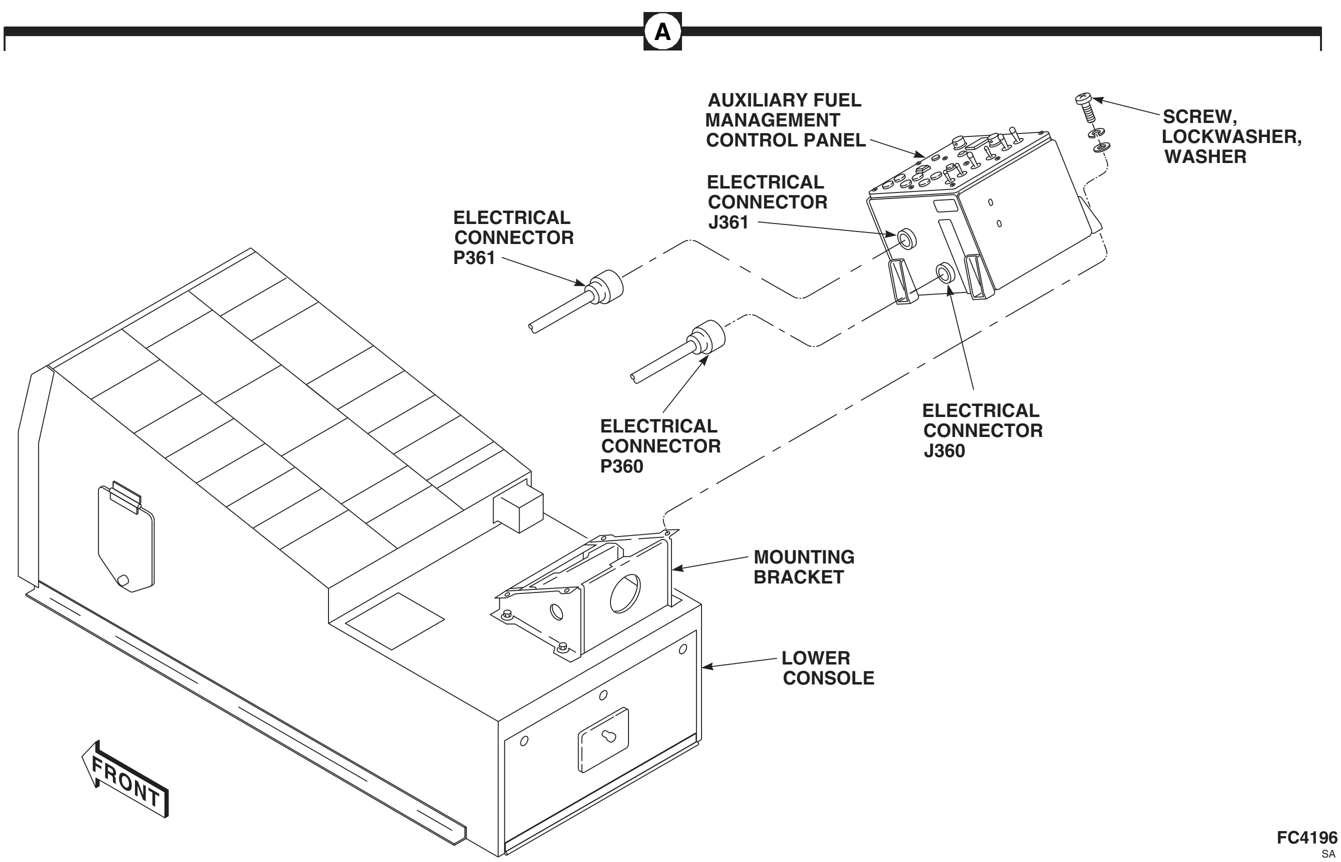


FIGURE 16-4-38-1. REPLACE ESSS AUXILIARY FUEL MANAGEMENT CONTROL PANEL

16-4-39 ESSS AUXILIARY FUEL MANAGEMENT CONTROL PANEL INFORMATION PLATE W/O AUX FUEL QTY .

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

References

- PARA 1-6-16
- PARA 9-2-5
- PARA 9-2-25

16-4-39.1 Replace ESSS Auxiliary Fuel Management Control Panel Information Plate.

16-4-39.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Loosen setscrews on two rotary switch knobs and remove knobs (Figure 16-4-39-1, Detail A).

c. Remove DEGRADED lens cap.

d. Remove PRESS TO TEST switch cover.

e. Loosen captive screws and remove information plate from auxiliary fuel management control panel.

16-4-39.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Position information plate on auxiliary fuel management control panel and secure with captive screws (Figure 16-4-39-1, Detail A).

c. Install rotary switch knobs on switch shafts. **TIGHTEN SETSCREWS.**

d. Screw DEGRADED lens cap into auxiliary fuel management control panel.

e. Install PRESS TO TEST switch cover.

f. Perform operational check of console lighting (PARA 9-2-5 and PARA 9-2-25).

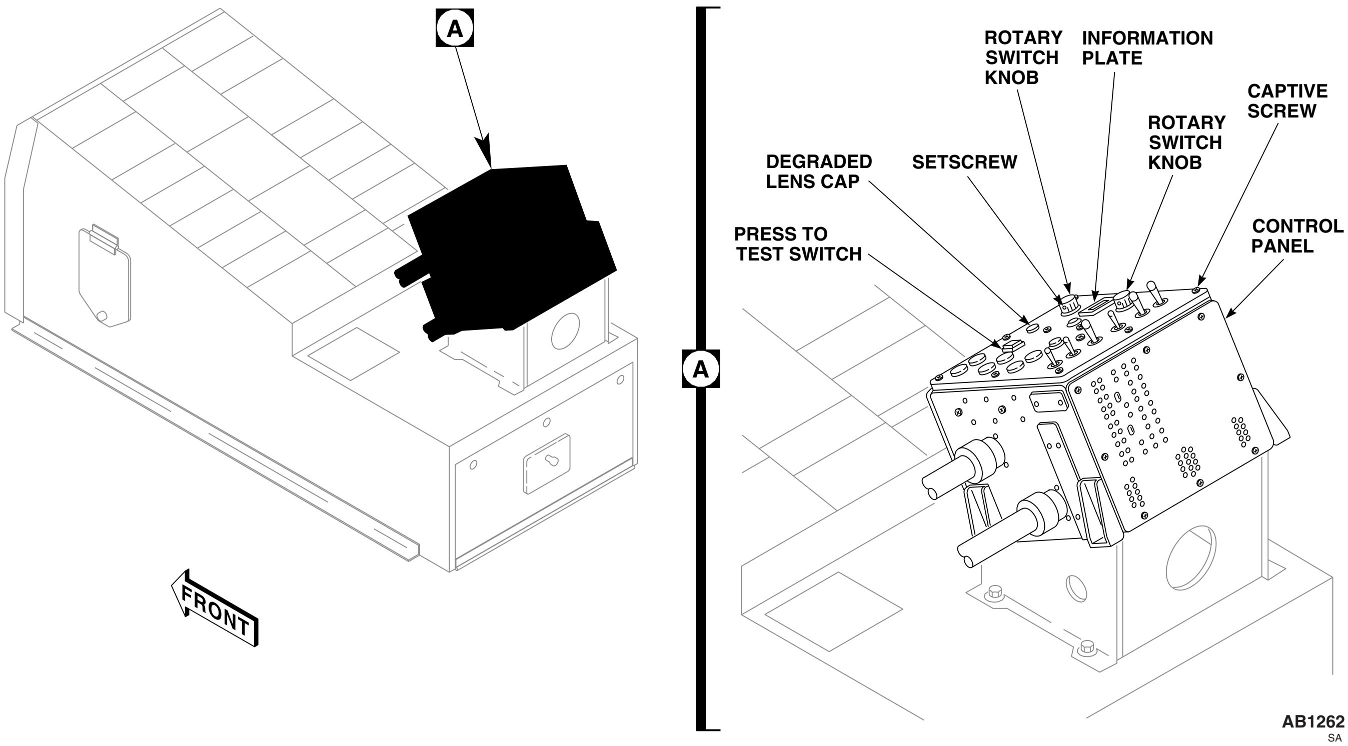


FIGURE 16-4-39-1. REPLACE ESSS AUXILIARY FUEL MANAGEMENT CONTROL PANEL INFORMATION PLATE

16-4-40 ESSS AUXILIARY FUEL MANAGEMENT CONTROL PANEL DIGITAL DISPLAY SEGMENT W/O AUX FUEL QTY .

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

References

- PARA 1-6-16

16-4-40.1 Replace ESSS Auxiliary Fuel Management Control Panel Digital Display Segment.

16-4-40.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove screws and bezel from auxiliary fuel management control panel (Figure 16-4-40-1, Detail A).

c. Pull out digital display segment from auxiliary fuel management control panel.

16-4-40.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Carefully push in replacement digital display segment in auxiliary fuel management control panel (Figure 16-4-40-1, Detail A).

c. Install bezel on auxiliary fuel management control panel with screws.

d. Turn on all electrical power (PARA 1-6-16).

e. Press TEST button and check that digital display indicates 8888.

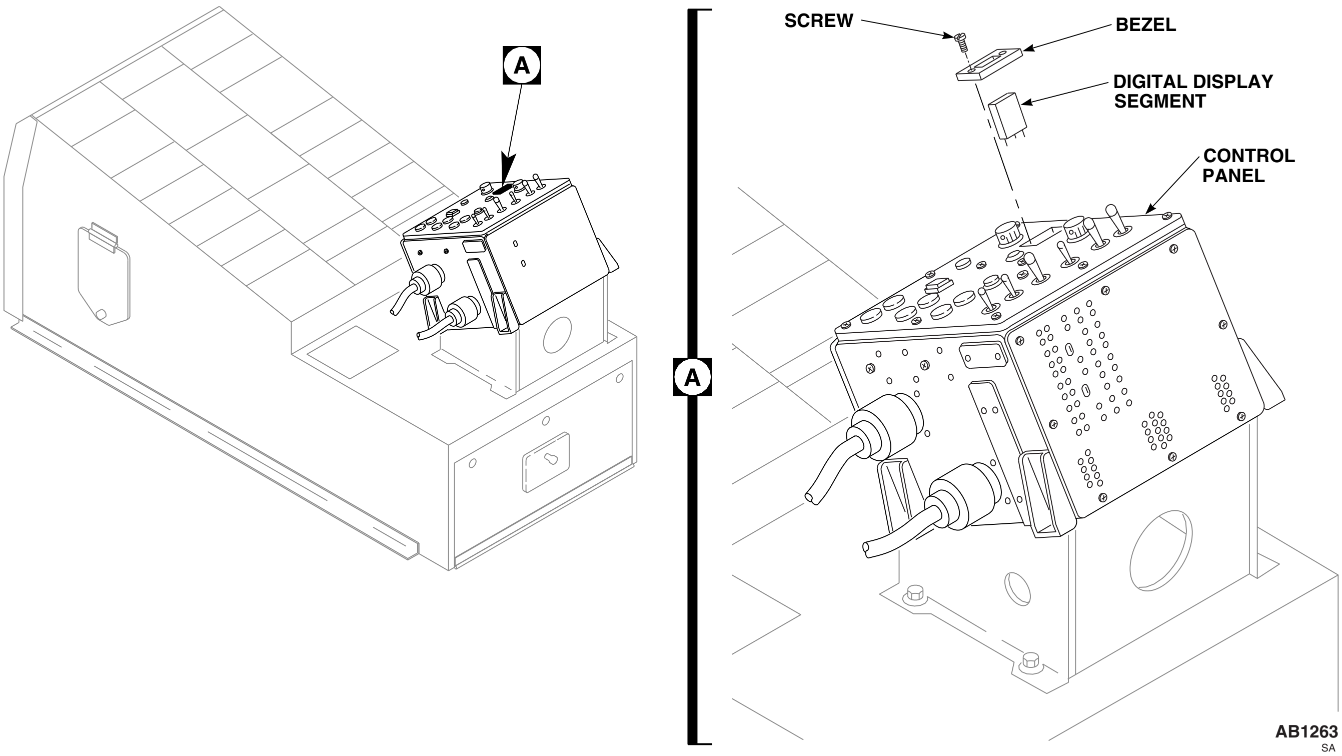


FIGURE 16-4-40-1. REPLACE ESSS AUXILIARY FUEL MANAGEMENT CONTROL PANEL DIGITAL DISPLAY SEGMENT

16-4-41 ESSS AUXILIARY FUEL MANAGEMENT CONTROL PANEL INDICATOR LAMPS W/O AUX FUEL QTY .

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

References

- PARA 1-6-16

16-4-41.1 Replace ESSS Auxiliary Fuel Management Control Panel Indicator Lamps.

NOTE

Replacement of all lamps is the same, except as noted.

16-4-41.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Except for DEGRADED light assembly, carefully remove lens cap or pushbutton assembly from auxiliary fuel management control panel. For DEGRADED light assembly, unscrew and remove DEGRADED lens cap from panel (Figure 16-4-41-1, Detail A).

c. Remove lamp from lens cap or pushbutton assembly.

16-4-41.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Insert new lamp into lens cap or pushbutton assembly (Figure 16-4-41-1, Detail A).

c. Except for DEGRADED light assembly, position lens cap or pushbutton assembly on auxiliary fuel management control panel and press fully into place. For DEGRADED light assembly, screw DEGRADED lens cap into control panel.

d. Turn on all electrical power (PARA 1-6-16).

e. On auxiliary fuel management control panel, press TEST button and check that indicator lamps light.

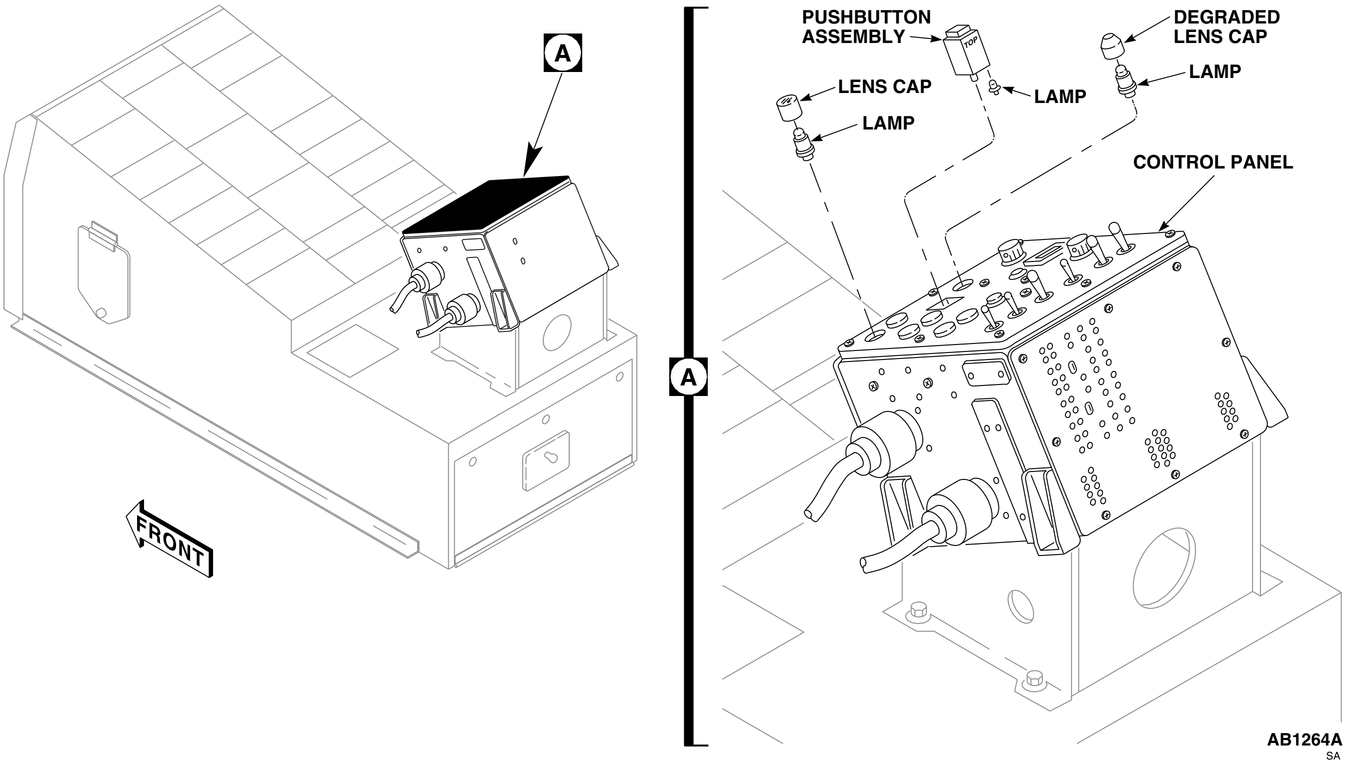


FIGURE 16-4-41-1. REPLACE ESSS AUXILIARY FUEL MANAGEMENT CONTROL PANEL INDICATOR LAMPS

16-4-42 ESSS AUXILIARY FUEL MANAGEMENT CONTROL PANEL DENSITY SWITCHES W/O AUX FUEL QTY .

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Plastic Alignment Tool

References

- PARA 1-6-16
- PARA 16-4-38

16-4-42.1 Adjust ESSS Auxiliary Fuel Management Control Panel Density Switches.

NOTE

If the following adjustment cannot be done, auxiliary fuel management control panel must be replaced (PARA 16-4-38).

a. Turn off all electrical power (PARA 1-6-16).

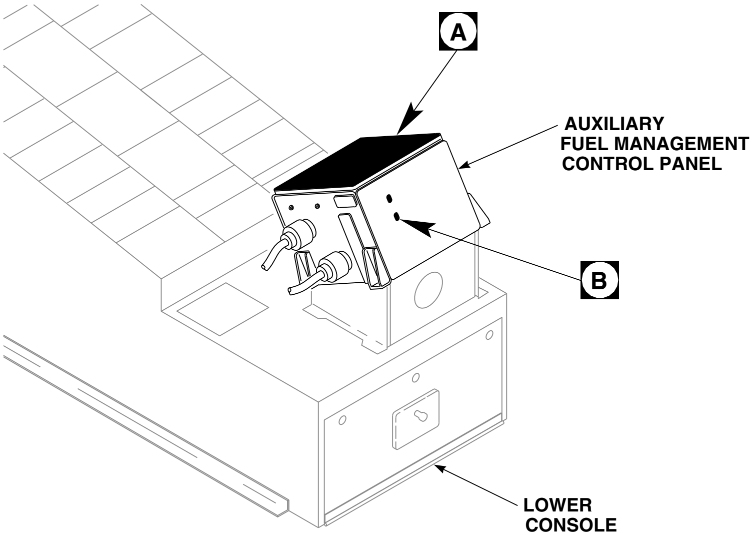
b. Using plastic alignment tool, set fuel density switches, to fuel type on helicopter (Figure 16-4-42-1, Detail B).

c. Turn on all electrical power (PARA 1-6-16).

d. Make certain that EXT FUEL RH, NO. 2 XFER CONTROL, EXT FUEL LH, and AUX FUEL QTY circuit breakers, on mission readiness circuit breaker panel, are in.

e. Place ESSS auxiliary fuel management control panel switches as follows:

Switch	Position
EXTERNAL FUEL TRANSFER	OFF
PRESSURE OUTBD	OFF
PRESSURE INBD	OFF
TANKS	OUTBD
MODE	OFF
MANUAL XFR RIGHT	OFF
MANUAL XFR LEFT	OFF
AUX FUEL QTY SWITCH	TOTAL



NOTE

FUEL TYPE	SWITCH SETTING			
	1	2	3	4
JP–4	RIGHT	RIGHT	RIGHT	RIGHT
JP–8	RIGHT	RIGHT	RIGHT	LEFT
JP–5	RIGHT	RIGHT	LEFT	RIGHT

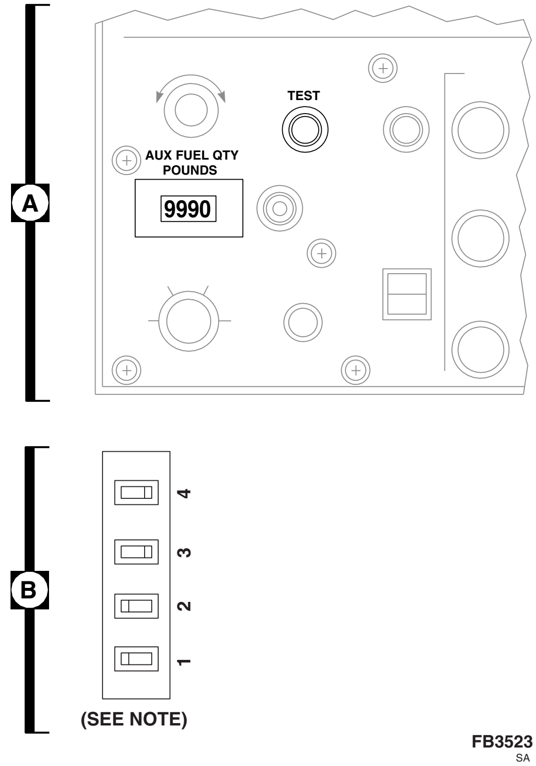


FIGURE 16-4-42-1. ADJUST ESSS AUXILIARY FUEL MANAGEMENT CONTROL PANEL DENSITY SWITCHES

f. Press and release ESSS auxiliary fuel management control panel TEST button, and observe this sequence of events (Figure 16-4-42-1, Detail A):

(1) With TEST button pressed, all ESSS auxiliary fuel management control panel indicators shall be on and digital display shall indicate 8888.

(2) When TEST button is released, digital display shall indicate the digit 8 in sequence from left to right three times.

(3) Then, digital display shall indicate GOOD for about 5 seconds.

(4) Then, digital display shall indicate preset fuel type in the left-most bit for about 3 seconds.

(5) Then, digital display shall indicate some value of auxiliary fuel remaining (unknown at this time).

g. Turn off all electrical power (PARA 1-6-16).

h. If the adjust procedure cannot be done, replace auxiliary fuel management control panel (PARA 16-4-38).

16-4-43 ESSS CABIN FUEL HARNESS W/O AUX FUEL QTY .

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Protective Caps and Plugs
- Tiedown Straps, Item 331, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 16-2-1
- PARA 16-4-21

16-4-43.1 Replace ESSS Cabin Fuel Harness.

16-4-43.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove upper and lower root fairings (PARA 16-4-21).

c. Disconnect electrical connectors, P360 from J360, and P361 from J361, on ESSS auxiliary fuel management control panel on lower console (Figure 16-4-43-1, Sheet 1, Detail A).

d. Install protective caps and plugs on electrical connectors.

e. Remove tiedown straps securing cabin fuel harness as required.

f. Remove clamps, screws, washers, and nuts from clamps on post. Remove cabin fuel harness from clamps (Figure 16-4-43-1, Sheet 1, Detail A and Sheet 2, Detail C).

g. Turn and release thumbscrew on clamps securing cabin fuel harness to seat support tubes. Remove cabin fuel harness from clamps (Figure 16-4-43-1, Sheet 2, Detail B and Detail D).

h. Turn and release screws of clamps securing cabin fuel harness to airframe. Remove cabin fuel harness from clamps (Figure 16-4-43-1, Sheet 2, Detail B and Detail E).

i. Disconnect electrical connector, P346 from J346 (left side), on range extension provisions panel (Figure 16-4-43-1, Sheet 2, Detail B).

j. Install protective caps and plugs on electrical connectors.

k. Remove doubler halves as follows (Figure 16-4-43-1, Sheet 2, Detail B):

(1) Remove bolts and washers securing lower halves of doublers to fuselage and remove lower halves.

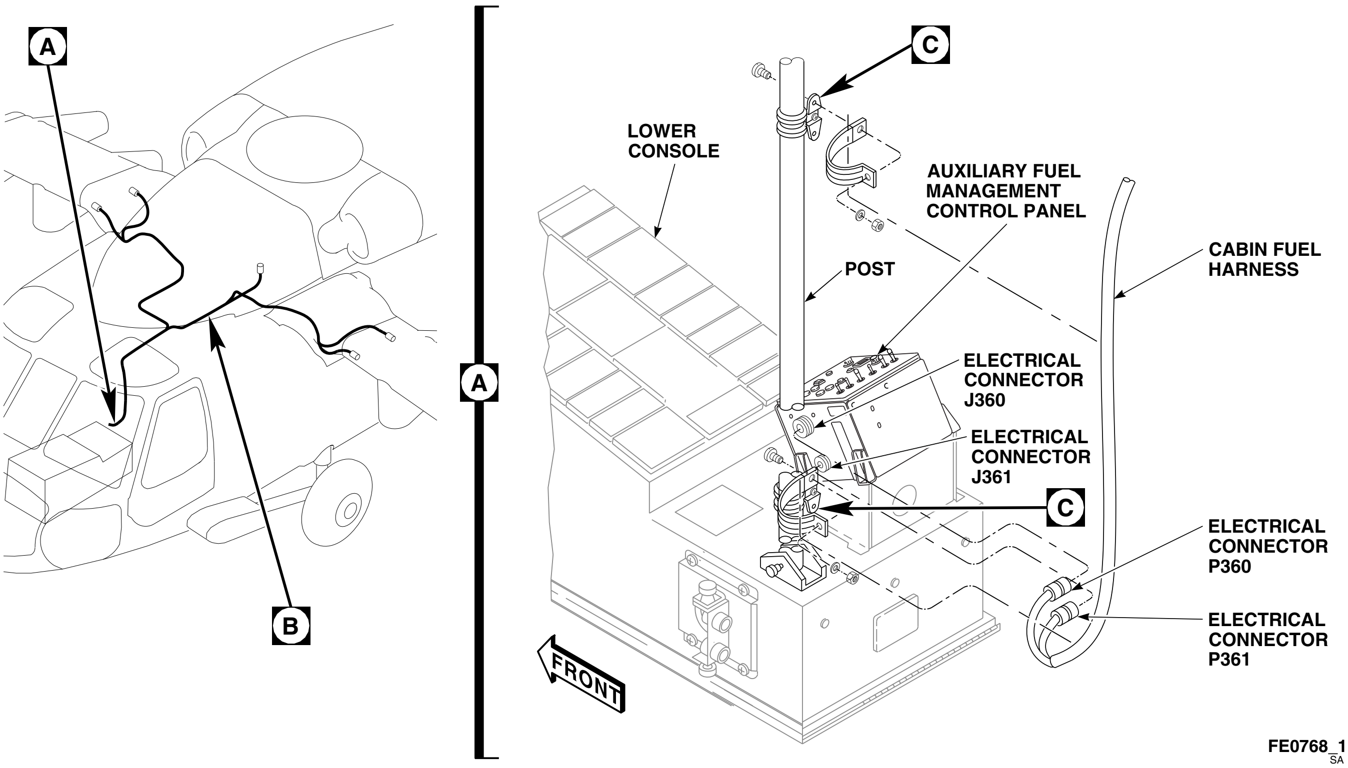


FIGURE 16-4-43-1. REPLACE ESSS CABIN FUEL HARNESS (SHEET 1 OF 2)

(2) Remove and slit grommet if required.

l. Turn and release screws of clamps securing cabin fuel harness to airframe. Remove cabin fuel harness from clamps (Figure 16-4-43-1, Sheet 2, Detail F).

m. Disconnect electrical connectors, P834 and P804 (right side), and P833 and P805 (left side), from HSS interconnects.

n. Install protective caps and plugs on electrical connectors.

o. Pull cabin fuel harness from access holes in fuselage.

p. Remove cabin fuel harness from helicopter.

NOTE

If replacement cabin fuel harness is to be installed, do not reinstall doubler halves.

q. If cabin fuel harness will not be installed, install doubler halves as follows (Figure 16-4-43-1, Sheet 2, Detail B):

(1) If necessary, install grommets in upper halves of doublers.

(2) Install doubler halves on fuselage using bolts and washers.

(3) **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

16-4-43.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove protective caps and plugs from electrical connectors.

c. Inspect and treat electrical connectors (PARA 1-7-20).

d. Connect electrical connectors, P360 to J360, and P361 to J361, to ESSS auxiliary fuel management control panel on lower console (Figure 16-4-43-1, Sheet 1, Detail A).

e. Install clamps around cabin fuel harness on side of post. Attach wiring harness clamps to clamps on post with screws, washers, and nuts

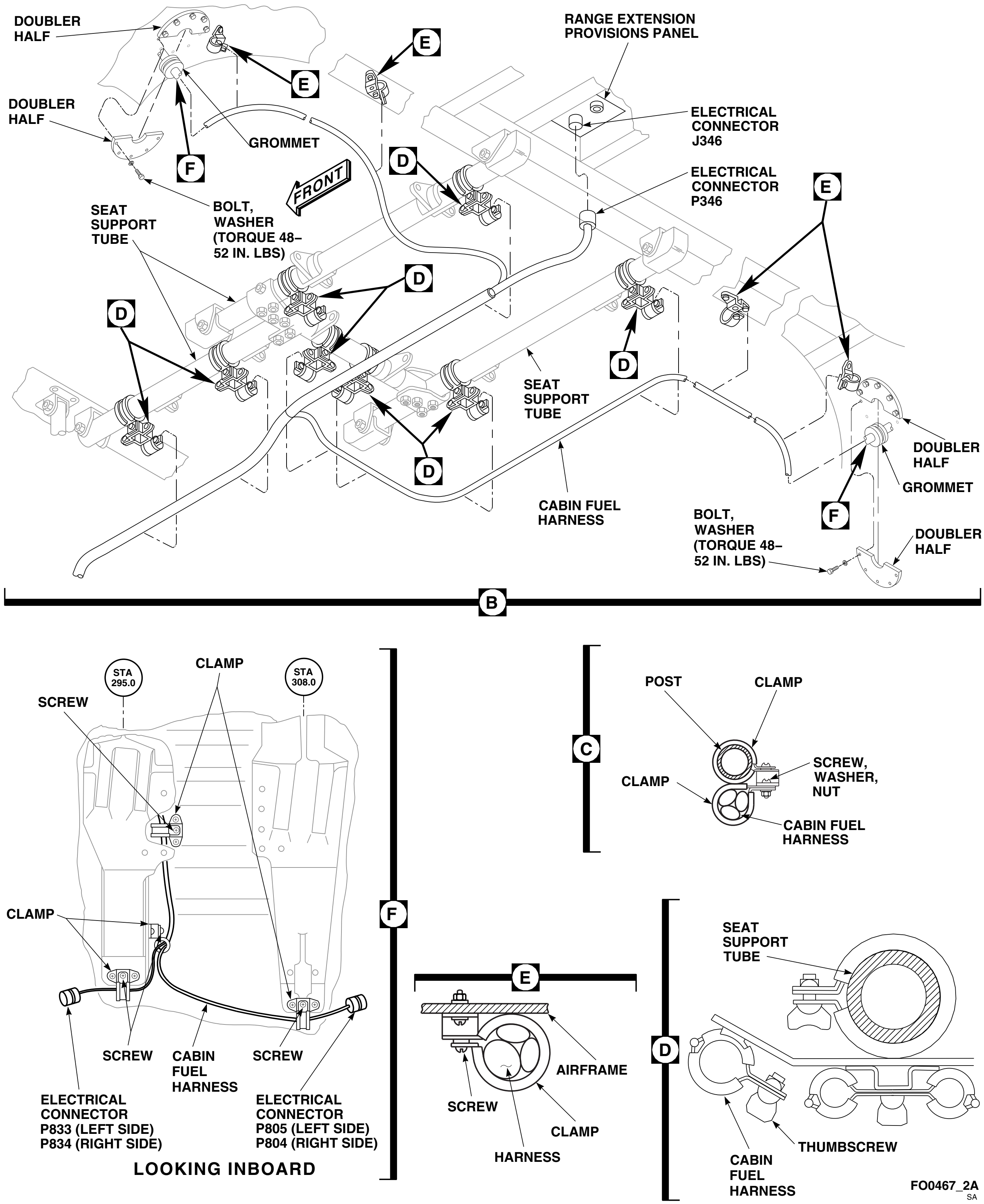


FIGURE 16-4-43-1. REPLACE ESSS CABIN FUEL HARNESS (SHEET 2 OF 2)

(Figure 16-4-43-1, Sheet 1, Detail A and Sheet 2, Detail C).

f. Install cabin fuel harness in clamps on seat support tubes. Install and turn thumbscrew to secure clamps (Figure 16-4-43-1, Sheet 2, Detail B and Detail D).

g. Remove protective caps and plugs from electrical connectors.

h. Inspect and treat electrical connectors (PARA 1-7-20).

i. Connect electrical connector, P346 to J346 (left side), on range extension provisions panel (Figure 16-4-43-1, Sheet 2, Detail B).

j. Feed electrical connectors, P834 (right side), P804 (right side), P833 (left side), and P805 (left side), through access holes in fuselage (Figure 16-4-43-1, Sheet 2, Detail B and Detail F).

k. Remove protective caps and plugs from electrical connectors.

l. Inspect and treat electrical connectors (PARA 1-7-20).

m. Connect electrical connectors, P834 and P804 (right side), and P833 and P805 (left side), to HSS interconnects (Figure 16-4-43-1, Sheet 2, Detail F).

n. Install cabin fuel harness in clamps on airframe. Install and turn screws to secure clamps (Figure 16-4-43-1, Sheet 2, Detail B, Detail E, and Detail F).

o. Install doubler halves as follows (Figure 16-4-43-1, Sheet 2, Detail B):

(1) If necessary, install grommets in upper halves of doublers.

(2) Install doubler halves on fuselage using bolts and washers.

(3) **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

p. Install tiedown straps, Item 331, Appendix D, on cabin fuel harness as necessary.

q. Install upper and lower root fairings (PARA 16-4-21).

r. Perform operational check of fuel transfer system (PARA 16-2-1).

16-4-44 ESSS HORIZONTAL STORES SUPPORT (HSS) FUEL SYSTEM REAR ELECTRICAL HARNESS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 16-2-1
- PARA 16-2-2
- PARA 16-4-21

16-4-44.1 Replace ESSS Horizontal Stores Support (HSS) Fuel System Rear Electrical Harness.

16-4-44.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove inboard and outboard rear trailing edge fairings, inboard and outboard saddle assemblies, and upper and lower root fairings from horizontal stores support (HSS) (PARA 16-4-21).

c. Remove screws, washers, and clamps from harness (Figure 16-4-44-1).

d. On left HSS, remove harness as follows:

(1) Disconnect electrical connector, P840, from outboard fuel gate shutoff valve.

(2) Disconnect electrical connector, P839, from outboard bleed-air regulator valve.

(3) Disconnect electrical connector, P808, from inboard bleed-air regulator valve.

(4) Disconnect electrical connector, P809, from inboard fuel gate shutoff valve.

(5) Disconnect electrical connector, P805 from J805, on HSS root fitting.

(6) Install protective caps and plugs on electrical connectors.

(7) Remove harness from HSS.

(8) Remove screws, washers, and nuts, and remove electrical connector, J805, from HSS root fitting.

e. On right HSS, remove harness as follows:

(1) Disconnect electrical connector, P838, from outboard fuel gate shutoff valve.

(2) Disconnect electrical connector, P837, from outboard bleed-air regulator valve.

(3) Disconnect electrical connector, P811, from inboard bleed-air regulator valve.

(4) Disconnect electrical connector, P810, from inboard fuel gate shutoff valve.

(5) Disconnect electrical connector, P804 from J804, on HSS root fitting.

(6) Install protective caps and plugs on electrical connectors.

(7) Remove harness from HSS.

(8) Remove screws, washers, and nuts, and remove electrical connector, J804, from HSS root fitting.

16-4-44.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove protective caps and plugs from electrical connectors.

c. Inspect and treat electrical connectors (PARA 1-7-20).

d. On left HSS, install harness (Figure 16-4-44-1) as follows:

(1) Install electrical connector, J805, to HSS root fitting with screws, washers, and nuts.

(2) Connect electrical connector, P805 to J805, on HSS root fitting.

(3) Connect electrical connector, P809, to inboard fuel gate shutoff valve.

(4) Connect electrical connector, P808, to inboard bleed-air regulator valve.

(5) Connect electrical connector, P839, to outboard bleed-air regulator valve.

(6) Connect electrical connector, P840, to outboard fuel gate shutoff valve.

(7) Clamp harness to HSS with clamps, screws, and washers.

e. Remove protective caps and plugs from electrical connectors.

f. Inspect and treat electrical connectors (PARA 1-7-20).

g. On right HSS, install harness as follows:

(1) Install electrical connector, J804, to HSS root fitting with screws, washers, and nuts.

(2) Connect electrical connector, P810, to inboard fuel gate shutoff valve.

(3) Connect electrical connector, P811, to inboard bleed-air regulator valve.

(4) Connect electrical connector, P837, to outboard bleed-air regulator valve.

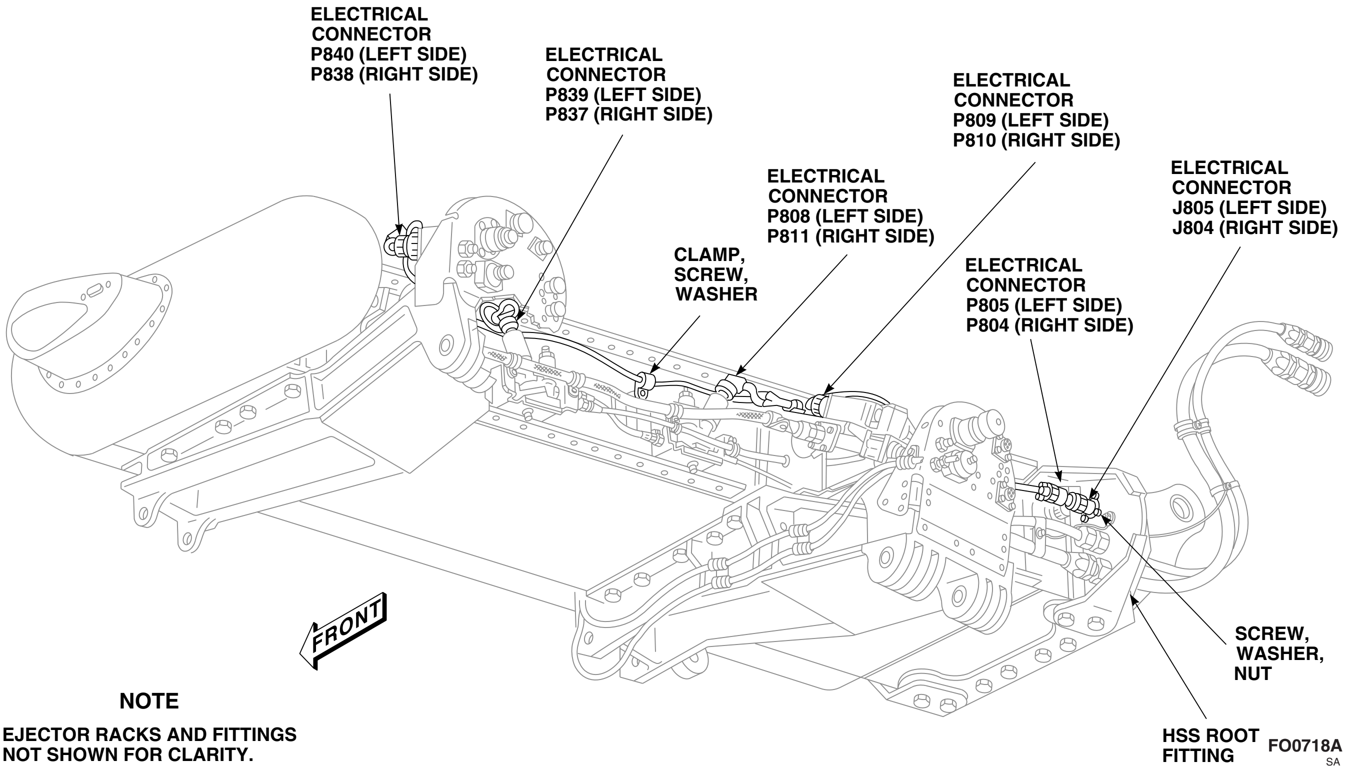
(5) Connect electrical connector, P838, to outboard fuel gate shutoff valve.

(6) Connect electrical connector, P804 to J804, on HSS root fitting.

(7) Clamp harness to HSS with clamps, screws, and washers.

h. Perform operational check of ESSS range extension system (PARA 16-2-1 or PARA 16-2-2).

i. Install inboard and outboard rear trailing edge fairings, inboard and outboard saddle assemblies, and upper and lower root fairings from HSS (PARA 16-4-21).



NOTE

EJECTOR RACKS AND FITTINGS NOT SHOWN FOR CLARITY.

FIGURE 16-4-44-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) FUEL SYSTEM REAR ELECTRICAL HARNESS

16-4-45 ESSS HORIZONTAL STORES SUPPORT (HSS) FUEL SYSTEM FRONT ELECTRICAL HARNESS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 16-2-1
- PARA 16-2-2
- PARA 16-4-21

16-4-45.1 Replace ESSS Horizontal Stores Support (HSS) Fuel System Front Electrical Harness.

16-4-45.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove leading edge fairing, inboard and outboard saddle assemblies, and inboard and outboard rear VSP fairings from HSS (PARA 16-4-21).

c. Remove screws, washers, nuts, and clamps from harness (Figure 16-4-45-1, Sheet 1).

d. On left HSS, remove harness as follows:

(1) Disconnect electrical connector, P833 from J833, on HSS root fitting.

(2) Disconnect electrical connector, P2/J835 from J835, on inboard VSP fuel arm extension (Figure 16-4-45-1, Sheet 2, Detail A).

(3) Disconnect electrical connector, P2/J849 from J849, on outboard VSP fuel arm extension (Figure 16-4-45-1, Sheet 2, Detail B).

(4) Install protective caps and plugs on electrical connectors.

(5) Remove screws, washers, and nuts, and remove connector receptacles, J835 and J849, from brackets on inboard and outboard VSP fuel arm extensions (Figure 16-4-45-1, Sheet 2, Detail A and Detail B).

(6) Remove harness from HSS.

(7) Remove screws, washers, and nuts, and remove electrical connector, J833, from HSS root fitting (Figure 16-4-45-1, Sheet 1).

e. On right HSS, remove harness as follows:

(1) Disconnect electrical connector, P834 from J834, on HSS root fitting (Figure 16-4-45-1, Sheet 1).

(2) Disconnect electrical connector, P2/J836 from J836, on inboard VSP fuel arm extension (Figure 16-4-45-1, Sheet 2, Detail A).

(3) Disconnect electrical connector, P2/J850 from J850, on outboard VSP fuel arm extension (Figure 16-4-45-1, Sheet 2, Detail B).

(4) Install protective caps and plugs on electrical connectors.

(5) Remove screws, washers, and nuts, and remove connector receptacles, J836 and J850, from brackets on inboard and outboard VSP fuel arm extensions (Figure 16-4-45-1, Sheet 2, Detail A and Detail B).

(6) Remove harness from HSS.

(7) Remove screws, washers, and nuts, and remove electrical connector, J834, from HSS root fitting (Figure 16-4-45-1, Sheet 1).

f. If drop tanks are installed, remove VSP fuel arm extension harness as follows:

(1) Disconnect electrical connector, P1 from J1, on VSP fuel arm extension (Figure 16-4-45-1, Sheet 2, Detail A and Detail B).

(2) Install protective caps and plugs on electrical connectors.

(3) Remove screws, washers, and nuts, and remove electrical connector, J1, from VSP fuel arm extension.

(4) Unfasten loop clamps, and remove harness from VSP fuel arm extension.

16-4-45.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. If drop tanks are installed, install VSP fuel arm extension harness as follows:

(1) Remove protective caps and plugs from electrical connectors.

(2) Inspect and treat electrical connectors (PARA 1-7-20).

(3) Install electrical connector, J1, to VSP fuel arm extension with screws, washers, and nuts (Figure 16-4-45-1, Sheet 2, Detail A and Detail B).

(4) Connect electrical connector, P1 to J1, on VSP fuel arm extension.

(5) Install harness to VSP fuel arm extension with loop clamps.

c. On left HSS, install harness as follows:

(1) Remove protective caps and plugs from electrical connectors.

(2) Inspect and treat electrical connectors (PARA 1-7-20).

(3) Install electrical connector, J833, to HSS root fitting with screws, washers, and nuts (Figure 16-4-45-1, Sheet 1).

(4) Connect electrical connector, P833 to J833, on HSS root fitting.

(5) Install connector receptacles, J835, and J849, to inboard and outboard HSS brackets with screws, washers, and nuts (Figure 16-4-45-1, Sheet 2, Detail A and Detail B).

(6) Connect electrical connector, P2/J835 to J835, on inboard VSP fuel arm extension (Figure 16-4-45-1, Sheet 2, Detail A).

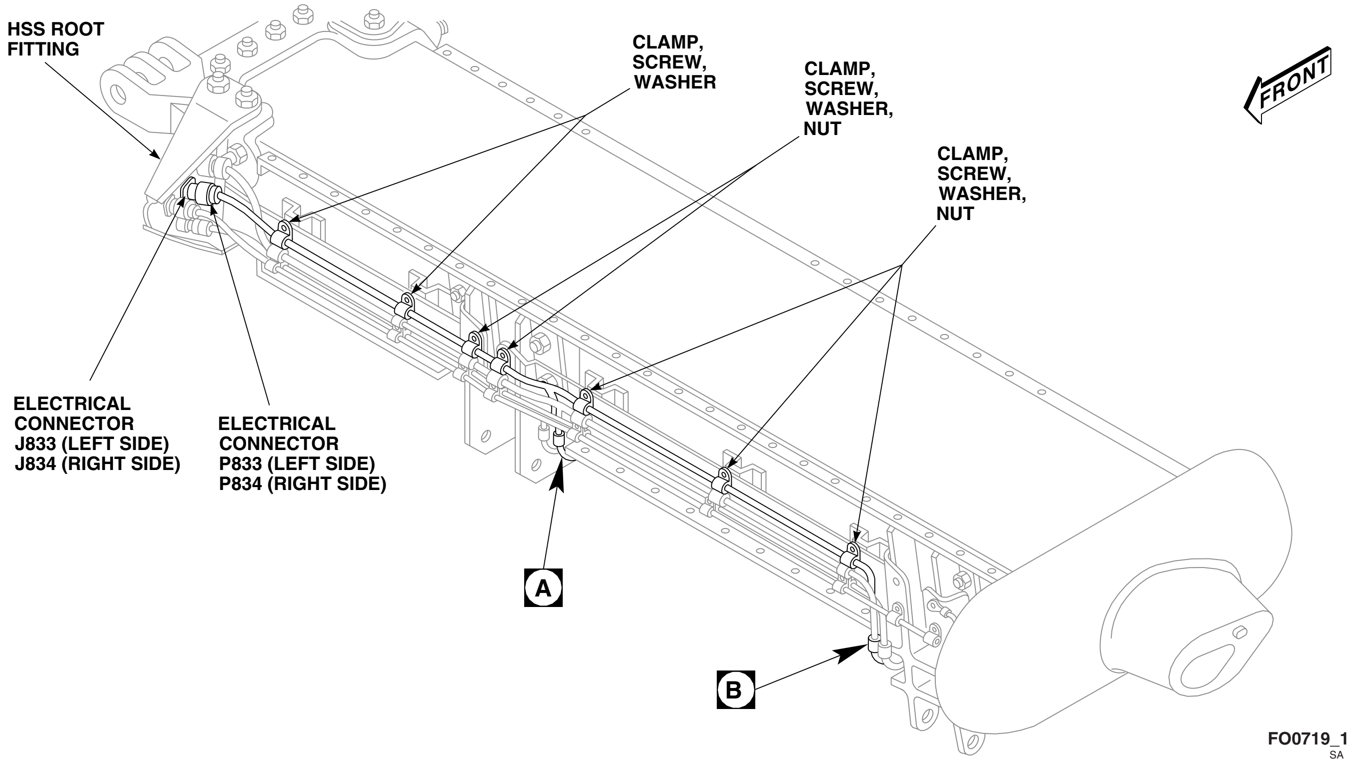


FIGURE 16-4-45-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) FUEL SYSTEM FRONT ELECTRICAL HARNESS (SHEET 1 OF 2)

(7) Connect electrical connector, P2/J849 to J849, on outboard VSP fuel arm extension (Figure 16-4-45-1, Sheet 2, Detail B).

(8) Secure harness to HSS with screws, washers, nuts, and clamps (Figure 16-4-45-1, Sheet 1).

d. On right HSS, install harness as follows:

(1) Remove protective caps and plugs from electrical connectors.

(2) Inspect and treat electrical connectors (PARA 1-7-20).

(3) Install electrical connector, J834, to HSS root fitting with screws, washers, and nuts (Figure 16-4-45-1, Sheet 1).

(4) Connect electrical connector, P834 to J834, on HSS root fitting.

(5) Install connector receptacles, J836 and J850, to brackets on inboard and outboard VSP fuel arm extensions with screws, washers, and nuts (Figure 16-4-45-1, Sheet 2, Detail A and Detail B).

(6) Connect electrical connector, P2/J836 to J836, on inboard VSP fuel arm extension (Figure 16-4-45-1, Sheet 2, Detail A).

(7) Connect electrical connector, P2/J850 to J850, on outboard VSP fuel arm extension (Figure 16-4-45-1, Sheet 2, Detail B).

(8) Secure harness to HSS with screws, washers, nuts, and clamps (Figure 16-4-45-1, Sheet 1).

e. Install leading edge fairing, inboard and outboard saddle assemblies, and inboard and outboard rear VSP fairings from HSS (PARA 16-4-21).

f. Perform operational check of ESSS range extension system (PARA 16-2-1 or PARA 16-2-2).

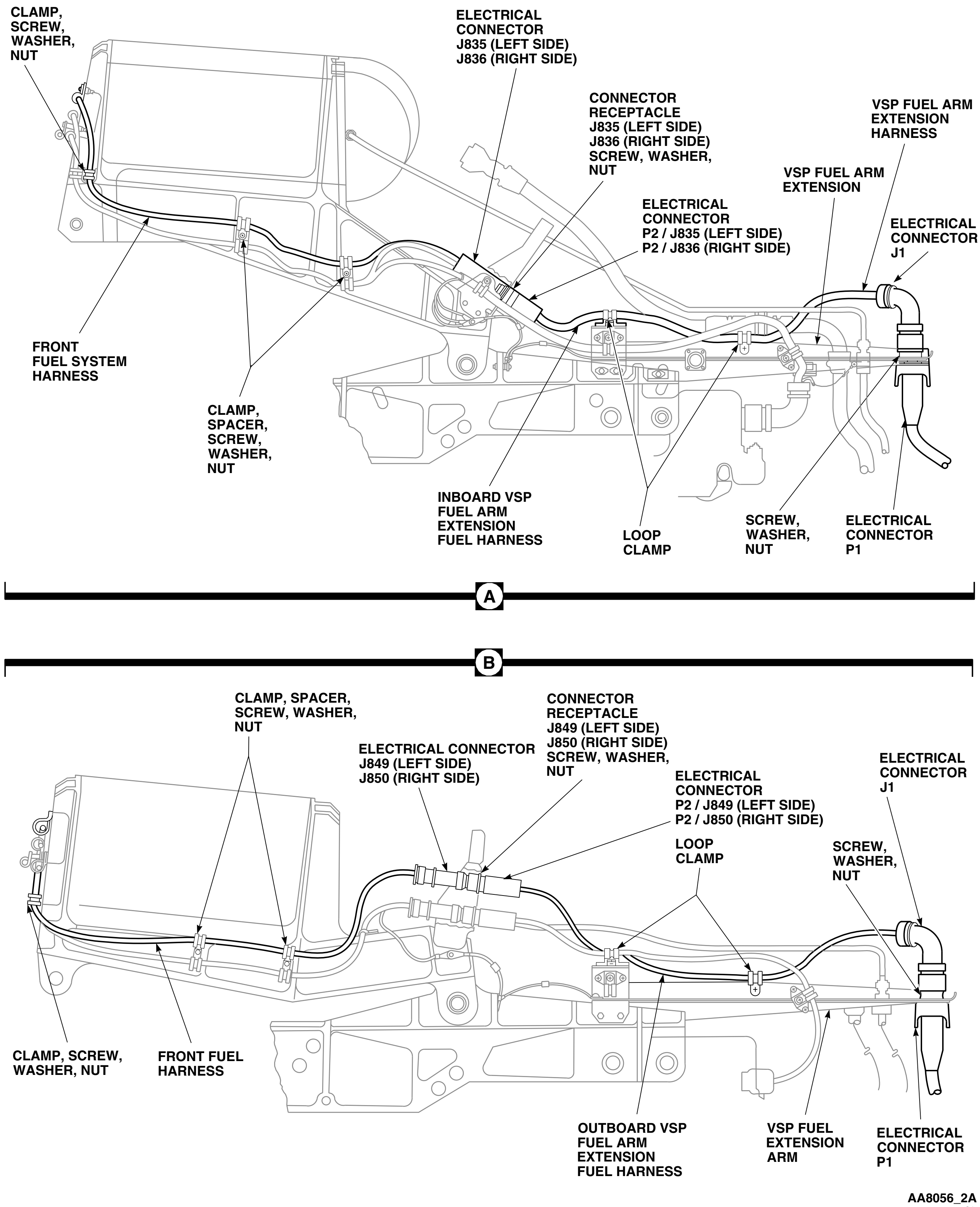


FIGURE 16-4-45-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) FUEL SYSTEM FRONT ELECTRICAL HARNESS (SHEET 2 OF 2)

16-4-46 ESSS HORIZONTAL STORES SUPPORT (HSS) POSITION LIGHT HARNESS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-14
- PARA 16-4-21

16-4-46.1 Replace ESSS Horizontal Stores Support (HSS) Position Light Harness.

16-4-46.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove leading edge fairing (PARA 16-4-21).

c. Remove tip fairing (PARA 16-4-21).

d. On left HSS, remove screws, washers, and nuts and remove electrical connectors, J714 and J719, from bulkheads (Figure 16-4-46-1).

e. On right HSS, remove screws, washers, and nuts and remove electrical connectors, J715 and J720, from bulkheads

f. On left HSS, remove electrical connector, P682 from J682. Remove screws and nuts, and remove electrical connector, J682, from root end bulkhead.

g. On right HSS, remove electrical connector, P681 from J681. Remove screws and nuts, and remove electrical connector, J681, from root end bulkhead.

h. Remove clamps, screws, washers, spacers, nuts, and harness from HSS.

i. Install protective caps and plugs on electrical connectors.

16-4-46.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove protective caps and plugs from electrical connectors.

c. Inspect and treat electrical connectors (PARA 1-7-20).

d. On left HSS outboard bulkhead, install electrical connectors labeled, J714 and J719, on bulkhead with screws, washers, and nuts (Figure 16-4-46-1).

e. On right HSS outboard bulkhead, install electrical connectors labeled, J715 and J720, on bulkhead with screws, washers, and nuts.

f. Remove protective caps and plugs from electrical connectors.

g. Inspect and treat electrical connectors (PARA 1-7-20).

h. On left HSS, install electrical connector, J682, on root end bulkhead with screws and nuts. Connect electrical connector, P682 to J682.

i. Remove protective caps and plugs from electrical connectors.

j. Inspect and treat electrical connectors (PARA 1-7-20).

k. On right HSS, install electrical connector, J681, on root end bulkhead with screws and nuts. Connect electrical connector, P681 to J681.

l. Secure harness to HSS with clamps, screws, washers, nuts, and spacers (Figure 16-4-46-1, Detail A).

m. Install leading edge fairing from HSS (PARA 16-4-21).

n. Install tip fairing from HSS (PARA 16-4-21).

o. Perform operational check of position light system (PARA 9-2-14).

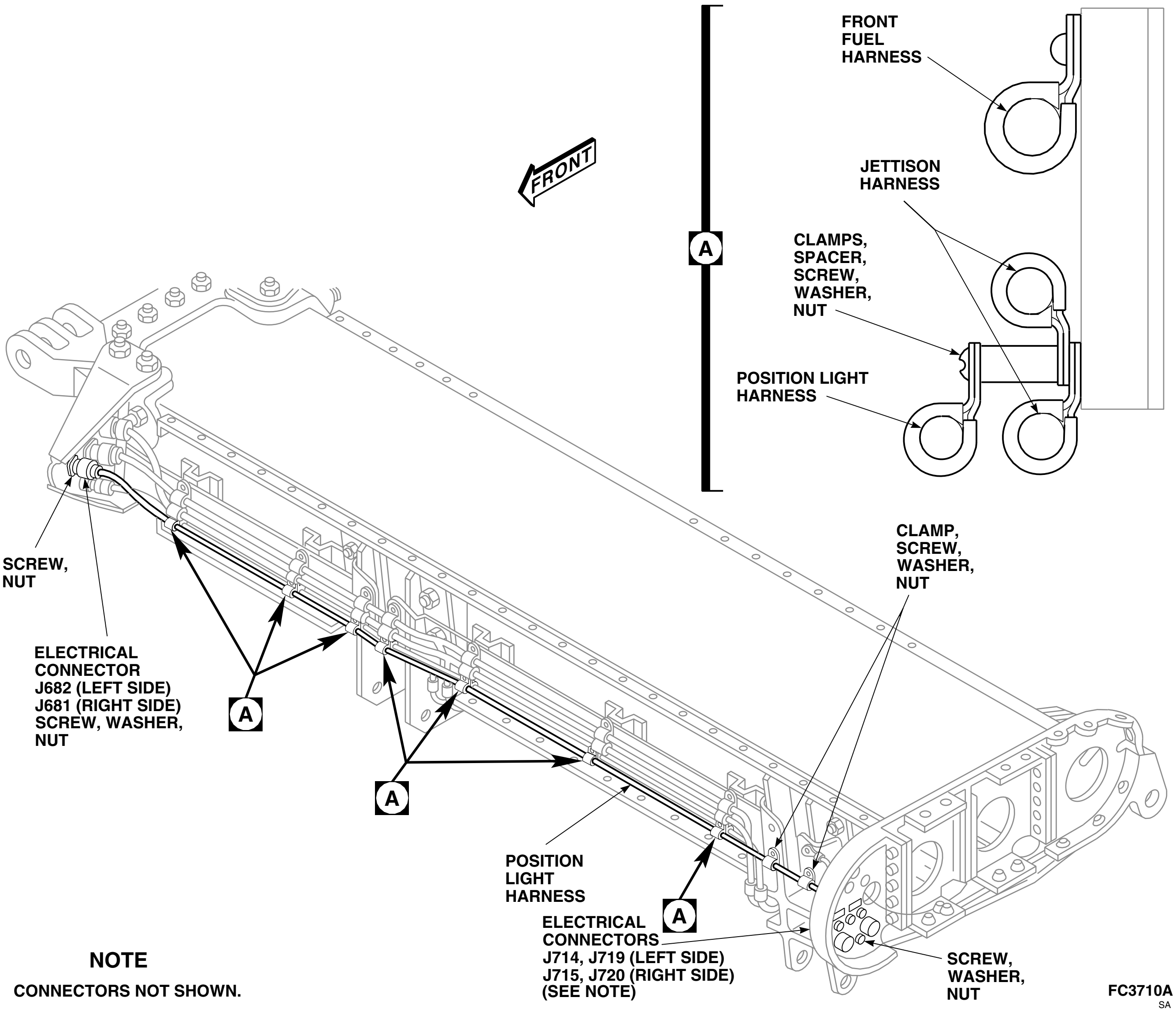


FIGURE 16-4-46-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) POSITION LIGHT HARNESS

16-4-47 ESSS FUEL OVERFLOW SENSOR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-45
- PARA 2-4-69
- PARA 16-2-1
- PARA 16-2-2

16-4-47.1 Replace ESSS Fuel Overflow Sensor.

16-4-47.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove passenger's seats from rear cabin (PARA 2-4-45).

c. Remove soundproofing panels from right and left bulkhead (PARA 2-4-69).

d. Remove main fuel tank access panel and remove components as required to reach rear fuel vent tube elbow.

e. Disconnect wiring from clamps on helicopter wiring harness.

f. Disconnect electrical connector, P376 from J376 (Figure 16-4-47-1, Detail A).

g. Install protective caps and plugs on electrical connectors.

h. Remove ESSS fuel overflow sensor and packing from fuel vent tube elbow. Discard packing.

16-4-47.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Clean surface around all mounting holes with dry-cleaning solvent, Item 124, Appendix D, for good electrical ground contact.

c. Lubricate packing with petrolatum, Item 234, Appendix D.

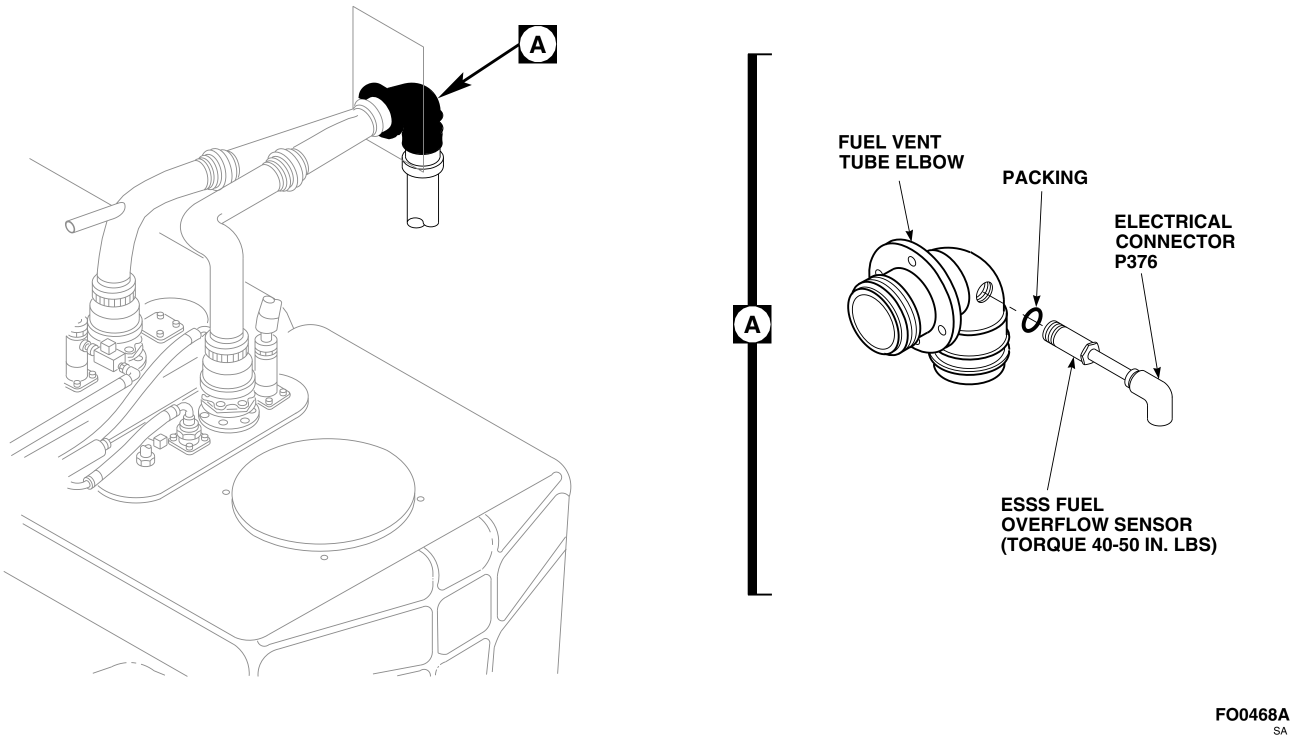


FIGURE 16-4-47-1. REPLACE ESSS FUEL OVERFLOW SENSOR

d. Install packing on end of ESSS fuel overflow sensor (Figure 16-4-47-1, Detail A).

e. Apply antiseize compound, Item 74, Appendix D, to the threads of ESSS fuel overflow sensor.

f. Screw ESSS fuel overflow sensor into fuel vent tube elbow. **TORQUE ESSS FUEL OVERFLOW SENSOR TO 40 - 50 INCH-POUNDS.**

g. Remove protective caps and plugs from electrical connectors.

h. Inspect and treat electrical connectors (PARA 1-7-20).

i. Connect electrical connector, P376 to J376.

j. Secure wiring to helicopter wiring harness with existing clamps.

k. Make sure area is clean and free of foreign material.

l. Perform operational check of fuel transfer system and check for leaks (PARA 16-2-1 or PARA 16-2-2).

m. Install main fuel tank access panel and install removed components as required.

n. Install soundproofing panels on right and left rear bulkhead (PARA 2-4-69).

o. Install passenger's seats in rear cabin (PARA 2-4-45).

16-4-48 ESSS SIGNAL CONDITIONER AUX FUEL QTY .

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Resistant Coating, Item 117, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-43
- PARA 16-2-2

16-4-48.1 Replace ESSS Signal Conditioner.

16-4-48.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove cockpit floor (PARA 2-4-43).

c. Disconnect electrical connectors, P360 from J1, and P361 from J2, from ESSS signal conditioner (Figure 16-4-48-1, Detail A).

d. Install protective caps and plugs on electrical connectors.

e. Remove screws, lockwashers, and washers securing ESSS signal conditioner to mounting bracket. Remove signal conditioner.

16-4-48.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Prepare mating surfaces for electrical bond as follows:

(1) Clean paint from around mounting holes using abrasive cloth, Item 8, Appendix D.

(2) Clean area with clean cheesecloth, Item 90, Appendix D, moistened with dry-cleaning solvent, Item 124, Appendix D.

(3) Apply corrosion resistant coating, Item 117, Appendix D, to contacting area.

c. Install ESSS signal conditioner on mounting bracket and secure with screws, lockwashers, and washers (Figure 16-4-48-1, Detail A).

d. Remove protective caps and plugs from electrical connectors.

e. Inspect and treat electrical connectors (PARA 1-7-20).

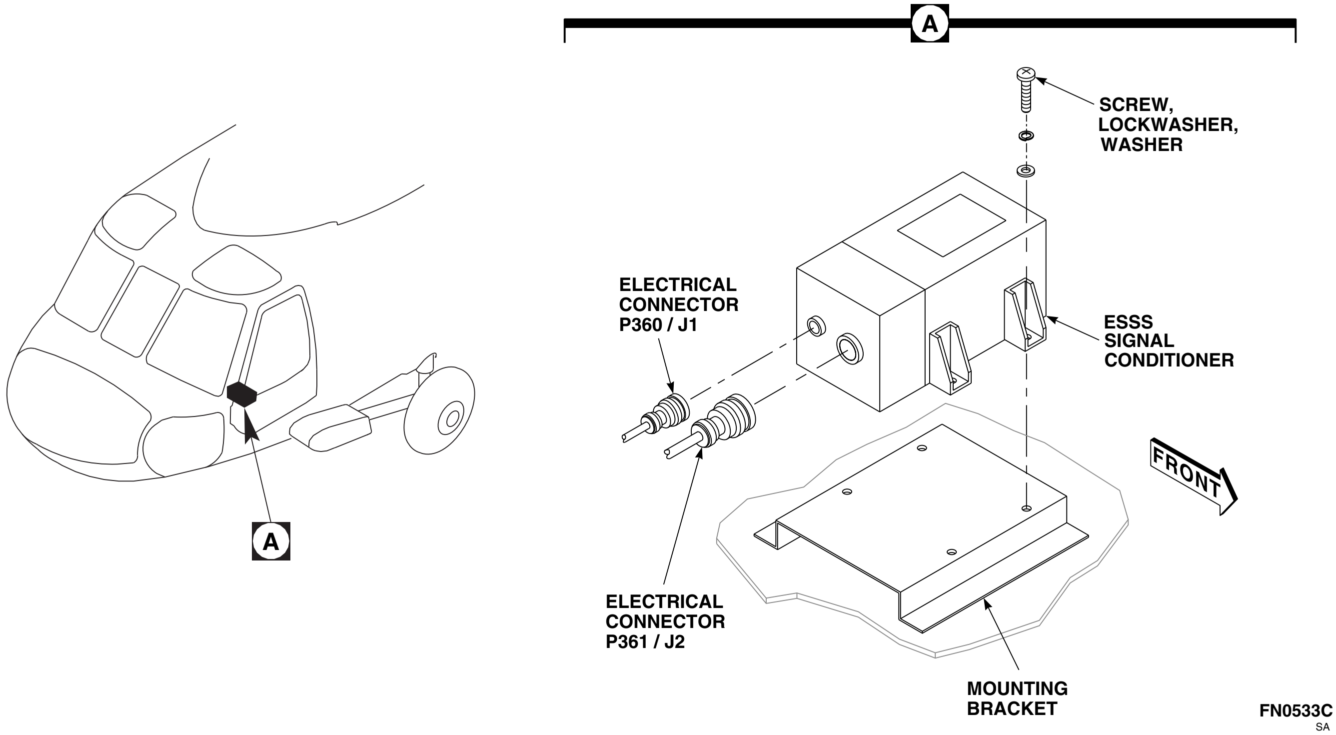


FIGURE 16-4-48-1. REPLACE ESSS SIGNAL CONDITIONER

f. Connect electrical connectors, P360 to J1, and P361 to J2, on ESSS signal conditioner.

g. Make sure area is clean and free of foreign material.

h. Install cockpit floor (PARA 2-4-43).

i. Perform operational check of ESSS range extension system (PARA 16-2-2).

16-4-49 ESSS CONTROL DISPLAY PANEL AUX FUEL QTY .

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 16-2-2

16-4-49.1 Replace ESSS Control Display Panel.

16-4-49.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

NOTE

Location of external stores support system (ESSS) control display panel may vary.

b. Loosen fasteners and lift ESSS control display panel from lower console until electrical connectors are exposed (Figure 16-4-49-1, Detail A).

c. Disconnect electrical connectors, P3001 from J1, and P3002 from J2, on ESSS control display panel. Remove panel.

d. Install protective caps and plugs on electrical connectors.

16-4-49.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove protective caps and plugs from electrical connectors.

c. Inspect and treat electrical connectors (PARA 1-7-20).

d. Connect electrical connectors, P3001 to J1, and P3002 to J2, on ESSS control display panel (Figure 16-4-49-1, Detail A).

e. Make sure area is clean and free of foreign material.

f. Install ESSS control display panel in lower console and secure with fasteners.

g. Perform operational check of ESSS range extension system (PARA 16-2-2).

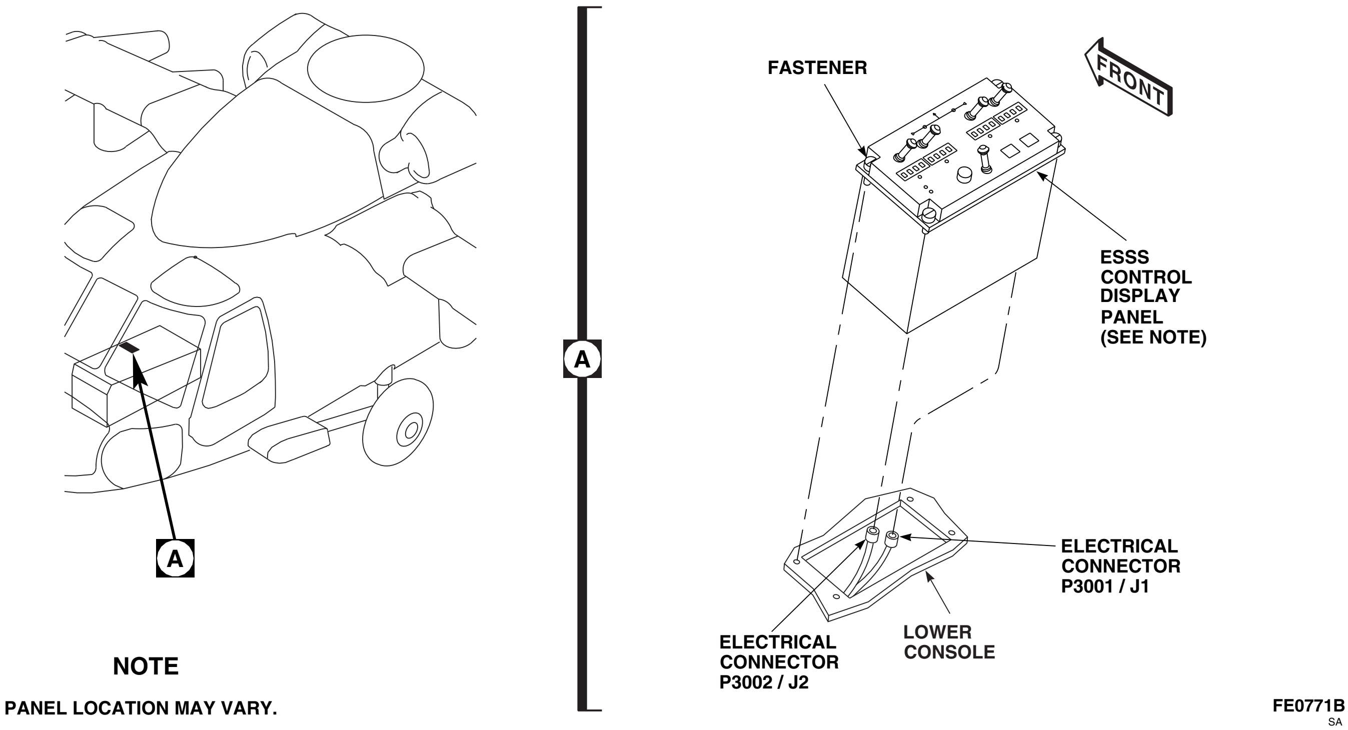


FIGURE 16-4-49-1. REPLACE ESSS CONTROL DISPLAY PANEL

16-4-50 ESSS STORES JETTISON CONTROL PANEL.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 16-2-4

16-4-50.1 Replace ESSS Stores Jettison Control Panel.

16-4-50.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Loosen fasteners and lift ESSS stores jettison control panel from lower console until electrical connectors are exposed (Figure 16-4-50-1, Detail A).

c. Disconnect electrical connectors, P645 from J1, and P646 from J2, on ESSS stores jettison control panel. Remove panel.

d. Install protective caps and plugs on electrical connectors.

16-4-50.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove protective caps and plugs from electrical connectors.

c. Inspect and treat electrical connectors (PARA 1-7-20).

d. Connect electrical connectors, P645 to J1, and P646 to J2, on ESSS stores jettison control panel (Figure 16-4-50-1, Detail A).

e. Make sure area is clean and free of foreign material.

f. Install ESSS stores jettison control panel in lower console and secure with fasteners.

g. Perform operational check of ESSS stores jettison system (PARA 16-2-4).

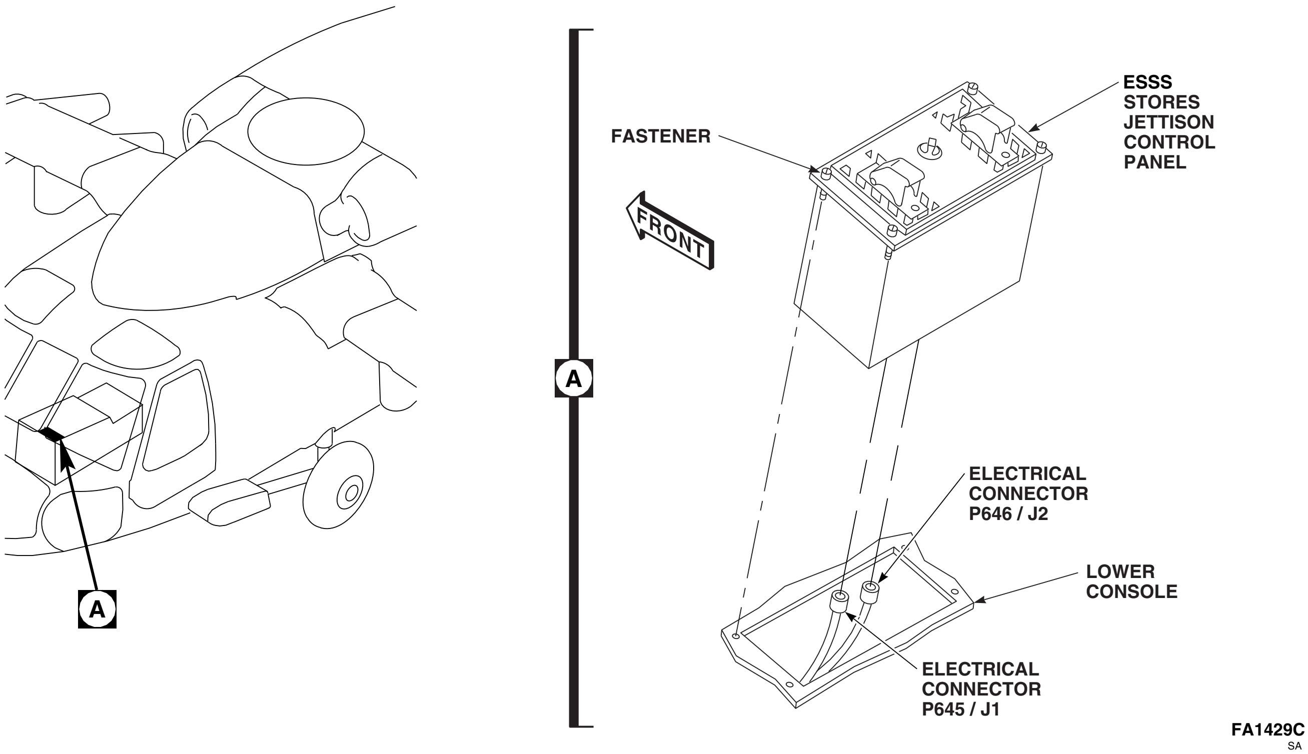


FIGURE 16-4-50-1. REPLACE ESSS STORES JETTISON CONTROL PANEL

16-4-51 ESSS STORES JETTISON CONTROL PANEL INFORMATION PLATE/LAMPS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

Materials/Parts

- Lockwire, Item 195, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 16-2-4

16-4-51.1 Replace ESSS Stores Jettison Control Panel Information Plate/Lamps.

16-4-51.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Loosen setscrews and remove rotary switch knob from ESSS stores jettison control panel on lower console (Figure 16-4-51-1, Detail A).

c. Remove screws and switch guards. Remove information plate from ESSS stores jettison control panel.

d. Remove and discard moisture seal from back of information plate.

e. Remove screws and remove lamp from information plate.

16-4-51.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Position lamp on information plate and secure with screws (Figure 16-4-51-1, Detail A).

c. Install new moisture seal over lamp.

d. Position information plate and switch guards on ESSS stores jettison control panel and secure with screws.

e. Position rotary switch knob on switch shaft. **TIGHTEN SETSCREWS.**

f. Lockwire, Item 195, Appendix D, to breakaway, EMER JETT ALL switch guard.

g. Perform operational check of ESSS stores jettison system (PARA 16-2-4).

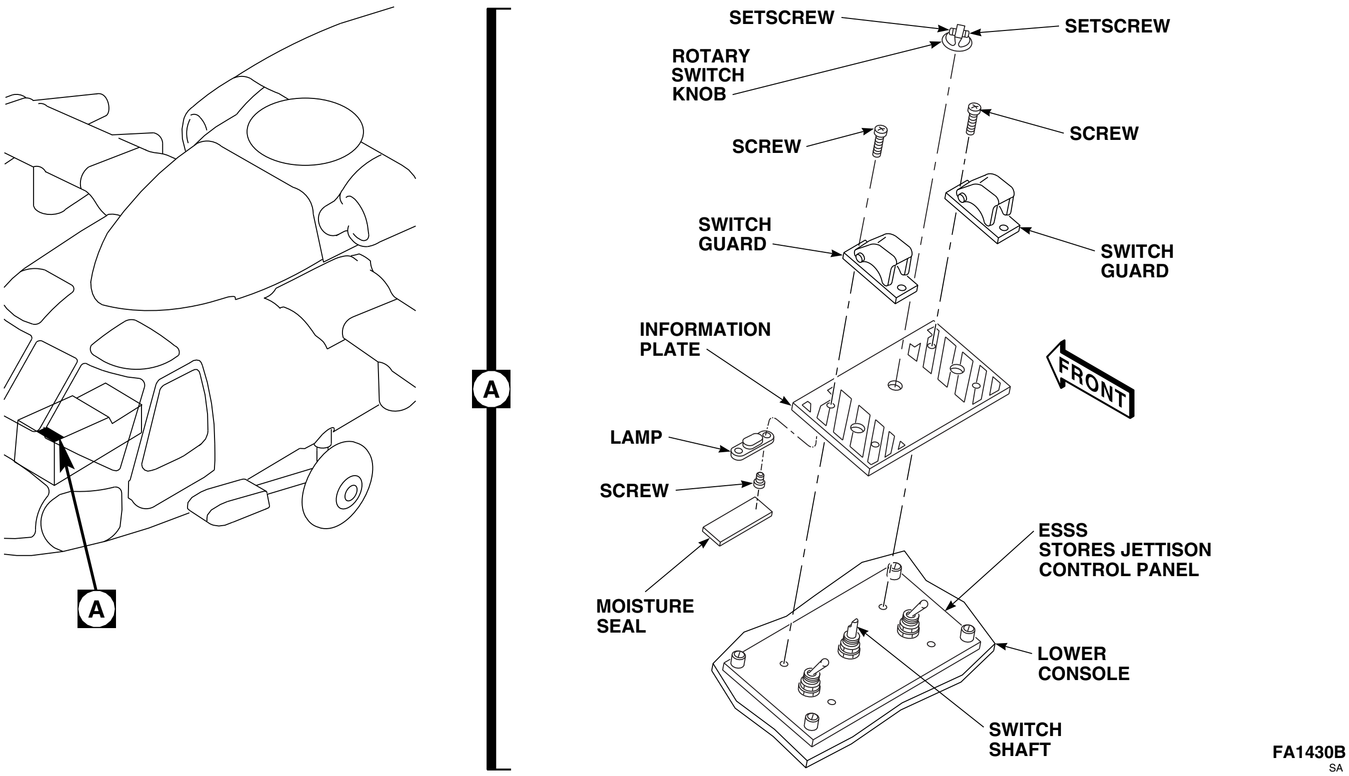


FIGURE 16-4-51-1. REPLACE ESSS STORES JETTISON CONTROL PANEL INFORMATION PLATE/LAMPS

16-4-52 ESSS CABIN JETTISON HARNESS W/O AUX FUEL QTY .

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Protective Caps and Plugs
- Tiedown Strap, Item 331, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 16-2-4
- PARA 16-4-21

16-4-52.1 Replace ESSS Cabin Jettison Harness.

16-4-52.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove upper and lower root fairings (PARA 16-4-21).

c. Disconnect electrical connectors, P660 from J660, and P661 from J661, on left side of lower console (Figure 16-4-52-1, Sheet 1, Detail A).

d. Install protective caps and plugs on electrical connectors.

e. Remove tiedown straps securing external stores support system cabin jettison harness as required.

f. Remove clamps, screws, washers, and nuts from clamps on post. Remove cabin jettison harness from clamps (Figure 16-4-52-1, Sheet 1, Detail A and Sheet 2, Detail C).

g. Turn and release thumbscrew on clamps securing cabin jettison harness to seat support tubes. Remove cabin jettison harness from clamps (Figure 16-4-52-1, Sheet 2, Detail B and Detail D).

h. Turn and release screws of clamps securing cabin jettison harness to airframe. Remove cabin jettison harness from clamps (Figure 16-4-52-1, Sheet 2, Detail B and Detail E).

i. Remove doubler halves as follows (Figure 16-4-52-1, Sheet 2, Detail B):

(1) Remove bolts and washers securing lower halves of doublers to fuselage and remove lower halves.

(2) If required, remove and slit grommet.

j. Turn and release screws of clamps securing cabin jettison harness to airframe. Remove cabin jettison harness from clamps (Figure 16-4-52-1, Sheet 2, Detail F).

k. Disconnect electrical connectors, P678 and P680 (left side), and P677 and P679 (right side), from horizontal stores support interconnects.

l. Install protective caps and plugs on electrical connectors.

m. Pull cabin jettison harness from access holes in fuselage.

n. Remove cabin jettison harness from helicopter.

NOTE

If replacement cabin fuel harness is to be installed, do not reinstall doubler halves.

o. If cabin fuel harness will not be installed, install doubler halves as follows (Figure 16-4-52-1, Sheet 2, Detail B):

(1) If necessary, install grommets in upper halves of doublers.

(2) Install doubler halves on fuselage using bolts and washers.

(3) **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

16-4-52.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove protective caps and plugs from electrical connectors.

c. Inspect and treat electrical connectors (PARA 1-7-20).

d. Connect electrical connectors, P660 to J660, and P661 to J661, on left side of lower console (Figure 16-4-52-1, Sheet 1, Detail A).

e. Install clamps around external stores support system (ESSS) cabin jettison harness on side of post. Secure harness clamps to clamps on posts with screws, washers, and nuts (Figure 16-4-52-1, Sheet 1, Detail A and Sheet 2, Detail C).

f. Install cabin jettison harness in clamps on seat support tubes. Install and turn thumbscrews to secure clamps (Figure 16-4-52-1, Sheet 2, Detail B and Detail D).

g. Feed electrical connectors, P678 and P680 (left side), and P677 and P679 (right side), through access holes in fuselage (Figure 16-4-52-1, Sheet 2, Detail B and Detail F).

h. Remove protective caps and plugs from electrical connectors.

i. Inspect and treat electrical connectors (PARA 1-7-20).

j. Connect electrical connectors, P678 and P680 (left side), and P677 and P679 (right side), to horizontal stores support interconnects (Figure 16-4-52-1, Sheet 2, Detail F).

k. Install cabin jettison harness in clamps on airframe. Install and turn screws to secure clamps (Figure 16-4-52-1, Sheet 2, Detail B, Detail E, and Detail F).

l. Install doubler halves as follows (Figure 16-4-52-1, Sheet 2, Detail B):

(1) If necessary, install grommets in upper halves of doublers.

(2) Install doubler halves on fuselage using bolts and washers.

(3) **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

m. Install tiedown straps, Item 331, Appendix D, on cabin jettison harness as necessary.

n. Install upper and lower root fairings (PARA 16-4-21).

o. Perform operational check of ESSS jettison system (PARA 16-2-4).

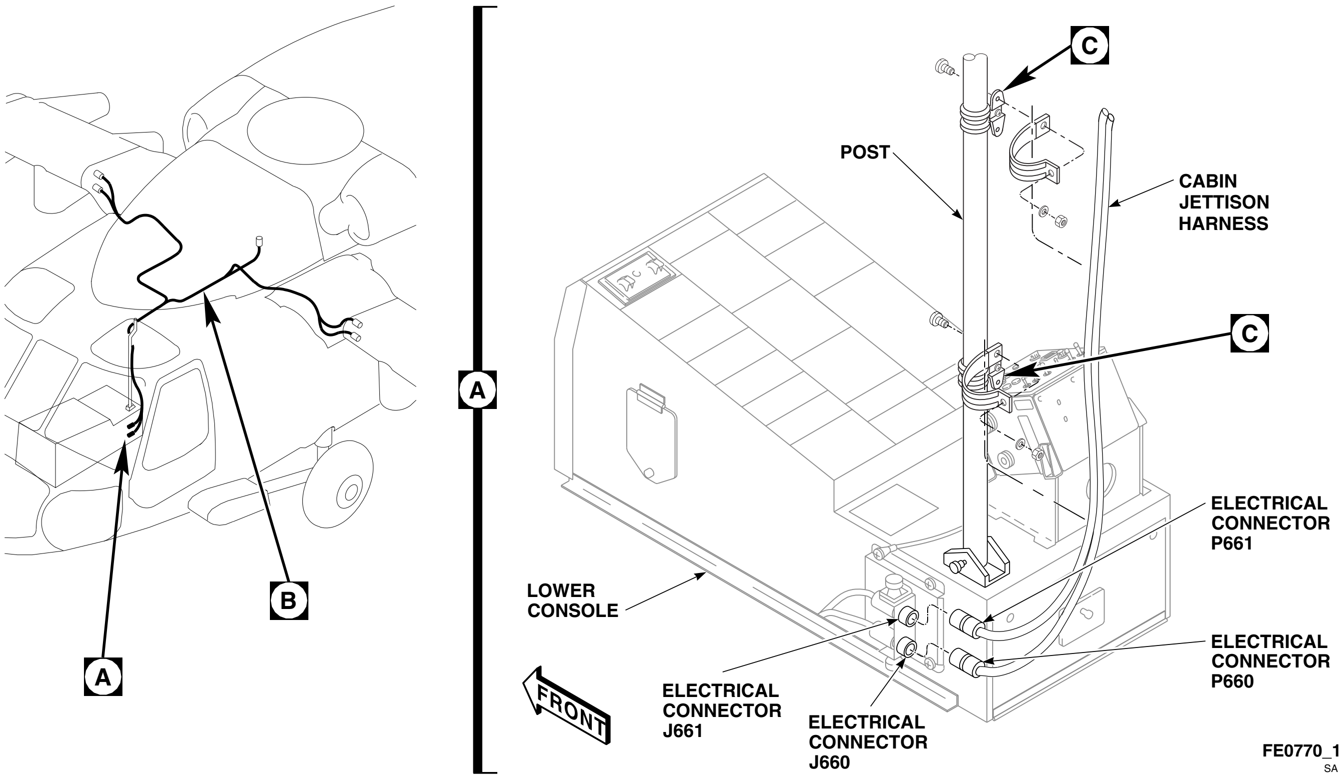


FIGURE 16-4-52-1. REPLACE ESSS CABIN JETTISON HARNESS (SHEET 1 OF 2)

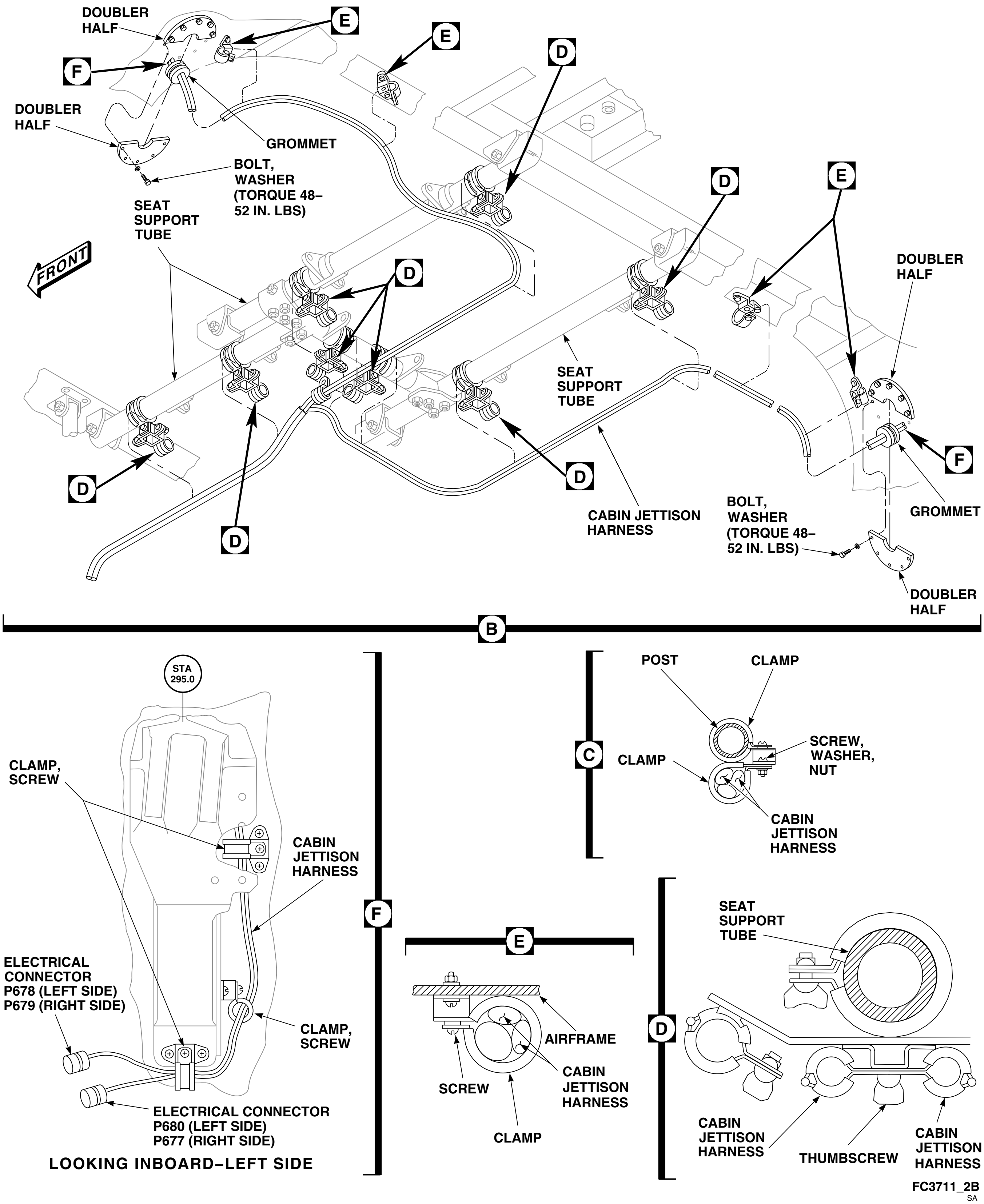


FIGURE 16-4-52-1. REPLACE ESSS CABIN JETTISON HARNESS (SHEET 2 OF 2)

16-4-53 ESSS HORIZONTAL STORES SUPPORT (HSS) JETTISON SYSTEM HARNESS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 16-2-4
- PARA 16-4-21

16-4-53.1 Replace ESSS Horizontal Stores Support (HSS) Jettison System Harness.

16-4-53.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

NOTE

If helicopter is to be flown without ejector racks, BRU-22A/A, installed, remove jettison firing cable from helicopter.

b. Remove front and rear vertical stores pylon (VSP) fairings from HSS (PARA 16-4-21).

c. Remove inboard and outboard saddle assemblies (PARA 16-4-21).

d. Remove leading edge fairing from HSS (PARA 16-4-21).

e. Remove jettison harness on outboard VSP as follows (Figure 16-4-53-1, Sheet 2, Detail B):

(1) Remove screw, washer, lockwasher, and nut securing ground wire to outboard VSP.

(2) Remove electrical connectors, P1/J710, and J710, from connector receptacle, J710, on right side. Remove electrical connectors, J705, and P1/J705, from connector receptacle, J705, on left side.

(3) Install protective caps and plugs on electrical connectors.

(4) Remove screws, washers, and nuts, and remove connector receptacle, J710 (right side), or J705 (left side), from VSP bracket.

(5) Remove clamps, screws, washers, nuts, spacers, and loop clamps securing jettison harness to outboard VSP.

f. Remove jettison harness on inboard VSP as follows (Figure 16-4-53-1, Sheet 2, Detail C):

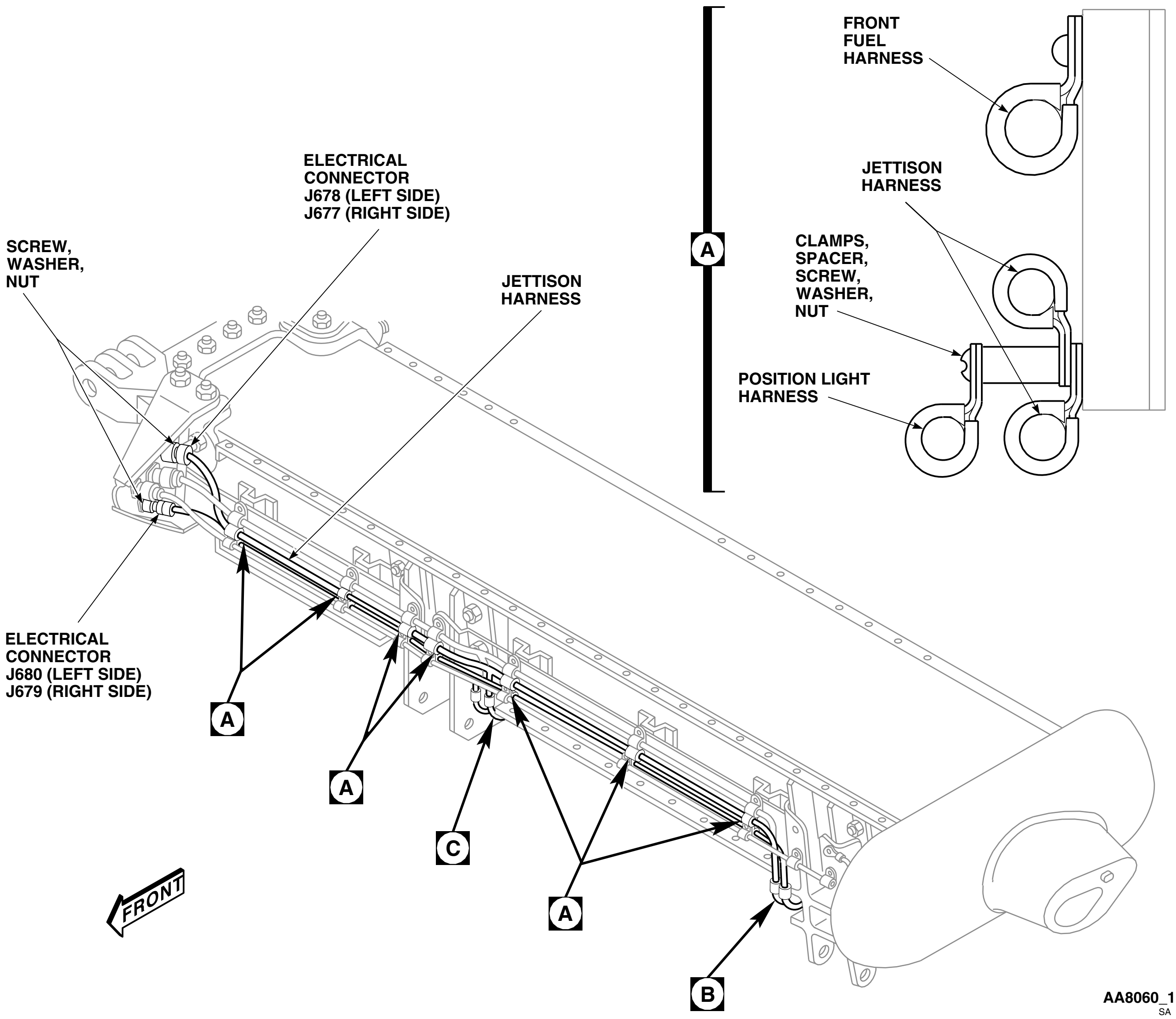


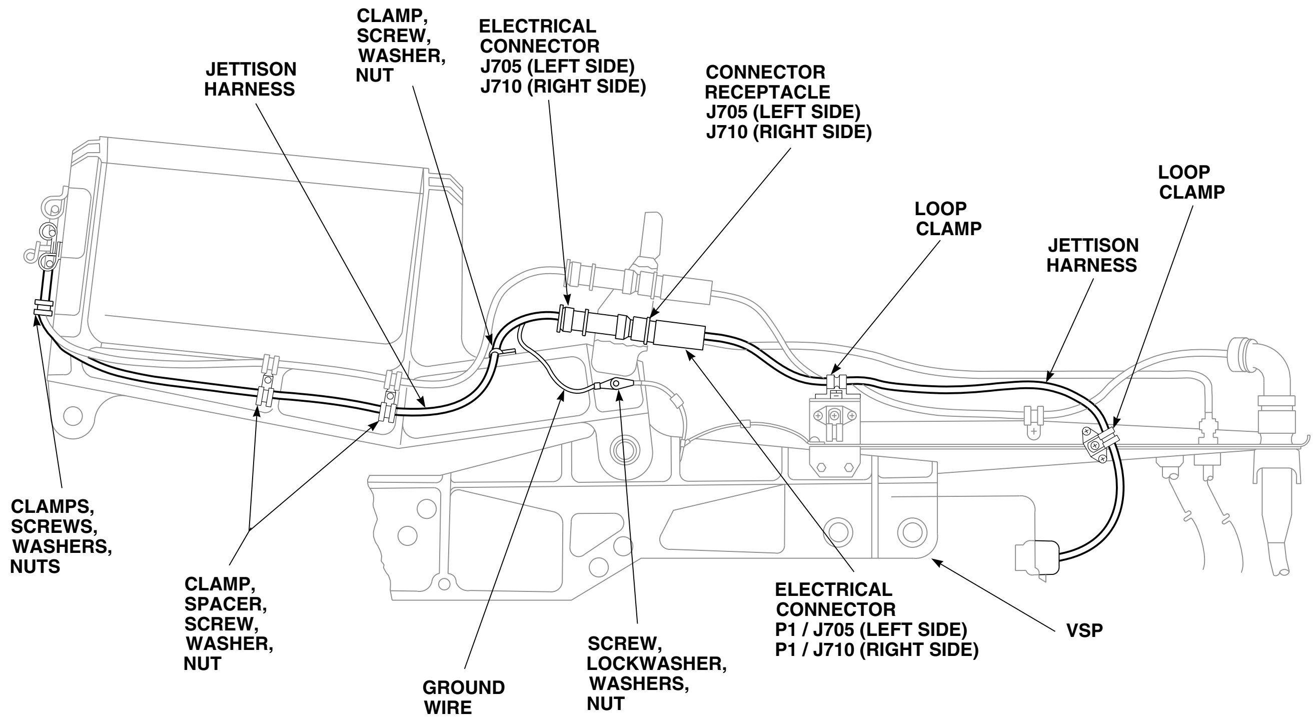
FIGURE 16-4-53-1. REPLACE HORIZONTAL STORES SUPPORT (HSS) JETTISON SYSTEM HARNESS (SHEET 1 OF 2)

(1) Remove screw, washers, and nut, and free ground wires.

(2) On left side, remove electrical connectors, J707 and P1/J707, from connector receptacle, J707, and P709, from either dummy receptacle or jettison rack, if installed. On right side, remove electrical connectors, J708 and P1/J708, from connector receptacle, J708, and P706, from either dummy receptacle or jettison rack, if installed.

(3) Install protective caps and plugs on electrical connectors.

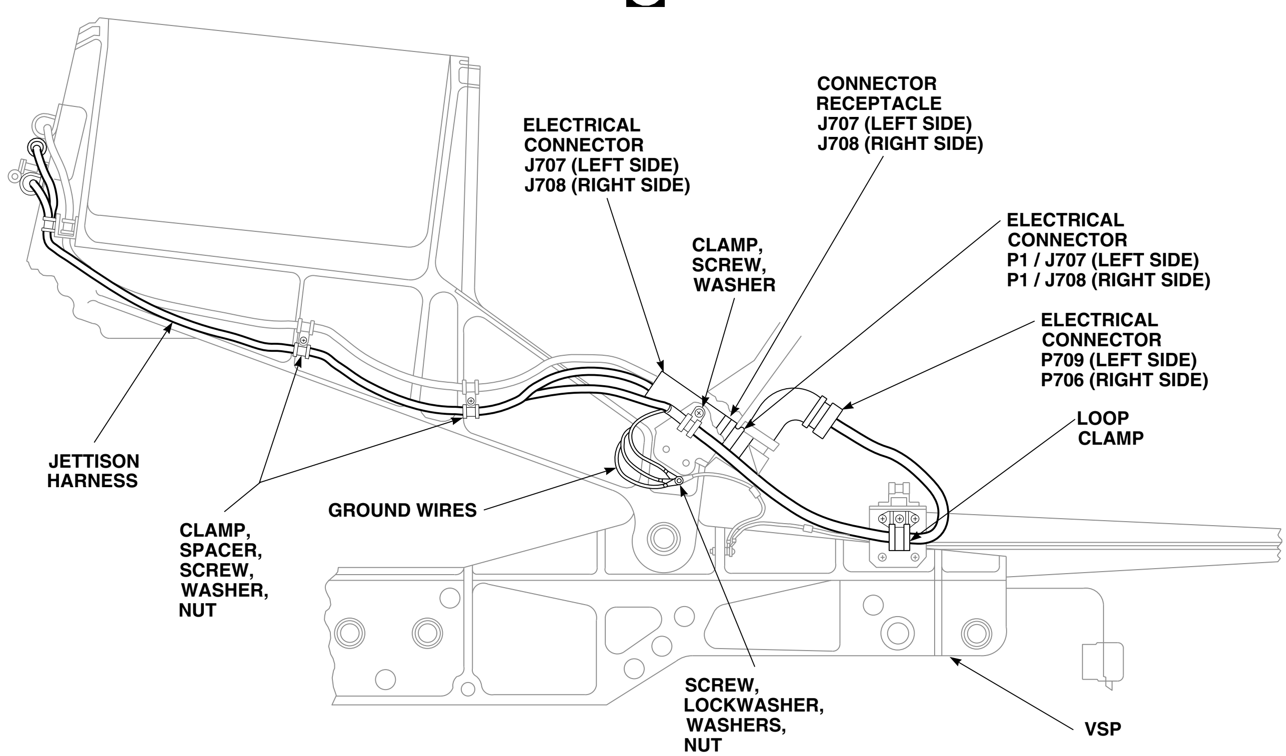
(4) Remove screws, washers, and nuts, and remove connector receptacle, J707 (left side), or J708 (right side), from VSP bracket.



OUTBOARD VERTICAL SUPPORT PYLON

B

C



INBOARD VERTICAL SUPPORT PYLON

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SA

FIGURE 16-4-53-1. REPLACE HORIZONTAL STORES SUPPORT (HSS) JETTISON SYSTEM HARNESS (SHEET 2 OF 2)

(5) Remove screws, washers, nuts, spacers, clamps, and loop clamp securing jettison harness to inboard VSP.

g. On HSS root fitting, remove electrical connectors, J678 and J680 (left side), or J677 and J679 (right side). Remove screws, washers, nuts, and electrical connectors (Figure 16-4-53-1, Sheet 1).

h. Install protective caps and plugs on electrical connectors.

i. Remove screws, washers, nuts, spacers, and clamps securing jettison harness to HSS (Figure 16-4-53-1, Sheet 1, Detail A).

16-4-53.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove protective caps and plugs from electrical connectors.

c. Inspect and treat electrical connectors (PARA 1-7-20).

NOTE

If helicopter is to be flown without ejector racks, BRU-22A/A, installed, remove jettison firing cable from helicopter.

d. On HSS root fitting, install electrical connectors, J678 and J680 (left side), or J677 and J679 (right side). Install screws, washers, and nuts securing connectors (Figure 16-4-53-1, Sheet 1).

e. Install jettison harness on inboard VSP as follows (Figure 16-4-53-1, Sheet 2, Detail C):

(1) Install connector receptacle, J707 (left side), or J708 (right side), on HSS bracket with screws, washers, and nuts.

(2) Remove protective caps and plugs from electrical connectors.

(3) Inspect and treat electrical connectors (PARA 1-7-20).

(4) Connect electrical connector, P709 (left side), or P706 (right side), to dummy connector, or jettison rack, if installed.

(5) Remove protective caps and plugs from electrical connectors.

(6) Inspect and treat electrical connectors (PARA 1-7-20).

(7) Connect electrical connector, P1/J707, on connector receptacle J707 and J1, to jettison rack as required on left side or connect electrical connector, P1/J708, on connector receptacle J708 and J1, to jettison rack as required on right side.

(8) Install harness ground wires with screw, lockwasher, washers, and nuts.

(9) Secure jettison harness to inboard VSP with clamps, screws, washers, nuts, spacers and loop clamp.

f. Install jettison harness on outboard VSP as follows (Figure 16-4-53-1, Sheet 2, Detail B):

(1) Install connector receptacle, J705, left side or J710, right side on HSS bracket with screws, washers, and nuts.

(2) Remove protective caps and plugs from electrical connectors.

(3) Inspect and treat electrical connectors (PARA 1-7-20).

(4) Connect electrical connectors, J705, and P1/J705, on connector receptacle, J705 (left side), or J710, and P1/J710, on connector receptacle, J710 (right side).

(5) Install harness ground wire on adapter with screw, lockwasher, washers, and nut.

(6) Secure jettison harness to outboard VSP with clamps, screws, washers, nuts, spacers and loop clamps.

g. Secure jettison harness to HSS with clamps, screws, washers, nuts, and spacers (Figure 16-4-53-1, Sheet 1, Detail A).

h. Install leading edge fairing on HSS (PARA 16-4-21).

i. Install inboard and outboard saddle assemblies (PARA 16-4-21).

j. Install front and rear vertical stores pylon fairings (PARA 16-4-21).

k. Perform operational check of ESSS jettison system (PARA 16-2-4).

16-4-54 ESSS EXTERNAL 230 GALLON FUEL TANKS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Lifting Sling
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-3-24
- PARA 1-3-25
- PARA 1-3-26
- PARA 1-6-16
- PARA 1-7-20
- PARA 1-7-35
- PARA 16-2-1
- PARA 16-2-2
- PARA 16-4-21
- PARA 16-4-35
- PARA 16-4-58
- PARA 16-4-59
- PARA 16-4-60

16-4-54.1 Replace ESSS External 230 Gallon Fuel Tanks.

NOTE

Replacement of all external stores support system (ESSS) external 230 gallon fuel tanks is the same.

16-4-54.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Make sure helicopter is properly grounded.

c. Remove ESSS vertical stores support (VSP) rear fairing (PARA 16-4-21).

d. Defuel ESSS external 230 gallon fuel tank (PARA 1-3-25).

NOTE

If helicopter is to be flown without fuel tanks, pull AUX FUEL QTY 115 VAC circuit breaker and NO. 2 XFER CONT 28 VDC circuit breaker.

e. If required, on mission readiness circuit breaker panel, pull AUX FUEL QTY 115 VAC circuit breaker and NO. 2 XFER CONT 28 VDC circuit breaker.

f. On upper console circuit breaker panel DC ESNTL BUS, pull ESSS JTSN OUTBD and ESSS JTSN INBD circuit breakers. On copilot's circuit breaker panel NO. 1 DC PRI BUS, pull ESSS JTSN INBD and ESSS JTSN OUTBD circuit breakers.

g. On upper console, make sure NO. 1 and NO. 2 generator switches and BATT switch are OFF.

h. Make sure jettison control panel switches are OFF.

i. Remove ejector rack, BRU 22A/A, explosive cartridge (PARA 16-4-35).

j. Loosen jamnut on sway brace screws and adjust screws upward (Figure 16-4-54-1, Sheet 1, Detail A).

k. Disconnect electrical connector, P1 from J1, on VSP fuel arm extension by pulling down on lanyards.

l. Install protective caps and plugs on electrical connectors.

WARNING

Injury to personnel and damage to equipment will result if fuel tanks are mishandled or dropped. Make sure four or more assistants are used when removing fuel tank.

CAUTION

Damage to fuel tank will result if lifted by tail cone fairing. Do not lift fuel tank by tail cone fairing.

m. Position two assistants at rear of fuel tank and two assistants with lifting sling on front end of tank.

n. Unlock ejector rack, BRU-22A/A, as follows:

(1) Set safety lever on ejector rack to UNLOCKED position (flag down) (Figure 16-4-54-1, Sheet 2, Detail E).

(2) Using manual release lever, open ejector rack hooks (Figure 16-4-54-1, Sheet 2, Detail B and Detail C).

o. Carefully lower fuel tank.

p. Remove ESSS external 230 gallon fuel tank electrical interface cable (PARA 16-4-58).

q. Remove ESSS external 230 gallon fuel tank air interface (PARA 16-4-60).

r. Remove ESSS external 230 gallon fuel tank fuel interface (PARA 16-4-59).

s. Install protective caps and plugs on exposed electrical, fuel, and air interface connections.

t. If required, preserve fuel tank (PARA 1-3-26).

16-4-54.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Make sure helicopter is properly grounded.

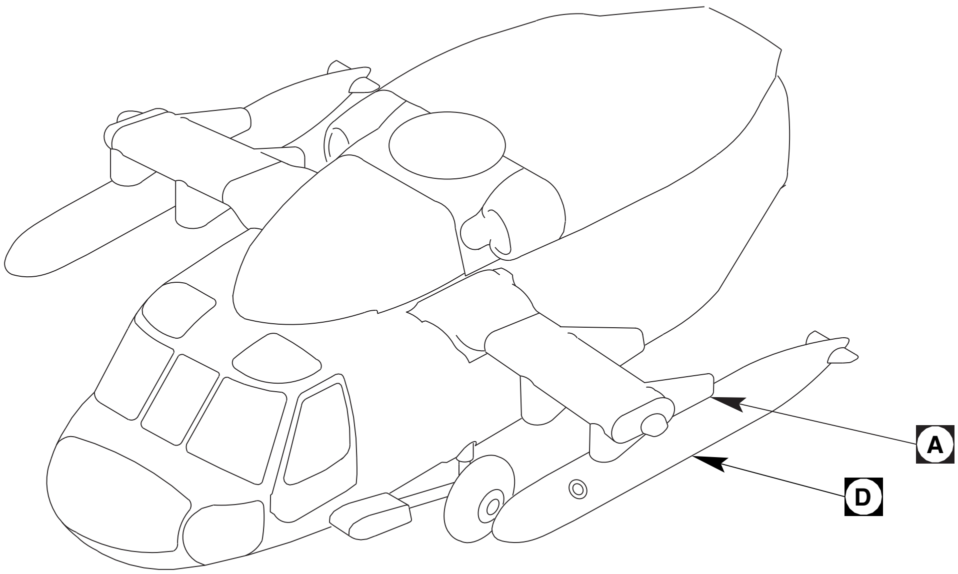
NOTE

Make sure fuel tanks are depreserved prior to installation.

c. If required, depreserve fuel tank (PARA 1-3-26).

d. Make sure fuel tank is empty.

e. On upper console circuit breaker panel DC ESNTL BUS, pull ESSS JTSN OUTBD and ESSS JTSN INBD circuit breakers. On copilot's circuit breaker panel NO. 1 DC PRI BUS, pull ESSS JTSN INBD and ESSS JTSN OUTBD circuit breakers.



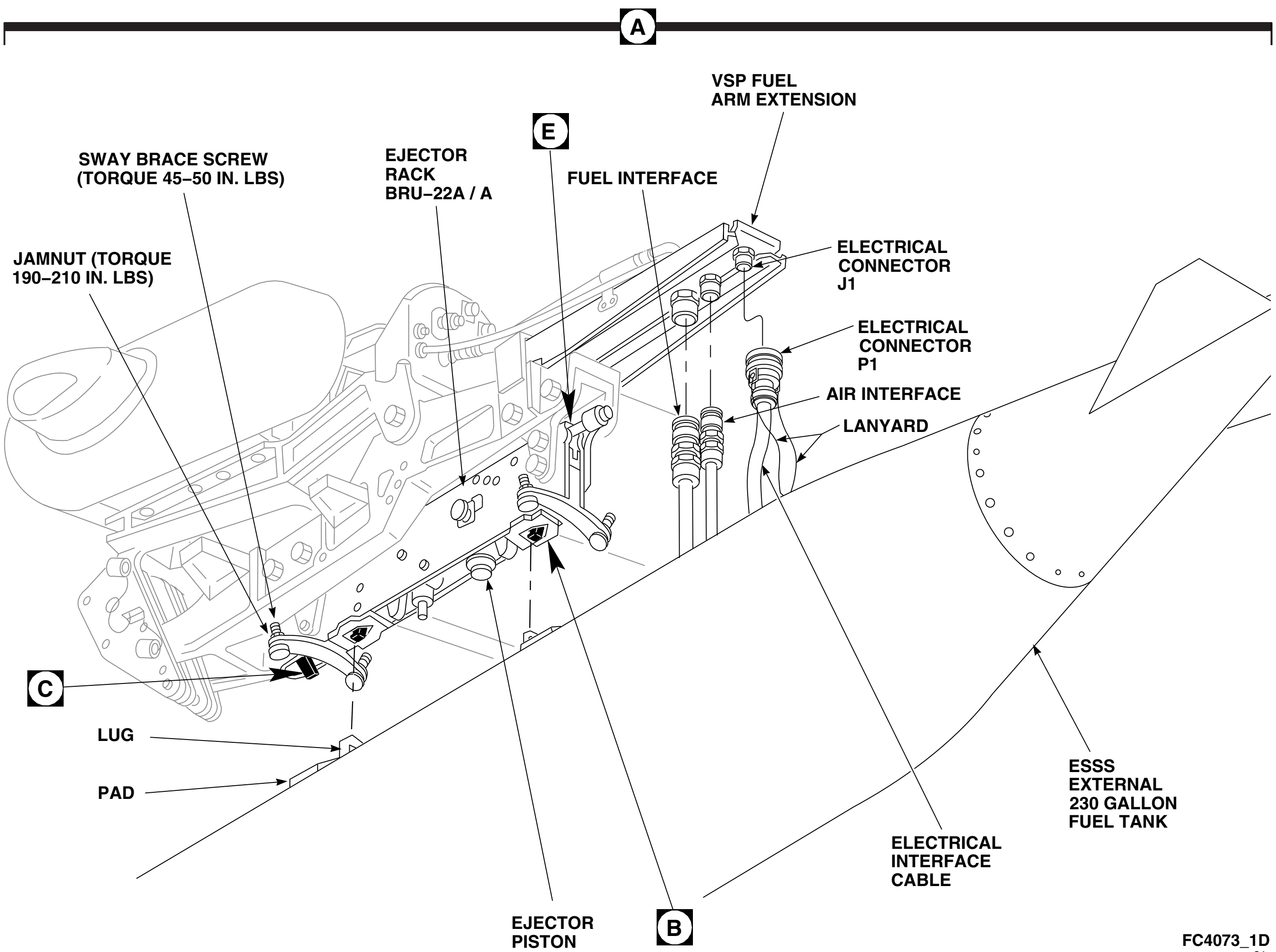


FIGURE 16-4-54-1. REPLACE ESSS EXTERNAL 230 GALLON FUEL TANKS (SHEET 1 OF 2)

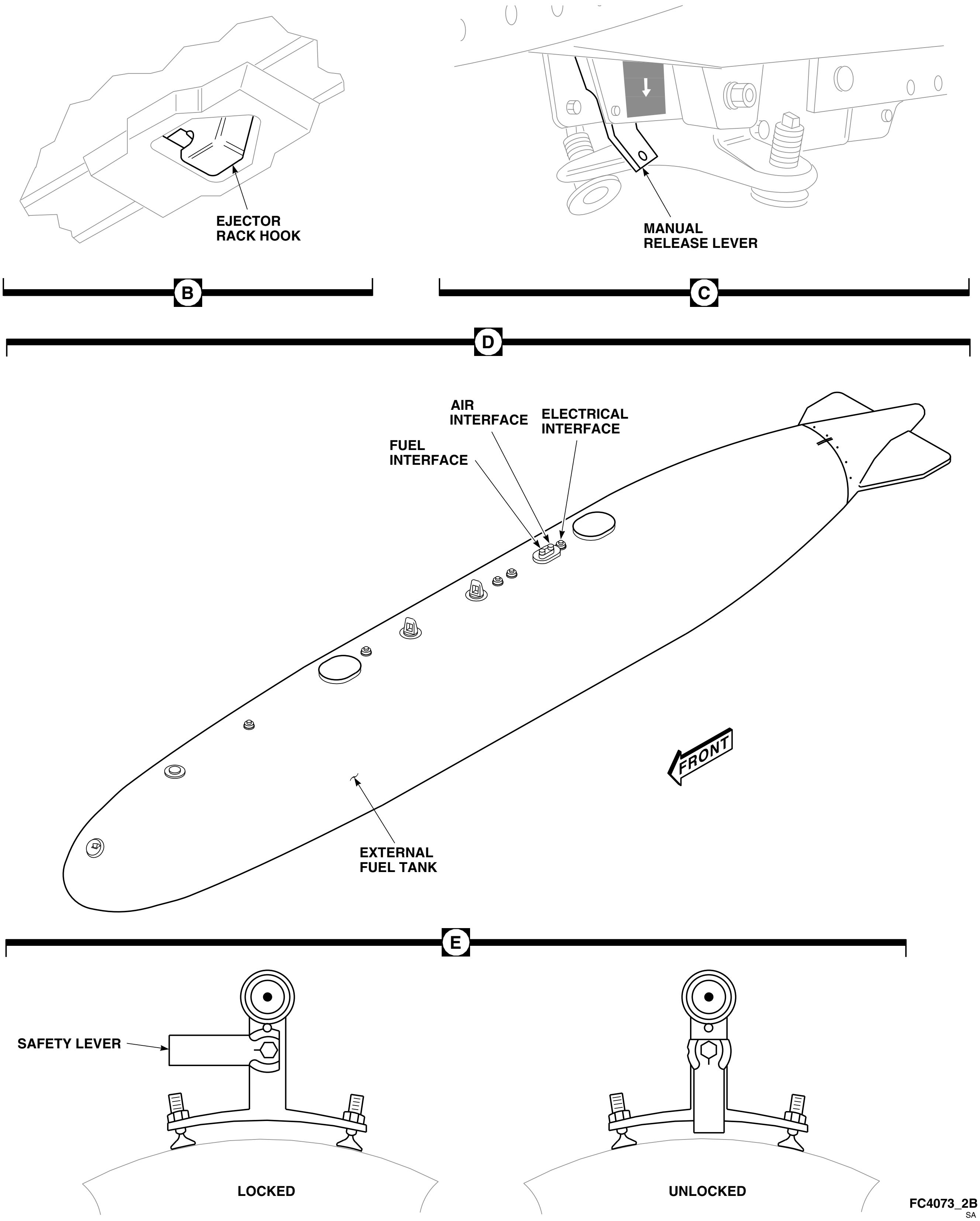


FIGURE 16-4-54-1. REPLACE ESSS EXTERNAL 230 GALLON FUEL TANKS (SHEET 2 OF 2)

NOTE

If helicopter was flown without fuel tanks, reset AUX FUEL QTY 115 VAC circuit breaker and NO. 2 XFER CONT 28 VDC circuit breaker.

f. If pulled, reset AUX FUEL QTY 115 VAC and NO. 2 XFER CONT 28 VDC circuit breakers on mission readiness circuit breaker panel.

g. On upper console, make sure NO. 1 and NO. 2 generator switches and BATT switch are OFF.

h. Make sure jettison control panel switches are OFF.

i. Remove ejector rack, BRU-22A/A, explosive cartridge (PARA 16-4-35).

j. Inspect ejector rack, BRU-22A/A (PARA 1-7-35).

k. Loosen jamnuts on sway brace screws and adjust screws upward (Figure 16-4-54-1, Sheet 1, Detail A).

l. Set safety lever on ejector rack to UNLOCKED position (flag down) (Figure 16-4-54-1, Sheet 2, Detail E).

m. Set ejector pistons full up (Figure 16-4-54-1, Sheet 1, Detail A).

n. Install ESSS external 230 gallon fuel tank fuel interface (PARA 16-4-59).

o. Install ESSS external 230 gallon fuel tank air interface (PARA 16-4-60).

p. Install ESSS external 230 gallon fuel tank electrical interface cable (PARA 16-4-58).

WARNING

Injury to personnel and damage to equipment will result if fuel tank is mishandled or dropped. Make sure four or more assistants are used when installing fuel tank.

q. Position two assistants at rear of fuel tank (one assistant by electrical, fuel, and air interfaces) and two assistants with lifting sling on front end of tank (Figure 16-4-54-1, Sheet 2, Detail D).

WARNING

Injury to personnel and damage to equipment will result if ejector rack hooks and lugs are not fully engaged and locked. Fuel tank will disconnect from ejector rack. Make sure ejector rack hooks are in locked position prior to releasing tank.

CAUTION

Damage to fuel tank will result if lifted by tail cone fairing. Do not lift fuel tank by tail cone fairing.

r. Carefully raise fuel tank up and connect fuel and air interfaces with fittings on VSP fuel arm extension. Continue lifting until lugs on tank are fully inserted in ejector rack hook slots (Figure 16-4-54-1, Sheet 1, Detail A and Sheet 2, Detail B).

s. Set safety lever on ejector rack to LOCKED position (yellow and black decal visible) (Figure 16-4-54-1, Sheet 2, Detail E).

t. With assistants still holding fuel tank, rock tank to ensure positive latching of hooks and lugs. If positive latching has not occurred, adjust suspension lugs as required.

u. Center fuel tank on ejector rack and thread front sway brace screws down evenly until screws contact pads on fuel tank (Figure 16-4-54-1, Sheet 1, Detail A).

v. Thread rear sway brace screws down evenly until screws contact pads on fuel tank.

CAUTION

Damage to equipment will result if sway brace screws are over torqued. Ejector rack may not function properly. Do not over torque sway brace screws.

w. **TORQUE SWAY BRACE SCREWS TO 45 - 50 INCH-POUNDS IN THE FOLLOWING SEQUENCE: FRONT INBOARD, REAR OUTBOARD, FRONT OUTBOARD, REAR INBOARD.**

x. Remove protective caps and plugs from electrical connectors.

y. Inspect and treat electrical connectors (PARA 1-7-20).

z. Connect electrical connector, P1 to J1, on VSP fuel arm extension.

aa. Refuel ESSS external 230 gallon fuel tank (PARA 1-3-24).

ab. **RETORQUE SWAY BRACE SCREWS TO 45 - 50 INCH-POUNDS IN THE FOLLOWING SEQUENCE: FRONT INBOARD, REAR OUTBOARD, FRONT OUTBOARD, REAR INBOARD.**

ac. **TORQUE JAMNUTS TO 190 - 210 INCH-POUNDS.**

ad. Pull ejector pistons down until they contact fuel tank.

ae. On upper console circuit breaker panel DC ESNTL BUS, push in ESSS JTSN OUTBD and ESSS JTSN INBD circuit breakers. On copilot's circuit breaker panel NO. 1 DC PRI BUS, push in ESSS JTSN OUTBD and ESSS JTSN INBD circuit breakers.

af. If only one tank has been replaced, perform operational check on either outboard or inboard fuel tank (PARA 16-2-1 or PARA 16-2-2). Check for leaks.

ag. Install ejector rack, BRU-22A/A, explosive cartridge (PARA 16-4-35).

ah. Install ESSS vertical stores support (VSP) rear fairing (PARA 16-4-21).

16-4-55 ESSS EXTERNAL 230 GALLON FUEL TANK DRAIN VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Petrolatum, Item 234, Appendix D

References

- Appendix D
- PARA 1-3-25

16-4-55.1 Replace ESSS External 230 Gallon Fuel Tank Drain Valve.

NOTE

Replacement of drain valve on all external stores support system (ESSS) external 230 gallon fuel tanks is the same.

16-4-55.1.1 Remove.

a. Defuel ESSS external 230 gallon fuel tank (PARA 1-3-25).

b. Loosen drain valve by turning counterclockwise (Figure 16-4-55-1, Detail A).

c. Remove drain valve and packing. Discard packing.

16-4-55.1.2 Install.

a. Lubricate packing with petrolatum, Item 234, Appendix D. Install packing on drain valve body (Figure 16-4-55-1, Detail A).

b. Install drain valve. **TIGHTEN VALVE.**

c. Make sure drain poppet is flush with surface of drain valve.

16-4-55.2 Repair ESSS External 230 Gallon Fuel Tank Drain Valve.

16-4-55.2.1 Remove.

NOTE

Due to similar method used when sampling fuel, care must be taken not to push on drain poppet during removal.

Turn drain poppet counterclockwise until it pops out (Figure 16-4-55-2, Detail A). Remove and discard packing.

16-4-55.2.2 Install.

a. Lubricate packing with petrolatum, Item 234, Appendix D. Install packing on drain poppet (Figure 16-4-55-2, Detail A).

b. Push drain poppet carefully into drain valve until flush with surface and turn clockwise to seal.

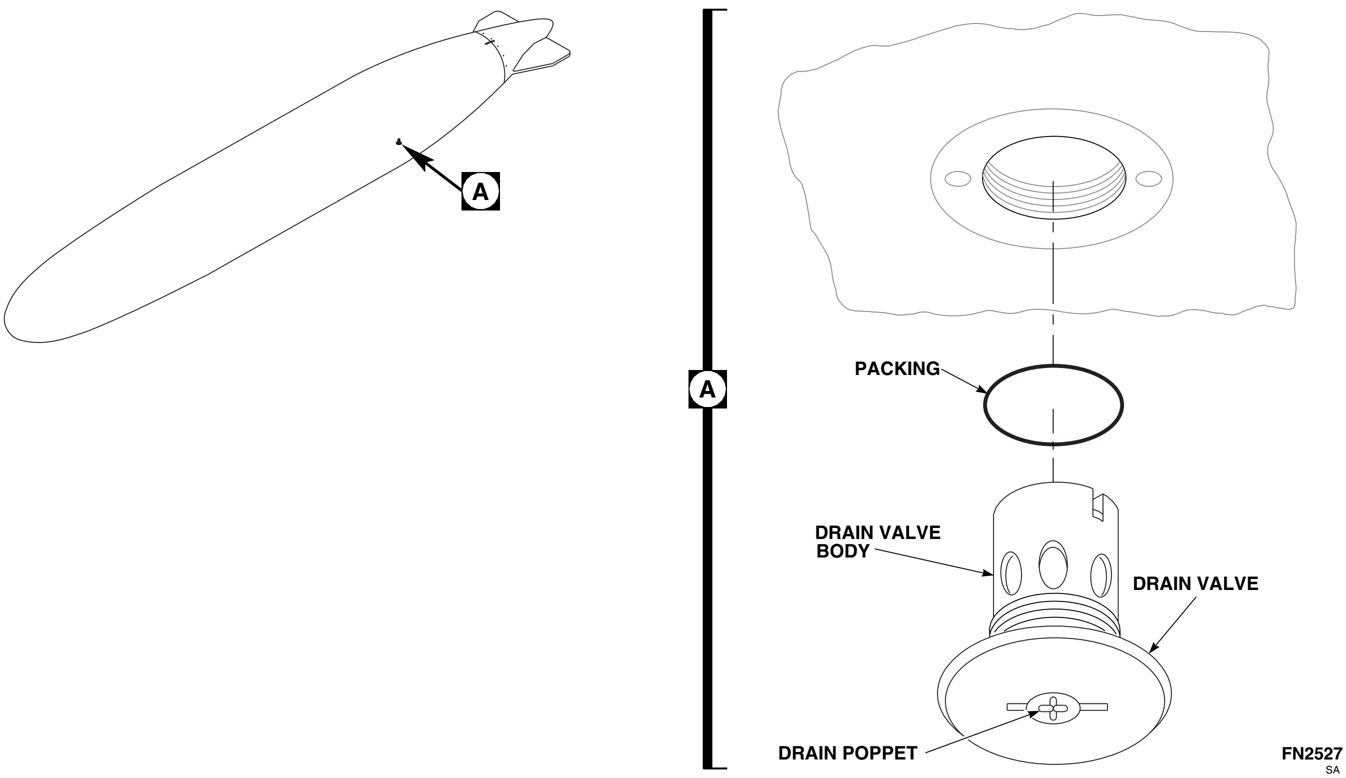


FIGURE 16-4-55-1. REPLACE ESSS EXTERNAL 230 GALLON FUEL TANK DRAIN VALVE

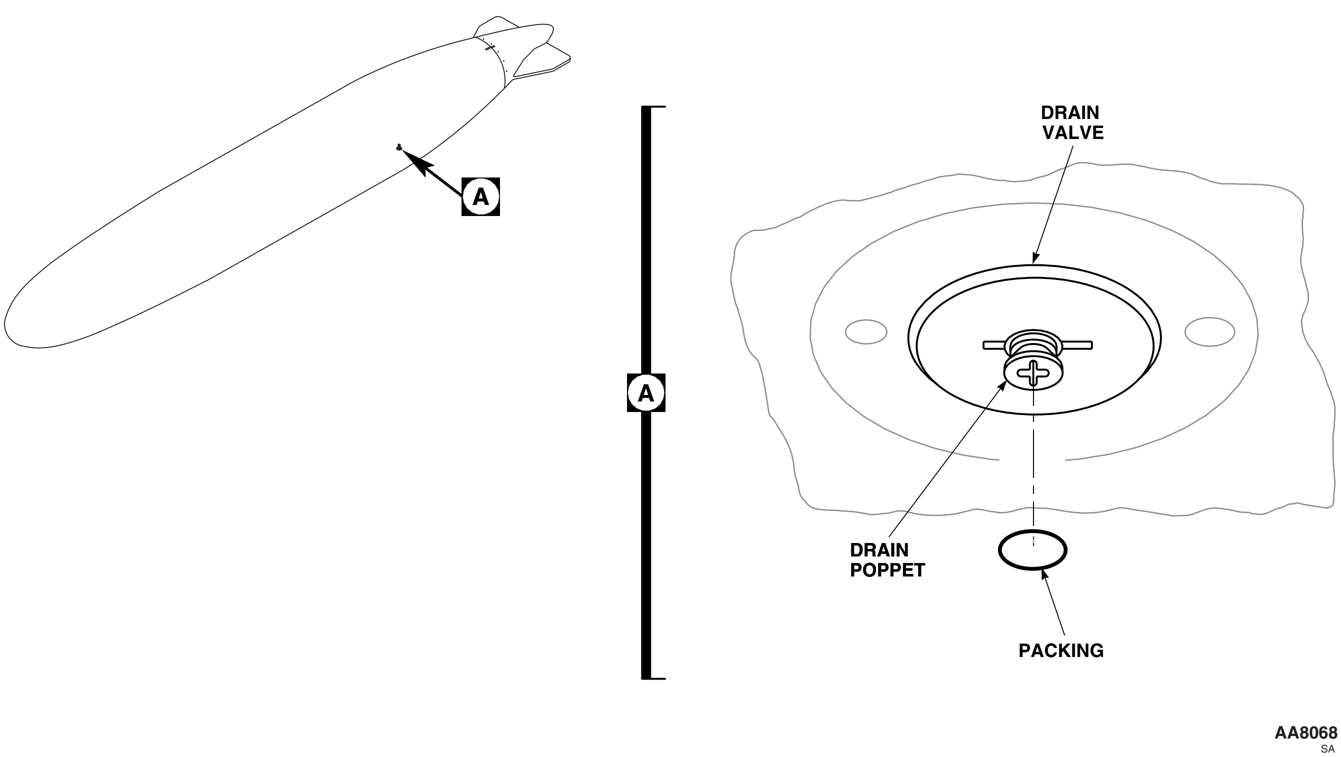


FIGURE 16-4-55-2. REPAIR ESSS EXTERNAL 230 GALLON FUEL TANK DRAIN VALVE

16-4-56 ESSS EXTERNAL 230 GALLON FUEL TANK GROUNDING JACK.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

16-4-56.1 Replace ESSS External 230 Gallon Fuel Tank Grounding Jack.

NOTE

Replacement of grounding jack on all external stores support system (ESSS) external 230 gallon fuel tanks is the same.

16-4-56.1.1 Remove.

a. Loosen grounding jack housing by turning counterclockwise (Figure 16-4-56-1, Detail A).

b. Remove grounding jack housing from ESSS external 230 gallon fuel tank.

c. Remove nut and star washer from top of grounding jack housing, and remove grounding jack and star washer from bottom of housing.

16-4-56.1.2 Install.

a. Discard flat washer from new grounding jack housing.

b. Cut solder point off of new star washer.

c. Slide star washer over new grounding jack (Figure 16-4-56-1, Detail A).

d. Slide grounding jack into grounding jack housing and secure with new star washer and nut.

e. Install housing in external stores support system external 230 gallon fuel tank. **TIGHTEN HOUSING.**

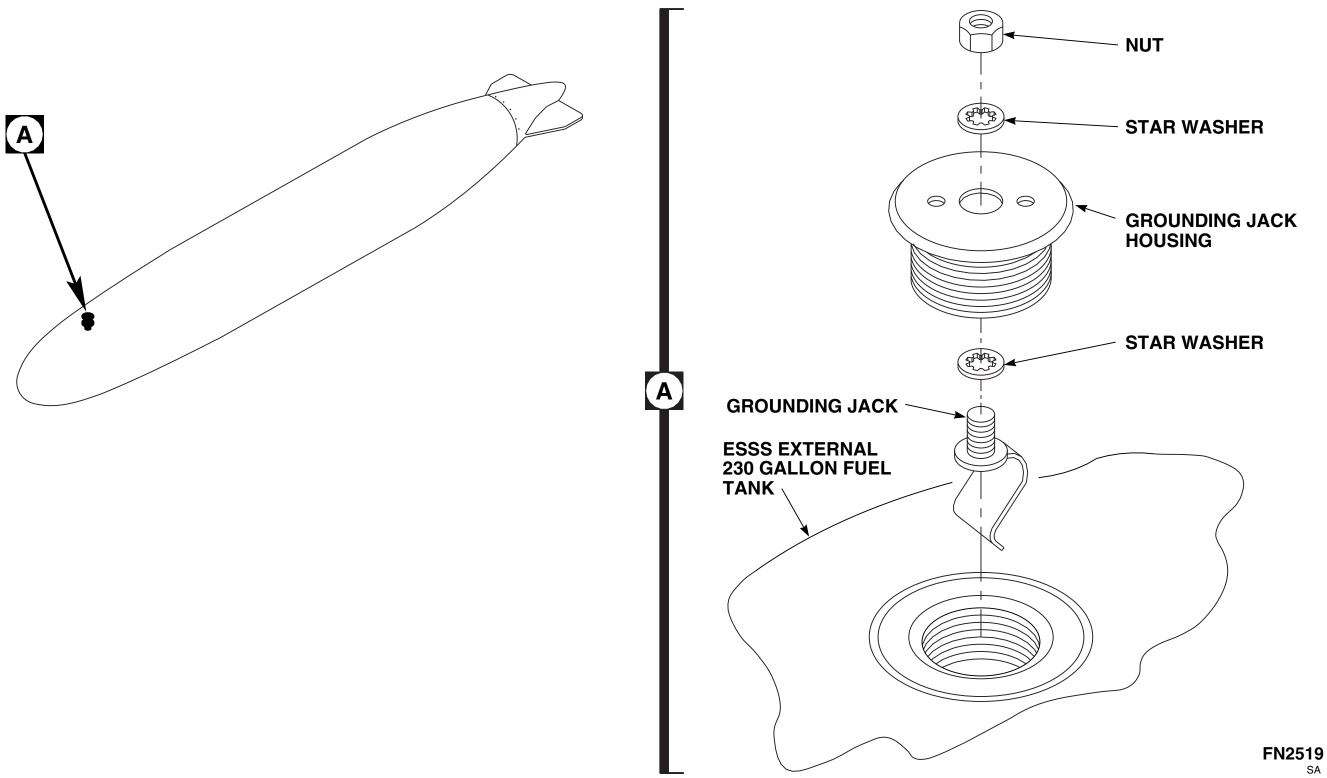


FIGURE 16-4-56-1. REPLACE ESSS EXTERNAL 230 GALLON FUEL TANK GROUNDING JACK

16-4-57 ESSS EXTERNAL 230 GALLON FUEL TANK FILLER CAP.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 1-6-16

16-4-57.1 Replace ESSS External 230 Gallon Fuel Tank Filler Cap.

NOTE

Replacement of filler cap on all external stores support system (ESSS) external 230 gallon fuel tanks is the same.

16-4-57.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

CAUTION

Injury to personnel will result if tank pressure is not removed prior to removing filler cap. Before turning thumb lock, listen for escaping air and make sure no pressure remains in tank.

b. Release thumb lock on top of filler cap by pulling up on thumb lock (Figure 16-4-57-1, Detail A).

c. Turn thumb lock counterclockwise until it stops.

d. Remove filler cap.

e. Remove cotter pin and disconnect lanyard from filler cap.

f. If replacing lanyard, remove lanyard from lanyard tie on tank.

16-4-57.1.2 Install.

a. If lanyard was removed, install lanyard on lanyard tie on ESSS external 230 gallon fuel tank (Figure 16-4-57-1, Detail A).

b. Open thumb lock on filler cap and turn counterclockwise until it stops.

c. Connect lanyard to filler cap with cotter pin.

d. Place filler cap in housing and align forward-point arrow.

e. Turn thumb lock clockwise to lock cap.

f. Snap thumb lock closed to seal cap.

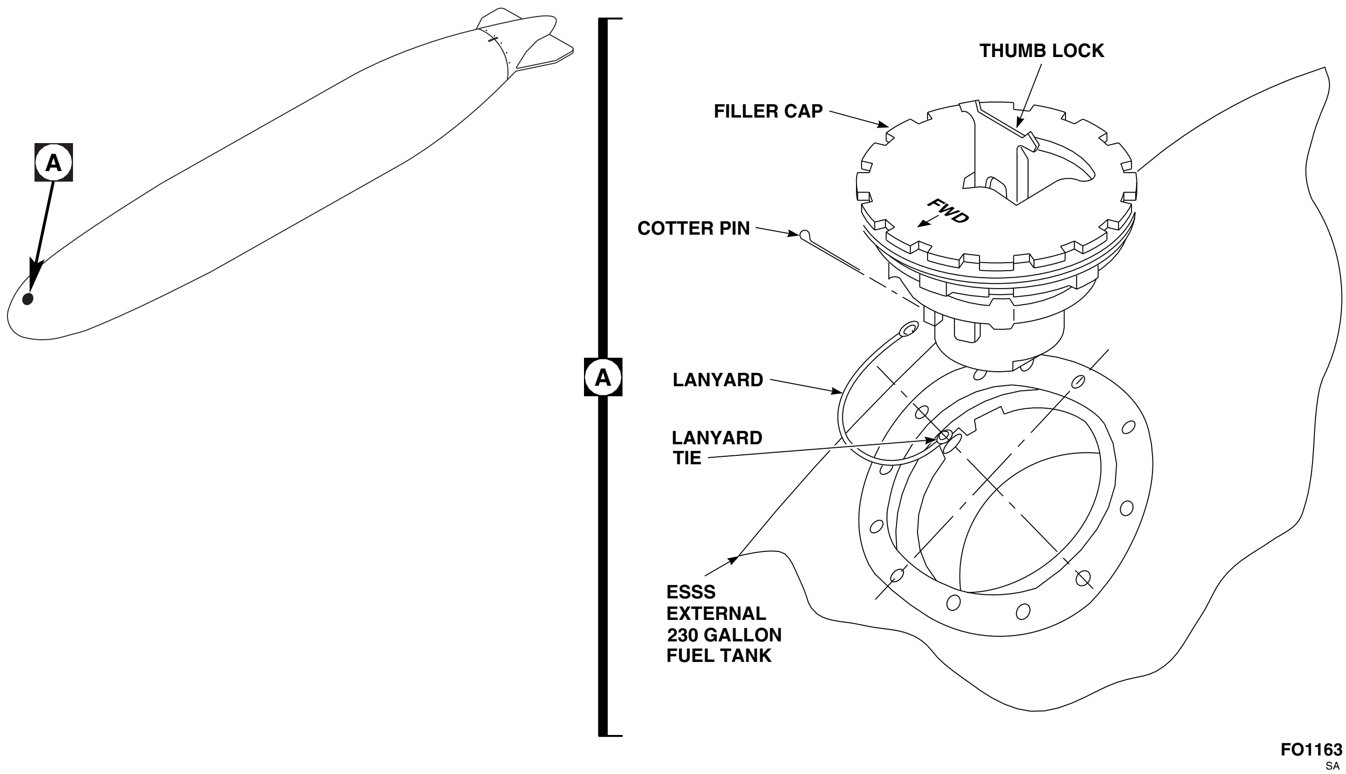


FIGURE 16-4-57-1. REPLACE ESSS EXTERNAL 230 GALLON FUEL TANK FILLER CAP

16-4-58 ESSS EXTERNAL 230 GALLON FUEL TANK ELECTRICAL INTERFACE CABLE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Lockwire, Item 194, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 16-4-35

16-4-58.1 Replace ESSS External 230 Gallon Fuel Tank Electrical Interface Cable.

NOTE

Replacement of electrical interface cable on all external stores support system (ESSS) external 230 gallon fuel tanks is the same.

16-4-58.1.1 Remove.

a. Remove ejector rack, BRU-22A/A, explosive cartridge (PARA 16-4-35).

b. Disconnect electrical connector, P1, on electrical interface cable from J1, on vertical support pylon fuel arm extension fitting by pulling down on lanyards (Figure 16-4-58-1, Detail A).

c. Remove capscrew(s) and retaining clip(s) from insert(s) and release lanyard(s) (Figure 16-4-58-1, Detail B).

d. W/O AUX FUEL QTY ▶ Disconnect electrical connector on electrical interface cable from electrical connector on ESSS external 230 gallon fuel tank by pushing down and turning counterclockwise.◀

e. AUX FUEL QTY ▶ Disconnect electrical connectors, P2, from electrical connector on external fuel tank, and P3, from electrical connector on rear access door by pushing down and turning counterclockwise.◀

f. Install protective caps and plugs on electrical connectors.

16-4-58.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove protective caps and plugs from electrical connectors.

c. Inspect and treat electrical connectors (PARA 1-7-20).

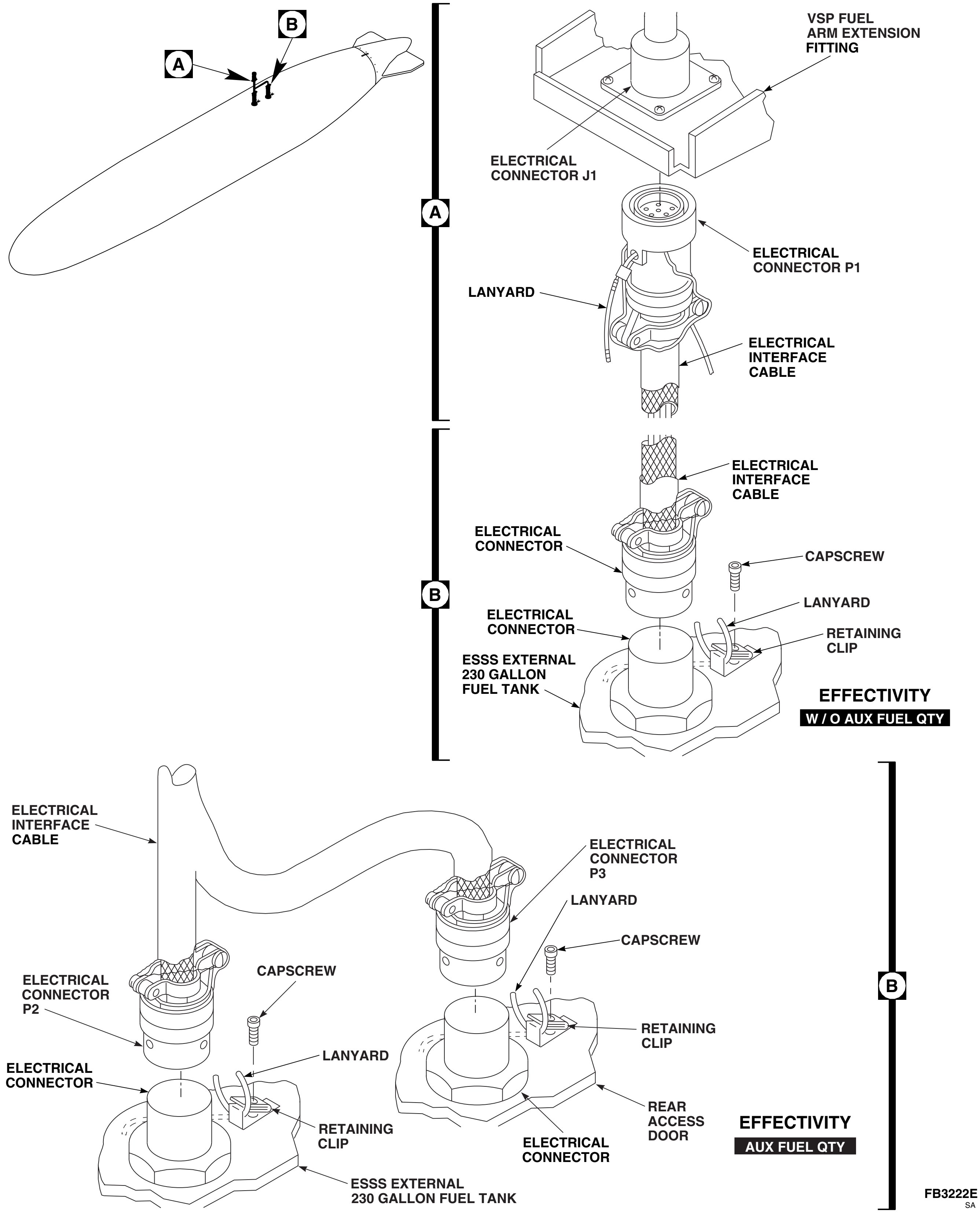


FIGURE 16-4-58-1. REPLACE ESSS EXTERNAL 230 GALLON FUEL TANK ELECTRICAL INTERFACE CABLE

d. **W/O AUX FUEL QTY** Align new electrical interface cable connector with electrical connector on external fuel tank, push down, and turn clockwise until it locks in place (Figure 16-4-58-1, Detail B).

e. **AUX FUEL QTY** Align new electrical cable interface connectors, P2, with electrical connector on external fuel tank, and P3, with electrical connector on rear access door, push down, and turn clockwise until it locks in place (Figure 16-4-58-1, Detail B).

f. Align keyway of electrical connector, P1, on electrical interface cable with electrical connector, J1, on vertical support pylon fuel arm extension fitting, and push up until it locks into place (Figure 16-4-58-1, Detail A).

g. Slip lanyard(s) into retaining clip(s) (Figure 16-4-58-1, Detail B).

h. Secure retaining clip(s) to external fuel tank with capscrew(s).

i. Lockwire, Item 194, Appendix D, retaining clip(s) to jamnut(s).

j. Install ejector rack, BRU-22A/A, explosive cartridge (PARA 16-4-35).

16-4-59 ESSS EXTERNAL 230 GALLON FUEL TANK FUEL INTERFACE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Petrolatum, Item 234, Appendix D

References

- Appendix D
- PARA 16-4-54

16-4-59.1 Replace ESSS External 230 Gallon Fuel Tank Fuel Interface.

NOTE

Replacement of fuel interfaces on all external stores support system (ESSS) external 230 gallon fuel tanks is the same.

16-4-59.1.1 Remove.

a. Remove ESSS external 230 gallon fuel tank (PARA 16-4-54).

b. Turn upper half of fuel interface counterclockwise and remove from vertical support pylon fuel arm extension fuel fitting (Figure 16-4-59-1, Detail A).

c. Remove washer and packing from fuel fitting. Discard packing.

d. Turn lower half of fuel interface counterclockwise and remove from fitting on ESSS external 230 gallon fuel tank.

e. Remove packings from upper and lower fuel interface halves. Discard packings.

16-4-59.1.2 Install.

a. Install washer onto fuel fitting (Figure 16-4-59-1, Detail A).

b. Lubricate packings with petrolatum, Item 234, Appendix D. Install packing onto vertical support pylon (VSP) fuel arm extension fuel fitting.

c. Install packings onto upper and lower halves of fuel interface.

d. Insert upper half of fuel interface in VSP fuel arm extension fuel fitting and turn clockwise to secure.

e. Insert lower half of fuel interface in ESSS external 230 gallon fuel tank fitting and turn clockwise to secure.

f. **TIGHTEN FUEL INTERFACE HALVES.**

g. Make sure area is clean and free of foreign material.

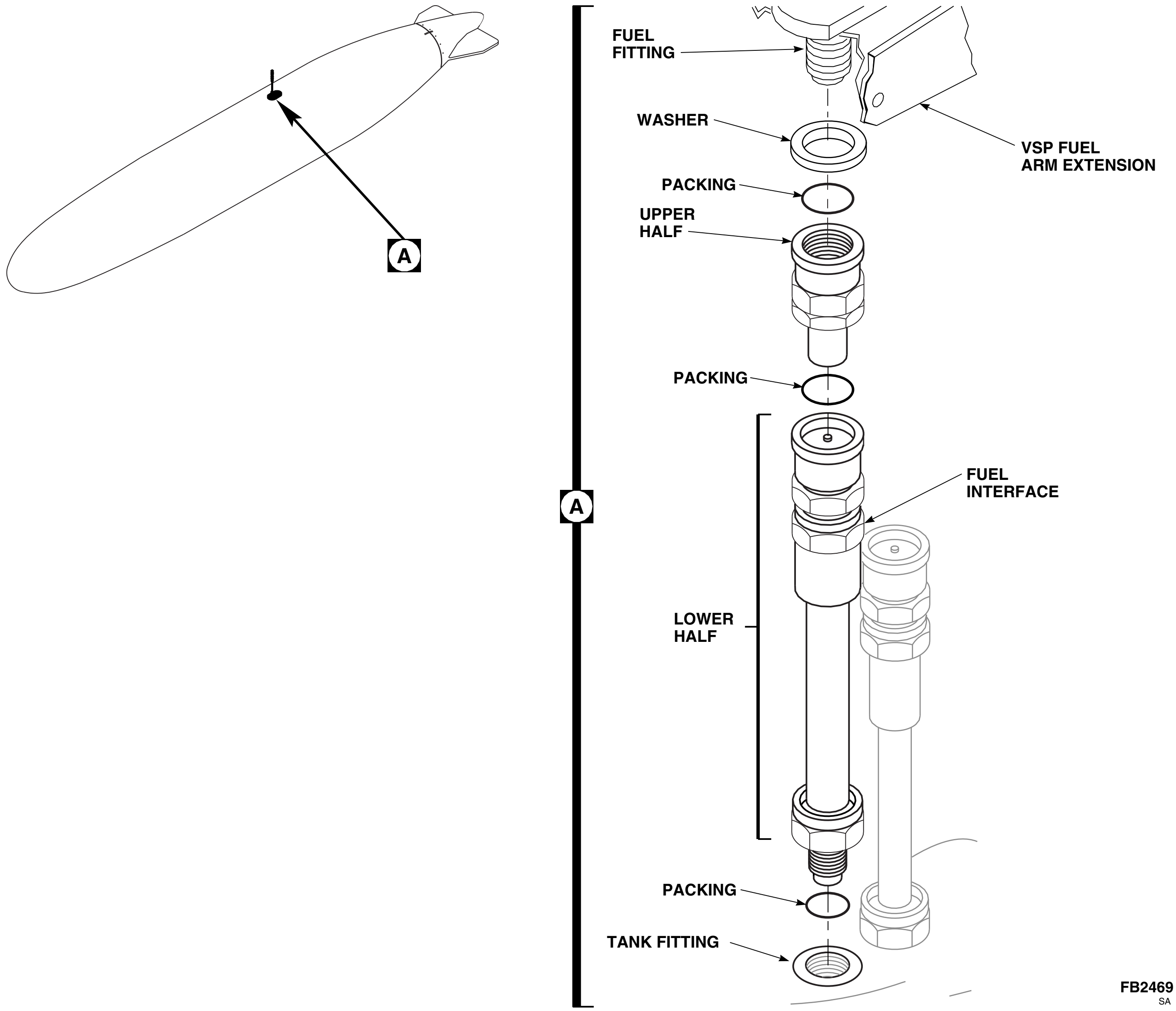


FIGURE 16-4-59-1. REPLACE ESSS EXTERNAL 230 GALLON FUEL TANK FUEL INTERFACE

h. Install ESSS external 230 gallon fuel tank (PARA 16-4-54).

16-4-60 ESSS EXTERNAL 230 GALLON FUEL TANK AIR INTERFACE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Petrolatum, Item 234, Appendix D

References

- Appendix D
- PARA 16-4-54

16-4-60.1 Replace ESSS External 230 Gallon Fuel Tank Air Interface.

NOTE

Replacement of air interfaces on all external stores support system (ESSS) external 230 gallon fuel tanks is the same.

16-4-60.1.1 Remove.

a. Remove ESSS external 230 gallon fuel tank (PARA 16-4-54).

b. Turn upper half of air interface counterclockwise and remove from vertical support pylon fuel arm extension air fitting (Figure 16-4-60-1, Detail A).

c. Remove packing and nut from air fitting. Discard packings.

d. Turn lower half of air interface counterclockwise and remove from fitting on ESSS external 230 gallon fuel tank.

e. Remove packings from upper and lower air interface halves. Discard packings.

16-4-60.1.2 Install.

a. Install nut onto air fitting (Figure 16-4-60-1, Detail A). **TIGHTEN NUT HANDTIGHT.**

b. Lubricate packings with petrolatum, Item 234, Appendix D. Install packing onto vertical support pylon (VSP) fuel arm extension air fitting.

c. Install packings onto upper and lower halves of air interface.

d. Insert upper half of air interface into VSP fuel arm extension air fitting and turn clockwise to secure.

e. Insert lower half of air interface in ESSS external 230 gallon fuel tank fitting and turn clockwise to secure.

f. **TIGHTEN AIR INTERFACE HALVES.**

g. Make sure area is clean and free of foreign material.

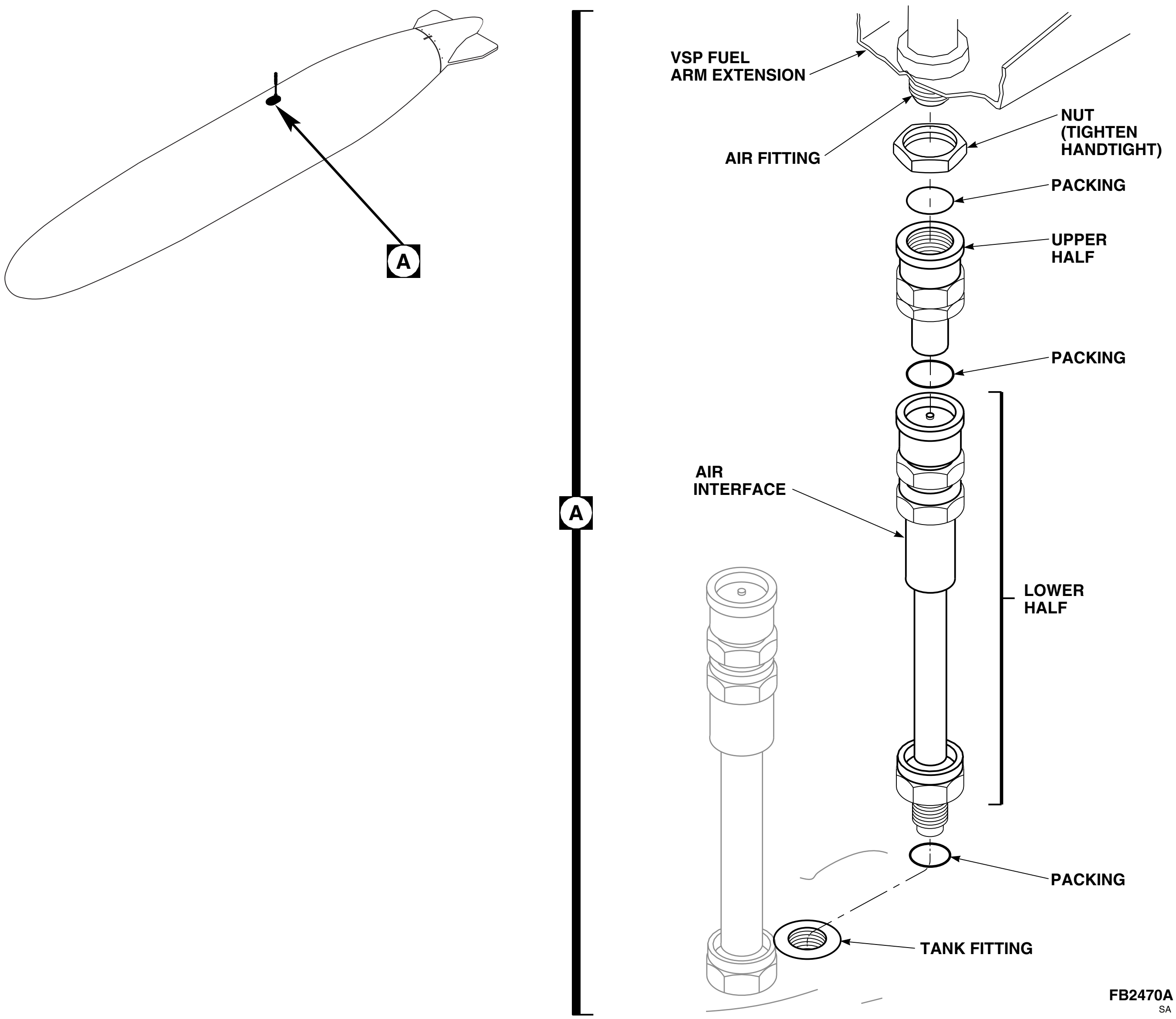


FIGURE 16-4-60-1. REPLACE ESSS EXTERNAL 230 GALLON FUEL TANK AIR INTERFACE

h. Install ESSS external 230 gallon fuel tank (PARA 16-4-54).

16-4-61 ESSS EXTERNAL 230 GALLON FUEL TANK SUSPENSION LUG.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Antiseize Compound, Item 74, Appendix D

References

- Appendix D
- PARA 16-4-54

16-4-61.1 Replace ESSS External 230 Gallon Fuel Tank Suspension Lug.

NOTE

Replacement of suspension lugs on all external stores support system (ESSS) external 230 gallon fuel tanks is the same.

16-4-61.1.1 Remove.

a. Remove ESSS external 230 gallon fuel tank (PARA 16-4-54).

b. Turn suspension lug counterclockwise and remove from tank (Figure 16-4-61-1, Detail A).

16-4-61.1.2 Install.

a. Coat threads of suspension lug with antiseize compound, Item 74, Appendix D.

b. Insert suspension lug in bushing opening and turn clockwise until it stops (Figure 16-4-61-1, Detail A).

c. Back lug counterclockwise until lug eyelet is aligned with tank centerline.

d. Install ESSS external 230 gallon fuel tank (PARA 16-4-54).

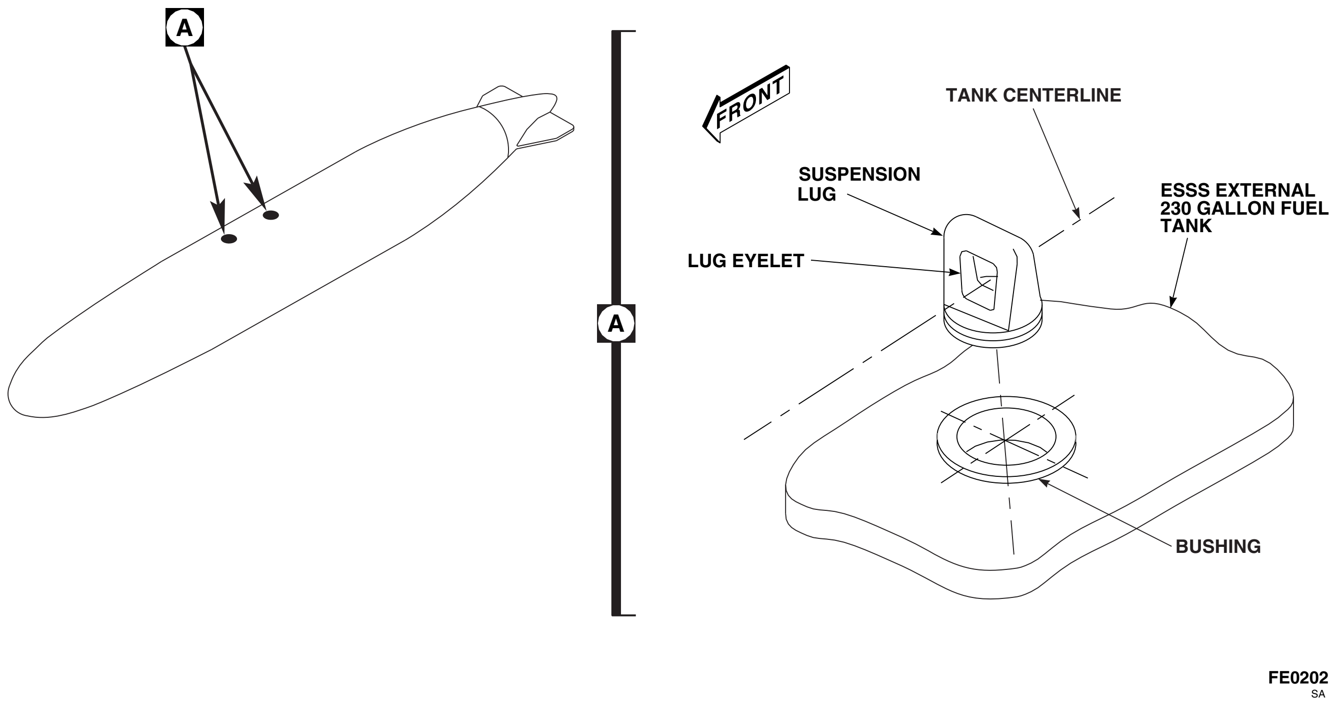


FIGURE 16-4-61-1. REPLACE ESSS EXTERNAL 230 GALLON FUEL TANK SUSPENSION LUG

16-4-62 ESSS EXTERNAL 230 GALLON FUEL TANK STABILIZER FIN.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 16-4-54
- PARA 16-4-71

16-4-62.1 Replace ESSS External 230 Gallon Fuel Tank Stabilizer Fin.

NOTE

Replacement of stabilizer fin on all external stores support system (ESSS) external 230 gallon fuel tanks is the same.

16-4-62.1.1 Remove.

a. Remove screws securing stabilizer fin (Figure 16-4-62-1, Detail A).

b. Slide stabilizer fin off rear of ESSS external 230 gallon fuel tank.

16-4-62.1.2 Inspect.

a. Inspect stabilizer fin mounting area for damaged mounting inserts (PARA 16-4-71). Proceed as follows:

(1) If damage exceeds limits, replace ESSS external 230 gallon fuel tank (PARA 16-4-54).

(2) If damage does not exceed limits, repair mounting inserts using class VII-A repair (PARA 16-4-71).

16-4-62.1.3 Install.

a. Slide stabilizer fin assembly onto rear of ESSS external 230 gallon fuel tank (Figure 16-4-62-1, Detail A).

b. Position stabilizer fin to alignment mark.

c. **INSTALL AND TIGHTEN SCREWS.**

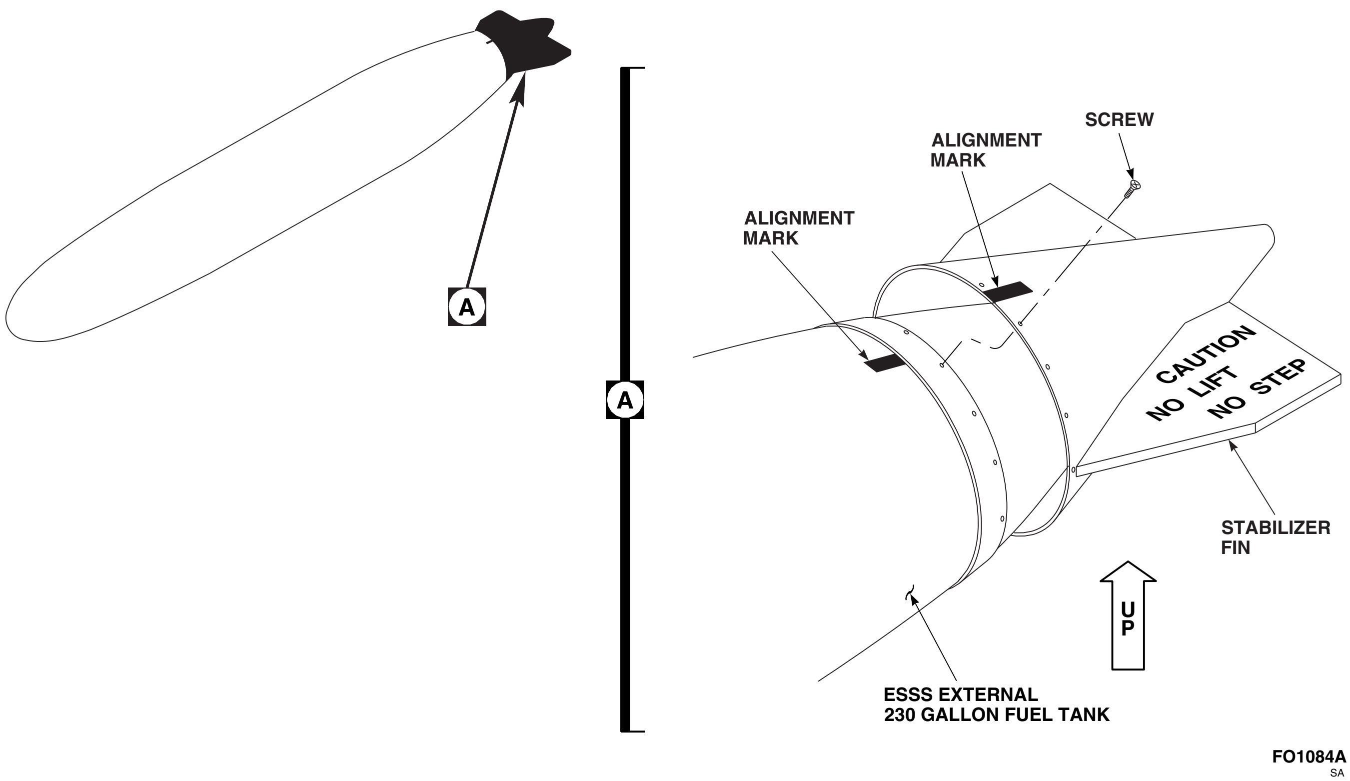


FIGURE 16-4-62-1. REPLACE ESSS EXTERNAL 230 GALLON FUEL TANK STABILIZER FIN

16-4-63 ESSS EXTERNAL 230 GALLON FUEL TANK ACCESS DOORS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Protective Cap
- Sealing Compound, Item 273, Appendix D
- Tags

References

- Appendix D
- PARA 1-7-20
- PARA 16-4-54
- PARA 16-4-64
- PARA 16-4-70

16-4-63.1 Replace ESSS External 230 Gallon Fuel Tank Access Doors.

NOTE

Replacement of front and rear external stores support system (ESSS) external 230 gallon fuel tank access doors is the same, except as noted.

16-4-63.1.1 Remove.

a. Remove ESSS external 230 gallon fuel tank (PARA 16-4-54).

b. Loosen, but do not remove, screws at perimeter of access door (Figure 16-4-63-1, Detail A).

c. Remove one screw and thread into hole in center of access door.

d. While holding onto screw installed in center of access door, remove remaining screws from perimeter.

e. **W/O AUX FUEL QTY** Lower door into tank, rotate as necessary, and remove from tank.

f. **AUX FUEL QTY** Remove access door as follows:

(1) Lower access door, with probe attached, into tank.

(2) Tag electrical wires. Remove screws securing electrical wires to probe.

(3) For rear access door, remove locknut securing electrical connector to access door.

(4) Install protective cap on electrical connector.

(5) Carefully rotate access door with probe, as necessary, to remove from tank.

(6) Remove gasket from top of access door.

(7) Remove screw from access door.

g. AUX FUEL QTY Remove ESSS external 230 gallon fuel tank probe (PARA 16-4-64).

16-4-63.1.2 Install.

a. AUX FUEL QTY Install ESSS external 230 gallon fuel tank probe (PARA 16-4-64).

b. Prior to installing fuel door, make sure area is clean and free of foreign material.

c. Coat both sides of gaskets with thin layer of sealing compound, Item 273, Appendix D.

d. Install gasket on access door with smooth side against door (Figure 16-4-63-1, Detail A).

e. Thread one screw into hole in center of access door.

f. W/O AUX FUEL QTY While holding onto screw in center of access door, turn door until it slips down through access port.

g. AUX FUEL QTY Connect electrical connectors as follows:

(1) While holding onto screw in center of access door, carefully lower door, with probe attached, into tank.

(2) Remove tags and secure electrical wires to probe with screws.

(3) Remove protective cap from electrical connector.

(4) Inspect and treat electrical connector (PARA 1-7-20).

(5) For rear access door, secure electrical connector to access door with locknut.

h. While holding onto screw in center of access door, position door inside access port and install screws around perimeter.

i. Remove screw from center of access door and install at perimeter.

j. **TORQUE SCREWS TO 50 - 70 INCH-POUNDS.**

k. Make sure area is clean and free of foreign material.

l. Pressure test ESSS external 230 gallon fuel tank (PARA 16-4-70).

m. Install ESSS external 230 gallon fuel tank (PARA 16-4-54).

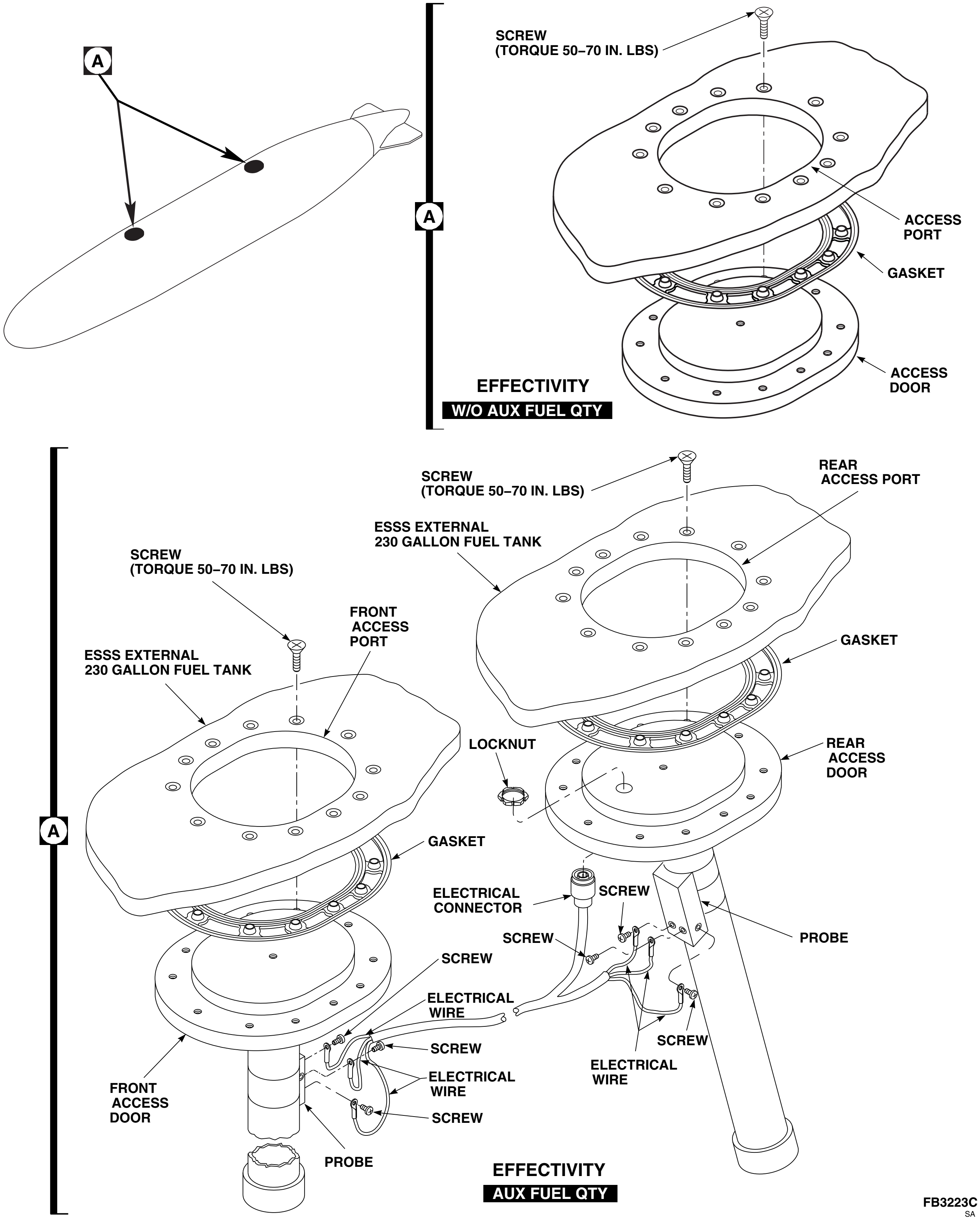


FIGURE 16-4-63-1. REPLACE ESSS EXTERNAL 230 GALLON FUEL TANK ACCESS DOORS

16-4-64 ESSS EXTERNAL 230 GALLON FUEL TANK PROBES AUX FUEL QTY .

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 196, Appendix D

References

- Appendix D
- PARA 1-3-21
- PARA 16-4-54
- PARA 16-4-58
- PARA 16-4-63

16-4-64.1 Replace ESSS External 230 Gallon Fuel Tank Probes.

NOTE

Replacement of front and rear external stores support system (ESSS) external 230 gallon fuel tank probes is the same.

16-4-64.1.1 Remove.

a. Remove ESSS external 230 gallon fuel tank (PARA 16-4-54).

b. Purge ESSS external 230 gallon fuel tank (PARA 1-3-21).

c. Remove ESSS external 230 gallon fuel tank electrical interface cable (PARA 16-4-58).

d. Remove front and rear ESSS external 230 gallon fuel tank access doors (PARA 16-4-63).

e. Remove bolts and washers securing probe to access door (Figure 16-4-64-1, Detail A).

f. Remove screws securing spacers to access door.

16-4-64.1.2 Install.

a. Install spacers on access door with screws (Figure 16-4-64-1, Detail A).

NOTE

Make sure front and rear fuel tank probes are installed in correct direction.

b. Install probe on access door with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

c. Using lockwire, Item 196, Appendix D, lockwire bolts.

d. Install front and rear ESSS external 230 gallon fuel tank access doors (PARA 16-4-63).

e. Install ESSS external 230 gallon fuel tank electrical interface cable (PARA 16-4-58).

f. Install ESSS external 230 gallon fuel tank (PARA 16-4-54).

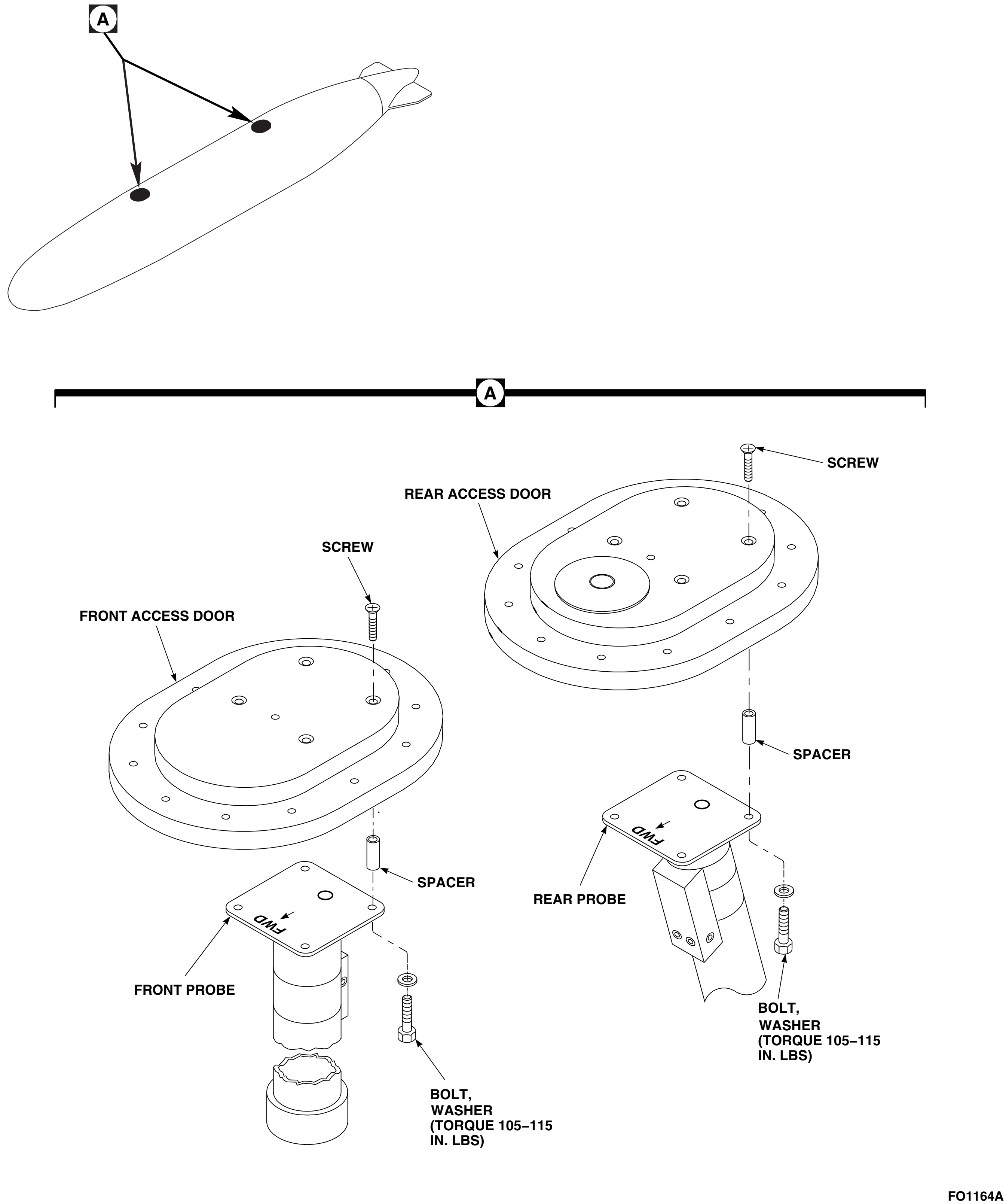


FIGURE 16-4-64-1. REPLACE ESSS EXTERNAL 230 GALLON FUEL TANK PROBES

16-4-65 ESSS EXTERNAL 230 GALLON FUEL TANK FUEL QUANTITY SENSOR HARNESS W/O AUX FUEL QTY .

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Lockwire, Item 196, Appendix D
- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-26
- PARA 1-7-20
- PARA 16-4-54
- PARA 16-4-58
- PARA 16-4-63

16-4-65.1 Replace ESSS External 230 Gallon Fuel Tank Fuel Quantity Sensor Harness.

NOTE

Replacement of external stores support system (ESSS) external 230 gallon fuel tank fuel quantity sensor harnesses is the same.

16-4-65.1.1 Remove.

a. Remove ESSS external 230 gallon fuel tank (PARA 16-4-54).

b. Purge ESSS external 230 gallon fuel tank (PARA 1-3-26).

c. Remove ESSS external 230 gallon fuel tank electrical interface cable (PARA 16-4-58).

d. Remove rear ESSS external 230 gallon fuel tank access door (PARA 16-4-63).

e. Loosen nut securing float switch to bracket at bottom of fuel tank (Figure 16-4-65-1, Detail A).

f. Slide float switch free of bracket.

g. Holding fuel quantity sensor harness inside fuel tank, remove jamnut and keywasher securing electrical connector to fuel tank.

h. Remove capscrew securing clamp to nut clip.

i. Remove fuel quantity sensor harness and packing through access port. Discard packing. Remove clamp from fuel quantity sensor harness.

j. Install protective caps and plugs on electrical connectors.

16-4-65.1.2 Install.

a. Lubricate packing with petrolatum, Item 234, Appendix D. Install packing onto electrical connector of fuel quantity sensor harness (Figure 16-4-65-1, Detail A).

b. Install clamp on fuel quantity sensor harness. Position fuel quantity sensor harness inside ESSS external 230 gallon fuel tank.

c. Secure clamp to nut clip with capscrew. Do not tighten capscrew at this time.

d. Position float switch into bracket by holding upright and sliding into bracket. Nut and washer should be outside bracket.

e. **TIGHTEN NUT TO SECURE FLOAT SWITCH IN BRACKET.**

f. Remove protective caps and plugs from electrical connectors.

g. Inspect and treat electrical connectors (PARA 1-7-20).

h. Hold fuel quantity harness inside fuel tank and slide electrical connector through bulkhead. Secure harness with keywasher and jamnut.

i. **TIGHTEN BULKHEAD JAMNUT.**

j. **TIGHTEN CAPSCREW.**

k. On outside of fuel tank, using lockwire, Item 196, Appendix D, lockwire keywasher.

l. Install rear ESSS external 230 gallon fuel tank access door (PARA 16-4-63).

m. Install ESSS external 230 gallon fuel tank electrical interface cable (PARA 16-4-58).

n. Install ESSS external 230 gallon fuel tank (PARA 16-4-54).

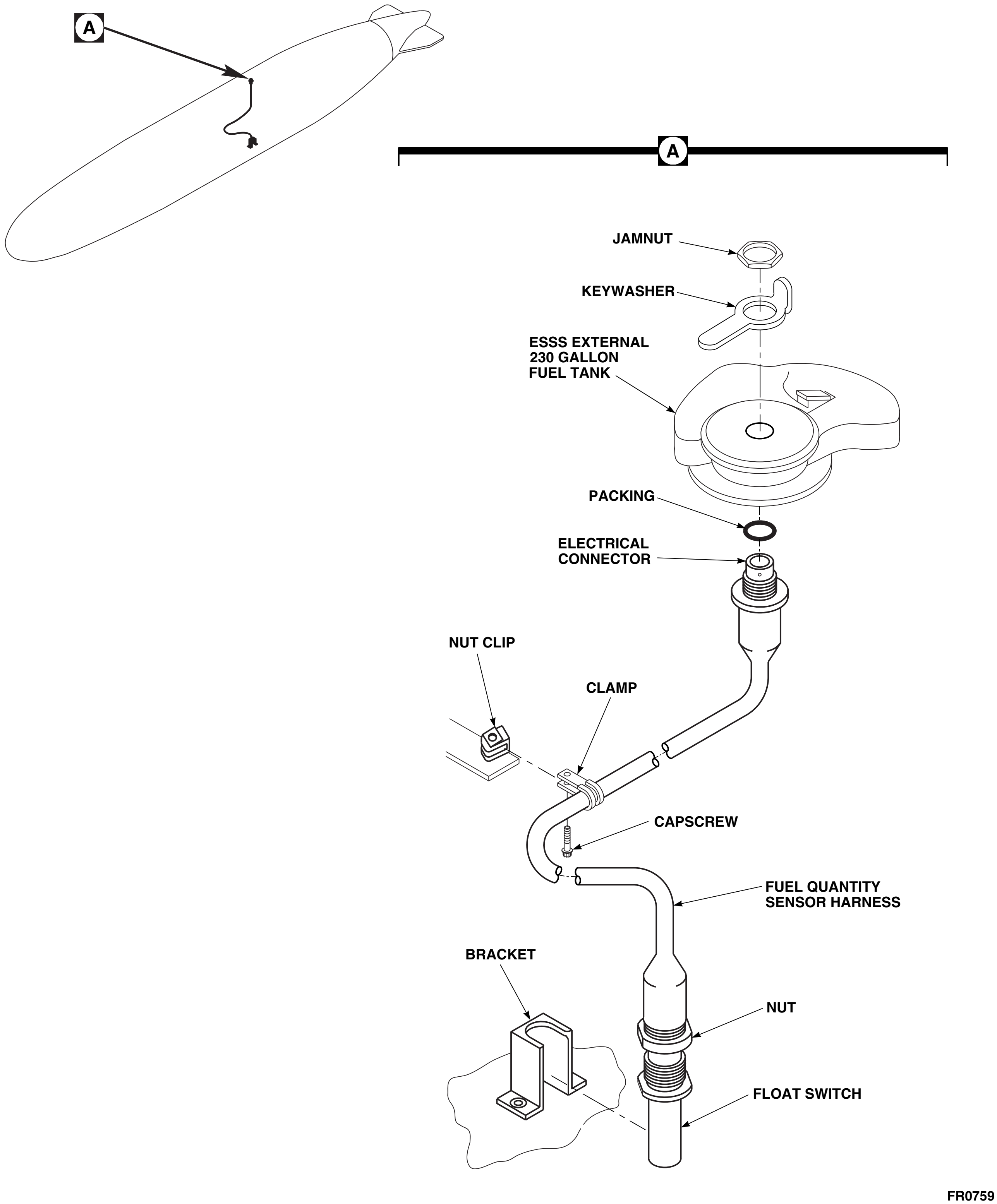


FIGURE 16-4-65-1. REPLACE ESSS EXTERNAL 230 GALLON FUEL TANK FUEL QUANTITY SENSOR HARNESS

16-4-66 ESSS EXTERNAL 230 GALLON FUEL TANK PROBE WIRING HARNESS AUX FUEL QTY .

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 1-3-21
- PARA 16-4-54
- PARA 16-4-58
- PARA 16-4-63

16-4-66.1 Replace ESSS External 230 Gallon Fuel Tank Probe Wiring Harness.

16-4-66.1.1. Remove.

a. Remove tank (PARA 16-4-54).

b. Purge tank (PARA 1-3-21).

c. Remove electrical interface cable (PARA 16-4-58).

d. Remove front and rear access doors (PARA 16-4-63).

e. Disconnect clamps attaching harness to baffles in tank and remove harness (Figure 16-4-66-1, Detail A).

16-4-66.1.2. Install.

a. Position probe wiring harness inside tank between the access ports.

b. Install probe wiring harness to baffles with existing harness clamps (Figure 16-4-66-1, Detail A).

c. Install front and rear access doors (PARA 16-4-63).

d. Install electrical interface cable (PARA 16-4-58).

e. Install tank (PARA 16-4-54).

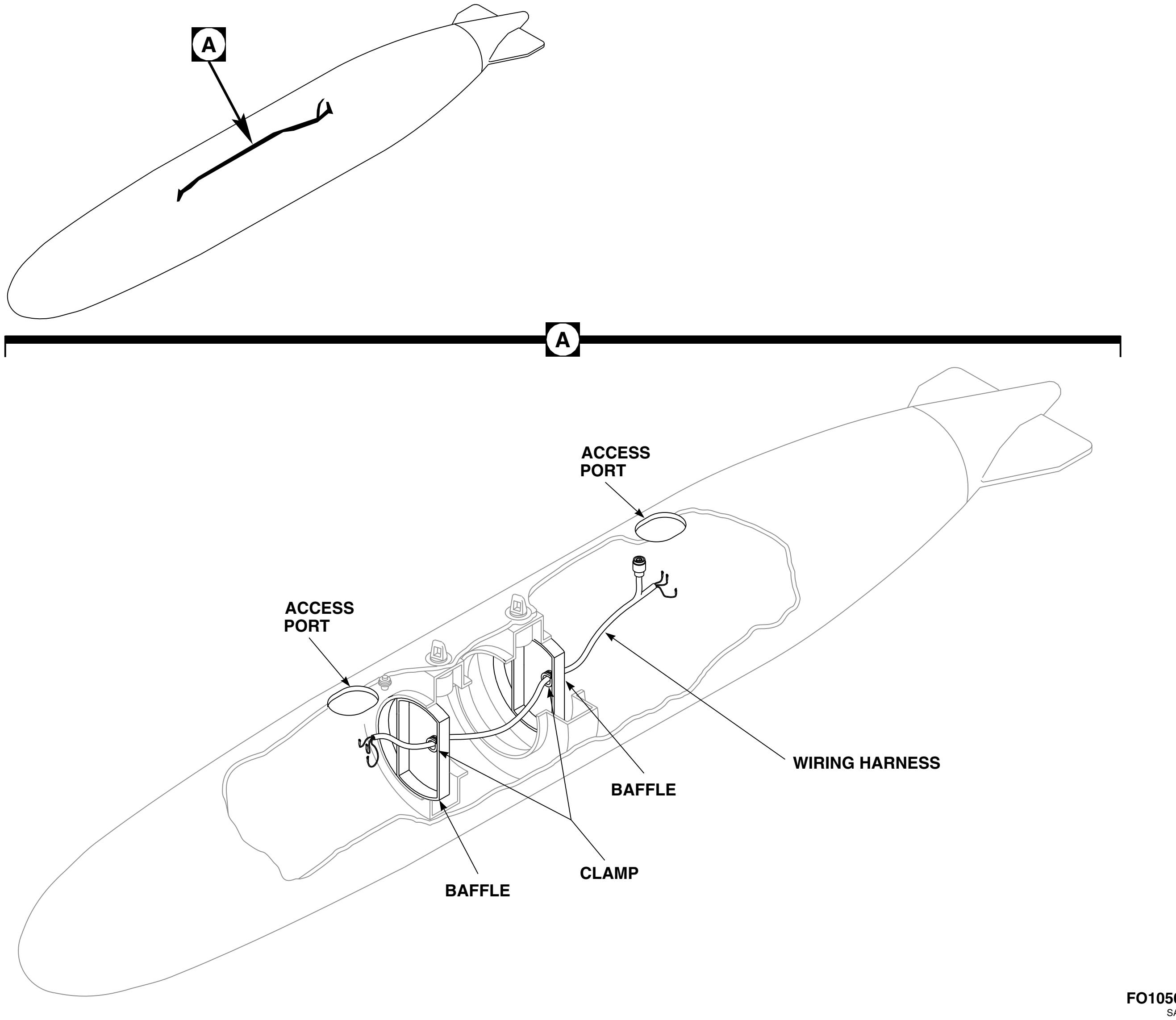


FIGURE 16-4-66-1. REPLACE ESSS EXTERNAL 230 GALLON FUEL TANK PROBE WIRING HARNESS

16-4-67 ESSS EXTERNAL 230 GALLON FUEL TANK FUEL VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 1-3-21
- PARA 16-4-54
- PARA 16-4-63

16-4-67.1 Replace ESSS External 230 Gallon Fuel Tank Fuel Valve.

NOTE

Replacement of external stores support system (ESSS) external 230 gallon fuel tank fuel valves is the same.

16-4-67.1.1 Remove.

a. Remove ESSS external 230 gallon fuel tank (PARA 16-4-54).

b. Purge ESSS external 230 gallon fuel tank (PARA 1-3-21).

c. Remove rear ESSS external 230 gallon fuel tank access door (PARA 16-4-63).

d. Inside tank, loosen B-nut on fuel supply tube at bulkhead fitting (Figure 16-4-67-1, Detail A).

e. Disconnect fuel supply tube from fuel valve and rotate supply fuel tube away from fuel valve.

f. Remove capscrews securing fuel valve to tank. Remove fuel valve.

16-4-67.1.2 Install.

a. Install fuel valve on bottom of ESSS external 230 gallon fuel tank and secure with capscrews (Figure 16-4-67-1, Detail A).

b. Connect lower end of fuel supply tube to fuel valve and start B-nut.

c. **TIGHTEN B-NUT ON BOTH ENDS OF FUEL SUPPLY TUBE.**

d. Install rear ESSS external 230 gallon fuel tank access door (PARA 16-4-63).

e. Install ESSS external 230 gallon fuel tank (PARA 16-4-54).

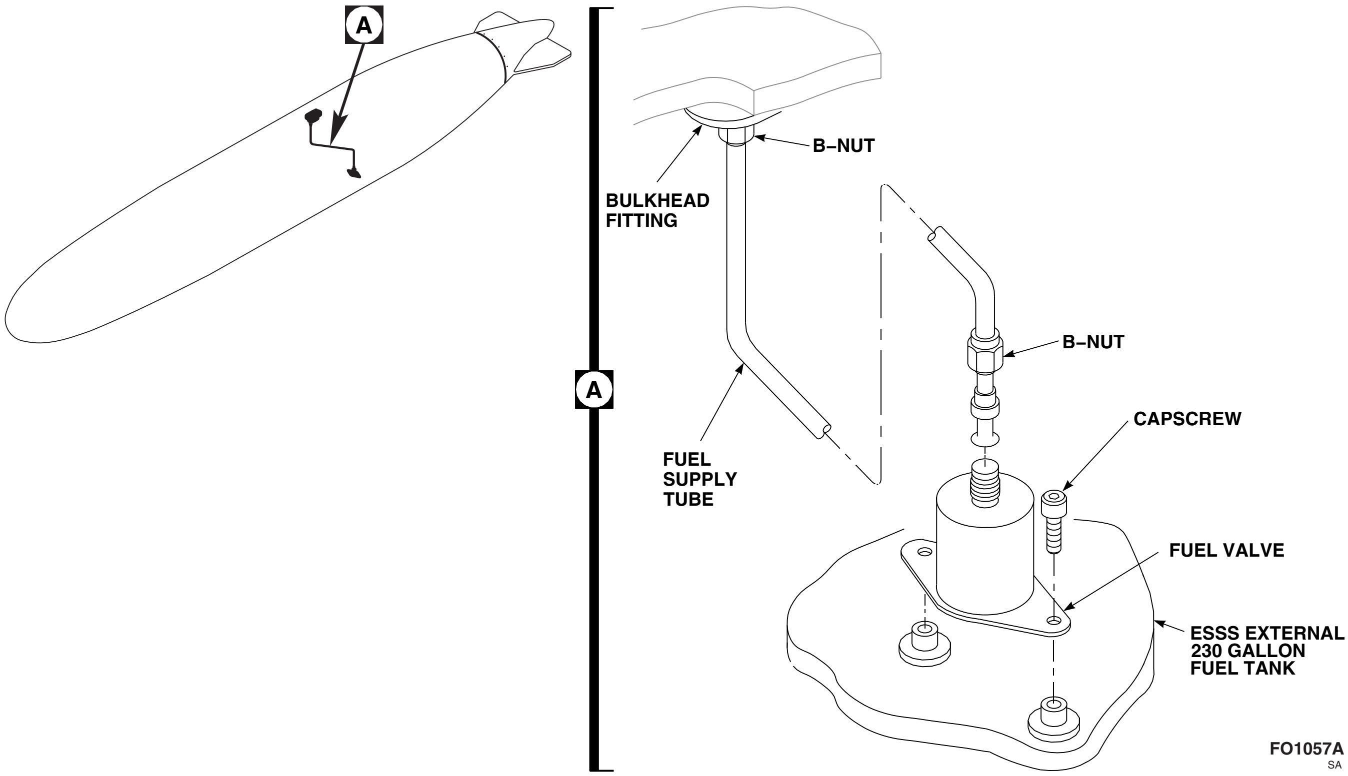


FIGURE 16-4-67-1. REPLACE ESSS EXTERNAL 230 GALLON FUEL TANK FUEL VALVE

16-4-68 ESSS EXTERNAL 230 GALLON FUEL TANK FUEL SUPPLY TUBE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 1-3-21
- PARA 16-4-54
- PARA 16-4-63

16-4-68.1 Replace ESSS External 230 Gallon Fuel Tank Fuel Supply Tube.

NOTE

Replacement of external stores support system (ESSS) external 230 gallon fuel tank fuel supply tube on all fuel tanks is the same.

16-4-68.1.1 Remove.

a. Remove ESSS external 230 gallon fuel tank (PARA 16-4-54).

b. Purge ESSS external 230 gallon fuel tank (PARA 1-3-21).

c. Remove rear ESSS external 230 gallon fuel tank access door (PARA 16-4-63).

d. From inside fuel tank, disconnect fuel supply tube from bulkhead fitting (Figure 16-4-68-1, Detail A).

e. Disconnect fuel supply tube from fuel valve.

f. Remove fuel supply tube.

16-4-68.1.2 Install.

a. Position fuel supply tube inside ESSS external 230 gallon fuel tank.

b. Connect lower end of fuel supply tube to fuel valve and start B-nut (Figure 16-4-68-1, Detail A).

c. Connect upper end of fuel supply tube to bulkhead fitting and start B-nut.

d. **TIGHTEN B-NUT ON BOTH ENDS OF FUEL SUPPLY TUBE.**

e. Install rear ESSS external 230 gallon fuel tank access door (PARA 16-4-63).

f. Install ESSS external 230 gallon fuel tank (PARA 16-4-54).

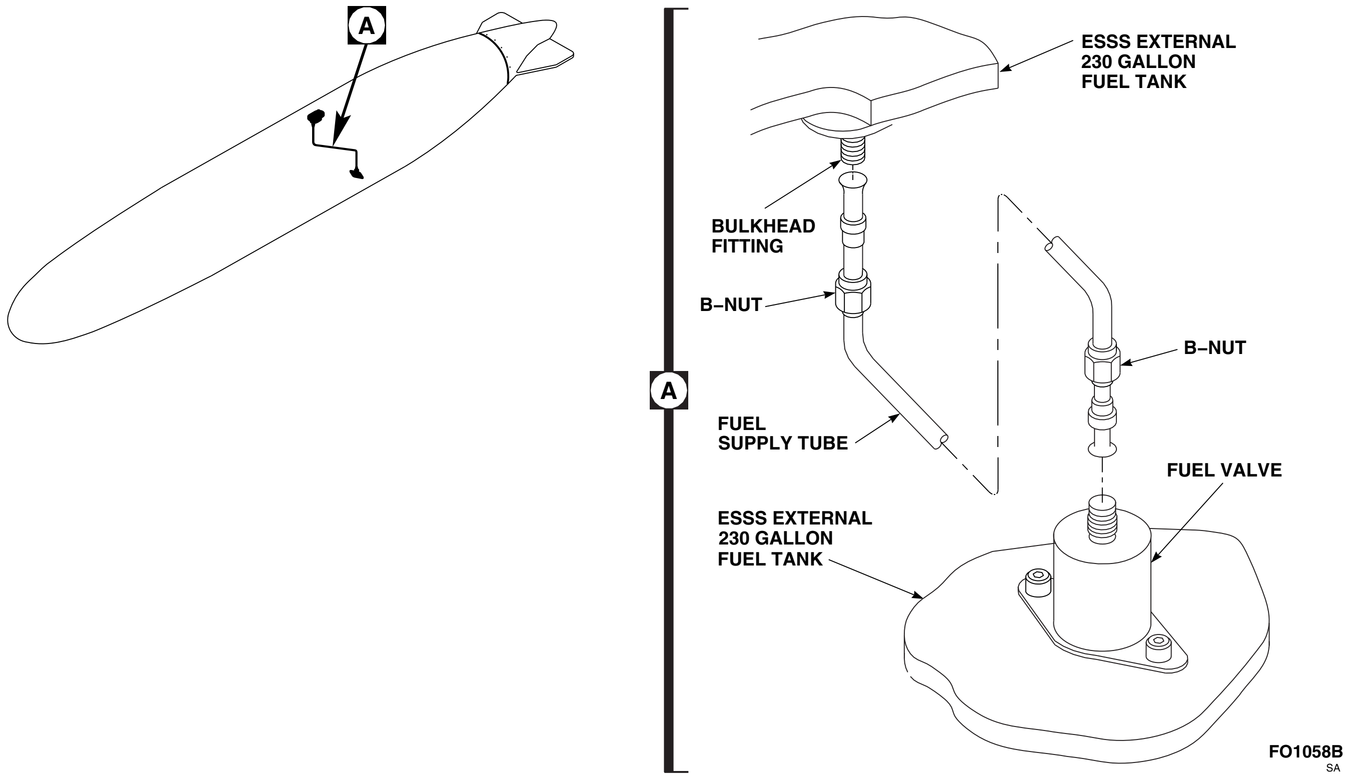


FIGURE 16-4-68-1. REPLACE ESSS EXTERNAL 230 GALLON FUEL TANK FUEL SUPPLY TUBE

16-4-69 ESSS EXTERNAL 230 GALLON FUEL TANK INLET AIR TUBE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 1-3-21
- PARA 16-4-54
- PARA 16-4-63

16-4-69.1 Replace ESSS External 230 Gallon Fuel Tank Inlet Air Tube.

NOTE

Replacement of all external stores support system (ESSS) external 230 gallon fuel tank inlet air tubes is the same.

16-4-69.1.1 Remove.

a. Remove ESSS external 230 gallon fuel tank (PARA 16-4-54).

b. Purge ESSS external 230 gallon fuel tank (PARA 1-3-21).

c. Remove ESSS external 230 gallon fuel tank access doors (PARA 16-4-63).

d. Remove ¼-inch tube, short ½-inch tube, and tee fitting, as an assembly, as follows:

(1) Working through rear access, disconnect short ½-inch tube from air interface fitting (Figure 16-4-69-1, Detail A).

(2) Disconnect ¼-inch tube from bulkhead fitting.

(3) Disconnect long ½-inch tube from tee fitting.

(4) Remove ¼-inch, short ½-inch tube, and tee fitting through rear access.

e. Disconnect short ½-inch tube from tee fitting.

f. Disconnect ¼-inch tube from reducer on tee fitting.

g. Remove coupling nut from reducer.

h. Remove reducer from tee fitting.

i. Loosen two nylon hexhead bolts securing front end of long ½-inch tube into bulkhead fitting.

j. Remove capscrews and clamps securing tube to nut clips.

k. Pull long ½-inch tube toward front until end reaches front access, then remove through access.

16-4-69.1.2 Inspect.

a. Visually inspect tubes for damage. **DENTS EXCEEDING 25% OF TUBE DIAMETER NOT ALLOWED.** If damaged, replace tube(s).

b. Inspect all welds on tubes for cracks. **NONE ALLOWED.** If cracked at welds, replace tube(s).

c. Inspect tubes and connections for signs of leakage or damage. **NONE ALLOWED.** If damaged or leaking, replace tube(s) and/or connection(s).

16-4-69.1.3 Install.

NOTE

Tube connections should not be tightened until installation is complete.

a. Assemble ¼-inch tube, short ½-inch tube, and tee fitting as follows (Figure 16-4-69-1, Detail A):

(1) Install reducer on tee fitting.

(2) Install coupling nut on reducer.

(3) Connect ¼-inch tube to end of reducer.

(4) Connect short ½-inch tube to middle fitting of tee fitting.

b. Install assembly inside ESSS external 230 gallon fuel tank and connect ¼-inch tube to bulkhead fitting and short ½-inch tube to air interface fitting so tee fitting is on right side of fuel tank.

NOTE

Long ½-inch tube is installed so that plastic ring is towards front of tank.

c. Install long ½-inch tube into tank through front access.

d. Slide long ½-inch tube through opening on right side of baffles.

e. Connect front end of tube with plastic ring into bulkhead fitting. **TIGHTEN NYLON HEX-HEAD BOLTS HANDTIGHT.**

f. Connect long ½-inch tube on tee fitting.

g. Install clamps to nut clips and secure with capscrews.

h. **TIGHTEN ALL B-NUTS AND CAPSCREWS.**

i. Install front and rear ESSS external 230 gallon fuel tank access doors (PARA 16-4-63).

j. Install ESSS external 230 gallon fuel tank (PARA 16-4-54).

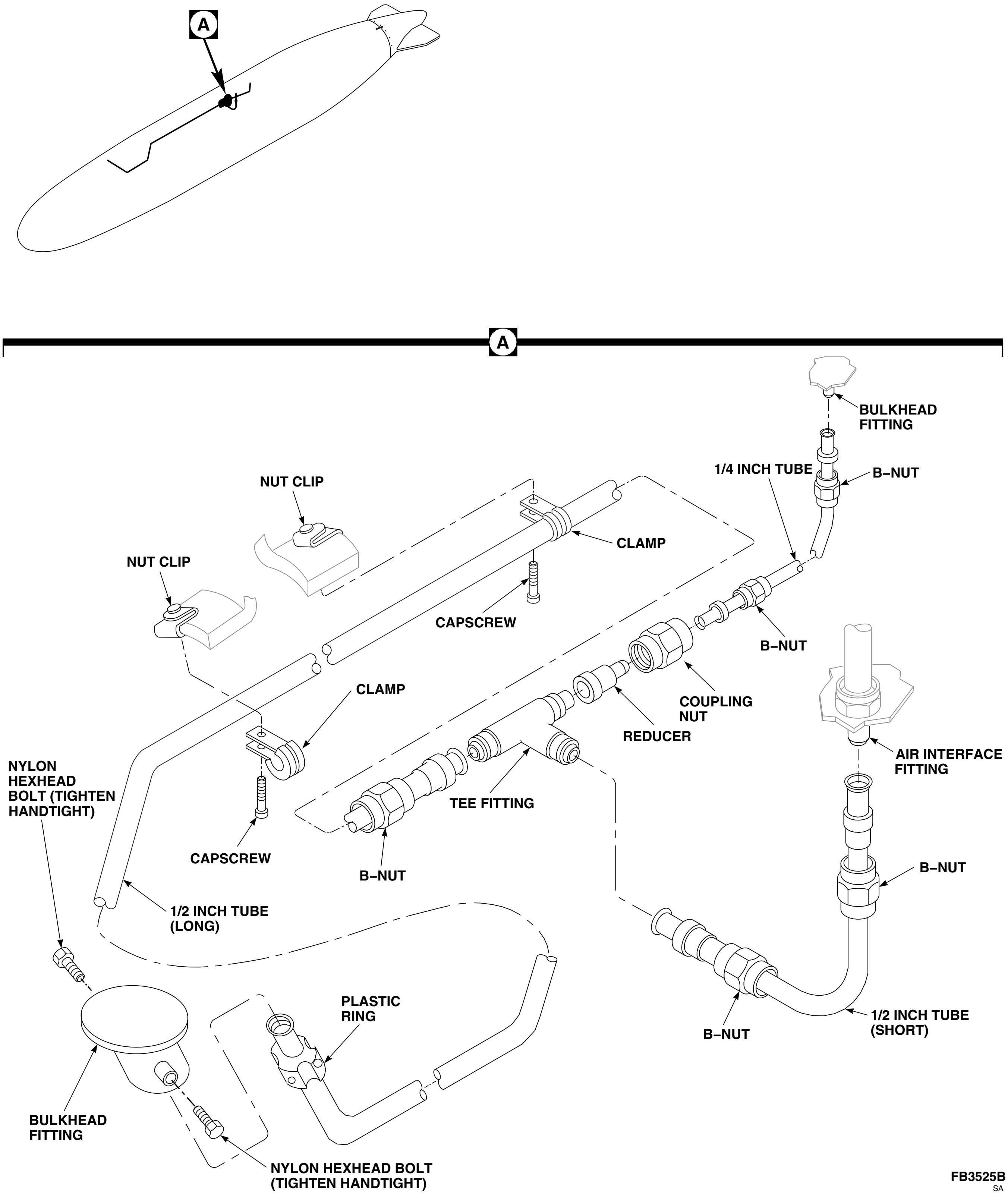


FIGURE 16-4-69-1. REPLACE ESSS EXTERNAL 230 GALLON FUEL TANK INLET AIR TUBE

16-4-70 ESSS EXTERNAL 230 GALLON FUEL TANK PRESSURE TEST.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Pneumatic Adapter (ESSS), Locally-Made, Figure H-151, Appendix H

Materials/Parts

- Leak Detection Fluid, Item 189, Appendix D

References

- Appendix D
- Appendix H
- PARA 16-4-60
- PARA 16-4-71

16-4-70.1 ESSS External 230 Gallon Fuel Tank Pressure Test.

WARNING

Injury to personnel and damage to equipment will result if pressure test is not performed carefully and safely. Personnel performing test should wear eye protection when working around pressurized tank. Air pressure must be regulated and gaged.

a. Locally make pneumatic adapter (Figure H-151, Appendix H).

b. Remove air interface (PARA 16-4-60).

c. Install pneumatic adapter into air interface bulkhead fitting on tank.

d. Move tank away from personnel and apply pressure to 30 psig for operational test or 66 psig for structural test for 15 minutes.

e. Apply leak detection fluid, Item 189, Appendix D, to fittings and repaired areas and check for leakage. **NO LEAKAGE ALLOWED.**

f. Turn off air pressure.

g. Slowly bleed off air pressure and disconnect air line.

h. Remove pneumatic adapter from tank.

i. Repair any leaks (PARA 16-4-71).

j. Install air interface (PARA 16-4-60).

16-4-71 ESSS EXTERNAL 230 GALLON FUEL TANK - REPAIR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Clamping Plates
- Heat Gun
- Heat Lamp
- Hole Saw
- Locating Tool, Locally-Made
- Protection, Eye, Skin, and Breathing
- Router (with Depth Control)
- Router Bits, 0.125-inch and 0.250-inch
- Vacuum Bags
- Vacuum Cleaner, A1625
- Vacuum Pump

Materials/Parts

- Abrasive Cloth, Item 4, Appendix D
- Abrasive Cloth, Item 6, Appendix D
- Acetone, Item 25, Appendix D
- Adhesive, Item 32, Appendix D
- Adhesive, Item 35, Appendix D
- Bleeder Cloth
- Body Filler

- Brush, Acid, Item 84, Appendix D
- Caul Sheet
- Cheesecloth, Item 90, Appendix D
- Chromate Seal
- Fiberglass Fabric, Style 7781
- Hytrel Material
- Masking Tape, 1-inch, Item 212, Appendix D
- Nomex Honeycomb
- Nylon Film, 0.002-inch thick
- Peel Ply
- Perma Plug
- Petrolatum, Item 234, Appendix D
- Plaster (Alternate for Body Filler)
- Threaded Insert, Access Door
- Threaded Insert, Tail Fin Mounting
- Stirring Stick, Beverage, Item 315C, Appendix D

References

- AMS-C-9084
- Appendix D
- PARA 1-3-21
- PARA 2-6-3
- PARA 2-6-5
- PARA 16-4-63
- PARA 16-4-70

16-4-71.1 General.

CAUTION

The following repair procedures are not to be used for rotor blade repair.

16-4-71.1.1 Equipment and Supplies.

a. The recommended glass fabric used for reinforced plastic repairs shall conform to AMS-C-9084, Class 2.

NOTE

Equipment and material used in repair of reinforced plastic should duplicate, as closely as possible, those used in the manufacture of the original part.

b. Use vacuum bags or clamping plates to provide pressure during cure.

c. Use heat lamps or heat guns, whenever possible, to cure repair areas.

d. Assemble equipment for storing resins, measuring, weighing, cutting fabric, cutting cured laminates, sanding, and finishing.

CAUTION

Injury to personnel will result if safety equipment is not worn when working around plastic dust. Reinforced plastic dust is a health hazard that requires eye, respiratory, and exposed skin protection. The dust is very abrasive and may damage other equipment and/or contaminate parts with which it comes in contact.

e. Make sure effective exhaust system is present to remove reinforced plastic dust from atmosphere.

f. Make sure solvents free of contaminants, such as acetone, are available for cleaning equipment and parts.

g. Control storage temperature and shelf life of resins to ensure resin reliability. Resins should be stored at normal room temperature of 75°F (24°C), unless otherwise indicated on containers.

h. Class IV through Class VI-A damage repairs must be followed with structural pressure test.

16-4-71.2 Vacuum Bag Procedure.

16-4-71.2.1 Install.

a. Cover repair area with peel ply at least 1-inch past edge of repaired area.

b. Apply layer of nylon film or pin prick over peel ply.

c. Apply layer of bleeder cloth over nylon film or pin prick, unless using caul sheet. If using caul sheet, apply over pin prick and overlay with bleeder cloth.

d. Surround repair area with strip of chromate seal about 2-inches past edge of repair area.

e. Wrap end of vacuum hose with 3 x 6-inch piece of bleeder cloth.

f. Lay vacuum hose on repair area so end of hose rests on bleeder cloth.

g. Apply layer of pink vacuum bag material over repair area and seal edges with chromate seal.

h. Apply vacuum of at least 15-inches of mercury. Check for leaks, which are indicated by hissing sounds.

i. Cure repair per cure cycle requirements.

j. Allow repair to cool prior to removing section.

16-4-71.2.2 Remove.

a. Turn off vacuum pump.

b. Remove vacuum bag materials.

c. Remove peel ply and verify that no traces remain.

d. Using abrasive cloth, Item 6, Appendix D, sand repaired area to blend with contour of tank in preparation for replacing finish.

16-4-71.3 Adhesive Systems.

WARNING

- Injury to personnel will result if fire safety is not used while working with adhesives. Use caution when using or mixing adhesives; fumes from these materials are a fire and health hazard. Post no smoking signs in area, and keep materials away from sparks and flames. Use properly grounded and explosion-proof electrical equipment in working area.
- Wash hands before eating or smoking.

Refer to Table 16-4-71-1 for adhesive information.

Table 16-4-71-1. Fuel Tank Adhesives

Adhesive	Curing Agent	Mix Ratio	Pot Life at Room Temperature (70°F (21°C)) in Minutes	Cure Cycle	Type of Structure That Can Be Repaired
Adhesive, Item 32, Appendix D, Part A	Adhesive, Item 32, Appendix D, Part B	100 : 58	30	200°F (93°C) for 1 hr or 21°F (6°C) for 5 days	Graphite/ Fiberglass/ Epoxy Fiberglass/ Epoxy
Hysol 9309.3 Part A	Hysol 9309.3 Part B	100 : 22	35	120°F (49°C) for 1 hr or 70°F (21°C) for 6 hrs	Hytrel Liner

16-4-71.4 Types of Damage.

16-4-71.4.1 Negligible Damage.

a. Dents less than 3-inches in diameter and show no cracks or breaks in surface require no repair.

b. Delaminations less than 3-inches in diameter require no repair.

16-4-71.4.2 Repairable Damage.

a. Repairable damage is damage that can be permanently repaired with no adverse effects on structural integrity, flight characteristics, or helicopter safety.

b. Repairable damage can be repaired using the procedures in this manual.

16-4-71.4.3 Non-Repairable Damage.

a. Any hole 4-inches or more in diameter or length.

b. Any break, crack, delamination, or dent 5-inches or more in diameter or length.

c. Any Class III through VI damage less than 2-inches from opening or attachment.

d. No two Class III through VI damaged areas within 3-inches of each other.

e. No Class V damage that exceeds 2.5-inches over frames.

f. No Class VI or Class VI-A damage repair that would extend into frame.

g. Total tank damages that exceed 25% of tank surface.

16-4-71.5 Classes of Damage.

16-4-71.5.1 General.

a. When two or more areas of Class III through VI damage are within 3-inches of each other, repair is not authorized.

b. When total tank damage exceeds 25% of tank surface area, repair is not authorized.

16-4-71.5.2 Class I Damage.

Class I damage is damage to the surface finish such as scratches, pits, erosion, or abrasions which does not penetrate outer skin.

16-4-71.5.3 Class II Damage.

a. Class II damage is damage to the outer skin such as cuts, scratches, pits, erosion, or abrasions which does not penetrate outer skin (Figure 16-4-71-1).

b. Any area which does not exceed 25% of tank surface area.

16-4-71.5.4 Class III Damage.

a. Class III damage is damage such as delamination, cuts, pits, erosion, or abrasions which penetrates the outer skin but does not damage honeycomb core or exceed 5-inches in length or diameter (Figure 16-4-71-2).

b. Distances from edge of damage to attachment or opening no less than 2-inches.

16-4-71.5.5 Class IV Damage.

a. Class IV damage is damage such as cuts, scratches, pits, erosion, or abrasions which penetrates the outer skin and honeycomb core or exceeds 5-inches in length or diameter (Figure 16-4-71-3).

b. Dents, with crushed honeycomb core, which do not exceed 5-inches in diameter.

c. Damage that does not penetrate inner skin.

d. Distance from edge of damage to attachment or opening is no less than 2-inches.

16-4-71.5.6 Class V Damage.

a. Class V damage is damage such as cuts, scratches, pits, erosion, or abrasions which penetrates the outer skin, honeycomb core, and inner skin or exceeds 4-inches in length or diameter and does not exceed 2.5-inches between tank STA 70.88 and STA 75.88 and between STA 86.13 and STA 91.13 (Figure 16-4-71-4).

b. Damage does not penetrate Hytrel liner.

c. Distance from edge of damage to attachment or opening is no less than 2-inches.

16-4-71.5.7 Class VI Damage.

a. Class VI damage is damage which penetrates the outer skin, honeycomb core, inner skin, and Hytrel liner but is no greater than 4-inches in length or diameter (Figure 16-4-71-5).

b. Damage or repair does not contact composite frames between tank STA 70.88 and STA 75.88 and between STA 86.13 and STA 91.13.

c. Distance from edge of damage to attachment or opening is no less than 2-inches.

16-4-71.5.8 Class VI-A Damage.

a. Class VI-A damage is damage which penetrates the outer skin, honeycomb core, inner skin, any Hytrel liner that cannot be reached by hand through the access door for repair and damage is no greater than 4-inches in length or diameter (Figure 16-4-71-6).

b. Damage or repair does not contact composite frames between tank STA 70.88 and STA 75.88 and between STA 86.13 and STA 91.13.

c. Distance from edge of damage to attachment or opening is no less than 2-inches.

16-4-71.5.9 Class VII Damage.

Class VII damage is damage to the access door mounting inserts. No access door inserts may be out of service at any time.

16-4-71.5.10 Class VII-A Damage.

a. Class VII-A damage is damage to the tail fin mounting inserts with no more than four tail fin inserts out of service at the same time.

b. No more than two adjacent tail fin inserts may be out of service at the same time.

16-4-71.5.11 Class VIII Damage.

Class VIII damages are classified as leaks around bulkhead fittings which pass through tank wall.

16-4-71.6 Class I Repair.

a. Determine class of damage (PARA 16-4-71.5).

b. Mask around damaged area using 1-inch masking tape, Item 212, Appendix D.

c. Lightly sand damaged area to blend with contour of tank using abrasive cloth, Item 6, Appendix D.

d. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

e. Verify damage classification is correct. If not, determine class of damage (PARA 16-4-71.5).

f. Remove masking tape.

g. Touch up paint finish (PARA 2-6-5).

16-4-71.7 Class II Repair.

a. Determine class of damage (PARA 16-4-71.5).

b. Mask around damaged area using 1-inch masking tape, Item 212, Appendix D (Figure 16-4-71-1).

c. Sand damaged area using abrasive cloth, Item 4, Appendix D.

d. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

e. Verify damage classification is correct. If not, determine class of damage (PARA 16-4-71.5).

f. Prepare enough adhesive, Item 35, Appendix D, to fill damaged area in skin.

g. Apply adhesive, Item 35, Appendix D, to damaged area in skin. Level adhesive to surface contour of tank.

h. Remove masking tape.

i. Cover repair with peel ply material.

CAUTION

Damage to tank and repair area will result if heat gun is not adjusted correctly when curing a repair. Incorrect positioning and temperature during cure cycle will result in a weak repair. Pay strict attention to temperature range, and adjust distance between heat source and surface as necessary (PARA 2-6-3).

j. Cure repair (PARA 2-6-3).

k. Using abrasive cloth, Item 6, Appendix D, remove peel ply and sand repaired area in preparation for replacing finish.

l. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

m. Touch up paint finish (PARA 2-6-5).

16-4-71.8 Class III Repair.

a. Determine class of damage (PARA 16-4-71.5).

b. Mask around damaged area using 1-inch masking tape, Item 212, Appendix D (Figure 16-4-71-2).

c. Fabricate round or oval template slightly larger than damage to be removed (Figure 16-4-71-9).

d. Overlay damaged area with template and trace perimeter.

NOTE

Use cutting tools with controllable cutting depth to avoid excessive damage.

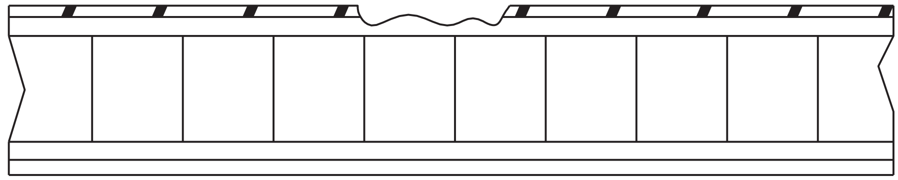
e. Cut and remove damaged outer skin the size of template (honeycomb may be undercut 0.020-inch maximum).

f. Verify damage classification is correct. If not, determine class of damage (PARA 16-4-71.5).

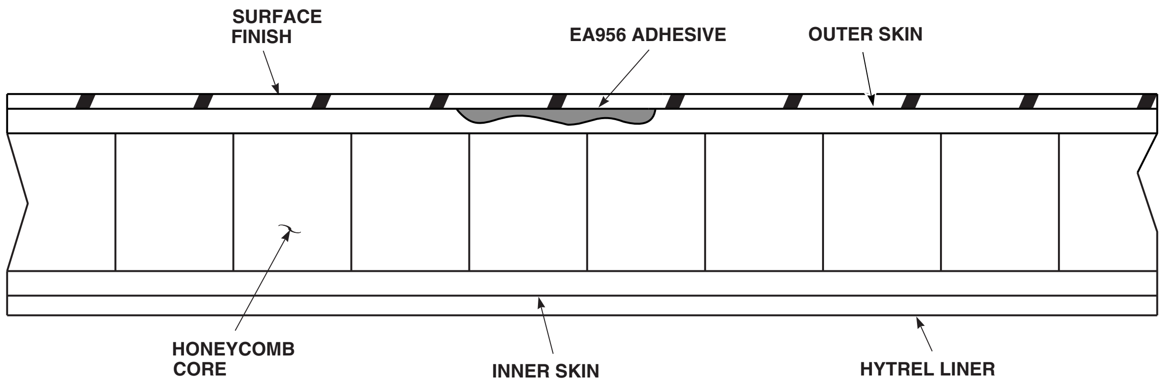
g. Sand off surface finish at least 2-inches around damaged area using abrasive cloth, Item 4, Appendix D.

h. Vacuum dust from repair area.

i. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.



CLASS II DAMAGE



CLASS II REPAIR

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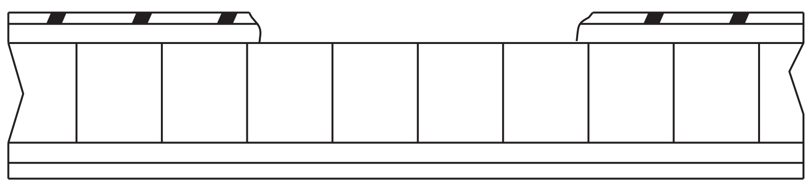
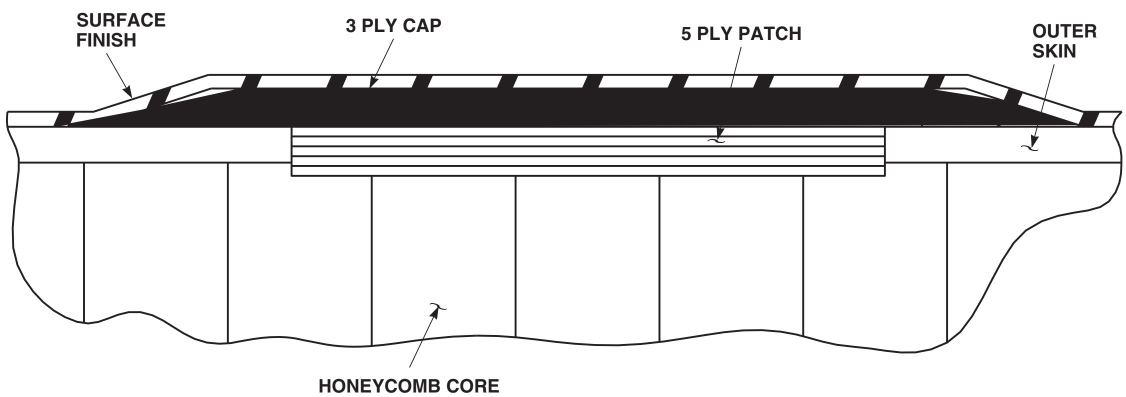
FIGURE 16-4-71-1. CLASS II DAMAGE AND REPAIR



CLASS III DAMAGE



CLASS III REPAIR

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FIGURE 16-4-71-2. CLASS III DAMAGE AND REPAIR

NOTE

Cut fiberglass fabric on a clean surface. Keep free of dust, oil, water, and other contaminants.

j. Cut one piece of fiberglass fabric large enough to contain six pieces 1.5-inches larger in radius than template.

k. Cut two pieces of nylon film at least 6-inches larger than fiberglass fabric.

l. Place fiberglass fabric on one piece of nylon film.

m. Prepare adhesive, Item 35, Appendix D, for repair (PARA 16-4-71.3).

n. Impregnate fiberglass fabric. Resin content of fiberglass shall be about 50% by weight when thoroughly impregnated.

o. Apply layer of nylon film on top of impregnated cloth and remove air bubbles.

p. Cut impregnated fiberglass fabric into eight pieces as follows:

(1) Five pieces the size of template.

(2) One piece 0.5-inch larger in radius than template.

(3) One piece 1-inch larger in radius than template.

(4) One piece 1.5-inches larger in radius than template.

q. Apply thin coat of adhesive, Item 35, Appendix D, using acid brush, Item 84, Appendix D, over honeycomb core in damaged cavity.

r. Remove nylon film from five template-sized pieces and stack them to form 5-ply laminate in damaged cavity.

NOTE

If 5-ply laminate patch does not fill cavity to top, prepare extra layers of fabric to fill cavity as required.

s. Remove nylon film from fabric piece that is 0.5-inch larger than template and overlay damage. Center fabric as closely as possible.

t. Remove nylon film from two remaining pieces of oversized fabric and overlay damage, smallest piece first.

u. Remove masking tape.

v. Perform vacuum bag procedure (PARA 16-4-71.2).

CAUTION

Damage to tank and repair area will result if heat gun is not adjusted correctly when curing a repair. Incorrect positioning and temperature during cure cycle will result in a weak repair. Pay strict attention to temperature range, and adjust distance between heat source and surface as necessary (PARA 2-6-3).

w. Cure repair (PARA 2-6-3).

x. Remove vacuum bag materials (PARA 16-4-71.2).

y. Using abrasive cloth, Item 6, Appendix D, sand repaired area to blend with contour in preparation for replacing finish.

z. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

aa. Touch up paint finish (PARA 2-6-5).

16-4-71.9 Class IV Repair.

a. Determine class of damage (PARA 16-4-71.5).

b. Mask around damaged area using 1-inch masking tape, Item 212, Appendix D (Figure 16-4-71-3).

c. Fabricate round or oval template slightly larger than damage to be removed (Figure 16-4-71-9).

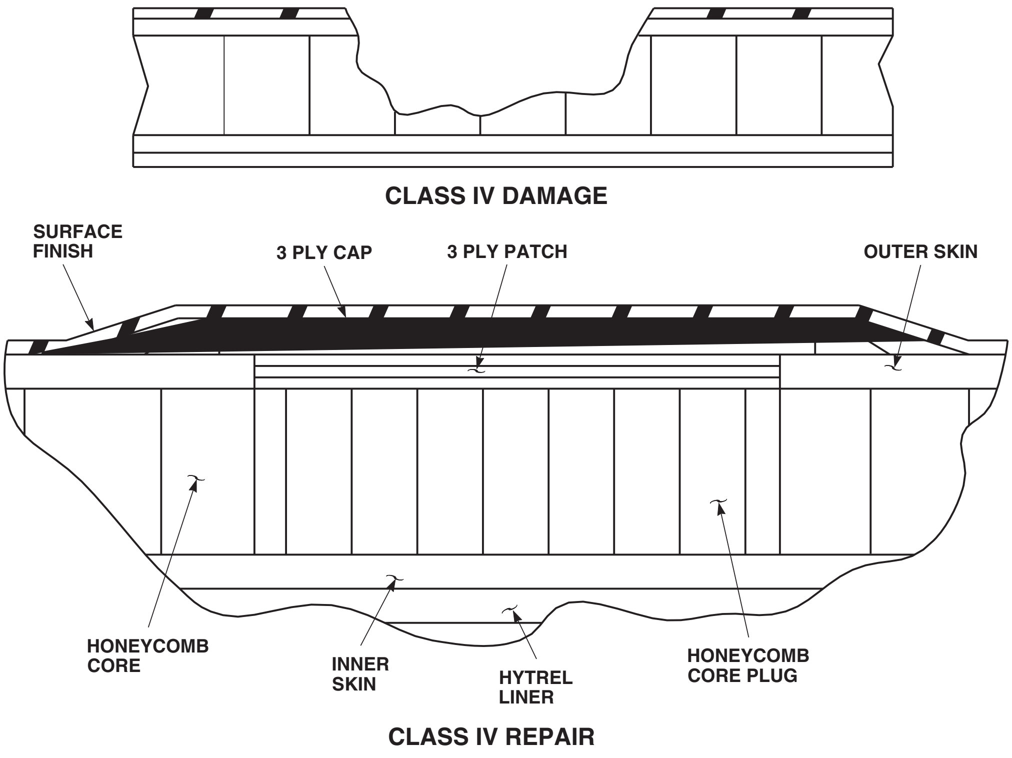


FIGURE 16-4-71-3. CLASS IV DAMAGE AND REPAIR

d. Overlay damaged area with template and trace perimeter.

NOTE

Cutting tools having a controllable cutting depth are suggested to be used to avoid excessive damage.

e. Cut and remove damaged outer skin and honeycomb core down to inner skin. Take care not to damage inner skin.

f. Verify damage classification is correct. If not, determine class of damage (PARA 16-4-71.5).

g. Sand off surface finish at least 2-inches around damaged area using abrasive cloth, Item 4, Appendix D.

h. Vacuum dust from repair area.

i. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

NOTE

Cut Nomex honeycomb and fiberglass fabric on a clean surface. Keep free of dust, oil, water, and other contaminants.

j. Cut one piece of Nomex honeycomb 0.125-inch larger than template.

k. Cut one piece of fiberglass fabric large enough to contain six pieces 1.5-inches larger in radius than template.

l. Cut two pieces of nylon film at least 6-inches larger than fiberglass fabric.

m. Place fiberglass fabric on one piece of nylon film.

n. Prepare adhesive, Item 35, Appendix D, for repair (PARA 16-4-71.3).

o. Impregnate fiberglass fabric. Resin content of fiberglass fabric shall be about 50% by weight when thoroughly impregnated.

p. Apply layer of nylon film on top of impregnated cloth and remove air bubbles.

q. Cut impregnated fiberglass fabric into six pieces as follows:

(1) Three pieces the size of template.

(2) One piece 0.5-inch larger in radius than template.

(3) One piece 1-inch larger in radius than template.

(4) One piece 1.5-inches larger in radius than template.

r. Apply thin layer of adhesive, Item 35, Appendix D, using acid brush, Item 84, Appendix D, to inner skin surface in damaged cavity.

s. Apply adhesive to all surfaces of honeycomb core and place in damaged cavity. Core must be slightly compressed to fit into position.

t. Remove nylon film for three template-sized pieces of fabric and stack them to form 3-ply laminate in damaged cavity.

NOTE

If 3-ply laminate patch does not fill cavity to top, prepare extra layers of fabric to fill cavity as required.

u. Remove nylon film from fabric piece that is 0.5-inch larger than template and overlay damage. Center fabric as closely as possible.

v. Remove nylon film from two remaining pieces of oversized fabric and overlay damage, using smallest piece first.

w. Remove masking tape.

x. Perform vacuum bag procedure (PARA 16-4-71.2).

CAUTION

Damage to tank and repair area will result if heat gun is not adjusted correctly when curing a repair. Incorrect positioning and temperature during cure cycle will result in a weak repair. Pay strict attention to temperature range, and adjust distance between heat source and surface as necessary (PARA 2-6-3).

y. Cure repair (PARA 2-6-3).

z. Remove vacuum bag material (PARA 16-4-71.2).

aa. Sand repaired area in preparation for replacing finish using abrasive cloth, Item 6, Appendix D.

ab. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

ac. Touch up paint finish (PARA 2-6-5).

ad. Perform pressure test procedure (PARA 16-4-70).

16-4-71.10 Class V Repair.

a. Determine class of damage (PARA 16-4-71.5).

b. Mask around damaged area using 1-inch masking tape, Item 212, Appendix D (Figure 16-4-71-4).

c. Fabricate round or oval template slightly larger than damage to be removed (Figure 16-4-71-9).

d. Overlay damaged area with template and trace perimeter.

NOTE

Cutting tools having a controllable cutting depth are suggested to be used to avoid excessive damage.

e. Cut and remove damaged outer skin and honeycomb core down to inner skin.

f. Remove honeycomb residue from inner skin.

g. Verify damage classification is correct. If not, determine class of damage (PARA 16-4-71.5).

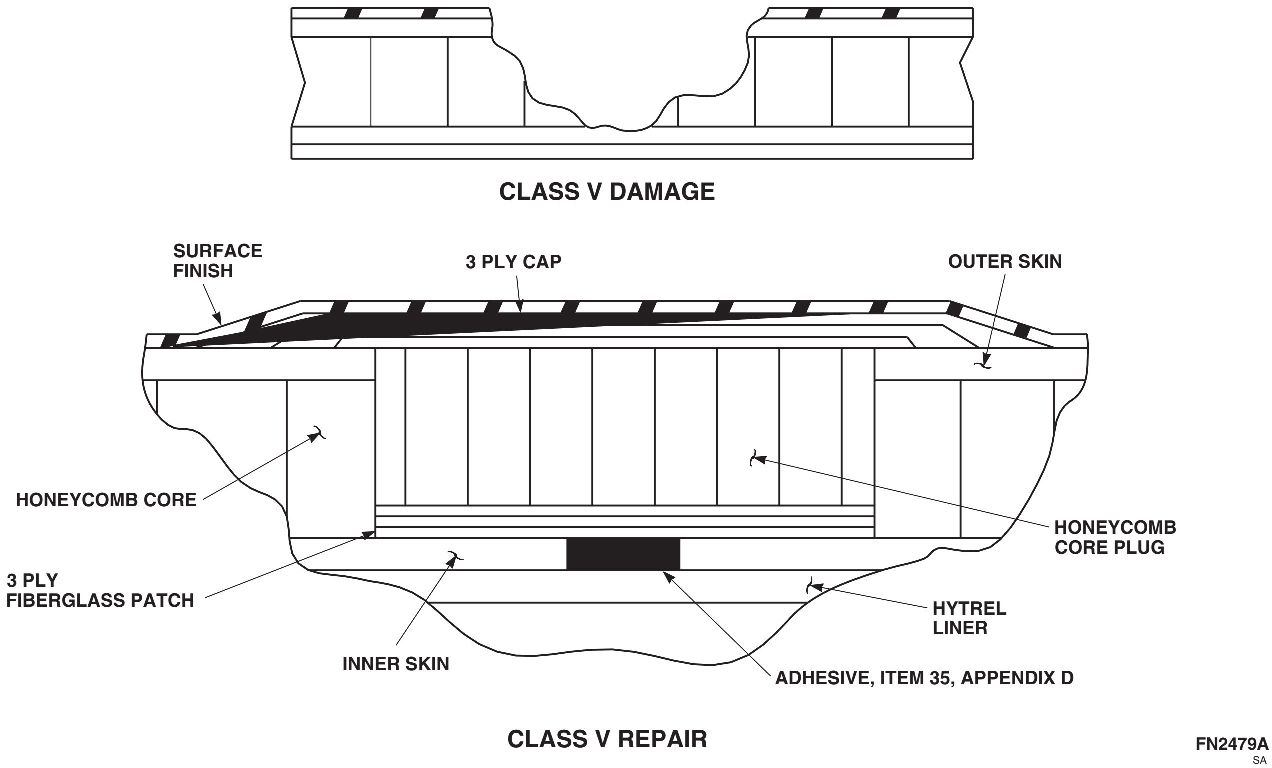


FIGURE 16-4-71-4. CLASS V DAMAGE AND REPAIR

h. Sand off surface finish at least 2-inches around damaged area using abrasive cloth, Item 4, Appendix D.

i. Vacuum dust from repair area.

j. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

NOTE

Cut Nomex honeycomb and fiberglass fabric on a clean surface. Keep free of dust, oil, water, and other contaminants.

k. Cut one piece of Nomex honeycomb 0.125-inch larger than template.

l. Cut one piece of fiberglass fabric large enough to contain six pieces 1.5-inches larger in radius than template.

m. Cut two pieces of nylon film at least 6-inches larger than fiberglass fabric.

n. Place fiberglass fabric on one piece of nylon film.

o. Prepare adhesive, Item 35, Appendix D, for repair (PARA 16-4-71.3).

p. Impregnate fiberglass fabric. Resin content of fiberglass fabric shall be about 50% by weight when thoroughly impregnated.

q. Apply layer of nylon film on top of impregnated cloth and remove air bubbles.

r. Cut impregnated fiberglass fabric into six pieces as follows:

(1) Three pieces the size of template.

(2) One piece 0.5-inch larger in radius than template.

(3) One piece 1-inch larger in radius than template.

(4) One piece 1.5-inches larger in radius than template.

s. Fill void in inner skin with adhesive, using beverage stirring stick, Item 315C, Appendix D, and apply thin layer to inner skin surface, using acid brush, Item 84, Appendix D.

t. Remove nylon film from three template-sized pieces of fiberglass fabric and stack them to form 3-ply laminate in damaged cavity.

u. Apply layer of adhesive to all sides of honeycomb core and place in damaged cavity. Core must be slightly compressed to fit into position.

NOTE

If honeycomb core level does not fill cavity to the top, cut layers of impregnated fiberglass fabric and add as required.

v. Remove nylon film from fabric piece 0.5-inch larger than template and overlay damage. Center fabric as closely as possible.

w. Remove nylon from two remaining plies of oversized fabric and overlay damage, smallest piece first.

x. Remove masking tape.

y. Perform vacuum bag procedure (PARA 16-4-71.2).

CAUTION

Damage to tank and repair area will result if heat gun is not adjusted correctly when curing a repair. Incorrect positioning and temperature during cure cycle will result in a weak repair. Pay strict attention to temperature range, and adjust distance between heat source and surface as necessary (PARA 2-6-3).

z. Cure repair (PARA 2-6-3).

aa. Remove vacuum bag materials (PARA 16-4-71.2).

ab. Sand repaired area in preparation for replacing finish using abrasive cloth, Item 6, Appendix D.

ac. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

ad. Touch up paint finish (PARA 2-6-5).

ae. Perform pressure test procedure (PARA 16-4-70).

16-4-71.11 Class VI Repair.

a. Purge ESSS external 230 gallon fuel tank (PARA 1-3-21).

b. Remove ESSS external 230 gallon fuel tank access door nearest damage (PARA 16-4-63).

c. Determine class of damage (PARA 16-4-71.5).

NOTE

If damage is in either end of the tank where it can not be reached by hand through the access door for repair, do not attempt to prepare a Hytrel liner patch as described in the following paragraphs. Instead, proceed to Class VI-A repair instructions and follow the instructions for installing a perma plug.

d. Mask 2-inches around damaged area using 1-inch masking tape, Item 212, Appendix D (Figure 16-4-71-5).

e. Determine extent of damage in Hytrel liner, then cut and remove tank wall and Hytrel liner to conform to damage.

f. Cut and remove laminate material down to inner skin to conform to size of outer skin damage.

NOTE

Outer skin and honeycomb must be cut out minimum of 0.5-inch larger in radius than Hytrel liner damage.

g. Remove honeycomb residue from inner skin.

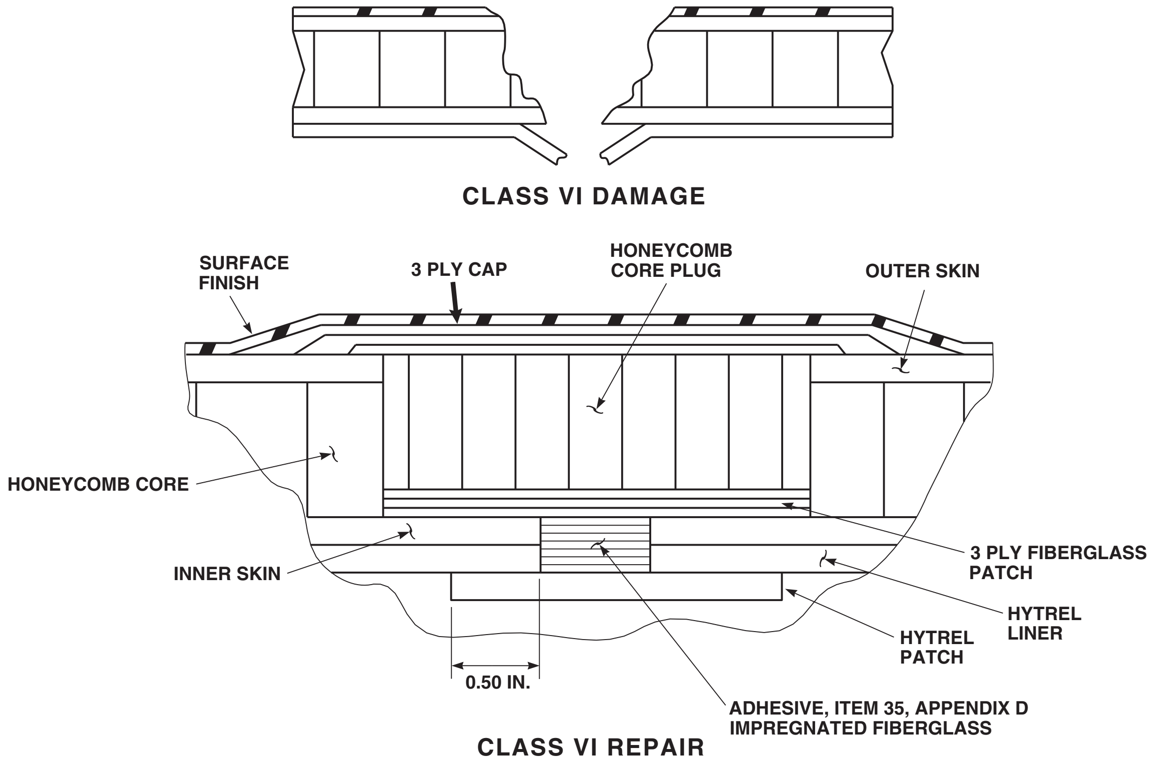


FIGURE 16-4-71-5. CLASS VI DAMAGE AND REPAIR

h. Verify damage classification is correct. If not, determine class of damage (PARA 16-4-71.5).

i. Fabricate template to match contour of hole in outer skin.

j. Sand off surface finish at least 2-inches around damaged area using abrasive cloth, Item 4, Appendix D.

k. Vacuum dust from repair area.

l. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

m. Cut one piece of Hytrel liner material 0.5-inch in radius larger than area cut out in Hytrel liner for patch.

n. Clean rough surface of Hytrel patch with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

o. Lightly sand inner surface of tank liner 1-inch in radius around edge of hole inside tank using abrasive cloth, Item 6, Appendix D.

p. Clean sanded surfaces with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

q. Prepare adhesive, Item 32, Appendix D, for repair (PARA 16-4-71.3).

r. Coat 0.5-inch of sanded surface around liner hole with adhesive, using acid brush, Item 84, Appendix D.

s. Coat rough side of Hytrel liner patch with adhesive 0.5-inch wide around outer edge.

t. Turn fuel tank face down so damaged area is on bottom and apply patch over hole.

u. Remove excessive adhesive with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D. Leave bead of adhesive around edge of patch both inside and outside of tank.

NOTE

A piece of peel ply may be applied over patch to prevent adhesive bonding weight to inside surface of fuel tank. Remove peel ply after repair is cured.

v. Apply weight to Hytrel patch with minimum of 1 pound per square-inch.

CAUTION

Damage to tank and repair area will result if heat gun is not adjusted correctly when curing a repair. Incorrect positioning and temperature during cure cycle will result in a weak repair. Pay strict attention to temperature range, and adjust distance between heat source and surface as necessary (PARA 2-6-3).

w. Cure repair (PARA 2-6-3).

x. Remove weight and turn fuel tank over so repair is on top.

y. Sand excess adhesive inside cavity using abrasive cloth, Item 6, Appendix D.

z. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

NOTE

Cut Hytrel, Nomex honeycomb, and fiberglass fabric on a clean surface. Keep free of dust, oil, water, and other contaminants.

aa. Cut one piece of Nomex honeycomb 0.125-inch larger than template.

ab. Cut one piece of fiberglass fabric large enough to contain six pieces 1.5-inches larger in radius than template.

ac. Cut two pieces of nylon film at least 6-inches larger than fiberglass fabric.

ad. Place fiberglass fabric on one piece of nylon film.

ae. Prepare adhesive, Item 35, Appendix D, for repair (PARA 16-4-71.3).

af. Impregnate fiberglass fabric. Resin content of fiberglass fabric shall be about 50% by weight when thoroughly impregnated.

ag. Apply layer of nylon film on top of impregnated cloth and remove air bubbles.

ah. Cut impregnated fiberglass fabric into six pieces as follows:

(1) Three pieces the size of template.

(2) One piece 0.5-inch larger in radius than template.

(3) One piece 1-inch larger in radius than template.

(4) One piece 1.5-inches larger in radius than template.

NOTE

If damage to liner and inner skin exceeds 0.5-inch diameter, cut layers of impregnated fiberglass to fill void. Otherwise, fill void in inner skin and Hytrel with adhesive, and apply a thin layer to inner skin surface.

ai. Remove nylon film from three template-sized pieces of fiberglass fabric, and stack to form 3-ply laminate in cavity.

aj. Apply layer of adhesive to all sides of honeycomb core and place in cavity. Core must be slightly compressed to fit.

NOTE

If honeycomb core level does not fill cavity to the top, cut layers of impregnated fiberglass fabric and add as required.

ak. Remove nylon film from fabric piece 0.5-inch larger than template and overlay damage. Center fabric as closely as possible.

al. Remove nylon from two remaining plies of oversized fabric and overlay damage, smallest piece first.

am. Remove masking tape.

an. Perform vacuum bag procedure (PARA 16-4-71.2).

CAUTION

Damage to tank and repair area will result if heat gun is not adjusted correctly when curing a repair. Incorrect positioning and temperature during cure cycle will result in a weak repair. Pay strict attention to temperature range, and adjust distance between heat source and surface as necessary (PARA 2-6-3).

ao. Cure repair (PARA 2-6-3).

ap. Remove vacuum bag materials (PARA 16-4-71.2).

aq. Sand repaired area in preparation for replacing finish using abrasive cloth, Item 6, Appendix D.

ar. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

as. Touch up paint finish (PARA 2-6-5).

at. Perform pressure test procedure (PARA 16-4-70).

16-4-71.12 Class VI-A Repair.

a. Purge ESSS external 230 gallon fuel tank (PARA 1-3-21).

b. Remove access door nearest damage (PARA 16-4-63).

c. Determine class of damage (PARA 16-4-71.5).

NOTE

If Hytrel liner patch cannot be installed because of damage being in an inaccessible area in fuel tank, install perma plug as described in the following instructions.

d. Mask 2-inches around damaged area using 1-inch masking tape, Item 212, Appendix D (Figure 16-4-71-6).

e. Determine extent of damage in Hytrel liner, then cut and remove tank wall and Hytrel liner to conform to damage.

NOTE

For installation of perma plug, there must be at least 0.5-inch minimum hole cut in the liner.

f. Cut and remove outer skin and honeycomb down to inner skin to conform to size of outer skin damage.

NOTE

Outer skin and honeycomb must be cut out minimum of 0.5-inch larger in radius than Hytrel liner damage.

g. Remove honeycomb residue from inner skin.

h. Verify damage classification is correct. If not, determine class of damage (PARA 16-4-71.5).

i. Fabricate template to match contour of hole in outer skin.

j. Sand off surface finish at least 2-inches around damaged area using abrasive cloth, Item 4, Appendix D.

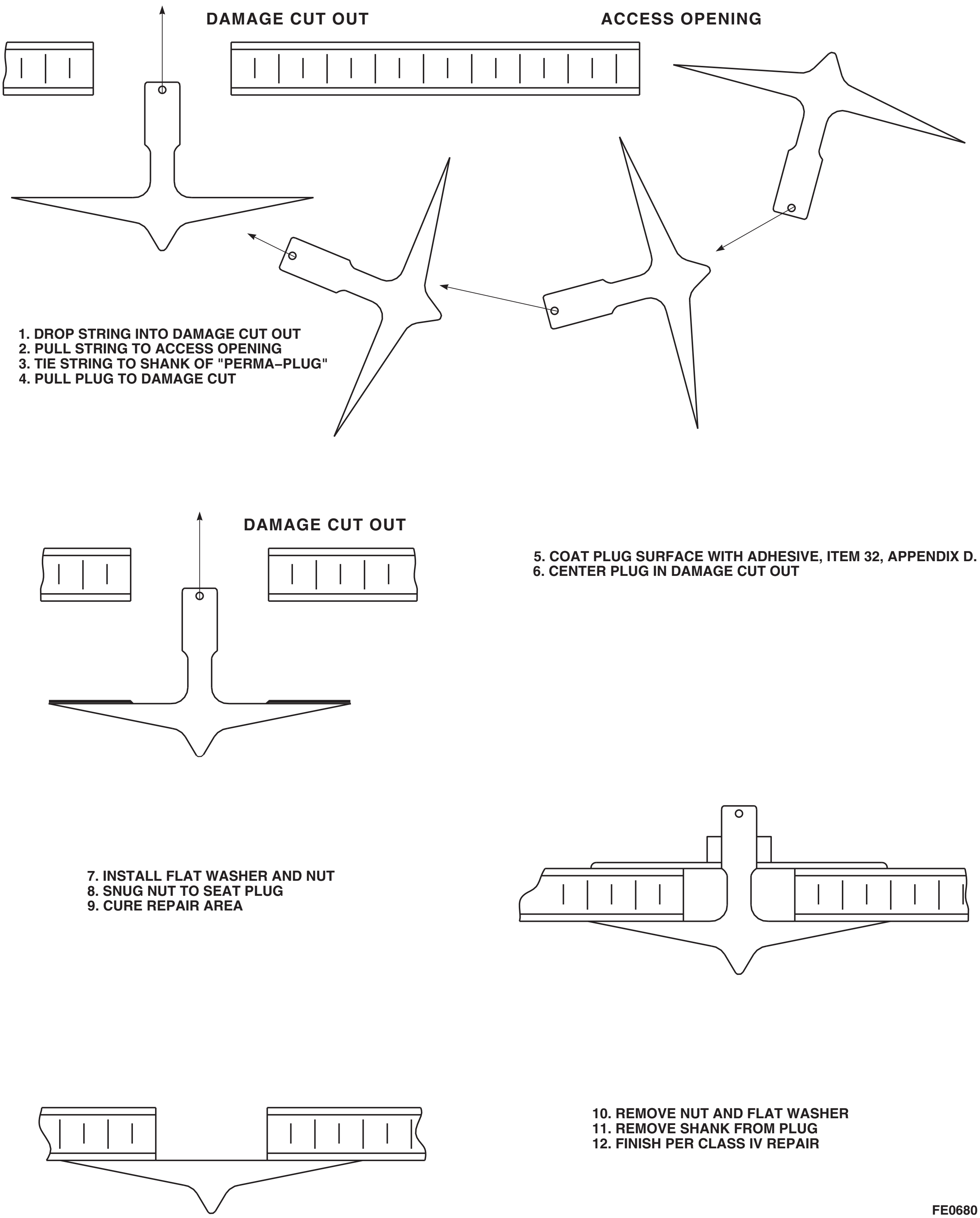


FIGURE 16-4-71-6. CLASS VI-A PERMA PLUG REPAIR

k. Vacuum dust from repair area.

l. If possible, lightly sand inner surface of tank liner 1.5-inches in radius around edge of hole using abrasive cloth, Item 6, Appendix D.

m. Clean sanded surfaces with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

n. Obtain perma plug 1-inch larger in radius than Hytrel damage. Remove nut and washer.

o. Sand bonding surface of perma plug using abrasive cloth, Item 6, Appendix D.

p. Clean bonding surface of perma plug with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

q. Attach cord to hole in end of threaded shaft of perma plug.

r. Feed end of cord into damaged hole and out access door.

s. Insert threaded shaft into perma plug.

t. Prepare adhesive, Item 32, Appendix D (PARA 16-4-71.3).

u. Apply adhesive to bonding surface of perma plug, using acid brush, Item 84, Appendix D.

v. If possible, coat 1-inch of sanded surface around hole with adhesive.

w. Pull cord through damaged hole to position perma plug up against damage hole inside tank.

NOTE

Make sure inner surface of plug makes full contact with surface of liner and is not distorted from being pulled in too tight.

x. Install washer and nut in shaft of perma plug. Secure plug in place.

NOTE

Make sure plug is centered in damage cut out area.

y. Cure repair (PARA 2-6-3).

CAUTION

Damage to tank and repair area will result if heat gun is not adjusted correctly when curing a repair. Incorrect positioning and temperature during cure cycle will result in a weak repair. Pay strict attention to temperature range, and adjust distance between heat source and surface as necessary (PARA 2-6-3).

z. Remove washer and nut from shaft of perma plug.

aa. Remove shaft from perma plug.

ab. Sand excess adhesive inside cavity using abrasive cloth, Item 6, Appendix D.

ac. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

NOTE

Cut Hytrel, Nomex honeycomb, and fiberglass fabric on a clean surface. Keep free of dust, oil, water, and other contaminants.

ad. Cut one piece of Nomex honeycomb 0.125-inch larger than template.

ae. Cut one piece of fiberglass fabric large enough to contain six pieces 1.5-inches larger in radius than template.

af. Cut two pieces of nylon film at least 6-inches larger than fiberglass fabric.

ag. Place fiberglass fabric on one piece of nylon film.

ah. Prepare adhesive, Item 35, Appendix D, for repair (PARA 16-4-71.3).

ai. Impregnate fiberglass fabric. Resin content of fiberglass fabric shall be about 50% by weight when thoroughly impregnated.

aj. Apply layer of nylon film on top of impregnated cloth and remove air bubbles.

ak. Cut impregnated fiberglass fabric into six pieces as follows:

(1) Three pieces the size of template.

(2) One piece 0.5-inch larger in radius than template.

(3) One piece 1-inch larger in radius than template.

(4) One piece 1.5-inches larger in radius than template.

al. Fill hole in perma plug with adhesive, using beverage stirring stick, Item 315C, Appendix D.

am. If damage to liner and inner skin exceeds 0.5-inch diameter, cut layers of impregnated fiberglass to fill void. Otherwise, fill void in inner skin and Hytrel with adhesive, and apply thin layer to inner skin surface.

an. Remove nylon film from three template-sized pieces of fiberglass fabric, and stack to form 3-ply laminate in cavity.

ao. Apply layer of adhesive to all sides of honeycomb core and place in cavity. Core must be slightly compressed to fit.

NOTE

If honeycomb core level does not fill cavity to the top, cut layers of impregnated fiberglass fabric and add as required.

ap. Remove nylon film from fabric piece 0.5-inch larger than template and overlay damage. Center fabric as closely as possible.

aq. Remove nylon from two remaining plies of oversized fabric and overlay damage, smallest piece first.

ar. Remove masking tape.

as. Perform vacuum bag procedure (PARA 16-4-71.2).

CAUTION

Damage to tank and repair area will result if heat gun is not adjusted correctly when curing a repair. Incorrect positioning and temperature during cure cycle will result in a weak repair. Pay strict attention to temperature range, and adjust distance between heat source and surface as necessary (PARA 2-6-3).

at. Cure repair (PARA 2-6-3).

au. Remove vacuum bag materials (PARA 16-4-71.2).

av. Sand repaired area in preparation for replacing finish using abrasive cloth, Item 6, Appendix D.

aw. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

ax. Touch up paint finish (PARA 2-6-5).

ay. Perform pressure test procedure (PARA 16-4-70).

16-4-71.13 Class VII Repair.

a. Determine class of damage (PARA 16-4-71.5).

NOTE

When removing molding compound, take care not to drill through to other side.

b. Using small router bit or hole saw, remove molding compound from around damaged insert (Figure 16-4-71-7, Step A).

c. Remove damaged insert.

d. Locate three serviceable inserts directly opposite of damaged insert for use in developing insert locating tool.

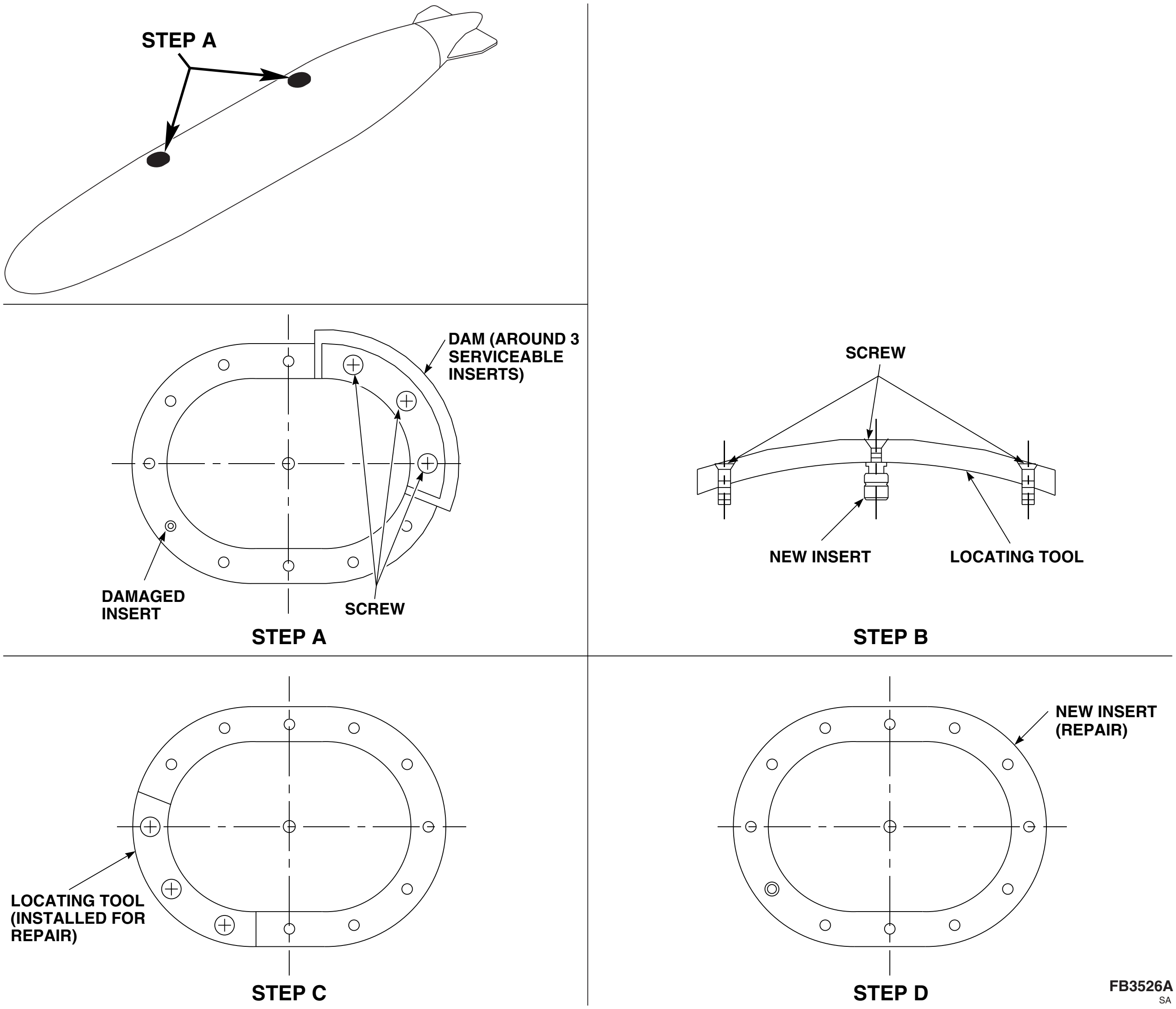


FIGURE 16-4-71-7. CLASS VII REPAIR, ACCESS DOOR INSERT REPAIR

e. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

f. Spread film of petrolatum, Item 234, Appendix D, around all surfaces near three serviceable inserts.

g. Spread film of petrolatum, Item 234, Appendix D, on three mounting screws. Insert screws into three serviceable inserts and turn in at least two threads.

h. Construct dam around three screws to provide mold for developing locating tool.

i. Fill mold with body filler or plaster up to taper on mounting screwheads, using beverage stirring stick, Item 315C, Appendix D.

j. Cure filler material per manufacturer's specifications to produce locating tool.

k. Unscrew three screws from access door, then remove locating tool from access door.

l. Spread film of petrolatum, Item 234, Appendix D, around locating tool and three screws.

m. Insert three screws into locating tool, and screw new insert onto middle screw (Figure 16-4-71-7, Step B).

n. Prepare adhesive, Item 35, Appendix D, for repair (PARA 16-4-71.3).

o. Fill damaged insert hole half full of adhesive.

p. Place locating tool with insert in position on damaged insert area (Figure 16-4-71-7, Step C).

q. Tighten outside screws of locating tool and secure in place.

CAUTION

Damage to tank and repair area will result if heat gun is not adjusted correctly when curing a repair. Incorrect positioning and temperature during cure cycle will result in a weak repair. Pay strict attention to temperature range, and adjust distance between heat source and surface as necessary (PARA 2-6-3).

r. Cure repair (PARA 2-6-3).

s. Remove locating tool.

t. Sand off excessive adhesive using abrasive cloth, Item 6, Appendix D.

u. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D (Figure 16-4-71-7, Step D).

16-4-71.14 Class VII-A Repair.

a. Determine class of damage (PARA 16-4-71.5).

NOTE

When removing molding compound, take care not to drill through to other side.

b. Using small router bit or hole saw, remove molding compound from around damaged insert. Take care not to drill through to other side (Figure 16-4-71-8, Step A).

c. Remove damaged insert.

d. Locate three to seven adjacent serviceable inserts for use in developing insert locating tool (Figure 16-4-71-8, Step B).

e. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

f. Spread film of petrolatum, Item 234, Appendix D, around all surfaces near three serviceable inserts.

g. Spread film of petrolatum, Item 234, Appendix D, on three to six mounting screws. Insert screws into three to six serviceable inserts and turn in at least two threads.

h. Construct dam around three to seven screws to provide mold for developing locating tool.

i. Fill mold with body filler or plaster up to taper on mounting screwheads, using beverage stirring stick, Item 315C, Appendix D.

j. Cure filler material per manufacturer's specifications to produce locating tool.

k. Unscrew three to seven screws from stabilizer fin, then remove locating tool from tail fin.

l. Spread film of petrolatum, Item 234, Appendix D, around locating tool and three to seven screws.

m. Insert three to seven screws into locating tool, and screw new insert onto middle screw (Figure 16-4-71-8, Step C).

n. Prepare adhesive, Item 35, Appendix D, for repair (PARA 16-4-71.3).

o. Fill damaged insert hole half full of adhesive.

p. Place locating tool with insert in position on damaged insert area.

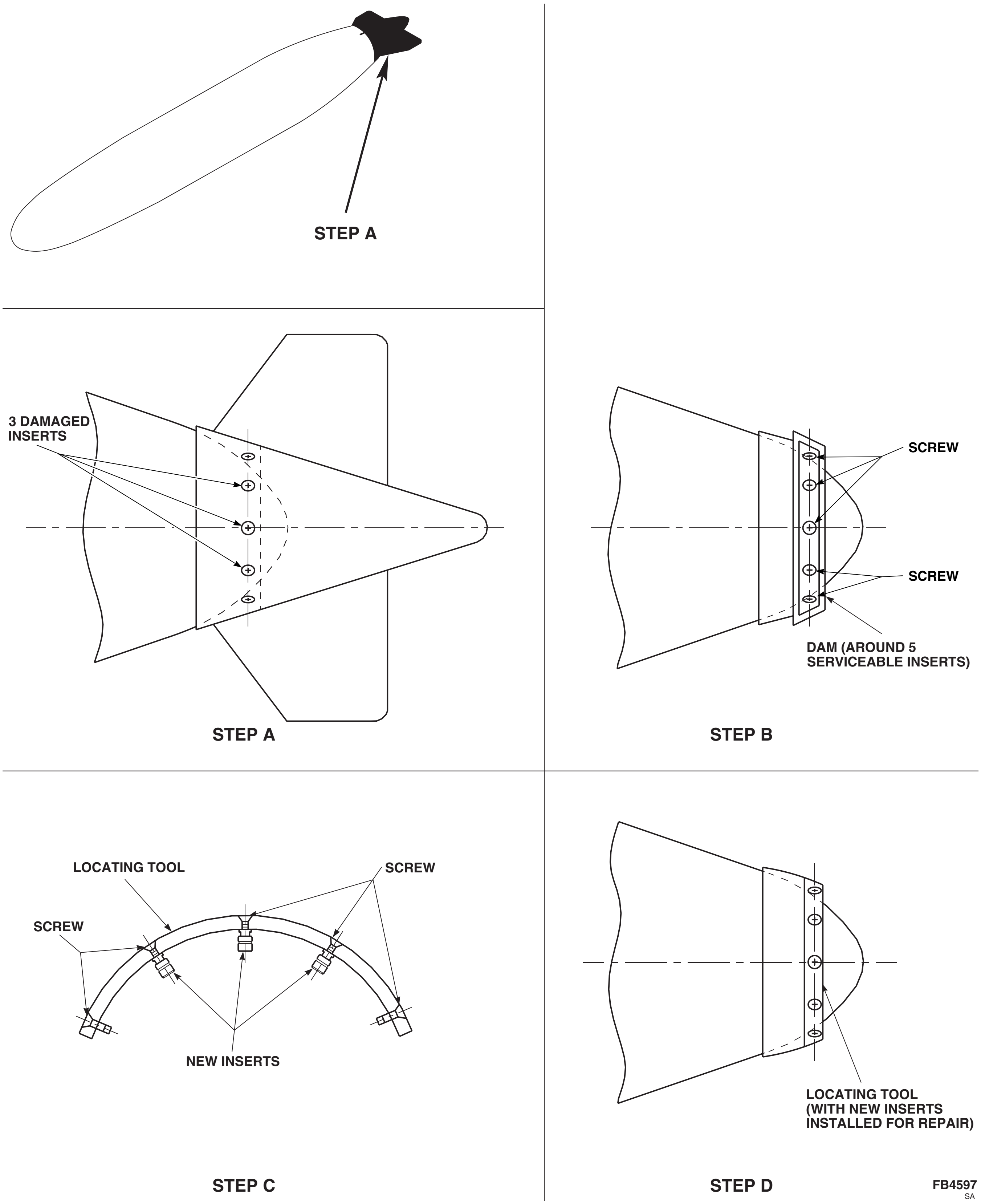


FIGURE 16-4-71-8. CLASS VII-A REPAIR, STABILIZER FIN MOUNTING INSERT

q. Tighten outside screws of locating tool and secure in place.

CAUTION

Damage to tank and repair area will result if heat gun is not adjusted correctly when curing a repair. Incorrect positioning and temperature during cure cycle will result in a weak repair. Pay strict attention to temperature range, and adjust distance between heat source and surface as necessary (PARA 2-6-3).

r. Cure repair (PARA 2-6-3).

s. Remove locating tool.

t. Sand off excess adhesive using abrasive cloth, Item 6, Appendix D.

u. Wipe area clean with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D (Figure 16-4-71-8, Step D).

16-4-71.15 Class VIII Repair.

a. Prepare adhesive, Item 32, Appendix D, for repair (PARA 16-4-71.3).

b. Inside tank, place bead of adhesive around bond joint of liner-to-bulkhead fitting.

c. Orient tank, as required, to prevent adhesive from running.

CAUTION

Damage to tank and repair area will result if heat gun is not adjusted correctly when curing a repair. Incorrect positioning and temperature during cure cycle will result in a weak repair. Pay strict attention to temperature range, and adjust distance between heat source and surface as necessary (PARA 2-6-3).

d. Cure repair (PARA 2-6-3).

16-4-71.16 Repair Bullet Hole.

INSERT QUICK-FIX PLUG IN DAMAGE HOLE AND TIGHTEN SNUG (Figure 16-4-71-11).

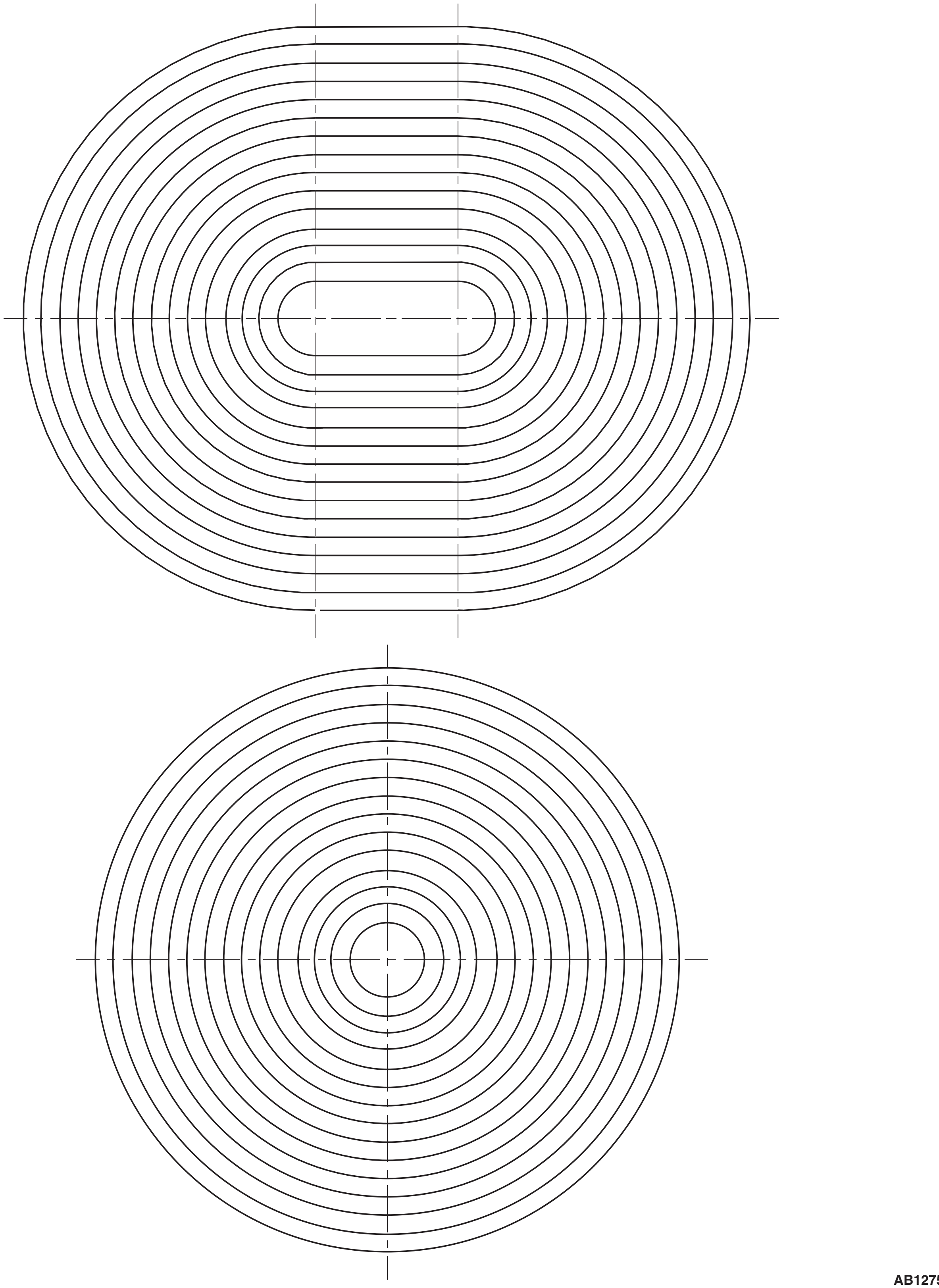
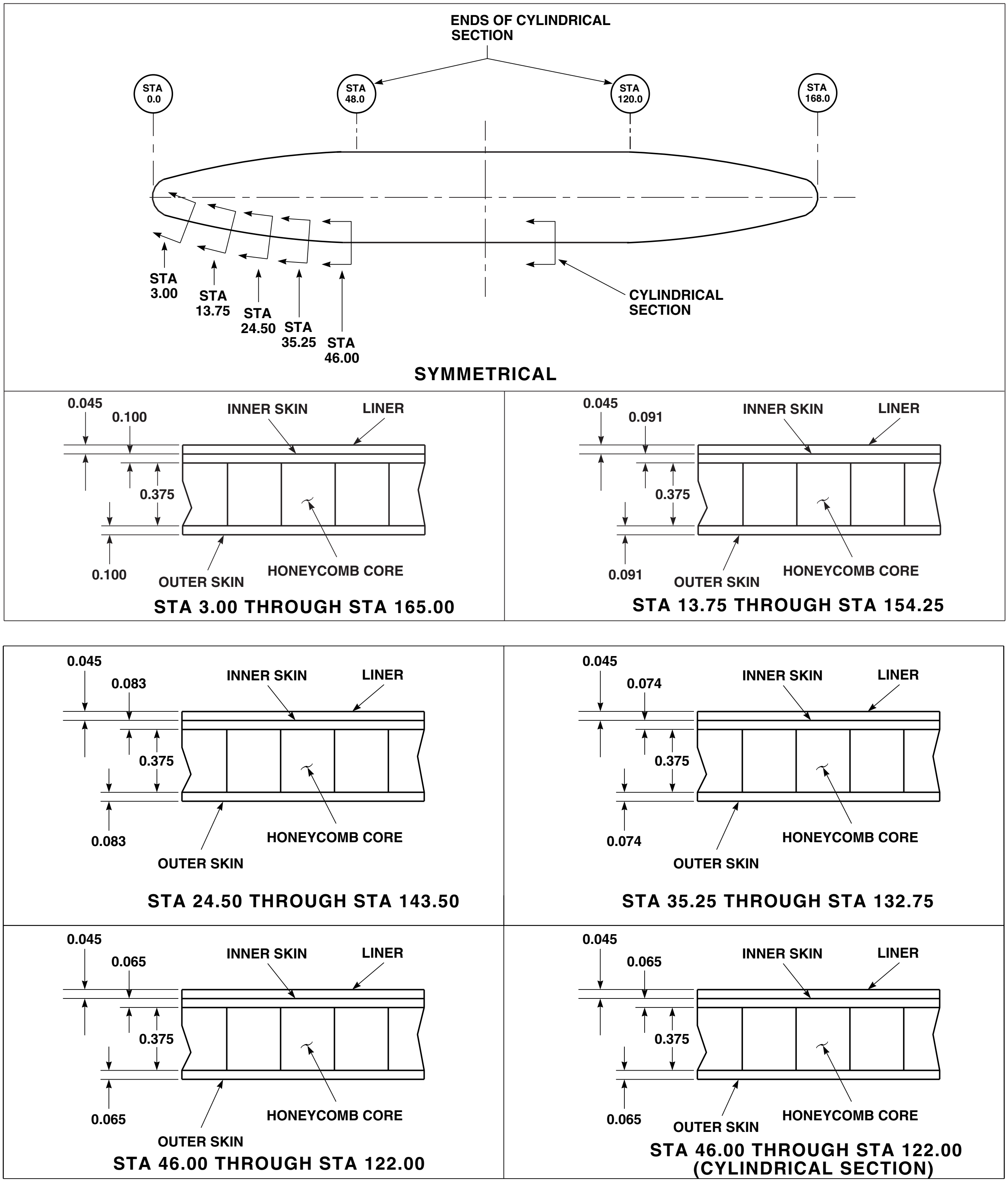


FIGURE 16-4-71-9. TANK REPAIR TEMPLATES



NOTE

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

FN2220A
SA

FIGURE 16-4-71-10. EXTERNAL 230 GALLON FUEL TANK DIMENSIONS

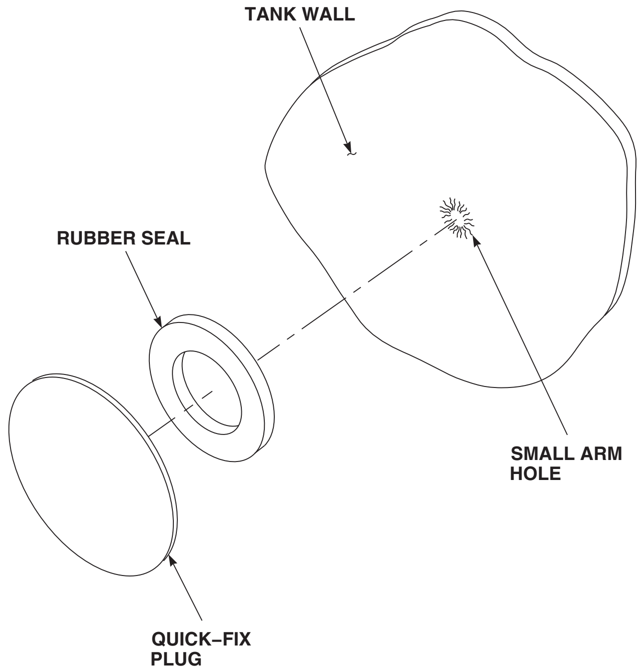


FIGURE 16-4-71-11. REPAIR BULLET HOLE

16-4-72 APU ENGINE AIR PARTICLE SEPARATOR (EAPS) AIR PARTICLE SEPARATOR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Electrical Repairer's Toolkit
- Powerplant Repairer's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 54, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-110
- PARA 7-4-4
- PARA 15-4-1
- PARA 15-4-2
- PARA 16-4-73

16-4-72.1 Replace APU Engine Air Particle Separator (EAPS) Air Particle Separator.

16-4-72.1.1. Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-16 and PARA 1-6-15).

b. Open APU compartment doors.

c. Remove APU access door beam (PARA 2-4-110).

d. Disconnect APU exhaust duct from rear of APU (PARA 15-4-2).

e. Remove collector box (PARA 16-4-73).

f. Remove APU (PARA 15-4-1).

g. Remove bolts and washers securing upper panel half to upper particle separator (Figure 16-4-72-1, Detail A).

h. Remove screws and washers securing upper panel half to shroud. Remove upper panel half.

i. Remove bolts and washers securing lower panel half to lower particle separator.

j. Remove screws and washers securing lower panel half to shroud. Allow lower panel half to hang freely from bleed air outlet flange.

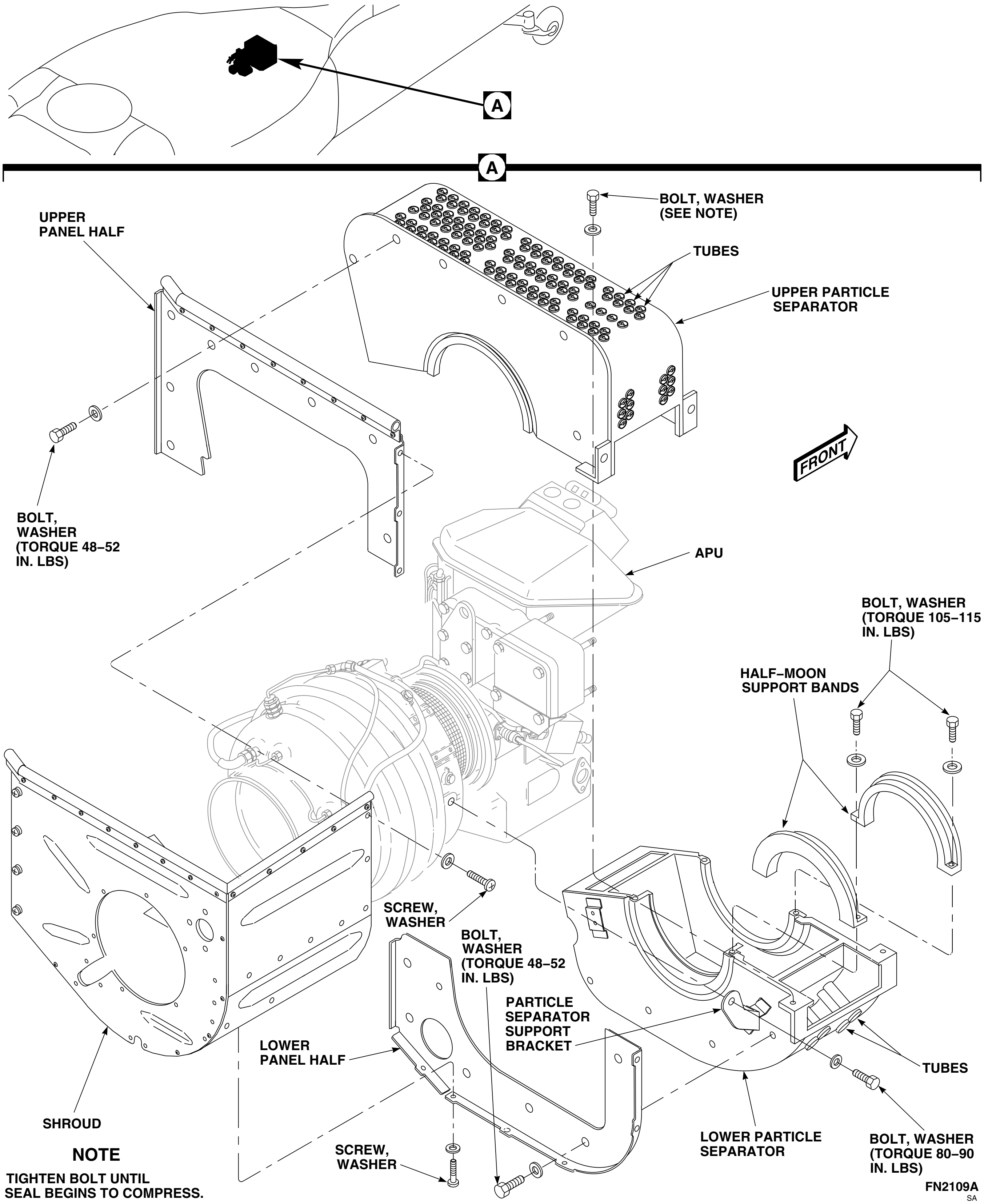


FIGURE 16-4-72-1. REPLACE APU ENGINE AIR PARTICLE SEPARATOR (EAPS) AIR PARTICLE SEPARATOR

k. Remove bolts and washers securing upper particle separator to lower particle separator. Remove upper particle separator.

l. Remove bolts and washers securing large half-moon support band to lower particle separator. Remove support band.

m. Remove bolts and washers securing small half-moon support band to lower particle separator. Remove support band.

n. Remove bolts and washers securing particle separator support brackets and lower particle separator to APU. Remove lower particle separator.

16-4-72.1.2. Inspection of Particle Separator Inlet and Outlet Tubes.

NOTE

Damaged or missing tubes must be distributed over a wide area of the particle separator. If damaged or missing tubes are concentrated in one area, the particle separator **MUST BE REPLACED.**

Inspect the inlet and outlet tubes in the upper and lower particle separator (Figure 16-4-72-1, Detail A). **THE MAXIMUM NUMBER OF DAMAGED OR MISSING TUBES SHALL NOT EXCEED 10% OF THE TOTAL NUMBER OF TUBES IN UPPER OR LOWER PARTICLE SEPARATOR. IF DAMAGED OR MISSING TUBES EXCEEDS 10%, REPLACE UPPER OR LOWER PARTICLE SEPARATOR.**

16-4-72.1.3. Install.

a. Position lower particle separator under APU and install particle separator support bracket and particle separator to APU with bolts and washers (Figure 16-4-72-1, Detail A). **TORQUE BOLTS TO 80 - 90 INCH-POUNDS.**

b. Install small half-moon support band over front of APU inlet with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

c. Install large half-moon support band over rear of APU inlet with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

d. Position upper particle separator over APU and install to lower particle separator with bolts and washers. **TIGHTEN BOLTS UNTIL SEAL BEGINS TO COMPRESS.**

e. Apply a light coat of adhesive, Item 54, Appendix D, to mating surfaces of lower panel half and lower particle separator.

f. Install lower panel half to lower particle separator with bolts and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

g. Secure lower panel half to shroud with screws and washers.

h. Apply a light coat of adhesive, Item 54, Appendix D, to mating surfaces of upper panel half and upper particle separator.

i. Install upper panel half to upper particle separator with bolts and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

j. Secure upper panel half to shroud with screws and washers.

k. Inspect APU check valve (PARA 7-4-4).

l. Install APU (PARA 15-4-1).

m. Install collector box (PARA 16-4-73).

n. Connect APU exhaust duct (PARA 15-4-2).

o. Install APU access door beam (PARA 2-4-110).

p. Make sure area is clean and free of foreign material.

q. Close and latch APU compartment doors.

16-4-73 APU ENGINE AIR PARTICLE SEPARATOR (EAPS) COLLECTOR BOX.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Powerplant Repairer's Toolkit

References

- PARA 1-6-15
- PARA 1-6-16

16-4-73.1 Replace APU Engine Air Particle Separator (EAPS) Collector Box.

16-4-73.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open APU compartment doors.

c. Loosen clamp securing flex coupling to rear of collector box (Figure 16-4-73-1, Detail A).

d. Remove screw and washer securing upper ejector support bracket to ejector.

e. Remove screw and washer securing lower ejector support bracket to ejector.

f. Loosen nuts securing bleed-air fitting assembly to unions on front of collector box. Remove fitting assembly from collector box.

g. Remove bolts and washers securing collector box to upper and lower particle separators. Remove collector box.

16-4-73.1.2. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Position collector box over upper and lower particle separators with ejector facing rear. Secure collector box to separators with bolts and washers (Figure 16-4-73-1, Detail A). **TIGHTEN BOLTS UNTIL SEAL BEGINS TO COMPRESS.**

c. Install bleed-air fitting assembly on unions on front of collector box and tighten nuts.

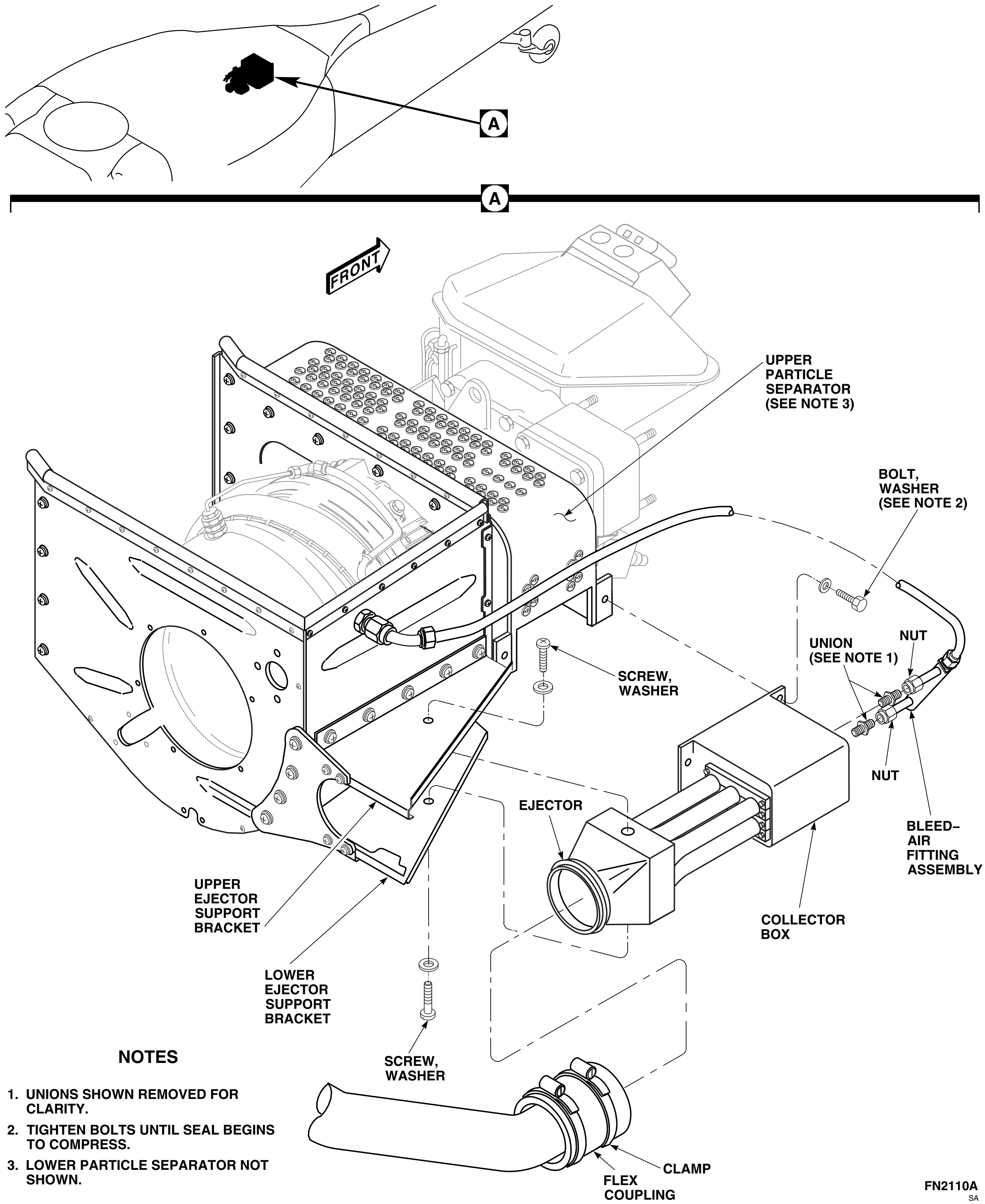
d. Secure lower ejector support bracket to ejector with screw and washer.

e. Secure upper ejector support bracket to ejector with screw and washer.

f. Slide flex coupling over ejector and secure coupling to ejector with clamp.

g. Make sure area is clean and free of foreign material.

h. Close and latch APU compartment doors.



NOTES

1. UNIONS SHOWN REMOVED FOR CLARITY.
2. TIGHTEN BOLTS UNTIL SEAL BEGINS TO COMPRESS.
3. LOWER PARTICLE SEPARATOR NOT SHOWN.

FIGURE 16-4-73-1. REPLACE APU ENGINE AIR PARTICLE SEPARATOR (EAPS) COLLECTOR BOX

16-4-74 APU ENGINE AIR PARTICLE SEPARATOR (EAPS) SCAVENGE EXHAUST DUCT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Powerplant Repairer's Toolkit
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 1-6-15
- PARA 1-6-16
- PARA 15-4-2
- PARA 16-4-73

16-4-74.1 Replace APU Engine Air Particle Separator (EAPS) Scavenge Exhaust Duct.

16-4-74.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Disconnect APU exhaust duct from rear of APU (PARA 15-4-2).

c. Remove collector box (PARA 16-4-73).

d. Loosen clamp securing scavenge exhaust duct to flex coupling. Remove flex coupling (Figure 16-4-74-1, Detail A).

e. Check flex coupling for cracks, tears, and signs of deterioration. **NONE ALLOWED.** If damaged, replace flex coupling.

f. Remove bolts, washers, and nuts securing scavenge exhaust duct to scavenge exhaust duct flange.

g. Remove scavenge exhaust duct from flange and remove from helicopter.

16-4-74.1.2. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Position APU scavenge exhaust duct in scavenge exhaust duct flange and secure to flange with bolts, washers, and nuts (Figure 16-4-74-1, Detail A). **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

c. Position flex coupling on scavenge exhaust duct. **SLIDE CLAMP OVER COUPLING AND SECURE COUPLING TO DUCT BY TIGHTENING CLAMP.**

d. Install collector box (PARA 16-4-73).

e. Connect APU exhaust duct to rear of APU (PARA 15-4-2).

f. Make sure area is clean and free of foreign material.

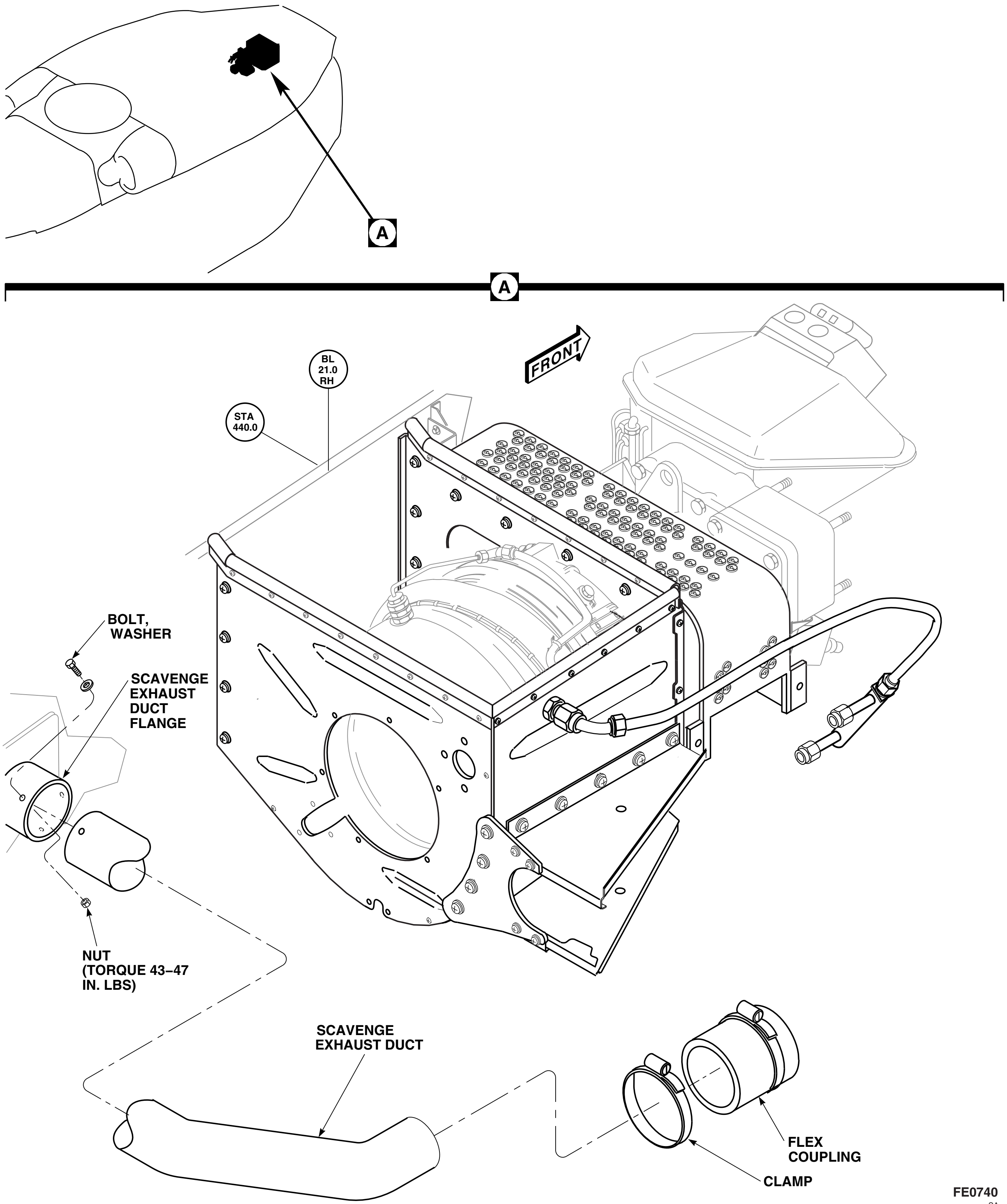


FIGURE 16-4-74-1. REPLACE APU ENGINE AIR PARTICLE SEPARATOR (EAPS) SCAVENGE EXHAUST DUCT

16-4-75 APU ENGINE AIR PARTICLE SEPARATOR (EAPS) EJECTOR SUPPORT BRACKETS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Powerplant Repairer's Toolkit

References

- PARA 1-6-15
- PARA 1-6-16

16-4-75.1 Replace APU Engine Air Particle Separator (EAPS) Ejector Support Brackets.

16-4-75.1.1 Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Open auxiliary power unit compartment doors.

c. Remove screws and washers securing upper/lower ejector support bracket to shroud and rear ejector support bracket (Figure 16-4-75-1, Detail A).

d. Remove screws and washers securing upper/lower support brackets to ejector. Remove upper/lower support brackets.

e. Remove screws and washers securing rear ejector support bracket to rear panel. Remove rear support bracket.

16-4-75.1.2 Install.

a. Install upper/lower ejector support bracket to shroud with screws and washers (Figure 16-4-75-1, Detail A).

b. Secure upper/lower support brackets to ejector with screws and washers.

c. Install rear ejector support bracket to rear panel with screws and washers.

d. Secure rear support bracket to upper and lower support brackets with screws and washers.

e. Make sure area is clean and free of foreign material.

f. Close auxiliary power unit compartment doors.

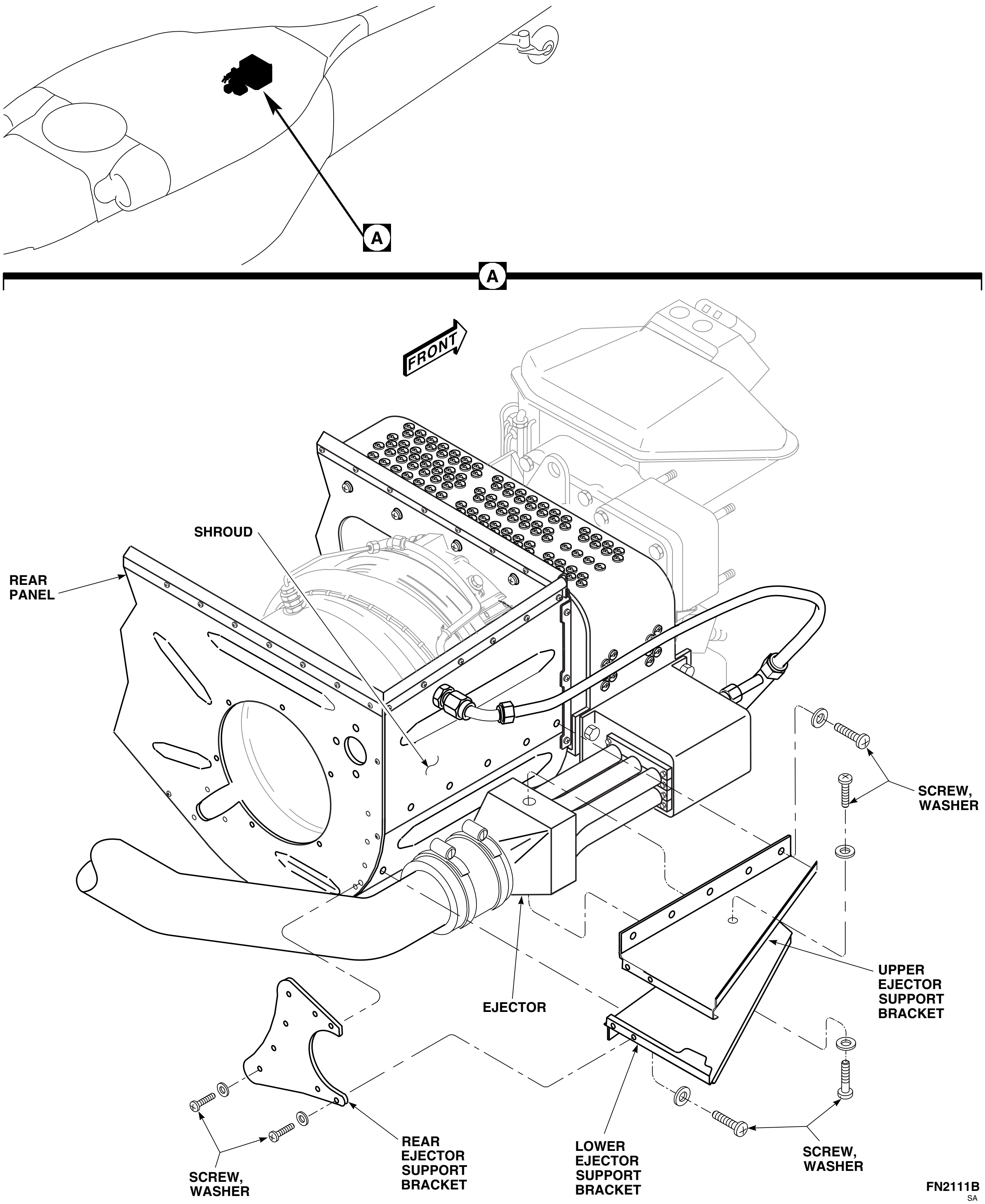


FIGURE 16-4-75-1. REPLACE APU ENGINE AIR PARTICLE SEPARATOR (EAPS) EJECTOR SUPPORT BRACKETS

SECTION 16-5

MAINTENANCE PROCEDURES (BENCH)

SECTION OVERVIEW

16-5-1 ESSS STORES JETTISON CONTROL PANEL ROTARY SELECTOR SWITCH (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 16-2-5
- PARA 16-4-51

16-5-1.1 Replace ESSS Stores Jettison Control Panel Rotary Selector Switch (BENCH).

16-5-1.1.1. Remove.

a. Remove information plate (PARA 16-4-51).

b. Remove screws and cover from stores jettison control panel (Figure 16-5-1-1).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

c. Tag wiring. Using soldering iron, unsolder wiring from rotary selector switch.

d. Remove nut, lockwasher, and washer securing rotary selector switch to stores jettison control panel. Remove switch.

16-5-1.1.2. Install.

a. Place rotary selector switch through opening in front of stores jettison control panel and fasten with nut, lockwasher, and washer (Figure 16-5-1-1).

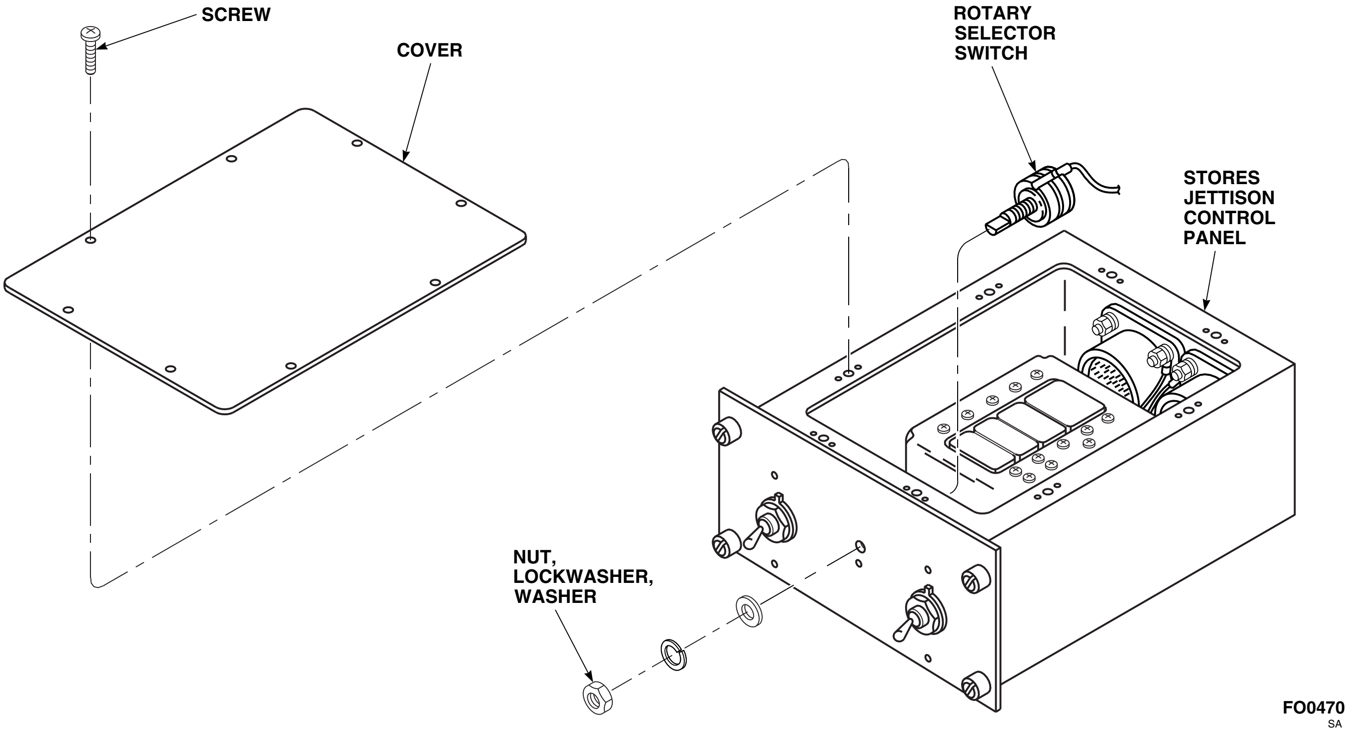


FIGURE 16-5-1-1. REPLACE ESSS STORES JETTISON CONTROL PANEL ROTARY SELECTOR SWITCH (BENCH)

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

b. Using soldering iron, solder, Item 313, Appendix D, wiring to rotary selector switch. Remove tags.

c. Make sure area is clean and free of foreign material.

d. Install cover on stores jettison control panel and secure with screws.

e. Install information plate (PARA 16-4-51).

f. Perform operational check of stores jettison control panel (PARA 16-2-5).

16-5-2 ESSS STORES JETTISON CONTROL PANEL TOGGLE SWITCHES (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

Materials/Parts

- Tags

References

- PARA 16-2-5
- PARA 16-4-51

16-5-2.1 Replace ESSS Stores Jettison Control Panel Toggle Switches (BENCH).

NOTE

Replacement of either toggle switch is the same.

16-5-2.1.1. Remove.

a. Remove information plate (PARA 16-4-51).

b. Remove screws and cover from stores jettison control panel (Figure 16-5-2-1).

c. Remove nut, lockwasher, and lockring securing toggle switch to stores jettison control panel. Remove switch.

d. Tag wiring. Remove screws and lockwashers securing wiring to toggle switch terminals.

16-5-2.1.2. Install.

a. Connect wiring to toggle switch terminals with screws and lockwashers (Figure 16-5-2-1). Remove tags.

b. Place toggle switch through opening in front of stores jettison control panel and fasten with nut, lockwasher, and lockring.

c. Make sure area is clean and free of foreign material.

d. Install cover on stores jettison control panel and secure with screws.

e. Install information plate (PARA 16-4-51).

f. Perform operational check of stores jettison control panel (PARA 16-2-5).

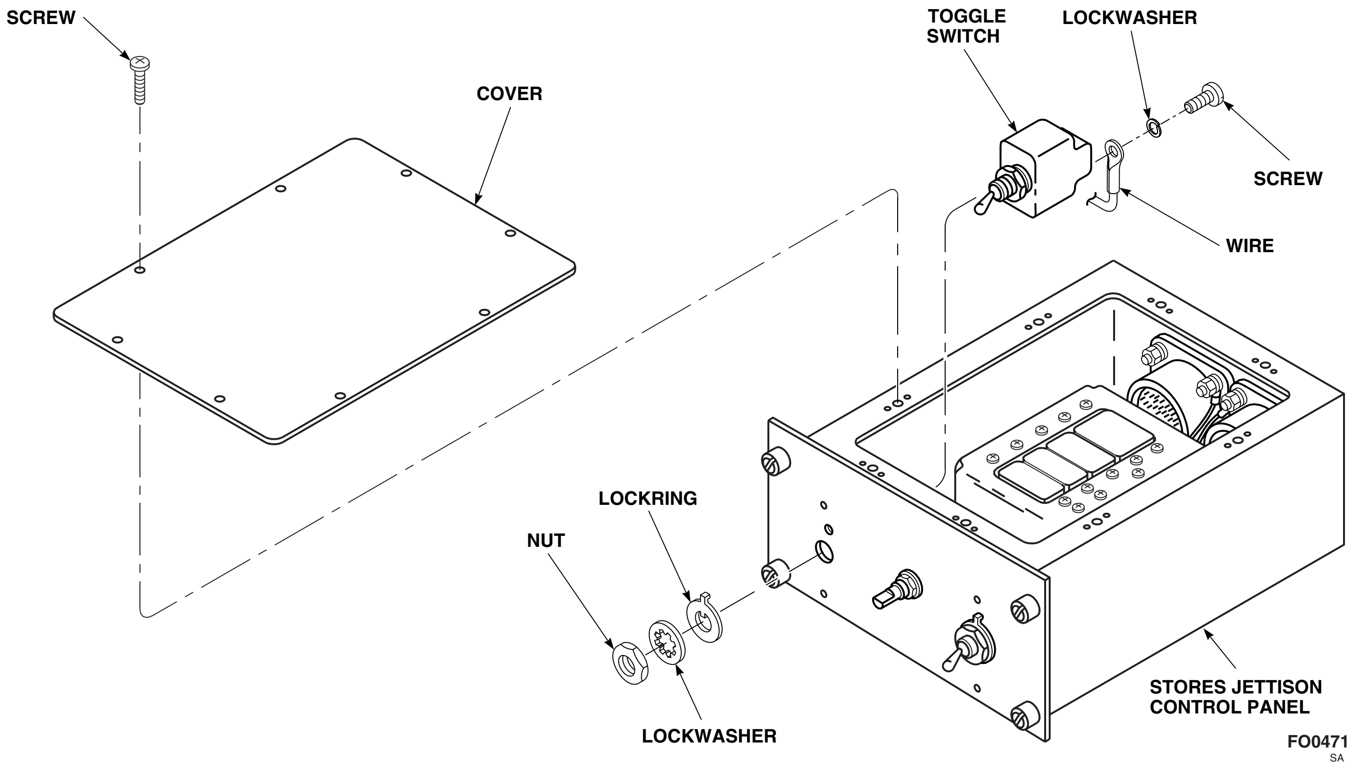


FIGURE 16-5-2-1. REPLACE ESSS STORES JETTISON CONTROL PANEL TOGGLE SWITCHES (BENCH)

16-5-3 ESSS STORES JETTISON CONTROL PANEL RELAYS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Heat Gun
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Insulation Sleeving, Heat-Shrinkable, Item 161, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 16-2-5

16-5-3.1 Replace ESSS Stores Jettison Control Panel Relays (BENCH).

NOTE

Replacement procedure for relays, K1 through K7, is the same.

16-5-3.1.1. Remove.

a. Remove screws and cover from stores jettison control panel (Figure 16-5-3-1).

b. Remove screws and washers securing relay bracket assembly to stores jettison control panel.

c. Slide insulation sleeving back from relay terminals.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

d. Tag wiring. Using soldering iron, unsolder and remove wiring from relay terminals (Figure 16-5-3-1, Detail A).

e. Remove screws, washers, and nuts securing relay to relay bracket assembly. Remove relay.

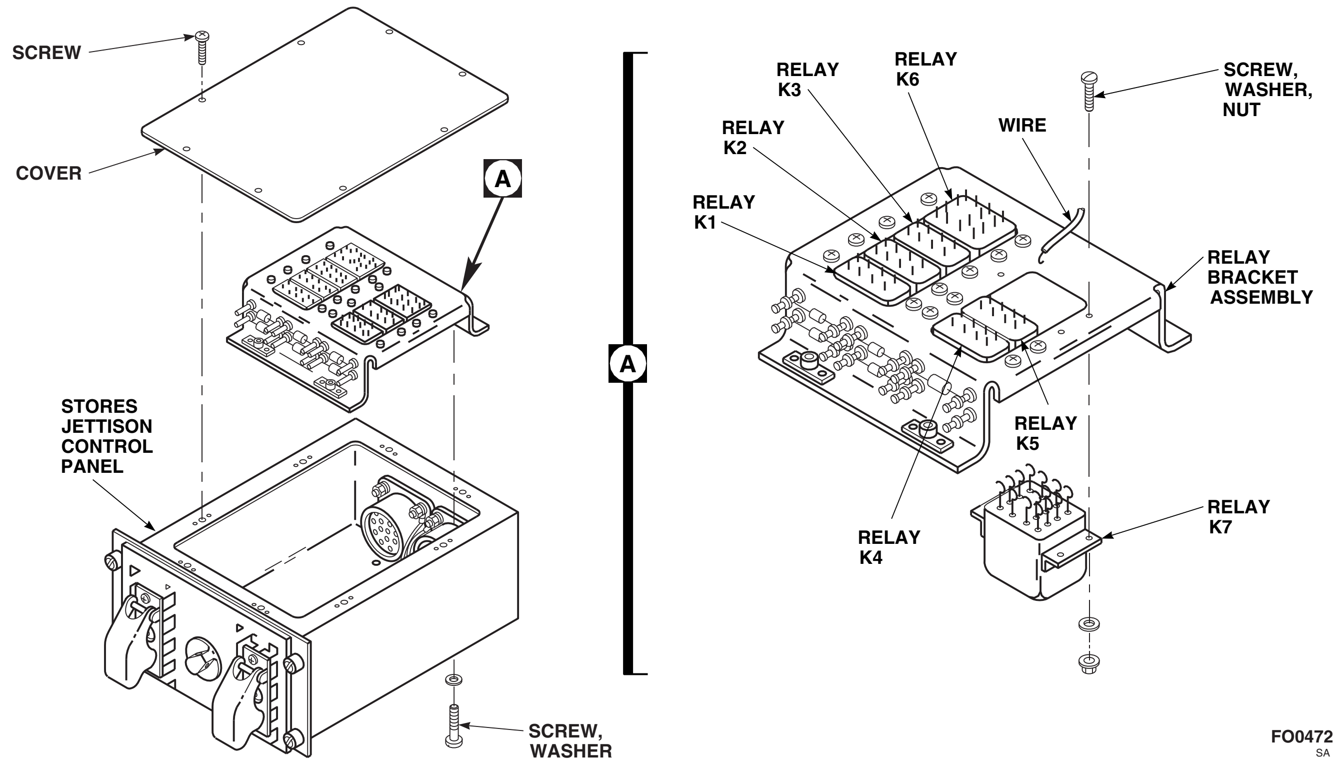


FIGURE 16-5-3-1. REPLACE ESSS STORES JETTISON CONTROL PANEL RELAYS (BENCH)

16-5-3.1.2. Install.

a. Position relay on relay bracket assembly and secure with screws, washers, and nuts (Figure 16-5-3-1, Detail A).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

b. Using soldering iron, solder, Item 313, Appendix D, wiring to relay terminals. Remove tags.

c. Slide heat-shrinkable insulation sleeving, Item 161, Appendix D, over relay terminals.

d. Using heat gun, shrink insulation sleeving.

e. Install relay bracket assembly in stores jettison control panel and secure with screws and washers (Figure 16-5-3-1).

f. Make sure area is clean and free of foreign material.

g. Install cover on stores jettison control panel and secure with screws.

h. Perform operational check of stores jettison control panel (PARA 16-2-5).

16-5-4 ESSS STORES JETTISON CONTROL PANEL DIODES (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D

References

- Appendix D
- PARA 16-2-5

16-5-4.1 Replace ESSS Stores Jettison Control Panel Diodes (BENCH).

NOTE

Replacement procedure for diodes, CR1 through CR8, is the same.

16-5-4.1.1. Remove.

a. Remove screws and cover from stores jettison control panel (Figure 16-5-4-1).

b. Remove screws and washers securing relay bracket assembly to stores jettison control panel.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

c. Using soldering iron, unsolder diode leads from terminals and remove diode (Figure 16-5-4-1, Detail A).

16-5-4.1.2. Install.

a. Install diode between terminals with diode cathode positioned as shown on relay bracket assembly (Figure 16-5-4-1, Detail A).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

b. Using soldering iron, solder, Item 313, Appendix D, diode leads to terminals.

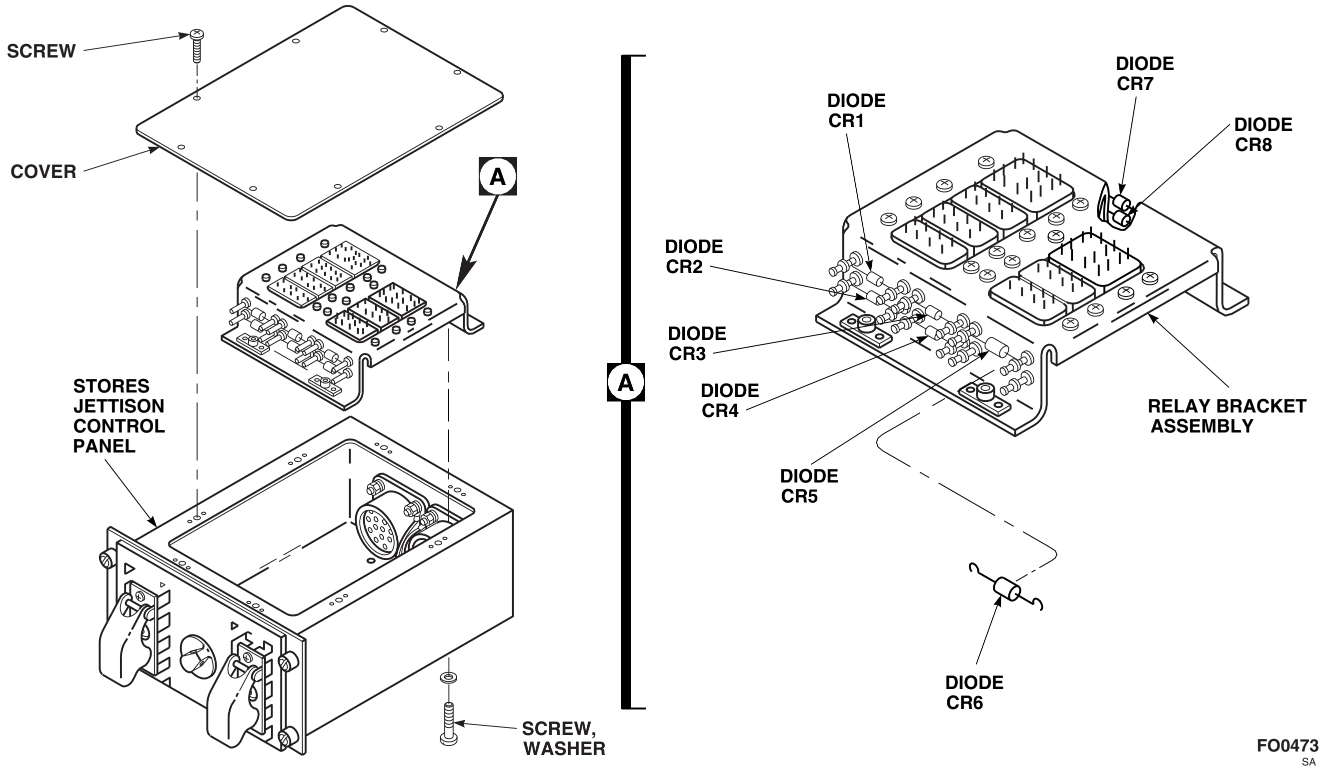


FIGURE 16-5-4-1. REPLACE ESSS STORES JETTISON CONTROL PANEL DIODES (BENCH)

c. Install relay bracket assembly in stores jettison control panel and secure with screws and washers (Figure 16-5-4-1).

d. Make sure area is clean and free of foreign material.

e. Install cover on stores jettison control panel and secure with screws.

f. Perform operational check of stores jettison control panel (PARA 16-2-5).

16-5-5 ESSS STORES JETTISON CONTROL PANEL ELECTRICAL CONNECTORS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Insert/Extract Tool

Materials/Parts

- Tags

References

- PARA 1-7-20
- PARA 16-2-5

16-5-5.1 Replace ESSS Stores Jettison Control Panel Electrical Connectors (BENCH).

NOTE

Replacement procedure for either electrical connector, J1 or J2, is the same.

16-5-5.1.1. Remove.

a. Remove screws and cover from stores jettison control panel (Figure 16-5-5-1).

b. Tag wiring. Using insert/extract tool, remove wires from electrical connector.

c. Remove screws, washers, lockwashers, and nuts securing electrical connector and ground wire to stores jettison control panel. Remove electrical connector.

16-5-5.1.2. Install.

a. Using insert/extract tool, insert wires into electrical connector. Remove tags.

b. Inspect and treat electrical connector (PARA 1-7-20).

c. Install electrical connector and ground wire to stores jettison control panel and secure with screws, washers, lockwasher, and nuts (Figure 16-5-5-1).

d. Make sure area is clean and free of foreign material.

e. Install cover on stores jettison control panel and secure with screws.

f. Perform operational check of stores jettison control panel (PARA 16-2-5).

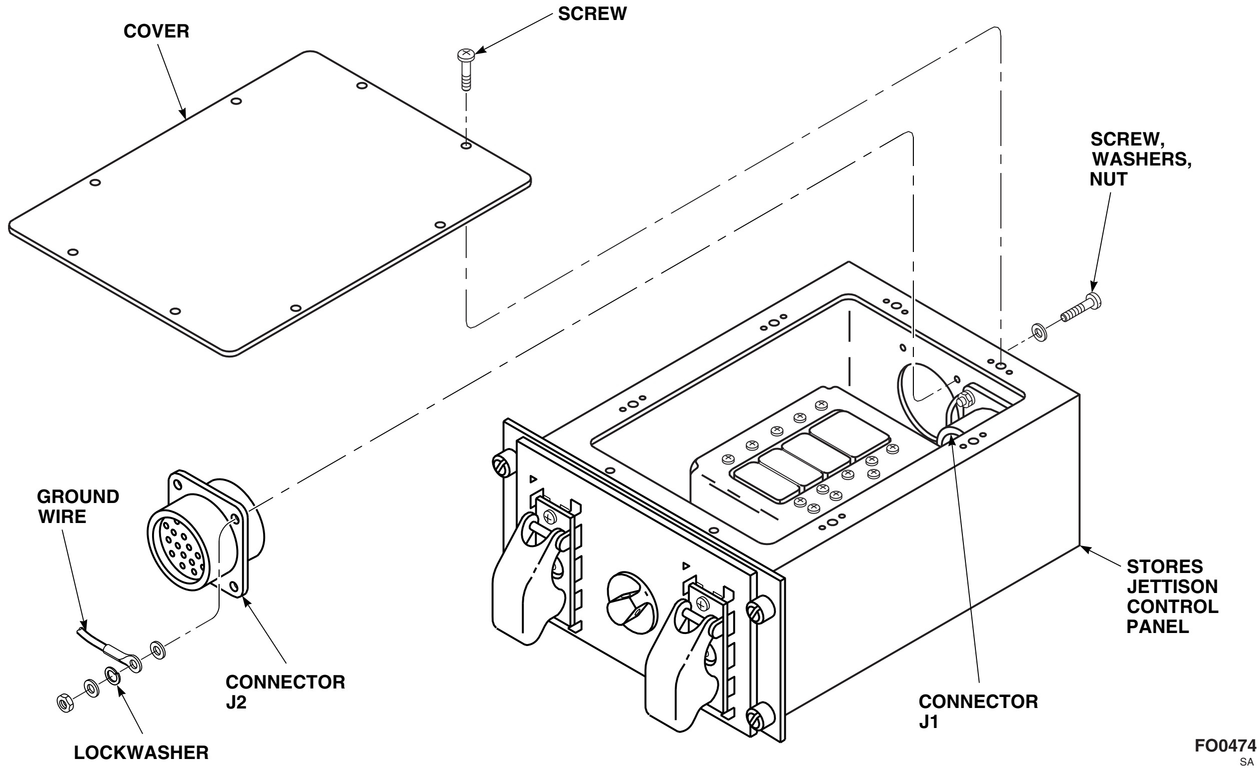


FIGURE 16-5-5-1. REPLACE ESSS STORES JETTISON CONTROL PANEL ELECTRICAL CONNECTORS (BENCH)

16-5-6 EJECTOR RACK, BRU-22A/A, CLEANING AND SURFACE TREATMENT (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Protection, Eye
- Steel Bristle Brush

Materials/Parts

- Coating Kit, Rain Erosion Resisting, Item 106B, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lacquer, Item 182C, Appendix D

References

- Appendix D
- PARA 16-4-36

16-5-6.1 Ejector Rack, BRU-22A/A, Cleaning and Surface Treatment (BENCH).

a. Remove ejector rack, BRU-22A/A, gun and breech (PARA 16-4-36).

b. Remove ejector rack, BRU-22A/A, mechanical arming units (PARA 16-4-36).

c. Using steel bristle brush, clean inside and outside of ejector rack in tank of dry-cleaning solvent, Item 124, Appendix D. Areas around holes and beneath washers and pin heads require particular attention.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

d. Using clean, dry, compressed air, blow dry ejector rack.

e. Spray nonelectrical parts with thin coating of corrosion preventive compound, Item 114, Appendix D.

f. Install ejector rack, BRU-22A/A, mechanical arming units (PARA 16-4-36).

g. Install ejector rack, BRU-22A/A, gun and breech (PARA 16-4-36).

h. Treat reworked metal surfaces with corrosion resistant coating, Item 118, Appendix D.

CAUTION

Damage to ejector rack will result if functional parts are painted. Do not paint bearing surfaces, working surfaces, wiring, or where application will interfere with function of assembly.

i. Treat all untreated interior surfaces with rain erosion resisting coating kit, Item 106B, Appendix D.

j. Touch up exterior surfaces with epoxy primer coating kit, Item 135, Appendix D. Finish with lacquer, Item 182C, Appendix D.

16-5-7 EJECTOR RACK, BRU-22A/A, WIRING HARNESS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Clamp
- Digital Multimeter
- Electrical Repairer's Toolkit

Materials/Parts

- Adhesive, Item 51, Appendix D
- Cleaning Compound, Item 96, Appendix D
- Cleaning Compound, Solvent, Item 98, Appendix D
- Tape, Insulation, Electrical, Item 320A, Appendix D

References

- Appendix D

16-5-7.1 Repair Ejector Rack, BRU-22A/A, Wiring Harness (BENCH).

NOTE

- The following procedure is to repair cracks or breaks in electrical connector molded shell.
- Repair of all ejector rack, BRU-22A/A, wiring harnesses is the same.

a. Using digital multimeter, run continuity check on electrical wires prior to crack repair.

b. Using cleaning compound, Item 96, Appendix D, clean crack or break in molded shell. Rinse shell using cleaning compound solvent, Item 98, Appendix D.

c. Mix adhesive, Item 51, Appendix D and apply to both sides of break.

d. Clamp break until adhesive dries.

e. Using two layers of electrical insulation tape, Item 320A, Appendix D, wrap entire molded section of cable assembly.

f. Apply light coating of adhesive, Item 51, Appendix D, over tape to prevent windstream removal. Allow to dry.

16-5-8 EJECTOR RACK, BRU-22A/A, TEST (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Digital Multimeter
- Electrical Repairer's Toolkit
- Hi Potential Tester, Model 411A1 (or equivalent)
- Spring Scale
- Test Tank, 230-Gallon
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 150 - 750 in. lbs

References

- PARA 16-4-54

16-5-8.1 Ejector Rack, BRU-22A/A, Test (BENCH).

16-5-8.1.1 Proof Load Test.

a. Provide fixture to support ejector rack.

b. Verify freedom of movement of swaybrace screws through full adjustment range. Adjust brace evenly for attachment to store.

NOTE

230-gallon test tank must have MK 6 MOD 0 suspension lugs installed.

c. Open ejector rack hooks, and secure 230-gallon test tank (PARA 16-4-54).

d. After hooks are latched, move safety stop lever to locked position.

e. Adjust swaybraces evenly until swaybraces are fingertight.

f. Move safety stop lever to unlocked position.

g. **TORQUE SWAYBRACES, ¼ TURN AT A TIME TO 250 INCH-POUNDS.** If either hook unlatches, reject ejector rack.

h. Back off swaybraces. **TORQUE SWAYBRACES EVENLY TO 20 INCH-POUNDS.**

i. Release ejector unit manually. Using spring scale applied parallel to top of ejector rack, measure release force. **RELEASE FORCE MUST BE BETWEEN 20 - 90 POUNDS.**

16-5-8.1.2 Breech Cap Dielectric Test.

a. Using Hi potential tester (or equivalent), apply 900 to 1100 volts (RMS), 60 Hz, for 1 minute between pins A and B and between pin A and conduit. Any arc-over or breakdown constitutes a failure.

b. Using digital multimeter, measure leakage current. Leakage current in excess of 4 milliampere at this test voltage requires retest at 5 milliampere. Any arc-over, breakdown, or leakage current in excess of 5 milliampere at this test voltage shall constitute a failure.

16-5-8.1.3 Ejector Rack Dielectric Test.

a. Using Hi potential tester, apply 450 to 550 volts (RMS), 60 Hz, for 1 minute between pin F of receptacle on top of ejector rack and ejector rack structure.

b. Using digital multimeter, measure leakage current. Any arc-over, breakdown, or change in leakage current in excess of 4 milliampere at this test voltage shall constitute a failure.

SECTION 16-6

MISSION KIT INSTALLATION AND REMOVAL PROCEDURES

SECTION OVERVIEW

16-6-1 WINTERIZATION KIT INSTALLATION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Bolt, ¼-inch
- Container, 1-Gallon
- Pneumatic Drill
- Rivnut Key Cutter
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Twist Drill Set

Materials/Parts

- Abrasive Cloth, Item 5, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Lockwire, Item 196, Appendix D
- Lockwire, Item 197, Appendix D
- Primer, Item 258, Appendix D
- Protective Caps and Plugs
- Tiedown Strap, Item 329, Appendix D

References

- Appendix D
- PARA 1-3-10
- PARA 1-3-11
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-45
- PARA 2-4-69
- PARA 7-4-13
- PARA 9-4-2
- PARA 13-5-3
- PARA 13-5-6
- PARA 15-2-1
- PARA 15-2-2
- PARA 15-2-3

16-6-1.1 Install Winterization Kit.

16-6-1.1.1 Prepare to Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Gain access to main fuel cell compartment as follows:

(1) Remove rear passenger's seats as required (PARA 2-4-45).

(2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).

(3) Remove cargo netting from stowage compartments.

(4) If air conditioning is installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

c. Unlock fasteners securing left and right panel enclosure on top of main fuel cells. Remove panels.

d. Remove screws and washers and remove center panel and top panel.

16-6-1.1.2 Install Second Accumulator.

a. Remove rear screws and washers from brackets in overhead above fuel cells. Move brackets outward.

b. On upper console, make sure APU CONTR switch is placed OFF.

c. Disconnect electrical connector, P290, from start valve (Figure 16-6-1-1, Sheet 1, Detail A).

d. Install protective caps and plugs on electrical connectors.

e. Remove auxiliary power unit (APU) exhaust plug.

f. Manually operate APU start valve lever to motor APU, and release hydraulic charge slowly.

g. Remove accumulator nitrogen servicing valve cap on pressure gage.

h. Loosen locknut on servicing valve and slowly release nitrogen charge from accumulator.

i. Make sure tape indicator on accumulator and nitrogen pressure gage show zero.

j. Place 1-gallon container under backup pump drain line.

k. Open main rotor pylon sliding cover.

l. Drain backup pump reservoir (PARA 7-4-13).

m. Place 1-gallon container under APU start valve to catch hydraulic fluid.

n. Remove screws and plate from start valve. Save for later installation (Figure 16-6-1-1, Sheet 1, Detail B).

NOTE

A ¼-inch-bolt may be used to remove plugs.

o. Remove plugs from ports in start valve. Save plugs for later installation.

p. Install second accumulator brackets and strap hangers to airframe with bolts and washers. **TORQUE BOLTS TO 43 - 47 INCH-POUNDS** (Figure 16-6-1-1, Sheet 2, Detail C).

q. Remove protective covers from second accumulator transfer tubes and nitrogen gas line fitting (Figure 16-6-1-1, Sheet 1, Detail A).

r. Install backup retainers and packings on transfer tubes on second accumulator.

s. Disconnect nitrogen gas line from elbow between pressure gage and first accumulator.

t. Remove elbow and nut from airframe. Retain elbow for use when winterization kit is removed.

u. Install tee in airframe, position middle fitting down and install jamnut. **TORQUE JAMNUT 85 - 105 INCH-POUNDS.**

v. Connect nitrogen gas lines between pressure gage and tee, and between tee and first accumulator. **TORQUE NITROGEN GAS LINE B-NUTS 135 - 145 INCH-POUNDS.**

w. Position long nitrogen gas line adjacent to rigid tubing along side of first accumulator. Install nitrogen gas line on tee, do not tighten now. Loosely install clamps, screw, washers, and nut securing nitrogen gas line to rigid tubing.

x. Loosen strap hangers securing first accumulator.

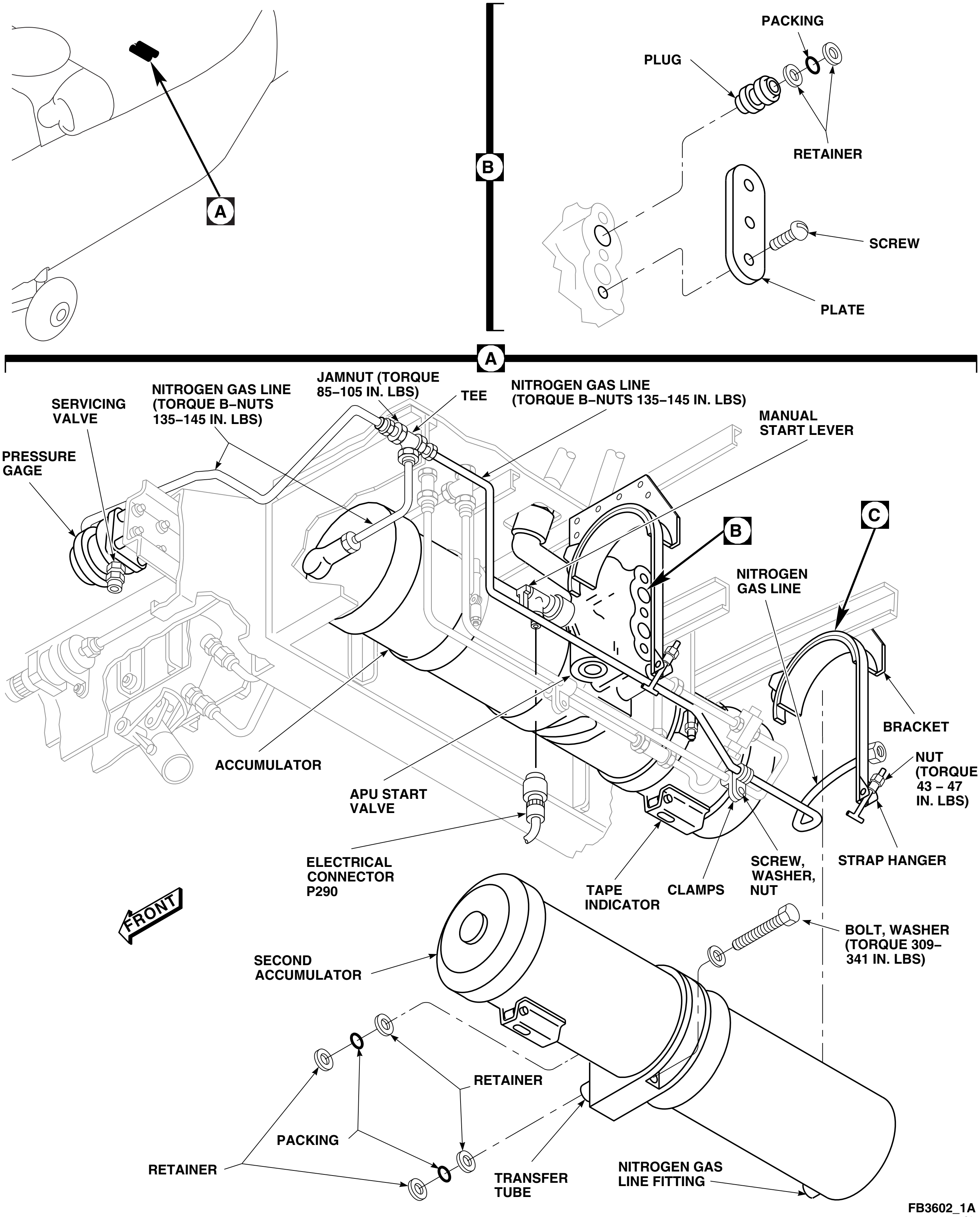


FIGURE 16-6-1-1. INSTALL SECOND ACCUMULATOR (SHEET 1 OF 2)

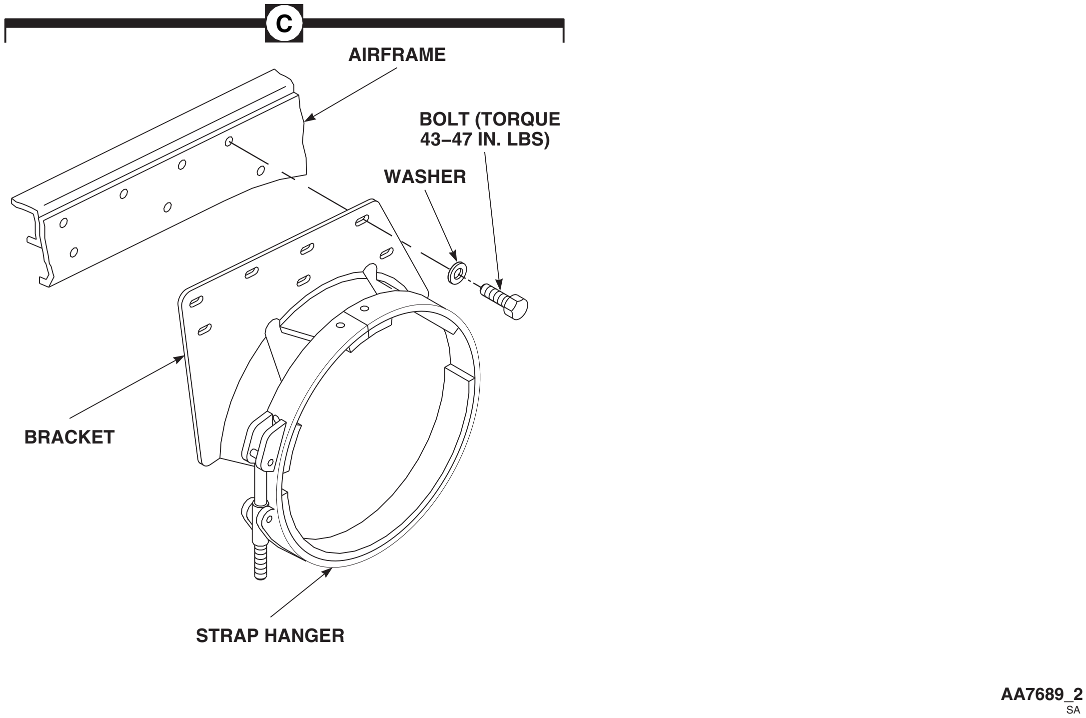


FIGURE 16-6-1-1. INSTALL SECOND ACCUMULATOR (SHEET 2 OF 2)

y. Place second accumulator in position and install transfer tubes in ports of start valve.

z. Secure accumulator to start valve with bolts and washers. **TORQUE BOLTS TO 309 - 341 INCH-POUNDS.** Lockwire, Item 197, Appendix D, boltheads together.

aa. Place strap hangers around second accumulator.

ab. **TIGHTEN NUTS ON STRAP HANGARS ON FIRST AND SECOND ACCUMULATORS, SECURING ACCUMULATORS TO BRACKETS. TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

ac. Install nitrogen gas line to second accumulator. **TORQUE NITROGEN GAS LINE B-NUTS TO 135 - 145 INCH-POUNDS.**

ad. **TIGHTEN CLAMPS SECURING NITROGEN GAS LINE TO RIGID TUBING.**

16-6-1.1.3 Install Harness Assembly.

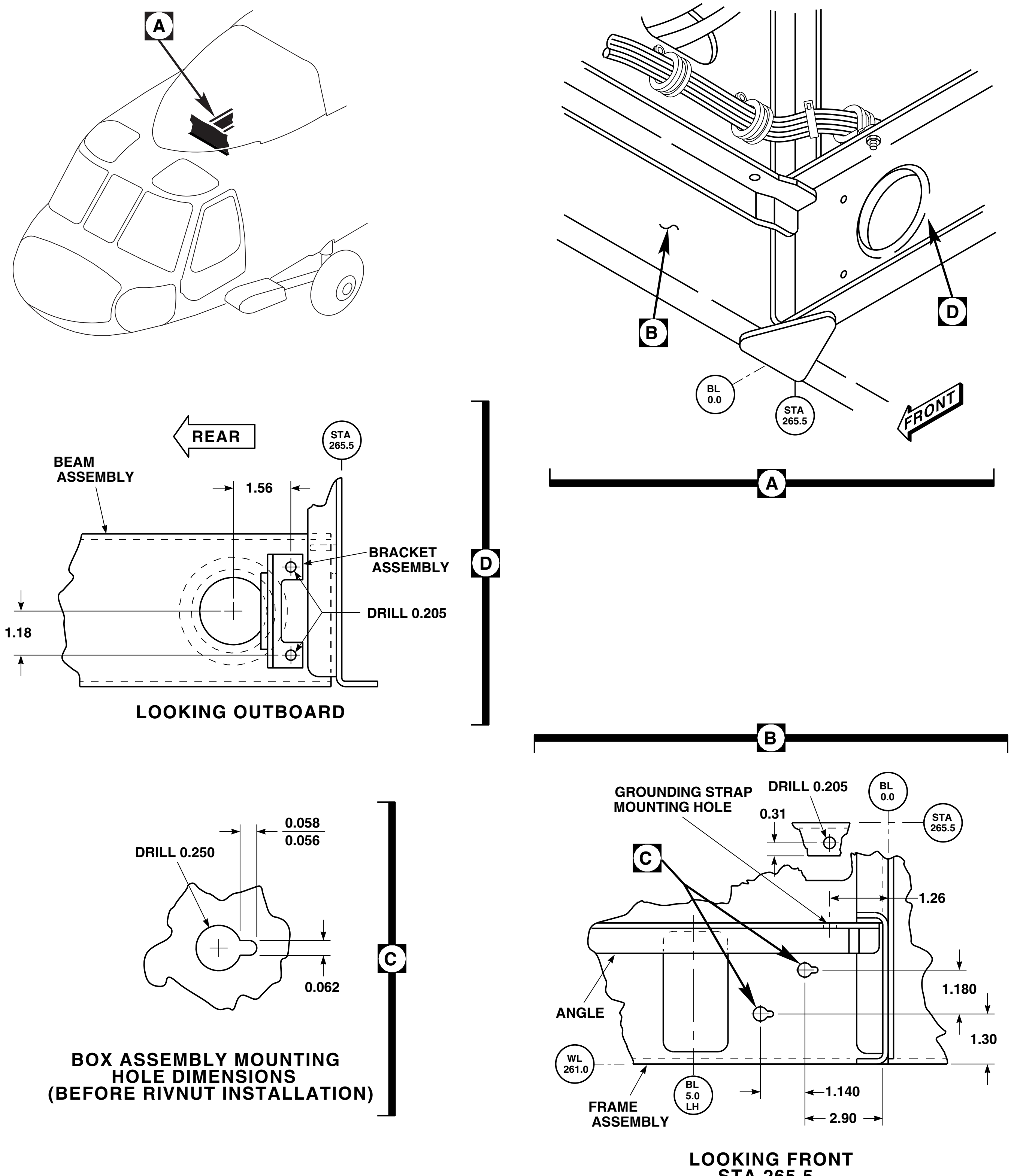
a. Measure and mark position of box assembly mounting holes on frame assembly at STA 265.5 (Figure 16-6-1-2, Detail B).

CAUTION

Damage to helicopter wiring and components will result if mounting holes are not drilled carefully. Examine area to be drilled, move wiring or limit penetration of bit to avoid damaging wiring and components.

b. Using pneumatic drill, rivnut key cutter, and twist drill set, drill box assembly mounting holes to 0.250-inch diameter (Figure 16-6-1-2, Detail C).

c. Install rivnuts in box assembly mounting holes (Figure 16-6-1-3, Detail A).



NOTE

ALL DIMENSIONS ARE IN INCHES.

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FIGURE 16-6-1-2. INSTALLATION HARNESS MOUNTING

d. Position bracket assembly on beam assembly at BL 0.0 (Figure 16-6-1-2, Detail D).

e. Using bracket assembly as a template, mark position of mounting holes.

f. Drill bracket assembly mounting holes to 0.205-inch diameter.

g. Measure and mark position of grounding jumper mounting hole on angle mounted on frame assembly at STA 265.5 (Figure 16-6-1-2, Detail B).

h. Using pneumatic drill and twist drill set, drill hole to 0.205-inch diameter.

i. Disconnect electrical connector, P244, from right relay panel (PARA 9-4-2).

j. Remove screw, washer, nut, and clamp from electrical connector, P244, wiring harness on beam at BL 0.0. Retain clamp and hardware for use when winterization kit is removed (Figure 16-6-1-3, Detail A).

k. Remove screws and washers and cover of box assembly.

l. While supporting box assembly, route electrical connector, P244B, wiring harness through hole in beam at BL 0.0.

CAUTION

Damage to harness will result if contact occurs with screwdriver when installing box assembly. Position harness to avoid contact when installing box assembly.

m. Secure box assembly to frame at STA 265.5 with screws, lockwashers, and washers.

n. Install box assembly cover with screws and washers.

o. Inspect and treat electrical connectors (PARA 1-7-20).

p. Connect electrical connector, P244, to J244A, on box assembly.

q. Inspect and treat electrical connectors (PARA 1-7-20).

r. Connect electrical connector, P244B, from box assembly to electrical connector, J244, on right relay panel.

s. Lightly sand area where grounding strap is to be installed with abrasive cloth, Item 5, Appendix D, until base metal is present.

t. Wipe area clean with clean cloth moistened with dry-cleaning solvent, Item 124, Appendix D.

u. Apply corrosion resistant coating, Item 118, Appendix D, to base metal.

NOTE

Steel washer is to be located between lockwasher and grounding strap. Untreated aluminum washer is to be located between grounding strap and angle. Treated aluminum washer is to be located between angle and nut.

v. Connect grounding strap to angle at STA 265.5 with screw, steel lockwasher, steel washer, and aluminum washers, and nut.

w. Touch up area with primer, Item 258, Appendix D.

x. Install bracket assembly with screws, washers, and nuts.

y. Secure wire harness for electrical connector, P244B, to bracket assembly with tiedown strap, Item 329, Appendix D.

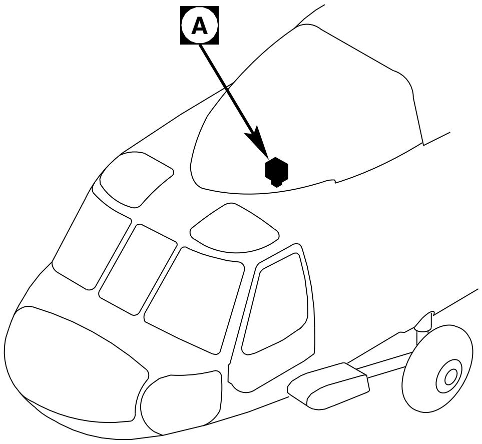
z. Inspect installation for security.

16-6-1.1.4 Complete Installation.

a. Remove protective caps and plugs from electrical connectors.

b. Inspect and treat electrical connectors (PARA 1-7-20).

c. Connect electrical connector, P290, to start valve. Lockwire, Item 196, Appendix D, connector (Figure 16-6-1-1, Sheet 1, Detail A).



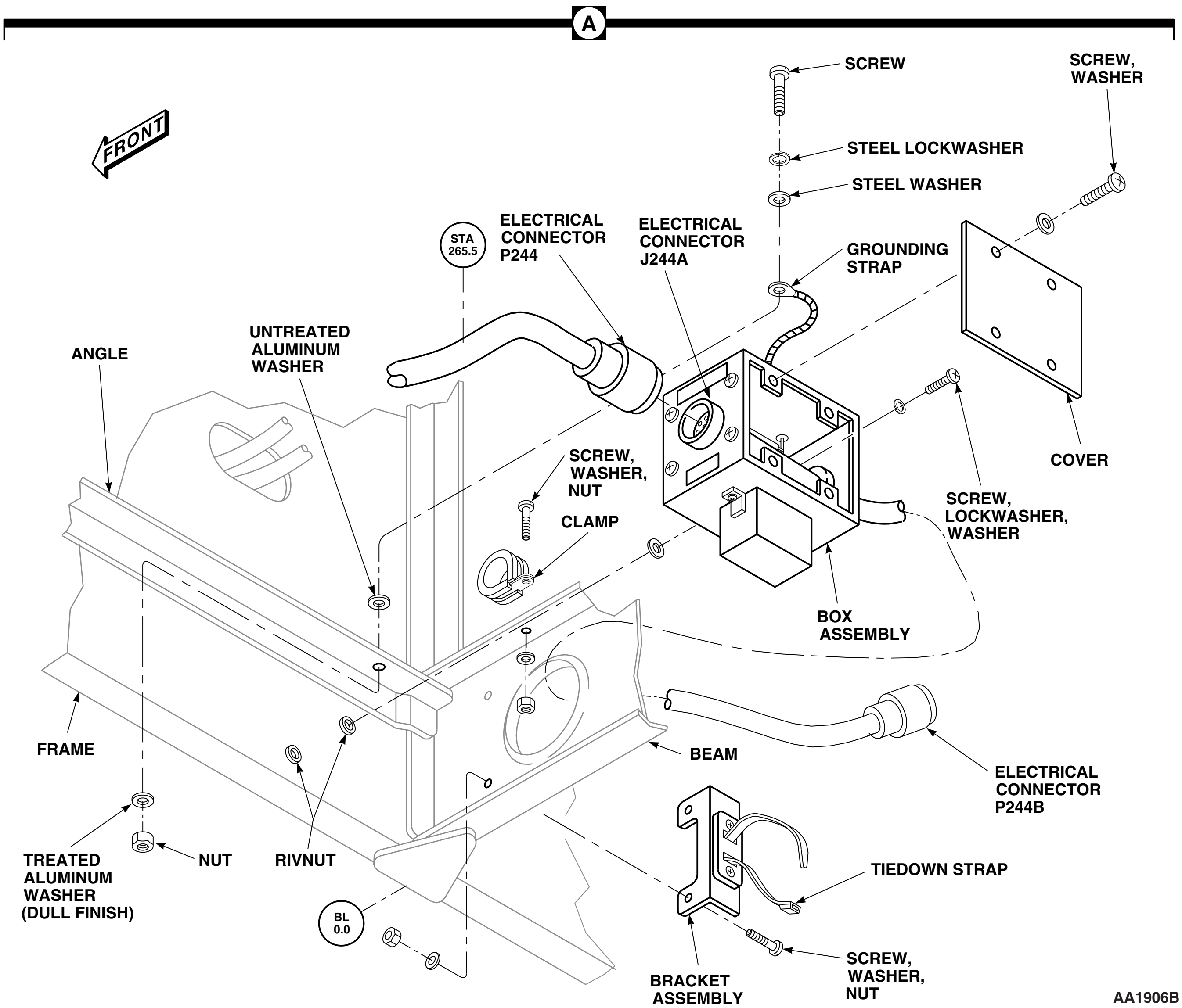


FIGURE 16-6-1-3. INSTALL SECOND ACCUMULATOR BOX ASSEMBLY

d. Service APU accumulators (PARA 1-3-10 and PARA 1-3-11).

e. Perform operational check of APU (PARA 15-2-1, PARA 15-2-2, or PARA 15-2-3).

f. Secure brackets on overhead above fuel cells with screws and washers.

g. Make sure area is clean and free of foreign material.

h. Install top panel and center front panel with screws and washers.

i. Install left and right panel enclosure on top of main fuel cells. Lock fasteners.

j. Close fuel cell compartment as follows:

(1) If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

(2) Install cargo netting.

(3) Install soundproofing panels (PARA 2-4-69).

(4) Install rear passenger's seats (PARA 2-4-45).

CAUTION

To prevent damage to hydraulic hand-pump handle when closing main rotor pylon sliding cover, make sure handle is full down before closing cover.

k. Close main rotor pylon sliding cover and latch.

l. Install APU exhaust plug.

16-6-2 WINTERIZATION KIT REMOVAL.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 1-Gallon
- Protective Covers
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 196, Appendix D
- Lockwire, Item 197, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-10
- PARA 1-6-5
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-45
- PARA 2-4-69
- PARA 2-6-5
- PARA 7-4-13
- PARA 9-4-2
- PARA 13-5-3
- PARA 13-5-6
- PARA 15-2-1
- PARA 15-2-2
- PARA 15-2-3

16-6-2.1 Remove Winterization Kit.

16-6-2.1.1 Prepare to Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Gain access to main fuel cell compartment as follows:

(1) Remove rear passenger's seats as required (PARA 2-4-45).

(2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).

(3) Remove cargo netting from stowage compartments.

(4) If air conditioning is installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

c. Unlock fasteners securing left and right panel enclosure on top of main fuel cells. Remove panels.

d. Remove screws and washers and remove center panel and top panel.

16-6-2.1.2 Remove Harness Assembly.

NOTE

Keep and identify removed items, with attaching hardware, as a kit for reinstallation.

a. Disconnect electrical connector, P244B, from right relay panel (PARA 9-4-2).

b. Install protective caps and plugs on electrical connectors.

c. Cut tiedown strap securing electrical connector, P244B, wiring harness to bracket assembly (Figure 16-6-2-1, Detail A).

d. Remove screws, washers, and nuts securing bracket assembly to beam at BL 0.0. Remove bracket.

e. Remove screw, lockwasher, washers, and nut securing grounding strap to angle at STA 265.5.

f. Disconnect electrical connector, P244, from electrical connector, J244A, on box assembly.

g. Install protective caps and plugs on electrical connectors.

h. Remove screws and washers securing cover to box assembly. Remove cover.

i. Remove screws, lockwashers, and washers securing box assembly to frame at STA 265.5.

j. Remove box assembly by carefully pulling electrical connector, P244B, wiring harness through hole in beam at BL 0.0.

k. Route electrical connector, P244, and wiring harness over beam at BL 0.0, and secure wiring harness to beam with clamp, screw, washer, and nut.

l. Inspect and treat electrical connectors (PARA 1-7-20).

m. Touch up paint as required for exposed metal (PARA 2-6-5).

16-6-2.1.3 Remove Second Accumulator.

a. Remove rear screws and washers from brackets in overhead above fuel cells. Move brackets outward.

b. On upper console, make sure APU CONTR switch is placed OFF.

c. Disconnect electrical connector, P290, from APU start valve (Figure 16-6-2-2, Sheet 1, Detail A).

d. Install protective caps and plugs on electrical connectors.

e. Remove APU exhaust plug (PARA 1-6-5).

f. Manually operate APU start valve lever to motor APU and slowly release hydraulic charge.

g. Remove accumulator nitrogen servicing valve cap on pressure gage.

h. Loosen locknut on servicing valve, and slowly release nitrogen charge from accumulator.

i. Make sure tape indicators on accumulators and nitrogen gage show zero.

j. Place 1-gallon container under backup pump drain line.

k. Open main rotor pylon sliding cover.

l. Drain backup pump reservoir (PARA 7-4-13).

m. Remove screw, washer, nut and clamps securing nitrogen gas line.

n. Remove nitrogen gas line from tee and from fitting on second accumulator.

o. Disconnect short nitrogen gas line from first accumulator at tee.

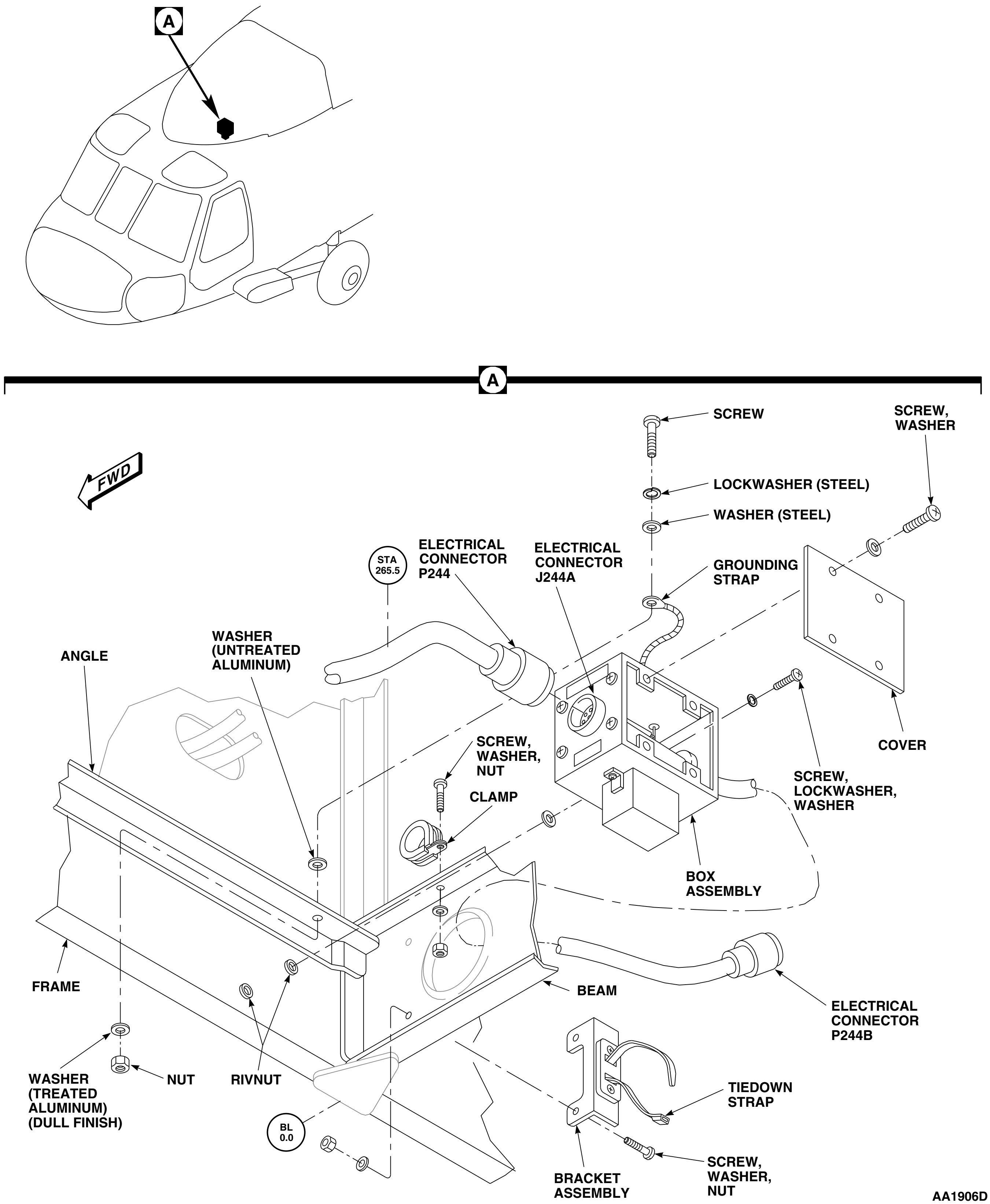


FIGURE 16-6-2-1. REMOVE SECOND ACCUMULATOR BOX ASSEMBLY

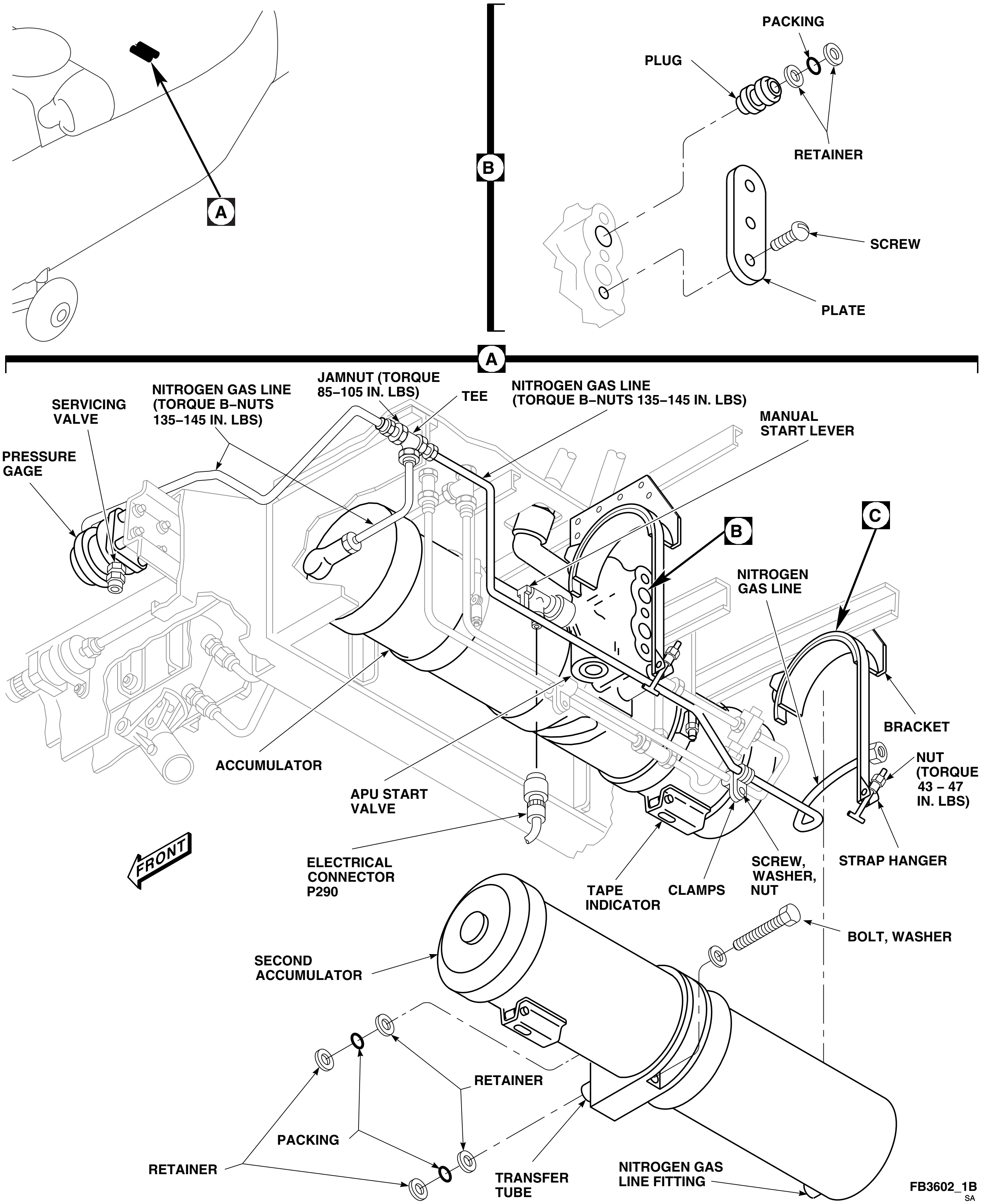


FIGURE 16-6-2-2. REMOVE SECOND ACCUMULATOR (SHEET 1 OF 2)

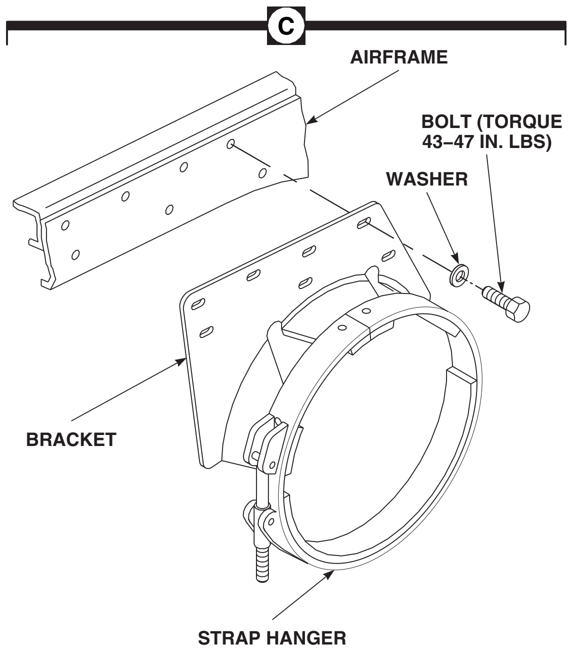


FIGURE 16-6-2-2. REMOVE SECOND ACCUMULATOR (SHEET 2 OF 2)

p. Disconnect nitrogen gas line from pressure gage at tee.

q. Remove jamnut and remove tee from airframe.

r. Install elbow and jamnut in airframe. Position elbow down. **TORQUE JAMNUT 85 - 105 INCH-POUNDS.**

s. Install nitrogen gas lines from pressure gage and from first accumulator on elbow. **TORQUE NITROGEN GAS LINE B-NUTS 135 - 145 INCH-POUNDS.**

t. Loosen strap hangers securing first and second accumulators.

u. Place 1-gallon container under APU start valve to catch hydraulic fluid.

v. Remove bolts and washers securing second accumulator to APU start valve.

w. Disconnect strap hangers from around second accumulator.

x. Remove second accumulator from APU start valve and remove backup retainers and packings from transfer tubes. Discard packings. Install protective covers on transfer tubes.

y. Remove bolts, washers, second accumulator brackets, and strap hangers from airframe (Figure 16-6-2-2, Sheet 2, Detail C).

z. **TIGHTEN STRAP HANGARS ON FIRST ACCUMULATOR. TORQUE NUTS TO 43 - 47 INCH-POUNDS** (Figure 16-6-2-2, Sheet 1, Detail A).

aa. Connect short nitrogen gas line to first accumulator.

ab. Install backup retainers and packings on plugs and install plugs in APU start valve (Figure 16-6-2-2, Sheet 1, Detail B).

ac. Install plate and screws on APU start valve. Using lockwire, Item 197, Appendix D, lockwire screws.

ad. Remove protective caps and plugs from electrical connector.

ae. Inspect and treat electrical connectors (PARA 1-7-20).

af. Connect electrical connector, P290, to APU start valve. Using lockwire, Item 196, Appendix D, lockwire electrical connector.

16-6-2.1.4 Complete Removal.

a. Service APU accumulator (PARA 1-3-10).

b. Perform operational check of APU (PARA 15-2-1, PARA 15-2-2, or PARA 15-2-3).

c. Secure brackets on overhead above fuel cells with screws and washers.

d. Make sure area is clean and free of foreign material.

e. Install top panel and center front panel with screws and washers.

f. Install left and right panel enclosure on top of main fuel cells. Lock fasteners.

g. Close fuel cell compartment as follows:

(1) If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

(2) Install cargo netting.

(3) Install soundproofing panels (PARA 2-4-69).

(4) Install rear passenger's seats (PARA 2-4-45).

h. Close main rotor pylon sliding cover.

i. Install APU exhaust plug (PARA 1-6-5).

16-6-3 COCKPIT/CABIN BLACKOUT KIT INSTALLATION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

16-6-3.1 Install Cockpit/Cabin Blackout Curtain.

16-6-3.1.1. Install.

a. Expose hook fastener tape across top of cabin.

b. Starting across top, install crew chief's cockpit/cabin blackout curtain by pressing pile fastener tape on curtain to hook fastener tape on ceiling panels (Figure 16-6-3-1, Detail A and Detail B).

c. Press side and top pile and hook fastener tape strips together from both front and rear to make sure of a neat and tight attachment. Make sure strips are in full contact and curtain is positioned properly at sides to join side pile and hook fastener tape strips on left and right side control enclosure cover.

d. Check fit of curtain around first aid kit, battery cover, and crew chief's seat upper support to prevent rips and binding.

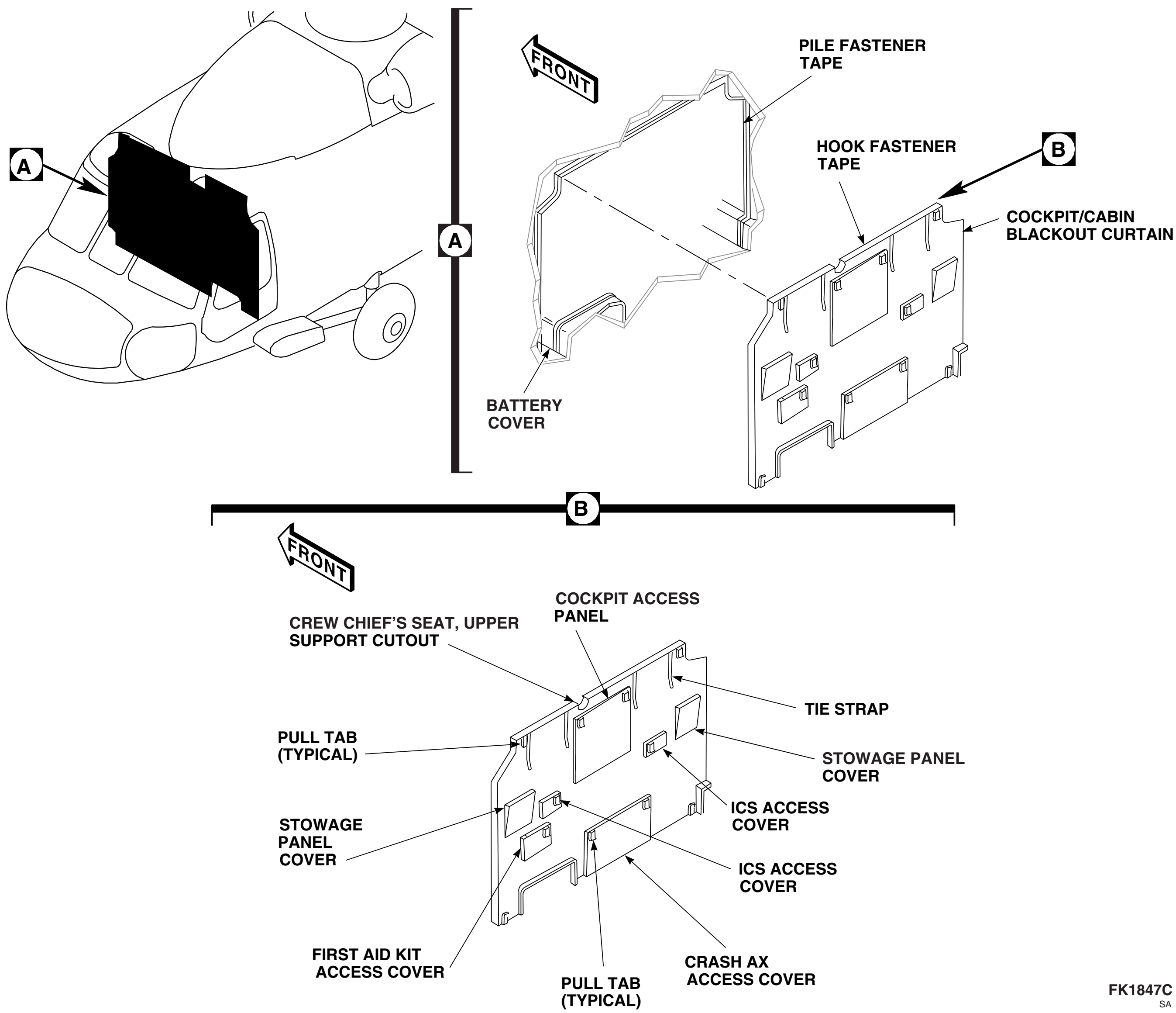


FIGURE 16-6-3-1. INSTALL COCKPIT/CABIN BLACKOUT CURTAIN

16-6-4 COCKPIT/CABIN BLACKOUT KIT REMOVAL.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

16-6-4.1 Remove Cockpit/Cabin Blackout Curtain.

16-6-4.1.1. Remove.

a. Start at bottom right side and pull gently on pull tab, separating pile and hook fastener tape strips, until right side of cockpit/cabin blackout curtain is pulled free to top (Figure 16-6-4-1, Detail A and Detail B).

b. Repeat step a. for left side of curtain.

c. Carefully roll curtain to top and using tie strap, tie curtain in place.

d. If curtain is to be stored, use the tab provided and gently pull the top of the curtain loose and fold for storage in stowage compartment.

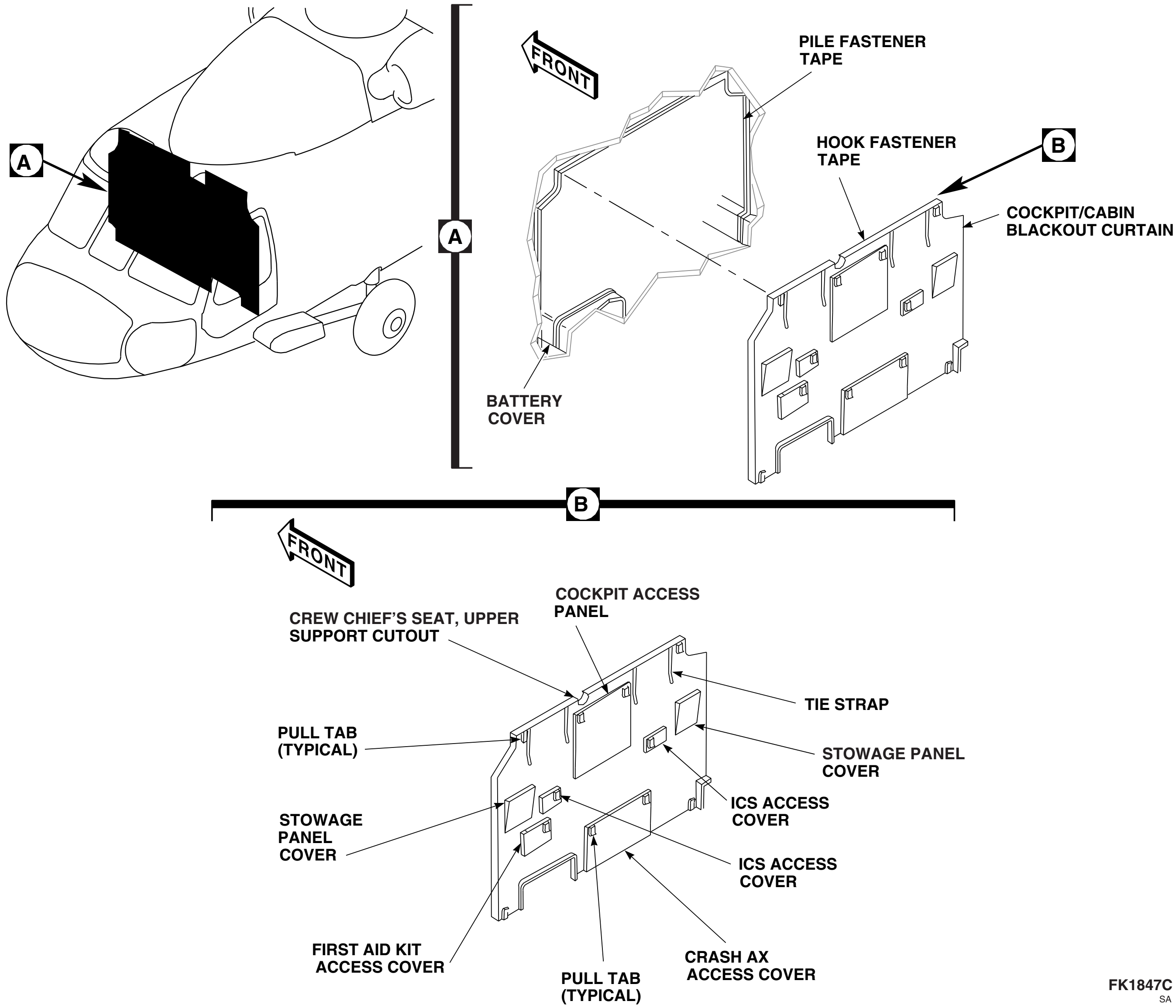


FIGURE 16-6-4-1. REMOVE COCKPIT/CABIN BLACKOUT CURTAIN

16-6-5 CREW CHIEF'S AND CABIN DOOR WINDOWS BLACKOUT KIT INSTALLATION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

16-6-5.1 Install Window Blackout Curtain.

NOTE

- Do not open crew chief's window with curtain installed.
- Install procedure for left and right crew chief's and cabin door window blackout curtains is the same.

16-6-5.1.1. Install.

a. With pull tab on bottom and starting across top, install curtain by pressing pile fastener tape and hook fastener tape together, and line up curtain to sides (Figure 16-6-5-1, Detail A).

b. Press thumb to Velcro and run around edge of curtain.

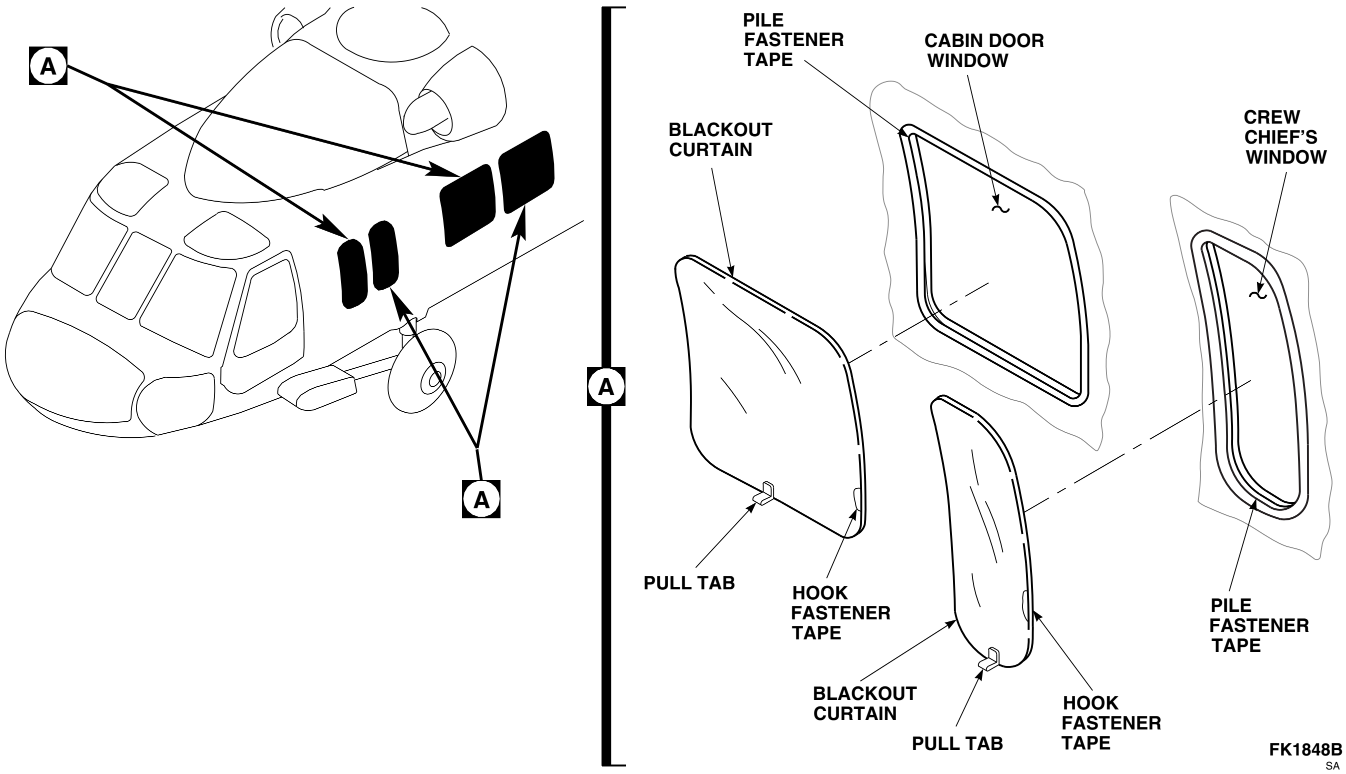


FIGURE 16-6-5-1. INSTALL WINDOW BLACKOUT CURTAIN

16-6-6 CREW CHIEF'S AND CABIN DOOR WINDOWS BLACKOUT KIT REMOVAL.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

16-6-6.1 Remove Window Blackout Curtain.

16-6-6.1.1. Remove.

NOTE

- Do not open crew chief's window with curtain installed.
- Remove procedure for left and right crew chief's and cabin window blackout curtains is the same.

a. Pull up gently on pull tab at bottom of curtain until hook fastener tape on curtain is free from pile fastener tape (Figure 16-6-6-1, Detail A).

b. Fold carefully and store in stowage compartment.

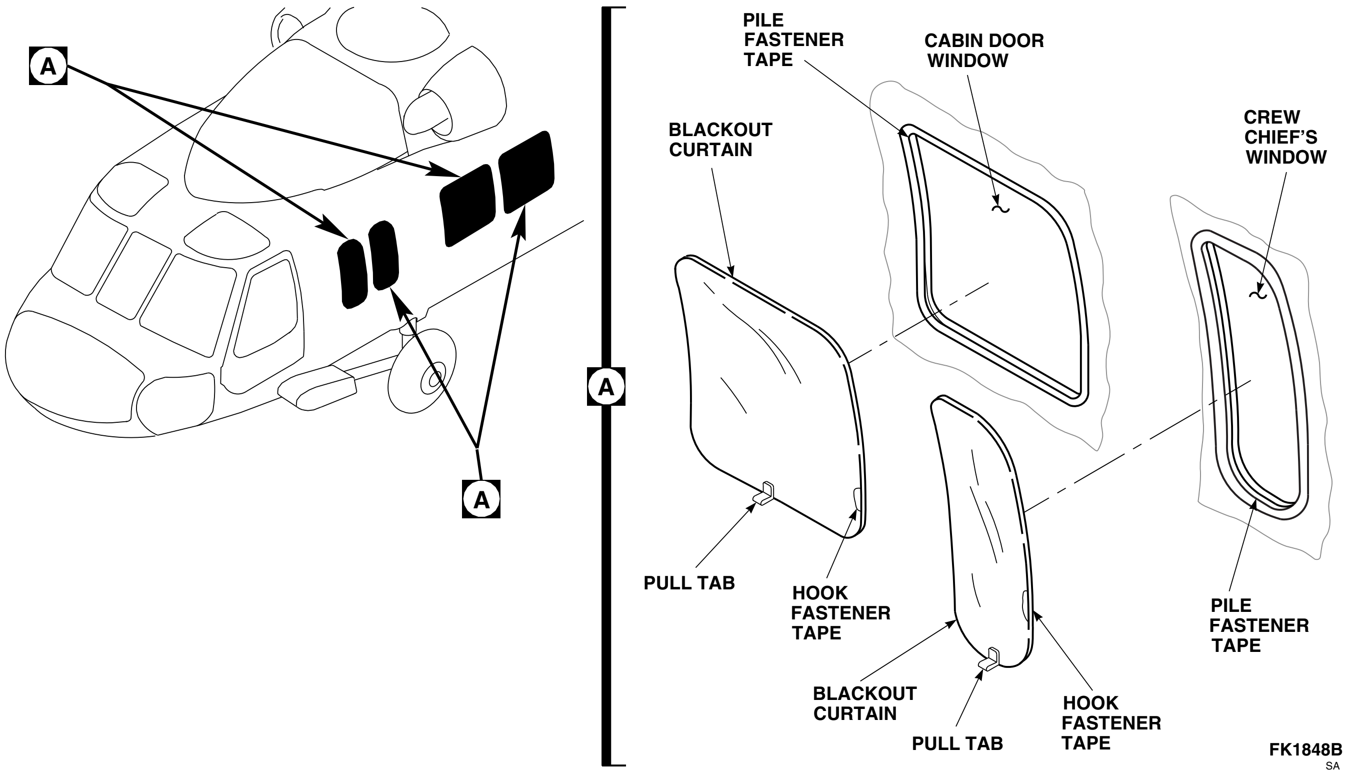


FIGURE 16-6-6-1. REMOVE WINDOW BLACKOUT CURTAIN

16-6-7 EXTERNAL STORES SUPPORT SYSTEM (ESSS) KIT INSTALLATION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Digital Multimeter
- Electrical Repairer's Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Wire Brush

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Antiseize Compound, Item 74, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix F
- PARA 1-3-2
- PARA 1-3-4
- PARA 1-3-12
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-43
- PARA 2-4-45
- PARA 2-4-69
- PARA 2-6-5
- PARA 6-4-19
- PARA 13-5-3

- PARA 13-5-6
- PARA 16-2-1
- PARA 16-2-2
- PARA 16-2-4
- PARA 16-3-1
- PARA 16-4-21
- PARA 16-4-33
- PARA 16-4-34
- PARA 16-4-37
- PARA 16-4-43
- PARA 16-4-52
- PARA 16-4-54

16-6-7.1 Install External Stores Support System (ESSS) Kit.

16-6-7.1.1 Prepare to Install.

CAUTION

Damage to main transmission will result if ESSS is installed without performing the following inspection. Main transmission could overheat. Inspection and adjustment of oil pumps is required on helicopters with main transmission, 70351-08100-060, and earlier installed.

a. Visually inspect main transmission to make sure oil pumps, 70351-08144-103, are installed.

b. If oil pumps, 70351-08144-103, are not installed and are not available, perform the following:

(1) Visually inspect input modules to make sure input modules, 70351-08001-046, 70351-08001-047, or 70351-08001-048 are installed.

(2) Ground run helicopter and record main transmission oil pressure. If main transmission oil pressure is between 50 - 55 psi, no further inspections required.

(3) If main transmission oil pressure is above or below limits, adjust pressure regulators to obtain 50 - 55 psi range (PARA 6-4-19).

c. Service main transmission (PARA 1-3-12).

d. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

e. Make sure helicopter is properly grounded.

f. Service main landing gear shock struts for ESSS configuration (PARA 1-3-2).

g. Service tail landing gear shock strut for ESSS configuration (PARA 1-3-4).

h. Remove rear passenger's seats as required (PARA 2-4-45).

i. Remove soundproofing panel from rear bulkhead covering left or right stowage compartment (PARA 2-4-69).

j. Remove cargo netting from stowage compartment.

k. If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

l. Unlock fasteners securing left and right enclosure panels on top of main fuel cells. Remove panels.

m. Position post in bracket on lower console and in bracket on upper console. Secure with quick-release pins attached to post (Figure 16-6-7-1, Detail A).

n. Unlock studs and remove upper and lower cap fairings from upper and lower ESSS fittings (Figure 16-6-7-2, Detail A and Detail B).

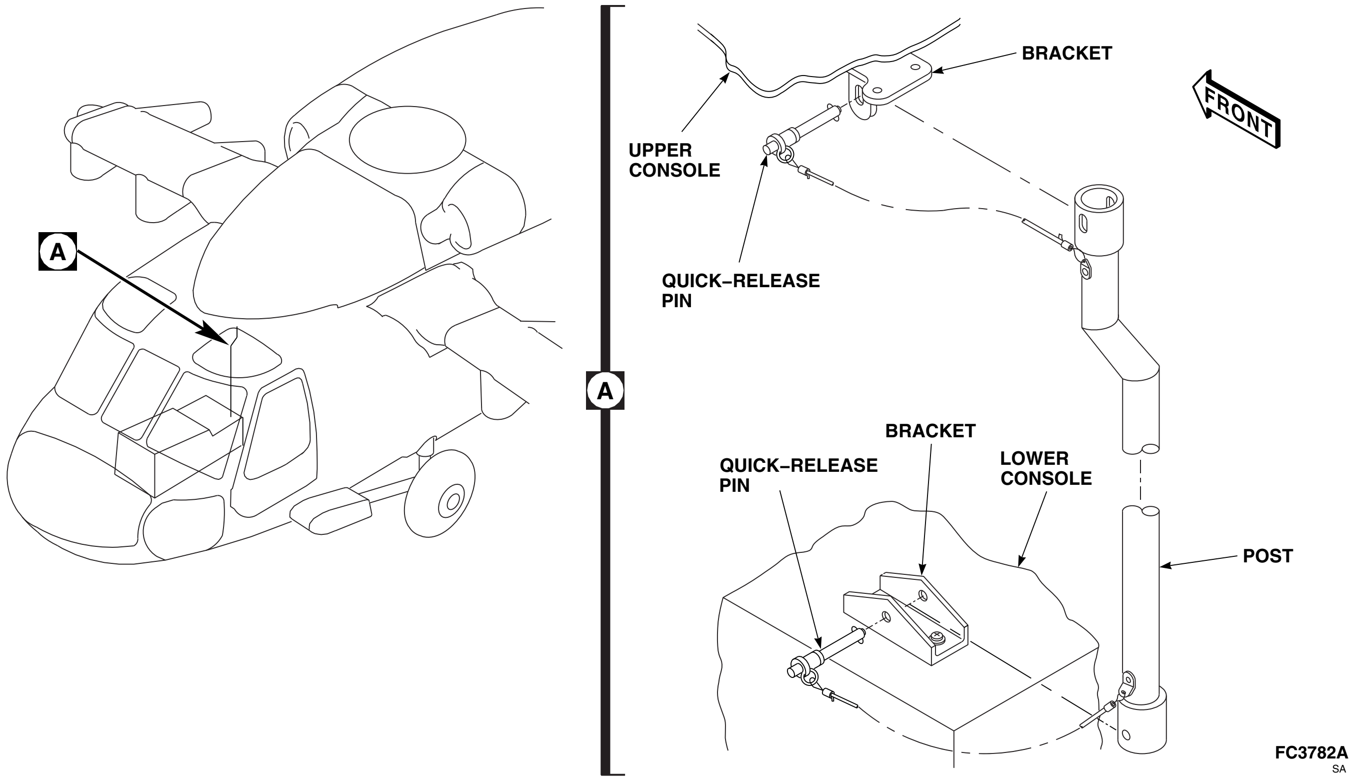


FIGURE 16-6-7-1. INSTALL POST

16-6-7.1.2 Install Auxiliary Fuel Management Control Panel W/O AUX FUEL QTY .

a. Prepare mating surfaces (lower console/ mounting bracket and mounting bracket/ control panel) for electrical bond as follows:

(1) Clean paint from around mounting holes using abrasive cloth, Item 8, Appendix D.

(2) Clean area with clean cheesecloth, Item 90, Appendix D, moistened with dry-cleaning solvent, Item 124, Appendix D.

(3) Apply corrosion resistant coating, Item 118, Appendix D, to contacting area.

b. Remove screw, lockwasher, and washer securing upper end of bonding jumper between mounting bracket and top surface of lower console (Figure 16-6-7-3, Detail A).

c. Install mounting bracket and bonding jumpers on lower console with screws, lockwashers, and washers.

d. Install auxiliary fuel management control panel on mounting bracket on lower console and secure with screws, lockwashers, and washers.

e. Remove protective caps and plugs from electrical connectors.

f. Inspect and treat electrical connectors (PARA 1-7-20).

g. Connect electrical connectors, P360 to J360, and P361 to J361, on auxiliary fuel management control panel.

h. Make sure area is clean and free of foreign material.

i. Touch up paint as required (PARA 2-6-5).

16-6-7.1.3 Install Signal Conditioner AUX FUEL QTY .

a. Remove cockpit floor (PARA 2-4-43).

b. Prepare mating surfaces for electrical bond as follows:

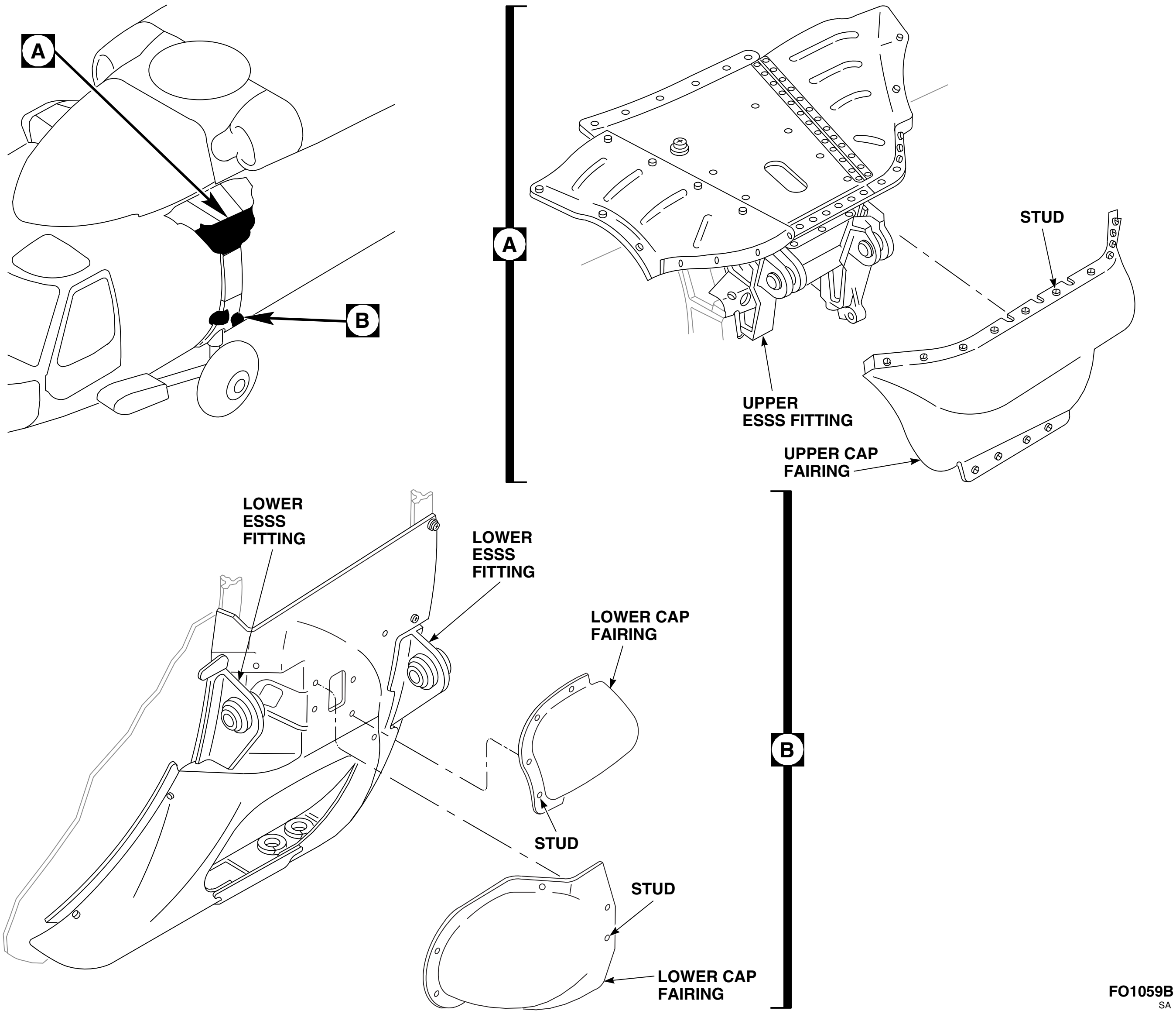


FIGURE 16-6-7-2. REMOVE UPPER AND LOWER CAP FAIRINGS

(1) Clean paint from around mounting holes using abrasive cloth, Item 8, Appendix D.

(2) Clean area with clean cheesecloth, Item 90, Appendix D, moistened with dry-cleaning solvent, Item 124, Appendix D.

(3) Apply corrosion resistant coating, Item 118, Appendix D, to contacting area.

c. Install signal conditioner on mounting bracket and secure with screws, lockwashers, and washers (Figure 16-6-7-4, Detail A).

d. Remove protective caps and plugs from electrical connectors.

e. Inspect and treat electrical connectors (PARA 1-7-20).

f. Connect electrical connectors, P360 to J1, and P361 to J2, on signal conditioner.

g. Make sure area is clean and free of foreign material.

h. Install cockpit floor (PARA 2-4-43).

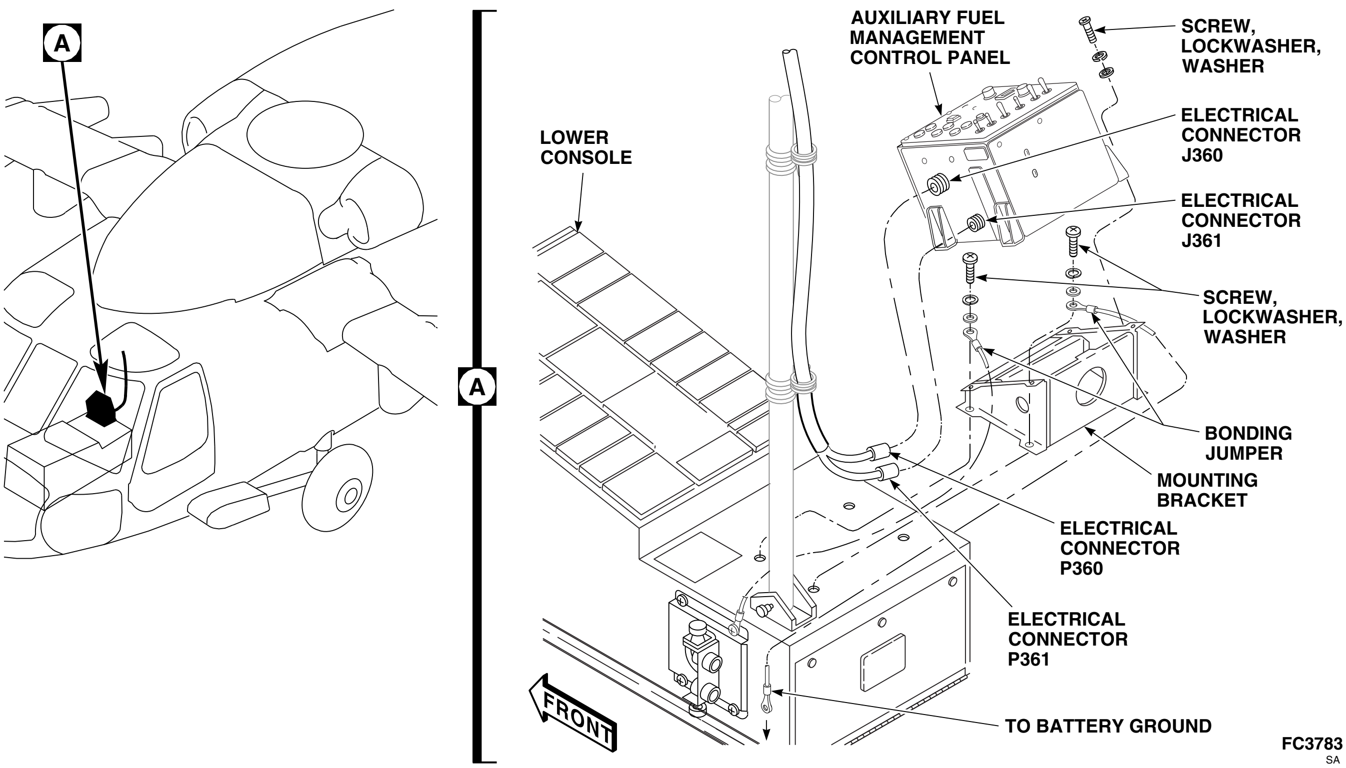


FIGURE 16-6-7-3. REMOVE AUXILIARY FUEL MANAGEMENT CONTROL PANEL

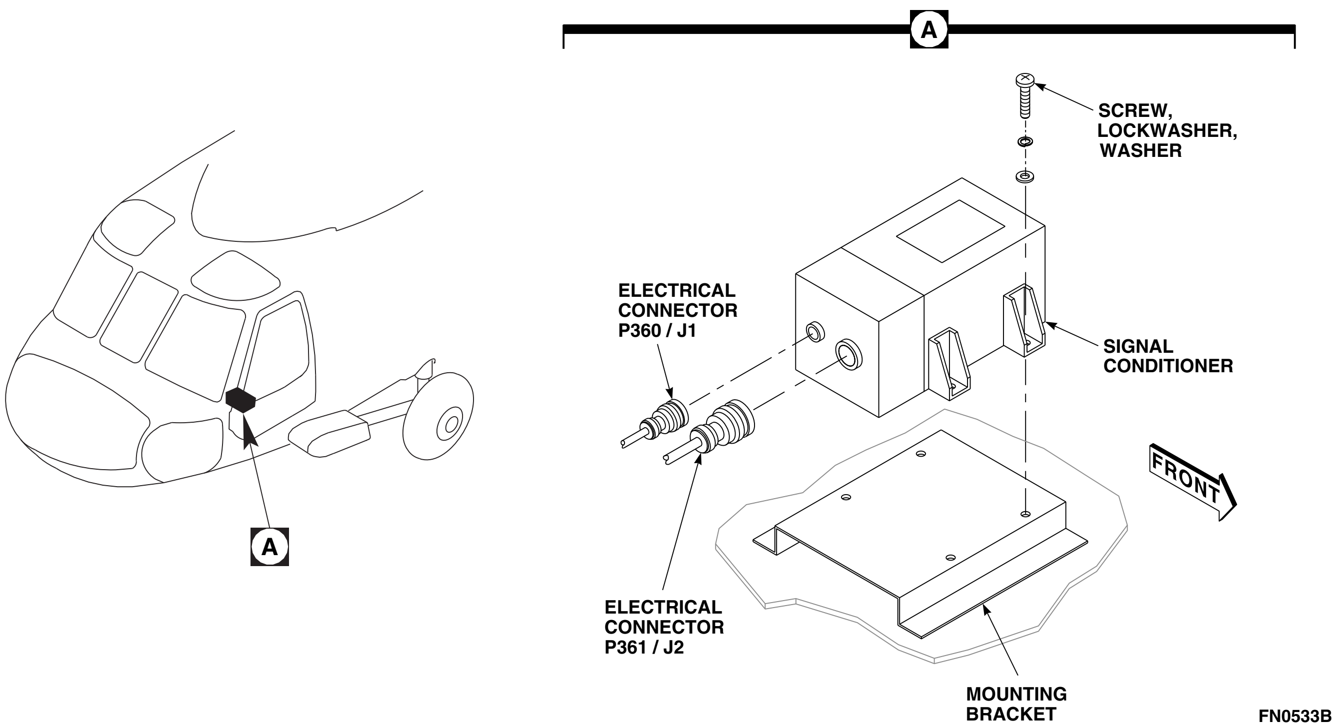


FIGURE 16-6-7-4. INSTALL SIGNAL CONDITIONER

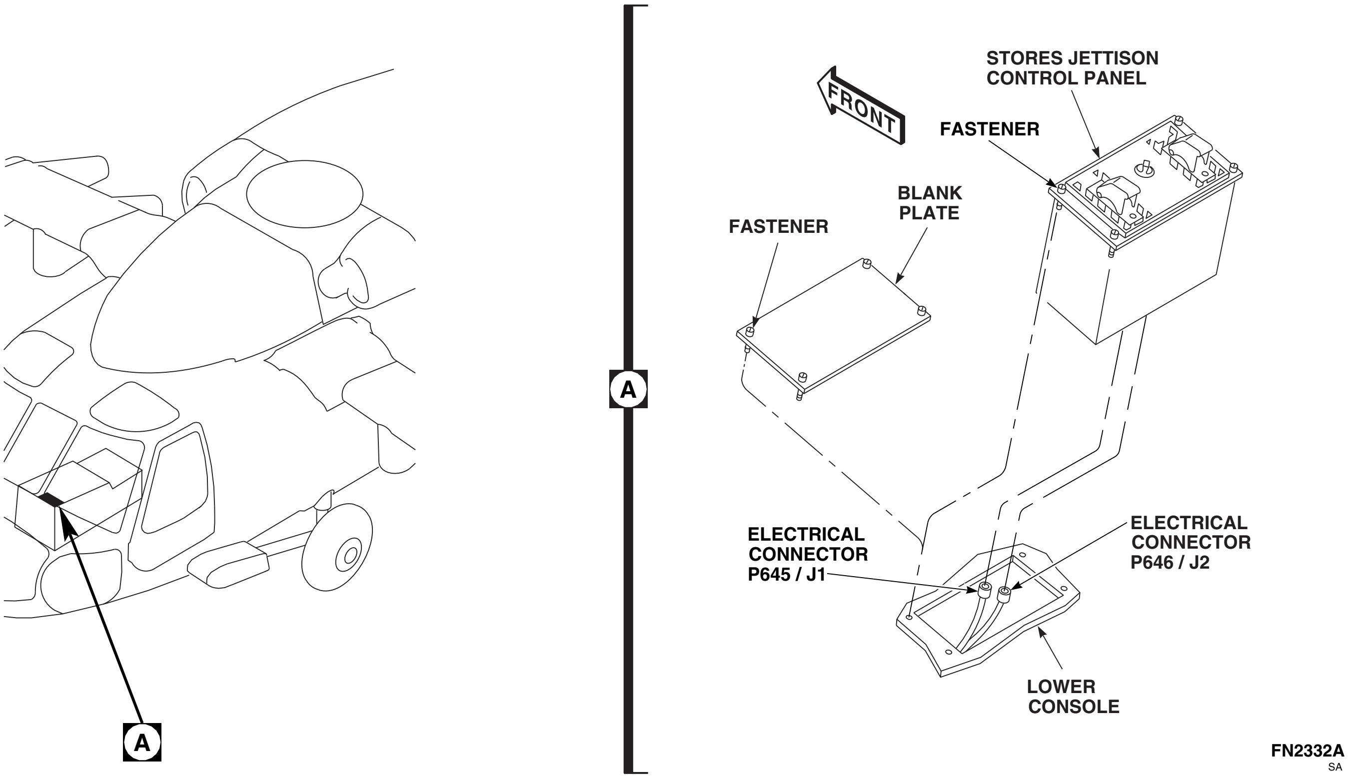


FIGURE 16-6-7-5. INSTALL STORES JETTISON CONTROL PANEL

16-6-7.1.4 Install Stores Jettison Control Panel.

a. Loosen fasteners and lift blank plate from lower console until electrical connectors are exposed (Figure 16-6-7-5, Detail A).

b. Disconnect electrical connectors, P645 and P646, from blank plate. Remove blank plate.

c. Inspect and treat electrical connectors (PARA 1-7-20).

d. Connect electrical connectors, P645 to J1, and P646 to J2, on stores jettison control panel.

e. Make sure area is clean and free of foreign material.

f. Install stores jettison control panel in lower console and secure with fasteners.

16-6-7.1.5 Install Cabin Fuel Harness W/O AUX FUEL QTY .

Install cabin fuel harness (PARA 16-4-43).

16-6-7.1.6 Install Control Display Panel AUX FUEL QTY .

a. Loosen fasteners and lift blank plate from lower console (Figure 16-6-7-6, Detail A).

b. Remove protective caps and plugs from electrical connectors.

c. Inspect and treat electrical connectors (PARA 1-7-20).

d. Connect electrical connectors, P3001 to J1, and P3002 to J2, on ESSS control display panel.

e. Make sure area is clean and free of foreign material.

f. Install control display panel in lower console and secure with fasteners.

16-6-7.1.7 Install Cabin Jettison Harness W/O AUX FUEL QTY .

Install cabin jettison harness (PARA 16-4-52).

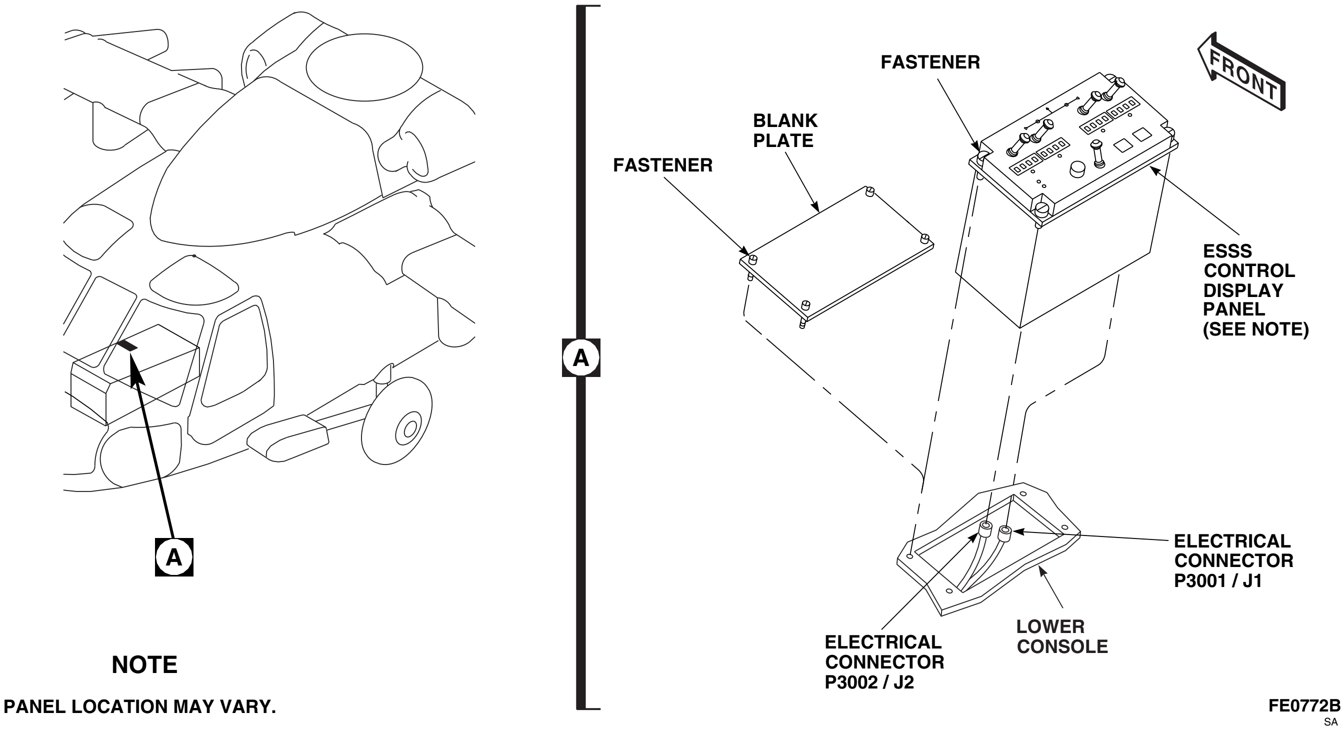


FIGURE 16-6-7-6. INSTALL CONTROL DISPLAY PANEL AUX FUEL QTY

16-6-7.1.8 Install Fuel Vent Tube Elbow and ESSS Fuel Overflow Sensor.

a. Remove fuel vent tube elbow, without ESSS fuel overflow sensor, as follows (Figure 16-6-7-7, Detail A):

(1) Loosen fitting on rear vent tube connector.

(2) Loosen fitting on vent tube Y connector.

(3) Remove screw, lockwasher, washers, bonding jumpers, and nuts. Remove fuel vent tube elbow from panel.

(4) Remove and discard packings from end of vent tube Y and rear vent tube.

b. Install fuel vent tube elbow, with ESSS fuel overflow sensor, as follows:

(1) Lubricate packings using petrolatum, Item 234, Appendix D. Install packings on end of vent tube Y and rear vent tube.

(2) Coat threads of fuel vent tube elbow using antiseize compound, Item 74, Appendix D.

(3) Install fuel vent tube elbow on panel using screws, washers, lockwasher, bonding jumpers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

(4) Seal bonding jumpers using sealing compound, Item 292, Appendix D.

(5) Screw fittings on rear vent tube and vent tube Y connectors to fuel vent tube elbow. **TIGHTEN FITTINGS FINGERTIGHT.**

(6) Lubricate packing using petrolatum, Item 234, Appendix D. Install packing on end of ESSS fuel overflow sensor (Figure 16-6-7-7, Detail B).

(7) Coat threads of overflow sensor using antiseize compound, Item 74, Appendix D.

(8) Screw overflow sensor into fuel vent tube elbow. **TORQUE OVERFLOW SENSOR TO 40 - 50 INCH-POUNDS.**

(9) Remove protective caps and plugs from electrical connectors.

(10) Inspect and treat electrical connectors (PARA 1-7-20).

(11) Connect electrical connector, P376 to J376.

(12) Secure wiring to helicopter wiring harness with existing clamps.

(13) Secure wiring to existing clip at STA 443.5, LBL 17.0 with clamp, screw, washer, and nut.

16-6-7.1.9 Install Flow Sensor Panel.

a. Install flow sensor panel on bracket and secure with fasteners (Figure 16-6-7-8, Detail A).

b. Connect quick-disconnect bonding jumper between flow sensor panel and bracket.

c. Disconnect electrical connector, P347, from receptacle on flow sensor panel. Install cap on receptacle. Remove cap from electrical connector, J347, located on intercostal.

d. Inspect and treat electrical connectors (PARA 1-7-20).

e. Connect electrical connector, P347 to J347, on intercostal. Install clamp on wire harness.

f. Disconnect flow sensor fuel lines from receptacles on panel. Install caps on receptacles.

g. Loosen B-nuts and remove quick-disconnects from flow sensor fuel lines.

h. Insert flow sensor fuel lines through holes in intercostal and install fuel lines in clips on airframe.

i. Install quick-disconnects on flow sensor fuel lines. **TORQUE B-NUTS TO 330 - 360 INCH-POUNDS.**

j. Remove caps from quick-disconnects at STA 398.0 and connect flow sensor fuel line quick-disconnects. Install caps on fittings.

k. Remove cap from extended range breakaway valve and store on dummy receptacle. Remove and discard packing.

l. Record K-FACTOR, stamped on flow transmitter, for use in calibrating auxiliary fuel management control panel.

m. Remove protective plug from main fuel hose.

CAUTION

Do not preload extended range breakaway valve when connecting flowmeter fitting on extended range breakaway valve. Frangible section of extended range breakaway valve is sensitive to bending and tension overload.

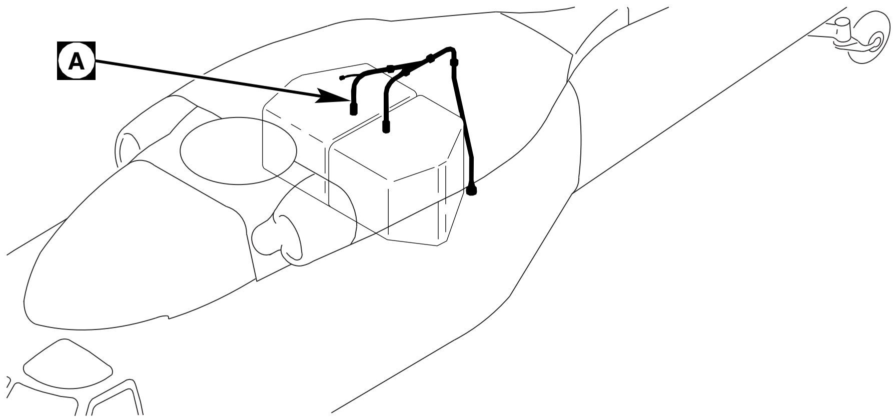
n. Connect flowmeter fitting, on main fuel hose, to extended range breakaway valve. **TIGHTEN GAMAH NUT ON FLOWMETER FITTING HANDTIGHT.**

o. Remove protective caps and plugs from electrical connectors.

p. Inspect and treat electrical connectors (PARA 1-7-20).

q. Connect electrical connector, P523, to flow transmitter on No. 2 main fuel cell.

r. Make sure area is clean and free of foreign material.



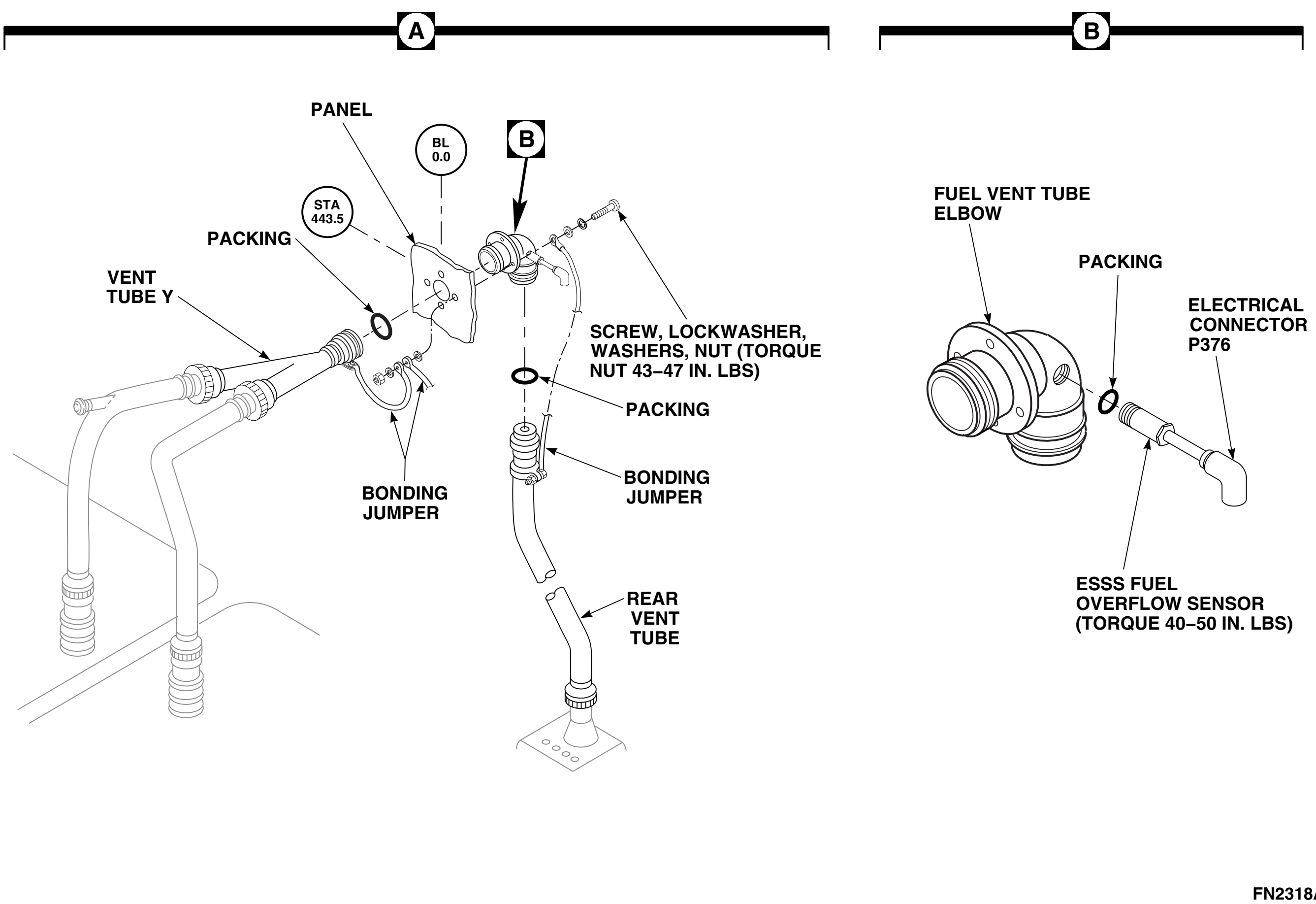


FIGURE 16-6-7-7. INSTALL FUEL VENT TUBE ELBOW AND ESSS FUEL OVERFLOW SENSOR

s. Make sure flow sensor fuel lines do not chafe on main fuel cell vent lines.

16-6-7.1.10 Install Horizontal Stores Support (HSS).

CAUTION

Damage to composite material will result if HSS or support struts are dropped or mishandled. Composite material may break, become warped, or be overstressed. Take care when handling HSS or support struts.

NOTE

- Retirement time of main rotor shafts, 70351-08131-044 through -046, is reduced from 1300 hours to 1000 hours if any type of external loads were installed on external stores support system helicopter hard points. Upon addition of loads to hard points, main modules with main rotor shafts, 70351-08131-044 through -046, installed must be reidentified with an ″E″ following main module serial number. There is no change in retirement life for main rotor shafts, 70351-08131-047 and -048.
- Installation of HSS is the same for left and right side, except where noted.

a. Inspect pneumatic check valve and fuselage fixed provisions for cleanliness.

NOTE

- Resistance test of fuel shutoff gate valves and associated wire harness is the same for left and right side.
- Gate valve resistance test can be performed with ESSS wing on or off helicopter.
- Allow approximately 20 seconds for capacitors to charge to attain greater than 20 megohm reading.

b. Using digital multimeter, perform resistance test of fuel shutoff gate valve and associated wire harness as follows:

(1) Measure resistance of right inboard fuel shutoff valve by connecting positive lead to ESSS wing disconnect electrical receptacle, J804, pin -3, and common lead to ground (ESSS structure). Repeat procedure at pins -4 and -5.

(2) Measure resistance of right outboard fuel shutoff valve by connecting positive lead to ESSS wing disconnect electrical receptacle, J804, pin -8, and common lead to ground (ESSS structure). Repeat procedure at pins -9 and -10.

(3) Measure resistance of left inboard fuel shutoff valve by connecting positive lead to ESSS wing disconnect electrical receptacle, J805, pin -3, and common lead to ground (ESSS structure). Repeat procedure at pins -4 and -5.

(4) Measure resistance of left outboard fuel shutoff valve by connecting positive lead to ESSS wing disconnect electrical receptacle, J805, pin -8, and common lead to ground (ESSS structure). Repeat procedure at pins -9 and -10.

(5) **IF MEASURED RESISTANCE BETWEEN ESSS WING DISCONNECTS, J804 AND J805, AND RESPECTIVE FUEL SHUTOFF VALVES IN STEP b. (1) THROUGH STEP b. (4) IS GREATER THAN 20 MEGOHMS ON ALL FOUR VALVES, RESISTANCE TEST IS ACCEPTABLE.**

(6) **IF MEASURED RESISTANCE BETWEEN ESSS WING DISCONNECTS, J804 AND J805, AND RESPECTIVE FUEL SHUTOFF VALVES IN STEP b. (1) THROUGH STEP b. (4) IS LESS THAN 20 MEGOHMS ON ANY VALVE, RESISTANCE TEST IS NOT ACCEPTABLE. TEST RESISTANCE OF EACH INBOARD AND OUTBOARD SHUTOFF VALVE** (PARA 16-4-33).

(7) If measured resistance on each fuel shutoff valve is acceptable, inspect and repair wire harness (Appendix F).

(8) If all shutoff valves measure greater than 20 megohms, inspect and repair wire harness (Appendix F).

c. Inspect fuel hose dummy electrical receptacle base on HSS fitting for corrosion. **NONE ALLOWED.** If corrosion is found, remove as follows:

(1) Mask off orange painted area and gold anodized areas of electrical receptacle using 1-inch masking tape, Item 212, Appendix D.

(2) Remove corrosion from electrical receptacle using wire brush.

(3) Clean surface using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D. Allow to dry.

(4) Using epoxy primer coating kit, Item 135, Appendix D, apply one coat of epoxy primer to electrical receptacle base and around edge. Allow to dry.

(5) Apply bead of sealing compound, Item 292, Appendix D, to edges around base.

(6) Remove masking tape from electrical receptacle.

d. Remove caps from pneumatic and fuel hoses, and electrical connectors on HSS. Store caps on dummy electrical receptacles on HSS.

e. Disconnect electrical connector, P682 (left side), and P681 (right side), from electrical receptacles, J684 (left side), and J683 (right side) (Figure 16-6-7-9, Sheet 2, Detail C).

f. Remove caps from dummy receptacles next to electrical receptacles and install caps on electrical receptacles, J684 (left side), and J683 (right side).

NOTE

- Expandable pins must be installed with hinged portion facing rear on rear strut and facing front on front strut. Improper installation may cause damage to main landing gear strut.
- Refer to decals on HSS and fuselage for installation of expandable pins.

g. Install fixed (nonadjustable) support strut on lower rear ESSS fitting and secure with expandable pin. Do not lock pin at this time (Figure 16-6-7-9, Sheet 1, Detail A).

h. With assistance, lift and install HSS with HSS fitting joined to upper ESSS fittings on fuselage (Figure 16-6-7-9, Sheet 2, Detail B).

i. Install fixed (nonadjustable) support strut on HSS fitting and secure with expandable pin. Do not lock pin at this time.

NOTE

Alternately raise and lower HSS to ease installation of expandable pins.

j. Secure HSS to upper ESSS fitting with expandable pins. Lock all pins at this time. Make sure handles are locked over tapered ends on pins, and that expanding ring is visible (Figure 16-6-7-9, Sheet 2, Detail B and Detail C). Inspect expandable pins (PARA 16-3-1). Apply torque stripe (PARA 2-6-5).

k. Install adjustable support strut on HSS and lower ESSS fitting. If necessary, loosen jamnut and adjust end fitting (Figure 16-6-7-9, Sheet 1, Detail A). Inspect expandable pins (PARA 16-3-1). **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

CAUTION

- Make sure adjustable (front) support strut end fitting lines up with lower fuselage end fitting before installing expandable pin. Improper alignment of fittings will prevent proper installation of expandable pin. Expandable pins shall be inserted with hand pressure only.
- Expandable pins must be installed with hinged portion facing rear on rear strut and facing front on front strut. Improper installation may cause damage to main landing gear strut.

l. Install expandable pins in fittings. Make sure handles are locked over nuts on ends of pins, and that expanding rings are visible. Inspect expandable pins (PARA 16-3-1). Apply torque stripe (PARA 2-6-5).

m. Connect quick-disconnect bonding jumper between struts and lower ESSS fittings.

n. Remove bolt, lockwasher, and washer located between ESSS fittings and install bonding jumpers from HSS. **TORQUE BOLTS TO 106 - 115 INCH-POUNDS.**

o. Remove protective caps and plugs from electrical connectors.

p. Inspect and treat electrical connectors (PARA 1-7-20).

q. Install electrical connectors, P682 to J682 (left side), and P681 to J681 (right side).

r. Remove dummy plugs from ends of fuel and pneumatic fixed provisions.

s. Install fuel and pneumatic lines on fuselage fuel and pneumatic quick-disconnect fittings.

t. Remove protective caps and plugs from electrical connectors.

u. Inspect and treat electrical connectors (PARA 1-7-20).

v. Connect fuel system wiring harness electrical connectors, located on airframe, to HSS fitting as follows:

(1) P805 to J805, and P833 to J833 (left side).

(2) P804 to J804, and P834 to J834 (right side).

w. Remove protective caps and plugs from electrical connectors.

x. Inspect and treat electrical connectors (PARA 1-7-20).

y. Connect jettison system wiring harness electrical connectors, located on airframe, to HSS fitting as follows:

(1) P678 to J678, and P680 to J680 (left side).

(2) P677 to J677, and P679 to J679 (right side).

z. Install electrical connector breakaway cables in channel attached to fitting with quick-release pin.

16-6-7.1.11 Install Vertical Support Pylons (VSP).

Install ESSS vertical support pylon (VSP) (PARA 16-4-37).

16-6-7.1.12 Install Ejector Racks, BRU-22A/A.

If external fuel tanks are to be installed, install ejector racks, BRU-22A/A (PARA 16-4-34).

16-6-7.1.13 Install External 230 Gallon Fuel Tanks.

NOTE

Make sure front vertical support pylon (VSP) fairings are installed before installing tanks.

a. If necessary, install front VSP fairings (PARA 16-4-21).

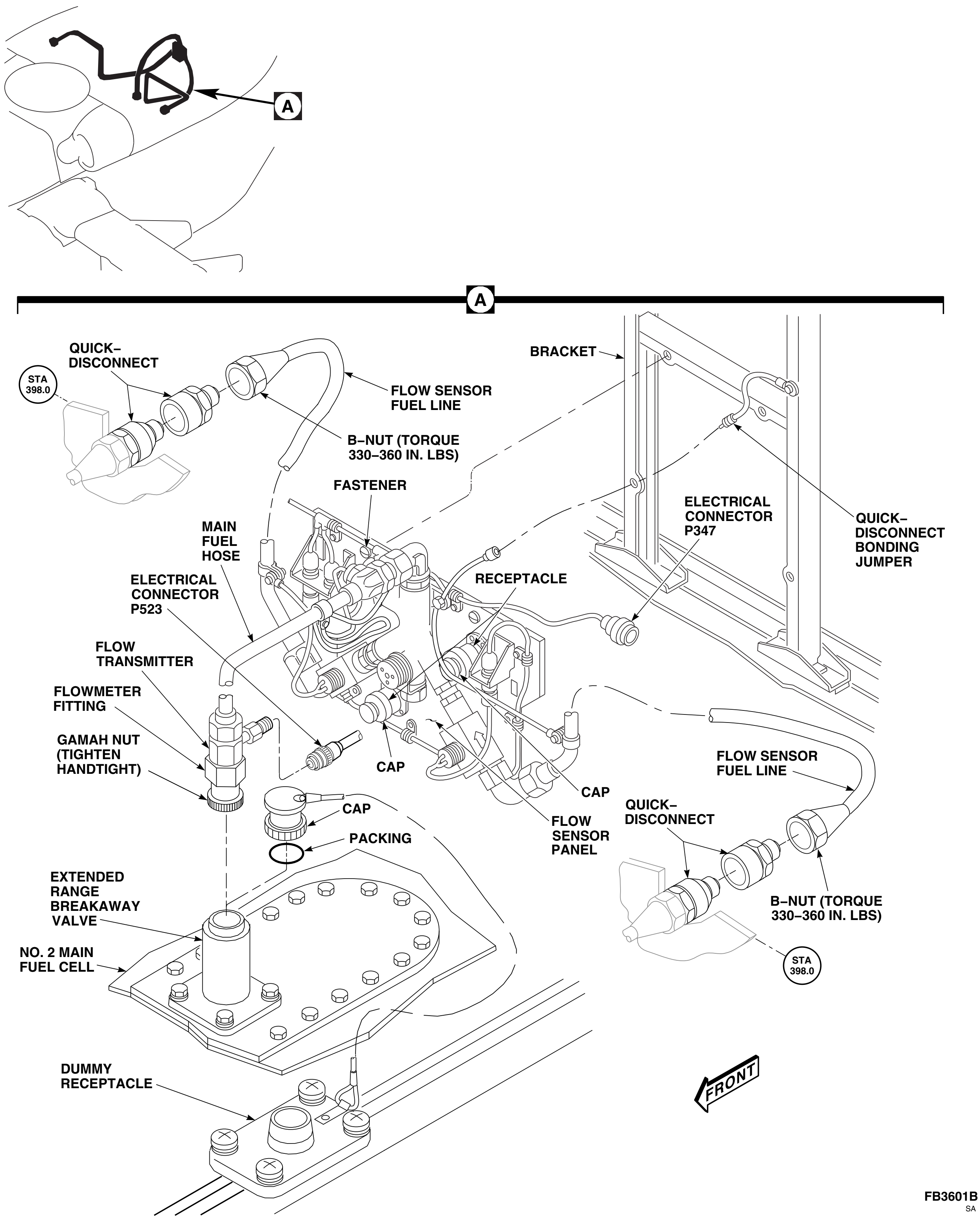
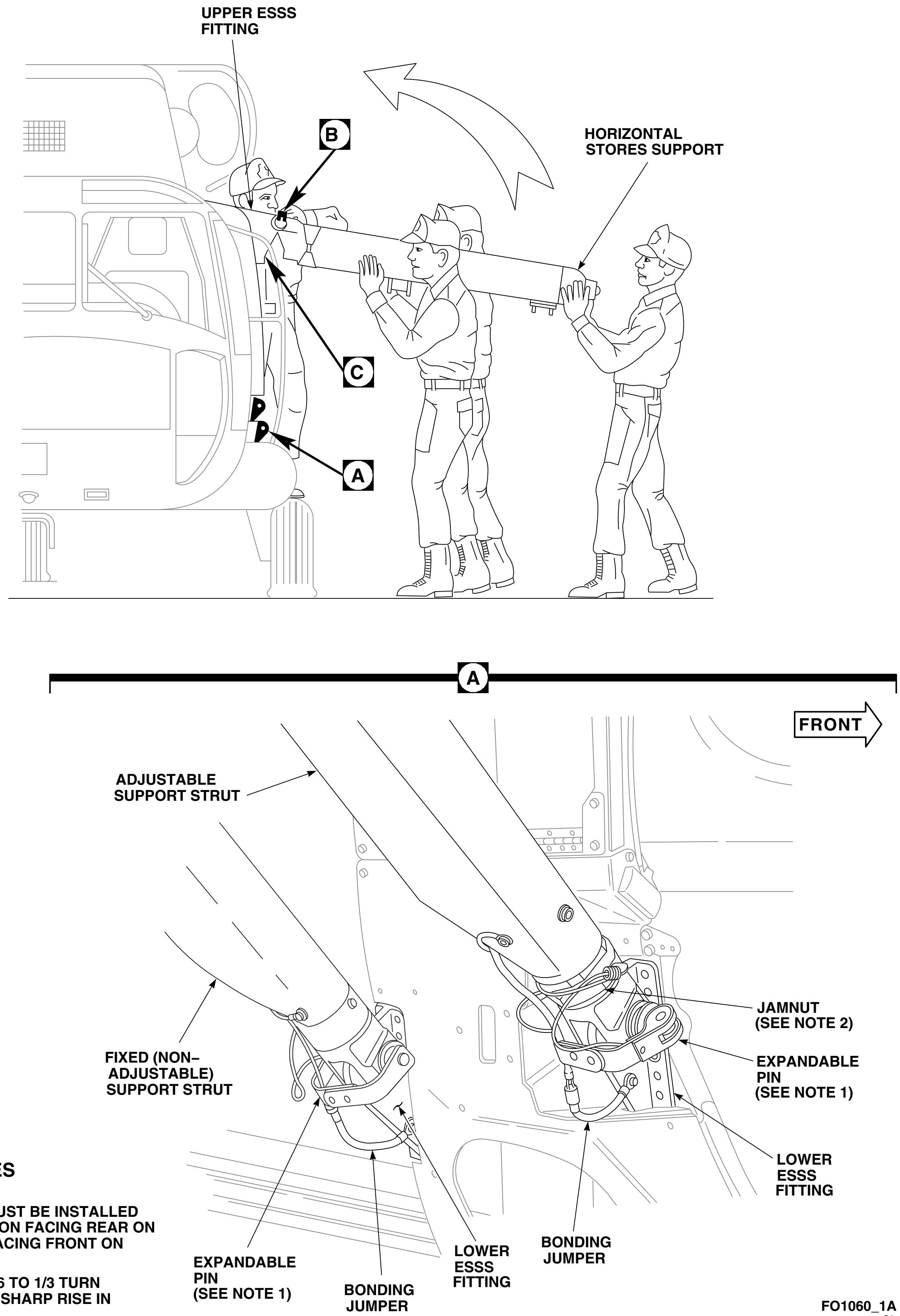


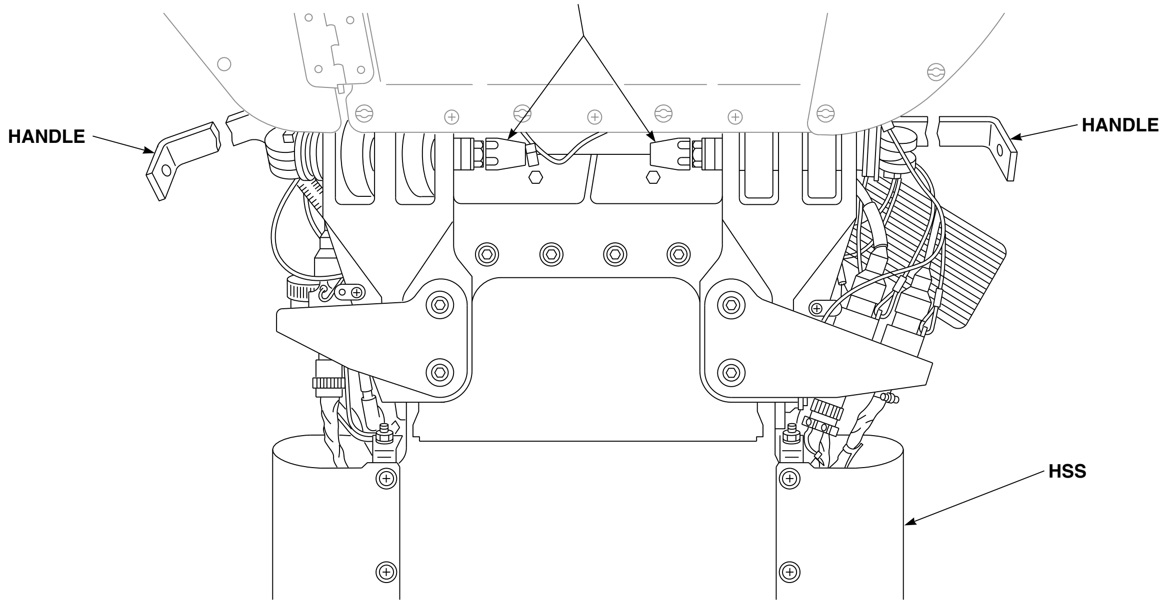
FIGURE 16-6-7-8. INSTALL FLOW SENSOR PANEL



NOTES

1. EXPANDABLE PIN MUST BE INSTALLED WITH HINGED PORTION FACING REAR ON REAR STRUT AND FACING FRONT ON FRONT STRUT.
2. TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

FIGURE 16-6-7-9. INSTALL HORIZONTAL STORES SUPPORT (HSS) (SHEET 1 OF 2)



TOP VIEW LOOKING DOWN AT HSS AND FUSELAGE CLEVIS
– PINS INSTALLED – HANDLES OPEN

B

C

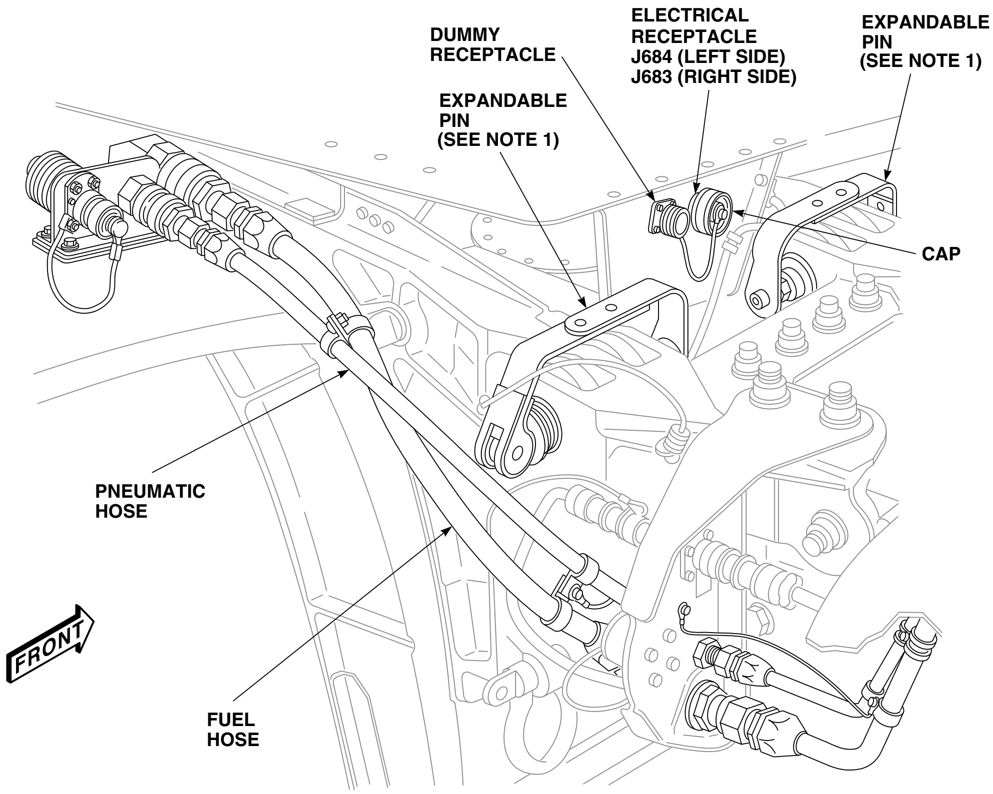


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FIGURE 16-6-7-9. INSTALL HORIZONTAL STORES SUPPORT (HSS) (SHEET 2 OF 2)

b. Install ESSS external 230 gallon fuel tanks (PARA 16-4-54).

16-6-7.1.14 Complete Installation.

a. Install rear vertical support pylon fairings (PARA 16-4-21).

b. Perform operational check of external stores support system (ESSS) range extension system and check for fuel leaks (PARA 16-2-1 or PARA 16-2-2).

c. Perform operational check of ESSS jettison system (PARA 16-2-4).

d. Install upper and lower root fairings (PARA 16-4-21).

e. Make sure fuel cell area is clean and free of foreign material.

f. If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

g. Install left and right enclosure panels on top of main fuel cells. Lock fasteners.

h. Install cargo netting.

i. Install soundproofing panels (PARA 2-4-69).

j. Install rear passenger's seats (PARA 2-4-45).

16-6-8 EXTERNAL STORES SUPPORT SYSTEM (ESSS) KIT REMOVAL.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 1-Gallon
- Electrical Repairer's Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lockwire, Item 196, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-2
- PARA 1-3-4
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-43
- PARA 2-4-45
- PARA 2-4-69
- PARA 9-2-14
- PARA 10-4-30
- PARA 13-5-3
- PARA 13-5-6
- PARA 16-4-21
- PARA 16-4-34
- PARA 16-4-37
- PARA 16-4-43
- PARA 16-4-52
- PARA 16-4-54

16-6-8.1 Remove External Stores Support System (ESSS) Kit.

16-6-8.1.1 Prepare to Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Make sure helicopter is properly grounded.

c. Remove upper and lower root fairings and inboard and outboard front and rear vertical support pylon fairings (PARA 16-4-21).

d. Remove rear passenger's seats as required (PARA 2-4-45).

e. Remove soundproofing panel from rear bulkhead covering left or right stowage compartment (PARA 2-4-69).

f. Remove cargo netting from stowage compartment.

g. If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

h. Unlock fasteners securing left and right enclosure panels on top of main fuel cells. Remove panels.

16-6-8.1.2 Remove External 230 Gallon Fuel Tanks.

Remove ESSS external 230 gallon fuel tanks (PARA 16-4-54).

16-6-8.1.3 Remove Ejector Racks, BRU-22A/A.

If installed, remove ejector racks, BRU-22A/A (PARA 16-4-34).

16-6-8.1.4 Remove Vertical Support Pylons (VSP).

Remove vertical support pylons (VSP) (PARA 16-4-37).

16-6-8.1.5 Remove Horizontal Stores Support (HSS).

a. Remove fuel system wiring harness electrical connectors, P805 and P833 (left side), and P804 and P834 (right side), from root end of HSS. Install dummy connectors.

b. Remove jettison system wiring harness electrical connectors, P678 and P680 (left side), and P677 and P679 (right side), from root end of HSS. Install dummy connectors.

c. Remove position light wiring harness electrical connectors, P682 (left side), and P681 (right side), from root end of HSS.

d. Remove bolts and washers securing bonding jumpers on HSS fitting to helicopter fuselage. Install bolts and washers on helicopter fuselage. **TORQUE BOLTS TO 106 - 115 INCH-POUNDS.**

e. Remove fuel and pneumatic quick-disconnects from fuselage provisions (Figure 16-6-8-1, Sheet 1, Detail B). Install dummy plugs on fuselage provisions.

f. Disconnect bonding quick-disconnects on jumper between ESSS fittings and support struts (Figure 16-6-8-1, Sheet 1, Detail A).

WARNING

Injury to personnel and damage to equipment will result if HSS is not supported prior to removal of expandable pins and support struts. HSS will drop. Make sure HSS is supported prior to removal of expandable pins and support struts.

CAUTION

Damage to composite material will result if HSS or support struts are dropped or mishandled. Composite

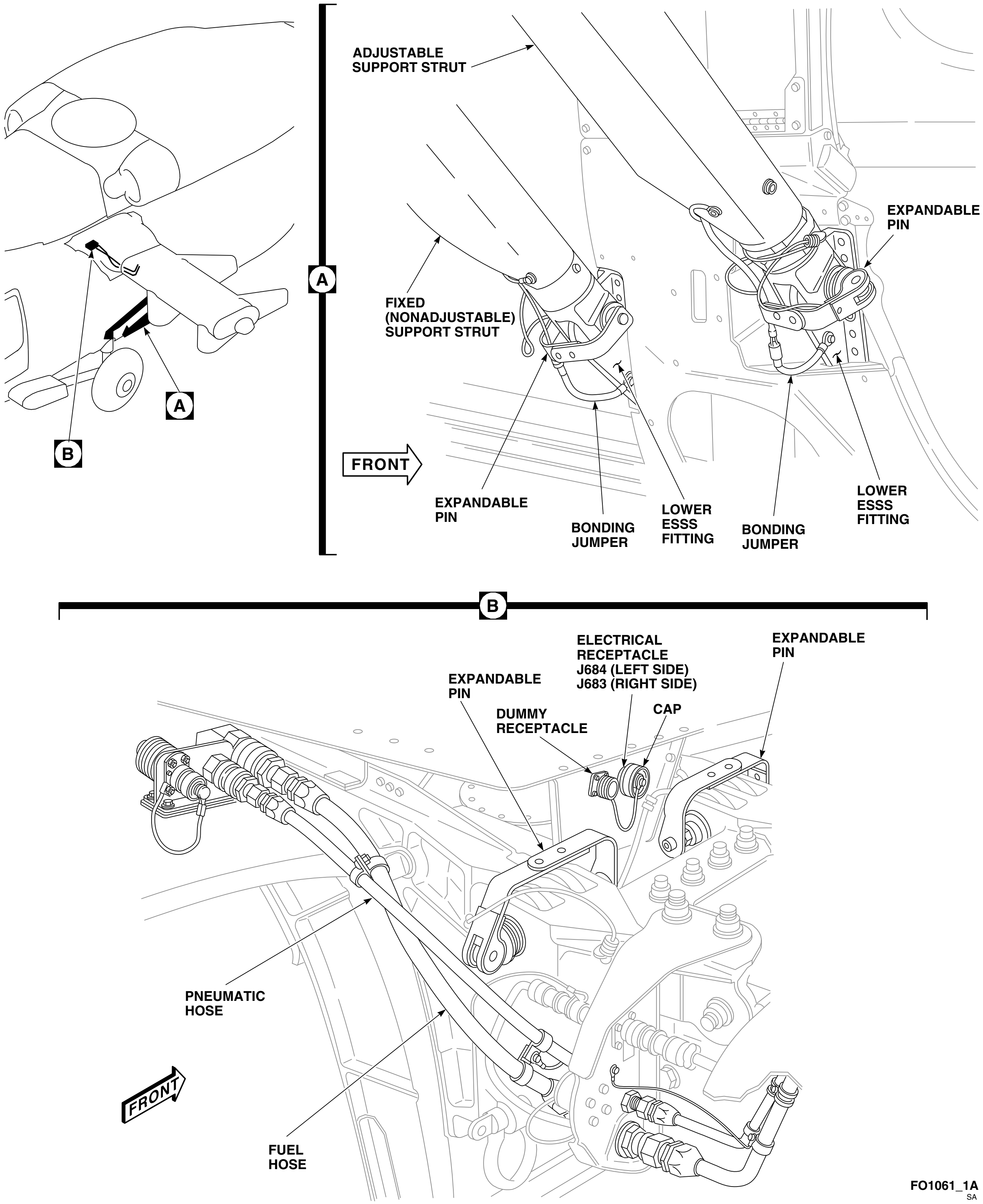


FIGURE 16-6-8-1. REMOVE HORIZONTAL STORES SUPPORT (HSS) (SHEET 1 OF 2)

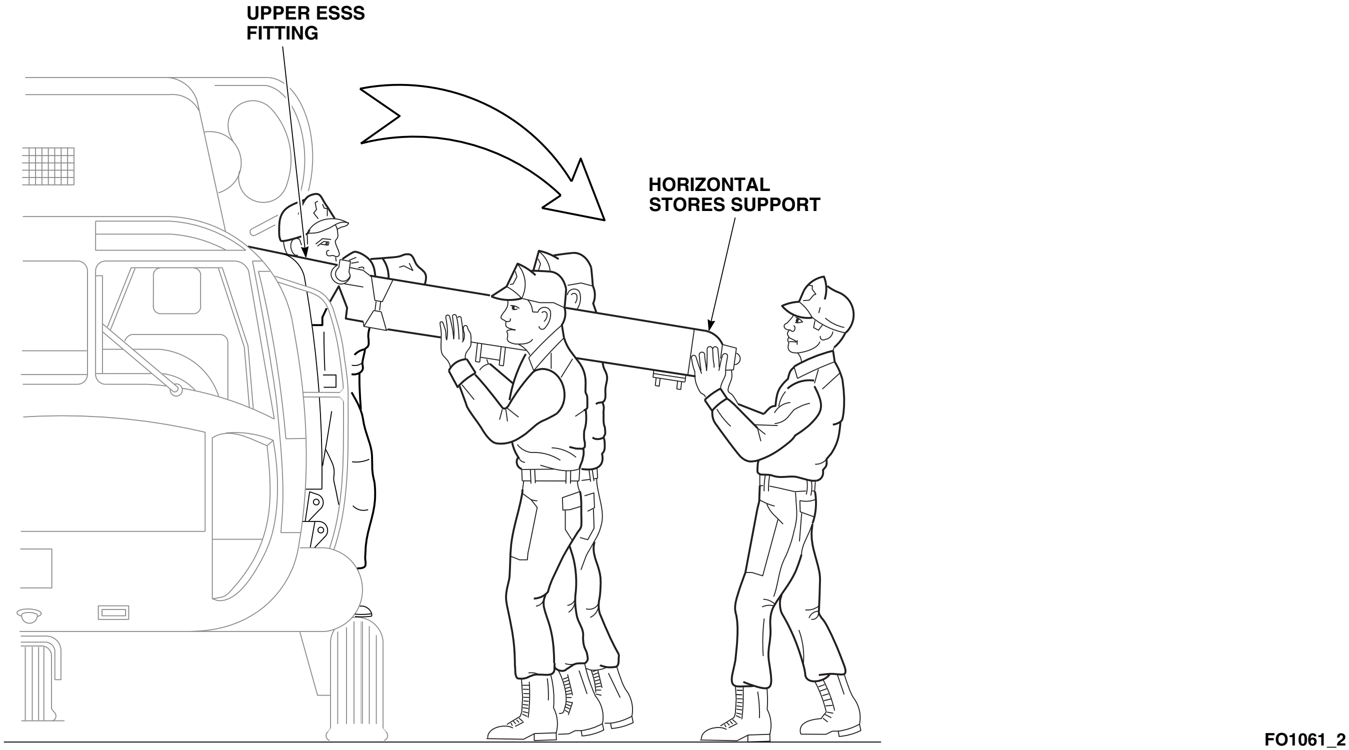


FIGURE 16-6-8-1. REMOVE HORIZONTAL STORES SUPPORT (HSS) (SHEET 2 OF 2)

material may break or become warped and/or overstressed. Take care when handling HSS or support struts.

g. Unlock and remove expandable pins securing struts to lower ESSS fittings. Remove struts from fuselage.

h. Unlock and remove expandable pins securing struts to HSS. Remove struts from HSS.

i. While assistants hold HSS, unlock and remove expandable pins securing support to upper ESSS fittings (Figure 16-6-8-1, Sheet 1, Detail B and Sheet 2).

j. Lower support to ground.

k. Remove caps from electrical connectors, J684 (left side), and J683 (right side). Install caps on dummy plugs next to connectors (Figure 16-6-8-1, Sheet 1, Detail B).

l. Inspect and treat electrical connectors (PARA 1-7-20).

m. Connect electrical connector, P682 (left side), and P681 (right side), to electrical connectors, J684 and J683.

n. Install fuel and pneumatic quick-disconnects on dummy plugs on HSS fitting.

NOTE

When storing HSS for extended period of time (greater than one year), do not install and close expandable pins in its respective part.

o. Tie unexpanded pins to part with lockwire, Item 196, Appendix D, when storing HSS for extended periods.

16-6-8.1.6 Remove Flow Sensor Panel.

a. Disconnect electrical connector, P523, from flow transmitter on No. 2 main fuel cell (Figure 16-6-8-2, Detail A).

b. Install protective caps and plugs on electrical connectors.

c. Disconnect flowmeter fitting, on main fuel hose, from extended range breakaway valve.

d. Install protective plug in flowmeter fitting.

CAUTION

Do not preload extended range breakaway valve when installing cap. Frangible section of extended range breakaway valve is sensitive to bending and tension overload.

e. Install packing on top of extended range breakaway valve, remove cap from dummy receptacle and install on valve.

f. Position 1-gallon container to catch fuel.

g. Disconnect flow sensor fuel lines from quick-disconnect ends at STA 398.0. Remove protective caps from fittings and install on quick-disconnect ends at STA 398.0.

h. Loosen B-nuts and remove quick-disconnects from flow sensor fuel lines.

i. Remove flow sensor fuel lines from clips on airframe and pull lines through hole in intercostal.

j. Install quick-disconnects on flow sensor fuel line. **TORQUE B-NUTS TO 330 - 360 INCH-POUNDS.**

k. Remove caps from receptacles on panel and connect fuel lines to receptacles.

l. Disconnect electrical connector, P347 from J347, on intercostal.

m. Install protective caps and plugs on electrical connectors.

n. Remove clamp from electrical harness.

o. Disconnect quick-disconnect bonding jumper between flow sensor panel and bracket.

p. Unlock fasteners and remove flow sensor panel from brackets by sliding panel to right and removing through opening.

q. Install protective cover on flowmeter fitting.

r. Remove protective caps and plugs from electrical connectors.

s. Inspect and treat electrical connectors (PARA 1-7-20).

t. Connect electrical connector, P347, to receptacle on flow sensor panel.

16-6-8.1.7 Remove Fuel Vent Tube Elbow and ESSS Fuel Overflow Sensor.

a. Remove fuel vent tube elbow, with external stores support system fuel overflow sensor, as follows:

(1) Remove clamp, screw, washer, and nut from wiring at STA 443.5 and LBL 17.0.

(2) Disconnect wiring from existing clamp on helicopter wiring harness.

(3) Disconnect electrical connector, P376 from J376 (Figure 16-6-8-3, Detail B).

(4) Install protective caps and plugs on electrical connectors.

(5) Remove overflow sensor and packing from fuel vent tube elbow. Discard packing.

(6) Loosen fitting on rear vent tube connector (Figure 16-6-8-3, Detail A).

(7) Loosen fitting on vent tube Y connector.

(8) Remove screws, washers, lockwasher, bonding jumpers, and nuts and remove fuel vent tube elbow.

(9) Remove and discard packings from end of vent tube Y and rear vent tube.

b. Install fuel vent tube elbow without overflow sensor (PARA 10-4-30).

16-6-8.1.8 Remove Cabin Fuel Harness W/O AUX FUEL QTY .

Remove cabin fuel harness (PARA 16-4-43).

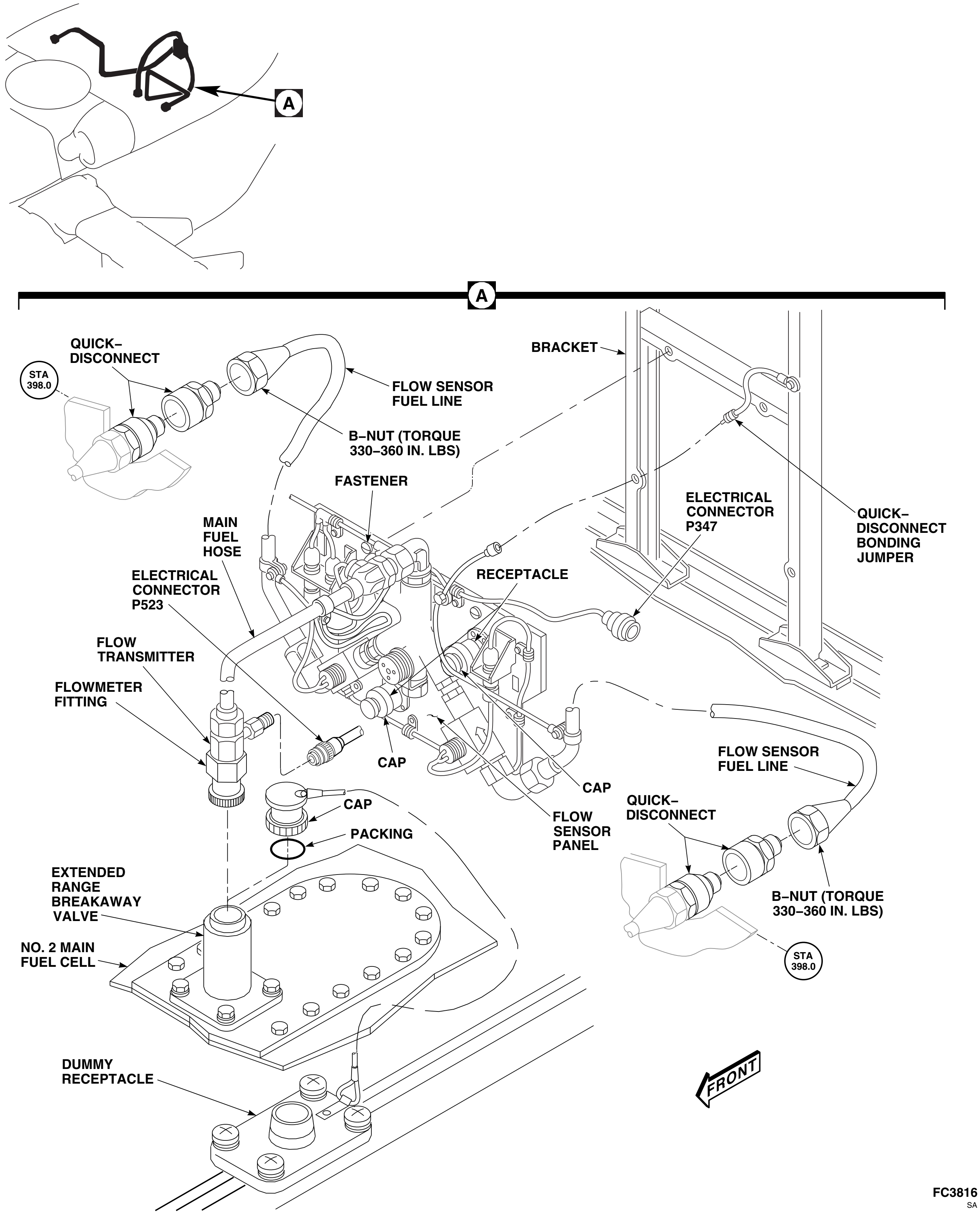


FIGURE 16-6-8-2. REMOVE FLOW SENSOR PANEL

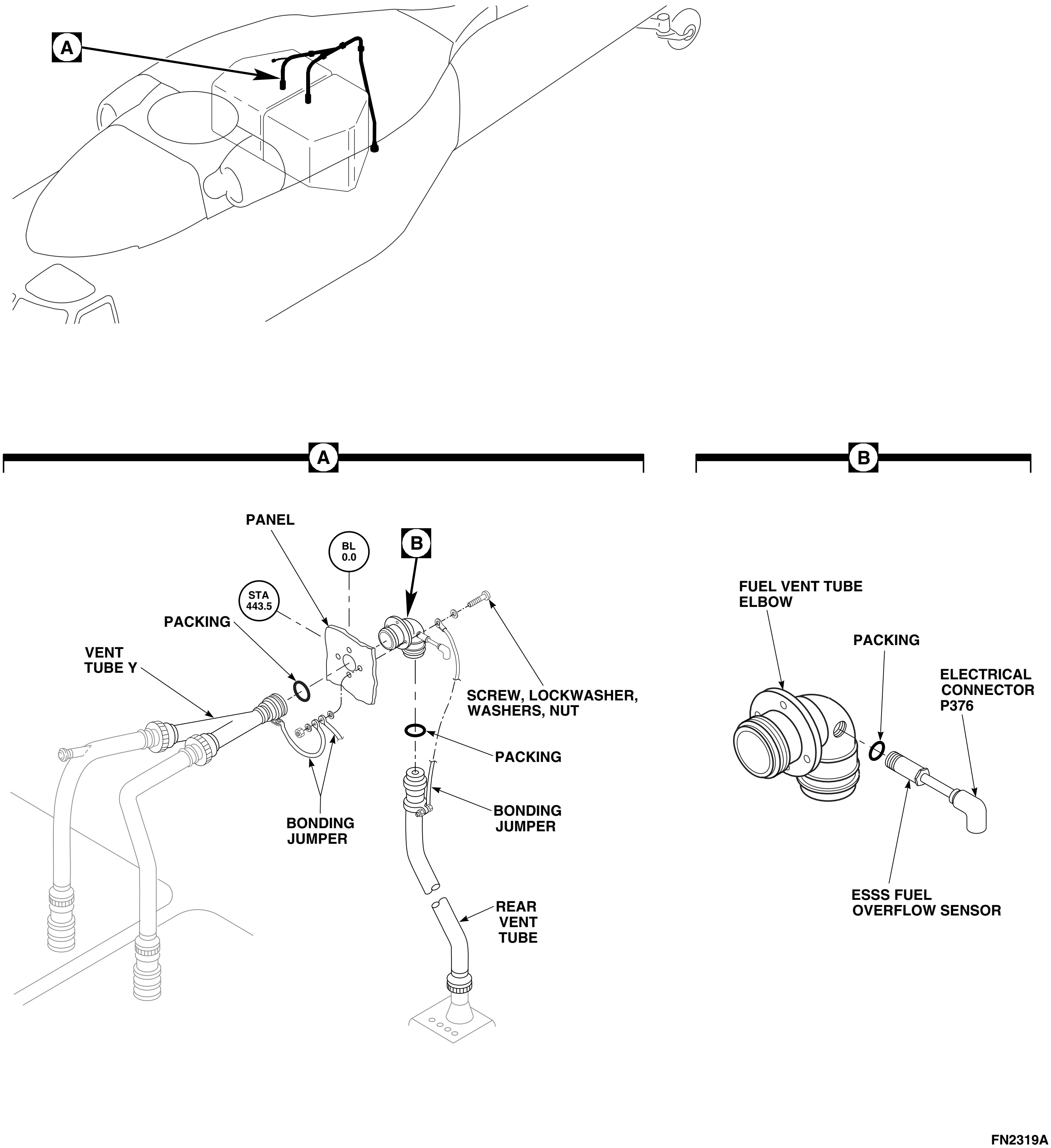


FIGURE 16-6-8-3. REMOVE FUEL VENT TUBE ELBOW AND ESSS FUEL OVERFLOW SENSOR

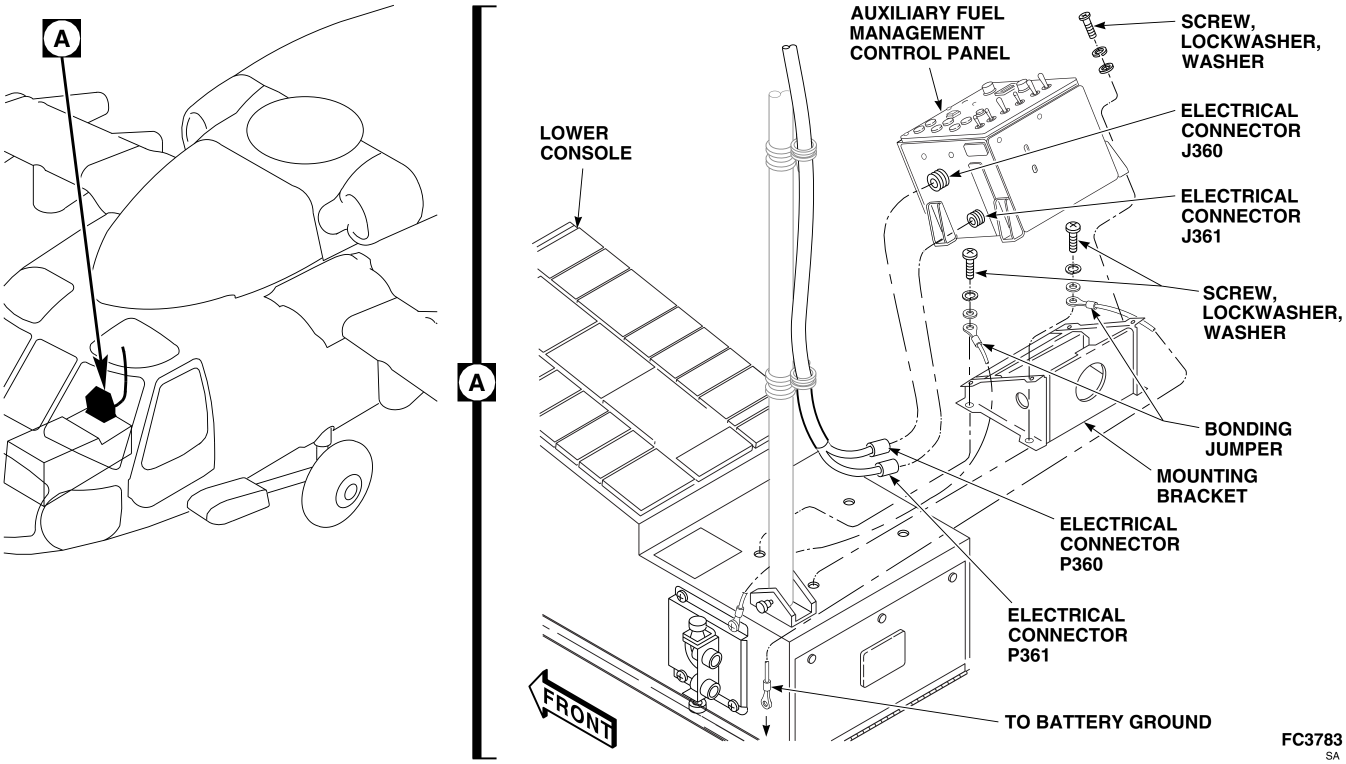


FIGURE 16-6-8-4. REMOVE AUXILIARY FUEL MANAGEMENT CONTROL PANEL

16-6-8.1.9 Remove Cabin Jettison Harness W/O AUX FUEL QTY .

Remove cabin jettison harness (PARA 16-4-52).

16-6-8.1.10 Remove Auxiliary Fuel Management Control Panel W/O AUX FUEL QTY .

a. Disconnect electrical connectors, P360 from J360, and P361 from J361, on auxiliary fuel management control panel (Figure 16-6-8-4, Detail A).

b. Install protective caps and plugs on electrical connectors.

c. Remove screws, lockwashers, and washers and remove auxiliary fuel management control panel from mounting bracket on lower console.

d. Remove screws, lockwashers, washers, upper ends of bonding jumpers, and mounting bracket from lower console.

16-6-8.1.11 Remove Signal Conditioner AUX FUEL QTY .

a. Remove cockpit floor (PARA 2-4-43).

b. Disconnect electrical connectors, P360 from J1, and P361 from J2, from signal conditioner (Figure 16-6-8-5, Detail A).

c. Install protective caps and plugs on electrical connectors.

d. Remove screws, lockwashers, washers and signal conditioner from mounting bracket.

e. Install cockpit floor (PARA 2-4-43).

16-6-8.1.12 Remove Control Display Panel AUX FUEL QTY .

a. Loosen fasteners and lift control display panel from lower console until electrical connectors are exposed (Figure 16-6-8-6, Detail A).

b. Disconnect electrical connectors, P3001 from J1, and P3002 from J2, from control display panel.

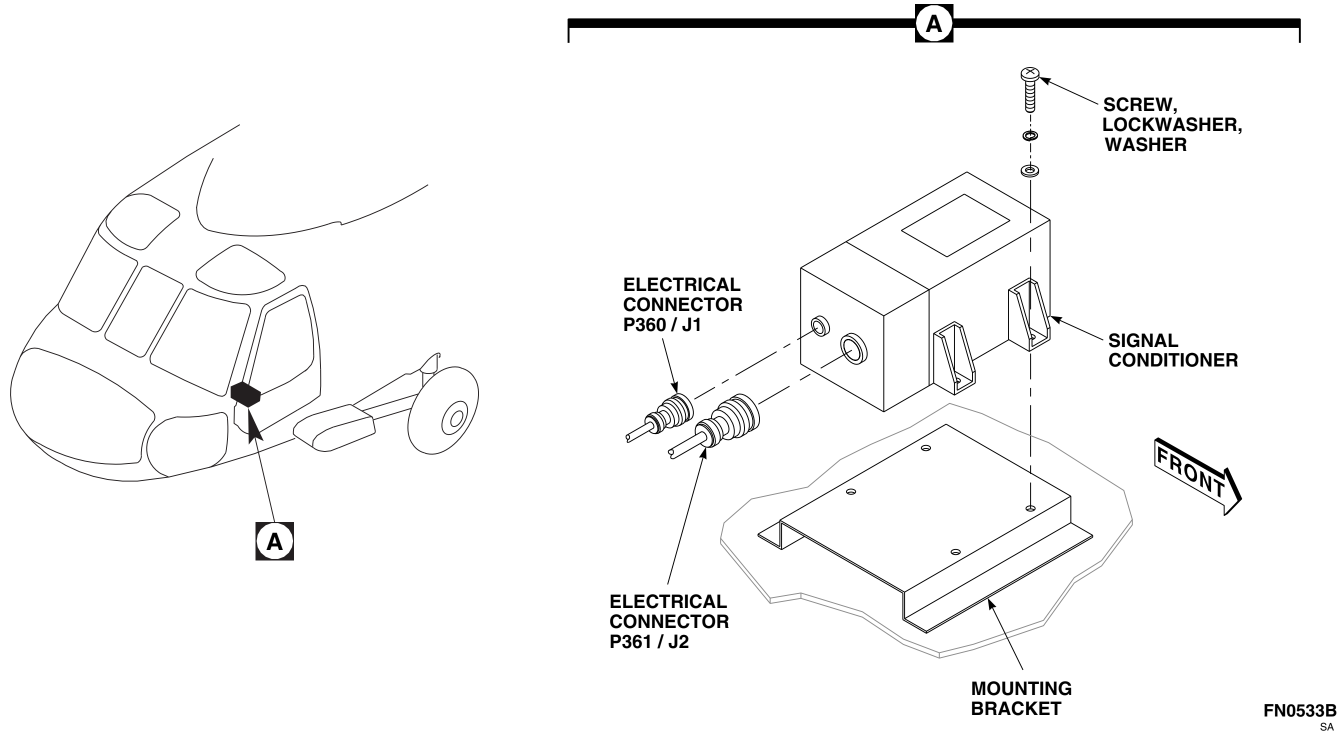


FIGURE 16-6-8-5. REMOVE SIGNAL CONDITIONER

c. Install protective caps and plugs on electrical connectors.

d. Make sure area is clean and free of foreign material.

e. Install blank plate on lower console and secure fasteners.

16-6-8.1.13 Remove Stores Jettison Control Panel.

a. Loosen fasteners and lift stores jettison control panel from lower console until electrical connectors are exposed (Figure 16-6-8-7, Detail A).

b. Disconnect electrical connectors, P645 from J1, and P646 from J2, from stores jettison control panel.

c. Inspect and treat electrical connectors (PARA 1-7-20).

d. Connect electrical connectors, P645 and P646, to blank plate.

e. Install blank plate on lower console and secure fasteners.

16-6-8.1.14 Complete Removal.

a. Press and release button on each quick-release pin securing post to brackets on upper and lower console. Remove quick-release pins and post (Figure 16-6-8-8, Detail A).

b. Install upper and lower cap fairings on upper and lower external stores support system (ESSS) fittings and secure studs (Figure 16-6-8-9, Detail A and Detail B).

c. Make sure fuel system area is clean and free of foreign material.

d. If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

e. Install left and right enclosure panels on top of main fuel cells. Lock studs.

f. Install cargo netting.

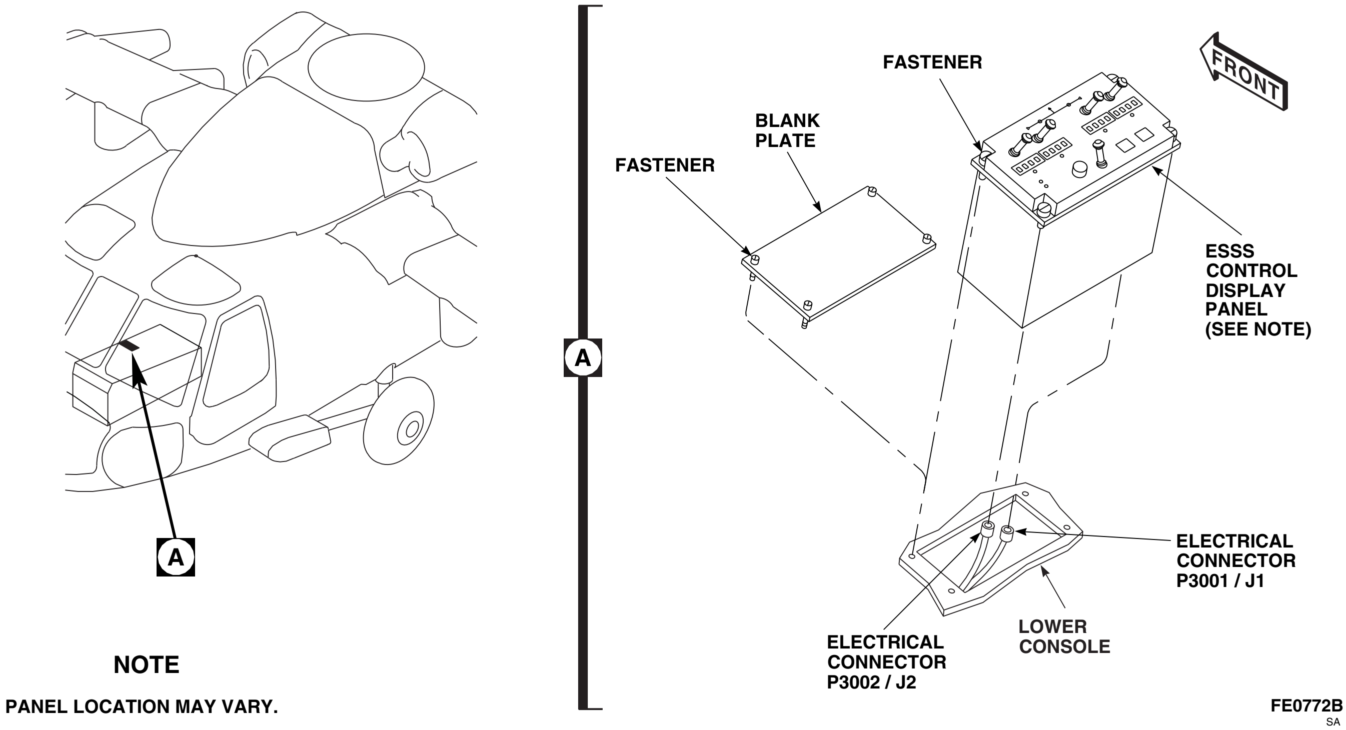


FIGURE 16-6-8-6. REMOVE CONTROL DISPLAY PANEL

g. Install soundproofing panels (PARA 2-4-69).

h. Install rear passenger's seats (PARA 2-4-45).

i. Service main landing gear shock struts (PARA 1-3-2).

j. Service tail landing gear shock strut (PARA 1-3-4).

k. Perform operational check of position lights (PARA 9-2-14).

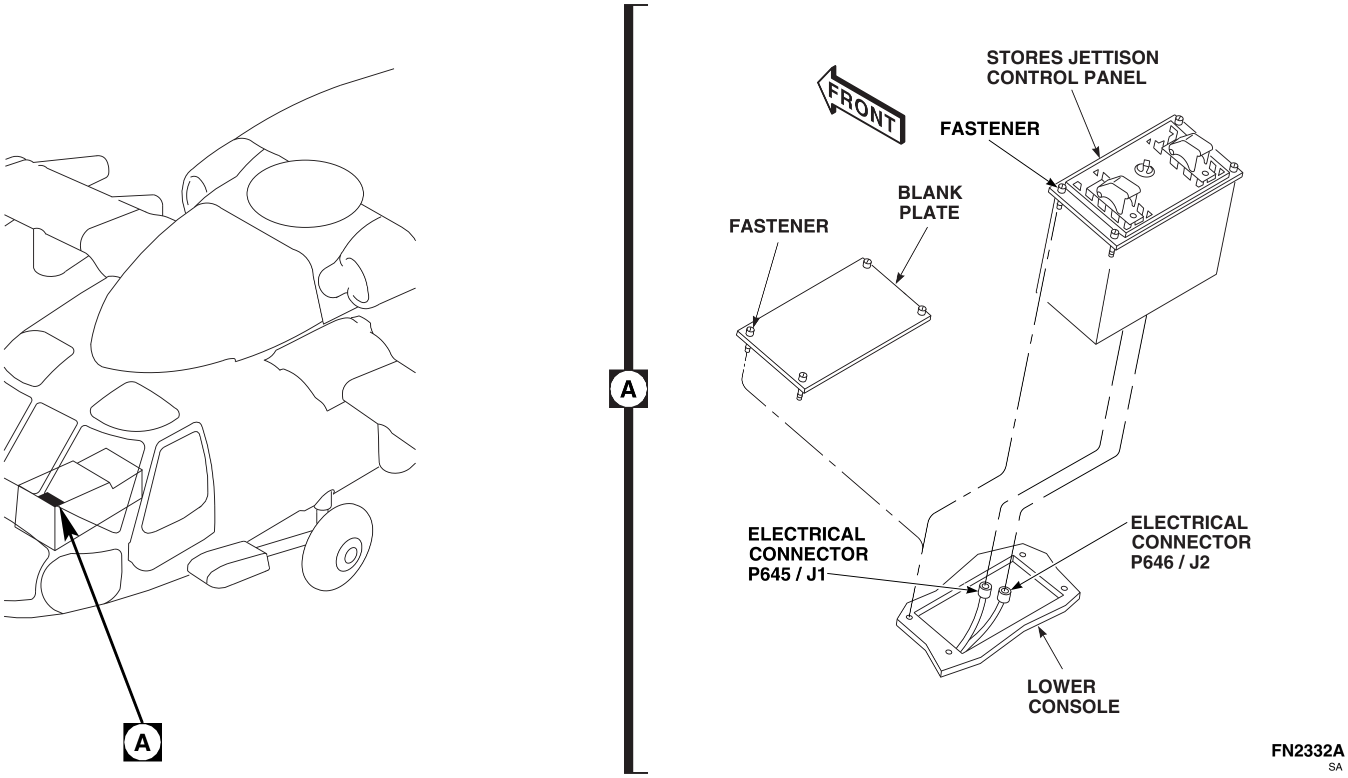


FIGURE 16-6-8-7. REMOVE STORES JETTISON CONTROL PANEL

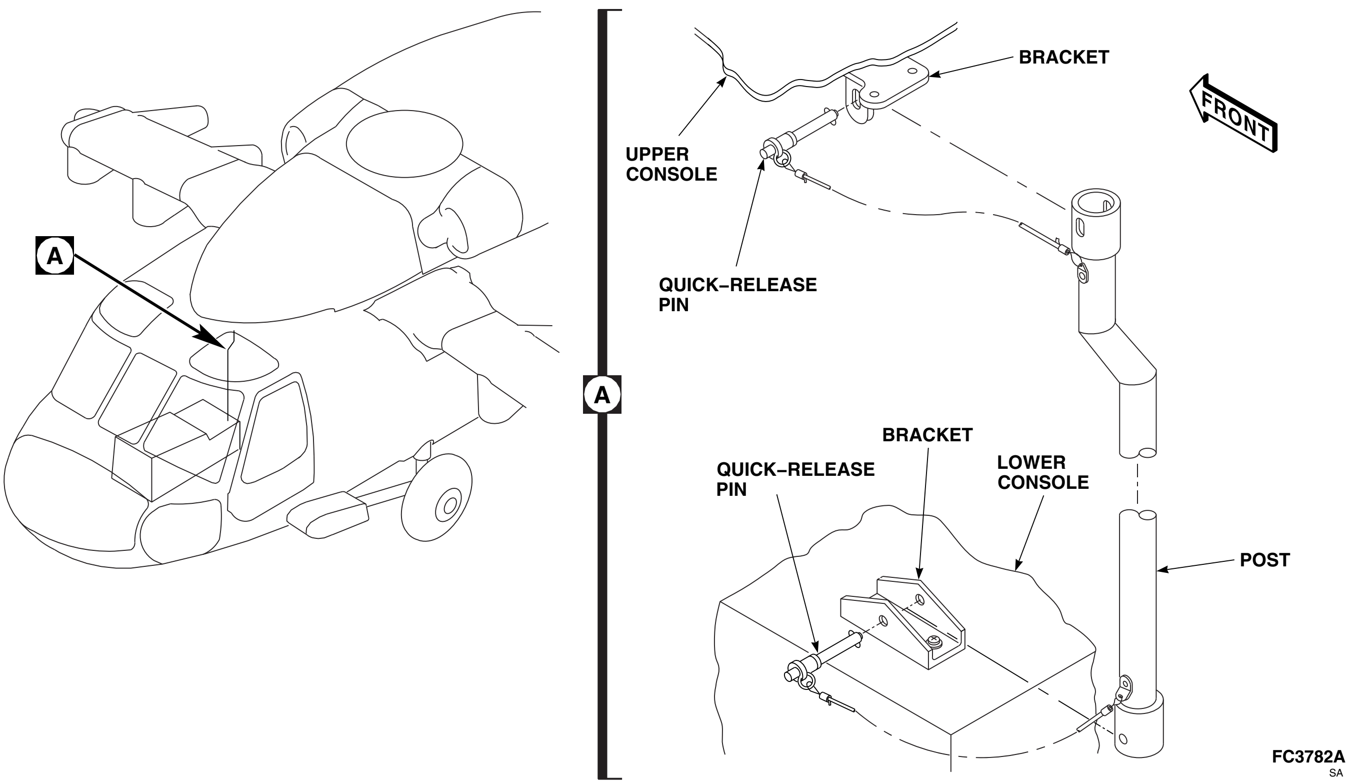


FIGURE 16-6-8-8. REMOVE POST

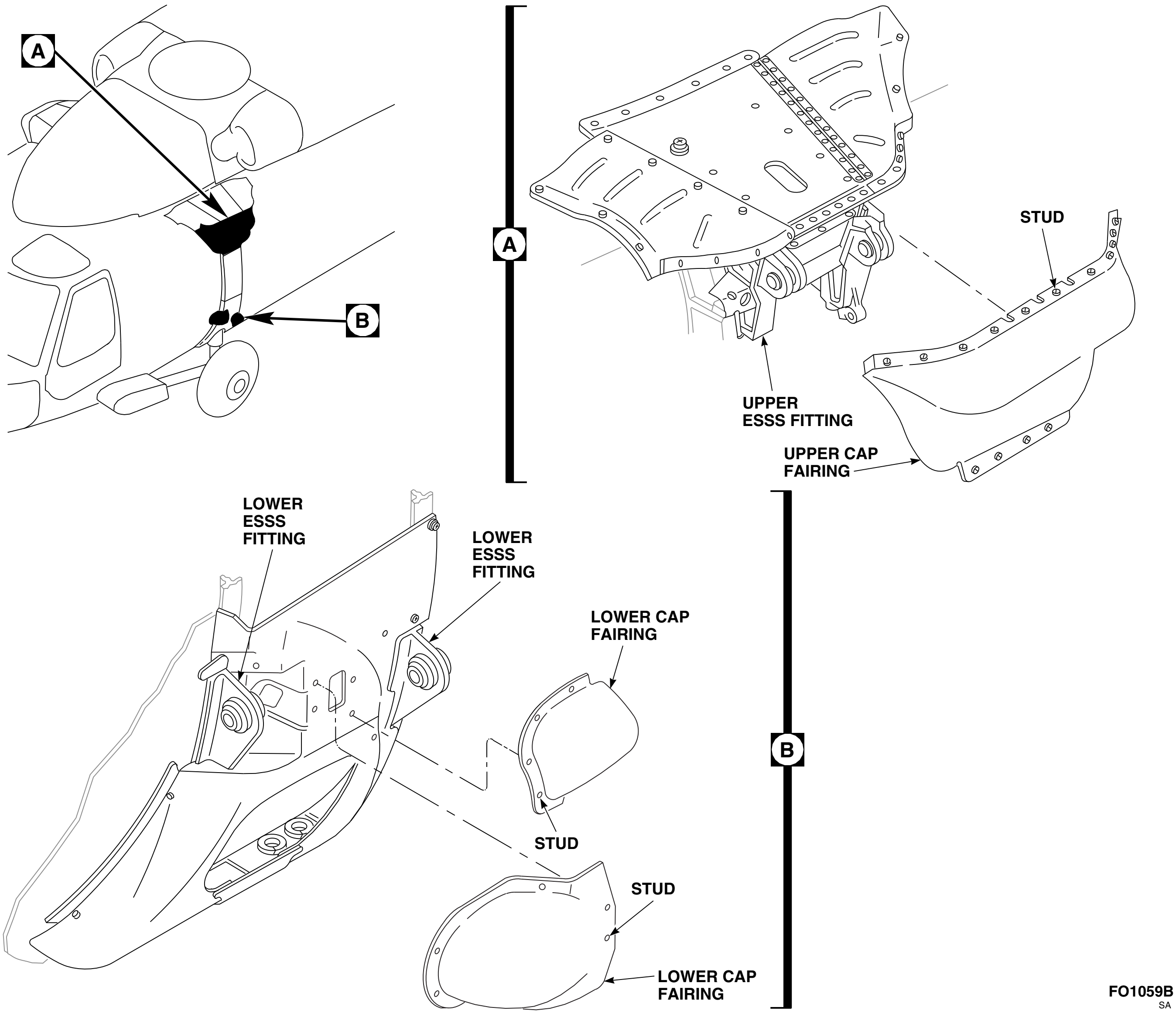


FIGURE 16-6-8-9. INSTALL UPPER AND LOWER CAP FAIRINGS

16-6-9 MAIN ROTOR BLADE TIP CAP EROSION PROTECTION KIT POLYURETHANE TAPE INSTALLATION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Container, 1-Quart
- Spray Bottle

Materials/Parts

- Abrasive Cloth, Item 4, Appendix D
- Abrasive Paper, Item 19, Appendix D
- Adhesion Promoter, Item 25A, Appendix D
- Adhesive, Item 32, Appendix D
- Alcohol, Denatured, Item 70, Appendix D
- Brush, Paint, Item 86, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Dishwashing Compound, Item 122, Appendix D
- Masking Tape, 2-inches, Item 214, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Pressure Sensitive Adhesive Tape, Item 243, Appendix D
- Pressure Sensitive Adhesive Tape, Item 244, Appendix D
- Stirring Stick, Beverage, Item 315C, Appendix D

References

- Appendix D
- PARA 1-6-4
- PARA 1-6-16
- PARA 2-6-5
- PARA 5-4-31
- PARA 16-6-11
- TM 1-70-VIB

16-6-9.1 Apply Polyurethane Tape.

16-6-9.1.1. Prepare to Install.

NOTE

When installing main rotor blade erosion protection kit, all main rotor blades must have polyurethane tapes applied.

a. Turn off all electrical power (PARA 1-6-16).

b. Move helicopter to dry, wind-free area.

NOTE

If initially installing erosion protection kit, main rotor blades must be removed. If replacing damaged section of polyurethane tape, blades may remain installed.

c. Remove or tie down main rotor blades (PARA 5-4-31 and PARA 1-6-4).

NOTE

Installation of 8 inch wide and 2 inch wide polyurethane tape is the same for all main rotor blades.

d. Mask off main rotor blade using 2-inch masking tape, Item 214, Appendix D. Mask 1 inch beyond area where 8 inch wide polyurethane tape will be applied (Figure 16-6-9-1, Detail A).

e. Remove any loose nickel on blade. **DO NOT GOUGE OR DAMAGE SURFACE.**

f. Smooth out high or rough spots using abrasive paper, Item 19, Appendix D.

g. Fill in low spots with adhesive, Item 32, Appendix D.

h. Sand smooth any high spots and adjoining surfaces with abrasive paper, Item 19, Appendix D.

NOTE

If replacing damaged sections of polyurethane tape, prepare only as many tape sections as needed.

i. Prepare polyurethane tape as follows:

(1) For 2-inch wide polyurethane tape, cut five pieces of pressure sensitive adhesive tape, Item 244, Appendix D, 36 inches long and one piece 12 inches long.

(2) For 8-inch wide polyurethane tape, cut five pieces of pressure sensitive adhesive tape, Item 243, Appendix D, 36 inches long and one piece 16.5 inches long.

j. Clean masked area of blade using cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.

k. Scuff masked area of blade using abrasive paper, Item 19, Appendix D. Sand in spanwise direction only.

l. Clean masked area of blade using cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean cleaning cloth.

m. Position 8 inch wide, 16.5 inch long polyurethane tape along leading edge and mark end of tape that overlaps tip cap (Figure 16-6-9-1, Detail B).

n. Cut a notch where tape overlaps tip cap.

o. Mix adhesion promoter, Item 25A, Appendix D, diluted in dishwashing compound, Item 122, Appendix D, using beverage stirring stick, Item 315C, Appendix D, and 1-quart container.

p. Apply thin coat of mixed adhesion promoter to masked area where 8 inch wide polyurethane tape is to be applied using paint brush, Item 86, Appendix D. Apply only as much mixed adhesion promoter as will be covered by length of polyurethane tape. **MINIMUM DRYING TIME 5 MINUTES. TAPE MUST BE APPLIED TO BLADE WITHIN 1 HOUR.**

q. Place leading edge alignment marks on polyurethane tape pieces (Figure 16-6-9-1, Detail A).

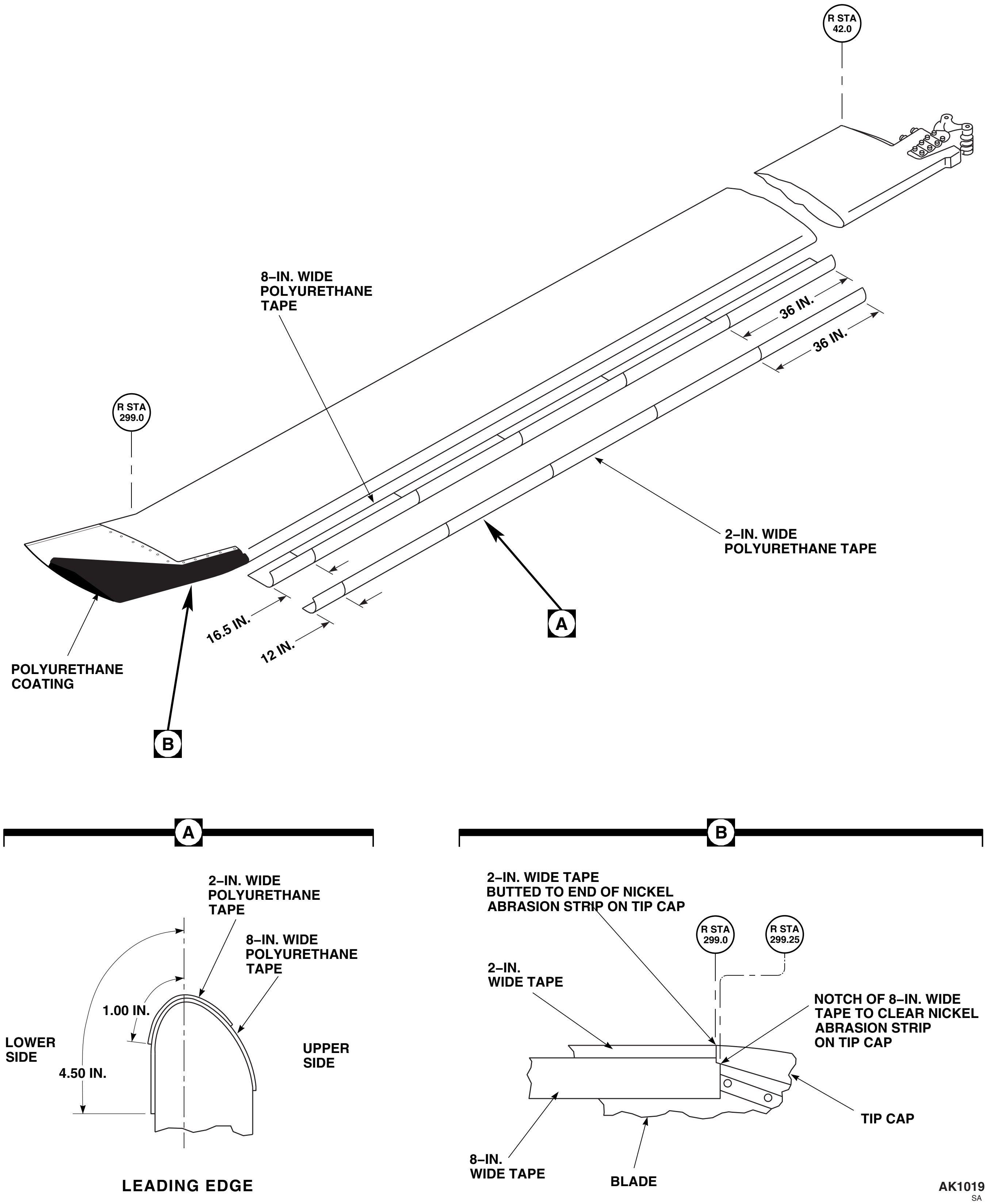


FIGURE 16-6-9-1. APPLY POLYURETHANE TAPE

16-6-9.1.2. Install Method A.

NOTE

Installation of polyurethane tape is faster using Method A. Method B requires 24 hours drying time but eliminates the possibility of trapped air. If Method B is to be used proceed to PARA 16-6-9.1.3.

a. Apply 8-inch wide polyurethane tape as follows:

NOTE

If initially installing erosion protection kit, apply 16.5 inch long polyurethane tape first.

(1) Peel fifth backing strip from polyurethane tape back 3 inches.

(2) Position polyurethane tape on leading edge of blade. Do not allow exposed portion of tape to contact adhesion promoter until tape is properly aligned (Figure 16-6-9-1, Detail A).

(3) When polyurethane tape is aligned, press down on exposed portion of tape. Slowly peel remaining backing strip from tape while rubbing vigorously to remove trapped air. Start at leading edge and work toward trailing edge of blade. Repeat until all backing strips are removed. **BUBBLES GREATER THAN 0.2 SQUARE INCH AREA WITHIN 2 INCHES OF EACH OTHER NOT ALLOWED.**

(4) Install remaining 8 inch wide polyurethane tapes.

b. Apply 2 inch wide polyurethane tape as follows:

(1) Scuff 8 inch wide polyurethane tape with abrasive cloth, Item 4, Appendix D, to area where 2 inch wide polyurethane tape will be applied in spanwise direction only.

(2) Clean scuffed area using cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean cleaning cloth.

(3) Apply thin coat of mixed adhesion promoter to scuffed area where 2 inch wide polyurethane tape is to be applied using paint brush, Item 86, Appendix D. Apply only as much mixed adhesion promoter as will be covered by length of polyurethane tape.

NOTE

If initially installing erosion protection kit, apply 12 inch long polyurethane tape first.

(4) Peel fifth backing strip from polyurethane tape back 3 inches.

(5) Position polyurethane tape on leading edge of blade. Do not allow exposed portion of tape to contact adhesion promoter until tape is properly aligned.

(6) When polyurethane tape is aligned, press down on exposed portion of tape. Slowly peel remaining backing strip from tape while rubbing vigorously to remove trapped air. Start at leading edge and work toward trailing edge. **BUBBLES GREATER THAN 0.2 SQUARE INCH AREA WITHIN 2 INCHES OF EACH OTHER NOT ALLOWED.**

(7) Install remaining 2 inch wide polyurethane tapes.

c. **MINIMUM DRYING TIME 24 HOURS.** Carefully remove masking tape when drying is complete.

d. Remove masking tape from remaining areas on blade.

e. Touch up paint as required (PARA 2-6-5).

f. If required, install main rotor blade (PARA 5-4-31).

g. If initially installing erosion protection kit, apply polyurethane coating to tip cap (PARA 16-6-11).

h. Check main rotor blade track and balance (TM 1-70-VIB).

16-6-9.1.3. Install Method B.

a. Mix a wetting solution consisting of 75% water, 25% denatured alcohol, Item 70, Appendix D, and 20 drops per liter of dishwashing compound, Item 122, Appendix D.

b. Apply 8 inch wide polyurethane tape as follows:

NOTE

If initially installing erosion protection kit, apply 16.5 inch long polyurethane tape first.

(1) Remove backing strips from polyurethane tape.

(2) Using spray bottle, coat surface of blade and adhesive side of polyurethane tape with wetting solution.

(3) Apply polyurethane tape to blade. Align tape by sliding it in position. Remove trapped air, strained areas, and excess wetting solution by rubbing from leading edge and work toward trailing edge.

(4) If required, tape edges of polyurethane tape using 2-inch masking tape, Item 214, Appendix D.

c. Apply 2 inch wide polyurethane tape as follows:

(1) Perform PARA 16-6-9.1.2, step b. (1) through step b. (3).

NOTE

If initially installing erosion protection kit, apply 12 inch long polyurethane tape first.

(2) Remove backing strip from polyurethane tape.

(3) Using spray bottle, coat surface of installed 8 inch wide polyurethane tape and adhesive side of 2 inch wide polyurethane tape with wetting solution.

(4) Apply polyurethane tape to blade. Align tape by sliding it in position. Remove trapped air, strained areas, and excess wetting solution by rubbing from leading edge to trailing edge.

(5) If required, tape edges of polyurethane tape using 2-inch masking tape, Item 214, Appendix D.

d. **MINIMUM DRYING TIME 24 HOURS.** Carefully remove masking tape when drying is complete.

e. Remove masking tape from remaining areas on blade.

f. Touch up paint as required (PARA 2-6-5).

g. If required, install main rotor blade (PARA 5-4-31).

h. If initially installing erosion protection kit, apply polyurethane coating to tip cap (PARA 16-6-11).

i. Check main rotor blade track and balance (TM 1-70-VIB).

16-6-10 MAIN ROTOR BLADE TIP CAP EROSION PROTECTION KIT POLYURETHANE TAPE REMOVE/REPAIR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Nonmetallic Scraper, Locally-Made
- Soldering Iron

Materials/Parts

- Cloth, Cleaning, Item 102, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 1-6-4
- PARA 1-6-16
- PARA 2-6-5
- PARA 16-6-9

16-6-10.1 Remove/Repair Polyurethane Tape.

16-6-10.1.1. Remove/Repair.

NOTE

- If replacing a damaged section of polyurethane tape, entire section must be removed. Replace only as many damaged sections as required. Blades do not require removal.
- Removal of 2 inch wide and 8 inch wide polyurethane tape is the same for all main rotor blades.

a. Turn off all electrical power (PARA 1-6-16).

b. Tie down main rotor blades (PARA 1-6-4).

c. Remove 2 inch wide polyurethane tape by slowly peeling away from leading edge of blade using nonmetallic scraper.

CAUTION

Damage to rotor blade will result if soldering iron is used outside prescribed area. Do not use soldering iron on blade more than 2 inches from leading edge on top side or more than 3 inches from trailing edge on bottom side of blade.

d. Remove 8 inch wide polyurethane tape by slitting or melting tape into strips using soldering iron.

e. Remove strips by peeling tape back 180°. Do not pull at right angle to blade.

f. Remove remaining 8 inch wide polyurethane tape by slowly peeling away from blade using nonmetallic scraper.

NOTE

If paint touchup is not necessary, adhesive residue may be left on blade.

g. If paint touchup is required, do this:

(1) Remove tape adhesive from leading edge of blade using cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean cleaning cloth.

(2) Touch up paint (PARA 2-6-5).

h. If replacing damaged section of polyurethane tape, install replacement tape (PARA 16-6-9).

16-6-11 MAIN ROTOR BLADE TIP CAP EROSION PROTECTION KIT POLYURETHANE COATING INSTALLATION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Container, 1-Quart
- Knife
- Protection, Eye and Skin
- Spray Bottle

Materials/Parts

- Abrasive Paper, Item 15, Appendix D
- Brush, Paint, Item 86, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Coating Kit, Rain Erosion Resisting, Item 106B, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Pressure Sensitive Adhesive Tape, Item 242, Appendix D
- Primer Coating, Item 258A, Appendix D
- Thinner, Item 324A, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-6-5
- PARA 5-4-31
- TM 1-70-VIB

16-6-11.1 Apply Polyurethane Coating.

16-6-11.1.1 Apply.

NOTE

When installing main rotor blade erosion protection kit, all main rotor blade tip caps must have polyurethane coating applied.

a. Turn off all electrical power (PARA 1-6-16).

b. Move helicopter to dry, wind-free area.

NOTE

Tip cap may be removed for ease of coating application.

c. If required, remove main rotor blade tip cap (PARA 5-4-31).

NOTE

Do not scuff inspection area (Figure 16-6-11-1, Detail B).

d. Using abrasive paper, Item 15, Appendix D, scuff area of tip cap to be covered by coating (Figure 16-6-11-1, Detail A).

WARNING

Wash contacted skin with soap and water. If solvent contacts eyes, flush them with clean water and get immediate medical help.

e. Clean surface several times using cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry using clean cleaning cloth.

f. Cut pressure sensitive adhesive tape, Item 242, Appendix D, in 1-inch wide strips. Use to mask area not being coated. Make sure to use two layers of tape on masked area.

g. Apply primer coating to tip cap as follows:

NOTE

- Use equal amounts of primer on all tip caps to minimize balance adjustments to main rotor.
- Do not apply coating on inspection area (Figure 16-6-11-1, Detail B).

(1) Mix primer coating, Item 258A, Appendix D.

NOTE

Primer coating, Item 258A, Appendix D, may be applied using paint brush, Item 86, Appendix D, or spray bottle.

(2) If spraying, thin primer coating using thinner, Item 324A, Appendix D (four parts coating to one part thinner).

(3) Apply heavy coat of primer coating, Item 258A, Appendix D, to area.

(4) Allow primer coating to cure for a minimum of 1 hour at 70°F (21°C) or above.

(5) If surface becomes contaminated, lightly wipe contaminated area using cleaning cloth, Item 102, Appendix D, moistened with thinner, Item 324A, Appendix D.

h. Apply polyurethane coating (rain erosion resisting coating) to tip cap as follows:

NOTE

- Use equal amounts of coating on all tip caps to minimize balance adjustments to main rotor.
- Do not apply coating on inspection area.

(1) Mix rain erosion resisting coating kit, Item 106B, Appendix D, part A and part B, in equal parts into 1-quart container.

NOTE

- Rain erosion resisting coating kit, Item 106B, Appendix D, may be applied using paint brush, Item 86, Appendix D, or spray bottle.
- Brushed coats tend to result in 0.002 to 0.003-inch thickness. Apply 15 - 16 coats when brushing on coating.
- Sprayed coats tend to result in 0.0007 to 0.001-inch thickness. Apply 35 - 40 coats when spraying on coating.

(2) If spraying, thin rain erosion resistant coating using thinner, Item 324A, Appendix D (four parts coating to one part thinner).

NOTE

- Taper coat thickness per Figure 16-6-11-1, Detail A, applying thicker coats at leading edge and progressively less toward rear edges.
- Use witness pieces to determine appropriate coating thickness.

(3) If brushing, apply first layer to tip cap using paint brush, Item 86, Appendix D. Allow 5 minutes drying time between coats until desired thickness is reached.

(4) If spraying, apply first layer to tip cap using spray bottle. Allow 2 minutes drying time between coats until desired thickness is reached.

CAUTION

- Do not allow masking to remain on blade while coating cures.
- Damage to tip cap and/or main rotor blade can occur while removing tape from masked area of blade. Hold cutting knife parallel to blade surface and slit coating at ridge created by the double layer of tape. Do not scratch or gouge blade surface. Remove tape.

(5) If brushing, remove masking after five coats are brushed on and apply remaining coats by eye.

(6) If spraying, remove masking after completion.

(7) Allow coating to cure for a minimum of 24 hours or 4 hours at 140 - 160°F (60 - 71°C).

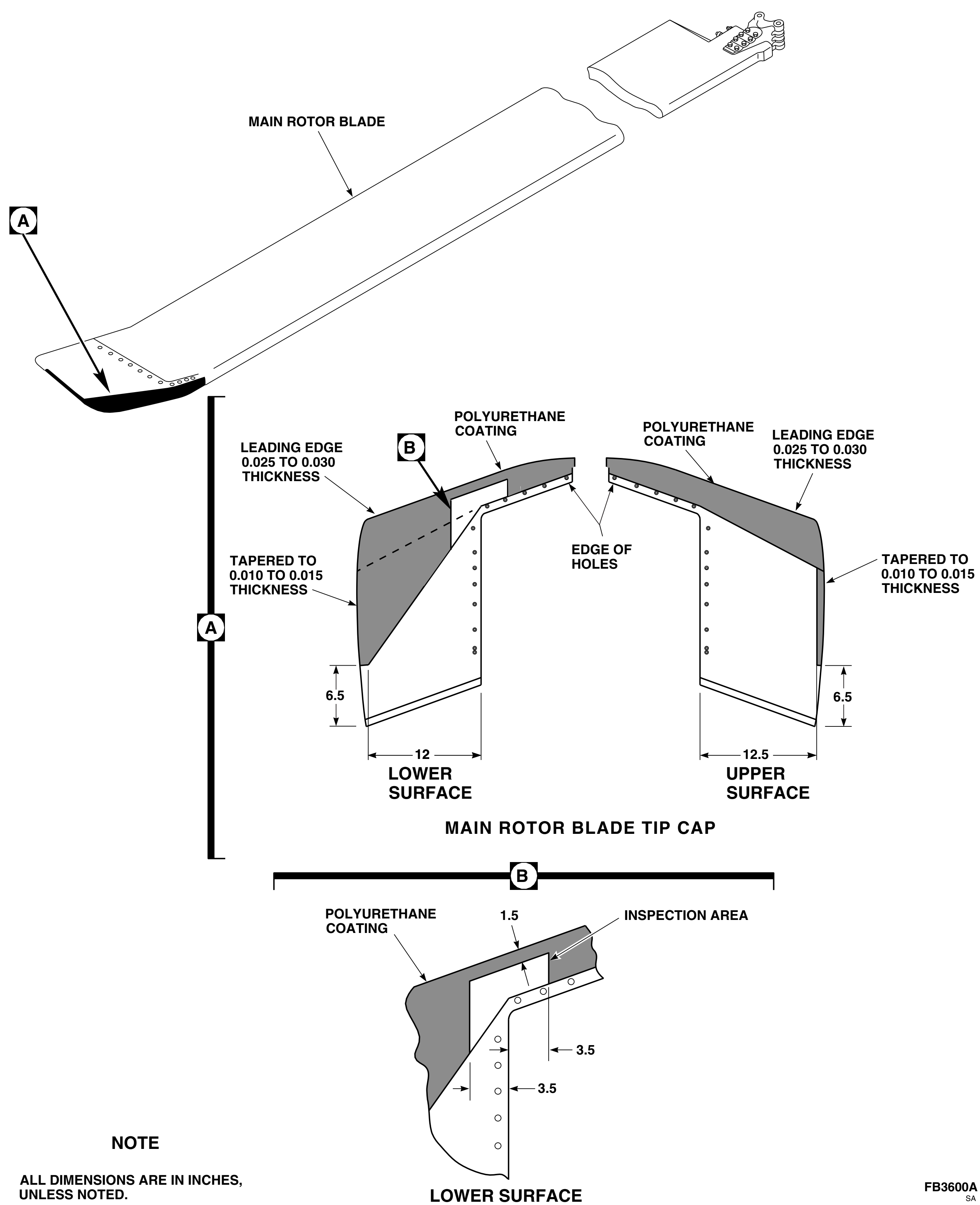
i. Touch up paint as required (PARA 2-6-5).

NOTE

Main rotor blade tip cap must be installed with new screws.

j. If required, install main rotor blade tip cap (PARA 5-4-31).

k. Check main rotor blade track and balance (TM 1-70-VIB).



NOTE

ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

FIGURE 16-6-11-1. APPLY POLYURETHANE COATING

16-6-12 MAIN ROTOR BLADE TIP CAP EROSION PROTECTION KIT POLYURETHANE COATING REMOVE/REPAIR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Brush
- Container, 1-Quart
- Nonmetallic Scraper, Locally-Made
- Protection, Eye and Skin
- Spray Bottle

Materials/Parts

- Abrasive Paper, Item 15, Appendix D
- Brush, Paint, Item 86, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Coating Kit, Rain Erosion Resisting, Item 106B, Appendix D
- Primer Coating, Item 258A, Appendix D
- Thinner, Item 324A, Appendix D

References

- Appendix D
- PARA 5-4-31
- PARA 16-6-11

16-6-12.1 Remove/Repair Polyurethane Coating.

16-6-12.1.1 Remove/Repair.

NOTE

Tip cap may be removed for ease of coating application.

a. If required, remove main rotor blade tip cap (PARA 5-4-31).

NOTE

To remove coating, proceed to step b. To repair coating, proceed to step c.

b. Remove polyurethane coating (rain erosion resisting coating) as follows:

WARNING

Wash contacted skin with soap and water. If solvent contacts eyes, flush them with clean water and get immediate medical help.

NOTE

If inspecting tip cap for damage, remove polyurethane coating from lower surface, only in area indicated, if inspection area was accidentally covered (Figure 16-6-12-1, Detail A and Detail B).

(1) Using thinner, Item 324A, Appendix D, saturate coating surface until coating is softened.

(2) Using nonmetallic scraper, remove coating.

(3) Wipe surface using cleaning cloth, Item 102, Appendix D, moistened with thinner, Item 324A, Appendix D. Wipe dry.

c. Repair polyurethane coating as follows:

(1) Carefully remove disbonded material.

(2) Using abrasive paper, Item 15, Appendix D, lightly scuff damaged area.

(3) Clean surface using cleaning cloth, Item 102, Appendix D, moistened with Item 324A, Appendix D, Appendix D. Wipe dry using clean cleaning cloth.

(4) If blade surface is exposed, apply primer coating, Item 258A, Appendix D, as follows:

NOTE

Do not apply coating on inspection area (Figure 16-6-12-1, Detail B).

(a) Mix primer coating, Item 258A, Appendix D.

NOTE

Primer coating, Item 258A, Appendix D, may be applied using paint brush, Item 86, Appendix D, or spray bottle.

(b) If spraying, thin primer coating using thinner, Item 324A, Appendix D (four parts coating to one part thinner).

(c) Apply heavy coat of primer coating, Item 258A, Appendix D, to area.

(d) Allow primer coating to cure for a minimum of 1 hour at 70°F (21°C) or above.

(e) If surface becomes contaminated, lightly wipe contaminated area using cleaning cloth, Item 102, Appendix D, moistened with thinner, Item 324A, Appendix D.

(5) Apply polyurethane coating (rain erosion resisting coating) as follows:

NOTE

Do not apply coating on inspection area.

(a) Mix rain erosion resisting coating kit, Item 106B, Appendix D, part A and part B, in equal parts into 1-quart container.

NOTE

Rain erosion resisting coating kit, Item 106B, Appendix D, may be applied using paint brush, Item 86, Appendix D, or spray bottle.

(b) If spraying, thin rain erosion resistant coating using thinner, Item 324A, Appendix D (four parts coating to one part thinner).

(c) If brushing, apply coating to tip cap using paint brush, Item 86, Appendix D. Allow 5 minutes drying time between coats until desired thickness is reached.

(d) If spraying, apply first layer to tip cap using spray bottle. Allow 2 minutes drying time between coats until desired thickness is reached.

(e) Allow coating to cure for a minimum of 24 hours or 4 hours at 140 - 160°F (60 - 71°C).

NOTE

Main rotor blade tip cap must be installed with new screws.

d. If required, install main rotor blade tip cap (PARA 5-4-31).

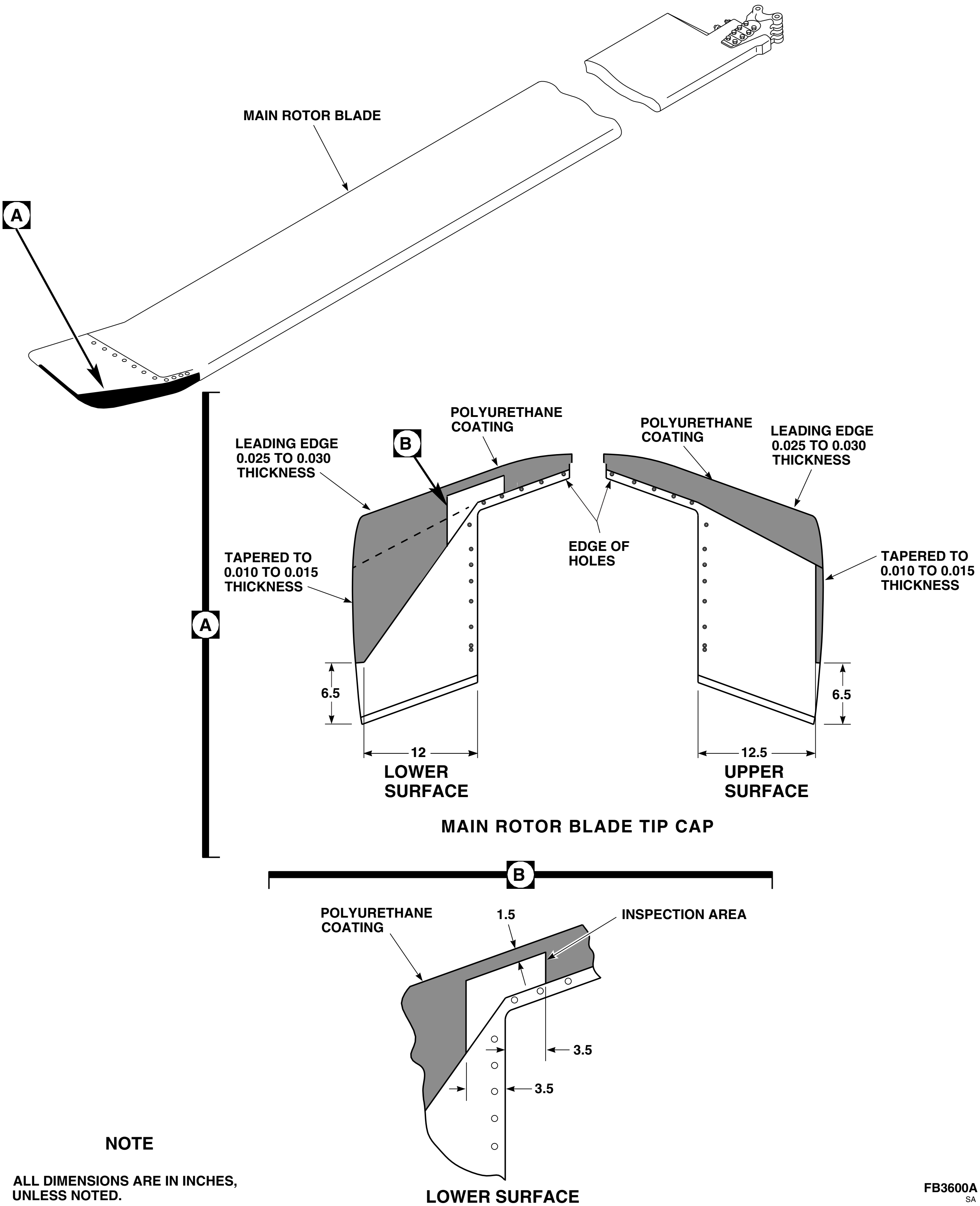


FIGURE 16-6-12-1. REMOVE/REPAIR POLYURETHANE COATING

16-6-13 MAIN ROTOR BLADE TIP CAP EROSION PROTECTION KIT BOOT-TO-TIP CAP INSTALLATION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Spray Bottle

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Abrasive Paper, Item 15, Appendix D
- Adhesion Promoter, Item 25A, Appendix D
- Alcohol, Ethyl, Item 71, Appendix D
- Alcohol, Isopropyl, Item 72, Appendix D
- Brush, Paint, Item 86, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Dishwashing Compound, Item 122, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 5-4-31

16-6-13.1 Install Boot-to-Tip Cap.

16-6-13.1.1 Install.

NOTE

When installing tail rotor blade erosion protection kit, all tail rotor blades must have polyurethane boot applied.

a. Turn off all electrical power (PARA 1-6-16).

b. Move helicopter to dry, wind-free area.

NOTE

Main rotor tip cap may be removed for ease of boot application.

c. If required, remove tip cap (PARA 5-4-31).

d. Clean surface using cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry using clean cleaning cloth.

e. Scuff nickel abrasion strip using abrasive paper, Item 15, Appendix D.

f. Scuff area of tip cap to be covered by boot, using abrasive paper, Item 15, Appendix D.

g. Clean surface using cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Repeat this several times.

h. Mix a wetting solution consisting of 75% water, 25% isopropyl alcohol, Item 72, Appendix D, and 20 drops of dishwashing compound, Item 122, Appendix D.

i. Remove backing on boot. Using spray bottle, apply wetting solution on exposed adhesive.

j. Using spray bottle, spray wetting solution on tip cap.

k. Apply boot-to-tip cap in correct position. Remove trapped air, strained areas, and excess wetting solution by rubbing vigorously from leading edge and work toward trailing edge.

l. **MINIMUM DRYING TIME 24 HOURS.**

m. Cut polyurethane tape as follows:

(1) Cut 4-inch wide polyurethane tape.

(2) Cut 2-inch wide polyurethane tape.

NOTE

Polyurethane boot must be scuffed until dull or tape will not adhere.

n. Scuff area of polyurethane boot to be covered by 4-inch wide polyurethane tape using abrasive paper, Item 15, Appendix D.

o. Clean surface using cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Repeat this several times.

p. Apply a thin coat of abrasive cloth, Item 10, Appendix D, to area to be covered with 4-inch wide tape using paint brush, Item 86, Appendix D. **MINIMUM DRYING TIME 5 MINUTES. TAPE MUST BE APPLIED TO TIP CAP WITHIN 1 HOUR.**

q. Apply 4-inch wide polyurethane tape as follows:

(1) Locate inboard end first. Peel one center backing strip from polyurethane tape back 3-inches.

(2) Position polyurethane tape on leading edge of tip cap. Do not allow exposed portion of tape to contact adhesion promoter until tape is properly aligned.

(3) When polyurethane tape is aligned, press down on exposed portion of tape. Slowly peel remaining backing strip from tape while rubbing vigorously to remove trapped air. Start at leading edge and work toward trailing edge of tip cap. Repeat until all backing strips are removed. **BUBBLES GREATER THAN 0.2 SQUARE INCH AREA WITHIN 2-INCHES OF EACH OTHER NOT ALLOWED.**

r. Trim 4-inch wide polyurethane tape along tip cap edge.

s. Scuff area of polyurethane tape to be covered by 2-inch wide polyurethane tape using abrasive paper, Item 15, Appendix D.

t. Clean surface using cleaning cloth, Item 102, Appendix D, moistened with ethyl alcohol, Item 71, Appendix D.

u. Apply a thin coat of adhesion promoter, Item 25A, Appendix D, to area to be covered with 2-inch wide tape using paint brush, Item 86, Appendix D. **MINIMUM DRYING TIME 5 MINUTES. TAPE MUST BE APPLIED TO TIP CAP WITHIN 1 HOUR.**

v. Apply 2-inch wide polyurethane tape as follows:

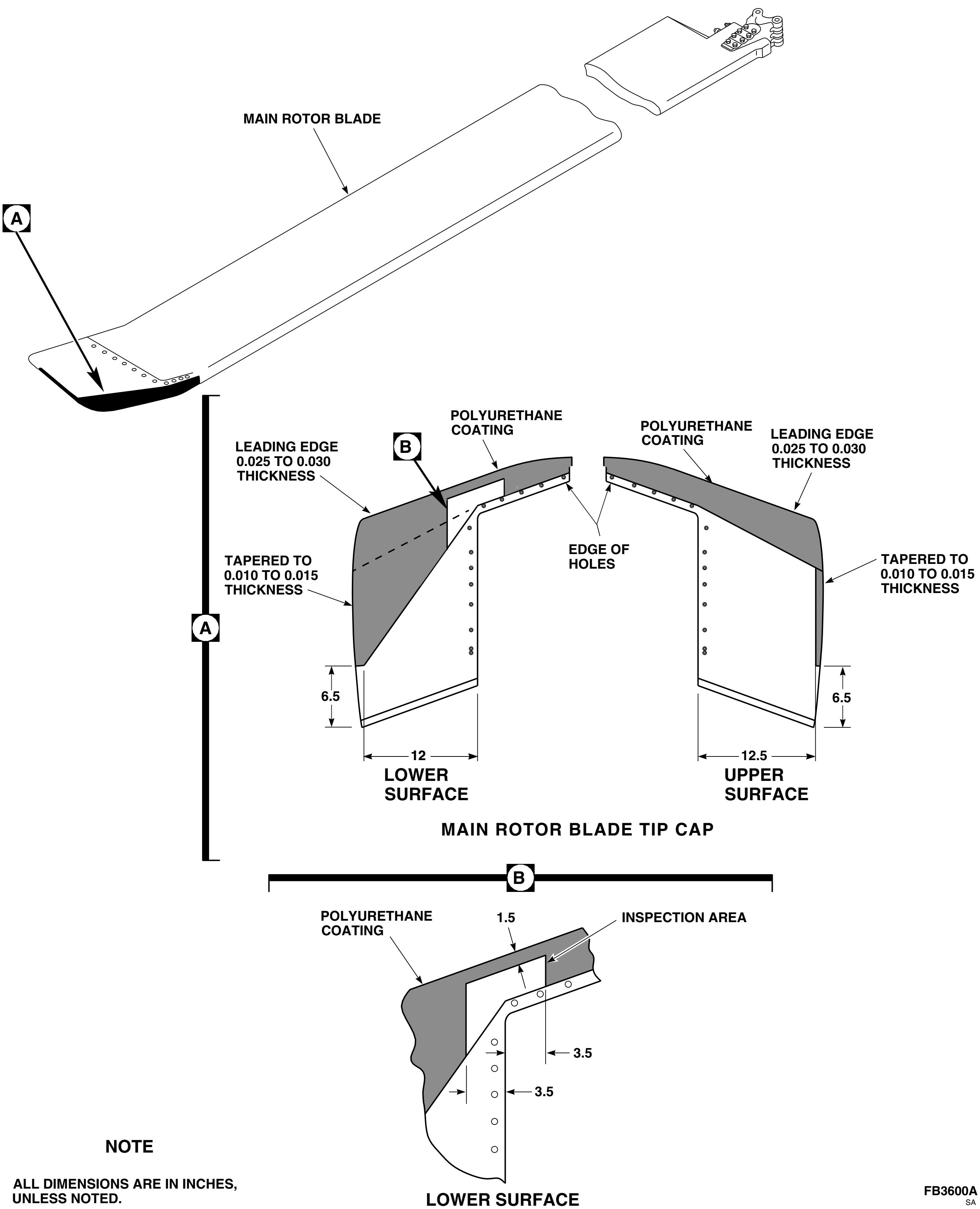
(1) Locate inboard end first. Peel one backing strip from polyurethane tape back 3-inches.

(2) Position polyurethane tape on leading edge of tip cap. Do not allow exposed portion of tape to contact adhesion promoter until tape is properly aligned.

(3) When polyurethane tape is aligned, press down on exposed portion of tape. Slowly peel remaining backing strip from tape while rubbing vigorously to remove trapped air. Start at leading edge and work toward trailing edge. **BUBBLES GREAT-**

ER THAN 0.2 SQUARE INCH AREA WITHIN 2 INCHES OF EACH OTHER NOT ALLOWED.

w. If required, install main rotor blade tip cap (PARA 5-4-31).



NOTE

ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

FIGURE 16-6-13-1. INSTALL BOOT-TO-TIP CAP

16-6-14 MAIN ROTOR BLADE TIP CAP EROSION PROTECTION KIT BOOT-TO-TIP CAP REMOVE/REPAIR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Cloth, Cleaning, Item 102, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 16-6-13

16-6-14.1 Remove/Repair Boot-to-Tip Cap.

16-6-14.1.1. Remove/Repair.

NOTE

A damaged polyurethane tip cap boot must be replaced.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove polyurethane tape by slowly peeling away from leading edge of blade using nonmetallic scraper.

c. Remove polyurethane boot by slowly peeling away from trailing edge of blade using nonmetallic scraper.

d. Remove boot adhesive from tip cap using cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean cleaning cloth.

e. If replacing damaged polyurethane boot, install replacement boot (PARA 16-6-13).

16-6-15 TAIL ROTOR BLADE EROSION PROTECTION KIT POLYURETHANE TAPE INSTALLATION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Container, 1-Quart
- Spray Bottle

Materials/Parts

- Abrasive Paper, Item 15, Appendix D
- Adhesion Promoter, Item 25A, Appendix D
- Alcohol, Isopropyl, Item 72, Appendix D
- Brush, Paint, Item 86, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Dishwashing Compound, Item 122, Appendix D
- Masking Tape, 2-inches, Item 214, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Pressure Sensitive Adhesive Tape, Item 242, Appendix D
- Stirring Stick, Beverage, Item 315C, Appendix D
- Toluene, Item 339, Appendix D

References

- Appendix D
- PARA 1-6-4
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-6-5
- PARA 16-6-17
- TM 1-70-VIB

16-6-15.1 Apply Polyurethane Tape.

16-6-15.1.1. Prepare to Install.

NOTE

When installing tail rotor blade erosion protection kit, all tail rotor blades must have polyurethane tapes applied.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Move helicopter to dry, wind-free area.

c. Tie down main rotor blades (PARA 1-6-4).

NOTE

Installation of polyurethane tape is the same for all tail rotor blades.

d. Mask off tail rotor blade using 2-inch masking tape, Item 214, Appendix D. Mask 1 inch beyond area where polyurethane tape will be applied (Figure 16-6-15-1, Detail A).

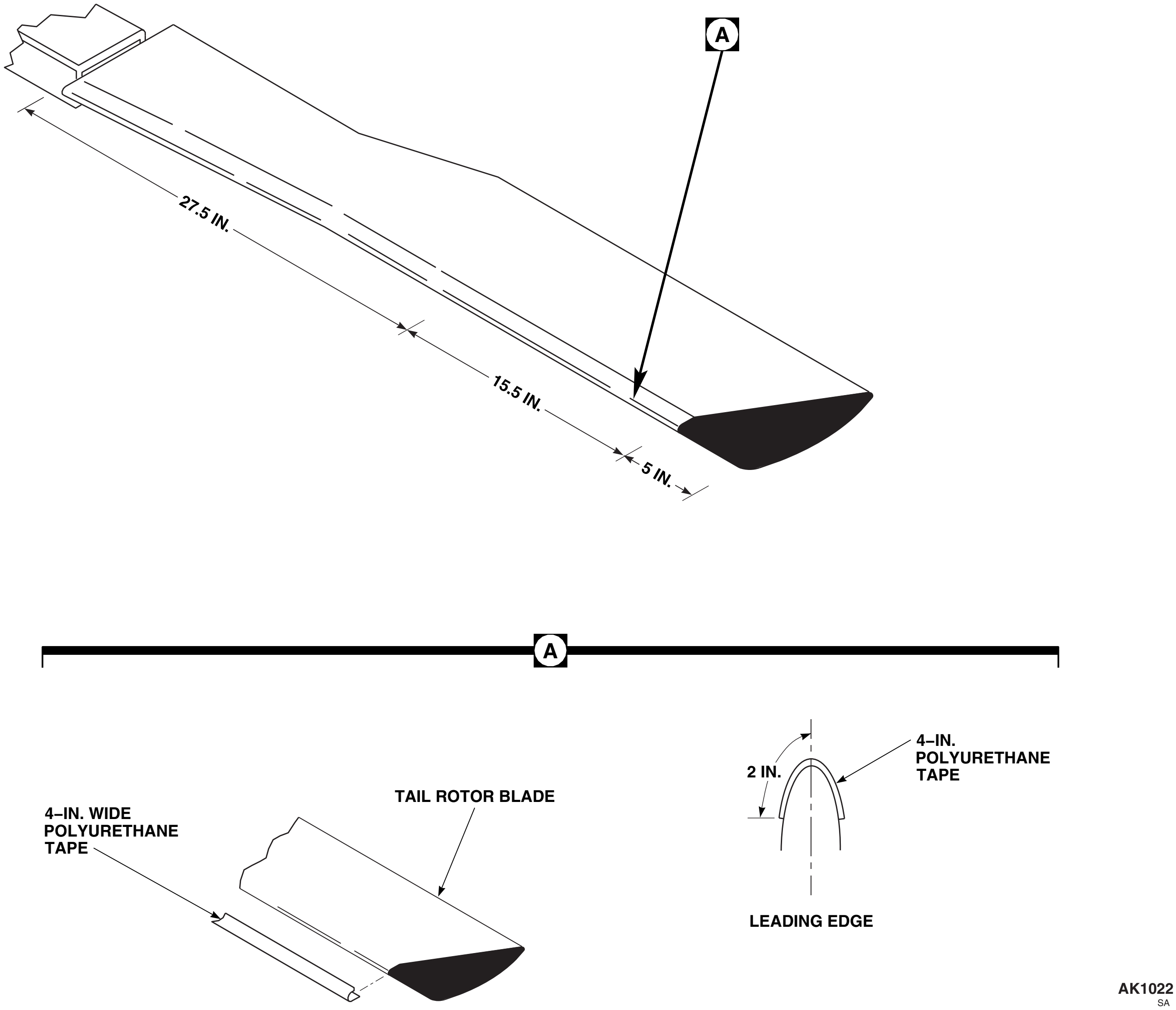


FIGURE 16-6-15-1. APPLY POLYURETHANE TAPE

NOTE

If replacing damaged sections of polyurethane tape, prepare only as many tape sections as is needed.

e. Prepare polyurethane tape as follows:

(1) Using 4-inch wide pressure sensitive adhesive tape, Item 242, Appendix D, cut four pieces 27.5-inches long.

(2) Using 4-inch wide pressure sensitive adhesive tape, Item 242, Appendix D, cut four pieces 15.5-inches long.

f. Clean masked area of blade using cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.

g. Scuff masked area of blade using abrasive paper, Item 15, Appendix D. Sand in spanwise direction only.

h. Clean masked area of blade using cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean cleaning cloth.

i. Mix adhesion promoter, Item 25A, Appendix D, diluted in equal parts with toluene, Item 339, Appendix D, using beverage stirring stick, Item 315C, Appendix D, and 1-quart container.

j. Apply a thin coat of mixed adhesion promoter, Item 25A, Appendix D, to masked area where 4 inch wide polyurethane tape is to be applied using paint brush, Item 86, Appendix D. Apply only as much mixed adhesion promoter as will be covered by length of polyurethane tape. **MINIMUM DRYING TIME 5 MINUTES. TAPE MUST BE APPLIED TO BLADE WITHIN 1 HOUR.**

k. Place leading edge alignment marks on polyurethane tape pieces.

16-6-15.1.2. Install Method A.

NOTE

Installation of polyurethane tape is faster using Method A. Method B requires 24 hours drying time but eliminates the possibility of trapped air. If Method B is used proceed to PARA 16-6-15.1.3.

a. Apply 15.5-inch long, 4 inch wide polyurethane tape as follows:

(1) Peel fifth backing strip from polyurethane tape back 3 inches.

(2) Position polyurethane tape on leading edge of blade. Do not allow exposed portion of tape to contact adhesion promoter until tape is properly aligned (Figure 16-6-15-1, Detail A).

(3) When polyurethane tape is aligned, press down on exposed portion of tape. Slowly peel remaining backing strip from tape while rubbing vigorously to remove trapped air. Start at leading edge and work toward trailing edge of blade. Repeat until all backing strips are removed. **BUBBLES GREATER THAN 0.2-SQUARE INCH AREA WITHIN 2 INCHES OF EACH OTHER NOT ALLOWED.**

b. Apply 27.5-inch long, 4 inch wide polyurethane tape as follows:

(1) Peel fifth backing strip from polyurethane tape back 3 inches.

(2) Position polyurethane tape on leading edge of blade. Do not allow exposed portion of tape to contact adhesion promoter until tape is properly aligned.

(3) When polyurethane tape is aligned, press down on exposed portion of tape. Slowly peel remaining backing strip from tape while rubbing vigorously to remove trapped air. Start at leading edge and work toward trailing edge. **BUBBLES GREATER THAN 0.2-SQUARE INCH AREA WITHIN 2 INCHES OF EACH OTHER NOT ALLOWED.**

c. **MINIMUM DRYING TIME 24 HOURS.** Carefully remove masking tape when drying is complete.

d. Remove masking tape from repairing areas on blade.

e. Touch up paint as required (PARA 2-6-5).

f. If initially installing erosion kit, apply polyurethane coating to tip cap (PARA 16-6-17).

g. Check tail rotor blade track and balance (TM 1-70-VIB).

16-6-15.1.3. Install Method B.

CAUTION

To prevent damage to tail rotor components do not use this method to apply 4 inch wide, 27.5-inch long polyurethane tape. Use Method A to apply 27.5-inch long tape.

a. Mix a wetting solution consisting of 75% water, 25% isopropyl alcohol, Item 72, Appendix D, and 20 drops per liter of dishwashing compound, Item 122, Appendix D, in a 1-quart container.

b. Apply 15.5-inch long, 4 inch wide polyurethane tape as follows:

(1) Remove backing strips from polyurethane tape.

(2) Using a spray bottle, coat surface of blade and adhesive side of polyurethane tape with wetting solution.

(3) Apply polyurethane tape to blade. Align tape by sliding it in position. Remove trapped air, strained areas, and excess wetting solution by rubbing from leading edge and work toward trailing edge (Figure 16-6-15-1, Detail A).

(4) If required, tape edges of polyurethane tape using 2-inch masking tape, Item 214, Appendix D.

c. **MINIMUM DRYING TIME 24 HOURS.** Carefully remove masking tape when drying is complete.

d. Apply 27.5-inch long, 4 inch wide polyurethane tape per PARA 16-6-15.1.2, step b.

e. Remove masking tape from repairing areas on blade.

f. Touch up paint as required (PARA 2-6-5).

g. If initially installing erosion kit, apply polyurethane coating to tip cap (PARA 16-6-17).

h. Check tail rotor blade track and balance (TM 1-70-VIB).

16-6-16 TAIL ROTOR BLADE EROSION PROTECTION KIT POLYURETHANE TAPE REMOVE/REPAIR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Cloth, Cleaning, Item 102, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 1-6-4
- PARA 1-6-15
- PARA 1-6-16
- PARA 16-6-15

16-6-16.1 Remove Polyurethane Tape.

16-6-16.1.1. Remove/Repair.

NOTE

If replacing a damaged section of polyurethane tape, entire section must be removed. Replace only as many damaged sections as required. Blades do not require removal.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Tie down main rotor blades (PARA 1-6-4).

c. Remove polyurethane tape by slowly peeling away from leading edge of blade using nonmetallic scraper.

d. Remove tape adhesive from leading edge of blade using cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean cleaning cloth.

e. If replacing damaged section of polyurethane tape, install replacement tape (PARA 16-6-15).

16-6-17 TAIL ROTOR BLADE TIP CAP EROSION PROTECTION KIT POLYURETHANE COATING INSTALLATION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 1-Quart
- Knife
- Protection, Eye and Skin
- Spray Bottle

Materials/Parts

- Abrasive Paper, Item 15, Appendix D
- Brush, Paint, Item 86, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Coating Kit, Rain Erosion Resisting, Item 106B, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer Coating, Item 258A, Appendix D
- Thinner, Item 324A, Appendix D

References

- Appendix D
- PARA 1-6-4
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-6-5
- TM 1-70-VIB

16-6-17.1 Apply Polyurethane Coating.

16-6-17.1.1 Apply.

NOTE

When installing tail rotor blade erosion protection kit, all tail rotor blade tip caps must have polyurethane coating applied.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Move helicopter to dry, wind-free area.

c. Tie down main rotor blades (PARA 1-6-4).

d. Using abrasive paper, Item 15, Appendix D, scuff area of tip cap to be covered by coating.

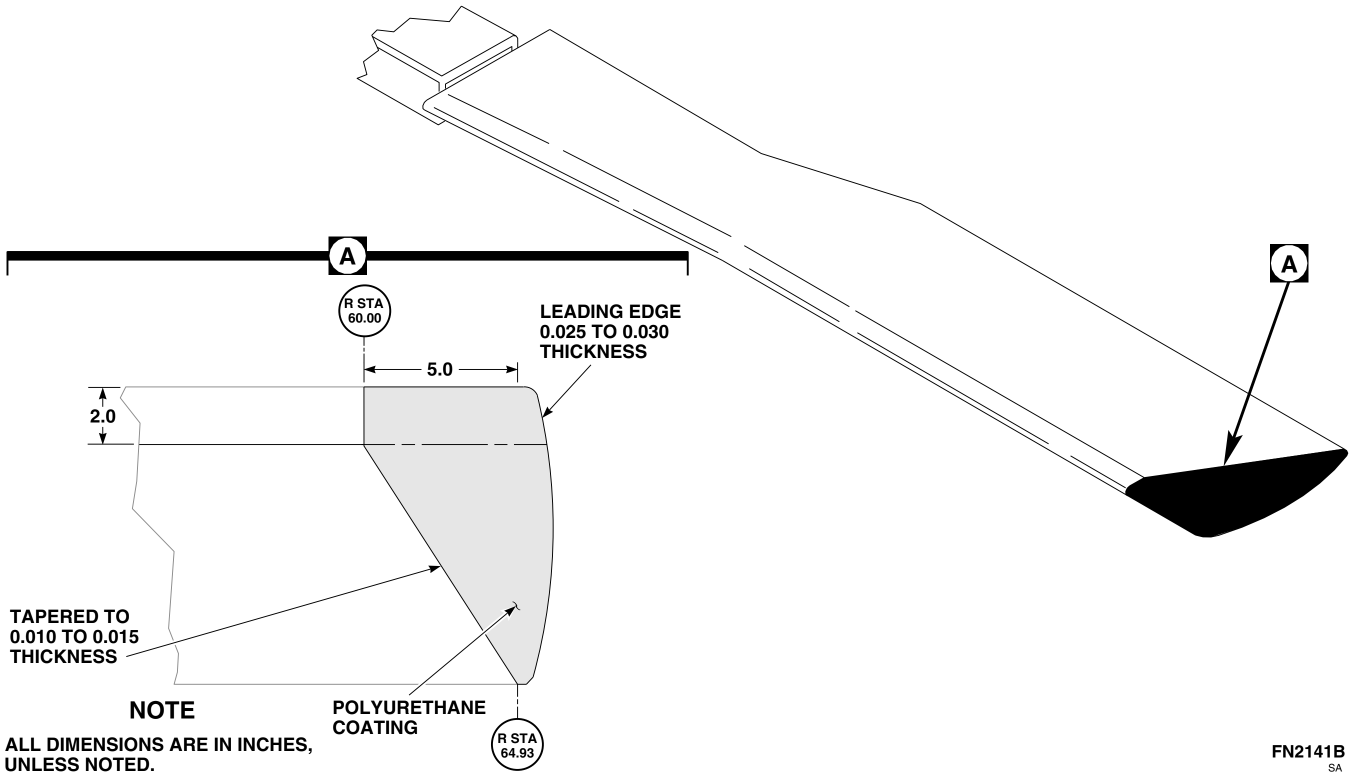


FIGURE 16-6-17-1. APPLY POLYURETHANE COATING

WARNING

Wash contacted skin with soap and water. If solvent contacts eyes, flush them with clean water and get immediate medical help.

e. Clean surface several times using cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry using clean cleaning cloth.

f. Cut excess polyurethane tape in 1-inch wide strips. Use to mask area not being coated (Figure 16-6-17-1, Detail A). Make sure to use two layers of tape on masked area.

g. Apply primer coating to tip cap as follows:

NOTE

Use equal amounts of primer on all tip caps to minimize balance adjustments to main rotor.

(1) Mix primer coating, Item 258A, Appendix D.

NOTE

Primer coating, Item 258A, Appendix D, may be applied using paint brush, Item 86, Appendix D, or spray bottle.

(2) If spraying, thin primer coating using thinner, Item 324A, Appendix D (four parts coating to one part thinner).

(3) Apply heavy coat of primer coating, Item 258A, Appendix D, to area.

(4) Allow primer coating to cure for a minimum of 1 hour at 70°F (21°C) or above.

(5) If surface becomes contaminated, lightly wipe contaminated area using cleaning cloth, Item 102, Appendix D, moistened with thinner, Item 324A, Appendix D.

h. Apply polyurethane coating (rain erosion resisting coating) as follows:

NOTE

Use equal amounts of coating on all tip caps to minimize balance adjustments to main rotor.

(1) Mix rain erosion resisting coating kit, Item 106B, Appendix D, part A and part B, in equal parts into 1-quart container.

NOTE

- Rain erosion resisting coating kit, Item 106B, Appendix D, may be applied using paint brush, Item 86, Appendix D, or spray bottle.
- Brushed coats tend to result in 0.002 to 0.003-inch thickness. Apply 15 - 16 coats when brushing on coating.
- Sprayed coats tend to result in 0.0007 to 0.001-inch thickness. Apply 35 - 40 coats when spraying on coating.

(2) If spraying, thin rain erosion resistant coating using thinner, Item 324A, Appendix D (four parts coating to one part thinner).

NOTE

- Taper coat thickness per Figure 16-6-11-1, Detail A, applying thicker coats at leading edge and progressively less toward rear edges.
- Use witness pieces to determine appropriate coating thickness.

(3) If brushing, apply first layer to tip cap using paint brush, Item 86, Appendix D. Allow 5 minutes drying time between coats until desired thickness is reached.

(4) If spraying, apply first layer to tip cap using spray bottle. Allow 2 minutes drying time between coats until desired thickness is reached.

CAUTION

- Do not allow masking to remain on blade while coating cures.
- Damage to tip cap and/or main rotor blade can occur while removing tape from masked area of blade. Hold cutting knife parallel to blade surface and slit coating at ridge created by the double layer of tape. Do not scratch or gouge blade surface. Remove tape.

(5) If brushing, remove masking after five coats are brushed on and apply remaining coats by eye.

(6) If spraying, remove masking after completion.

(7) Allow coating to cure for a minimum of 24 hours or 4 hours at 140 - 160°F (60 - 71°C).

i. Touch up paint as required (PARA 2-6-5).

j. Check tail rotor blade track and balance (TM 1-70-VIB).

16-6-18 TAIL ROTOR BLADE TIP CAP EROSION PROTECTION KIT POLYURETHANE COATING REMOVE/REPAIR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Nonmetallic Scraper, Locally-Made
- Protection, Eye and Skin
- Spray Bottle

Materials/Parts

- Abrasive Paper, Item 15, Appendix D
- Brush, Paint, Item 86, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Coating Kit, Rain Erosion Resisting, Item 106B, Appendix D
- Primer Coating, Item 258A, Appendix D
- Thinner, Item 324A, Appendix D

References

- Appendix D
- PARA 16-6-17

16-6-18.1 Remove/Repair Polyurethane Coating.

16-6-18.1.1 Remove/Repair.

NOTE

To remove coating, proceed to step b. To repair coating, proceed to step c.

a. Remove polyurethane coating (rain erosion resisting coating) as follows:

WARNING

Wash contacted skin with soap and water. If solvent contacts eyes, flush them with clean water and get immediate medical help.

(1) Using thinner, Item 324A, Appendix. saturate coating surface until coating is softened.

(2) Using nonmetallic scraper, remove coating.

(3) Wipe surface using cleaning cloth, Item 102, Appendix D, moistened with thinner, Item 324A, Appendix D. Wipe dry.

b. Repair polyurethane coating as follows:

(1) Carefully remove disbonded material.

(2) Using abrasive paper, Item 15, Appendix D, lightly scuff damaged area Figure 16-6-18-1, Detail A).

(3) Clean surface using cleaning cloth, Item 102, Appendix D, moistened with Item

324A, Appendix D, Appendix D. Wipe dry using clean cleaning cloth.

(4) If blade surface is exposed, apply primer coating, Item 258A, Appendix D, as follows:

(a) Mix primer coating, Item 258A, Appendix D.

NOTE

Primer coating, Item 258A, Appendix D, may be applied using paint brush, Item 86, Appendix D, or spray bottle.

(b) If spraying, thin primer coating using thinner, Item 324A, Appendix D (four parts coating to one part thinner).

(c) Apply heavy coat of primer coating, Item 258A, Appendix D, to area.

(d) Allow primer coating to cure for minimum of 1 hour at 70°F (21°C) or above.

(e) If surface becomes contaminated, lightly wipe contaminated area using cleaning cloth, Item 102, Appendix D, moistened with thinner, Item 324A, Appendix D.

(5) Apply polyurethane coating (rain erosion resisting coating) as follows:

(a) Mix rain erosion resisting coating kit, Item 106B, Appendix D, part A and part B, in equal parts into 1-quart container.

NOTE

Rain erosion resisting coating kit, Item 106B, Appendix D, may be applied using paint brush, Item 86, Appendix D, or spray bottle.

(b) If spraying, thin rain erosion resistant coating using thinner, Item 324A, Appendix D (four parts coating to one part thinner).

(c) If brushing, apply coating to tip cap using paint brush, Item 86, Appendix D. Allow 5 minutes drying time between coats until desired thickness is reached.

(d) If spraying, apply first layer to tip cap using spray bottle. Allow 2 minutes drying time between coats until desired thickness is reached.

(e) Allow coating to cure for a minimum of 24 hours or 4 hours at 140 - 160°F (60 - 71°C).

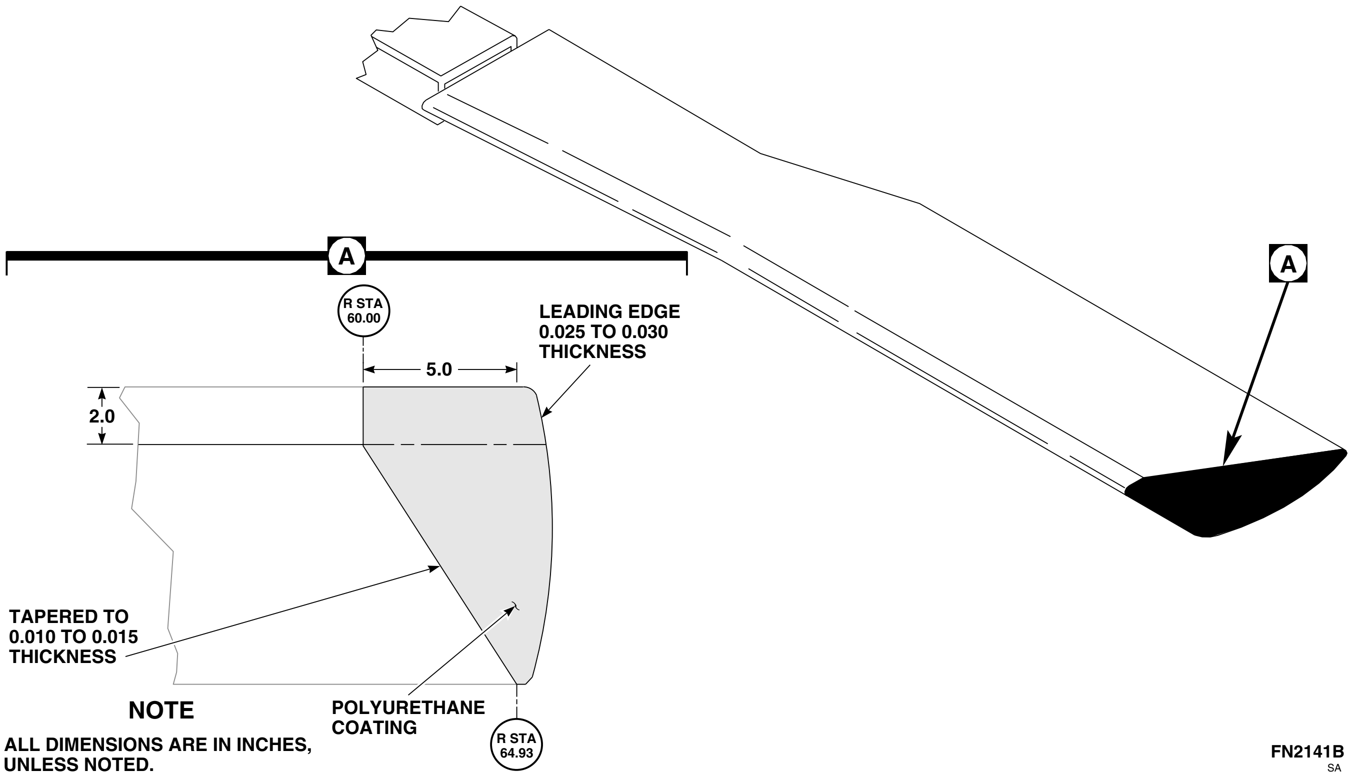


FIGURE 16-6-18-1. REMOVE/REPAIR POLYURETHANE COATING

16-6-19 APU ENGINE AIR PARTICLE SEPARATOR (EAPS) KIT INSTALLATION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- APU Engine Air Particle Separator (EAPS) Kit, 70073-30003-011
- Container, 1-Gallon
- Drill, 90°
- Drill Bit, 13/64-inch diameter
- Drill Bit, No. 1
- Drill Bit, No. 8
- Electrical Repairer's Toolkit
- Powerplant Repairer's Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 100 - 500 in. lbs

Materials/Parts

- Adhesive, Item 59, Appendix D
- Antiseize Compound, Item 74, Appendix D
- Lockwire, Item 196, Appendix D
- Sealing Compound, Item 292, Appendix D
- Tags

References

- Appendix D
- Chapter 15
- PARA 1-3-9
- PARA 1-7-20
- PARA 2-4-110
- PARA 7-4-4
- PARA 15-4-1
- PARA 15-4-2

16-6-19.1 Install APU Engine Air Particle Separator (EAPS) Kit.

NOTE

- APU EAPS kit can only be installed on helicopters with APU, 116305-100, 116305-200, and 116305-300 series.
- Tag and identify all items removed during this modification for future use when EAPS kit is removed.
- Tag and identify any excess kit components for future use.

16-6-19.1.1 Remove Original Components.

NOTE

When removing original APU engine components, refer to Chapter 15 for component locations.

a. Remove APU (PARA 15-4-1).

b. Remove fire extinguisher fitting and compartment drain from APU shroud. Tag and save for future use when installing APU EAPS kit.

c. Remove flame detector from APU rear panel. Tag and save for future use when installing APU EAPS kit.

d. Remove APU access door beam (PARA 2-4-110).

e. Remove electrical connectors from APU.

NOTE

APU start counter is only found on 116305-300 series APU.

f. Remove bolts securing APU start counter to electrical connector mounting bracket. Remove counter.

g. Remove remaining bolts securing electrical connector bracket and remove bracket.

h. Remove screws, lockwashers, and nuts securing terminal board to electrical connector mounting bracket. Tag and save hardware for use when installing APU EAPS kit.

i. Remove screws and washers securing electrical connector to electrical connector mounting bracket. Remove nutplate collar from electrical connector. Tag and save electrical connector mounting hardware and nutplate collar for use when installing APU EAPS kit.

j. Remove high tension ignition cable from underside of exciter.

k. Remove high tension ignition cable from igniter located on combustor.

l. Remove thermocouple probe from exhaust of APU.

m. Remove clamps securing electrical harness to APU. Remove harness.

n. On 116305-300 series APU, tag and remove start counter wires from terminal block. Remove start counter (with wiring) from harness and save for future use when installing APU EAPS kit.

o. Remove APU fuel enclosure cover.

p. Disconnect PCD air tube from inside of fuel containment box to gain access to main start fuel lines.

q. Disconnect and remove main start fuel tubes and PCD air tube from combustor.

r. Disconnect main and start fuel tubes and PCD air tube from fuel containment box.

s. Remove bolts securing accessory drive cover (with exciter mounting bracket) to APU. Remove cover. Remove exciter from mounting bracket.

t. Remove bolts and washers securing ejector tube to start bypass valve. Remove tube. Tag and save for future use when installing APU EAPS kit.

u. Remove bolts and washers securing start bypass valve to combustor. Remove gasket and valve.

v. Remove drain plug from underside of gear box and drain oil from APU into 1-gallon container.

w. Remove magnetic chip detector plug from front of gear box and discard packing. Install magnetic chip detector plug and packing (from APU EAPS kit) to underside of gear box. **TORQUE PLUG TO 138 - 152 INCH-POUNDS.** Lockwire, Item 196, Appendix D, plug.

x. Remove high oil temperature switch from rear of gear box and install packing and plug, AN814-5DL, from APU EAPS kit. Lockwire, Item 196, Appendix D, plug.

y. Remove oil cooler plug.

16-6-19.1.2 Install Kit Components.

NOTE

The installed kit weight is 21.0 pounds at STA 433.0.

a. Check APU inlet screen to make sure front and rear packings are installed.

b. If front or rear APU inlet screen packings are not installed, install as follows:

NOTE

Installation of front and rear APU inlet packings is the same.

(1) Cut packing at 15° to 30° scarf angle.

(2) Install packing around APU inlet and secure cut ends together with adhesive, Item 59, Appendix D.

c. Install packing on standoff, 164488-100, and install standoff in oil cooler plug hole. **TORQUE TO 72 - 78 INCH-POUNDS** (Figure 16-6-19-1, Sheet 5, Detail F).

d. Install gasket and accessory drive cover, 36051-1, to APU with bolts, MS9915-12, and washers, NAS1149C0432R, in top two holes. **TORQUE BOLTS TO 100 - 130 INCH-POUNDS.**

e. Install exciter mounting bracket, 164487-100, over accessory drive cover and install cover and bracket with bolts, MS9915-13, and washers, NAS1149C0432R, in bottom two holes (Figure 16-6-19-1, Sheet 5, Detail E). **TORQUE BOLTS TO 100 - 130 INCH-POUNDS.**

f. Install exciter bracket to standoff with bolt, AN3CH3A, and washer, NAS1149C0332R. **TORQUE BOLT TO 48 - 52 INCH-POUNDS.**

g. Install exciter to mounting bracket with electrical connector at 4 o'clock position with bolts, MS9914-06, washers, NAS1149F0332P, and nuts, MS21045-3. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

h. Install support brackets, 70303-03210-101 and 70303-03210-102, to bottom half of particle separator with bolts, AN3C3A, and washers, NAS1149C0332R. **DO NOT TORQUE BOLTS AT THIS TIME** (Figure 16-6-19-1, Sheet 4, Detail D).

i. Position lower panel half, 70303-03201-044, over bleed air outlet flange and allow to hang freely.

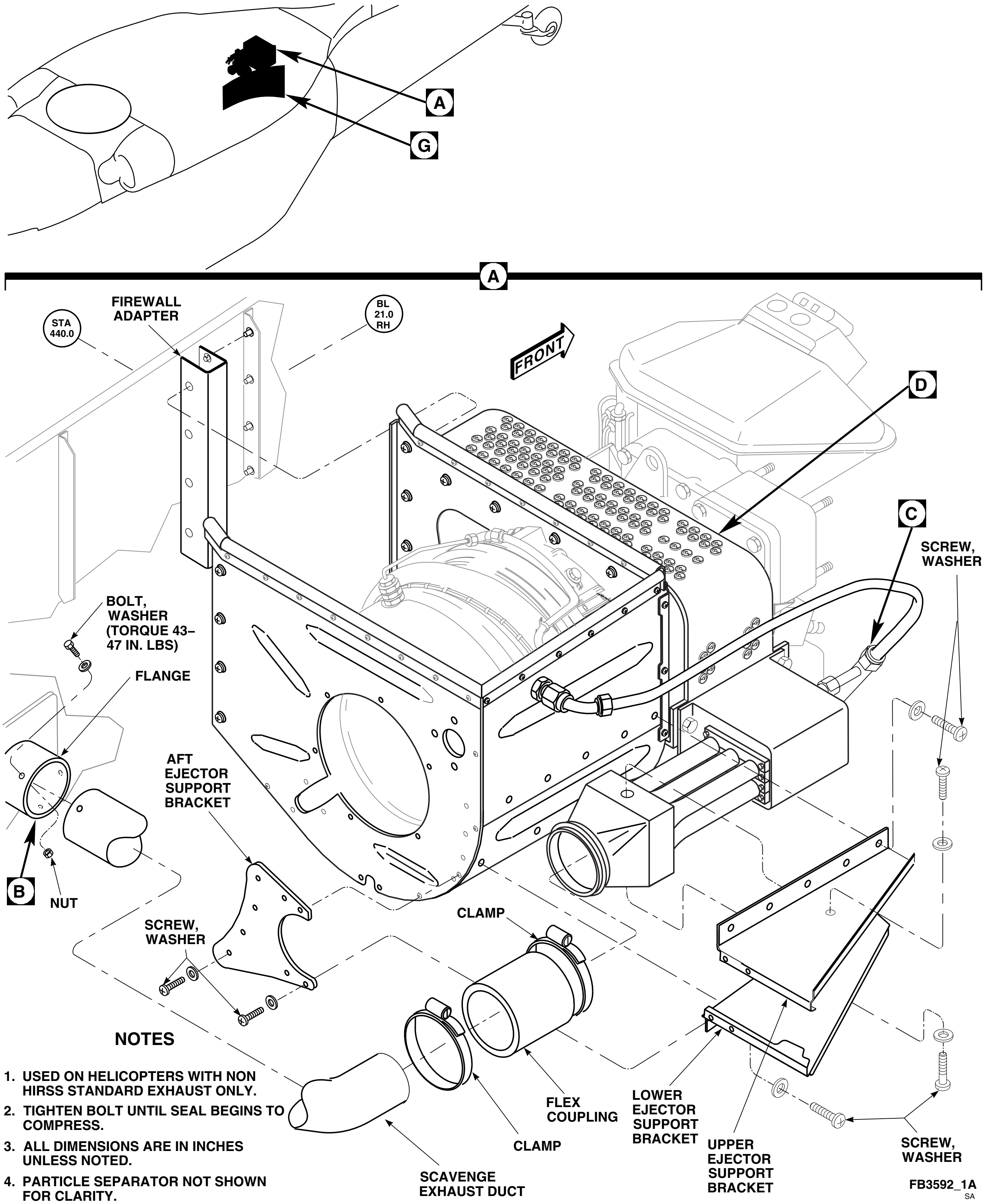
j. Position lower particle separator, 164095-100, under APU and install using bolts, AN5H4A, and washers, NAS1149C0563R, through two mounting brackets. **DO NOT TORQUE BOLTS AT THIS TIME.**

k. Install large half-moon support band, 164082-100, over rear side of inlet and onto particle separator with bolts, AN4C7A. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

l. Install small half-moon support band, 164084-100, over front side of inlet and onto particle separator with bolts, AN4C7A. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

m. **TORQUE BOLTS SECURING SUPPORT BRACKETS TO APU TO 80 - 90 INCH-POUNDS.** Lockwire, Item 196, Appendix D, bolts to brackets.

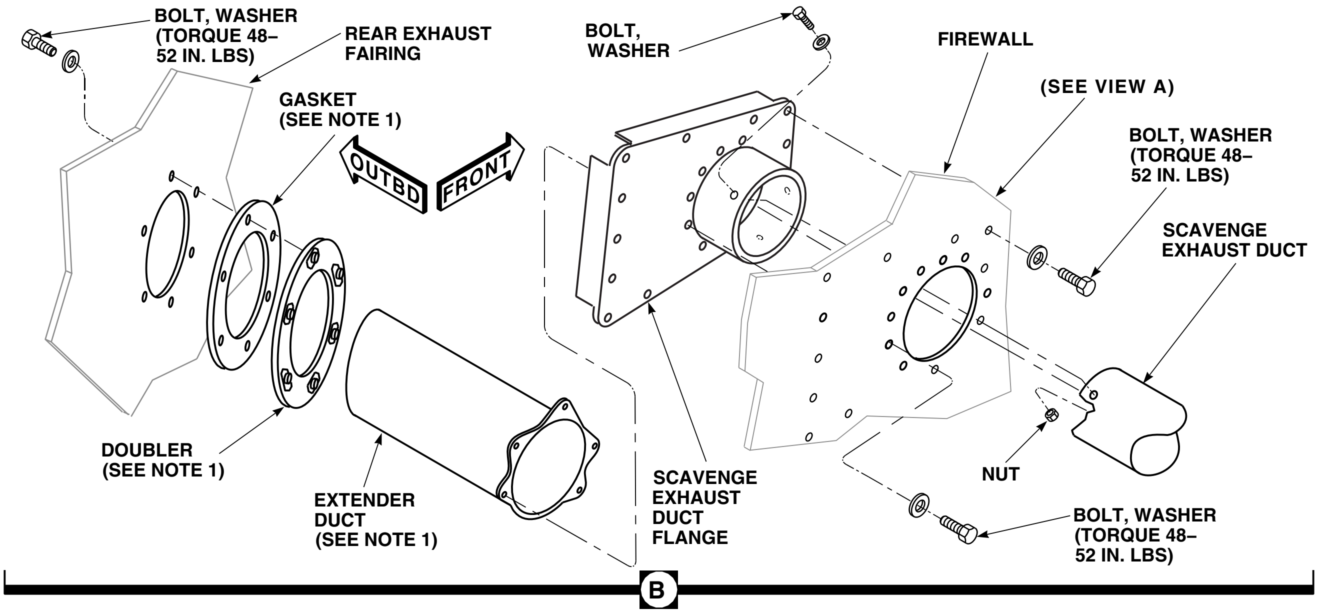
n. **TORQUE BOLTS SECURING SUPPORT BRACKETS TO PARTICLE SEPARATORS TO 48 - 52 INCH-POUNDS.**

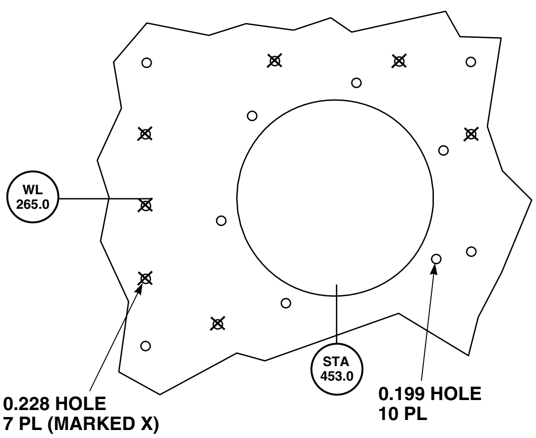


NOTES

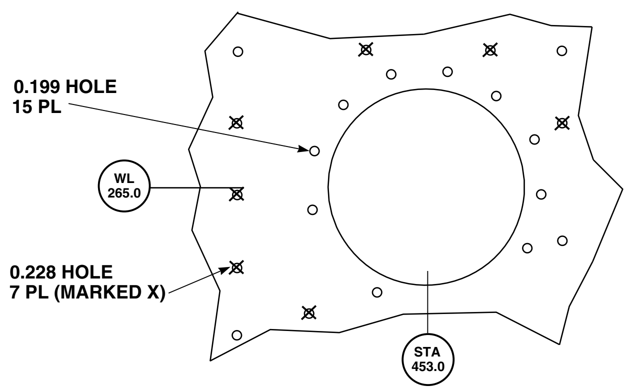
1. USED ON HELICOPTERS WITH NON HIRSS STANDARD EXHAUST ONLY.
2. TIGHTEN BOLT UNTIL SEAL BEGINS TO COMPRESS.
3. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.
4. PARTICLE SEPARATOR NOT SHOWN FOR CLARITY.

FIGURE 16-6-19-1. INSTALL APU ENGINE AIR PARTICLE SEPARATOR (EAPS) KIT (SHEET 1 OF 8)





VIEW A
FIREWALL HOLE PATTERN
(LOOKING OUTBOARD)
HELICOPTERS
WITH HIRSS



VIEW A
FIREWALL HOLE PATTERN
(LOOKING OUTBOARD)
HELICOPTERS WITHOUT
HIRSS STANDARD EXHAUST

FB3592_2
SA

FIGURE 16-6-19-1. INSTALL APU ENGINE AIR PARTICLE SEPARATOR (EAPS) KIT
(SHEET 2 OF 8)

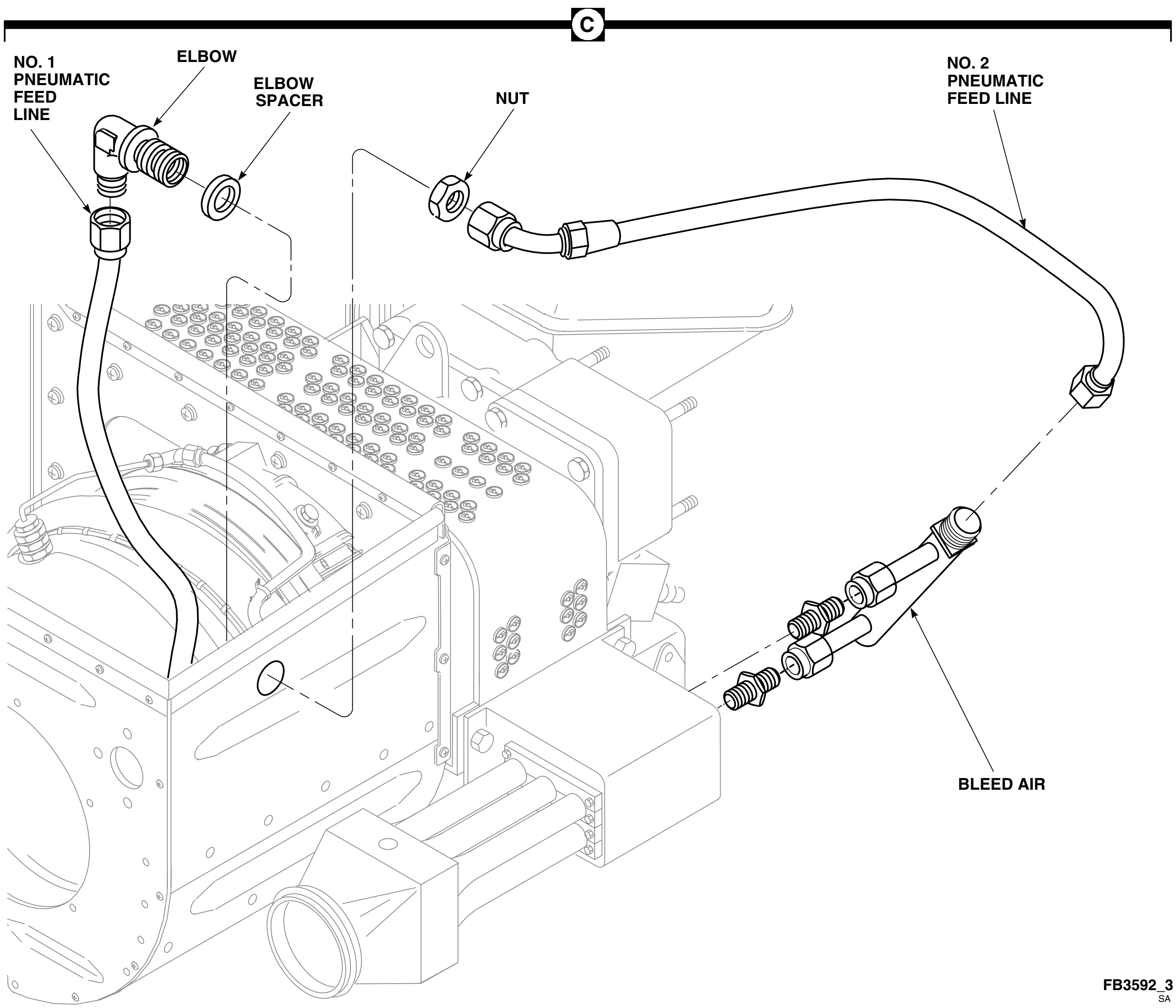


FIGURE 16-6-19-1. INSTALL APU ENGINE AIR PARTICLE SEPARATOR (EAPS) KIT
(SHEET 3 OF 8)

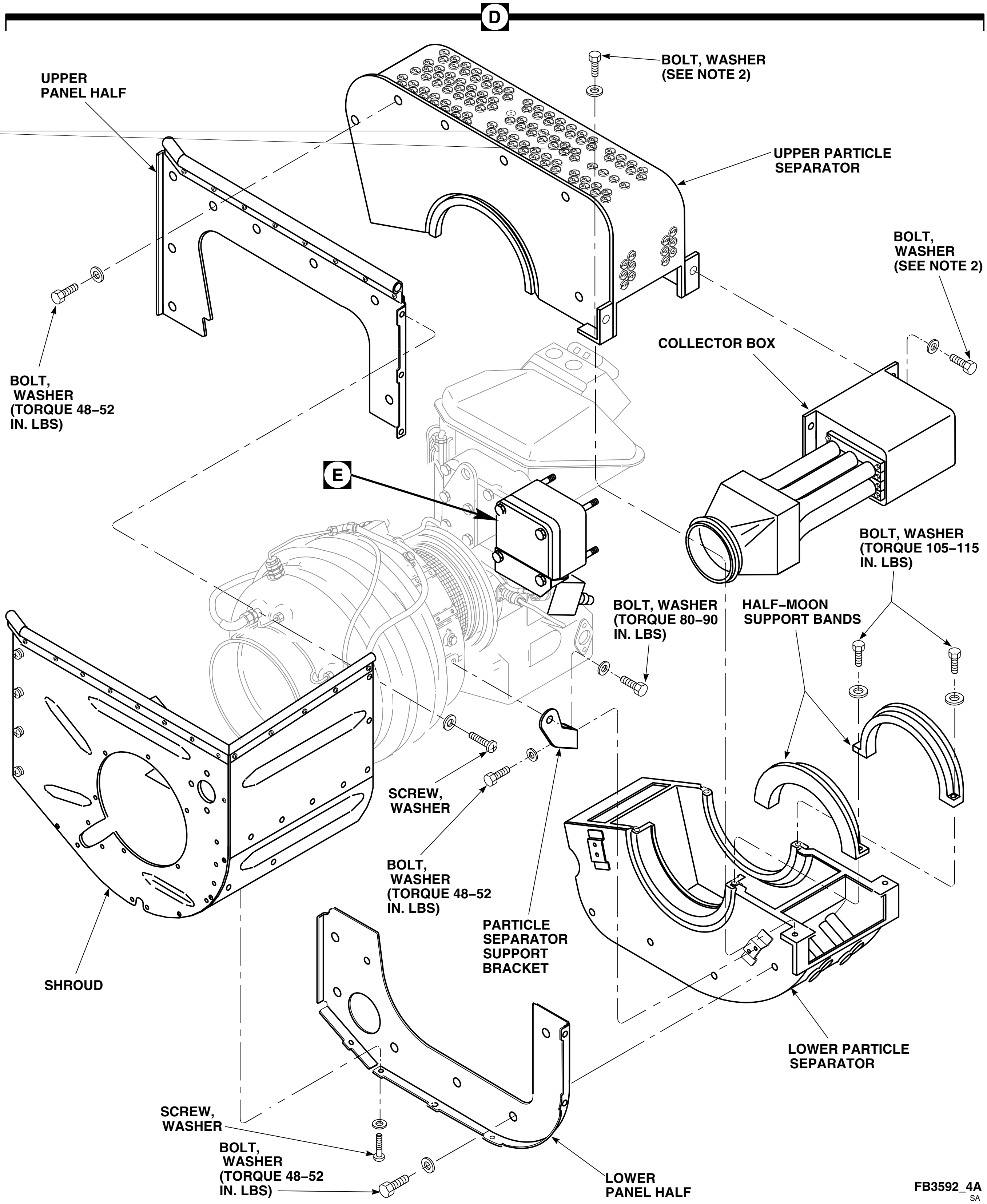


FIGURE 16-6-19-1. INSTALL APU ENGINE AIR PARTICLE SEPARATOR (EAPS) KIT (SHEET 4 OF 8)

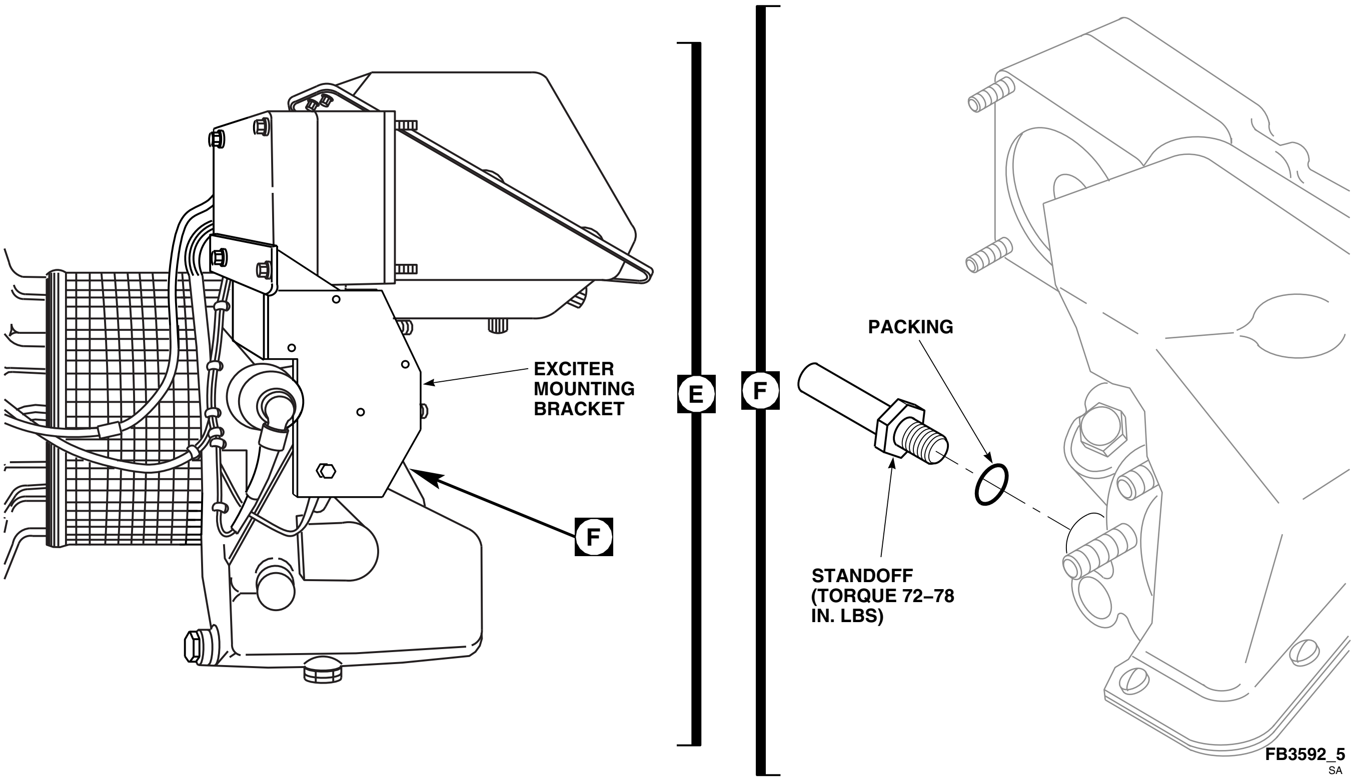


FIGURE 16-6-19-1. INSTALL APU ENGINE AIR PARTICLE SEPARATOR (EAPS) KIT (SHEET 5 OF 8)

o. Position upper particle separator, 164095-100, over APU and install to lower separator with bolts, AN4C6A, and washers, NAS1149C0432R. **TIGHTEN BOLTS UNTIL SEAL BEGINS TO COMPRESS.**

p. Position gasket and start bypass valve adapter, 164091-100, on bleed air duct with nozzle at 4 o'clock position (looking forward). Secure adapter to duct with Allen head screws, MS16998-28L. **TORQUE SCREWS TO 48 - 52 INCH-POUNDS.**

q. Install gasket and start bypass valve to adapter with bolts and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

r. Remove screws and washers holding solenoid to top of start bypass valve. Remove solenoid and gasket. Do not discard screws and washers.

s. Position gasket and solenoid on start bypass valve with connector facing away from APU.

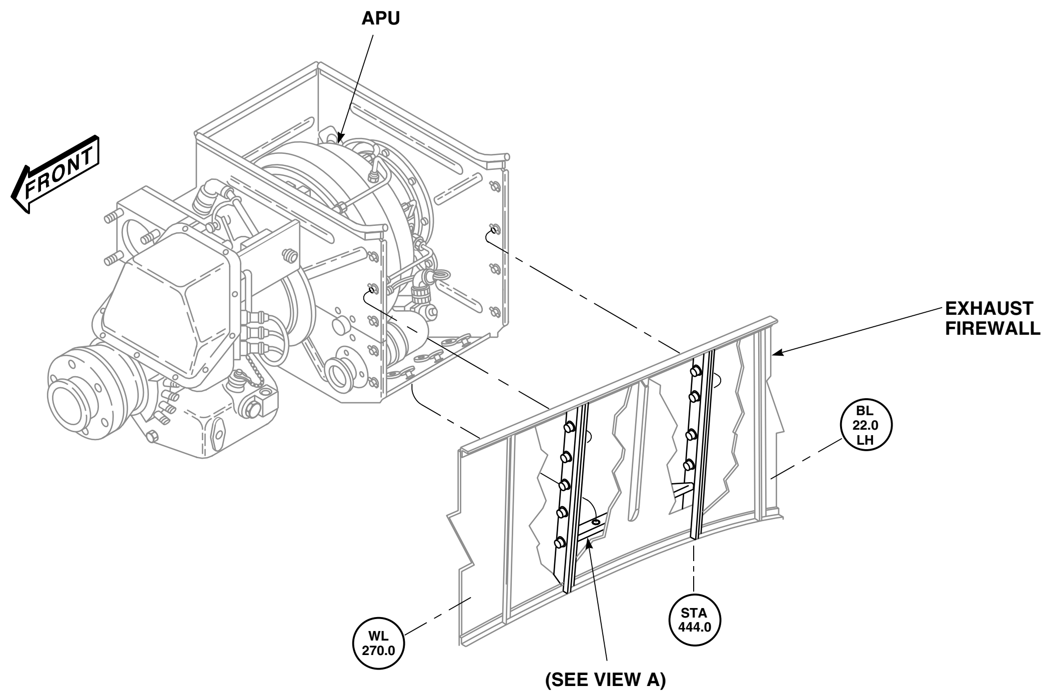
t. Install gasket and solenoid to valve with screws and washers. Lockwire, Item 196, Appendix D, screws together.

u. Install ejector tube, 70303-03204-041, to start bypass valve with ejector opening at 2 o'clock position (looking forward) with bolts and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.** Lockwire, Item 196, Appendix D, bolts together.

v. Connect 45° end of No. 1 pneumatic feed line, SS40C6D155000, to start bypass valve adapter (Figure 16-6-19-1, Sheet 3, Detail C).

w. Apply a light coat of antiseize compound, Item 74, Appendix D, to mating surfaces of lower panel half and lower particle separator.

x. Install lower panel half to lower particle separator with bolts, AN3C3A, and washers, NAS1149C0363R. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS** (Figure 16-6-19-1, Sheet 4, Detail D).



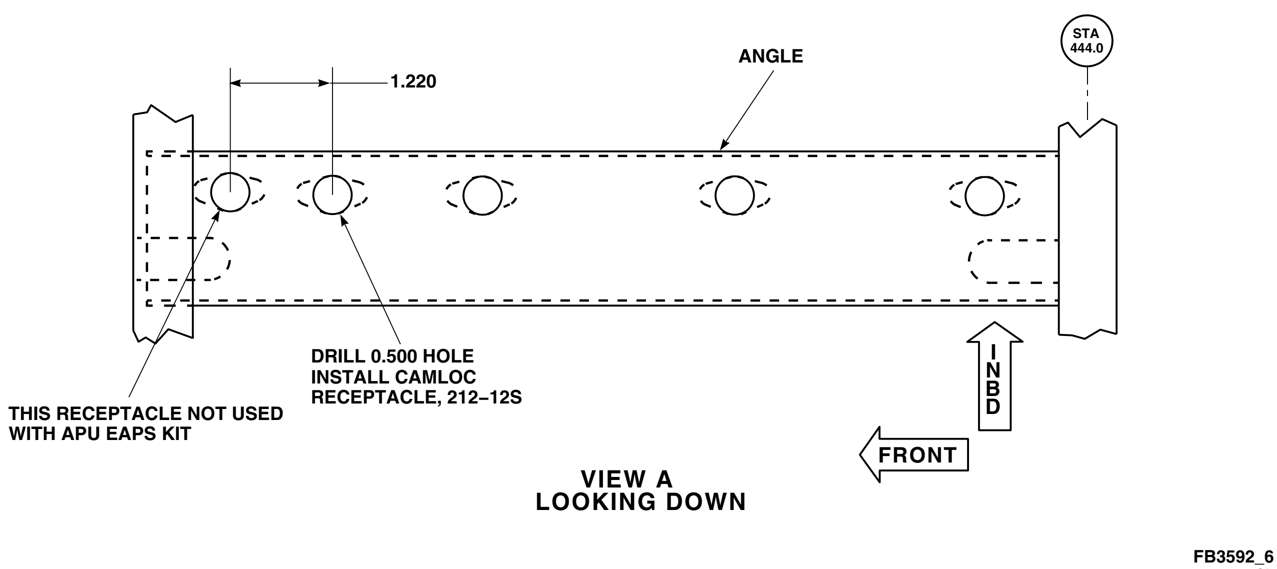


FIGURE 16-6-19-1. INSTALL APU ENGINE AIR PARTICLE SEPARATOR (EAPS) KIT (SHEET 6 OF 8)

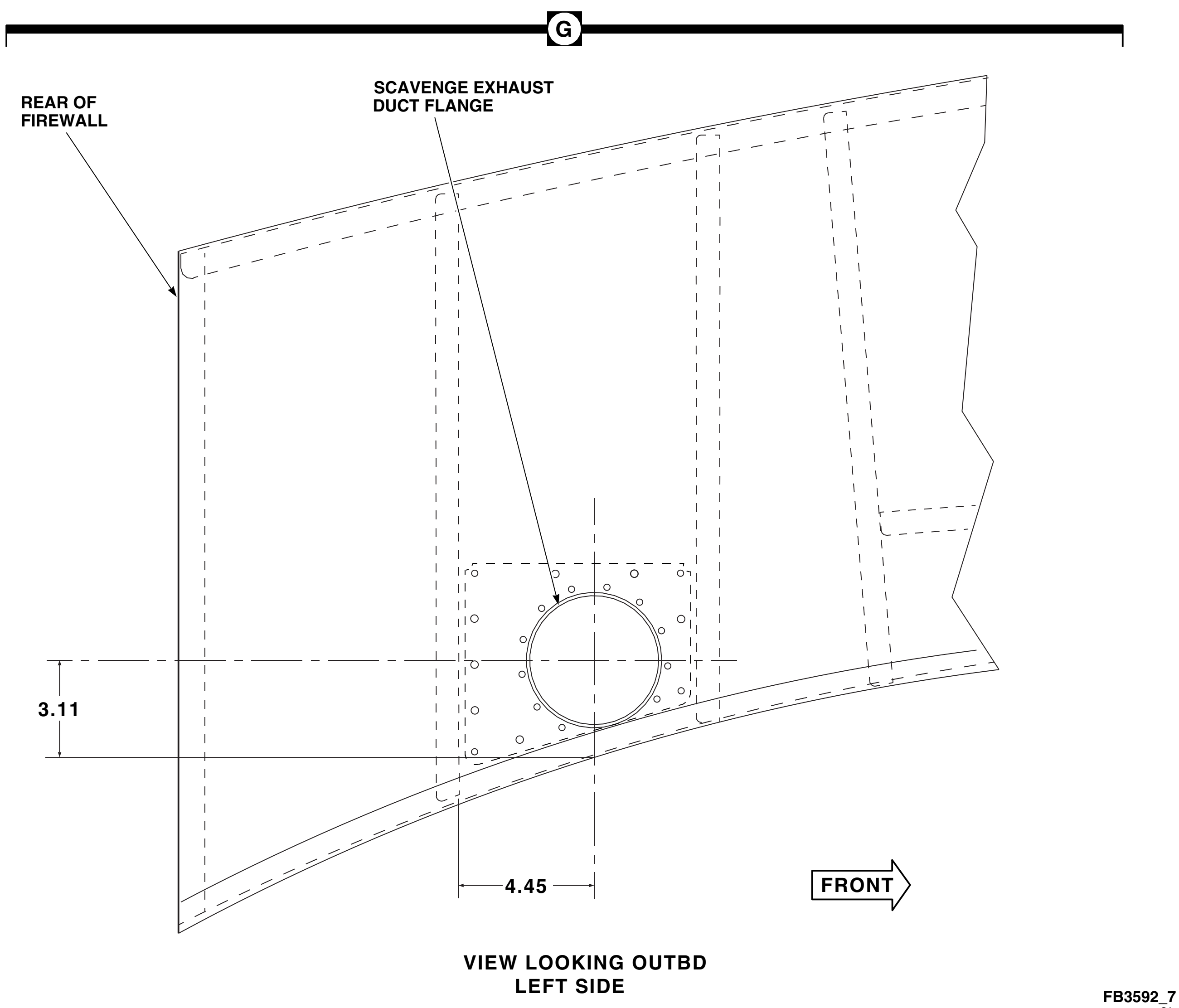


FIGURE 16-6-19-1. INSTALL APU ENGINE AIR PARTICLE SEPARATOR (EAPS) KIT (SHEET 7 OF 8)

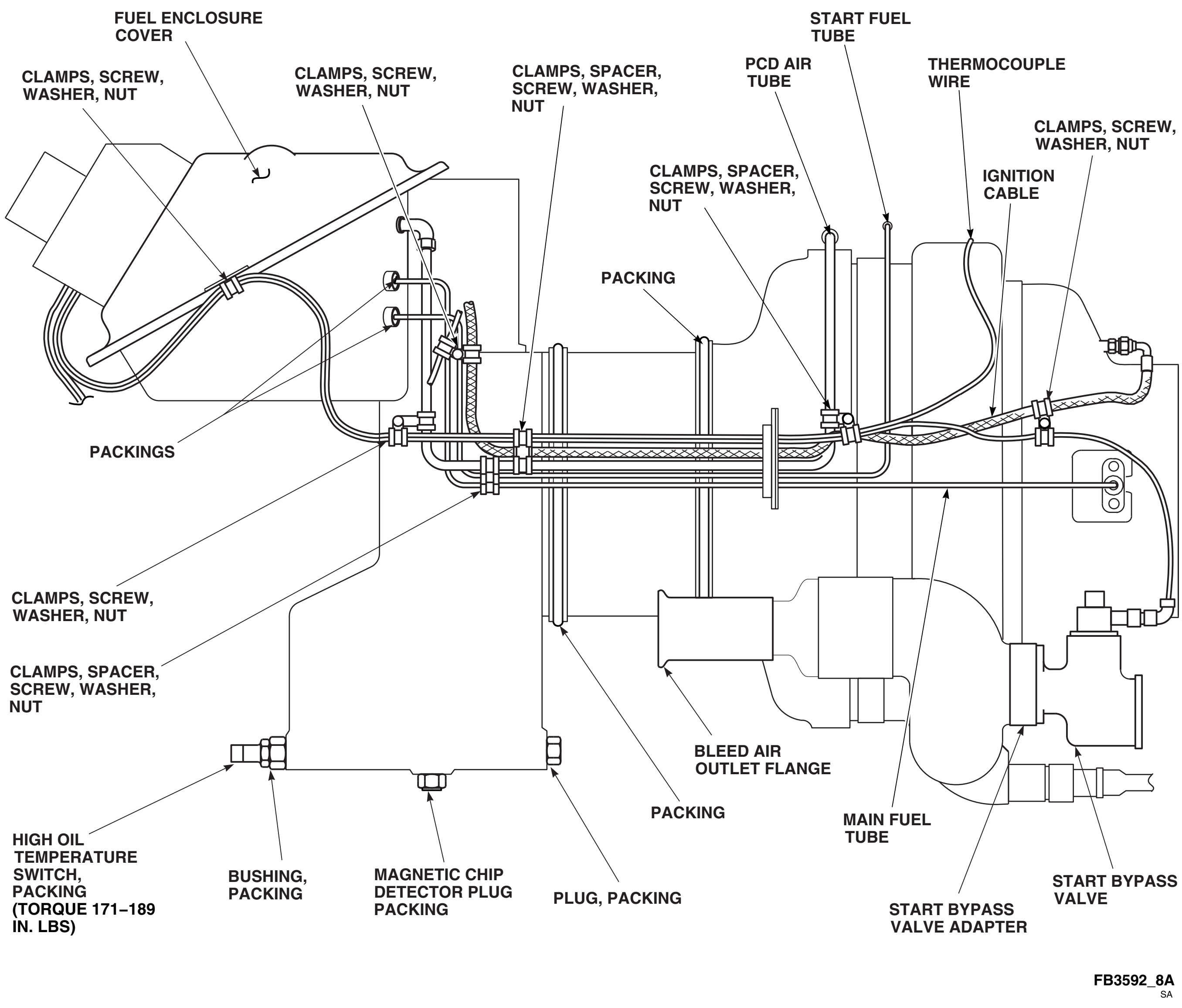


FIGURE 16-6-19-1. INSTALL APU ENGINE AIR PARTICLE SEPARATOR (EAPS) KIT (SHEET 8 OF 8)

y. Install ignition cable, 179505-2, to igniter plug. **TORQUE NUT ON CABLE TO 138 - 152 INCH-POUNDS** (Figure 16-6-19-1, Sheet 8).

z. Install packings to main fuel tube.

aa. On 116305-100 series APU's install start fuel tube, 172371-100, main fuel tube, 172365-100, and PCD air tube bundle, 172372-100.

ab. On 116305-200/116305-300 series APU's, install start fuel tube, 172364-100, main fuel tube, 172365-100, and PCD air tube bundle, 172366-100.

ac. Connect PCD air tube located inside fuel containment box.

ad. Install fuel enclosure cover, 164097-100.

ae. Install bushing and packing to front of gear box.

af. Install packing to high oil temperature switch.

ag. Install high oil temperature switch to bushing located on front of gear box. **TORQUE SWITCH TO 171 - 189 INCH-POUNDS.**

ah. Install nutplate collar to electrical connector, J301.

ai. Inspect and treat electrical connector (PARA 1-7-20).

aj. Install electrical connector, J301, to inside of bracket, 164092-1, with screws and washers from original electrical connector.

ak. On 116305-300 series APU, install start counter wires to terminal block and remove tags.

al. Install terminal block, MS18029-1L-6, to inside of bracket with screws, lockwashers, and nuts from original terminal block.

am. Position bracket, 164092-1, on fuel enclosure cover with large opening facing down. Install bracket to cover with bolts, MS9914-05, and washer, NAS1149F0332P. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

an. On 116305-300 series APU, install start counter to bracket on fuel enclosure cover with bolts, MS9914-05, washers, NAS1149F0332P, and nuts, MS21045-3. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

ao. Inspect and treat electrical connector (PARA 1-7-20).

ap. Connect electrical connector, P305, to start fuel solenoid valve.

aq. Inspect and treat electrical connector (PARA 1-7-20).

ar. Connect electrical connector, P306, to main fuel solenoid valve.

as. Inspect and treat electrical connector (PARA 1-7-20).

at. Connect electrical connector, P307, to maximum fuel solenoid valve.

au. Install ground wire.

av. Inspect and treat electrical connector (PARA 1-7-20).

aw. Connect electrical connector, P303, to bottom of exciter box.

ax. Connect electrical connector, P302, to magnetic pickup located on bottom side of accessory drive case.

ay. Inspect and treat electrical connector (PARA 1-7-20).

az. Connect electrical connector, P304, to low oil pressure switch located under accessory drive case.

ba. Inspect and treat electrical connector (PARA 1-7-20).

bb. Connect electrical connector, P310, to high oil temperature switch located on front of gear box.

bc. Install ignition cable to exciter box. **TORQUE NUT ON CABLE TO 138 - 152 INCH-POUNDS.**

bd. Inspect and treat electrical connector (PARA 1-7-20).

be. Connect electrical connector, P308, to start bypass valve.

bf. Apply a light coat of antiseize compound, Item 74, Appendix D, to threads of thermocouple. Install thermocouple to APU. **TORQUE THERMOCOUPLE TO 95 - 105 INCH-POUNDS.**

bg. Apply a light coat of antiseize compound, Item 74, Appendix D, to mating surfaces of upper panel half, 70303-03201-043, and upper particle separator.

bh. Install upper panel half to particle separator with bolts, AN3C3A, and washers, NAS1149C0332R. **DO NOT TORQUE BOLTS AT THIS TIME** (Figure 16-6-19-1, Sheet 4, Detail D).

bi. Position tube bundle firewall seal, 169126-2, on rear side of panel halves and secure backing plate, 164086-100, to seal and panel halves with bolts, AN3CH3A, and washers, NAS1149C0332R. **TIGHTEN BOLTS UNTIL SEAL BEGINS TO COMPRESS.**

bj. Install clamps as required to secure APU electrical harness to APU (Figure 16-6-19-1, Sheet 8).

bk. Install fire extinguisher fitting to APU shroud, 70303-03202-041, with inside fitting at 10 o'clock position.

bl. Install compartment drain to bottom of APU shroud.

bm. Install shroud to APU with screws, MS27039C0806, and washers, NAS1149CN816R.

bn. **TORQUE BOLTS SECURING UPPER PANEL HALF TO PARTICLE SEPARATOR TO 48 - 52 INCH-POUNDS.**

bo. Install fire detector to APU rear panel, 70303-03201-041.

bp. Install rear panel to APU with screws, MS27039C0806, and washers, NAS1149CN816R.

bq. Install rear mounting lug. **TORQUE 133 - 147 INCH-POUNDS.**

br. Install lower ejector support bracket, 70303-03205-041, to shroud using screws, MS27039C0808, and washers, NAS1149CN816R (Figure 16-6-19-1, Sheet 1).

bs. Install upper ejector support bracket, 70303-03205-042, to shroud using screws, MS27039C0808, and washers, NAS1149CN816R.

bt. Install aft ejector support bracket, 70303-03205-103, to shroud using screws, MS27039C0808 (Figure 16-6-19-1, Sheet 1, Detail A).

bu. Install elbow spacer, 70303-03207-101, and elbow, AN833-8J, to shroud with nut, AN924-8J. Connect No. 1 pneumatic feed line to elbow.

bv. Attach firewall adapter, 70303-03203-041, to firewall APU angle (Figure 16-6-19-1, Sheet 1, Detail C).

bw. If firewall to shroud mounting angle has not been previously modified by adding additional camloc receptacle, perform the following (Figure 16-6-19-1, Sheet 6, View A):

(1) Mark location of hole to be cut 1.220 inches rear of first receptacle.

(2) Cut 0.500 inch hole in angle. Remove burrs from hole.

(3) Install camloc receptacle, 212-12S, to angle with rivets, NAS1097AD3.

bx. If firewall has not been previously modified with scavenge exhaust duct flange mounting holes, perform the following:

(1) Position scavenge exhaust duct flange, 70303-03208-041, on firewall and mark

location of flange mounting holes and 4.20 inch duct hole (Figure 16-6-19-1, Sheet 7, Detail G). Remove flange from firewall.

(2) Drill mounting holes in firewall using No. 1 and No. 8 drill bits (Figure 16-6-19-1, Sheet 2, Detail A).

(3) Cut out duct hole 4.20 inches in diameter.

(4) Remove burrs from all holes.

by. Install flange to outboard side of firewall with bolts, AN3C3A, and washers, NAS1149C0332R. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS** (Figure 16-6-19-1, Sheet 2, Detail B).

bz. Locate APU exhaust duct in position under APU panel.

ca. Install flexible coupling, CA030-4.00-16, to front of scavenge exhaust duct, 70303-03206-042, and locate duct under APU exhaust duct.

cb. Inspect APU check valve (PARA 7-4-4).

cc. Install APU (PARA 15-4-1).

cd. Service APU (PARA 1-3-9).

ce. If flange and duct have not been previously modified with flange to duct mounting holes, perform the following:

(1) Temporarily install collector box to particle separators with bolts, AN4C3A, and washers, NAS1149C0432R.

(2) Position scavenge exhaust duct in flange and onto collector box.

(3) Mark location of duct to flange mounting holes on duct.

(4) Remove flexible coupling and scavenge exhaust duct from collector. Remove collector from particle separator.

(5) Remove scavenge exhaust duct.

(6) Drill holes through flange and duct using 13/64-inch diameter drill bit.

(7) Remove burrs from all holes.

(8) Reposition scavenge exhaust duct under APU exhaust duct and install to flange with bolts, AN3C3A, washers, NAS1149C0332R, and nuts, MS21044C3. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.** Cover nuts with sealing compound, Item 292, Appendix D.

(9) Install collector box to flexible coupling and scavenge exhaust duct with clamps, MB9325T52S412V (Figure 16-6-19-1, Sheet 1, Detail A).

(10) Install collector box to particle separator with bolts, AN4C5A, and washers, NAS1149C0463R. **TIGHTEN BOLTS UNTIL SEAL BEGINS TO COMPRESS.**

(11) Install upper and lower ejector support brackets to collector box with screws, MS27039C1-06, and washers, NAS1149C0332R.

cf. Install bleed air fitting assembly to front of collector box (Figure 16-6-19-1, Sheet 3, Detail C).

cg. Connect No. 2 pneumatic feed line, SS40C28D204050, to elbow on firewall and bleed air fitting assembly (Figure 16-6-19-1, Sheet 3, Detail C).

ch. Install APU exhaust (PARA 15-4-2).

ci. Install stringer to airframe at BL 0.0 using screws.

cj. Check to make sure upper APU firewall seals are slightly compressed when APU compartment doors are closed and latched.

ck. Make sure area is clean and free of foreign material.

cl. Close and latch APU compartment doors.

16-6-20 APU ENGINE AIR PARTICLE SEPARATOR (EAPS) KIT REMOVAL.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- APU Engine Air Particle Separator (EAPS) Kit, 70073-30003-011
- Container, 1-Gallon
- Electrical Repairer's Toolkit
- Powerplant Repairer's Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 300 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lockwire, Item 196, Appendix D
- Tags

References

- Appendix D
- Chapter 15
- PARA 1-3-9
- PARA 1-7-20
- PARA 15-4-1
- PARA 15-4-2

16-6-20.1 Remove APU Engine Air Particle Separator (EAPS) Kit.

16-6-20.1.1. Remove Kit Components.

NOTE

Tag and identify all kit items removed from APU for future use when EAPS kit is reinstalled.

a. Open APU compartment doors.

b. Remove screws securing stringer to airframe at BL 0.0.

c. Disconnect APU exhaust duct from rear of APU (PARA 15-4-2).

d. Remove clamps securing scavenge exhaust duct to flex coupling.

e. Remove bolts, washers, and nuts securing exhaust duct to scavenge exhaust duct flange.

f. Remove bolts and washers securing scavenge exhaust duct flange to firewall. Remove flange.

g. Remove APU (PARA 15-4-1).

h. Remove scavenge exhaust duct.

i. Remove flex coupling from rear of collector box.

j. Remove No. 1 and No. 2 pneumatic feed lines.

k. Remove elbow, elbow spacer, and nut from shroud.

l. Remove screws securing aft ejector support bracket to shroud. Remove bracket.

m. Remove screws securing upper ejector support bracket to shroud and collector box. Remove bracket.

n. Remove screws securing lower ejector support bracket to shroud and collector box. Remove bracket.

o. Remove bolts and washers securing collector box to particle separator. Remove collector.

p. Remove screws from rear panel. Remove panel.

q. Remove flame detector from rear panel.

r. Remove rear mounting lug.

s. Remove shroud.

t. Remove compartment drain and fire extinguisher fitting.

u. Remove bolts from backing plate.

v. Remove backing plate and firewall seal from upper and lower panel half.

w. Remove bolts and washers from upper panel half. Remove panel half.

x. Remove bolts and washers from lower panel half. Allow panel half to hang on bleed air outlet flange.

y. Remove electrical connectors and ground wire from APU.

z. Remove thermocouple from rear of APU.

aa. On 116305-300 series APU, tag and remove APU start counter wires from terminal block.

ab. Remove start counter from fuel enclosure cover and remove counter from harness.

ac. Remove clamps as required to remove harness. Remove fuel enclosure cover and remove harness.

ad. Remove clamps as required to remove harness. Remove fuel enclosure cover and remove harness.

ae. Remove terminal block from bracket. Save hardware for use when installing original components.

af. Remove electrical connector from bracket. Save hardware and nutplate collar for use when installing original components.

ag. Disconnect ignition cable from igniter plug and exciter box. Remove ignition cable.

ah. Remove main and start fuel tubes and PCD air tube bundle.

ai. Loosen fitting on pneumatic feed line.

aj. Remove bolts securing ejector tube to start bypass valve. Remove tube.

ak. Remove screws and washers securing solenoid to top of start bypass valve. Remove solenoid and gasket.

al. Position gasket and solenoid on start bypass valve with connector facing rear.

am. Install gasket and solenoid to valve with screws and washers. Lockwire Item 196, Appendix D, screws together.

an. Remove bolts and washers securing start bypass valve to start bypass valve adapter. Remove valve and gasket.

ao. Remove Allen head bolts securing start bypass valve adapter to bleed air duct. Remove gasket and adapter.

ap. Remove bolts and washers securing upper particle separator to lower particle separator. Remove upper separator.

aq. Remove bolts securing large and small half-moon support bands. Remove support bands.

ar. Have assistant support lower particle separator and remove mounting bolts from bracket. Remove lower separator.

as. Remove bolts and washers securing exciter to exciter mounting bracket. Remove exciter.

at. Remove bolt and washer securing exciter mounting bracket to standoff.

au. Remove bolts and washers securing accessory drive cover and exciter mounting bracket to APU. Remove bracket, gasket, and cover.

av. Remove standoff from oil cooler plug hole.

16-6-20.1.2. Inspection of Particle Separator Inlet and Outlet Tubes.

NOTE

Damaged or missing tubes must be distributed over a wide area of the particle separator. If damaged or missing tubes are concentrated in one area, the particle separator **MUST BE REPLACED.**

Inspect the inlet and outlet tubes in the upper and lower particle separator. **THE MAXIMUM NUMBER OF DAMAGED OR MISSING TUBES SHALL NOT EXCEED 10% OF THE TOTAL NUMBER OF TUBES IN UPPER OR LOWER PARTICLE SEPARATOR. IF DAMAGED OR MISSING TUBES EXCEEDS 10%, REPLACE UPPER OR LOWER PARTICLE SEPARATOR.**

16-6-20.1.3. Install Original Components.

NOTE

When installing original APU engine components refer to Chapter 15 for component's original location.

a. Remove magnetic drain plug from underside of oil pan and drain APU oil into 1-gallon container. Install plug.

b. Remove plug from rear of oil cooler pan and reinstall in oil cooler plug hole. **TORQUE PLUG TO 138 - 152 INCH-POUNDS.**

c. Remove high oil temperature switch from front of oil pan.

d. Reinstall magnetic drain plug in front hole of oil pan. **TORQUE PLUG TO 138 - 152 INCH-POUNDS.**

e. Install packing to high oil temperature switch.

f. Install temperature switch to rear side of oil pan. **TORQUE SWITCH TO 178 - 189 INCH-POUNDS.**

g. Install gasket and start bypass valve to bleed air duct with bolts and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

h. Install ejector tube to start bypass valve with bolts and washers. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.** Lockwire, Item 196, Appendix D, bolts together.

i. Install accessory drive cover (with exciter mounting bracket) to APU with bolts and washers. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**

j. Install exciter box to bracket with bolts, washers, and nuts. **TORQUE NUTS TO 47 - 53 INCH-POUNDS.**

k. Lockwire, Item 196, Appendix D, all connectors which are not self-locking.

l. Install packing to main and start fuel tubes.

m. Install main and start fuel tubes, and PCD air tube bundle.

n. Install fuel enclosure cover.

o. Inspect and treat electrical connector (PARA 1-7-20).

p. Connect electrical connector, P305, to start fuel solenoid valve.

q. Inspect and treat electrical connector (PARA 1-7-20).

r. Connect electrical connector, P306, to main fuel solenoid valve.

s. Inspect and treat electrical connector (PARA 1-7-20).

t. Connect electrical connector, P307, to maximum fuel solenoid valve.

NOTE

APU start counter is found on 116305-300 series APU only.

u. Position APU start counter on counter bracket and install bracket and counter to APU with bolts and washers. **TORQUE BOLTS TO 43 - 47 INCH-POUNDS.**

v. Install terminal block and electrical connector to counter mounting bracket.

w. Inspect and treat electrical connector (PARA 1-7-20).

x. Connect electrical connector, P303, to exciter box.

y. Inspect and treat electrical connector (PARA 1-7-20).

z. Connect electrical connector, P302, to magnetic pickup located on bottom side of accessory drive case.

aa. Inspect and treat electrical connector (PARA 1-7-20).

ab. Connect electrical connector, P304, to low oil pressure switch located under accessory drive case.

ac. Inspect and treat electrical connector (PARA 1-7-20).

ad. Connect electrical connector, P310, to high oil temperature switch located on rear side of oil pan.

ae. Inspect and treat electrical connector (PARA 1-7-20).

af. Connect electrical connector, P308, to start bypass valve.

ag. Apply a light coat of antiseize compound, Item 74, Appendix D, to threads of thermocouple. Install thermocouple to rear of APU. **TORQUE THERMOCOUPLE TO 95 - 105 INCH-POUNDS.**

ah. Install high tension ignition cable to exciter box and igniter plug.

ai. Install APU (PARA 15-4-1).

aj. Service APU (PARA 1-3-9).

ak. Install stringer to airframe at BL 0.0 using screws.

al. Make sure area is clean and free of foreign material.

am. Check to make sure upper APU firewall seals are slightly compressed when APU compartment doors are closed and latched.

an. Close and latch APU compartment doors.

16-6-21 LITTER KIT INSTALLATION.

PARAGRAPH BREAKDOWN

		PAGE
16-6-21.1	Install Litter Kit	16-6-21-1

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 0 - 600 ft lbs

Materials/Parts

- Lockwire, Item 199, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-45

16-6-21.1 Install Litter Kit.

a. Turn off all electrical power (PARA 1-6-16).

NOTE

If six man litter configuration is used, the left seat must be removed.

b. If required, remove left seat (PARA 2-4-45).

c. Remove seats from helicopter as required (PARA 2-4-45).

d. Prepare cabin floor for litter kit installation as follows:

(1) For front and rear litter poles, mark studs for pole pintles in floor pans at STA 308.0, 344.0, and LBL 12.75 (Figure 16-6-21-1, Sheet 1, Detail A).

(2) For front litter strap tiedown quick disconnects, mark studs in floor pans at STA 308.0, RBL 11.40, and LBL 31.75 (Figure 16-6-21-1, Sheet 2, Detail B).

(3) For rear litter strap tiedown quick disconnects, mark studs in floor pans at STA 341.85, RBL 11.40, and LBL 31.75.

e. Prepare cabin ceiling for litter kit installation as follows:

(1) If required, install rear cabin ceiling support at STA 344.0 with bolts and washers from litter kit at LBL 16.50, and RBL 16.50. Do not torque bolts or lockwire now (Figure 16-6-21-1, Sheet 1, Detail A).

NOTE

Rear litter pole upper fitting stud is part of ceiling support at STA 344.0.

(2) For front and rear litter poles, mark studs for pole pintles in cabin ceiling at STA 308.0 and 344.0, LBL 12.75.

(3) For front litter strap safety hooks, mark retaining rings in cabin ceiling at STA 308.0, RBL 6.92, and LBL 32.50 (Figure 16-6-21-1, Sheet 2, Detail B).

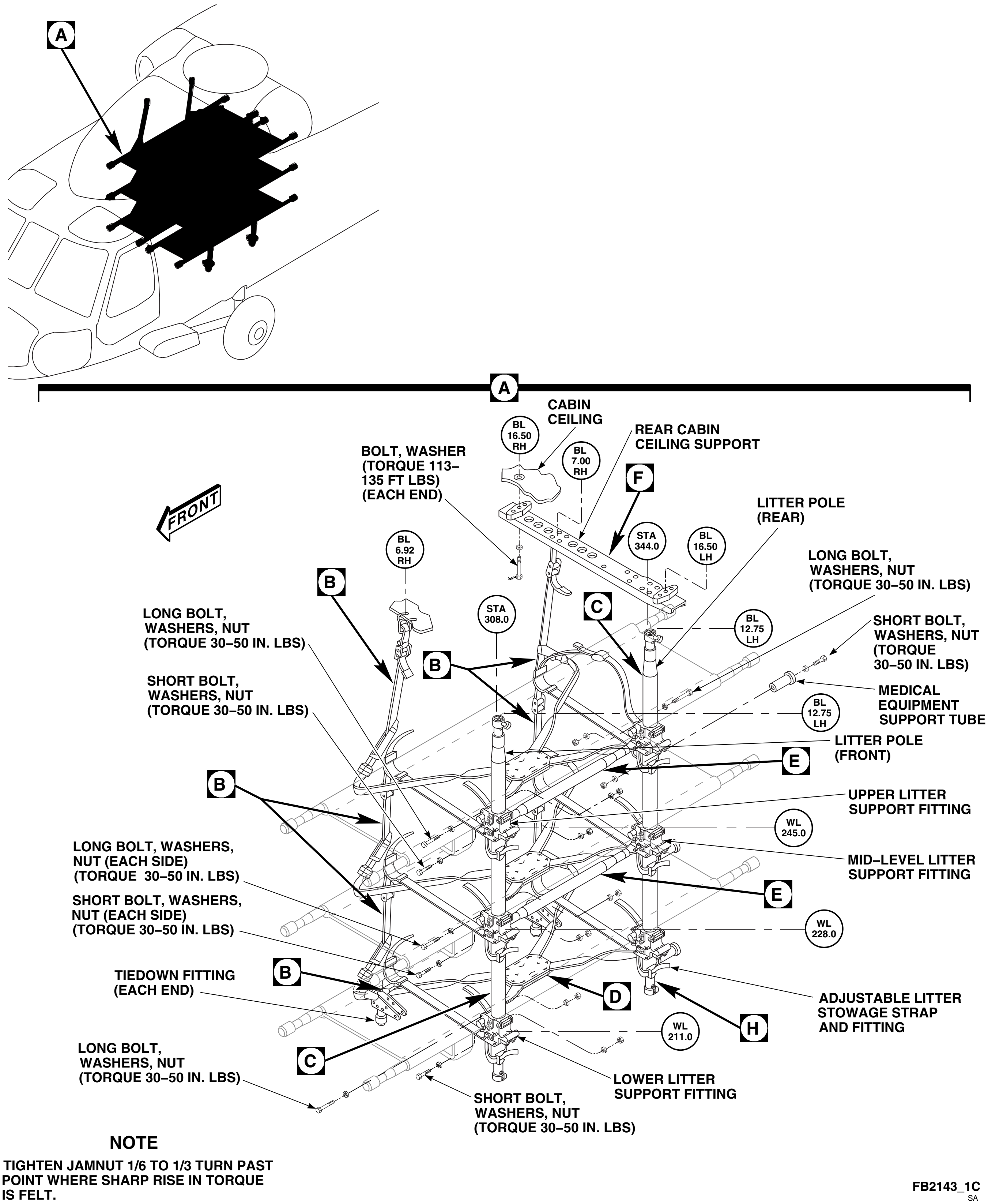


FIGURE 16-6-21-1. INSTALL SIX MAN LITTER KIT (SHEET 1 OF 5)

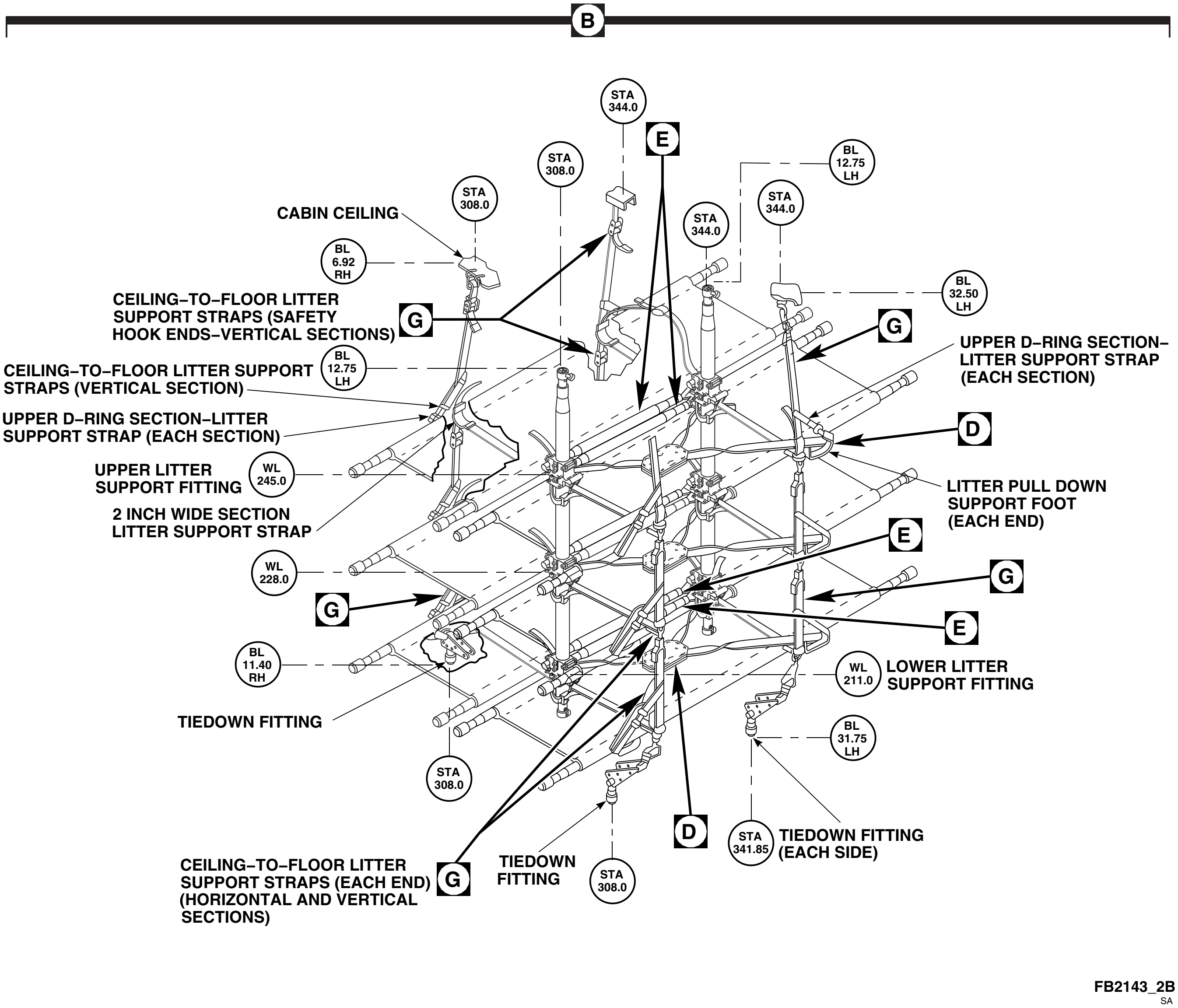


FIGURE 16-6-21-1. INSTALL SIX MAN LITTER KIT (SHEET 2 OF 5)

NOTE

Rear litter strap retaining ring at RBL 7.0 is part of ceiling support at STA 344.0.

(4) For rear litter strap safety hooks, install retaining ring in ceiling support at STA 344.0, RBL 7.00, and in cabin ceiling at STA 344.0, LBL 32.50.

(5) Verify that rear litter pole upper fitting stud in ceiling support is at LBL 12.75.

(6) **TIGHTEN BOLTS SECURING CEILING SUPPORT AT LBL 16.50, AND RBL 16.50. TORQUE BOLTS TO 113 - 135 FOOT POUNDS.** Lockwire, Item 199, Appendix D, boltheads to ceiling support (Figure 16-6-21-1, Sheet 1, Detail A).

f. Install rear litter pole in cabin as follows:

(1) Pull quick-release pin from upper pintle on litter pole and slide pintle collar downward (Figure 16-6-21-1, Sheet 3, Detail C).

(2) Pull quick-release pin from lower pintle on litter pole and lift and hold pintle collar upward.

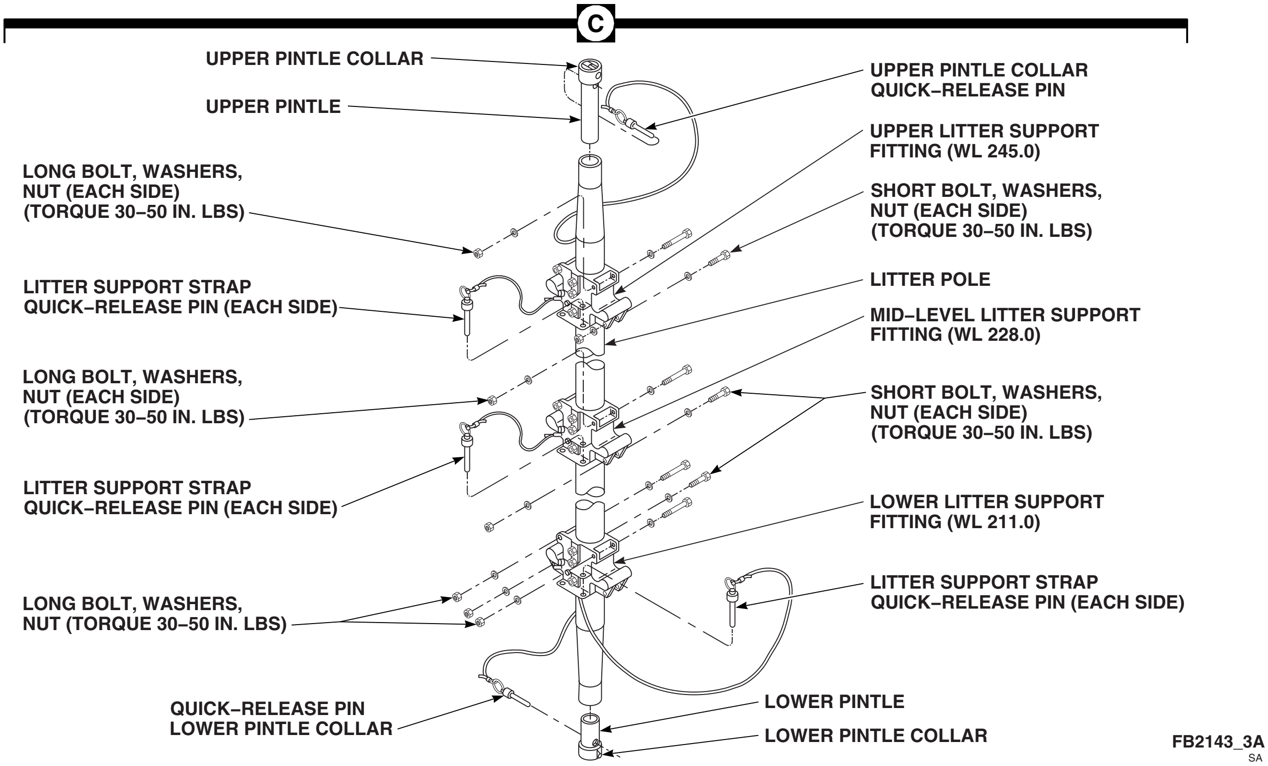


FIGURE 16-6-21-1. INSTALL SIX MAN LITTER KIT (SHEET 3 OF 5)

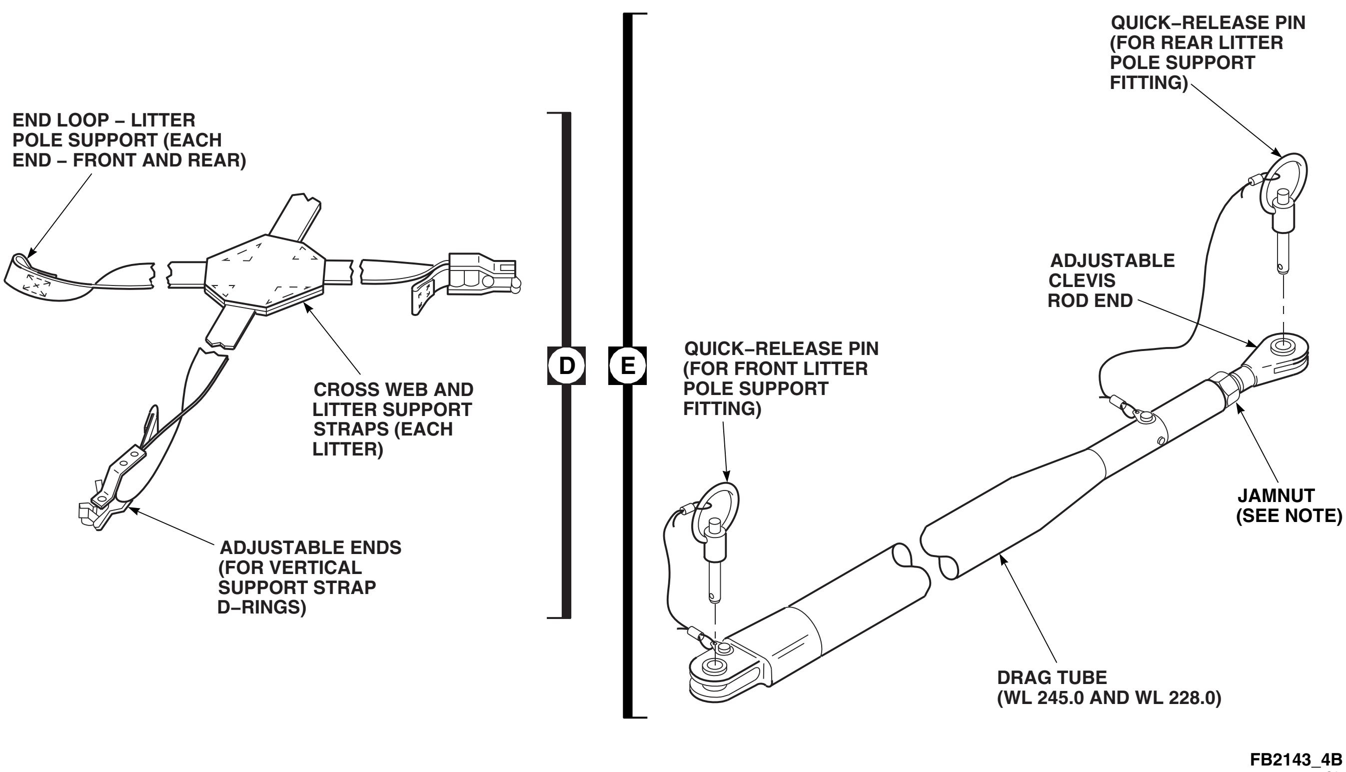


FIGURE 16-6-21-1. INSTALL SIX MAN LITTER KIT (SHEET 4 OF 5)

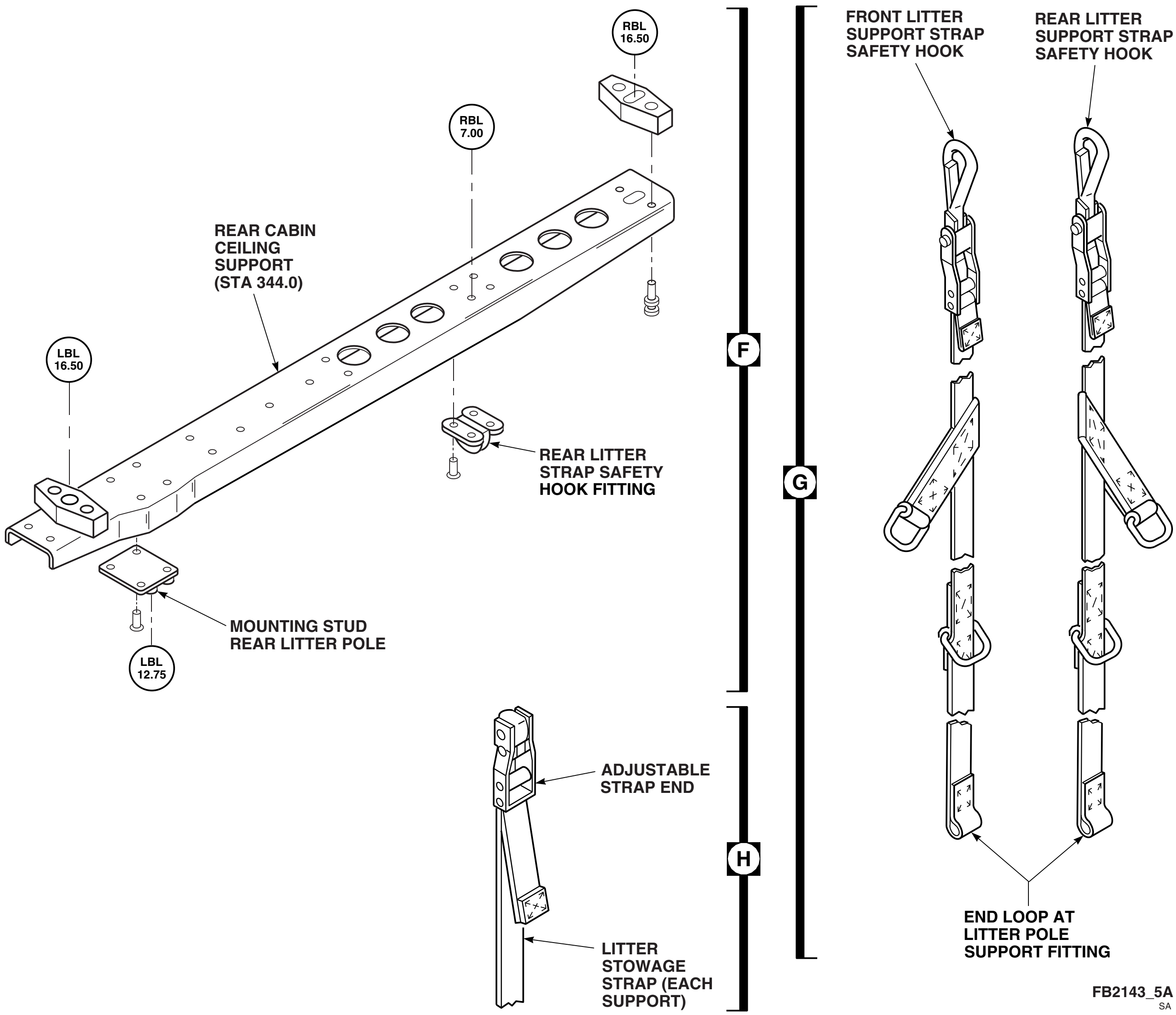


FIGURE 16-6-21-1. INSTALL SIX MAN LITTER KIT (SHEET 5 OF 5)

(3) Position lower pintle collar and rear litter pole on litter pole stud in cabin floor at STA 344.0, LBL 12.75, with quick-release pins in litter support fittings facing front (Figure 16-6-21-1, Sheet 1, Detail A and Sheet 3, Detail C).

(4) Insert quick-release pin in litter pole, just above pintle collar over floor stud support (Figure 16-6-21-1, Sheet 3, Detail C).

(5) Slide upper pintle collar on litter pole upward over litter pole stud in ceiling support at LBL 12.75. Maintain position of litter support fittings, with quick-release pins in fittings facing front, and insert quick-release pin in litter pole just below upper pintle collar in ceiling support at STA 344.0.

g. Install front litter pole in cabin as follows:

(1) Locate front litter pole upper fitting stud in cabin ceiling at STA 308.0, LBL 12.75 (Figure 16-6-21-1, Sheet 2, Detail B).

(2) Pull quick-release pin from upper pintle on remaining litter pole and slide pintle

collar downward (Figure 16-6-21-1, Sheet 3, Detail C).

(3) Pull quick-release pin from lower pintle on litter pole and lift and hold pintle collar upward.

(4) Position lower pintle collar on remaining litter pole over litter pole stud in cabin floor at STA 308.0, LBL 12.75, with quick-release pins in litter support fittings facing toward rear of helicopter (Figure 16-6-21-1, Sheet 1, Detail A and Sheet 3, Detail C).

(5) Insert quick-release pin in litter pole, just above pintle collar over floor stud support (Figure 16-6-21-1, Sheet 3, Detail C).

(6) Slide upper pintle collar on litter pole upward over litter pole stud in cabin ceiling at STA 308.0, LBL 12.75. Maintain position of litter support fittings, with quick-release pins in fittings facing toward rear of helicopter, and insert quick-release pin in litter pole just below upper pintle collar in cabin ceiling.

h. Install drag tubes between the front and rear litter pole stretcher supports as follows:

(1) Pull quick-release pin from each end of drag tube and slide clevis ends of tube over center flange bore of upper front and rear litter pole stretcher supports. Install front quick-release pin (Figure 16-6-21-1, Sheet 1, Detail A, Sheet 3, Detail C, and Sheet 4, Detail E).

NOTE

The nominal length between clevis bores of drag tubes is 31.44-inches.

(2) If rear-facing adjustable pushrod end of drag tube does not reach to center flange bore of rear litter support, loosen jamnut and adjust pushrod end of drag tube. Install rear quick-release pin.

(3) Check that quick-release pin at each end of drag tube inserts through tube ends, and center bores of front and rear litter supports. **TORQUE JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE A SHARP RISE IN TORQUE IS FELT.**

NOTE

The center flange of the upper front and rear litter pole stretcher supports should be at WL 245.0 (approximately) for proper litter mounting level.

(4) Check that upper litter mounting level is at WL 245.0 (approximately).

NOTE

The drag tube mounting flange of the front and rear litter pole support for the middle stretcher should be at WL 228.0 (approximately) to provide proper mounting level for the middle litter.

(5) Install the middle level drag tube between the front and rear litter pole stretcher support fittings. Check that middle litter mounting level is at WL 228.0 (approximately).

(6) Inspect rear-facing adjustable rod end of drag tube. If necessary, loosen jamnut and adjust rod end to reach center bore of rear litter support, and **TORQUE JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE A SHARP RISE IN TORQUE IS FELT.**

NOTE

- There is no drag tube between front and rear of lower litter support fittings.
- For proper mounting of lower litter, top of litter support quick-release pin bores should be at WL 211.0 (approximately).

(7) Inspect litter poles for quick-release pin location at WL 211.0 (approximately).

NOTE

Two quick-release pins are part of litter support fittings on litter poles. Release pins on front pole face rear of helicopter. Rear pole, at STA 344.0, was installed with fitting release pins facing front.

(8) Inspect litter support fittings for correct orientation of release pins.

i. Assemble the left side of the six-man pole-and-strap litter support configuration as follows:

(1) With keeper spring of safety hook facing inboard, clip (upper end) litter strap safety hook into retaining ring in rear cabin ceiling at STA 344.0, LBL 32.50 (Figure 16-6-21-1, Sheet 2, Detail B).

NOTE

The D-ring fitting nearest to letter strap hook is offset 45° from the vertical safety hook. D-ring section of the litter strap must trail toward rear of helicopter to pick up clip fitting at end of under-the-stretcher strap support webbing.

(2) Check that safety hook is clipped to retaining ring in rear cabin ceiling with upper D-ring section of litter strap facing toward rear of helicopter.

NOTE

In the stowed configuration, litters are installed with the inboard stretcher poles resting in the front and rear litter pole fitting grooves at WL 245.0, 228.0, and 211.0. Outboard litter poles have been folded back towards the inboard litter poles, and the outboard stretcher poles rest on top of the inboard stretcher poles. Folded stretchers are installed at litter support fittings by a 42-inch web strap having a length adjuster at one end (Figure 16-6-21-1, Sheet 5, Detail H).

(3) If litter kit has been in the three-man configuration, or has been stowed, loosen web strap length adjuster at bottom of front and rear litter pole fittings at WL 245.0, and remove 42-inch web straps and adjusters from upper litter support fittings and stretcher handles.

(4) With steel rule, locate 1/4-inch bolt (short bolt), with 1 9/16-inch grip (Figure 16-6-21-1, Sheet 1, Detail A).

(5) Locate 4-inch long medical equipment support tube. Insert bolt and washer into tube.

(6) With flange end of tube facing rear of helicopter, hold tube against lower outboard bore in upper rear litter pole fitting. Slide bolt through end loop of litter support strap clipped to ceiling at STA 344.0, and install equipment support tube and outboard litter strap to upper rear litter support fitting with bolt, washers, and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.**

(7) Check that D-ring in two-inch wide section of litter strap is facing downward and falls below the outboard stretcher edge at WL 245.0, LBL 33.0 (approximately).

(8) With steel rule, locate 1/4-inch bolt (long bolt) with 2 7/16-inch grip. Fold section of litter strap with length adjuster, inboard from two-inch wide section of strap toward upper bores in rear litter pole (STA 344.0) fitting at WL 245.0, LBL 12.75.

(9) Insert bolt with washer through rear face of upper litter pole fitting, and through end loop in one-inch wide section of litter strap with length adjuster. Install strap in upper rear pole fitting bore with bolt, washers, and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.**

j. Continue installation of the upper left litter at front end of helicopter (STA 308.0) as follows:

NOTE

Orientation of the upper D-ring section of litter support strap is different at front litter pole (STA 308.0) than at STA 344.0. The D-ring fitting nearest to litter strap safety hook is also offset 45° from the vertical safety hook. However, at STA 308.0 the D-ring section of litter strap must face toward front of helicopter to pick up clip fitting at front end of under-the-stretcher strap support webbing (Figure 16-6-21-1, Sheet 5, Detail G).

(1) Check that keeper spring of safety hook is facing inboard, and clip upper end of litter strap safety hook into retaining ring in front cabin ceiling at STA 308.0, LBL 32.50 (Figure 16-6-21-1, Sheet 2, Detail B).

(2) Check that upper D-ring section of litter support strap, nearest cabin ceiling, faces toward front of helicopter.

(3) With steel rule, locate ¼-inch (short) bolt with 1 7⁄16-inch grip. Place washer over lower outboard bore in rear face of upper litter support fitting at STA 308.0. Insert bolt in bore. Slide bolt through end loop of litter strap clipped to ceiling, and install outboard litter support strap to front litter pole fitting (lower bore) with bolt, washers, and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS** (Figure 16-6-21-1, Sheet 1, Detail A).

(4) Check that D-ring in two-inch wide section of litter strap is facing downward and falls below outboard stretcher edge at WL 245.0, LBL 33.0 (approximately).

(5) With steel rule, locate ¼-inch bolt (long bolt) with 2 7⁄16-inch grip. Fold section of litter strap with length adjuster, inboard from two-inch wide section of litter support strap toward upper bores in litter support fitting at WL 245.0, LBL 12.75.

(6) Place washer over upper outboard bore in rear face of upper litter support fitting at STA 308.0. Insert bolt in bore. Slide bolt through end loop in one-inch wide section of litter strap with length adjuster. Install strap in upper front support fitting with bolt, washers, and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.**

k. Install cross pad and litter support straps below upper left stretcher as follows:

(1) Remove left litter support strap quick-release pin from front face of upper litter support fitting at STA 344.0 (rear pole), WL 245.0 (Figure 16-6-21-1, Sheet 2, Detail B and Sheet 3, Detail C).

NOTE

- Installed front-and-rear litter strap support webbing will provide both vertical (ceiling-toward-floor) support, and horizontal (outboard stretcher handle-toward-litter pole fitting) support.
- To stabilize the stretcher against front or rearward motion during helicopter climb or descent, a "crisscross" litter strap is provided for use under each stretcher. The sewn loop ends of the crisscross webbing are fastened to the front and rear litter support fittings.
- The adjustable attachment fitting ends of the cross webbing are fed underneath and around stretcher support feet. Fitting ends are then clipped onto the 45° offset upper D-ring section of ceiling-toward-floor litter strap support webbing.

(2) At rear of cabin, install loop end of cross pad and support strap into front-facing flanges of upper litter support fitting at STA 344.0 WL 245.0, and reinsert left quick-release pin through loop end of litter support strap into fitting flange.

(3) Pull rear support foot on outboard edge of litter downward, and feed adjustable attachment fitting end of cross pad and support strap around rear edge of litter support foot, and clip fitting end onto the

rearward facing upper D-ring section (45° offset section) of the ceiling-toward-floor litter strap support webbing at STA 344.0.

(4) At front of cabin, remove left quick-release pin from rear face of upper litter support fitting at STA 308.0 (front pole), WL 245.0.

(5) Install front loop end of "crisscross" pad and support strap into rear-facing flanges of upper litter support fitting at STA 308.0, WL 245.0, and reinsert quick-release pin into top bore in fitting flange, through loop end of support strap, and through lower bore in fitting flange.

(6) At front of cabin, pull front support feet on outboard edge of stretcher downward, and feed front adjustable attachment fitting end of "crisscross" pad and support strap around front edge of support feet. Bend fitting end of support strap toward rear of helicopter, and clip fitting end onto the front facing upper D-ring section (45° offset section) of ceiling-toward-floor litter strap support webbing at STA 308.0 and adjust straps to tighten installation.

NOTE

This completes installation of upper left litter. Installation of all litters is similar, except for orientation and location of attachment fittings.

l. Install mid-level, left litter as follows:

(1) If litter kit has been in the three-man configuration, or has been stowed, loosen web strap length adjuster at bottom of front and rear litter pole fittings at WL 228.0, and remove 42-inch web straps and adjusters from mid level litter pole fittings and stretcher handles.

(2) Check that keeper springs of front and rear safety hooks are facing inboard, and clip hook ends (upper end) of litter straps are facing inboard into D-rings located below outboard edge of upper stretcher in the two-inch wide sections of upper litter support strap.

(3) Install mid-level litter support straps in same manner as upper support straps, and check that outboard level of mid-level litter stretcher handles, and litter pole fitting grooves are at WL 228.0 (Figure 16-6-21-1, Sheet 2, Detail B).

m. Install bottom level, left litter as follows:

(1) If litter kit has been in the three-man configuration, or has been stowed, loosen web strap length adjuster at bottom of front and rear litter support fittings at WL 211.0, and remove 42-inch web straps and adjusters from lower litter pole fittings and litter handles.

(2) Install bottom level litter support straps in same manner as upper support straps, and check that outboard level of bottom litter stretcher handles, and litter pole fitting grooves are at WL 211.0.

NOTE

There are four tiedown fittings in litter kit. Two rear cabin fittings secure D-rings located below left and right outboard litter edges to quick disconnect ends which fit over mounting studs in cabin floor at STA 341.85. Two front cabin fittings perform the same function at STA 308.0.

n. Install all three left litters from mounting supports in cabin ceiling, to mounting supports in cabin floor as follows:

(1) Lift up on tiedown fitting collar. Place quick disconnect collar over mounting stud in rear cabin floor pan at STA 341.85, LBL 31.75.

(2) Release tiedown fitting collar. Position opening in tiedown hook outboard, and 45° toward rear of helicopter to engage D-ring in two-inch wide section of litter support strap located below outboard left litter edge.

NOTE

Tiedown fittings have a three-position adjustable hook-and-keeper spring. Hook clips onto D-ring at the two-inch wide sections of litter support straps located below outboard litter edge. Adjustments are made by location of quick disconnect pin through appropriate fitting bores to position hook and secure D-ring to tiedown fitting.

(3) If necessary, remove quick-release pin from tiedown fitting and position hook within fitting to clip onto D-ring. Replace quick-release pin in tiedown fitting.

(4) Repeat procedure and install tiedown fitting in left cabin floor pan at STA 308.0, LBL 31.75. Position opening in tiedown hook outboard to engage D-ring located below outboard left litter edge at front of helicopter cabin.

(5) If necessary, repeat procedure for rear tiedown fitting to adjust and install hook in front cabin tiedown fitting.

(6) Clip hook to left front stretcher D-ring.

NOTE

Litters on left side of helicopter are now installed.

o. Install litters on right side of helicopter as follows:

(1) With keeper spring of safety hook facing inboard, clip (upper end) litter support strap safety hook into retaining ring in ceiling support at STA 344.0, RBL 7.0 (Figure 16-6-21-1, Sheet 1, Detail A).

(2) Repeat procedure used with left litter upper support for installation of four-inch long medical equipment support tube. Slide bolt through end loop of litter support strap clipped to ceiling support at STA 344.0, and install equipment support tube and outboard litter strap to rear litter support fitting with 1/4-inch bolt with 1 9/16-inch grip (short bolt). Bolt is installed in lower bore of upper rear litter pole fitting. **TORQUE NUT TO 30 - 50 INCH-POUNDS.**

(3) Check D-ring in two-inch wide section of litter strap for position below outboard stretcher edge at WL 245.0, RBL 11.0 (approximately).

(4) Repeat left litter upper support procedure for rear cabin installation of: end loop of right side litter strap length adjuster section at STA 344.0, and 1/4-inch bolt with 2 7/16-inch grip (long bolt). Bolt is installed in upper bore of rear litter pole fitting. **TORQUE NUT TO 30 - 50 INCH-POUNDS.**

p. With keeper spring of safety hook facing inboard, repeat left litter support procedure for front cabin installation of litter strap safety hook into retaining ring in cabin ceiling at STA 308.0, RBL 6.92.

(1) Check that upper D-ring section of litter support strap, nearest cabin ceiling, faces toward front of helicopter.

(2) Repeat procedure used with left litter upper support for installation of end loop of right side litter support strap clipped to ceiling support at STA 308.0, and 1/4-inch bolt with 1 7/16-inch grip. Bolt is installed in lower bore of front litter pole fitting. **TORQUE NUT TO 30 - 50 INCH-POUNDS.**

(3) Check D-ring in two-inch wide section of litter strap for position below outboard stretcher edge at WL 245.0, RBL 11.0 (approximately).

(4) Repeat left litter upper support procedure for front cabin installation of end loop of right side upper litter strap length adjuster section at STA 308.0, and 1/4-inch bolt with 2 7/16-inch grip (long bolt). Bolt is installed in upper bore of front litter support fitting at WL 245.0. **TORQUE NUT TO 30 - 50 INCH-POUNDS.**

q. Install cross pad and support straps below upper right litter (WL 245.0) as follows:

(1) Remove right quick-release pin from front face of upper litter support fitting at STA 344.0 (rear pole), WL 245.0 (Figure 16-6-21-1 Sheet 2, Detail B and Sheet 3, Detail C).

(2) At rear of cabin, install loop end of cross pad and support strap into front-facing flanges of upper litter support fitting at STA 344.0 WL 245.0, and reinsert right quick-release pin through loop end of strap into fitting flange.

(3) Pull rear support feet on outboard edge of right side litter downward, and feed adjustable attachment fitting end of cross pad and support strap around rear edge of support feet, and clip fitting end onto the rearward facing upper D-ring section (45° offset section) of the ceiling-toward-floor litter strap support webbing at STA 344.0.

(4) At front of cabin, remove right quick-release pin from rear face of upper litter support fitting at STA 308.0 (front pole), WL 245.0.

(5) Install front loop end of right side "crisscross" pad and support strap into rear-facing flanges of upper litter pole fitting at STA 308.0, WL 245.0, and reinsert quick-release pin into top bore in fitting flange, through loop end of support strap, and through lower bore in fitting flange.

(6) At front of cabin, pull front support feet on outboard edge of right stretcher downward, and feed front adjustable attachment fitting end of "crisscross" pad and support strap around front edge of litter support feet. Bend fitting end of support strap toward rear of helicopter, and clip fitting end onto the front facing upper D-ring section (45° offset section) of ceiling-toward-floor litter strap support webbing at STA 308.0, WL 245.0.

NOTE

This completes installation of upper right litter. Installation of all litters is similar, except for orientation and location of attachment fittings.

r. Install mid-level, right side litter as follows:

(1) If litter kit has been stowed, loosen web strap length adjuster at bottom of front and rear litter pole fittings at WL 211.0, and remove 42-inch web strap and adjusters from mid-level pole fittings and litter handles.

(2) Check that keeper springs of front and rear safety hooks are facing inboard, and clip hook ends (upper end) of litter straps are facing inboard into D-rings located below outboard edge of upper stretcher in the two-inch wide sections of upper litter support strap.

(3) Install mid-level litter support straps in same manner as upper support straps, and check that outboard level of mid-level litter stretcher handles, and litter pole fitting grooves are at WL 211.0 (Figure 16-6-21-1, Sheet 2, Detail B).

NOTE

This completes installation of upper right litter, installation of all litters is similar, except for orientation and location of attachment fittings.

s. Install bottom level, right litter as follows:

(1) If litter kit has been stowed, loosen web strap length adjuster at bottom of front and rear litter pole fittings at WL, 211.0, and remove 42-inch web strap and adjusters from mid-level pole fittings and litter handles.

(2) Install bottom level right litter support straps in same manner as upper support straps, and check that outboard level of bottom litter stretcher handles, and litter pole fitting grooves are at WL 211.0.

NOTE

Two tiedown fittings mount on front cabin and rear cabin D-rings located below right outboard litter edge to quick disconnect ends. The rear cabin fitting end fits over mounting stud in cabin floor at STA 341.85. Front cabin fitting end fits over mounting stud in floor at STA 308.0.

t. Install all three right litters from mounting supports in cabin ceiling, to mounting supports in cabin floor as follows:

(1) Lift up on tiedown fitting collar. Place quick-disconnect collar over mounting stud in rear cabin floor pan at STA 341.85, RBL 11.40.

(2) Release tiedown fitting collar. Position opening in tiedown hook inboard, and 45° toward rear of helicopter to engage D-ring in two-inch wide section of litter support strap located below outboard right litter edge.

NOTE

Tiedown fittings have three-position adjustable hook-and-keeper spring. Hook clips onto D-ring in the two-inch wide sections of litter support straps located below outboard stretcher edge. Adjustments are made by location of quick-disconnect pin through appropriate fitting bores to tiedown fitting.

(3) If necessary, remove quick-release pin from tiedown fitting and position hook within fitting to clip onto D-ring. Replace quick-release pin in tiedown fitting.

(4) Repeat procedure and install tiedown fitting in front cabin floor pan at STA 308.0, RBL 11.40. Position opening in tiedown hook inboard to engage D-ring located below outboard right litter edge at front of cabin.

(5) If necessary repeat procedure to adjust hook. Then, install hook in front cabin tiedown fitting.

NOTE

Litters on right side of helicopter are now installed, and six man litter kit installation is complete.

(6) Clip hook to right front lower stretcher D-ring.

16-6-22 LITTER KIT REMOVAL.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 1-6-16
- PARA 2-4-45

16-6-22.1 Remove Litter Kit.

a. Turn off all electrical power (PARA 1-6-16).

b. To remove litter kit from left side cabin floor, perform the following:

NOTE

To remove drag tube assemblies, litter pole assemblies, and rear cabin ceiling litter support, litter belts and support strap webbing must be removed.

(1) Lift up on quick-disconnect collar of litter strap tiedown fitting over mounting pad stud in rear cabin floor pan at STA 341.85, LBL 31.75 (Figure 16-6-21-1, Sheet 2, Detail B).

(2) Lift tiedown fitting from mounting stud, and disengage tiedown hook from D-ring in two-inch wide section of litter support strap (Figure 16-6-21-1, Sheet 2, Detail B and Sheet 5, Detail G) located below lower outboard left stretcher edge.

(3) If necessary, remove quick-release pin from tiedown fitting to unclip D-ring. Replace quick-release pin in tiedown fitting.

(4) Repeat procedure and remove tiedown fitting in front cabin floor pan at STA 308.0, LBL 31.75. Position opening in tiedown hook as required to unclip from D-ring located below outboard left lower stretcher edge at front of helicopter cabin. Replace quick-release pin in tiedown fitting.

c. To remove ″crisscross″ pad and support straps at lower (WL 211.0) left stretcher, perform the following:

(1) At outboard side of litter, unclip adjustable attachment fitting end of cross pad and support strap around rear edge of support feet from rear facing upper D-ring section (45° offset section) of the ceiling-toward-floor litter strap support webbing at STA 344.0, WL 211.0 (Figure 16-6-21-1, Sheet 2, Detail B, and Sheet 4, Detail D).

(2) Repeat procedure to unclip fitting end of cross pad and support strap from front facing D-ring section of ceiling-toward-floor support webbing at STA 308.0 (right cabin), WL 211.0.

(3) Fold up front and rear support feet on outboard edge of lower litter, and remove litter from support fitting.

(4) To remove inboard ends of cross pad and support strap, remove left quick-release

pin from front face of lower litter pole fitting at STA 344.0 (rear pole), WL 211.0 (Figure 16-6-21-1, Sheet 2, Detail B and Sheet 3, Detail C).

(5) At front cabin, repeat procedure and remove left quick-release pin from rear face of lower litter support fitting at STA 308.0, WL 211.0.

NOTE

Cross pad and support straps below lower left litter are now free, and attachment flanges on outboard sides of litter poles are accessible for removal of loop ends of vertical-horizontal litter support straps at each end of litter.

d. Replace quick-release pins in front, and in rear lower litter support fittings. Stow cross pad and support straps.

e. To remove lower inboard loop end of vertical-horizontal litter support strap at front of cabin (Figure 16-6-21-1, Sheet 5, Detail G), perform the following:

(1) Remove bolt, washers, and nut from lower outboard, left side flange bores of bottom litter pole fitting at LBL 12.75, WL 211.0, STA 308.0, and remove loop end of left litter support strap (Figure 16-6-21-1, Sheet 1, Detail A).

(2) Unclip safety hook at opposite end of litter support strap from D-ring located below outboard edge of mid-level stretcher in two-inch wide section of mid-level floor-to-ceiling litter support strap.

f. To remove (upper) left side loop end of length adjuster section of bottom litter floor-to-ceiling support strap at front of cabin, perform the following:

(1) Remove bolt, washers, and nut from upper outboard, left side bores in bottom litter support fitting at LBL 12.75, WL 211.0, STA 308.0.

(2) Remove upper loop end of length adjuster section of bottom litter support strap.

NOTE

Lower section of ceiling-to-floor litter support strap at front cabin is now free.

g. Replace bolts, washers, and nuts in upper and lower left side litter pole support bores at front cabin (STA 308.0). Stow front cabin, lower section, ceiling-to-floor litter support strap.

h. To remove lower loop end of vertical-horizontal litter support strap at rear of cabin, perform the following:

(1) Remove bolt, medical equipment support tube, washers, and nut from lower outboard left side flange bores of bottom litter pole fitting at LBL 12.75, WL 211.0, STA 308.0, and remove loop end of left side litter support strap.

(2) Unclip safety hook at opposite end of litter support strap from D-ring located below outboard edge of mid-level stretcher in two-inch wide section of mid-level floor-to-ceiling litter support strap.

i. Remove bolt, washers, and nut from upper outboard, left side bores in bottom litter support fitting at LBL 12.75, WL 211.0, STA 308.0, and remove upper loop end of length adjuster section of bottom litter support strap.

NOTE

Lower section of ceiling-to-floor litter support strap in rear cabin is now free.

j. Replace lower bore medical equipment support tube, and bolts, washers, and nuts in upper and lower left side litter pole support bores in rear cabin (STA 344.0). Stow rear cabin, lower section, ceiling-to-floor litter support strap.

k. To remove ″crisscross″ pad and support straps below mid-level (WL 228.0) left stretcher, repeat procedures for bottom-level stretcher. Stow cross pad and support straps.

l. To remove left side vertical-horizontal litter support straps, and length adjuster section of litter support straps, at front and rear cabin locations for mid-level (WL 228.0) left litter, repeat procedures for bottom-level stretcher. Stow ceiling-to-floor litter support straps.

m. To remove ″crisscross″ pad and support straps below upper (WL 245.0) left litter, repeat procedures for bottom-level stretcher. Stow cross pad and support straps.

n. To remove left side vertical-horizontal litter support straps, and length adjuster section of litter support straps, at front and rear cabin locations for upper (WL 245.0) left litter, repeat procedures for removal of end loops of support straps for bottom level litter.

o. Unclip adjustable safety hook end of front and rear left side litter support straps from ceiling supports at STA 308.0 and 344.0, LBL 32.50. Stow upper sections of ceiling-to-floor litter support straps.

NOTE

Left side of litter kit has now been removed.

p. To remove right side of litter kit from cabin floor, perform the following:

(1) Lift up on quick disconnect collar of litter strap tie down fitting over mounting stud in rear cabin floor pan at STA 341.85, RBL 11.40 (Figure 16-6-21-1, Sheet 2, Detail B).

(2) Lift tiedown fitting from mounting stud, and disengage tiedown hook from D-ring in 2-inch wide section of litter support strap located below lower outboard right stretcher edge.

(3) If necessary, remove quick-release pin from tiedown fitting to unclip D-ring. Replace quick-release pin in tiedown fitting.

(4) Repeat procedure and remove tiedown fitting in front cabin floor pan at STA 308.0, RBL 11.40. Position opening in tiedown hook as required to unclip from D-ring located below outboard right lower stretcher edge at front of helicopter cabin. Replace quick-release pin in tiedown fitting.

q. To remove ″crisscross″ pad and support straps below lower (WL 211.0) right stretcher, repeat procedure for lower left litter at STA 344.0 (rear cabin) and 308.0 (front cabin), WL 211.0. Stow cross pad and support straps.

r. To remove lower inboard loop ends of vertical-horizontal litter support straps at front and rear litter pole supports, repeat procedure for bottom left litter at STA 344.0 (right rear cabin) and 308.0 (right front cabin), WL 211.0.

s. To remove (upper) right side loop end of length adjuster section of bottom litter floor-to-ceiling support straps at front and rear litter pole supports, repeat procedure for bottom left stretcher at STA 344.0 (right rear cabin), and 308.0 (right front cabin), WL 211.0.

NOTE

Lower section of ceiling-to-floor litter support straps at front and rear of right side cabin are now free, and support straps and mounting hardware are stowed.

t. Remove mid-level (WL 228.0) right side litter in same manner as mid-level left litter.

u. Remove upper-level (WL 245.0) right side cross pad and support straps by repeating procedure for left litter.

v. Stow cross pad and support straps.

w. To remove loop ends of right side vertical-horizontal litter support straps, and length adjuster section of litter support straps at front and rear cabin locations for upper (WL 245.0) right litter, repeat procedures for removal of end loops of support straps for bottom left litter.

x. Unclip adjustable safety hook end of rear right support strap from ceiling support at STA 344.0, RBL 7.0.

y. Unclip safety hook end of right front side litter support strap from front cabin ceiling fitting at STA 308.0, RBL 6.92.

NOTE

Upper sections of ceiling-to-floor litter support straps at front and rear of right side cabin are now free.

z. Stow support straps and mounting hardware.

aa. Right side of litter kit has now been removed, to remove drag tube assemblies between the front and rear litter pole stretcher supports, perform the following:

NOTE

It may be necessary to loosen jamnut, at adjustable pushrod clevis end of drag tube at STA 344.0, to withdraw quick-release pin from litter pole fitting and clevis bores.

(1) If necessary, loosen jamnut on adjustable clevis end of upper drag tube at rear of helicopter cabin. Pull quick-release pin from each end of upper drag tube and slide each end of tube away from center flange bores of upper front and rear litter pole stretcher supports (Figure 16-6-21-1, Sheet 4, Detail E).

(2) Stow quick-release pin in bores at each end of upper litter drag tube, and stow drag tube assembly.

(3) Repeat procedure, and remove drag tube at mid-level litter position, WL 228.0. Stow quick-release pins in drag tube, and stow drag tube assembly.

NOTE

Rear litter pole upper fitting stud is part of ceiling support at STA 344.0.

ab. To remove rear cabin litter pole from ceiling support at STA 344.0, pull quick-release pin from top pintle on litter pole and slide pintle collar downward.

ac. Pull quick-release pin from lower pintle on litter pole and slide pintle collar upward from cabin floor.

ad. Remove rear cabin litter pole from mounting stud in ceiling support, and mounting stud in cabin floor pan.

ae. Slide pintle collars away from center of rear cabin litter pole, and stow quick-release pin in bores at each end of litter pole below pintle collars.

af. To remove front cabin litter pole at STA 308.0, repeat procedure performed for rear cabin litter pole.

ag. Stow quick-release pin in bores at each end of front cabin litter pole.

ah. To remove ceiling support assembly at rear cabin, STA 344.0, remove bolts and washers from ceiling support at RBL 16.50, and LBL 16.50 (Figure 16-6-21-1, Sheet 5, Detail F).

ai. Remove ceiling support tube, and preserve and store litter kit as required.

aj. If required, install passenger's seats (PARA 2-4-45).